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Christos A. Athanasiadis

Discrete Mathematics: A Combinatorial Approach

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Christos A. Athanasiadis

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Dedicated to the memory of the teacher
Gabriel Andreas Athanasiadis
(1890, Tsapik, district of Kagizman, Kars Province – 1972, Kilkis, Macedonia)

Preface

This book is intended as an undergraduate text for a semester-long first course in discrete mathematics, as well as a source of independent or self study for anyone interested in this subject. The main objectives are to familiarize the readers with structures of combinatorial and discrete mathematics which will be useful in their further study of mathematics and its applications and to make them appreciate and absorb the corresponding methods and ideas and develop their own combinatorial skills.

There is a minimum of prerequisites for this book, a fact which makes it accessible to a broad spectrum of readers with an interest in mathematics. However, some prior experience with mathematical proofs will be useful. Effort has been made that the exposition is clear and easy to follow, but free of unnecessary details and stories. The material covered is comparable to that encountered in introductory undergraduate mathematical courses with title "Discrete Mathematics," or "Combinatorics and Graph Theory" and, inevitably, reflects the author's personal preferences and interests as well. There is some emphasis on the fundamentals of enumeration and generating functions, so that readers can develop good basic skills on these important topics.

The first chapter is a review primarily of the proof techniques, as well as the background, with which the reader must be comfortable in order to understand the following chapters. The second chapter is a brief introduction to enumerative combinatorics. The basic principles of enumeration studied there serve as tools which are applied throughout the book. As a small novelty, the multiplication principle is not only expressed in popularized form but also in precise mathematical language, so that the reader understands to which exactly mathematical statement it corresponds and can learn to apply it correctly. Some emphasis is given to the proof of combinatorial identities and the computation of combinatorial sums, problems that are encountered in a broad spectrum of contemporary mathematics. Because of its importance, the principle of inclusion-exclusion is studied in a separate (the third) chapter. The following two chapters focus on some fundamental concepts and structures of discrete mathematics, such as set partitions, partial orders and graphs, and provide applications to problems of extremal combinatorics,

problems of combinatorial constructions, and elsewhere. The sixth chapter is an introduction to a prominent tool of enumerative combinatorics, whose importance is evident in several areas within undergraduate mathematics, the method of generating functions. The last chapter introduces the basics of the theory of discrete probability, relying upon and extending the material of previous chapters.

An integral part of the book are the exercises given at the end of each chapter, the aims and origins of which vary. Most of them aim at the understanding of the material at a good level and at the reader's active participation in the mathematical process. Some are parts of theory given in the form of an exercise, so that on the one hand readers who wish to do so can broaden and deepen their knowledge of the subject, and on the other the length of the book stays at a reasonable level. The instructor who will consult this book may choose to incorporate some of the exercises into the lectures, depending on the circumstances and the goals of the course. About the origin, some of the exercises have appeared before in excellent textbooks (such as Brualdi [6], Matousek–Nesetril [16], Stanley [23]) or problem collections (such as Lovász [13], Tomescu [28], Sevastyanov–Christyakov–Zubkov [21]), others are suitably modified problems that appeared in mathematical competitions or results from research articles and many of them are due to the author. The reader is strongly encouraged to try to find their own solutions, beginning with the simpler problems and working toward the more difficult, before they consult any hints or solutions provided.

Acknowledgments The book grew out of lecture notes that have been used by the author to teach discrete mathematics in the University of Athens and the University of Crete since 2004; it was first published in Greek in 2023 by Efaltirio Press. The present English version comes along with a few additions and adjustments. The author wishes to thank his colleague Dimitris Cheliotis for useful technical advice on Chapter 7, the anonymous reviewers for their valuable comments, and Elizabeth Loew, Executive Editor of Springer in Mathematics, whose work and encouragement made publication of this book possible.

Athens, Greece
October 2025

Christos A. Athanasiadis

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Chapter 1
Introduction



This chapter serves as a short introduction to some fundamental mathematical notions from elementary arithmetic and set theory and methods of proof, as essential preparation for the following chapters. An emphasis is given on the method of mathematical induction and concepts such as that of a map between sets, of a recursively defined sequence, and of divisibility in the set of natural numbers.

For extensive introductions to these topics, we recommend [17, 27]. We begin by explaining notation, which will be followed, and prerequisites, which are assumed.

Notation and Preliminaries
We will use the following notation:

$\varnothing$: the empty set
$\mathbb{N} = \{0, 1, 2, \dots\}$: the set of natural numbers
$\mathbb{Z}$: the ring of integers
$\mathbb{Z}_{>0} = \{1, 2, \dots\}$: the set of positive integers
$\mathbb{Q}$: the field of rational numbers
$\mathbb{R}$: the field of real numbers
$\mathbb{C}$: the field of complex numbers
$[n]$: the set $\{1, 2, \dots, n\}$,

where $[0] := \varnothing$.

We assume familiarity with basic notions of set theory, such as those of a set and its subsets and those of the union, the intersection, and the cartesian product of two sets. We write $x \in X$ if x belongs to the set X and $x \notin X$ otherwise, $X \subseteq Y$ if the set X is contained in the set Y (in which case X is a subset of Y) and $X \subset Y$ when

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$X \subseteq Y$ and equality is not an option (in which case X is a proper subset of Y). We denote by

$$X \cup Y = \{z : z \in X \text{ or } z \in Y\}$$
$$X \cap Y = \{z : z \in X \text{ and } z \in Y\}$$

the union and the intersection of the sets X and Y, respectively, with $X \setminus Y$ or $X - Y$ the difference $\{x \in X : x \notin Y\}$ of Y from X, with 2^X the set of all subsets (powerset) of the set X and with

$$X \times Y = \{(x, y) : x \in X,\ y \in Y\}$$

the cartesian product of the sets X and Y.

We also assume familiarity with basic properties of the ordered field of real numbers, the field of complex numbers and the ring of polynomials in one variable with integer, real or complex coefficients (some background on algebra is reviewed in Appendix A, for the convenience of the reader). Occasionally, a basic knowledge of calculus will be helpful.

1.1 Natural Numbers

This section reviews some basic properties of natural numbers, giving an emphasis on the method of mathematical induction. For a rigorous treatment of the construction of natural numbers (which is not an essential requirement for this book), the reader should consult the relevant bibliography (for instance, [10, Chapter 2] [17, Chapetr 5] [27, Chapter 8]).

1.1.1 Mathematical Induction

The principle of finite mathematical induction, which we state in two forms, is equivalent to the well-ordering principle for the natural numbers. It constitutes the main method of proof used in this book and, more generally, an especially important and useful method of proof in mathematics.

Proposition 1.1.1. *Let* $S \subseteq \mathbb{N}$.

(a) (Principle of finite Mathematical Induction, first form) *If*

- $0 \in S$ *and*
- $k \in S \Rightarrow k + 1 \in S$,

then $S = \mathbb{N}$.

(b) (Principle of finite Mathematical Induction, second form) *If*

- $0 \in S$ *and*
- $0, 1, \ldots, k \in S \Rightarrow k+1 \in S$,

then $S = \mathbb{N}$.

(c) (Well-ordering Principle) *If* $S \neq \varnothing$, *then* S *has a minimum element, that is there exists* $a \in S$ *such that* $a \leq x$ *for every* $x \in S$.

Before describing applications of Proposition 1.1.1, let us introduce some notation. The sum and the product of the complex numbers (more generally, the sum and the product of the elements of a ring with a unit, such as polynomials; see Section A.1) $a_1, a_2, \ldots, a_n$ is defined recursively by the equalities

$$a_1 + a_2 + \cdots + a_n = \begin{cases} 0, & \text{if } n = 0 \\ (a_1 + a_2 + \cdots + a_{n-1}) + a_n, & \text{if } n \geq 1 \end{cases} \tag{1.1}$$

and

$$a_1 a_2 \cdots a_n = \begin{cases} 1, & \text{if } n = 0 \\ (a_1 a_2 \cdots a_{n-1}) \cdot a_n, & \text{if } n \geq 1, \end{cases} \tag{1.2}$$

respectively (for a brief discussion of recursions, see Section 1.1.2). These definitions suggest that mathematical induction is a suitable method for proving statements about sums and products of several numbers. We will frequently use the notation

$$\sum_{i=1}^{n} a_i = a_1 + a_2 + \cdots + a_n \quad \text{and} \quad \prod_{i=1}^{n} a_i = a_1 a_2 \cdots a_n.$$

For example,

$$\sum_{i=1}^{n} i(i+1) = 1 \cdot 2 + 2 \cdot 3 + \cdots + n(n+1) \quad \text{and} \quad \prod_{i=1}^{n} i^2 = 1^2 \cdot 2^2 \cdots n^2$$

for every $n \in \mathbb{N}$. Two basic properties of the sum are given in Exercise 1.

Example 1.1.1. Let us prove that

$$(1 + a_1)(1 + a_2) \cdots (1 + a_n) \geq 1 + (a_1 + a_2 + \cdots + a_n) \tag{1.3}$$

for nonnegative real numbers $a_1, a_2, \ldots, a_n$ (this statement is known as *Weierstrass' inequality*).

We will apply the principle of mathematical induction in the form of Proposition 1.1.1 (a). Let us denote by S the set of numbers $n \in \mathbb{N}$ for which (1.3) is valid

for all real numbers $a_1, a_2, \ldots, a_n \geq 0$. Since (1.3) holds as an equality for $n = 0$ and $n = 1$, we have $0 \in S$ and $1 \in S$. We assume that $k \in S$ and consider numbers $a_1, a_2, \ldots, a_{k+1} \geq 0$. By our assumption,

$$\prod_{i=1}^{k}(1 + a_i) \geq 1 + a_1 + a_2 + \cdots + a_k.$$

Multiplying both sides of this inequality with $1 + a_{k+1}$, we find that

$$\begin{aligned}\prod_{i=1}^{k+1}(1 + a_i) &= \prod_{i=1}^{k}(1 + a_i) \cdot (1 + a_{k+1}) \\ &\geq (1 + a_1 + a_2 + \cdots + a_k)(1 + a_{k+1}) \\ &= 1 + a_1 + a_2 + \cdots + a_k + a_{k+1} + (a_1 + a_2 + \cdots + a_k)a_{k+1} \\ &\geq 1 + (a_1 + a_2 + \cdots + a_{k+1})\end{aligned}$$

and conclude that the statement holds for $n = k + 1$ as well, that is, $k + 1 \in S$. According to Proposition 1.1.1 (a) we must have $S = \mathbb{N}$, meaning that the statement holds for every $n \in \mathbb{N}$. □

REMARK. A close examination of the previous proof shows that (1.3) is also valid for real numbers $a_1, a_2, \ldots, a_n$ with $-1 \leq a_i \leq 0$ for every $i \in [n]$.

Example 1.1.2. Let us prove the identity

$$a^n - 1 = (a - 1)(a^{n-1} + a^{n-2} + \cdots + a + 1) \tag{1.4}$$

for every positive integer n and every $a \in \mathbb{C}$ (our proof will be valid for any element a of a ring with a unit, such as a polynomial or a square matrix; see Section A.1).

We note that (1.4) is trivial for $n = 1$ and assume that it holds for $n = k \in \mathbb{Z}_{>0}$, meaning that

$$a^k - 1 = (a - 1)(a^{k-1} + a^{k-2} + \cdots + a + 1)$$

for every $a \in \mathbb{C}$. Adding $a^{k+1} - a^k$ to both sides of this equality, we find that

$$\begin{aligned}a^{k+1} - 1 &= a^{k+1} - a^k + a^k - 1 \\ &= a^{k+1} - a^k + (a - 1)(a^{k-1} + a^{k-2} + \cdots + a + 1) \\ &= (a - 1)a^k + (a - 1)(a^{k-1} + a^{k-2} + \cdots + a + 1) \\ &= (a - 1)(a^k + a^{k-1} + \cdots + a + 1).\end{aligned}$$

Therefore, (1.4) holds for $n = k + 1$ as well.

Let now S denote the set of $n \in \mathbb{N}$ for which (1.4) holds for $n+1$ and every $a \in \mathbb{C}$. We have shown that $0 \in S$ and that $k \in S \Rightarrow k+1 \in S$. According to Proposition 1.1.1 (a) we must have $S = \mathbb{N}$, which means that (1.4) holds for every integer $n \geq 1$ and every $a \in \mathbb{C}$. □

As suggested by the previous examples, in order to prove by the method of mathematical induction that every natural number has a certain property, two steps are required. The first (usually the easiest) step verifies that zero has the desired property, in other words that $0 \in S$, where S is the set of natural numbers that have the desired property. The second step, called the *inductive step*, proves the implication $k \in S \Rightarrow k+1 \in S$, or that $0, 1, \ldots, k \in S \Rightarrow k+1 \in S$. That is, assuming that k, or that each one of $0, 1, \ldots, k$, has the desired property (this is the induction hypothesis), one shows that $k+1$ has the same property. We note that the first step cannot be omitted. For example, if S is the set of numbers $n \in \mathbb{N}$ such that $n = n+1$, then one certainly has that $k \in S \Rightarrow k+1 \in S$. Thus, omitting the first step from the proof leads one to the erroneous conclusion that $n = n+1$ for every $n \in \mathbb{N}$.

As the reader may have noticed, in Example 1.1.2, we showed by induction that some statement holds for every integer $n \geq 1$, rather than for every integer $n \geq 0$. The next proposition follows by applying Proposition 1.1.1 (a) to the set $\{n \in \mathbb{N} : n_0 + n \in S\}$.

Proposition 1.1.2 (Principle of Finite Mathematical Induction, Third Form). *Let $n_0 \in \mathbb{Z}$ and $S \subseteq \mathbb{Z}$ be such that*

- *$n_0 \in S$ and*
- *$k \in S \Rightarrow k+1 \in S$.*

Then, $\{n_0, n_0+1, n_0+2, \ldots\} \subseteq S$; that is, $n \in S$ for every $n \in \mathbb{Z}$ with $n \geq n_0$.

Part (b) of Proposition 1.1.1 can be generalized in a similar way. In the sequel, we will apply these generalizations without special reference. Besides, we will omit the definition of the set S, unless this is not clear from the context.

Example 1.1.3. Let us show that if each one of the integers $p_1, p_2, \ldots, p_n$ can be written as the sum of the squares of two integers, then so does their product $p_1 p_2 \cdots p_n$. For instance, since $2 = 1+1$, $5 = 1+4$, and $13 = 4+9$, the number $2^m \cdot 5^p \cdot 13^q$ can be written as the sum of the squares of two integers for all $m, p, q \in \mathbb{N}$.

The statement we wish to show is trivial for $n \in \{0, 1\}$. Let's assume that it holds for $n = k \geq 1$, and let's consider integers $p_1, p_2, \ldots, p_{k+1}$, every one of which can be written as the sum of the squares of two integers. By our hypotheses, we have $p_1 p_2 \cdots p_k = a^2 + b^2$ and $p_{k+1} = x^2 + y^2$ for some $a, b, x, y \in \mathbb{Z}$. These facts and the identity

$$(a^2+b^2)(x^2+y^2) = (ax+by)^2 + (ay-bx)^2$$

imply that $p_1p_2\cdots p_{k+1} = (p_1p_2\cdots p_k)\cdot p_{k+1}$ can also be written as the sum of the squares of two integers, namely $ax+by$ and $ay-bx$. Hence, the statement holds for $n = k+1$ as well. According to Proposition 1.1.1 (a), it must hold for every $n \in \mathbb{N}$. □

Example 1.1.4. Let's find all positive integers n such that $2^{n-1} \geq n^2$. One expects that this inequality will be valid for large enough n. Thus, by successive trials we find that it holds for $n = 1$ and $n = 7$, but for no $n \in \{2, 3, 4, 5, 6\}$. Moreover, assuming that it holds for $k \geq 7$, we find that

$$2^k = 2\cdot 2^{k-1} \geq 2k^2 = k^2 + k^2 \geq k^2 + 7k > k^2 + 2k + 1 = (k+1)^2$$

and hence, it holds for $k+1$ as well.

Proposition 1.1.2, applied for $n_0 = 7$, shows that the inequality holds for every integer $n \geq 7$. As a conclusion, for positive integers n we have $2^{n-1} \geq n^2$ if and only if $n = 1$ or $n \geq 7$. □

Example 1.1.5. Let x, y be positive real numbers such that $x + y \geq 1$. We will show that

$$x^n + y^n \geq \frac{1}{2^{n-1}} \tag{1.5}$$

for every $n \in \mathbb{N}$.

We will apply the principle of mathematical induction, in the form of Proposition 1.1.1 (a). We first note that (1.5) holds as an equality for $n = 0$. Let us assume that the claimed statement holds for $n = k \in \mathbb{N}$ and consider positive real numbers x, y such that $x + y \geq 1$. Because of our assumption, we have $x^k + y^k \geq 1/2^{k-1}$. We multiply this inequality with $x + y \geq 1$ and find that

$$(x+y)(x^k + y^k) \geq \frac{1}{2^{k-1}}.$$

To conclude that $x^{k+1} + y^{k+1} \geq 1/2^k$, it suffices to show that

$$2(x^{k+1} + y^{k+1}) \geq (x+y)(x^k + y^k).$$

Indeed, the latter inequality can be rewritten as $(x-y)(x^k - y^k) \geq 0$, which holds because $x - y$ and $x^k - y^k$ are either both nonnegative or both nonpositive real numbers (explain why).

We have shown that (1.5) holds for $n = k+1$ as well. Therefore, according to Proposition 1.1.1 (a), it holds for every $n \in \mathbb{N}$. □

Example 1.1.6. Let us prove that every positive integer n can be written in the form $n = 2^k \cdot q$ for some $k \in \mathbb{N}$ and some odd positive integer q (in fact, this representation of n is unique). For instance, for $n = 2024$, one may choose $k = 3$ and $q = 253$, since $2024 = 2^3 \cdot 253$.

Let us apply the well-ordering principle. We denote by S the set of positive integers, which *cannot* be written in this form, so we wish to show that $S = \varnothing$. Let us assume that, on the contrary, $S \neq \varnothing$. According to Proposition 1.1.1 (c), the set S has a minimum element, say a. Since every odd positive integer n can be written in the form in question as $n = 2^0 \cdot q$ with $q = n$, and hence does *not* belong to S, we must have $a = 2b$ for some $b \in \mathbb{Z}_{>0}$. Since $b = a/2 < a$ and a is the minimum element of S, we must have $b \notin S$. This means that $b = 2^k \cdot q$ for some $k \in \mathbb{N}$ and some odd $q \in \mathbb{Z}_{>0}$. As a result, $a = 2^{k+1} \cdot q \notin S$, a fact that contradicts our assumption that $a \in S$. This contradiction leads us to conclude that $S = \varnothing$. □

The proof we have just given is a typical example of a *proof by contradiction*. To prove that a certain statement holds, we assume that it is *not* true and derive logically a statement that is absurd, or that contradicts a hypothesis we have made. Such proofs will appear frequently in this book.

Example 1.1.7. Let

$$S_n = \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{n \cdot (n+1)} = \sum_{k=1}^{n} \frac{1}{k(k+1)}$$

for $n \in \mathbb{N}$ and let us prove that $S_n < 1$ for every $n \in \mathbb{N}$. We note that $S_0 = 0$ by Equation (1.1).

In this situation, it seems unclear how one could apply induction directly, so we make some computations and find that $S_0 = 0$, $S_1 = 1/2$, $S_2 = 2/3$, $S_3 = 3/4$, $S_4 = 4/5$. We then suspect that $S_n = n/(n+1)$ for every $n \in \mathbb{N}$ and attempt to prove this statement by induction on n. As already noticed $S_0 = 0$, so the statement holds for $n = 0$. Assuming that $n \geq 1$ and that $S_{n-1} = (n-1)/n$, we find that

$$S_n = \sum_{k=1}^{n} \frac{1}{k(k+1)} = S_{n-1} + \frac{1}{n(n+1)} = \frac{n-1}{n} + \frac{1}{n(n+1)}$$

$$= \frac{n^2}{n(n+1)} = \frac{n}{n+1}.$$

Thus, according to Proposition 1.1.1 (a), we have $S_n = n/(n+1)$ and, in particular, $S_n < 1$, for every $n \in \mathbb{N}$. □

Note that in Example 1.1.7, in order to show that $S_n < 1$ for $n \in \mathbb{N}$, we needed to invent a more suitable (and stronger) statement, namely that $S_n = n/(n+1)$ for $n \in \mathbb{N}$, for which we were able to apply induction. More examples that illustrate this point appear in the sequel.

The result of the following example is a special case of Turán's theorem in graph theory.

Example 1.1.8. During a meeting of n people, some shake hands (we assume that nobody shakes hands with himself and that any two persons shake hands at most once). Suppose that there are no three persons who have pairwise shaken hands. We will prove that, under these conditions, the total number of handshakes during that meeting is at most $n^2/4$.

We will apply the principle of mathematical induction, in the form of Proposition 1.1.1 (b). The claimed statement is trivial for $n \leq 2$. Let's assume that $n \geq 3$ and that the statement holds for every positive integer less than n, and let's consider a meeting of n people, as already described. We may certainly assume that at least one pair of persons, say a and b, have shaken hands. By the induction hypothesis, there have been at most $(n-2)^2/4$ handshakes among the $n-2$ persons other that a and b. Moreover, we know by assumption that each of these $n-2$ persons has shaken hands with at most one of a and b. Therefore, there has been a total of at most

$$1 + (n-2) + (n-2)^2/4 = n^2/4$$

handshakes during the meeting. This proves the claimed statement for n. According to Proposition 1.1.1 (b), the statement holds for every $n \in \mathbb{N}$. □

Example 1.1.9. Let us prove the *arithmetic-geometric mean inequality*

$$\frac{a_1 + a_2 + \cdots + a_n}{n} \geq (a_1 a_2 \cdots a_n)^{1/n} \tag{1.6}$$

for nonnegative real numbers $a_1, a_2, \ldots, a_n$ with the method of mathematical induction. Clearly, we may assume that all numbers $a_1, a_2, \ldots, a_n$ are nonzero.

We denote by P the right-hand side of (1.6) and set $b_i = a_i/P$ for $i \in [n]$. Then, $b_1 b_2 \cdots b_n = a_1 a_2 \cdots a_n / P^n = 1$ and (1.6) can be written as $b_1 + b_2 + \cdots + b_n \geq n$. Thus, it suffices to show that if the product of n positive real numbers is equal to 1, then their sum is greater than or equal to n. Actually, it is more convenient to prove the equivalent statement that if the product of n positive real numbers is greater than or equal to 1, then their sum is greater than or equal to n. This is the statement that we choose to prove by induction on n.

The claimed statement is trivial for $n = 1$. Let's assume that it holds for $n = k \geq 1$, and let's consider positive real numbers $b_1, b_2, \ldots, b_{k+1}$ such that $b_1 b_2 \cdots b_{k+1} \geq 1$. We will show that $b_1 + b_2 + \cdots + b_{k+1} \geq k+1$. Clearly, we must have $b_i \geq 1$ for at least one index $i \in [k+1]$, so we may assume that $b_{k+1} \geq 1$. Since the desired conclusion is trivial if $b_i \geq 1$ for every $i \in [k]$, we may also assume that $b_k \leq 1$. Then, $(b_k - 1)(b_{k+1} - 1) \leq 0$ and, equivalently, $b_k + b_{k+1} \geq b_k b_{k+1} + 1$. As a result,

$$b_1 \cdots b_{k-1}(b_k + b_{k+1} - 1) \geq b_1 \cdots b_{k-1}(b_k b_{k+1}) = b_1 b_2 \cdots b_{k+1} \geq 1.$$

Since the product of the k positive real numbers $b_1, \dots, b_{k-1}, b_k + b_{k+1} - 1$ is greater than or equal to 1, our induction hypothesis implies that their sum is greater than or equal to k, that is, $b_1 + b_2 + \cdots + b_{k+1} \geq k + 1$.

We have shown that the claimed statement holds for $n = k + 1$ as well. Thus, according to Proposition 1.1.1 (a), it holds for every positive integer n. □

1.1.2 Sequences and Recursions

A *sequence* $(x_n)_{n\in\mathbb{N}} = (x_0, x_1, x_2, \dots)$ of elements of a set X is a map $f : \mathbb{N} \to X$ with $f(n) = x_n$ for $n \in \mathbb{N}$ (maps of sets, more generally, are discussed in Section 1.2.2). By the construction of natural numbers, such a sequence can be defined recursively, meaning that one may specify first the initial term $x_0 \in X$ and then x_n as a function of $x_0, x_1, \dots, x_{n-1}$ for $n \geq 1$ (see [10, Chapter 6] [17, Chapter 5] [27, Chapter 8]). Such a formula for x_n is called a *recurrence*. For example, the natural number $n! = 1 \cdot 2 \cdots n$ can be defined as the n-th term of the sequence $(a_n)_{n\in\mathbb{N}}$ of elements of $\mathbb{N}$ for which

$$a_n = \begin{cases} 1, & \text{if } n = 0 \\ n a_{n-1}, & \text{if } n \geq 1 \end{cases}$$

for $n \in \mathbb{N}$. Thus, $a_0 = 1$, $a_1 = 1 \cdot a_0 = 1$, $a_2 = 2 \cdot a_1 = 2$, $a_3 = 3 \cdot a_2 = 6$, $a_4 = 4 \cdot a_3 = 24$, $a_4 = 5 \cdot a_4 = 120$, and so on. The uniqueness of this sequence is a consequence of the principle of mathematical induction. Indeed, if $(b_n)_{n\in\mathbb{N}}$ is also a sequence of elements of $\mathbb{N}$ such that

$$b_n = \begin{cases} 1, & \text{if } n = 0 \\ n b_{n-1}, & \text{if } n \geq 1 \end{cases}$$

for $n \in \mathbb{N}$, then a simple application of Proposition 1.1.1 (a) shows that $a_n = b_n$ for every $n \in \mathbb{N}$.

Example 1.1.10. Let us consider the sequence $(a_n)_{n\in\mathbb{N}}$ of rational numbers, which is defined recursively, setting $a_0 = a_1 = 1$ and

$$a_{n+1} = \frac{a_n^2 + (-1)^{n+1}}{a_{n-1}} \tag{1.7}$$

for $n \geq 1$. Thus,

$$a_2 = (a_1^2 + 1)/a_0 = 2,$$
$$a_3 = (a_2^2 - 1)/a_1 = 3,$$

$$a_4 = (a_3^2 + 1)/a_2 = 5,$$
$$a_5 = (a_4^2 - 1)/a_3 = 8,$$
$$a_6 = (a_5^2 + 1)/a_4 = 13$$

and so on. We will show that a_n is a positive integer for every $n \in \mathbb{N}$. For that reason, we plan to find a simpler recurrence for this sequence. Indeed, we observe that $a_2 = a_1 + a_0$, $a_3 = a_2 + a_1$, $a_4 = a_3 + a_2$, $a_5 = a_4 + a_3$, and so on. We will thus attempt to prove that

$$a_{n+1} = a_n + a_{n-1} \tag{1.8}$$

for every $n \geq 1$. This equality is valid for $n \in \{1, 2\}$. Let's assume that $n \geq 3$ and that (1.8) is valid for all positive integers less than n. Then, $a_n = a_{n-1} + a_{n-2}$ and hence, applying (1.7) twice, we find that

$$\begin{aligned} a_{n+1} &= \frac{a_n^2 + (-1)^{n+1}}{a_{n-1}} = \frac{(a_{n-1} + a_{n-2})^2 + (-1)^{n+1}}{a_{n-1}} \\ &= \frac{a_{n-1}^2 + 2a_{n-1}a_{n-2} + a_{n-2}^2 + (-1)^{n+1}}{a_{n-1}} \\ &= a_{n-1} + 2a_{n-2} + \frac{a_{n-2}^2 + (-1)^{n+1}}{a_{n-1}} = a_{n-1} + 2a_{n-2} + a_{n-3}. \end{aligned}$$

Since, by the induction hypothesis, $a_n = a_{n-1} + a_{n-2}$ and $a_{n-1} = a_{n-2} + a_{n-3}$, the identity we ended up with is equivalent to (1.8). Therefore, according to Proposition 1.1.1 (b), the recurrence (1.8) holds for every $n \geq 1$. Given that, a new application of Proposition 1.1.1 (b) yields that $a_n \in \mathbb{Z}_{>0}$ for every $n \in \mathbb{N}$.

The sequence (a_n) will appear frequently in the remainder of this book; it is called the *Fibonacci sequence*. The term a_{n-1} of this sequence, called a *Fibonacci number*, is often denoted by F_n, so $F_0 = 0$, $F_1 = 1$, and $F_n = F_{n-1} + F_{n-2}$ for $n \geq 2$.

Example 1.1.11. Let's consider the sequence $(a_n)_{n\in\mathbb{N}}$ of rational numbers, which is defined recursively, setting $a_0 = a_1 = a_2 = a_3 = 1$ and

$$a_{n+1} = \frac{a_n a_{n-2} + a_{n-1}^2}{a_{n-3}}$$

for $n \geq 3$. Thus,

$$a_4 = (a_3 a_1 + a_2^2)/a_0 = 2,$$
$$a_5 = (a_4 a_2 + a_3^2)/a_1 = 3,$$

$$a_6 = (a_5a_3 + a_4^2)/a_2 = 7,$$
$$a_7 = (a_6a_4 + a_5^2)/a_3 = 23,$$
$$a_8 = (a_7a_5 + a_6^2)/a_4 = 59,$$
$$a_9 = (a_8a_6 + a_7^2)/a_5 = 314,$$

$a_{10} = 1529$, $a_{11} = 8209$, $a_{12} = 83313$. Although this is not as easy to prove as in the case of Example 1.1.10, the number a_n is a positive integer for every $n \in \mathbb{N}$. The sequence (a_n) was discovered by Michael Somos around 1982 and belongs to a larger family of sequences with fascinating properties. Further information can be found in the articles [9] [25, Section 2] and the references given there. □

Example 1.1.12. For integers n, k with $0 \leq k \leq n$ we set

$$C(n,k) = \frac{n!}{k!(n-k)!}.$$

For example, $C(5,3) = 5!/3!2! = 10$. Clearly, $C(n,k)$ is a rational number. We will prove that $C(n,k)$ is a positive integer for all n, k.

Given that this seems difficult to establish directly, we attempt to find a recurrence satisfied by $C(n,k)$. For that reason, we observe that

$$n! = n \cdot (n-1)! = (n-k)(n-1)! + k(n-1)!$$

and hence that

$$\begin{aligned} C(n,k) &= \frac{n!}{k!(n-k)!} = \frac{(n-k)(n-1)!}{k!(n-k)!} + \frac{k(n-1)!}{k!(n-k)!} \\ &= \frac{(n-1)!}{k!(n-k-1)!} + \frac{(n-1)!}{(k-1)!(n-k)!} \\ &= C(n-1,k) + C(n-1,k-1) \end{aligned} \tag{1.9}$$

for $1 \leq k \leq n-1$.

We may now prove by induction on n that $C(n,k) \in \mathbb{Z}_{>0}$ for every $k \in \{0,1,\ldots,n\}$. The latter holds for $k \in \{0,n\}$, since $C(n,0) = C(n,n) = 1$; in particular, our claim holds for $n = 0$ and $n = 1$. Let's assume that our claim holds for $n-1$, where $n \geq 2$. Then, for every $k \in [n-1]$, the induction hypothesis guarantees that $C(n-1,k)$ and $C(n-1,k-1)$ are positive integers. This fact and (1.9) imply that $C(n,k)$ is also a positive integer, as a sum of two positive integers. Therefore, our claim holds for n and thus, by Proposition 1.1.1 (a), for every $n \in \mathbb{N}$. □

REMARK. The number $C(n,k)$ is equal to the binomial coefficient $\binom{n}{k}$, which will be introduced and studied in Section 2.3.4 of Chapter 2. The equality (1.9) is the basic recurrence satisfied by these coefficients. A refinement of the numbers $C(n,k)$ appears in Exercise 21.

1.1.3 Divisibility

Let us now review some concepts from elementary arithmetic, which will appear often in this book. A fundamental notion is that of divisibility. Given integers m, n, we say that m *divides* n (or that m is a *divisor* of n, or that n is *divisible* by m), and that n is an *integer multiple* of m, if there exists $q \in \mathbb{Z}$ such that $n = mq$. We then write $m \mid n$.

For example, $14 \mid 70$ because $70 = 14 \cdot 5$. Moreover, the identity (1.4) implies that $a - 1 \mid a^n - 1$ for all $a \in \mathbb{Z}$ and $n \in \mathbb{Z}_{>0}$ and the result of Example 1.1.12 implies that $k!(n-k)! \mid n!$ for all integers $0 \le k \le n$.

Example 1.1.13. Let us show that $2^n \mid (n+1)(n+2)\cdots(2n)$ for every $n \in \mathbb{Z}_{>0}$. We set $a(n) = (n+1)(n+2)\cdots(2n)$ and observe that

$$\begin{aligned} a(n) &= \frac{(2n)!}{n!} = \frac{1}{n!} \cdot 2 \cdot 4 \cdot 6 \cdots (2n) \cdot 1 \cdot 3 \cdot 5 \cdots (2n-1) \\ &= 2^n \cdot q(n), \end{aligned}$$

where $q(n) = 1 \cdot 3 \cdot 5 \cdots (2n-1)$ is an odd integer. We conclude that $2^n \mid a(n)$ and, in fact, that 2^n is the largest power of 2, which divides $a(n)$, for every $n \in \mathbb{Z}_{>0}$. □

Divisibility of integers has the following basic properties.

Proposition 1.1.3. *For $a, b, c \in \mathbb{Z}$:*

(a) $a \mid a$,
(b) *if $a \mid b$ and $b \mid a$, then $|a| = |b|$,*
(c) *if $a \mid b$ and $b \mid c$, then $a \mid c$,*
(d) *if $a \mid b$ and $a \mid c$, then $a \mid bx + cy$ for all $x, y \in \mathbb{Z}$.*

Proof. Part (a) holds since $a = a \cdot 1$ for every $a \in \mathbb{Z}$.

For part (b), let's assume that $a \mid b$ and $b \mid a$. Then, $b = ap$ and $a = bq$ for some $p, q \in \mathbb{Z}$. If a, b are nonzero, then the same must be true for p, q and, since these numbers are integers, $|p| \ge 1$ and $|q| \ge 1$. Hence, $|b| = |a| \cdot |p| \ge |a|$ and $|a| = |b| \cdot |q| \ge |b|$, whence $|a| = |b|$. Otherwise, we must have $a = b = 0$. So, in either case, $|a| = |b|$.

For part (c), let's assume that $a \mid b$ and $b \mid c$. Then, $b = ap$ and $c = bq$ for some $p, q \in \mathbb{Z}$ and hence, $c = apq = at$ with $t = pq \in \mathbb{Z}$. Therefore, $a \mid c$.

For part (d), let's assume that $a \mid b$ and $a \mid c$. Then, $b = ap$ and $c = aq$ for some $p, q \in \mathbb{Z}$ and hence, $bx + cy = a(px + qy)$. Since $px + qy \in \mathbb{Z}$, it follows that $a \mid bx + cy$. □

For a nontrivial application of these properties, see Exercise 22 (d) and its solution.

Divisibility appears in a more general context in the following fundamental proposition.

Proposition 1.1.4 (Euclidean Division). *Given $n \in \mathbb{Z}$ and $m \in \mathbb{Z}_{>0}$, there exist unique integers q, r such that $n = mq + r$ and $0 \leq r < m$. Moreover, m divides n if and only if $r = 0$.*

The integers q and r are called the *quotient* and the *remainder* of the Euclidean division of n by m, respectively. The following statement is a consequence of Proposition 1.1.4. Equation (1.10), which appears there, is called the *m-ary representation* of n. The numbers a_i are the *digits* of n in base m.

Proposition 1.1.5. *Let $m, k \in \mathbb{N}$, with $m \geq 2$. Then, every natural number $n < m^{k+1}$ can be written uniquely in the form*

$$n = a_0 + a_1 m + a_2 m^2 + \cdots + a_k m^k \tag{1.10}$$

with $a_0, a_1, \ldots, a_k \in \{0, 1, \ldots, m-1\}$.

Since the previous two propositions do not play a central role in this book, we refer to the exercises of this chapter for their proofs.

Example 1.1.14. Which terms of the Fibonacci sequence $(F_n)_{n\geq 0}$, with $F_0 = 0$, $F_1 = 1$, and $F_n = F_{n-1} + F_{n-2}$ for $n \geq 2$, are divisible by 3? Applying the recurrence three times, we find that

$$F_n = F_{n-1} + F_{n-2} = 2F_{n-2} + F_{n-3} = 3F_{n-3} + 2F_{n-4}$$

and, in view of Proposition 1.1.3 (d), we conclude that $3 \mid F_n \Leftrightarrow 3 \mid F_{n-4}$ for $n \geq 4$. More generally, we find that $3 \mid F_n \Leftrightarrow 3 \mid F_{n-4m}$ for all $m, n \in \mathbb{N}$ with $n \geq 4m$ by induction on m. Since $F_0 = 0$, $F_1 = F_2 = 1$ and $F_3 = 2$, it follows that 3 divides F_{4m} but none of F_{4m+1}, F_{4m+2} and F_{4m+3}, for $m \in \mathbb{N}$. Since (by Proposition 1.1.4) every $n \in \mathbb{N}$ can be written uniquely as $n = 4m + r$ with $m \in \mathbb{N}$ and $r \in \{0, 1, 2, 3\}$, we conclude that $3 \mid F_n \Leftrightarrow 4 \mid n$ for $n \in \mathbb{N}$. A partial generalization of this statement appears in Exercise 26 (c). □

An integer $p \geq 2$ is called a *prime number* if 1 and p are the only positive divisors of p. Thus, 2, 3, 5, 7, 11, 13, 17, 19, 23, 29 are the smallest prime numbers. Every integer $n \geq 2$ can be expressed as a product of (one or more) prime numbers (see Exercise 27). This factorization is essentially unique (the proof of uniqueness requires the concept of the greatest common divisor). For thorough introductions to elementary number theory, we recommend [4, 7].

1.2 Maps and Relations

This section first discusses binary relations between sets and their compositions and then focuses on maps, one-to-one correspondences, and their properties. The reader is assumed to have some familiarity with these important notions.

1.2.1 Binary Relations

A *binary relation* (or, simply, relation) is a triple $R = (X, Y, \Gamma)$, where X and Y are sets and Γ is a subset of $X \times Y$. For $(x, y) \in \Gamma$ we write xRy and say that $x \in X$ is *related* to $y \in Y$ under R. Thus, given such a relation R, every element of X may be related to none, or one, or more than one elements of Y. We say that R is a binary relation from the set X to the set Y and, in case $X = Y$, that it is a binary relation on X. The set $\Gamma = \Gamma_R$ is called that *graph* of R.

Example 1.2.1.

(a) The *identity* binary relation id_X is defined on any set X by declaring that x is related to y if and only if $x = y$, for $x, y \in X$.
(b) Let R be the binary relation on the set $[n]$ for which xRy if and only if $x < y$. For $n = 4$ we have $\Gamma_R = \{(1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4)\}$; two ways to draw this relation are shown on Figure 1.1. Under R, 1 is related to all other elements of $[n]$, while n is not related to any element.

□

Given sets X, Y, Z and binary relations R from X to Y and S from Y to Z, the *composition* $R \circ S$ is defined as the binary relation from X to Z for which $x \in X$ is related to $z \in Z$ if there exists $y \in Y$ such that xRy and ySz. Composition of binary relations has the following basic properties.

Proposition 1.2.1. *Let X, Y, Z, W be sets.*

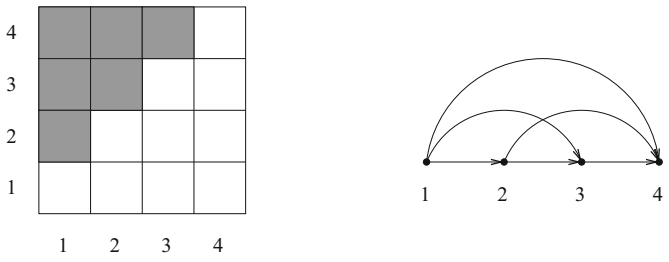


Figure 1.1 A binary relation on $\{1, 2, 3, 4\}$

(a) *The associative law* $(R \circ S) \circ T = R \circ (S \circ T)$ *holds for any binary relations* R *from* X *to* Y, S *from* Y *to* Z *and* T *from* Z *to* W.
(b) *We have* $\mathrm{id}_X \circ R = R \circ \mathrm{id}_Y = R$ *for every binary relation* R *from* X *to* Y.

Proof. To verify part (a), we note that $R \circ S$ is a binary relation from X to Z and that $(R \circ S) \circ T$ is a binary relation from X to W. Moreover, under the latter, $x \in X$ is related to $w \in W$ if and only if there exists $z \in Z$ such that $x(R \circ S)z$ and zTw. That happens if and only if there exist $y \in Y$ and $z \in Z$ such that xRy, ySz and zTw. A similar reasoning yields the same result for $R \circ (S \circ T)$ and shows that $(R \circ S) \circ T = R \circ (S \circ T)$. The verification of part (b) is left to the reader. □

EXAMPLE. Let R be the relation of Example 1.2.1 (b) on the set $[n]$ and $x, y \in [n]$. Then, x is related to y under $R \circ R$ if and only if $x + 1 < y$.

1.2.2 Maps

A binary relation $f = (X, Y, \Gamma)$ is called a *map* if for every $x \in X$ there exists a unique $y \in Y$ such that xfy. In simpler terms, a map f is determined by two sets X, Y and a rule Γ, which assigns to every element $x \in X$ a unique element $y \in Y$. We write $f : X \to Y$ and for $x \in X$, we denote by $f(x)$ the unique element of Y to which x is related, under f; see Figure 1.2. The sets X and Y are called the *domain* and the *range* of f, respectively, and $f(x) \in Y$ is called the *image* of $x \in X$.

Example 1.2.2.

(a) For every set X, the identity relation id_X of Example 1.2.1 (a) is a map $\mathrm{id}_X : X \to X$ and $\mathrm{id}_X(x) = x$ for every $x \in X$.
(b) According to the well-ordering principle, we may define a map $f : 2^{\mathbb{N}} \setminus \{\varnothing\} \to \mathbb{N}$ by declaring that $f(S)$ is the minimum element of S, for every nonempty $S \subseteq \mathbb{N}$. For instance, $f(S) = 2$ if S is the set of prime numbers. □

Example 1.2.3. Given a set Ω, a map $\circ : \Omega \times \Omega \to \Omega$ is called an *operation* on Ω. We often denote by $a \circ b$ the image of $(a, b) \in \Omega \times \Omega$ under $\circ$, instead of $\circ(a, b)$. For instance, addition and multiplication

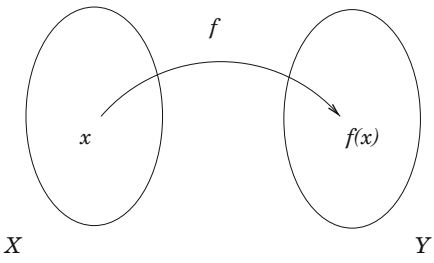


Figure 1.2 A map $f : X \to Y$

$$+ : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$$
$$\times : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$$

are operations on the set $\mathbb{N}$ of natural numbers (similarly for the sets $\mathbb{Z}_{>0}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$). Under these operations, a pair $(a, b) \in \mathbb{N} \times \mathbb{N}$ of natural numbers is mapped to the sum $a + b \in \mathbb{N}$ and the product $ab \in \mathbb{N}$ of a and b, respectively.

Similarly, the union and intersection

$$\bigcup : 2^X \times 2^X \to 2^X$$
$$\bigcap : 2^X \times 2^X \to 2^X$$

are operations on the powerset 2^X of a set X. Under these operations, a pair $(A, B) \in 2^X \times 2^X$ of subsets of X is mapped to the union $A \cup B \in 2^X$ and the intersection $A \cap B \in 2^X$ of A and B, respectively.

For the definition and main properties of the union and intersection of two or more subsets of a set, see Exercises 32 and 33. □

Given a map $f : X \to Y$, the *image* of $A \subseteq X$ and the *inverse image* of $B \subseteq Y$ are defined as

$$f(A) = \{f(x) : x \in A\},$$
$$f^{-1}(B) = \{x \in X : f(x) \in B\},$$

respectively. The subset $f(X)$ of Y is called the *image* of f. For simplicity, we will sometimes denote by $f^{-1}(y)$ the inverse image $f^{-1}(\{y\}) = \{x \in X : f(x) = y\}$ of a one-element subset $\{y\}$ of Y.

The restriction of the map $f : X \to Y$ to $A \subseteq X$ is defined as the map $f|_A : A \to Y$ for which $f|_A(x) = f(x)$ for every $x \in A$. More generally, the restriction

$$f|_{A,B} : A \to B$$

of f is defined for $A \subseteq X$ and $B \subseteq Y$, provided that $f(A) \subseteq B$.

EXAMPLE. Let $f : 2^{\mathbb{N}} \setminus \{\varnothing\} \to \mathbb{N}$ be the map of Example 1.2.2 (b). Then, the inverse image $f^{-1}(\{3, 4\})$ consists of all subsets of $\mathbb{N}$, which contain 3 or 4 (or both), but none of 0, 1, 2.

EXAMPLE. Let X be a nonempty set. Since the union of any two nonempty subsets of X is again a nonempty subset of X, the operation $\cup : 2^X \times 2^X \to 2^X$ of Example 1.2.3 restricts to an operation on the set of nonempty subsets of X. However, this statement is no longer valid if union is replaced by intersection, unless X has a single element (explain why). □

The composition of two maps $f : X \to Y$ and $g : Y \to Z$, as defined in Section 1.2.1, is again a map and is denoted by $g \circ f : X \to Z$ (instead of $f \circ g : X \to Z$). Thus, we have $(g \circ f)(x) = g(f(x))$ for every $x \in X$. We now recall the following basic definition.

Definition 1.2.1. *A map $f : X \to Y$ is called*

(a) *injective (or one-to-one, or an injection) if for every element $y \in Y$ there exists at most one element $x \in X$, such that $f(x) = y$,*
(b) *surjective (or onto, or a surjection) if for every element $y \in Y$ there exists at least one element $x \in X$, such that $f(x) = y$,*
(c) *bijective (or one-to-one and onto, or a one-to-one correspondence, or a bijection) if for every element $y \in Y$, there exists exactly one element $x \in X$, such that $f(x) = y$.*

Thus, $f : X \to Y$ is a bijection if and only if it is both an injection and a surjection. Here are two equivalent ways (each statement is the contrapositive of the other) to express the injectivity of $f : X \to Y$:

- $x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2)$, for $x_1, x_2 \in X$,
- $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$, for $x_1, x_2 \in X$.

Note also that $f : X \to Y$ is surjective if and only if $f(X) = Y$.

Example 1.2.4.

(a) Let $f : \mathbb{N} \to \mathbb{Z}$ be the map defined by setting

$$f(x) = \begin{cases} x/2, & \text{if } x \text{ is even} \\ -(x+1)/2, & \text{if } x \text{ is odd} \end{cases}$$

for $x \in \mathbb{N}$. For $y \in \mathbb{Z}$, solving the equation $f(x) = y$ we find the unique solution

$$x = \begin{cases} 2y, & \text{if } y \in \mathbb{N} \\ -2y - 1, & \text{otherwise} \end{cases}$$

and conclude that f is bijective.

(b) Let $g : \mathbb{N} \to \mathbb{N}$ and $h : \mathbb{N} \cup \{-1\} \to \mathbb{N}$ be the maps for which $g(x) = x + 1$ for $x \in \mathbb{N}$ and $h(x) = |x|$ for $x \in \mathbb{N} \cup \{-1\}$, respectively. Then, g is one-to-one since for every $y \in \mathbb{N}$, the equation $g(x) = y$ has at most one solution $x \in \mathbb{N}$. On the other hand, g is not onto since this equation has no solution $x \in \mathbb{N}$ for $y = 0$. Similarly, h is onto since for every $y \in \mathbb{N}$, the equation $h(x) = y$ has the solution $x = y$. On the other hand, h is not one-to-one since this equation has two solutions $x \in \mathbb{N} \cup \{-1\}$ for $y = 1$. □

We note (explain why) that the composition of two injections (respectively, surjections) is also an injection (respectively, surjection). As a result, the composition of two bijections is also a bijection. We will often apply the following criterion to prove that a given map is bijective.

Proposition 1.2.2. *A map $f : X \to Y$ is bijective if and only if there exists a map $g : Y \to X$ such that $g(f(x)) = x$ for every $x \in X$ and $f(g(y)) = y$ for every $y \in Y$; in other words, the composition $g \circ f$ is the identity map on X and $f \circ g$ is the identity map on Y.*

Proof. If f is bijective then, given $y \in Y$, we define $g(y)$ to be the unique element $x \in X$ such that $f(x) = y$ and observe that the resulting map $g : Y \to X$ has the required properties. The converse follows from Exercise 35. □

Each one of f, g in the statement of Proposition 1.2.2 is called the *inverse map* of the other.

EXAMPLE. The inverse map of $f : \mathbb{N} \to \mathbb{Z}$ in Example 1.2.4 (a) is the map $g : \mathbb{Z} \to \mathbb{N}$ with

$$g(y) = \begin{cases} 2y, & \text{if } y \in \mathbb{N} \\ -2y - 1, & \text{otherwise} \end{cases}$$

for $y \in \mathbb{Z}$.

1.3 Exercises

1. For complex numbers (more generally, for elements of an additive abelian group) a_i, b_i, and a_{ij}, prove that:
   (a) $\displaystyle\sum_{i=1}^{n}(a_i + b_i) = \left(\sum_{i=1}^{n} a_i\right) + \left(\sum_{i=1}^{n} b_i\right).$
   (b) $\displaystyle\sum_{i=1}^{m}\left(\sum_{j=1}^{n} a_{ij}\right) = \sum_{j=1}^{n}\left(\sum_{i=1}^{m} a_{ij}\right).$
2. Prove the following identities for $n \in \mathbb{N}$:
   (a) $\displaystyle\sum_{k=1}^{n} k = n(n+1)/2.$
   (b) $\displaystyle\sum_{k=1}^{n} k(k-1) = n(n^2-1)/3.$

(c) $\displaystyle\sum_{k=1}^{n} k^2 = n(n+1)(2n+1)/6.$

(d) $\displaystyle\sum_{k=1}^{n} (-1)^{k-1}k^2 = (-1)^{n-1}\sum_{k=1}^{n} k.$

(e) $\displaystyle\sum_{k=1}^{n} k^3 = n^2(n+1)^2/4.$

(f) $\displaystyle\sum_{k=1}^{n} \frac{(k-1)(k+2)}{k(k+1)} = \frac{n(n-1)}{n+1}.$

3. Find all positive integers n such that $2^{n-1} \geq n^3$. When does equality hold?
4. Prove that the representation $n = 2^k \cdot q$ of Example 1.1.6 is unique for every positive integer n.
5. Let $n \in \mathbb{N}$.
   (a) Use the well-ordering principle to prove that there exists a pair $(a, b) \in \mathbb{N} \times \mathbb{N}$ such that $a < b$ and $n = a + b(b-1)/2$. For example, $20 = 5 + 6 \cdot (6-1)/2$ and $21 = 0 + 7 \cdot (7-1)/2$.
   (b) Prove that such a pair (a, b) is unique.
6. Prove that

$$n^{n/2} \leq n! \leq 2(n/2)^n \tag{1.11}$$

   for every $n \in \mathbb{Z}_{>0}$.
7. Prove that

$$2^n (n!)^2 \leq (2n)! \leq 4^n (n!)^2 \tag{1.12}$$

   for every $n \in \mathbb{N}$. Additionally, find all natural numbers n for which each of the following inequalities holds:
   (a) $(2n)! \leq (n+1)(n!)^3$.
   (b) $(2n)! \leq (n!)^3$.
8. For $a \in \mathbb{C}$, prove that

$$na^{n+2} - (n+1)a^{n+1} + a = (a-1)^2 \cdot (a + 2a^2 + \cdots + na^n) \tag{1.13}$$

   for every $n \in \mathbb{N}$:
   (a) by induction on n,
   (b) by applying identity (1.4).

Moreover:

(c) Deduce a simple formula for the sum $\sum_{k=0}^{n} k2^k = 1\cdot 2+2\cdot 2^2+\cdots+n\cdot 2^n$.
(d) Prove the formula you found in part (c) by induction on n.

9. Let $p(n) = \dfrac{1}{4^n}\cdot\dfrac{(2n)!}{(n!)^2}$ for $n \in \mathbb{N}$.

(a) Prove that

$$(p(n))^2 = \frac{1}{2n+1}\left(1-\frac{1}{2^2}\right)\left(1-\frac{1}{4^2}\right)\cdots\left(1-\frac{1}{(2n)^2}\right)$$
$$= \frac{1}{4n}\left(1-\frac{1}{3^2}\right)^{-1}\left(1-\frac{1}{5^2}\right)^{-1}\cdots\left(1-\frac{1}{(2n-1)^2}\right)^{-1}$$

for every $n \in \mathbb{Z}_{>0}$.
(b) Deduce the stronger than (1.12) inequalities

$$\frac{1}{2\sqrt{n}} \le p(n) \le \frac{1}{\sqrt{2n+1}}$$

for every $n \in \mathbb{Z}_{>0}$.

10. Find all positive integers n with the following property: there exist positive integers $p_1 < p_2 < \cdots < p_n$ such that

$$1 = \frac{1}{p_1} + \frac{1}{p_2} + \cdots + \frac{1}{p_n}. \tag{1.14}$$

11. Consider the sequence $(a_n)_{n\in\mathbb{N}}$ of positive integers, which is defined by setting $a_0 = a_1 = 1$ and $a_n = (n-1)(a_{n-1}+a_{n-2})$ for $n \ge 2$.

(a) Complute a_n for $n \le 5$ and guess a simple formula for a_n.
(b) Prove this formula by induction on n.

12. Let

$$f(n) = \sum_{i=1}^{n} i! = 1! + 2! + \cdots + n!$$

for $n \in \mathbb{N}$ (so $f(0) = 0$). Find a recurrence for the sequence $(f(n))_{n\in\mathbb{N}}$ of the form $f(n) = p(n)f(n-1) + q(n)f(n-2)$, where $p(x)$ and $q(x)$ are polynomial functions of x.

13. Consider the Fibonacci sequence (F_n) of Example 1.1.10, with $F_1 = F_2 = 1$ and $F_{n+1} = F_n + F_{n-1}$ for $n \ge 2$. We have $F_3 = 2$, $F_4 = 3$, $F_5 = 5$, $F_6 = 8$, and so on. Prove that for $n \in \mathbb{Z}_{>0}$:

(a) $\displaystyle\sum_{k=1}^{n} F_k = F_{n+2} - 1.$

(b) $\displaystyle\sum_{k=1}^{n} F_{2k} = F_{2n+1} - 1.$

(c) $\displaystyle\sum_{k=1}^{n} F_{2k-1} = F_{2n}.$

(d) $\displaystyle\sum_{k=1}^{n} kF_k = (n-1)F_{n+2} - F_{n+1} + 2.$

(e) $\displaystyle\sum_{k=1}^{n} F_k^2 = F_n F_{n+1}.$

(f) $\displaystyle\sum_{k=1}^{n} F_k F_{k+1} = \begin{cases} F_{n+1}^2 - 1, & \text{if } n \text{ is even} \\ F_{n+1}^2, & \text{if } n \text{ is odd.} \end{cases}$

14. Prove that every natural number can be written as a sum of distinct terms of the Fibonacci sequence $(F_n)_{n\in\mathbb{N}}$, defined by $F_0 = 0$, $F_1 = 1$ and $F_{n+1} = F_n + F_{n-1}$ for $n \geq 1$.
15. Let $(G_n)_{n\in\mathbb{N}}$ be the sequence of rational numbers defined by setting $G_0 = G_1 = 1$ and

$$G_{n+1} = (G_n^2 + 1)/G_{n-1}$$

for $n \geq 1$.

(a) Prove that $G_{n+1} = 3G_n - G_{n-1}$ for $n \geq 1$.
(b) Deduce that $G_n \in \mathbb{N}$ for every $n \in \mathbb{N}$.
(c) Prove that $G_n = F_{2n-1}$ for $n \geq 1$, where (F_n) is the Fibonacci sequence of Exercise 13.

16. Let $(a_n)_{n\in\mathbb{N}}$ be the sequence of real numbers defined by setting $a_0 = a_1 = 1$ and $a_n = a_{n-1} + (-1)^n a_{n-2}$ for $n \geq 2$. Prove that a_n is a positive integer for every $n \in \mathbb{N}$.
17. Let $(a_n)_{n\in\mathbb{N}}$ be the sequence of rational numbers defined by setting $a_0 = a_1 = 1$ and $a_n = a_{n-1} - a_{n-2}/4$ for $n \geq 2$. Thus, $a_2 = 3/4$, $a_3 = 1/2$, $a_4 = 5/16$, $a_5 = 3/16$.

(a) Prove that $a_n = \dfrac{n+1}{2n}\, a_{n-1}$ for $n \geq 1$.
(b) Find a simple formula for a_n.

18. Let $(a_n)_{n\in\mathbb{N}}$ be the sequence of natural numbers defined by setting $a_0 = 1$, $a_1 = 0$ and $a_n = a_{n-1} + (n-1)^2 a_{n-2}$ for $n \geq 2$. Thus, $a_2 = a_3 = 1$, $a_4 = 10$, $a_5 = 26$.

(a) Find a simple recurrence for $b_n := a_n - na_{n-1}$, where $n \geq 1$.

(b) Find a simple formula for b_n.
(c) Find a simple formula for a_n. What is the limit of $a_n/n!$ as $n \mapsto \infty$?

19. Let $a_n = \left(\frac{1+\sqrt{5}}{2}\right)^n + \left(\frac{1-\sqrt{5}}{2}\right)^n$ for $n \in \mathbb{N}$.

(a) Prove that $a_n = a_{n-1} + a_{n-2}$ for $n \geq 2$.
(b) Compute the sum

$$\sum_{k=1}^{n} \frac{a_{k-1}a_{k+2}}{(a_k a_{k+1})^2}$$

for $n \in \mathbb{N}$. What is the limit as $n \mapsto \infty$?
(c) Compute the product

$$\prod_{k=1}^{n} \left(\frac{a_{k+1}}{a_k} - \frac{a_k}{a_{k+1}}\right)$$

for $n \in \mathbb{N}$. What is the limit as $n \mapsto \infty$?

20. The integers $a_{n,k}$ are defined for $n \in \mathbb{N}$ and $k \in \{0, 1, \ldots, 2n\}$ by the identity

$$(1 + x + x^2)^n = \sum_{k=0}^{2n} a_{n,k} x^k. \tag{1.15}$$

We set $a_{n,k} = 0$ for integers k outside the range $0 \leq k \leq 2n$.

(a) Prove that $a_{n,k} = a_{n-1,k} + a_{n-1,k-1} + a_{n-1,k-2}$ for all $n \in \mathbb{Z}_{>0}$ and $k \in \mathbb{Z}$.
(b) Using part (a), prove that

$$\begin{aligned} a_{n,0} + a_{n,3} + a_{n,6} + \cdots &= a_{n,1} + a_{n,4} + a_{n,7} + \cdots \\ &= a_{n,2} + a_{n,5} + a_{n,8} + \cdots \\ &= 3^{n-1} \end{aligned}$$

for every $n \in \mathbb{Z}_{>0}$.
(c) Solve part (b) by setting $x = 1$, $x = \omega$, and $x = \omega^2$ in Equation (1.15), where $\omega = (-1 + \mathrm{i}\sqrt{3})/2$ is a primitive third root of unity.
(d) Generalize to the coefficients of $(1 + x + x^2 + \cdots + x^m)^n$.

21. For integers $0 \leq k \leq n$ and a real number $x > 1$ we set

$$f_{n,k}(x) = \frac{(x-1)(x^2-1)\cdots(x^n-1)}{(x-1)(x^2-1)\cdots(x^k-1)\cdot(x-1)(x^2-1)\cdots(x^{n-k}-1)}.$$

For example,

$$f_{4,2}(x) = \frac{(x-1)(x^2-1)(x^3-1)(x^4-1)}{(x-1)(x^2-1)\cdot(x-1)(x^2-1)} = 1 + x + 2x^2 + x^3 + x^4.$$

(a) Prove that $f_{n,k}(x) = f_{n-1,k}(x) + x^{n-k} f_{n-1,k-1}(x)$ for $1 \le k \le n-1$.
(b) Deduce from (a) that $f_{n,k}(x)$ is a positive integer for every integer $x \ge 2$.
(c) Deduce from (a) the recurrence (1.9).

22. Prove the divisibility relations

(a) $a^2 + a + 1 \mid a^4 + a^2 + 1$
(b) $a^2 + a + 1 \mid a^8 + a^4 + 1$
(c) $a^2 + a + 1 \mid a^5 + a + 1$
(d) $a^4 + a^3 + a^2 + a + 1 \mid a^{24} + a^{18} + a^{12} + a^6 + 1$

for $a \in \mathbb{Z}$.

23. For positive integers a, n, prove that the following conditions are equivalent:

(i) $a + 1 \mid a^n + 1$
(ii) $1 + a + \cdots + a^{n-1} \mid 1 + a^2 + \cdots + a^{2n-2}$
(iii) either $a = 1$, or $a \ge 2$ and n is odd.

24. Let $a_n = 6^n - 1 - 5n$ for $n \in \mathbb{N}$.

(a) Find a recurrence for the sequence $(a_n)_{n\in\mathbb{N}}$.
(b) Prove that a_n is divisible by 25 for every $n \in \mathbb{N}$.
(c) For which $n \in \mathbb{N}$ is a_n divisible by 125?

25. For the Fibonacci sequence $(F_n)_{n\ge 0}$, with $F_0 = 0$, $F_1 = 1$, and $F_{n+1} = F_n + F_{n-1}$ for $n \ge 1$, prove that:

(a) $F_n \mid F_{2n}$ for every $n \in \mathbb{Z}_{>0}$.
(b) F_{2n+1} can be expressed as the sum of the squares of two integers for every $n \in \mathbb{N}$.

26. For the Fibonacci sequence $(F_n)_{n\ge 0}$, with $F_0 = 0$, $F_1 = 1$, and $F_{n+1} = F_n + F_{n-1}$ for $n \ge 1$:

(a) Prove that $F_{m+n} = F_{m+1}F_n + F_m F_{n-1}$ for $m, n \in \mathbb{Z}_{>0}$.
(b) For those who are familiar with matrix multiplication: prove that

$$\begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}^n \tag{1.16}$$

for every $n \in \mathbb{Z}_{>0}$ and deduce the identity of part (a).
(c) Prove that $m \mid n \Rightarrow F_m \mid F_n$ for $m, n \in \mathbb{N}$.

27. Prove that every integer $n \geq 2$ can be expressed as a product of (one or more) prime numbers. Deduce that every such integer has at least one prime divisor.
28. Prove Proposition 1.1.4.
29. Prove Proposition 1.1.5.
30. The sequence $f : \mathbb{N} \to \mathbb{N}$ is defined recursively by setting $f(0) = 0$ and

$$f(2n) = f(n),$$
$$f(2n+1) = f(n) + 1,$$

for $n \in \mathbb{N}$.

(a) Compute $f(2025)$.
(b) Find a nonrecursive definition of $f(n)$.
(c) Which $n \in \{1000, 1001, \ldots, 2000\}$ satisfy $f(n) = 2$?
(d) Which $n \in \mathbb{N}$ satisfy $f(n) = n/3$?

31. Given a binary relation R on a set X, we set $R^1 = R$ and $R^n = R^{n-1} \circ R$ for $n \geq 2$.

(a) Give an example such that $\Gamma_{R^1} \subset \Gamma_{R^2} \subset \cdots$.
(b) Give an example such that $\Gamma_{R^1} \supset \Gamma_{R^2} \supset \cdots$.

32. Let $\circ : \Omega \times \Omega \to \Omega$ be an operation on the set Ω (see Example 1.2.3) such that $(a \circ b) \circ c = a \circ (b \circ c)$ for $a, b, c \in \Omega$ (such an operation is called *associative*). For $a_1, a_2, \ldots, a_n \in \Omega$ we define

$$a_1 \circ a_2 \circ \cdots \circ a_n = \begin{cases} a_1, & \text{if } n = 1 \\ (a_1 \circ a_2 \circ \cdots \circ a_{n-1}) \circ a_n, & \text{if } n \geq 2 \end{cases} \tag{1.17}$$

and set $a_1 \circ a_2 \circ \cdots \circ a_n = a^n$ when $a_1 = a_2 = \cdots = a_n = a$.

(a) Prove that

$$a_1 \circ a_2 \circ \cdots \circ a_n = (a_1 \circ \cdots \circ a_i) \circ (a_{i+1} \circ \cdots \circ a_n) \tag{1.18}$$

for all $a_1, a_2, \ldots, a_n \in \Omega$ and every $i \in [n-1]$.
(b) Deduce that $a^m \circ a^n = a^{m+n}$ and $(a^m)^n = a^{mn}$ for all $m, n \in \mathbb{Z}_{>0}$ and every $a \in \Omega$.

33. For $A_1, A_2, \ldots, A_n \subseteq \Omega$ we set

$$\bigcup_{i=1}^{n} A_i = A_1 \cup A_2 \cup \cdots \cup A_n, \qquad \bigcap_{i=1}^{n} A_i = A_1 \cap A_2 \cap \cdots \cap A_n,$$

where the right-hand sides are defined by means of Exercise 32. For $A_1, A_2, \ldots, A_n \subseteq \Omega$ and $B \subseteq \Omega$ prove that:

(a) $\left(\bigcup_{i=1}^{n} A_i\right) \cap B = \bigcup_{i=1}^{n} (A_i \cap B)$.

(b) $\left(\bigcap_{i=1}^{n} A_i\right) \cup B = \bigcap_{i=1}^{n} (A_i \cup B)$.

(c) $\Omega \setminus \left(\bigcup_{i=1}^{n} A_i\right) = \bigcap_{i=1}^{n} (\Omega \setminus A_i)$.

(d) $\Omega \setminus \left(\bigcap_{i=1}^{n} A_i\right) = \bigcup_{i=1}^{n} (\Omega \setminus A_i)$.

Moreover, if $f : X \to Y$ is a map, $A_1, A_2, \ldots, A_n \subseteq X$ and $B_1, B_2, \ldots, B_n \subseteq Y$, prove that

(e) $f\left(\bigcup_{i=1}^{n} A_i\right) = \bigcup_{i=1}^{n} f(A_i)$.

(f) $f\left(\bigcap_{i=1}^{n} A_i\right) \subseteq \bigcap_{i=1}^{n} f(A_i)$.

(g) $f^{-1}\left(\bigcup_{i=1}^{n} B_i\right) = \bigcup_{i=1}^{n} f^{-1}(B_i)$.

(h) $f^{-1}\left(\bigcap_{i=1}^{n} B_i\right) = \bigcap_{i=1}^{n} f^{-1}(B_i)$.

34. In which of the following situations is the map $f : X \to Y$ defined?

(a) $X = Y = 2^{\mathbb{N}}$ and $f(S) = \mathbb{N} \setminus S$, for $S \subseteq \mathbb{N}$.
(b) $X = 2^{\mathbb{N}}$, $Y = \mathbb{N}$, and $f(S)$ is the minimum element of $\mathbb{N} \setminus S$, for $S \subseteq \mathbb{N}$.
(c) $X = Y = \mathbb{N}$ and $f(n) = n^2 - 1$, for $n \in \mathbb{N}$.
(d) $X = Y = \mathbb{N}$, $f(0) = 0$ and

$$f(2n) = (f(n))^2,$$
$$f(2n+1) = f(n) + 1,$$

for $n \in \mathbb{N}$.

(e) $X = Y = \mathbb{N}$,

$$f(n) = \begin{cases} f(n-2) + 1, & \text{if } 2 \mid n \\ f(n-3) + 2, & \text{if } 3 \mid n \end{cases}$$

for $n \in \mathbb{Z}_{>0}$ and $f(n) = 0$ for all other values of $n \in \mathbb{N}$.

35. Prove that a map $f : X \to Y$ is:

    (a) injective if and only if there exists a map $g : Y \to X$ such that $g(f(x)) = x$ for every $x \in X$,
    (b) surjective if and only if there exists a map $g : Y \to X$ such that $f(g(y)) = y$ for every $y \in Y$.

36. Let $X = \{x \in \mathbb{R} : x \leq 1\}$ and $f(x) = \dfrac{1}{2 - x}$, for $x \in X$.

    (a) Prove that $f : X \to X$ is a well-defined map. Is it one-to-one? Is it onto?
    (b) For $n \in \mathbb{N}$, guess and prove by induction a simple formula for $f^n(x)$, where $f^n : X \to X$ is the map defined (in agreement with Exercise 32) by setting $f^0 = \mathrm{id}_X$ and $f^n = f \circ f^{n-1}$ for $n \geq 1$.

Chapter 2
Elementary Enumerative Combinatorics



The main problem of enumerative combinatorics is to count the elements of a finite set. Problems of this type appear in a broad spectrum of contemporary mathematics. The methods that have been developed for their solution are especially important within discrete mathematics. This chapter introduces some basic principles of enumeration and provides applications to elementary problems of enumeration and of computation of combinatorial sums. The concept of a bijection (one-to-one correspondence) of finite sets will be of primary importance.

2.1 Finite Sets

Before studying techniques of enumeration, we should define precisely the concept of a finite set and that of the cardinality (number of elements) of such a set. We recall the notation $[m] := \{1, 2, \ldots, m\}$ for $m \in \mathbb{N}$, where $[m] := \varnothing$ for $m = 0$, and that the composition of two bijective maps is also bijective.

Definition 2.1.1. *A set S is called finite if there exists $m \in \mathbb{N}$ and a bijective map $f : S \to [m]$. The number m is called the cardinality, or number of elements, of S and is denoted by $|S|$, or by $\#S$.*

This number is unique for a given finite set S. Indeed, suppose that there exist bijective maps $f : S \to [m]$ and $g : S \to [n]$ for some $m, n \in \mathbb{N}$. Then, the composition of g with the inverse of f is a bijection from $[m]$ to $[n]$. The existence of such a map implies (see Exercise 1) that $m = n$.

The remainder of this chapter discusses methods to compute $\#S$. Sets that are not finite are called *infinite*.

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2.2 Basic Principles of Enumeration

We begin with two fundamental principles of enumeration. The first is a consequence of Definition 2.1.1; the second has already been applied silently (explain how) in the argument given in Example 1.1.8.

Proposition 2.2.1. *Let $\varphi : X \to Y$ be a bijective map of finite sets. Then, $\#X = \#Y$.*

Proof. Let $\#Y = m$. Then, by Definition 2.1.1, there exists a bijective map $f : Y \to [m]$. Since the composite map $f \circ \varphi : X \to [m]$ is also bijective, according to Definition 2.1.1, we have $\#X = m = \#Y$. □

Proposition 2.2.2 (Addition Principle). *Let $A_1, A_2, \ldots, A_n$ be pairwise disjoint (meaning that $A_i \cap A_j = \varnothing$ for $i \neq j$) finite sets. Then,*

$$\# \bigcup_{i=1}^{n} A_i = \sum_{i=1}^{n} \#A_i. \tag{2.1}$$

Proof. The case $n = 1$ is trivial. We let $n = 2$ and set $\#A_1 = m_1$ and $\#A_2 = m_2$. According to Definition 2.1.1, there exist bijective maps $f_1 : A_1 \to [m_1]$ and $f_2 : A_2 \to [m_2]$. We define the map $f : A_1 \cup A_2 \to [m_1 + m_2]$ by setting

$$f(x) = \begin{cases} f_1(x), & \text{if } x \in A_1 \\ f_2(x) + m_1, & \text{if } x \in A_2 \end{cases}$$

and leave it to the reader to verify that f is also bijective. As a result,

$$\#(A_1 \cup A_2) = m_1 + m_2 = (\#A_1) + (\#A_2).$$

To prove the statement in full generality, we proceed by induction on n. Assuming that the statement holds for $n - 1$, we consider pairwise disjoint finite sets $A_1, A_2, \ldots, A_n$ and write $\cup_{i=1}^{n} A_i = (\cup_{i=1}^{n-1} A_i) \cup A_n$. We observe that $(\cup_{i=1}^{n-1} A_i) \cap A_n = \varnothing$ and apply the special case $n = 2$ and the induction hypothesis to conclude that

$$\# \bigcup_{i=1}^{n} A_i = \left(\# \bigcup_{i=1}^{n-1} A_i \right) + (\#A_n) = \left(\sum_{i=1}^{n-1} \#A_i \right) + (\#A_n) = \sum_{i=1}^{n} \#A_i.$$

This completes the proof. □

We recall that given a set Ω and a subset $S \subseteq \Omega$, the *complement* of S in Ω is defined as $\Omega \setminus S = \{x \in \Omega : x \notin S\}$.

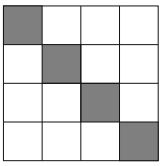


Figure 2.1 A 4×4 chessboard

Corollary 2.2.1. *Let Ω be a finite set and $S \subseteq \Omega$ be a subset. Then,*

$$\#(\Omega \setminus S) = (\#\Omega) - (\#S).$$

Proof. Since Ω is the disjoint union of S and $\Omega \setminus S$, we have $\#\Omega = (\#S) + \#(\Omega \setminus S)$ by Proposition 2.2.2 and the proof follows. □

Let's apply these principles to some specific problems.

Example 2.2.1. How many of the squares of an $n \times n$ chessboard lie above its main diagonal? For $n = 4$, we have shaded the squares *on* the main diagonal in Figure 2.1. There are six squares above those.

Since the chessboard has n rows and every row has n squares, according to the addition principle, the total number of squares of the chessboard is equal to $n + n + \cdots + n = n^2$. There are exactly n squares on the main diagonal and clearly, the squares above the main diagonal are in one-to-one correspondence with those below it. Therefore, if a_n is the number of squares of the chessboard above the main diagonal, then $n^2 = n + 2a_n$ by Propositions 2.2.1 and 2.2.2. As a result, $a_n = (n^2 - n)/2$ for every positive integer n.

Another application of the addition principle shows that $a_n = 1+2+\cdots+(n-1)$. Hence, our arguments yield the identity $1 + 2 + \cdots + (n - 1) = n(n - 1)/2$ of Exercise 2 (a) of Chapter 1. □

REMARK. In the previous example we proved the identity $1 + 2 + \cdots + (n - 1) = n(n - 1)/2$ by counting the elements of a finite set in two different ways. This method is usually referred to as *double counting*; it will be applied frequently in this book, in various forms.

Example 2.2.2. Let a_n be the number of subsets of $[n] = \{1, 2, \ldots, n\}$. We have $a_0 = 1$, $a_1 = 2$, $a_2 = 4$, and $a_3 = 8$. For instance, the four subsets of $\{1, 2\}$ are $\varnothing$, $\{1\}$, $\{2\}$, and $\{1, 2\}$. We will show that $a_n = 2^n$ for every $n \in \mathbb{N}$.

First Proof. Let A_n be the powerset (set of all subsets) of $[n]$. We observe that for $S \subseteq [n]$, either $n \notin S$ or $n \in S$. Therefore, setting $B_n = \{S \subseteq [n] : n \in S\}$, we have $A_n = A_{n-1} \cup B_n$ and $A_{n-1} \cap B_n = \varnothing$. The addition principle then implies that

$$a_n = \#A_n = a_{n-1} + (\#B_n).$$

Clearly, there is a one-to-one correspondence between the sets A_{n-1} and B_n, namely the map $\varphi : A_{n-1} \to B_n$ with $\varphi(S) = S \cup \{n\}$ for $S \in A_{n-1}$. Hence, in view of

Proposition 2.2.1, we have $\#B_n = \#A_{n-1} = a_{n-1}$. We conclude that $a_n = 2a_{n-1}$ for every positive integer n. From this recurrence and the initial condition $a_0 = 1$, we deduce the formula $a_n = 2^n$ by induction on n. □

Second Proof. Let A_n be the powerset of $[n]$, as before, and let $B_n = \{0, 1, \ldots, 2^n - 1\}$. For $S \subseteq [n]$ we set

$$\varphi_n(S) = \varepsilon_1 + \varepsilon_2 2 + \varepsilon_3 2^2 + \cdots + \varepsilon_n 2^{n-1},$$

where

$$\varepsilon_i = \begin{cases} 1, & \text{if } i \in S \\ 0, & \text{if } i \notin S. \end{cases}$$

For example, if $n = 5$ and $S = \{2, 4\}$, then $\varepsilon_1 = \varepsilon_3 = \varepsilon_5 = 0$ and $\varepsilon_2 = \varepsilon_4 = 1$ and hence $\varphi_n(S) = 0 + 2 + 0 + 2^3 + 0 = 10$.

Applying the identity (1.4) for $a = 2$ we find that $0 \leq \varphi_n(S) \leq 1 + 2 + \cdots + 2^{n-1} = 2^n - 1$, so that $\varphi_n(S) \in B_n$, for every $S \subseteq [n]$. Thus, we have defined a map $\varphi_n : A_n \to B_n$, which, for $n = 3$, is the following:

$$\begin{array}{rlrl} \varnothing &\mapsto 0 + 0 \cdot 2 + 0 \cdot 2^2 = 0 & \{3\} &\mapsto 0 + 0 \cdot 2 + 1 \cdot 2^2 = 4 \\ \{1\} &\mapsto 1 + 0 \cdot 2 + 0 \cdot 2^2 = 1 & \{1, 3\} &\mapsto 1 + 0 \cdot 2 + 1 \cdot 2^2 = 5 \\ \{2\} &\mapsto 0 + 1 \cdot 2 + 0 \cdot 2^2 = 2 & \{2, 3\} &\mapsto 0 + 1 \cdot 2 + 1 \cdot 2^2 = 6 \\ \{1, 2\} &\mapsto 1 + 1 \cdot 2 + 0 \cdot 2^2 = 3 & \{1, 2, 3\} &\mapsto 1 + 1 \cdot 2 + 1 \cdot 2^2 = 7 \end{array}$$

We claim that φ_n is bijective for every n. Indeed, by Proposition 1.1.5, every $y \in B_n$ can be written uniquely in the form $y = \varepsilon_1 + \varepsilon_2 2 + \cdots + \varepsilon_n 2^{n-1}$ with $\varepsilon_i \in \{0, 1\}$ for $i \in [n]$. As a result, there exists a unique $S \subseteq [n]$ such that $\varphi_n(S) = y$, namely $S = \{i \in [n] : \varepsilon_i = 1\}$. Given that $\varphi_n : A_n \to B_n$ is a bijection, Proposition 2.2.1 yields that $\#A_n = \#B_n = 2^n$ for every n. □

REMARK. The first proof (or a suitable application of Proposition 2.2.1) shows that the number of subsets of X is equal to 2^n for every n-element set X. □

The second proof in Example 2.2.2 applies directly the fundamental principle of Proposition 2.2.1 (but requires a higher level of creativity than the first); such a proof is called a *bijective proof*. Bijective proofs are always interesting and often give additional useful information about the enumeration problem in question. For example, the bijective proof in Example 2.2.2 shows that for every $k \in \{0, 1, \ldots, n\}$, the number of k-element subsets of the set $[n]$ is equal to the number of integers $x \in \{0, 1, \ldots, 2^n - 1\}$ whose binary representation has exactly k nonzero digits. We will give a bijective proof of an important result about the enumeration of trees in Section 5.3.2 (see Sections 2.3.6 and 2.3.7 and the exercises of this chapter for other applications).

The reasoning in the first proof in Example 2.2.2 leads to a principle of enumeration with wide applicability, namely the multiplication principle. To state

it precisely, we need to introduce the notion of an m-to-one correspondence, which generalizes that of a bijection (the special case $m = 1$).

Definition 2.2.1. *Let $m \in \mathbb{N}$. A map $f : X \to Y$ is said to be m-to-one if for every $y \in Y$, there exist exactly m elements $x \in X$ such that $f(x) = y$.*

Equivalently, $f : X \to Y$ is m-to-one if the inverse image $f^{-1}(y) = \{x \in X : f(x) = y\}$ has cardinality m for every $y \in Y$.

EXAMPLE. The map $f : \mathbb{Z} \to \mathbb{N}$ with $f(x) = |x|$ for $x \in \mathbb{Z}$ is not m-to-one for any $m \in \mathbb{N}$, because there are exactly two values of $x \in \mathbb{Z}$ such that $f(x) = 1$, while there exists a unique $x \in \mathbb{Z}$ such that $f(x) = 0$.

The restriction $g : \mathbb{Z}\setminus\{0\} \to \mathbb{Z}_{>0}$ of f, with $g(x) = |x|$ for $x \in \mathbb{Z}\setminus\{0\}$, is two-to-one, because for every $y \in \mathbb{Z}_{>0}$, there are exactly two values of $x \in \mathbb{Z}\setminus\{0\}$, namely $x = y$ and $x = -y$, such that $g(x) = y$. □

EXAMPLE. Let A_n be the powerset of $[n]$ for $n \in \mathbb{N}$, as in Example 2.2.2, and let $f : A_n \to A_{n-1}$ be the map defined by

$$f(S) = S\setminus\{n\} = \begin{cases} S, & \text{if } n \notin S \\ S\setminus\{n\}, & \text{if } n \in S \end{cases}$$

for $S \in A_n$ and $n \geq 1$. For example, for $n = 5$, we have $f(\{1, 3, 5\}) = f(\{1, 3\}) = \{1, 3\}$.

The map f is two-to-one because for every $T \in A_{n-1}$, there are exactly two elements $S \in A_n$ such that $f(S) = T$, namely $S = T$ and $S = T \cup \{n\}$. This property implies that A_n has twice as many elements as A_{n-1}. □

The following statement generalizes Proposition 2.2.1 and expresses the multiplication principle, in a simplified form.

Proposition 2.2.3. *Let $f : X \to Y$ be an m-to-one map of finite sets. Then, $\#X = m \cdot (\#Y)$.*

Proof. We observe that X can be written as the disjoint union

$$X = \bigcup_{y \in Y} f^{-1}(y) \tag{2.2}$$

of the sets $f^{-1}(y)$, for $y \in Y$. There are as many as $\#Y$ such sets and, according to our hypothesis, each one of them has exactly m elements. Thus, the result follows from the addition principle. □

REMARK. The proof of the previous proposition shows that the assumption that the set X is finite follows from the other assumptions and can thus be omitted from the statement.

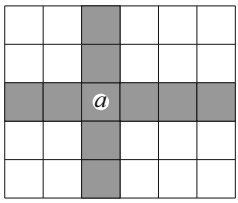


Figure 2.2 Squares in the same row or column with a

Example 2.2.3. Let $m, n \geq 2$ be integers. In how many ways can two distinct rooks be placed in the mn squares of an $m \times n$ chessboard (each rook in a single square), so that the two rooks lie in different rows and different columns of the chessboard?

Let Y be the set of squares of the chessboard and X be the set of (ordered) pairs (a, b), where $a, b \in Y$ lie in different rows and different columns. Clearly, $\#Y = mn$. To compute $\#X$, we consider the map $f : X \to Y$ with $f(a, b) = a$ for every such pair. We claim that this map is p-to-one for $p = (m-1)(n-1)$. Indeed, given any $a \in Y$, exactly $m + n - 1$ squares of Y lie in the same row or column with a and hence, exactly $mn - m - n + 1 = (m-1)(n-1)$ squares do not (see Figure 2.2 for an example with $m = 6$, $n = 5$). This means that there are exactly $(m-1)(n-1)$ pairs $x \in X$ such that $f(x) = a$. Proposition 2.2.3 allows us to conclude that $\#X = p(\#Y) = m(m-1)n(n-1)$.

In other words, there are mn ways to choose the square of the first rook and for every such choice, there are $(m-1)(n-1)$ ways to choose the square of the second, given that the latter has to lie in a different row and different column from the first. Therefore, there are $mn(m-1)(n-1)$ placements of the two rooks on the chessboard with the desired properties. □

Example 2.2.4. Let's count the pairs (S, a), where S is a subset of $[n]$ and a is an element of S. For $n = 2$, these pairs are the following: $(\{1\}, 1)$, $(\{2\}, 2)$, $(\{1, 2\}, 1)$, and $(\{1, 2\}, 2)$.

Let X be the set of pairs (S, a) such that $S \subseteq [n]$ and $a \in S$, and let $f : X \to [n]$ be the map defined by setting $f(S, a) = a$ for every pair $(S, a) \in X$. We observe that f is m-to-one for $m = 2^{n-1}$ since for every $a \in S$, exactly 2^{n-1} subsets of $[n]$ contain a (explain why). This fact and Proposition 2.2.3 imply that $\#X = mn = n2^{n-1}$.

In other words, to choose a pair (S, a) with the desired properties, there are n choices for $a \in [n]$ and for every such choice, there are 2^{n-1} choices of a subset S of $[n]$ such that $a \in S$. Therefore, the number of pairs we wish to compute equals $n2^{n-1}$. □

In the sequel, to avoid our exposition being too technical, we will employ arguments similar to the second argument given in either of the previous two examples when we apply Proposition 2.2.3.

Example 2.2.5. How many subsets of $[2n] = \{1, 2, \ldots, 2n\}$ contain at least one odd number?

The total number of subsets of $[2n]$ is equal to 2^{2n}. Those subsets that do not contain an odd number are exactly the subsets of $\{2, 4, \ldots, 2n\}$ and hence, are counted by 2^n. Corollary 2.2.1 then implies that the number of subsets we wish to compute is equal to $2^{2n} - 2^n = 2^n(2^n - 1)$. We find the same answer by applying Proposition 2.2.3. Indeed, every subset of $[2n]$, which contains at least one odd number, can be written uniquely as the disjoint union $S \cup T$, where S is a subset of $\{2, 4, \ldots, 2n\}$ and T is a nonempty subset of $\{1, 3, \ldots, 2n - 1\}$. There are 2^n ways to choose S and for each such choice, there are $2^n - 1$ ways to choose T. Therefore, there are exactly $2^n(2^n - 1)$ subsets of $[2n]$ with the required property. □

The multiplication principle of enumerative combinatorics, in its most general form, follows by repeated application of Proposition 2.2.3.

Proposition 2.2.4 (Multiplication Principle). *Let $X_1, X_2, \ldots, X_n$ be sets, $m_1, m_2, \ldots, m_n$ be positive integers and $f_i : X_i \to X_{i-1}$ be maps for $2 \leq i \leq n$. If f_i is m_i-to-one for $2 \leq i \leq n$ and $\#X_1 = m_1$, then $\#X_n = m_1 m_2 \cdots m_n$.*

Proof. Applying Proposition 2.2.3 to the maps $f_i : X_i \to X_{i-1}$ gives that $\#X_i = m_i(\#X_{i-1})$ for $2 \leq i \leq n$. Hence,

$$\begin{aligned} \#X_n &= m_n(\#X_{n-1}) = m_n m_{n-1}(\#X_{n-2}) = \cdots \\ &= m_n m_{n-1} \cdots m_2(\#X_1) = m_1 m_2 \cdots m_n \end{aligned}$$

and the proof follows. □

The multiplication principle can be expressed in less technical language as follows. Suppose that a set X_n in constructed by an n-stage process, with which $X_1, X_2, \ldots, X_n$ are constructed successively. Suppose also that there are positive integers $m_2, \ldots, m_n$ such that at each stage $i \geq 2$, every element of the set X_i is produced from some (any) element of X_{i-1} in m_i different ways. Then, $\#X_n = m_1 m_2 \cdots m_n$, where $m_1 = \#X_1$. This is the rule expressed by Proposition 2.2.4 in more precise mathematical language, by which the elements of X_i, which are produced from $y \in X_{i-1}$ are exactly the inverse images of y under $f_i : X_i \to X_{i-1}$. Let us emphasize that for the multiplication principle to be applicable, it is necessary that the number of choices m_i that are available at stage i of the process is independent of the choices that have been made at all previous stages.

EXAMPLE. Let's return to Example 2.2.2. To choose a subset S of the set $[n]$, one may decide whether each of $1, 2, \ldots, n$ belongs to S or not by an n-stage process. At the first stage, one has the two choices $1 \notin S$ or $1 \in S$. For every such choice, one then has the two choices $2 \notin S$ or $2 \in S$, and so on. At the last stage, regardless of the choices already made, one has the two choices $n \notin S$ or $n \in S$. Thus, according to the multiplication principle, there is a total of $2 \cdot 2 \cdots 2 = 2^n$ subsets of $[n]$. The process of choices is often represented by a tree diagram, such as that of Figure 2.3 in the case $n = 3$.

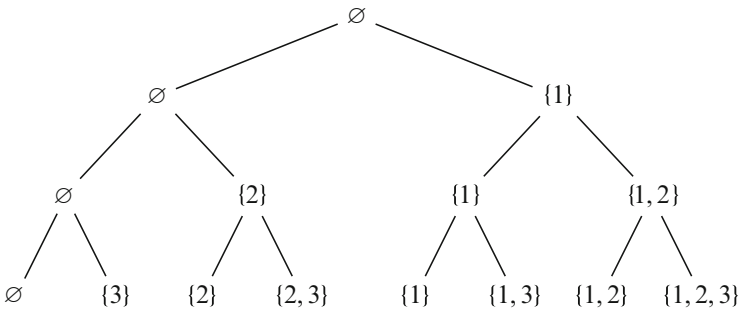


Figure 2.3 A tree diagram

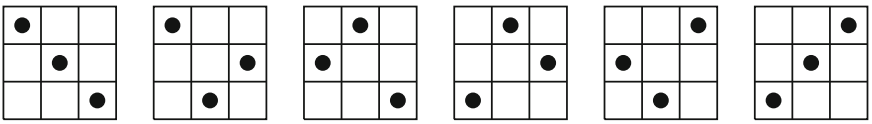


Figure 2.4 Nonattacking rook placements for $n = 3$

To express the previous argument in the language of Proposition 2.2.4, let X_n be the powerset of $[n]$. As already mentioned, the map $f_i : X_i \to X_{i-1}$ with $f_i(S) = S \setminus \{i\}$ for every $S \in X_i$ is two-to-one, for all $2 \leq i \leq n$. Thus, the assumptions of Proposition 2.2.4 for $f_2, \ldots, f_n$ are satisfied with $m_i = 2$ for every $i \in [n]$ and hence, $\#X_n = m_1 m_2 \cdots m_n = 2^n$.

Example 2.2.6. In how many ways can n identical rooks be placed in the squares of an $n \times n$ chessboard (each rook in a single square), so that no two rooks lie in the same row or column of the chessboard? Rooks placed with this property are called *nonattacking*. Figure 2.4 shows the six possible placements for $n = 3$.

We observe that every row of the chessboard must contain exactly one rook. Let's place the rooks successively in rows $1, 2, \ldots, n$, numbered from top to bottom. There are n ways to place a rook in the first row, which consists of n squares. Given this choice, there are $n - 1$ ways to place a rook in the second row, so that the square of the second row, which lies in the same column with the rook already placed in the first row, remains empty. Similarly, given our first two choices, there are $n - 2$ ways to place a rook in the third row, and so on. Finally, at the end of the process, there is only one spot available in the last row for a rook to be placed there. Thus, according to the multiplication principle, there are exactly $n(n-1)\cdots 2 \cdot 1 = n!$ such rook placements.

This argument translates in the formal language of Proposition 2.2.4 as follows. For $i \in [n]$, let X_i be the set the elements of which are sets of i squares of the $n \times n$ chessboard (the squares in which the rooks are to be placed), one in each of the rows $1, 2, \ldots, i$ and no two in the same column. We wish to count the elements of X_n. For $2 \leq i \leq n$, let $f_i : X_i \to X_{i-1}$ be the map for which $f_i(x)$ is the set

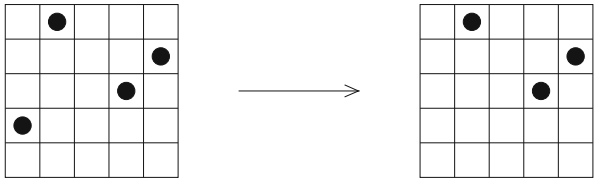


Figure 2.5 Example for the map f_i with $n = 5$ and $i = 4$

obtained from $x \in X_i$ by deleting the unique square of row i, which belongs to x (see Figure 2.5 for an example). The map f_i is m_i-to-one for $m_i = n - i + 1$ because for every $y \in X_{i-1}$, there are exactly $n - i + 1$ squares in row i, which do not lie in the same column with any of the $i - 1$ squares of y. Since $\#X_1 = n$, we may apply Proposition 2.2.4 and conclude that $\#X_n = (\#X_1)m_2 \cdots m_n = n(n-1) \cdots 2 \cdot 1 = n!$.

How many of these $n!$ nonattacking rook placements have no rook placed on the main diagonal of the chessboard (shown in Figure 2.1 for $n = 4$)? This question cannot be answered by a direct application of the multiplication principle. Our previous argument, for instance, does not work because the number of ways to place a rook in the second row now depends (explain how) on where we have chosen to place a rook in the first row. We will solve this problem with a different method in Section 3.2.3. □

The following corollary can be viewed as a special case of the multiplication principle.

Corollary 2.2.2. *For any finite sets $X_1, X_2, \ldots, X_n$,*

$$\#\, X_1 \times X_2 \times \cdots \times X_n = \prod_{i=1}^{n} \#X_i,$$

where $X_1 \times X_2 \times \cdots \times X_n = \{(x_1, x_2, \ldots, x_n) : x_1 \in X_1, x_2 \in X_2, \ldots, x_n \in X_n\}$.

Proof. Let $m_i = \#X_i$ for $i \in [n]$. To form an element $(x_1, x_2, \ldots, x_n)$ of $X_1 \times X_2 \times \cdots \times X_n$, there are m_1 ways to choose $x_1 \in X_1$. For every such choice, there are m_2 ways to choose $x_2 \in X_2$, and so on. From the multiplication principle, it follows that $\#X_1 \times X_2 \times \cdots \times X_n = m_1 m_2 \cdots m_n$.

Let's translate this argument in the more formal language of Proposition 2.2.4. We set $Y_i = X_1 \times X_2 \times \cdots \times X_i$ for every $i \in [n]$ and wish to show that $\#Y_n = m_1 m_2 \cdots m_n$. For $i \in \{2, 3, \ldots, n\}$, we define the projection map $f_i : Y_i \to Y_{i-1}$ by setting

$$f_i(x_1, x_2, \ldots, x_i) = (x_1, x_2, \ldots, x_{i-1})$$

for $(x_1, x_2, \ldots, x_i) \in Y_i$. We then observe that for every $y \in Y_{i-1}$, there are exactly m_i elements $x \in Y_i$ such that $f_i(x) = y$ (as many as the elements $x_i \in X_i$). This means that the map f_i is m_i-to-one for every i and the result follows from Proposition 2.2.4. □

More applications of the multiplication principle are given in the following sections and in the exercises. In the sequel, when applying the multiplication principle, we will use arguments similar to the first argument in Example 2.2.6 and leave to the reader their translation in the language of Proposition 2.2.4.

2.3 Applications

Let us apply the principles of Section 2.2 to a few more elementary enumeration problems. More applications can be found in the exercises of this chapter, as well as in subsequent chapters.

2.3.1 Maps

How many maps $f : [m] \to [n]$ are there?

Proposition 2.3.1. *Let X and Y be finite sets with cardinalities m and n, respectively. Then, the number of maps $f : X \to Y$ is equal to n^m.*

Proof. Since $\#X = m$, we may write $X = \{x_1, x_2, \ldots, x_m\}$ for some pairwise distinct elements $x_1, x_2, \ldots, x_m$ of X (explain why). To define a map $f : X \to Y$, there are n choices for $f(x_i) \in Y$ for every $i \in [m]$. It follows from the multiplication principle that there are exactly $n \cdot n \cdots n = n^m$ maps $f : X \to Y$. □

How many maps $f : [m] \to [n]$ are one-to-one? How many are onto? The answer to the second question will be given in Section 3.2.1.

Proposition 2.3.2. *Let X and Y be finite sets with cardinalities m and n, respectively.*

(a) *The number of injective maps $f : X \to Y$ is equal to $n(n-1)\cdots(n-m+1)$, if $m \leq n$, and zero otherwise.*
(b) *The number of bijective maps $f : X \to Y$ is equal to $n!$, if $m = n$, and zero otherwise.*

Proof. Let us write $X = \{x_1, x_2, \ldots, x_m\}$, as in the proof of Proposition 2.3.1. If $m > n$, then part (a) follows from Exercise 1 (b).

Assume that $m \leq n$. To define an injective map $f : X \to Y$, we need to choose pairwise distinct elements $f(x_1), f(x_2), \ldots, f(x_m)$ of Y. There are n ways to choose $f(x_1) \in Y$. For every such choice, there are $n-1$ ways to choose $f(x_2) \in Y$ so that $f(x_2) \neq f(x_1)$. Similarly, regardless of the choices of $f(x_1)$ and $f(x_2)$, there are $n-2$ ways to choose $f(x_3) \in Y$ so that $f(x_3) \neq f(x_1), f(x_2)$, and so on. Finally, there are $n-m+1$ ways to choose $f(x_m) \in Y$ so that $f(x_m) \neq f(x_1), \ldots, f(x_{m-1})$ and the result follows from the multiplication principle. We have proven part (a).

Part (b) is a special case of part (a), when $m = n$ (because of Exercise 3). For $m \neq n$, it follows from Proposition 2.2.1. □

EXAMPLE. There are $5^3 = 125$ maps $f : [3] \to [5]$ and $5 \cdot 4 \cdot 3 = 60$ of them are injective. There are $5^5 = 3125$ maps $g : [5] \to [5]$ and $5! = 120$ of them are bijective.

EXAMPLE. Four boys and seven girls meet at a party. The boys dance simultaneously, each with one girl. In how many ways can this be done? We wish to compute the number of injective maps from the set of four boys to the set of seven girls. According to Proposition 2.3.2, this number is equal to $7 \cdot 6 \cdot 5 \cdot 4 = 840$.

2.3.2 Permutations

Let Ω be a finite set. A sequence $(\sigma_1, \sigma_2, \ldots, \sigma_n)$ in which every element of Ω appears exactly once is called a *permutation* of Ω. For example, the set $[3] = \{1, 2, 3\}$ has the following six permutations: $(1, 2, 3)$, $(1, 3, 2)$, $(2, 1, 3)$, $(2, 3, 1)$, $(3, 1, 2)$, and $(3, 2, 1)$. The sequence $(4, 2, 5, 1, 3)$ is a permutation of the set $[5]$.

How many permutations of Ω are there? We have already answered this question in Proposition 2.3.2, since the permutations of a set Ω with n elements are exactly the bijective maps $f : [n] \to \Omega$. More generally, we say that a sequence $(\sigma_1, \sigma_2, \ldots, \sigma_k)$ of elements of Ω in which every element of Ω appears at most once is a *k-partial permutation* of Ω (this is nonstandard terminology). For example, the 2-partial permutations of the set $[3]$ are $(1, 2)$, $(1, 3)$, $(2, 1)$, $(2, 3)$, $(3, 1)$, $(3, 2)$, and the sequence $(3, 5, 2, 6, 8)$ is a 5-partial permutation of $[9]$. Since the k-partial permutations of Ω coincide with the injective maps $f : [k] \to S$, the following proposition is a consequence (essentially a restatement) of Proposition 2.3.2.

Proposition 2.3.3. *Let Ω be a finite set of cardinality n. The number of k-partial permutations of Ω is equal to $n(n-1)\cdots(n-k+1)$ for every $k \in [n]$. In particular, the number of permutations of Ω is equal to $n!$.*

EXAMPLE. There are $8 \cdot 7 \cdot 6 = 336$ ordered triples of pairwise distinct elements of the set $[8]$ and $7! = 5040$ permutations of the set $[7]$.

Remark 2.3.1. As a consequence of Proposition 2.3.3 and Example 2.2.6, there are as many permutations of the set $[n]$ as placements of n nonattacking identical rooks on an $n \times n$ chessboard. This statement can be deduced directly from the fundamental principle of Proposition 2.2.1 as follows: Let X_n and Y_n be the sets of permutations of $[n]$ and placements of n rooks, respectively, just described. Let's number the rows and columns of the chessboard as $1, 2, \ldots, n$ from top to bottom and from left to right, respectively. Given $\sigma = (\sigma_1, \sigma_2, \ldots, \sigma_n) \in X_n$, let $\varphi(\sigma)$ be the rook placement for which there is a rook in the square where row i meets column σ_i for every $i \in [n]$. For $n = 3$, the six permutations of $[3]$, in the order they were listed in the beginning of this section, are mapped under φ to the six

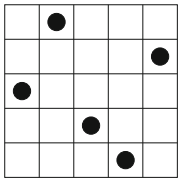


Figure 2.6 Five nonattacking rooks

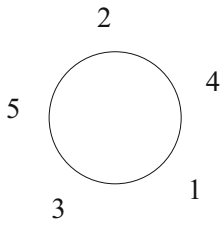


Figure 2.7 A cyclic permutation of [5]

rook placements of Figure 2.4, read from left to right. For $n = 5$, the permutation $(2, 5, 1, 3, 4)$ maps to the rook placement shown in Figure 2.6. It follows from the definition of permutation that $\varphi(\sigma) \in Y_n$ for every $\sigma \in X_n$ and that $\varphi : X_n \to Y_n$ is a bijection. Therefore, $\#X_n = \#Y_n$.

Example 2.3.1. A dancing group consists of n men and n women. In how many ways can the $2n$ dancers form a line to dance a line dance, if the first dancer must be a man and no two dancers of the same sex are allowed to stand next to each other?

According to Proposition 2.3.3, there are $n!$ ways to line up the men. For every such line arrangement, according to Proposition 2.3.2 (b), there are $n!$ ways to place the n women in the $n - 1$ spots between successive men in the line and in the spot right after the last man, so that exactly one woman is placed in each of these n spots. By the multiplication principle, the number of line arrangements of the $2n$ dancers we wish to compute equals $(n!)^2$.

2.3.3 Cyclic Permutations

In how many ways can n knights $1, 2, \ldots, n$ sit at a round table with n identical seats? We consider two such arrangements identical if one of them can be obtained from the other by rotation or, equivalently, if knight i is followed by the same knight in the clockwise order in the two arrangements, for every $i \in [n]$.

These configurations, which we wish to enumerate, are called *cyclic permutations* of the set $[n]$. Figure 2.7 shows one cyclic permutation for $n = 5$. We extend the definition of a cyclic permutation of any finite set in the obvious way.

Proposition 2.3.4. *The number of cyclic permutations of any n-element set is equal to $(n - 1)!$.*

Proof. We may assume that $[n]$ is the given set. Let A_n and C_n be the sets of all permutations and all cyclic permutations of $[n]$, respectively. By Propositions 2.2.1 and 2.3.3, it suffices to find a bijective map $\varphi : A_{n-1} \to C_n$. Given $\sigma = (\sigma_1, \ldots, \sigma_{n-1}) \in A_{n-1}$, we let $\varphi(\sigma) \in C_n$ be the cyclic permutation of $[n]$ such that (i) for every $i \in \{2, 3, \ldots, n-1\}$, σ_{i-1} is followed by σ_i in the clockwise order; and (ii) n follows σ_{n-1} and is followed by σ_1. For example, if $n = 5$ and $\sigma = (2, 4, 1, 3)$, then $\varphi(\sigma)$ is the cyclic permutation of Figure 2.7. The map $\varphi : A_{n-1} \to C_n$, which we have defined, is a bijection. Indeed, for every cyclic permutation $\tau \in C_n$, there exists a unique permutation $\sigma \in A_{n-1}$ such that $\varphi(\sigma) = \tau$; specifically, σ is the sequence that lists the elements of τ other than n in the clockwise order, beginning with the one that follows n. □

Example 2.3.2. A dancing group consists of n men and n women. In how many ways can the $2n$ dancers form a circle to dance a circle dance, if no two dancers of the same sex are allowed to stand next to each other?

According to Proposition 2.3.4, there are $(n-1)!$ cyclic permutations of the set of n men. For every such permutation, according to Proposition 2.3.2 (b), there are $n!$ ways to place the n women in the n spots between successive men, so that exactly one woman is placed in each spot. By the multiplication principle, the number of cyclic permutations of the $2n$ dancers we wish to compute equals $(n-1)!n!$. □

2.3.4 *Subsets and Binomial Coefficients*

Let Ω be an n-element set. Any subset of Ω of cardinality k will be referred to as a *k-subset* of Ω, or as a *k-combination* of elements of Ω. We will denote by $\binom{n}{k}$ the number of k-subsets of Ω. The numbers $\binom{n}{k}$ are called *binomial coefficients*.

EXAMPLE. We have $\binom{4}{2} = 6$ because there are six 2-subsets of the set $\{a, b, c, d\}$, namely $\{a, b\}$, $\{a, c\}$, $\{a, d\}$, $\{b, c\}$, $\{b, d\}$, and $\{c, d\}$.

Proposition 2.3.5. *For all $n \in \mathbb{N}$ and $k \in \{0, 1, \ldots, n\}$,*

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} = \frac{n(n-1)\cdots(n-k+1)}{k!}, \tag{2.3}$$

where $0! = 1$.

Proof. Let us denote by $A(n, k)$ the set of k-partial permutations of an n-element set Ω. We will count the elements of $A(n, k)$ in two different ways. First, we have $\#A(n, k) = n(n-1)\cdots(n-k+1)$ by Proposition 2.3.3. Second, we observe that there are $\binom{n}{k}$ ways to choose the set $S \subseteq \Omega$, the elements of which appear in a k-partial permutation σ of Ω and for every such choice, there are $k!$ ways to rearrange the elements of S so as to form σ. With this reasoning, it follows from the multiplication principle that $\#A(n, k) = k!\binom{n}{k}$. Our two expressions for $\#A(n, k)$

imply that

$$k!\binom{n}{k} = n(n-1)\cdots(n-k+1)$$

and the proof follows. □

The following statement explains the terminology "binomial coefficient" adopted for the numbers $\binom{n}{k}$. The assumption there that $xy = yx$ is essential (there are situations in mathematics where variables x, y do *not* commute; for instance, they might represent square matrices of the same size that do not commute).

Theorem 2.3.1 (Binomial Theorem). *For commuting variables x, y (meaning $xy = yx$),*

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k} \tag{2.4}$$

for every positive integer n.

Proof. Let $x_1, x_2, \ldots, x_n$ and $y_1, y_2, \ldots, y_n$ be pairwise commuting variables. For $S \subseteq [n]$, let x_S (respectively, y_S) denote the product of the variables x_i (respectively, y_i) for $i \in S$, where $x_\varnothing = y_\varnothing = 1$ by convention. By induction on n, we find (see Exercise 21) that

$$(x_1+y_1)(x_2+y_2)\cdots(x_n+y_n) = \sum_{S\subseteq[n]} x_S y_{\bar{S}}, \tag{2.5}$$

where $\bar{S} = [n] \setminus S$. For example,

$$(x_1+y_1)(x_2+y_2) = x_1x_2 + x_1y_2 + x_2y_1 + y_1y_2$$

for $n = 2$. We now set $x_1 = \cdots = x_n = x$ and $y_1 = \cdots = y_n = y$ in (2.5). We observe that $(x_1+y_1)\cdots(x_n+y_n) = (x+y)^n$, that $x_S y_{\bar{S}} = x^k y^{n-k}$ if $\#S = k$ and that there are exactly $\binom{n}{k}$ subsets S of $[n]$ with $\#S = k$ and Equation (2.4) follows. □

The following proposition lists some of the main properties of binomial coefficients.

Proposition 2.3.6. *For $n \in \mathbb{Z}_{>0}$:*

(a) $\binom{n}{k} = \binom{n}{n-k}$ *for* $k \in \{0, 1, \ldots, n\}$.

(b) $\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}$ *for* $k \in [n]$.

(c) $\displaystyle\sum_{k=0}^{n}\binom{n}{k}=2^n$.

(d) $\displaystyle\sum_{k=0}^{n}(-1)^k\binom{n}{k}=0$.

(e) $\displaystyle\binom{n}{0}<\binom{n}{1}<\cdots<\binom{n}{\lfloor n/2\rfloor}$.

First Proof. Applying the binomial theorem twice, we get

$$\sum_{k=0}^{n}\binom{n}{k}x^k y^{n-k}=(x+y)^n=(y+x)^n=\sum_{k=0}^{n}\binom{n}{k}y^k x^{n-k}$$

$$=\sum_{k=0}^{n}\binom{n}{n-k}x^k y^{n-k}.$$

Part (a) follows by comparing the coefficients of $x^k y^{n-k}$ in the extreme handsides of this string of equations. Similarly, applying the Binomial Theorem and the equality $(x+y)^n=(x+y)^{n-1}(x+y)$, we find that

$$\sum_{k=0}^{n}\binom{n}{k}x^k y^{n-k}=(x+y)^n=(x+y)\,(x+y)^{n-1}$$

$$=(y+x)\sum_{k=0}^{n-1}\binom{n-1}{k}x^k y^{n-k-1}$$

$$=\sum_{k=0}^{n-1}\binom{n-1}{k}x^k y^{n-k}+\sum_{k=0}^{n-1}\binom{n-1}{k}x^{k+1}y^{n-k-1}$$

$$=\sum_{k=0}^{n-1}\binom{n-1}{k}x^k y^{n-k}+\sum_{k=1}^{n}\binom{n-1}{k-1}x^k y^{n-k}$$

and conclude, as before, that (b) holds. Alternatively, parts (a) and (b) may be deduced from Proposition 2.3.5. Indeed,

$$\binom{n}{n-k}=\frac{n!}{(n-k)!k!}=\frac{n!}{k!(n-k)!}=\binom{n}{k}$$

for $k \in \{0, 1, \ldots, n\}$ and

$$\begin{aligned}\binom{n-1}{k} + \binom{n-1}{k-1} &= \frac{(n-1)!}{k!(n-k-1)!} + \frac{(n-1)!}{(k-1)!(n-k)!} \\ &= (n-k) \cdot \frac{(n-1)!}{k!(n-k)!} + k \cdot \frac{(n-1)!}{k!(n-k)!} = n \cdot \frac{(n-1)!}{k!(n-k)!} \\ &= \frac{n!}{k!(n-k)!} = \binom{n}{k}\end{aligned}$$

for $k \in [n]$.

Parts (c) and (d) follow from the Binomial Theorem by setting $x = y = 1$ and $x = -y = 1$, respectively. To prove part (e), we observe that for $k \geq 1$,

$$\frac{\binom{n}{k-1}}{\binom{n}{k}} = \frac{n!}{(k-1)!(n-k+1)!} \cdot \frac{k!(n-k)!}{n!} = \frac{k}{n-k+1}$$

and conclude that $\binom{n}{k} < \binom{n}{k+1}$ if $k \leq n-k$, that is, $k \leq n/2$. □

Second Proof. The following proof of parts (a)–(d) bypasses the Binomial Theorem and uses only the fundamental principles of enumeration (Propositions 2.2.1 and 2.2.2). For such a proof of part (e), see Exercise 18 of Chapter 4 or the solution of Exercise 46 of this chapter.

(a) The map that sends a k-subset S of $[n]$ to its complement $[n] \setminus S$ in $[n]$ is a one-to-one correspondence between the set of k-subsets of $[n]$ and that of $(n-k)$-subsets of $[n]$ (explain why). In view of Proposition 2.2.1, this proves part (a).

(b) The number of k-subsets of $[n]$ is equal to $\binom{n}{k}$ and for every such subset S we have either $n \notin S$ or $n \in S$. Moreover, the number of k-subsets S of $[n]$ with $n \notin S$ is equal to $\binom{n-1}{k}$ and the numbers of k-subsets S of $[n]$ with $n \in S$ is equal to $\binom{n-1}{k-1}$ (explain why). Part (b) follows from these observations and the addition principle.

(c) Let us count the subsets of the set $[n]$ in two different ways. On the one hand, according to Example 2.2.2, their number is equal to 2^n. On the other, by the addition principle, it is equal to the sum $\sum_{k=0}^{n} \binom{n}{k}$ since there are exactly $\binom{n}{k}$ subsets of $[n]$ with k elements for every $k \in \{0, 1, \ldots, n\}$. Thus, the identity of part (c) holds.

(d) The proposed identity may be rewritten as

$$\binom{n}{0} + \binom{n}{2} + \cdots = \binom{n}{1} + \binom{n}{3} + \cdots$$

or, equivalently, as $\#X_n = \#Y_n$, where X_n (respectively, Y_n) is the set of subsets of $[n]$ with an even (respectively, odd) number of elements. As a

result, it suffices to find a one-to-one correspondence $\varphi : X_n \to Y_n$. Such a correspondence is the map $\varphi : X_n \to Y_n$ defined by setting

$$\varphi(S) = \begin{cases} S \setminus \{n\}, & \text{if } n \in S \\ S \cup \{n\}, & \text{if } n \notin S \end{cases}$$

for $S \in X_n$. For example, for $n = 4$, φ is the map

$$\begin{array}{rr} \varnothing \mapsto \{4\} & \{1,4\} \mapsto \{1\} \\ \{1,2\} \mapsto \{1,2,4\} & \{2,4\} \mapsto \{2\} \\ \{1,3\} \mapsto \{1,3,4\} & \{3,4\} \mapsto \{3\} \\ \{2,3\} \mapsto \{2,3,4\} & \{1,2,3,4\} \mapsto \{1,2,3\}. \end{array}$$

Indeed, the inverse of φ is the map $\psi : Y_n \to X_n$ with

$$\psi(T) = \begin{cases} T \setminus \{n\}, & \text{if } n \in T \\ T \cup \{n\}, & \text{if } n \notin T \end{cases}$$

for $T \in Y_n$ and the proof follows. □

REMARK. The binomial coefficients are usually displayed in a triangular array of numbers, known as *Pascal's triangle*, as shown below. The numbers that appear in row $n + 1$, counting from the top, are the binomial coefficients $\binom{n}{k}$ for $k \in \{0, 1, \ldots, n\}$.

$$\begin{array}{ccccccccccc} & & & & & 1 & & & & & \\ & & & & 1 & & 1 & & & & \\ & & & 1 & & 2 & & 1 & & & \\ & & 1 & & 3 & & 3 & & 1 & & \\ & 1 & & 4 & & 6 & & 4 & & 1 & \\ 1 & & 5 & & 10 & & 10 & & 5 & & 1 \\ & 1 & & 6 & & 15 & & 20 & & 15 & & 6 & & 1 \\ 1 & & 7 & & 21 & & 35 & & 35 & & 21 & & 7 & & 1 \\ & & & & & \vdots & & & & & \end{array}$$

Thus, according to the Binomial Theorem, the fourth and fifth rows correspond to the identities

$$\begin{aligned} (x+y)^3 &= x^3 + 3x^2y + 3xy^2 + y^3 \\ (x+y)^4 &= x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^4, \end{aligned}$$

respectively. According to Proposition 2.3.6 (b), every number in the array is equal to the sum of the two numbers of the previous row that lie exactly to its left and right (where missing elements are assumed to be zero). □

Proposition 2.3.6 deals with some equalities of the form $a = b$, the sides a, b of which may be interpreted combinatorially. The proofs we have given follow two basic methods. The first expresses a, b as the coefficients of a certain monomial in the two sides of an equality between two polynomials and the second as the cardinalities of two finite sets between which a one-to-one correspondence is given (or as the cardinality of a finite set, the elements of which are counted in two different ways). In the sequel, we apply these methods to two more examples of identities. Further examples are given in Section 2.4 and in the exercises of this chapter.

Proposition 2.3.7. *We have*

$$\sum_{k=0}^{n} k\binom{n}{k} = n2^{n-1}$$

for every $n \in \mathbb{Z}_{>0}$.

First Proof. Differentiating Equation (2.4) with respect to x, we get

$$\sum_{k=1}^{n} k\binom{n}{k} x^{k-1} y^{n-k} = n(x+y)^{n-1}.$$

The proposed identity follows by setting $x = y = 1$. □

Second Proof. We observe that

$$k\binom{n}{k} = k \cdot \frac{n!}{k!(n-k)!} = n \cdot \frac{(n-1)!}{(k-1)!(n-k)!} = n\binom{n-1}{k-1}.$$

In view of Proposition 2.3.6 (c), we conclude that

$$\sum_{k=0}^{n} k\binom{n}{k} = \sum_{k=1}^{n} n\binom{n-1}{k-1} = n\sum_{i=0}^{n-1}\binom{n-1}{i} = n2^{n-1}.$$

□

Third Proof. The left-hand side of the proposed identity may be interpreted combinatorially as the number of ways, say m, that one can form a committee E out of a set of n students and appoint one member of E as the chairman. Indeed, there are $\binom{n}{k}$ ways to choose E, if it has k members, and for every such choice there are k

ways to choose the chairman of E. As a result, $m = \sum_{k=1}^{n} k\binom{n}{k}$. Alternatively, one may first choose the chairman of the committee out of the total of n students in n ways and then choose the other members of the committee in 2^{n-1} ways (as many as there are subsets of an $(n-1)$-element set). Therefore, $m = n2^{n-1}$. □

Example 2.3.3. How many ordered pairs (S, T) of subsets of $[n]$ are there? For how many of them we have $S \subseteq T$?

A direct application of the multiplication principle shows that the answer to the first question is $2^n \cdot 2^n = 4^n$. To answer the second question, we may reason as follows. There are $\binom{n}{k}$ ways to choose $T \subseteq [n]$, if T has k elements and, for every such choice, there are 2^k choices for $S \subseteq T$. Hence, the number of pairs (S, T) with $S \subseteq T \subseteq [n]$ is equal to

$$\sum_{k=0}^{n} \binom{n}{k} 2^k = (2+1)^n = 3^n.$$

Since the answer we found is 3^n, it is reasonable to expect that the result can be deduced directly from the multiplication principle. Indeed, for $S, T \subseteq [n]$ we have $S \subseteq T$ if and only if exactly one of the following conditions holds for every $i \in [n]$:

$$i \in S \text{ and } i \in T,$$
$$i \notin S \text{ and } i \in T,$$
$$i \notin S \text{ and } i \notin T.$$

The sets S and T are uniquely determined by choosing one of these conditions for every i. For example, if $n = 4$ and $1 \notin S, 1 \in T, 2 \notin S, 2 \notin T, 3 \in S, 3 \in T, 4 \notin S$, and $4 \in T$, then $S = \{3\}$ and $T = \{1, 3, 4\}$. There are three choices for every $i \in [n]$ and hence, there are 3^n pairs (S, T) such that $S \subseteq T \subseteq [n]$. For a generalization, see Exercise 14 (a). □

2.3.5 *Permutations of Multisets and Multinomial Coefficients*

This section generalizes the main results of Sections 2.3.2 and 2.3.4 as follows. Consider a set with r elements $x_1, x_2, \ldots, x_r$ and let $\mathcal{A}$ be the collection (also called a multiset) which consists of n_i copies of x_i for every $i \in [r]$. A *permutation* of $\mathcal{A}$ is defined as any sequence $(\sigma_1, \sigma_2, \ldots, \sigma_n)$ having $n = n_1 + n_2 + \cdots + n_r$ terms in which x_i appears exactly n_i times for every $i \in [r]$. For example, (x, y, x, z, z, x) is a permutation of the collection $\{x, x, x, y, z, z\}$.

Proposition 2.3.8. *Consider a set with r elements $x_1, x_2, \ldots, x_r$. The number of permutations of the collection that consists of n_i copies of x_i for every $i \in [r]$ is equal to*

$$\binom{n}{n_1, n_2, \ldots, n_r} := \frac{n!}{n_1! n_2! \cdots n_r!}, \tag{2.6}$$

where $n = n_1 + n_2 + \cdots + n_r$.

Proof. Let $\sigma = (\sigma_1, \sigma_2, \ldots, \sigma_n)$ be such a permutation and for $i \in [r]$, let S_i be the set of positions of σ in which x_i appears, that is, $S_i = \{j \in [n] : \sigma_j = x_i\}$. By our hypotheses, $S_1, S_2, \ldots, S_r$ are mutually disjoint subsets of $[n]$ with union equal to $[n]$ and $\#S_i = n_i$ for every $i \in [r]$. Hence, there are $\binom{n}{n_1}$ ways to choose S_1 and, for every such choice, there are $\binom{n-n_1}{n_2}$ ways to choose S_2, since this is an n_2-subset of $[n] \setminus S_1$, there are $\binom{n-n_1-n_2}{n_3}$ ways to choose S_3, and so on. We conclude that there is a total of

$$\binom{n}{n_1}\binom{n-n_1}{n_2} \cdots \binom{n-n_1-\cdots-n_{r-1}}{n_r} = \frac{n!}{n_1! n_2! \cdots n_r!}$$

choices for σ, where we have applied formula (2.3) to simplify the product of binomial coefficients on the left-hand side of the above equation. □

REMARK. Proposition 2.3.8 simultaneously generalizes Propositions 2.3.3 and 2.3.5. Indeed, if $r = k + 1$ and the collection $\mathcal{A}$ consists of one copy of each of $1, 2, \ldots, k$ and $n - k$ copies of zero, then the permutations of $\mathcal{A}$ are in one-to-one correspondence with the injective maps $f : [k] \to [n]$ (explain why), and hence, with the k-partial permutations of an n-element set, and

$$\binom{n}{n-k, 1, \ldots, 1} = \frac{n!}{(n-k)!1! \cdots 1!} = \frac{n!}{(n-k)!} = n(n-1) \cdots (n-k+1),$$

in agreement with Proposition 2.3.3. Similarly, if $r = 2$ and $\mathcal{A}$ consists of k copies of 1 and $n-k$ copies of 0, then the permutations of $\mathcal{A}$ are in one-to-one correspondence with the k-subsets of $[n]$ and

$$\binom{n}{k, n-k} = \frac{n!}{k!(n-k)!},$$

in agreement with Proposition 2.3.5. □

As suggested by the proof of Proposition 2.3.8, the integer (2.6) can be interpreted combinatorially as:

- the number of sequences $(S_1, S_2, \ldots, S_r)$ of mutually disjoint subsets of $[n]$ with union equal to $[n]$ such that $\#S_i = n_i$ for every $i \in [r]$,

- the number of ways that n distinct objects $\mu_1, \mu_2, \ldots, \mu_n$ can be distributed to r distinct boxes $\Delta_1, \Delta_2, \ldots, \Delta_n$ so that exactly n_i objects end up in box Δ_i for every $i \in [r]$.

The sequences $(S_1, S_2, \ldots, S_r)$ of mutually disjoint subsets of a set Ω with union equal to Ω are called *weak ordered partitions* of Ω with r parts (where the term *weak* may be omitted if the parts S_i are nonempty).

The integers (2.6) are called *multinomial coefficients* for reasons that are explained by the following generalization of the Binomial Theorem.

Theorem 2.3.2 (Multinomial Theorem). *For pairwise commuting variables $x_1, x_2, \ldots, x_r$ and for every positive integer n,*

$$(x_1 + x_2 + \cdots + x_r)^n = \sum \binom{n}{n_1, n_2, \ldots, n_r} x_1^{n_1} x_2^{n_2} \cdots x_r^{n_r}, \tag{2.7}$$

where the sum runs through all vectors $(n_1, n_2, \ldots, n_r) \in \mathbb{N}^r$ such that $n_1 + n_2 + \cdots + n_r = n$.

Proof. We first show by induction on n that

$$(x_1 + x_2 + \cdots + x_r)^n = \sum_{f:[n]\to[r]} x_{f(1)} x_{f(2)} \cdots x_{f(n)}, \tag{2.8}$$

where the sum runs through all maps $f : [n] \to [r]$ (see Exercise 35). We then observe that the coefficient of $x_1^{n_1} x_2^{n_2} \cdots x_r^{n_r}$ in the sum of the right-hand side is equal to the number of sequences $(f(1), f(2), \ldots, f(n))$ of elements of $[r]$ such that $x_{f(1)} x_{f(2)} \cdots x_{f(n)} = x_1^{n_1} x_2^{n_2} \cdots x_r^{n_r}$. These sequences are exactly the permutations of the collection which consists of n_i copies of i, for $i \in [r]$. According to Proposition 2.3.8, the number of these permutations is equal to the multinomial coefficient (2.6) and the proof follows. □

EXAMPLE. How many permutations of the letters of the word PARAPHRASE are there? We seek to compute the number of permutations of the collection that consists of three copies of A, one copy of E, one copy of H, two copies of P, two copies of R, and one copy of S. According to Proposition 2.3.8, this number is equal to $10!/3!\,1!\,1!\,2!\,2!\,1! = 151200$. The same number is equal to

- the number of ways that ten distinct balls can be distributed to six distinct boxes so that the first box gets three balls, the fourth and the fifth box get two balls each, and the remaining three boxes get one ball each,
- the coefficient of $x_1^3 x_2 x_3 x_4^2 x_5^2 x_6$ in the expansion of $(x_1+x_2+x_3+x_4+x_5+x_6)^{10}$.

EXAMPLE. The coefficient of $x^2 y^{n-1} z^{n-1}$ in the expansion of $(x+y+z)^{2n}$ is equal to

$$\binom{2n}{2, n-1, n-1} = \frac{(2n)!}{2((n-1)!)^2}.$$

EXAMPLE. The coefficient of $x_1x_2\cdots x_n$ in the expansion of $(x_1+x_2+\cdots+x_n)^n$ is equal to $\binom{n}{1,1,\ldots,1} = n!$.

2.3.6 Compositions

Let n be a positive integer. A *composition*, or *ordered partition*, of n is defined as any sequence $\rho = (r_1, r_2, \ldots, r_k)$ of positive integers, which sum to n. The integers r_i are called the *parts* of ρ. The composition ρ may also be presented as $r_1 + r_2 + \cdots + r_k$.

EXAMPLE. The compositions of $n = 4$ are

$$\begin{array}{llll} (4) & (1,3) & (2,1,1) & (1,1,2) \\ (3,1) & (2,2) & (1,2,1) & (1,1,1,1). \end{array} \tag{2.9}$$

The number $n = 5$ has 16 compositions, from which $(3,1,1)$, $(1,3,1)$, $(1,1,3)$, $(2,2,1)$, $(2,1,2)$, and $(1,2,2)$ have three parts. For $n = 1, 2, 3$, the number of compositions of n is equal to $1, 2, 4$, respectively. □

Compositions of integers appear often in mathematics. The proof of the following statement shows that the compositions of n are in one-to-one correspondence with the subsets of $[n-1]$.

Proposition 2.3.9. *The number of compositions of n is equal to 2^{n-1} for every positive integer n. The number of compositions of n with k parts is equal to $\binom{n-1}{k-1}$.*

Proof. Let X_n be the set of compositions of n, and let Y_n be the set of all subsets of $[n-1]$. Given $\rho = (r_1, r_2, \ldots, r_k) \in X_n$, we set

$$\varphi(\rho) = \{r_1, r_1 + r_2, \ldots, r_1 + r_2 + \cdots + r_{k-1}\}.$$

For example, if $n = 9$ and $\rho = (3,1,2,2,1)$, then $\varphi(\rho) = \{3,4,6,8\}$. Our assumption that $r_1, r_2, \ldots, r_k$ are positive integers implies that $\varphi(\rho) \in Y_n$. One then verifies that the map $\varphi : X_n \to Y_n$ has the inverse $\psi : Y_n \to X_n$ defined by setting

$$\psi(S) = (s_1, s_2 - s_1, \ldots, n - s_{k-1})$$

for every $S = \{s_1, s_2, \ldots, s_{k-1}\} \subseteq [n-1]$ with $s_1 < s_2 < \cdots < s_{k-1}$. Therefore, $\varphi : X_n \to Y_n$ is a bijection and hence, $\#X_n = \#Y_n = 2^{n-1}$ by Proposition 2.2.1 and Example 2.2.2. Furthermore, we observe that the composition $\rho \in X_n$ has k parts if and only if $\varphi(\rho)$ has $k-1$ elements. As a consequence, φ restricts to a bijection from the set of compositions of n with k parts to the set of $(k-1)$-subsets of $[n-1]$. Hence, the number of these compositions is equal to $\binom{n-1}{k-1}$. □

EXAMPLE. Under the map φ of the previous proof, the eight compositions (2.9) of $n = 4$ correspond to the subsets

$$\begin{array}{cccc} \varnothing & \{1\} & \{2,3\} & \{1,2\} \\ \{3\} & \{2\} & \{1,3\} & \{1,2,3\} \end{array}$$

of [3], respectively. □

For a nontrivial application of Proposition 2.3.9, see Exercise 42.

2.3.7 Combinations with Repetition

The binomial coefficient $\binom{n}{k}$ may be interpreted as the number of vectors $(a_1, a_2, \ldots, a_n) \in \{0,1\}^n$ such that $a_1 + a_2 + \cdots + a_n = k$. The number of vectors $(a_1, a_2, \ldots, a_n) \in \mathbb{N}^n$ with $a_1 + a_2 + \cdots + a_n = k$ is often referred to as the number of *k-combinations with repetition* of n objects $x_1, x_2, \ldots, x_n$; it is denoted by $\left(\!\binom{n}{k}\!\right)$.

The number $\left(\!\binom{n}{k}\!\right)$ may be interpreted combinatorially in various ways, for instance as:

- the number of monomials $x_1^{a_1} x_2^{a_2} \cdots x_n^{a_n}$ of degree k in n variables $x_1, x_2, \ldots, x_n$ (where the degree of $x_1^{a_1} x_2^{a_2} \cdots x_n^{a_n}$ is defined to be $a_1 + a_2 + \cdots + a_n$),
- the number of ways with which k identical objects can be distributed to n distinct boxes (where it is possible that some of the boxes end up empty),
- the number of vectors $(i_1, i_2, \ldots, i_k)$ with integer coordinates $1 \le i_1 \le \cdots \le i_k \le n$.

EXAMPLE. We have $\left(\!\binom{3}{2}\!\right) = 6$ because there are six triples of natural numbers which sum to 2, namely (2, 0, 0), (1, 1, 0), (1, 0, 1), (0, 2, 0), (0, 1, 1), and (0, 0, 2). The corresponding monomials are x_1^2, x_1x_2, x_1x_3, x_2^2, x_2x_3, and x_3^2, and the corresponding increasing vectors (i_1, i_2) are (1, 1), (1, 2), (1, 3), (2, 2), (2, 3), and (3, 3).

Proposition 2.3.10. *We have*

$$\left(\!\binom{n}{k}\!\right) = \binom{n+k-1}{k} \tag{2.10}$$

for every positive integer n and every $k \in \mathbb{N}$.

Proof. Let $A(n,k)$ denote the set of vectors $(a_1, a_2, \ldots, a_n) \in \mathbb{N}^n$ such that $a_1 + a_2 + \cdots + a_n = k$. We observe that if $x = (a_1, a_2, \ldots, a_n) \in A(n,k)$ and $b_i = a_i + 1$ for every $i \in [n]$, then the b_i are positive integers with sum $b_1 + b_2 + \cdots + b_n = n + k$. Therefore, $\varphi(x) = (b_1, b_2, \ldots, b_n)$ is a composition of $n+k$ with n parts. Moreover, for every composition $y = (b_1, b_2, \ldots, b_n)$ of $n+k$ there exists a unique vector $x \in$

$A(n,k)$ such that $\varphi(x) = y$, namely $x = (a_1, a_2, \ldots, a_n)$ with $a_i = b_i - 1$ for every $i \in [n]$. Thus, we have defined a one-to-one correspondence from $A(n,k)$ to the set of compositions of $n+k$ with n parts. The result then follows from Propositions 2.2.1 and 2.3.9 and from part (a) of Proposition 2.3.6. □

EXAMPLE. Under the correspondence of the previous proof, for $n = 3$ and $k = 2$, the triples $(2,0,0)$, $(1,1,0)$, $(1,0,1)$, $(0,2,0)$, $(0,1,1)$, and $(0,0,2)$ are mapped to the compositions $(3,1,1)$, $(2,2,1)$, $(2,1,2)$, $(1,3,1)$, $(1,2,2)$, and $(1,1,3)$ of 5 with three parts, respectively. □

For a nontrivial application of Proposition 2.3.10, see Exercise 43.

2.3.8 *Lattice Paths and Catalan Numbers*

Many interesting problems in enumerative combinatorics can be expressed as problems of enumeration of paths in the integer lattice $\mathbb{Z}^d$, called *lattice paths*. This section studies two main examples.

Let d be a positive integer and $a, b \in \mathbb{Z}^d$ be two points. A *path of length* k in $\mathbb{Z}^d$ with initial vertex a and terminal vertex b is defined as any sequence $p = (p_0, p_1, \ldots, p_k)$ of points of $\mathbb{Z}^d$ such that $p_0 = a$ and $p_k = b$. The vectors $p_i - p_{i-1}$ for $1 \le i \le k$ are the *steps* of p. We will be concerned with the enumeration of paths with given initial vertex a, terminal vertex b and steps in the set $\{\mathrm{e}_1, \mathrm{e}_2, \ldots, \mathrm{e}_d\}$, where $\mathrm{e}_i \in \mathbb{Z}^d$ is the vector with ith coordinate equal to 1 and all other coordinates equal to zero. For example, there are six paths in $\mathbb{Z}^2$ with initial vertex $a = (0,0)$, terminal vertex $b = (2,2)$ and steps $\mathrm{e}_1 = (1,0)$ or $\mathrm{e}_2 = (0,1)$. These paths are shown in Figure 2.8.

Proposition 2.3.11. *The number of paths in $\mathbb{Z}^d$ with initial vertex $(0,0,\ldots,0)$, terminal vertex $(n_1, n_2, \ldots, n_d) \in \mathbb{N}^d$ and steps in the set $\{\mathrm{e}_1, \mathrm{e}_2, \ldots, \mathrm{e}_d\}$ is equal to the multinomial coefficient*

$$\binom{n_1 + n_2 + \cdots + n_d}{n_1, n_2, \ldots, n_d} = \frac{(n_1 + n_2 + \cdots + n_d)!}{n_1! n_2! \cdots n_d!}.$$

In particular, the number of paths in $\mathbb{Z}^2$ with initial vertex $(0,0)$, terminal vertex $(m,n) \in \mathbb{N}^2$ and steps $(1,0)$ or $(0,1)$ is equal to the binomial coefficient $\binom{m+n}{n}$.

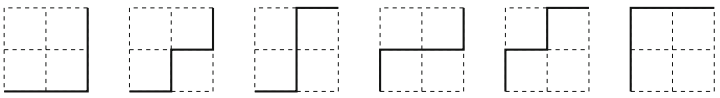


Figure 2.8 Six paths in $\mathbb{Z}^2$

Proof. We set $a = (0, 0, \ldots, 0)$ and $b = (n_1, n_2, \ldots, n_d)$ and denote by $\mathcal{P}(b)$ the set of paths in $\mathbb{Z}^d$ with initial vertex a, terminal vertex b and steps in the set $\{\mathrm{e}_1, \mathrm{e}_2, \ldots, \mathrm{e}_d\}$. Given $p = (p_0, p_1, \ldots, p_k) \in \mathcal{P}(b)$, we consider the sequence

$$\varphi(p) = (p_1 - p_0, p_2 - p_1, \ldots, p_k - p_{k-1})$$

of successive steps of p. Since the sum of the terms of $\varphi(p)$ equals $b - a = (n_1, n_2, \ldots, n_d)$, we have $k = n_1 + n_2 + \cdots + n_d$ and for every $i \in [d]$, exactly n_i terms of $\varphi(p)$ are equal to e_i. As a result, $\varphi(p)$ is a permutation of the collection which consists of n_i copies of e_i for every $i \in [d]$ and hence, we have defined a map $\varphi : \mathcal{P}(b) \to \mathcal{A}(b)$, where $\mathcal{A}(b)$ is the set of these permutations. For example, for $b = (2, 2)$, the paths of Figure 2.8, in the order in which they appear there, are mapped to the permutations $(\mathrm{e}_1, \mathrm{e}_1, \mathrm{e}_2, \mathrm{e}_2)$, $(\mathrm{e}_1, \mathrm{e}_2, \mathrm{e}_1, \mathrm{e}_2)$, $(\mathrm{e}_1, \mathrm{e}_2, \mathrm{e}_2, \mathrm{e}_1)$, $(\mathrm{e}_2, \mathrm{e}_1, \mathrm{e}_1, \mathrm{e}_2)$, $(\mathrm{e}_2, \mathrm{e}_1, \mathrm{e}_2, \mathrm{e}_1)$, and $(\mathrm{e}_2, \mathrm{e}_2, \mathrm{e}_1, \mathrm{e}_1)$.

We leave it to the reader to verify that for every $\sigma = (\sigma_1, \sigma_2, \ldots, \sigma_k) \in \mathcal{A}(b)$ there exists a unique path $p \in \mathcal{P}(b)$ such that $\varphi(p) = \sigma$ (namely, the path $p = (p_0, p_1, \ldots, p_k)$ with $p_i = \sigma_1 + \sigma_2 + \cdots + \sigma_i$ for $i \in \{0, 1, \ldots, k\}$). This implies that the map φ is a bijection and hence, according to Proposition 2.2.1, that $\#\mathcal{P}(b) = \#\mathcal{A}(b)$. The result now follows from this equality and Proposition 2.3.8. □

As a special case of Proposition 2.3.11, there are $\binom{2n}{n}$ paths in $\mathbb{Z}^2$ with initial vertex $(0, 0)$, terminal vertex (n, n), and steps in the set $\{\mathrm{e}_1, \mathrm{e}_2\}$, for every $n \in \mathbb{N}$. How many of these lie entirely in the half-plane $x \geq y$ (meaning, they stay below the line $x = y$) of $\mathbb{R}^2$? Such paths are often called *Dyck paths* of length $2n$ in the literature. Denoting their number by C_n, we have $C_0 = C_1 = 1$, $C_2 = 2$, $C_3 = 5$, $C_4 = 14$, $C_5 = 42$, and $C_6 = 132$. The five possible such paths for $n = 3$ are shown in Figure 2.9. The sequence $(C_n)_{n \in \mathbb{N}}$ (one of the most fascinating sequences that have been studied in mathematics) is determined in the following proposition.

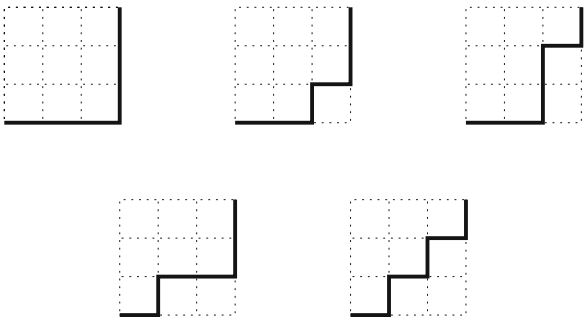


Figure 2.9 Five paths for $n = 3$

Proposition 2.3.12. *Let C_n be the number of paths in $\mathbb{Z}^2$ which lie in the half-plane $x \geq y$ of $\mathbb{R}^2$ and have initial vertex $(0,0)$, terminal vertex (n,n), and steps in the set $\{e_1, e_2\}$. Then,*

$$C_n = \frac{1}{n+1}\binom{2n}{n} \tag{2.11}$$

for every $n \in \mathbb{N}$. Moreover, the recursive formula

$$C_n = \sum_{i=0}^{n-1} C_i C_{n-1-i} \tag{2.12}$$

holds for $n \geq 1$, where $C_0 = 1$.

Proof. Following the notation of the proof of Proposition 2.3.11, we denote by $\mathcal{P}(m,n)$ the set of paths in $\mathbb{Z}^2$ with initial vertex $(0,0)$, terminal vertex (m,n), and steps in the set $\{e_1, e_2\}$. We also denote by $\mathcal{P}^+(n,n)$ the set consisting of those paths in $\mathcal{P}(n,n)$ whose vertices belong to the set $\{(x,y) \in \mathbb{Z}^2 : x \geq y\}$ and with $\mathcal{P}^-(n,n)$ the complement of $\mathcal{P}^+(n,n)$ in $\mathcal{P}(n,n)$. Then, we have $C_n = \#\mathcal{P}^+(n,n)$ and claim that

$$\#\mathcal{P}^-(n,n) = \#\mathcal{P}(n-1, n+1)$$

for $n \in \mathbb{N}$. From this claim, Corollary 2.2.1 and Proposition 2.3.11, it follows that

$$\begin{aligned} C_n &= \#\mathcal{P}(n,n) - \#\mathcal{P}^-(n,n) = \#\mathcal{P}(n,n) - \#\mathcal{P}(n-1,n+1) \\ &= \binom{2n}{n} - \binom{2n}{n-1} = \frac{1}{n+1}\binom{2n}{n}, \end{aligned}$$

which proves (2.11).

To prove our claim, it suffices to find a bijective map $\varphi : \mathcal{P}^-(n,n) \to \mathcal{P}(n-1, n+1)$. Let $p = (p_0, p_1, \ldots, p_{2n}) \in \mathcal{P}^-(n,n)$. We consider the smallest index $i \in [2n]$ such that p_i lies in the open half-plane $x < y$ (and necessarily on the line $y = x+1$) of $\mathbb{R}^2$ and set $\varphi(p) = (p_0, \ldots, p_i, p'_{i+1}, \ldots, p'_{2n})$, where a' is the orthogonal reflection of $a \in \mathbb{Z}^2$ with respect to the line $y = x+1$ (see Figure 2.10 for an example). We observe that $\varphi(p)$ is a path in $\mathbb{Z}^2$ with initial vertex $(0,0)$, terminal vertex $(n,n)' = (n-1, n+1)$, and steps in the set $\{e_1, e_2\}$ and hence $\varphi(p) \in \mathcal{P}(n-1, n+1)$. Thus, we have defined a map

$$\varphi : \mathcal{P}^-(n,n) \to \mathcal{P}(n-1, n+1).$$

To show that φ is a bijection, let $q = (q_0, q_1, \ldots, q_{2n}) \in \mathcal{P}(n-1, n+1)$. We consider the smallest index $i \in [2n]$ such that q_i lies on the line $y = x+1$ (explain

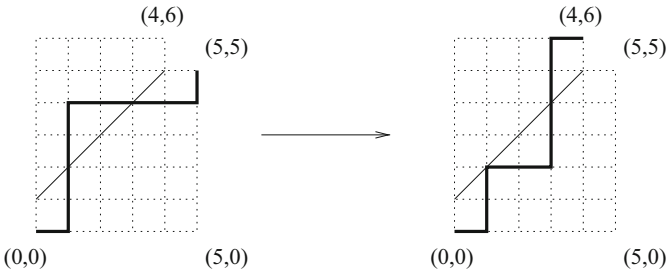


Figure 2.10 A path $p \in \mathcal{P}^-(5,5)$ on the left and $\varphi(p)$ on the right

why such an index exists) and set $\psi(q) = (q_0, \ldots, q_i, q'_{i+1}, \ldots, q'_{2n})$. We leave it to the reader to verify that $\psi(q) \in \mathcal{P}^-(n,n)$ and that the resulting map $\psi : \mathcal{P}(n-1, n+1) \to \mathcal{P}^-(n,n)$ is the inverse of φ.

We now prove (2.12). Given a path $p \in \mathcal{P}(n,n)$, we denote by $f(p)$ the smallest index $i \in [n]$ such that (i,i) is a vertex of p (informally, $f(p)$ is the vertex after $(0,0)$ at which p returns to the line $y = x$ for the first time). We claim that there are exactly $C_{i-1}C_{n-i}$ paths $p \in \mathcal{P}^+(n,n)$ with $f(p) = i$. Indeed, to choose such a path, there are C_{i-1} choices for the part of p of length $2i$ between points $(0,0)$ and (i,i) (since the first step of this part must be e_1 and the last e_2) and for every such choice, there are C_{n-i} choices for the part of p of length $2(n-i)$ between points (i,i) and (n,n). It follows from our claim and the addition principle that

$$C_n = \sum_{i=1}^{n} C_{i-1}C_{n-i},$$

which is exactly the recursive formula (2.12). □

The trick on which our proof of (2.11) was based in known as the *reflection principle*. The integer C_n, known as the nth Catalan number, admits several combinatorial interpretations other than the one offered by Proposition 2.3.12. Some of them will appear in the following chapters of this book. For more information on Catalan numbers, see [24, Chapter 6] [26].

2.4 Combinatorial Identities

Section 2.3.4 provides simple examples of identities for combinatorial sums, meaning sums that involve powers of integers, factorials, binomial coefficients, and other quantities, which can be interpreted combinatorially. The general problem to compute such sums is important, as it appears very often in mathematics. This section describes more examples, useful identities, and methods of computation.

Other examples appear in the exercises and in the following chapters. For thorough treatments of this topic, we refer the reader to the books [19, 29].

The methods we emphasize here are the interpretation of a combinatorial sum as the coefficient of a monomial in a certain polynomial, or as the cardinality of a finite set, and the computation of this coefficient or cardinality by algebraic or combinatorial techniques, respectively. Our first main identity is known as *Vandermonde convolution*.

Proposition 2.4.1. *For $m, n, k \in \mathbb{N}$,*

$$\binom{m+n}{k} = \sum_{i=0}^{k} \binom{m}{i}\binom{n}{k-i}.$$

First Proof. Applying the Binomial Theorem three times, we find that

$$\sum_{k=0}^{m+n} \binom{m+n}{k} x^k = (x+1)^{m+n} = (x+1)^m (x+1)^n$$
$$= \left(\sum_{i=0}^{m} \binom{m}{i} x^i\right)\left(\sum_{j=0}^{n} \binom{n}{j} x^j\right).$$

We now compare the coefficents of x^k on the two extreme sides of this string of equalities and deduce that

$$\binom{m+n}{k} = \sum_{i+j=k} \binom{m}{i}\binom{n}{j} = \sum_{i=0}^{k} \binom{m}{i}\binom{n}{k-i}.$$

□

Second Proof. Let S and T be disjoint sets with cardinalities m and n, respectively. Let us count the k-subsets of $S \cup T$ in two different ways. On the one hand, the number of these subsets is equal to $\binom{m+n}{k}$. On the other, every k-subset of $S \cup T$ is the disjoint union of an i-subset of S, which can be chosen in $\binom{m}{i}$ ways, and a $(k-i)$-subset of T, which can be chosen in $\binom{n}{k-i}$ ways, for some $0 \leq i \leq k$. Therefore, the number of k-subsets of $S \cup T$ is also equal to the sum on the right-hand side of the proposed identity. □

Corollary 2.4.1. *The following identities hold:*

(a) $\displaystyle\sum_{k=0}^{n} \binom{n}{k}^2 = \binom{2n}{n}$ *for every* $n \in \mathbb{N}$.

(b) $\displaystyle\sum_{k=0}^{n-1} \frac{1}{k+1}\binom{n}{k}\binom{n-1}{k} = \frac{1}{n+1}\binom{2n}{n}$ *for every* $n \in \mathbb{Z}_{>0}$.

Proof. Part (a) follows from Proposition 2.4.1 by setting $m = n = k$ and applying the identity $\binom{n}{n-i} = \binom{n}{i}$. For part (b), we apply the identity $\binom{n+1}{k+1} = \frac{n+1}{k+1}\binom{n}{k}$ and Proposition 2.4.1 to find that

$$\sum_{k=0}^{n-1} \frac{n+1}{k+1}\binom{n}{k}\binom{n-1}{k} = \sum_{k=0}^{n-1} \binom{n+1}{k+1}\binom{n-1}{k} = \sum_{k=0}^{n-1} \binom{n-1}{k}\binom{n+1}{n-k}$$
$$= \binom{2n}{n}.$$

We now divide by $n+1$ and the proof follows. □

Proposition 2.4.2. *For integers* $1 \leq k \leq n$,

$$\binom{n}{k} = \sum_{m=k}^{n} \binom{m-1}{k-1}.$$

First Proof. According to the Binomial Theorem, the right-hand side of the proposed equality is equal to the coefficient of x^{k-1} in the sum $\sum_{m=k}^{n}(x+1)^{m-1}$. Since

$$\sum_{m=k}^{n}(x+1)^{m-1} = (x+1)^{k-1}\sum_{m=0}^{n-k}(x+1)^m = (x+1)^{k-1}\cdot\frac{(x+1)^{n-k+1}-1}{x}$$
$$= \frac{(x+1)^n - (x+1)^{k-1}}{x},$$

the right-hand side of the proposed equality is also equal to the coefficient of x^k in the polynomial $(x+1)^n - (x+1)^{k-1}$. The latter is equal to $\binom{n}{k}$. □

Second Proof. By its definition, $\binom{n}{k}$ is equal to the number of all k-subsets of the set $[n]$. For $k \leq m \leq n$, exactly $\binom{m-1}{k-1}$ of these subsets have maximum element equal m (explain why). The desired formula for $\binom{n}{k}$ follows from these observations and the addition principle. □

Third Proof. This identity follows from the basic recursion $\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}$ for binomial coefficients by induction on n. The details are left to the reader. □

The following statement, for which we offer two proofs, generalizes the identity of Proposition 2.3.6 (d) (which is the special case $m = n$). A third proof, one that generalizes our second proof of Proposition 2.3.6 (d), is possible; see Exercise 60.

Proposition 2.4.3. *For integers $n \geq 1$ and $0 \leq m \leq n$,*

$$\sum_{k=0}^{m} (-1)^k \binom{n}{k} = (-1)^m \binom{n-1}{m}.$$

First Proof. We observe that $(-1)^k \binom{n}{k}$ is equal to the coefficient of x^m in $(1-x)^n x^{m-k}$. Hence, the left-hand side of the proposed equality is equal to the coefficient of x^m in the sum $\sum_{k=0}^{m} (1-x)^n x^{m-k}$, which can be expressed as

$$\begin{aligned} \sum_{k=0}^{m} (1-x)^n x^{m-k} &= (1-x)^n \frac{1-x^{m+1}}{1-x} = (1-x)^{n-1}(1-x^{m+1}) \\ &= (1-x)^{n-1} - x^{m+1} p(x) \end{aligned}$$

for some polynomial $p(x)$. As a result, the coefficient of x^m in the sum above is equal to $(-1)^m \binom{n-1}{m}$. □

Second Proof. A proof by induction on m, based on the recurrence $\binom{n}{m} = \binom{n-1}{m} + \binom{n-1}{m-1}$, is possible. Indeed, the proposed identity holds for $m = 0$ (explain why). Assuming that $m \geq 1$ and that it holds for $m-1$, we find that

$$\begin{aligned} \sum_{k=0}^{m} (-1)^k \binom{n}{k} &= \sum_{k=0}^{m-1} (-1)^k \binom{n}{k} + (-1)^m \binom{n}{m} \\ &= (-1)^{m-1} \binom{n-1}{m-1} + (-1)^m \binom{n}{m} \\ &= (-1)^m \left(\binom{n}{m} - \binom{n-1}{m-1} \right) = (-1)^m \binom{n-1}{m} \end{aligned}$$

and conclude that it holds for m. Therefore, the identity holds for every $m \in \mathbb{N}$. □

The following example provides an application of the identity of Proposition 2.4.2 (for another nontrivial application, see Exercise 33).

Example 2.4.1. Let's compute the sum

$$f(n) = \sum_{k=0}^{n} k \binom{2n-k-1}{n-1}$$

for $n \in \mathbb{Z}_{>0}$. Changing the index of summation from k to $n-k$, we find that

$$f(n) = \sum_{k=0}^{n} (n-k) \binom{n+k-1}{n-1}.$$

We now write the summand in the form

$$\begin{aligned}(n-k)\binom{n+k-1}{n-1} &= n\binom{n+k-1}{n-1} - k\binom{n+k-1}{n-1} \\ &= n\binom{n+k-1}{n-1} - k\cdot\frac{(n+k-1)!}{(n-1)!\,k!} \\ &= n\binom{n+k-1}{n-1} - n\cdot\frac{(n+k-1)!}{n!\,(k-1)!} \\ &= n\binom{n+k-1}{n-1} - n\binom{n+k-1}{n}.\end{aligned}$$

Suitably applying Proposition 2.4.2 twice, we conclude that

$$\begin{aligned}f(n) &= n\sum_{k=0}^{n}\binom{n+k-1}{n-1} - n\sum_{k=0}^{n}\binom{n+k-1}{n} = n\binom{2n}{n} - n\binom{2n}{n+1} \\ &= \frac{n}{n+1}\binom{2n}{n}\end{aligned}$$

for every $n \in \mathbb{Z}_{>0}$. For a different approach to this problem, see Exercise 46 and its solution. □

Example 2.4.2. Let us compute the sum

$$\sum_{k=0}^{n}(-1)^k\binom{n}{k}^2 = \binom{n}{0}^2 - \binom{n}{1}^2 + \cdots + (-1)^n\binom{n}{n}^2$$

for $n \in \mathbb{N}$. Since

$$(x+1)^n(x-1)^n = \left(\sum_{i=0}^{n}\binom{n}{i}x^i\right)\left(\sum_{j=0}^{n}(-1)^{n-j}\binom{n}{j}x^j\right),$$

the coefficient of x^n in the polynomial $(x+1)^n(x-1)^n$ is equal to

$$\sum_{i+j=n}\binom{n}{i}(-1)^{n-j}\binom{n}{j} = \sum_{k=0}^{n}(-1)^{n-k}\binom{n}{n-k}\binom{n}{k} = (-1)^n\sum_{k=0}^{n}(-1)^k\binom{n}{k}^2.$$

Since $(x+1)^n(x-1)^n = (x^2-1)^n$, this coefficient is also equal to zero, if n is odd, and to $(-1)^{n/2}\binom{n}{n/2}$, if n is even. We conclude that

$$\sum_{k=0}^{n}(-1)^k\binom{n}{k}^2 = \begin{cases} 0, & \text{if } n \text{ is odd} \\ (-1)^{n/2}\binom{n}{n/2}, & \text{if } n \text{ is even.} \end{cases} \tag{2.13}$$

For other proofs of this identity, see Exercises 50 and 58. □

2.5 Exercises

1. Let $m, n \in \mathbb{N}$. Prove the following, without using Proposition 2.2.1:
   (a) If there exists a bijective map $f : [m] \to [n]$, then $m = n$.
   (b) If there exists an injective map $f : [m] \to [n]$, then $m \leq n$.
   (c) If there exists a surjective map $f : [m] \to [n]$, then $m \geq n$.
2. Prove that the following conditions on a set S are equivalent:
   (a) S is infinite.
   (b) There exists an injective map $f : \mathbb{N} \to S$.
   (c) There exists a surjective map $g : S \to \mathbb{N}$.
3. Let X, Y be finite sets with $\#X = \#Y$. Prove that the following conditions on a map $f : X \to Y$ are equivalent:
   (a) f is injective.
   (b) f is surjective.
   (c) f is bijective.
4. A tennis tournament, in which n players compete, is run with knock out matches. Specifically, if n is even, then the players pair up at random and the winners qualify to the next round, while losers get disqualified. If n is odd, then one player, chosen at random, qualifies to the next round and the rest pair up. The process continues until only one player, the winner of the tournament, remains. What is the total number of matches played in the tournament?
5. Prove the identities
$$(x+y)^2 = x^2 + 2xy + y^2$$
$$(x+y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$$
for $x, y \in \mathbb{N}$ using only the addition and multiplication principles.
6. Let $m, n \geq 2$. Three rooks are placed in the mn squares of an $m \times n$ chessboard (at most one rook in each square), so that two of the rooks lie in the same row

of the chessboard and two of them lie in the same column. How many such placements are there in the following situations?

(a) The rooks are identical.
(b) The rooks are pairwise distinguishable.

7. Let $n \in \mathbb{Z}_{>0}$. We denote by $\max(S)$ and by $\min(S)$ the maximum and the minimum element of a nonempty subset S of $[n]$, respectively.

   (a) Given $k \in [n]$, how many nonempty subsets S of $[n]$ are such that $\max(S) = k$?
   (b) Given $k \in [n]$, how many nonempty subsets S of $[n]$ are such that $\min(S) = k$?

   How many nonempty subsets S of [9] have the following property?

   (c) $\max(S) \geq 4$.
   (d) $\min(S) \geq 3$ and $\max(S) \leq 8$.
   (e) $\max(S)$ is divisible by 3.

8. Let $n \in \mathbb{Z}_{>0}$.

   (a) Prove the identity $1 + 2 + 2^2 + \cdots + 2^{n-1} = 2^n - 1$ by counting the nonempty subsets of $[n]$ in two different ways.
   (b) More generally, prove the identity $(a-1)(1+a+a^2+\cdots+a^{n-1}) = a^n - 1$ for every integer $a \geq 2$ in a similar way.

9. Let $n \in \mathbb{Z}_{>0}$. How many pairs (S, T) of nonempty subsets of $[n]$ have the following property?

   (a) $\max(S) = \max(T)$.
   (b) $\max(S) = \max(T)$ and $\min(S) = \min(T)$.
   (c) $\max(S) = \min(T)$.
   (d) $\max(S) \leq \max(T)$.
   (e) $\max(S) \leq \min(T)$.

10. Let X and Y be finite sets with cardinalities m and n, respectively.

    (a) How many binary relations from X to Y are there?
    (b) Deduce from your answer to (a) that the number of binary relations from X to Y is equal to the number of binary relations from Y to X and give a bijective proof of this fact.

11. Let $n \in \mathbb{N}$. We consider matrices $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with $a, b, c, d \in \{0, 1, \ldots, n\}$.

    (a) Given $0 \leq k \leq n$, how many of these matrices are such that $a + d = b + c = k$?
    (b) How many of these matrices are such that $a + d = b + c$?

12. We consider the words with five letters that can be formed from the alphabet {A, B, C, D}, such as BAADB and CADCB.

(a) How many of these words contain at least one A?
(b) How many contain at most one A?
(c) How many contain two successive A's followed by a B?
(d) How many do not contain two successive A's?

13. Given $n \in \mathbb{Z}_{>0}$, we consider the maps $f : [2n] \to [2n]$ such that $f(x) \leq x$ for every $x \in [2n]$.

(a) How many such maps are there?
(b) How many have the property that $x \in 2\mathbb{N} \Rightarrow f(x) \in 2\mathbb{N}$?
(c) How many have the property that $x \in 2\mathbb{N} \Leftrightarrow f(x) \in 2\mathbb{N}$?

14. Let $n, k \in \mathbb{Z}_{>0}$.

(a) How many sequences $(S_1, S_2, \ldots, S_k)$ of subsets of $[n]$ are there, such that $S_1 \subseteq S_2 \subseteq \cdots \subseteq S_k$?
(b) How many sequences $(S_1, S_2, \ldots, S_k)$ of subsets of $[n]$ are there, such that $S_1 \cap S_2 \cap \cdots \cap S_k = \varnothing$?
(c) How many sequences $(S_1, S_2, \ldots, S_k)$ of subsets of $[n]$ are there, such that $S_1 \cup S_2 \cup \cdots \cup S_k = [n]$?

15. Consider the set of points $X_n = [n] \times [n]$ in the plane $\mathbb{R}^2$.

(a) Find the number of squares in the plane with vertices in X_n and sides parallel to one of the lines $x = 0$ and $y = 0$. There is one such square for $n = 2$ and five for $n = 3$.
(b) Find the total number of squares in the plane with vertices in X_n. For $n = 3$, there are four such squares with side lengths equal to 1, one with side length equal to 2, and one with side length equal to $\sqrt{2}$.

16. Find the number of permutations $(\sigma_1, \sigma_2, \ldots, \sigma_8)$ of the set $[8]$ such that:

(a) $\sigma_2 = \sigma_1 + 1$.
(b) $\sigma_3 = \sigma_2 + 1 = \sigma_1 + 2$.
(c) $\sigma_2 = \sigma_1 + 1$ and $\sigma_4 = \sigma_3 + 1$.
(d) $\sigma_i < \sigma_{i+3}$ for $1 \leq i \leq 5$.

17. Given $n \in \mathbb{Z}_{>0}$, how many permutations $(\sigma_1, \sigma_2, \ldots, \sigma_{2n})$ of the set $[2n]$ have the following property?

(a) $\sigma_1 < \sigma_2 < \cdots < \sigma_n$.
(b) $\sigma_1 < \sigma_3 < \cdots < \sigma_{2n-1}$ and $\sigma_2 > \sigma_4 > \cdots > \sigma_{2n}$.
(c) $\sigma_i \geq 2i$ for every $i \in [n]$.

18. A permutation $(\sigma_1, \sigma_2, \ldots, \sigma_n)$ of the set $[n]$ (more generally, a sequence of real numbers) is called *unimodal* if $\sigma_1 \leq \sigma_2 \leq \cdots \leq \sigma_j \geq \sigma_{j+1} \geq \cdots \geq \sigma_n$ for some index $i \in [n]$. For example, the unimodal permutations for $n = 3$ are $(1, 2, 3)$, $(1, 3, 2)$, $(2, 3, 1)$, and $(3, 2, 1)$. Find the number of unimodal permutations of $[n]$ for any positive integer n.

19. Let $n \geq 2$ be given.

    (a) How many ways are there to first choose a permutation $(\sigma_1, \sigma_2, \ldots, \sigma_n)$ of $[n]$ and then choose an index $i \in [n-1]$ such that $\sigma_i < \sigma_{i+1}$?
    (b) How many ways are there to first choose a permutation $(\sigma_1, \sigma_2, \ldots, \sigma_n)$ of $[n]$ and then choose a pair of indices $(i, j) \in [n] \times [n]$ such that $i < j$ and $\sigma_i < \sigma_j$?

20. Let $n \geq 3$. Knights $1, 2, \ldots, n$ sit at a round table, as in Section 2.3.3. How many of these arrangements have the following property?

    (a) Knights 1 and 2 are *not* sitting next to each other.
    (b) Knight 1 is sitting next to knight 2, who is sitting next to knight 3.
    (c) Knight i is sitting next to at least one of knights $1, 2, \ldots, i-1$ for all $1 < i \leq n$.

21. Prove the identity (2.5).
22. A diagonal of a convex n-gon Π is defined as a line segment that joins two vertices of Π and lies (except for its endpoints) in the interior of Π. Find the number of diagonals of a convex n-gon:

    (a) By applying Proposition 2.3.5.
    (b) By counting the elements of a finite set in two different ways.

23. Let m, n, k be positive integers such that $k \leq \min(m, n)$. We wish to place k identical rooks in the squares of an $m \times n$ chessboard (each rook in a single square). In how many ways can this be done, if:

    (a) No two rooks should lie in the same row?
    (b) No two rooks should lie in the same column?
    (c) No two rooks should lie in the same row or column?

24. Prove that $k\binom{n}{k} = (n-k+1)\binom{n}{k-1}$ for integers $1 \leq k \leq n$ as follows:

    (a) By applying Proposition 2.3.5.
    (b) By counting the elements of a finite set in two different ways.

25. Prove that $\binom{n}{i}\binom{n-i}{j} = \binom{n}{j}\binom{n-j}{i}$ for $n, i, j \in \mathbb{N}$ with $n \geq i + j$ as follows:

    (a) By applying Proposition 2.3.5.
    (b) By counting the elements of a finite set in two different ways.

26. Prove that $\binom{n}{k}^2 > \binom{n}{k-1}\binom{n}{k+1}$ for integers $1 \leq k \leq n-1$.
27. Prove that

$$\left(\frac{n}{k}\right)^k \leq \binom{n}{k} \leq \left(\frac{en}{k}\right)^k$$

    for integers $1 \leq k \leq n$, where e is the base for the natural logarithm.

28. Let

$$p_n = \prod_{i=1}^{n} \binom{i+1}{2} = \binom{2}{2}\binom{3}{2}\cdots\binom{n+1}{2}$$

for $n \in \mathbb{N}$, where $p_0 = 1$.

(a) Prove that

$$\frac{p_n}{p_k\, p_{n-k}} = \frac{1}{k+1}\binom{n}{k}\binom{n+1}{k}$$

for every $k \in \{0, 1, \ldots, n\}$.

(b) Prove that

$$\frac{p_n}{p_k\, p_{n-k}} = \binom{n}{k}\binom{n+2}{k+1} - \binom{n+1}{k}\binom{n+1}{k+1}$$

for every $k \in \{0, 1, \ldots, n\}$.

(c) Deduce that $p_n / p_k\, p_{n-k} \in \mathbb{N}$ for every $k \in \{0, 1, \ldots, n\}$.

29. Prove that

$$\sum_{k=0}^{\lfloor n/2 \rfloor} k\binom{n}{2k} = n2^{n-3}$$

for $n \geq 2$ as follows:

(a) By applying the Binomial Theorem.
(b) By applying the identity $k\binom{n}{2k} = \frac{n}{2}\binom{n-1}{2k-1}$.
(c) By counting the elements of a finite set in two different ways.

30. Compute the sums:

(a) $\displaystyle\sum_{k=0}^{n} k^2\binom{n}{k}$, (b) $\displaystyle\sum_{k=0}^{n} k^3\binom{n}{k}$

for $n \in \mathbb{N}$.

31. Let $1 \leq k \leq n$ be integers. In how many ways can n soldiers $1, 2, \ldots, n$ be arranged to form k nonempty vertical lines, listed from left to right, and aligned on the same horizontal line? For example, for $n = 3$ and $k = 2$, there are the following 12 such arrangements

$$\begin{array}{cccccccccccc} 1\,3 & & 2\,3 & & 1\,2 & & 3\,2 & & 2\,1 & & 3\,1 & & 1\,3 & & 2\,3 \\ 2 & & 1 & & 3 & & 1 & & 3 & & 2 & & 2 & & 1 \end{array}$$

$$\begin{array}{cccccccc} 1\,2 & & 3\,2 & & 2\,1 & & 3\,1 \\ 3 & & 1 & & 3 & & 2 \end{array}$$

whereas for $n = 10$ and $k = 4$, one such arrangement is $\begin{matrix} 4\,9\,6 & 2 \\ 8\,1 & 10 \\ 7 & 3 \\ 5 & \end{matrix}$.

32. Let n, k be integers such that $n \geq 2k - 1$. How many k-subsets of $[n]$ do not contain two consecutive integers? For $n = 6$ and $k = 3$, these subsets are $\{1, 3, 5\}$, $\{1, 3, 6\}$, $\{1, 4, 6\}$, and $\{2, 4, 6\}$.
33. Let $n \in \mathbb{N}$.

    (a) We are given n straight lines in the plane such that no two of them are parallel and no three pass through the same point. Prove that these lines dissect the plane into $1 + n + \binom{n}{2}$ regions.
    (b) We are given n straight planes in three-dimensional Euclidean space such that no two of them are parallel, no three intersect along a line, and no four pass through the same point. Prove that these planes dissect space into $1 + n + \binom{n}{2} + \binom{n}{3}$ three-dimensional regions.
    (c) For those who have the relevant background: Generalize to Euclidean spaces of arbitrary (finite) dimension.

34. Let $m, n \in \mathbb{Z}_{>0}$.

    (a) How many two-to-one maps $f : [2n] \to [n]$ are there?
    (b) More generally, how many m-to-one maps $f : [mn] \to [n]$ are there?

35. Prove the identity (2.8).
36. For $n \in \mathbb{N}$, compute the sums

    (a) $\sum \frac{(-1)^a}{a!b!c!}$, (b) $\sum \frac{a^2 + b^2 + c^2}{a!b!c!}$

    running over all triples $(a, b, c) \in \mathbb{N}$ such that $a + b + c = n$.
37. Find:

    (a) The number of solutions $(x, y, z, w) \in \mathbb{N}^4$ of the equation $x+y+z+w = 12$ such that $x \geq 2$, $z \geq 1$, and $w \geq 3$.
    (b) The number of solutions $(x, y, z, w) \in \mathbb{Z}^4$ of the equation $|x| + |y| + |z| + |w| = 2$.
    (c) The number of solutions $(x, y, z, w) \in \mathbb{Z}^4$ of the equation $|x| + |y| + |z| + |w| = 3$.

38. Let $n \in \mathbb{Z}_{>0}$. Find the number of ways that one can choose a composition ρ of n and color each part of ρ black or white:

    (a) By using the Binomial Theorem.
    (b) By using only basic principles of enumeration.

    For example, for $n = 2$, the possibilities are 2, $\bar{2}$, $1 + 1$, $\bar{1} + 1$, $1 + \bar{1}$, and $\bar{1} + \bar{1}$ (where a bar over a number stands for black color).

39. Let a_n be the number of compositions of n with parts equal to 1 or 2. For example $a_4 = 5$, because the compositions of 4 with parts 1 or 2 are $(2, 2)$, $(2, 1, 1)$, $(1, 2, 1)$, $(1, 1, 2)$, and $(1, 1, 1, 1)$. Prove that:

    (a) $a_n = a_{n-1} + a_{n-2}$ for $n \geq 2$, where $a_0 := 1$.
    (b) a_n is equal to the number of subsets of $[n-1]$, which do not contain two consecutive integers.
    (c) $a_n = \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n-k}{k}$.

    Note: It follows from (a) that (a_n) is the Fibonacci sequence of Example 1.1.10.

40. For $m \in \mathbb{N}$ and $n \in \mathbb{Z}_{>0}$, we let $\zeta(m, n)$ be the number of ways to arrange the numbers $1, 2, \ldots, n$ and m identical balls around a circle (where two such arrangements are considered identical if one can be obtained from the other by rotation). For example, $\zeta(2, 3) = 12$. Prove that $\zeta(m, n)$ equals the number of injective maps $f : [n-1] \to [m+n-1]$:

    (a) By combining Propositions 2.3.4 and 2.3.10.
    (b) By applying the multilpication principle and Proposition 2.3.2 (a).

41. For $m, n \in \mathbb{Z}_{>0}$, we let $\sigma(m, n)$ denote the number of solutions $(x_1, x_2, \ldots, x_n) \in \mathbb{N}^n$ of the inequality $x_1 + x_2 + \cdots + x_n \leq m$.

    (a) Prove that $\sigma(m, n) = \binom{m+n}{n}$ and deduce that $\sigma(m, n) = \sigma(n, m)$.
    (b) Give a bijective proof of the fact that $\sigma(m, n) = \sigma(n, m)$.

42. Let $n \geq 2k \geq 6$ be integers and Π be a convex n-gon. We consider convex polygons Π' with k vertices such that: (a) every vertex of Π' is a vertex of Π; and (b) every edge of Π' is a diagonal of Π (see Exercise 22).

    (a) How many such polygons Π' contain a given vertex v of Π?
    (b) How many such polygons Π' are there in total?

43. Consider m boys $a_1, a_2, \ldots, a_m$ and n girls $b_1, b_2, \ldots, b_n$. How many cyclic permutations of the boys and girls are such that the boys appear clockwise in the cyclic order $a_1, a_2, \ldots, a_m$ and the girls appear clockwise in the cyclic order $b_1, b_2, \ldots, b_n$? For example, there are m such cyclic permutations, if $n = 1$ and $m(m+1)$, if $n = 2$.

44. By applying Proposition 2.3.11 or by a different method, prove that

$$\binom{i+j}{j}\binom{m+n-i-j}{n-j} \leq \binom{m+n}{n}$$

    for all $m, n, i, j \in \mathbb{N}$ such that $i \leq m$ and $j \leq n$.

45. By applying Proposition 2.3.11 or by a different method, prove that

$$\binom{m+n}{n} = \sum_{j=0}^{n} \binom{i+j}{j}\binom{m+n-i-j-1}{n-j} \tag{2.14}$$

for all $m, n \in \mathbb{N}$ and $i \in \{0, 1, \ldots, m\}$. Which identity is obtained in the special case $i = 0$?

46. For $n, k \in \mathbb{N}$, the numbers $f(n, k)$ are defined recursively by setting

$$f(n, k) = f(n, k-1) + f(n-1, k) \tag{2.15}$$

for $1 \leq k \leq n$, $f(n, 0) = 1$ for $n \in \mathbb{N}$ and $f(n, k) = 0$ for $k > n$. Thus,

$$(f(n,k)) = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 2 & 2 & 0 & 0 & 0 \\ 1 & 3 & 5 & 5 & 0 & 0 \cdots \\ 1 & 4 & 9 & 14 & 14 & 0 \\ 1 & 5 & 14 & 28 & 42 & 42 \\ & & & \vdots & & \end{pmatrix}.$$

(a) Find a combinatorial interpretation of $f(n, k)$ and a simple formula for it.
(b) Prove that $f(n, n) = \frac{1}{n+1}\binom{2n}{n}$ for every $n \in \mathbb{N}$.
(c) Prove that $f(n, n) = \sum_{k=0}^{n-1} f(n-1, k)$ for $n \geq 1$. Which identity for the Catalan numbers does this equality yield?

47. Compute the sums:

(a) $\displaystyle\sum_{k=0}^{n} k\binom{n}{k}^2$, (b) $\displaystyle\sum_{k=0}^{n} k^2\binom{n}{k}^2$, (c) $\displaystyle\sum_{k=0}^{n} \frac{(-1)^k}{k+1}\binom{n}{k}$

for $n \in \mathbb{N}$.

48. Compute the sums

(a) $\displaystyle\sum_{i=0}^{k} \frac{n-2i+1}{n-i+1}\binom{n}{i}$, (b) $\displaystyle\sum_{i=0}^{n} \frac{n-2i+1}{n-i+1}\binom{n}{i}^2$

for integers $0 \leq k \leq n$.

49. Prove that

$$\binom{2n}{k} = \sum_{i=0}^{\lfloor k/2 \rfloor} \binom{n}{k-i}\binom{k-i}{i} 2^{k-2i}$$

as follows:

(a) By applying the Binomial Theorem.
(b) By counting the elements of a finite set in two different ways.

Deduce that

$$\binom{2n}{n} = \sum_{i=0}^{\lfloor n/2 \rfloor} \binom{n}{i}\binom{n-i}{i} 2^{n-2i}$$

for every $n \in \mathbb{N}$.

50. Prove the identity (2.13) of Example 2.4.2 using only basic principles of enumeration.
51. Let $n \in \mathbb{N}$.

(a) Prove the identity

$$\sum_{i=0}^{n} \frac{(-1)^i}{p+i}\binom{n}{i} = \frac{(p-1)!n!}{(p+n)!}$$

for every $p \in \mathbb{Z}_{>0}$.
(b) Compute the sum

$$\sum_{i=0}^{n} \frac{(-1)^i}{(p+i)(q-i)}\binom{n}{i}$$

for every $p \in \mathbb{Z}_{>0}$ and every integer $q \geq n+1$.

52. Compute the sum $\displaystyle\sum_{k=0}^{n} \frac{\binom{n}{k}}{\binom{2n+1}{k}}$ for $n \in \mathbb{N}$.
53. For $m, n \in \mathbb{N}$ we set

$$f(m,n) = \sum_{k=0}^{m} \binom{m}{k}\binom{n+k}{m}.$$

(a) Prove that

$$f(m,n) = \sum_{k=0}^{\min\{m,n\}} \binom{m}{k}\binom{n}{k} 2^k.$$

(b) Deduce that $f(m,n) = f(n,m)$.

54. For $m, n \in \mathbb{N}$, we denote by $\tau(m, n)$ the number of solutions $(x_1, x_2, \ldots, x_n) \in \mathbb{Z}^n$ of the inequality $|x_1| + |x_2| + \cdots + |x_n| \leq m$, where $\tau(m, 0) := 1$.

    (a) Prove that $\tau(m, n) = f(m, n)$, where $f(m, n)$ was defined in Exercise 53.
    (b) Deduce that $\tau(m, n) = \tau(n, m)$.

55. Prove that

$$\sum_{k=1}^{n} \frac{(-1)^{k-1}}{k} \binom{n}{k} = 1 + \frac{1}{2} + \cdots + \frac{1}{n}$$

    for $n \in \mathbb{Z}_{>0}$ as follows:

    (a) By induction on n.
    (b) By applying the Binomial Theorem.

56. Prove the identity

$$\sum_{k=0}^{m} \binom{m}{k} \binom{n}{k} x^{m-k} y^k = \sum_{k=0}^{m} \binom{m}{k} \binom{n+k}{k} (x - y)^{m-k} y^k$$

    for $m, n \in \mathbb{N}$. Deduce the formula

$$f(m, n) = (-1)^m \sum_{k=0}^{m} \binom{m}{k} \binom{n+k}{k} (-2)^k$$

    for the function $f(m, n)$ of Exercise 53.

57. Let $n \in \mathbb{N}$.

    (a) Prove that

$$\sum_{k=0}^{n} \binom{n}{k}^2 x^k = \sum_{k=0}^{n} \binom{n}{k} \binom{2n-k}{n} (x - 1)^k.$$

    (b) Deduce that

$$\sum_{k=0}^{n} (-1)^k \binom{n}{k} \binom{2n-k}{n} = 1.$$

58. Let $n \in \mathbb{N}$.

    (a) Prove that

$$\binom{n}{k}^2 = \sum_{i+j=k} \binom{n}{i} \binom{n-i}{i} \binom{n-2i}{j}$$

for $k \in \{0, 1, \ldots, n\}$ by counting the elements of a finite set in two different ways.

(b) Deduce that

$$\sum_{k=0}^{n} \binom{n}{k}^2 x^k = \sum_{i=0}^{\lfloor n/2 \rfloor} \binom{n}{i} \binom{n-i}{i} x^i (1+x)^{n-2i}.$$

What happens when we set $x = -1$ in this equality?

59. Prove that

$$\sum_{k=0}^{n} \frac{(-1)^k}{\binom{n}{k}} = \begin{cases} 0, & \text{if } n \text{ is odd} \\ \dfrac{n+1}{n/2+1}, & \text{if } n \text{ is even} \end{cases}$$

for $n \in \mathbb{N}$.

60. Prove Proposition 2.4.3 by suitably generalizing the second proof of part (d) of Proposition 2.3.6, given in the text.

Chapter 3
Inclusion–Exclusion



The principle of inclusion–exclusion is one of the most important principles of enumeration. This chapter states and proves this principle and provides applications to enumerative combinatorics (enumeration of surjective maps, permutations without fixed points and rook placements, proofs of combinatorial identities) and to elementary number theory.

3.1 Principle of Inclusion–Exclusion

How many of the numbers $1, 2, 3, \ldots, 3000$ are not divisible by 3? How many of these numbers are divisible by neither 3 nor 4? How many of these numbers are divisible by none of 3, 4, and 5? Setting $\Omega = \{1, 2, 3, \ldots, 3000\}$, we seek to count the elements of the sets

$\Omega - A_1$,
$\Omega - A_1 \cup A_2$,
$\Omega - A_1 \cup A_2 \cup A_3$,

where A_1, A_2, and A_3 are the subsets of Ω, which consist of the integer multiples of 3, 4, and 5, respectively, in this set. These questions motivate us to find a useful formula for the cardinality of the set difference

$$\Omega - \bigcup_{i=1}^{n} A_i,$$

where Ω is a finite set and $A_1, A_2, \ldots, A_n$ are subsets.

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Applying the addition principle (equivalently, Corollary 2.2.1), we find that

$$\#(\Omega - A_1) = \#\Omega - \#A_1 \tag{3.1}$$

and that

$$\#(A_1 \cup A_2) = \#A_1 + \#(A_2 - (A_1 \cap A_2)) = \#A_1 + \#A_2 - \#(A_1 \cap A_2),$$

whence

$$\#(\Omega - A_1 \cup A_2) = \#\Omega - \#A_1 - \#A_2 + \#(A_1 \cap A_2). \tag{3.2}$$

Here is a way to interpret this formula. To list the elements of the set difference $\Omega - A_1 \cup A_2$, each exactly once, one first includes all elements of Ω, then excludes the elements of A_1 and those of A_2 and finally includes the elements of the intersection $A_1 \cap A_2$, since each of these has been excluded twice. A similar reasoning leads to the formula

$$\begin{aligned}\#(\Omega - A_1 \cup A_2 \cup A_3) = \#\Omega - \#A_1 - \#A_2 - \#A_3 + \\ \#(A_1 \cap A_2) + \#(A_1 \cap A_3) + \#(A_2 \cap A_3) - \\ \#(A_1 \cap A_2 \cap A_3).\end{aligned} \tag{3.3}$$

The inclusion–exclusion principle, a cornerstone of enumerative combinatorics, generalizes the previous identities, as well as Proposition 2.2.2. Usually, it is applied when one wants to count the elements of a finite set Ω, which do *not* have any of n given (not desirable) properties and it is not hard to count the elements of the intersections $A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_k}$ for $1 \le i_1 < \cdots < i_k \le n$, where A_i is the set of elements of Ω, which have the ith property.

Theorem 3.1.1 (Principle of Inclusion–Exclusion). *For a finite set Ω and for any subsets $A_1, A_2, \ldots, A_n$, we have*

$$\#\left(\Omega - \bigcup_{i=1}^{n} A_i\right) = \sum_{k=0}^{n} (-1)^k \sum_{1 \le i_1 < i_2 < \cdots < i_k \le n} \#(A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_k}) \tag{3.4}$$

$$= \sum_{I \subseteq [n]} (-1)^{\#I} \#\bigcap_{i \in I} A_i, \tag{3.5}$$

where the internal sum of the right-hand side of (3.4) runs through all k-subsets $\{i_1, i_2, \ldots, i_k\}$ of $[n]$ and $A_{i_1} \cap \cdots \cap A_{i_k} := \Omega$ for $k = 0$.

Remark 3.1.1. The sum (3.5) has exactly 2^n terms, one for every subset I of $[n]$, and specializes to the right-hand sides of (3.1), (3.2), and (3.3) for $n = 1$, $n = 2$, and $n = 3$, respectively. For $n = 4$, we have

$$\begin{aligned}\#(\Omega - A_1 \cup A_2 \cup A_3 \cup A_4) = {} & \#\Omega - \#A_1 - \#A_2 - \#A_3 - \#A_4 + \\ & \#(A_1 \cap A_2) + \#(A_1 \cap A_3) + \#(A_1 \cap A_4) + \\ & \#(A_2 \cap A_3) + \#(A_2 \cap A_4) + \#(A_3 \cap A_4) - \\ & \#(A_1 \cap A_2 \cap A_3) - \#(A_1 \cap A_2 \cap A_4) - \\ & \#(A_1 \cap A_3 \cap A_4) - \#(A_2 \cap A_3 \cap A_4) + \\ & \#(A_1 \cap A_2 \cap A_3 \cap A_4).\end{aligned}$$

Equation (3.4) may be written equivalently as

$$\#\bigcup_{i=1}^{n} A_i = \sum_{k=1}^{n} (-1)^{k-1} \sum_{1 \le i_1 < \cdots < i_k \le n} \#(A_{i_1} \cap \cdots \cap A_{i_k}). \tag{3.6}$$

We give a few applications before proving Theorem 3.1.1.

Example 3.1.1.

(a) Let $\Omega = \{1, 2, \ldots, 3000\}$ and let A_1, A_2, A_3 consist of those elements of Ω, which are divisible by 3, 4, and 5, respectively, as in the beginning of this section.

Clearly, $\#\Omega = 3000$, $\#A_1 = 3000/3 = 1000$, $\#A_2 = 3000/4 = 750$, and $\#A_3 = 3000/5 = 600$ (explain why). Moreover, since $A_1 \cap A_2$, $A_1 \cap A_3$, $A_2 \cap A_3$, and $A_1 \cap A_2 \cap A_3$ consist of those elements of Ω, which are divisible by 12, 15, 20, and 60, respectively, we have $\#(A_1 \cap A_2) = 3000/12 = 250$, $\#(A_1 \cap A_3) = 3000/15 = 200$, $\#(A_2 \cap A_3) = 3000/20 = 150$, and $\#(A_1 \cap A_2 \cap A_3) = 3000/60 = 50$. From these computations and Equations (3.1), (3.2), and (3.3), we infer that

$$\begin{aligned}&\#(\Omega - A_1) = 3000 - 1000 = 2000,\\ &\#(\Omega - A_1 \cup A_2) = 3000 - 1000 - 750 + 250 = 1500,\\ &\#(\Omega - A_1 \cup A_2 \cup A_3) = 3000 - 1000 - 750 - 600 + 250 + 200 + 150 - 50 = 1200.\end{aligned}$$

Therefore, 2000 elements of Ω are not divisible by 3, 1500 elements are divisible by neither 3 nor 4, and 1200 elements are not divisible by any of 3, 4, and 5.

(b) Let us consider the cyclic permutations of eight people formed by four married couples $\{a_1, b_1\}$, $\{a_2, b_2\}$, $\{a_3, b_3\}$, and $\{a_4, b_4\}$. How many of these have the property that no two married persons sit next to each other?

Let Ω be the set of all cyclic permutations of the eight people and for $1 \le i \le 4$, let A_i be the subset of those cyclic permutations in which a_i and b_i sit next to each other. Then, we need to count the elements of the set $\Omega - A_1 \cup A_2 \cup A_3 \cup A_4$. We have $\#\Omega = 7!$ by Proposition 2.3.4. Moreover, $\#A_i = 2 \cdot 6!$ for $1 \le i \le 4$ because

to choose an element of A_i, there are two ways to choose the relative position of a_i and b_i, who are sitting next to each other, and 6! ways to permute cyclically the couple $\{a_i, b_i\}$ (considered as a single object) and the other six people. Similarly, we find that

$$\begin{aligned} \#(A_i \cap A_j) &= 2^2 \cdot 5! \text{ for all pairs } 1 \le i < j \le 4, \\ \#(A_i \cap A_j \cap A_k) &= 2^3 \cdot 4! \text{ for all triples } 1 \le i < j < k \le 4, \\ \#(A_1 \cap A_2 \cap A_3 \cap A_4) &= 2^4 \cdot 3!. \end{aligned}$$

From these computations and Theorem 3.1.1, we infer that

$$\#(\Omega - A_1 \cup A_2 \cup A_3 \cup A_4) = 7! - 4 \cdot 2 \cdot 6! + 6 \cdot 2^2 \cdot 5! - 4 \cdot 2^3 \cdot 4! + 2^4 \cdot 3! = 1488.$$

That is, there are exactly 1488 cyclic permutations with the desired properties. □

Example 3.1.2. Each of the nine squares of a 3×3 chessboard is colored either black or white. How many of these colorings are such that no row or column of the chessboard has all its squares colored black?

Let Ω be the set of all colorings of the squares of the chessboard with black or white color and for $1 \le i \le 3$, let A_i (respectively, B_i) be the subset of those colorings of Ω for which all squares of row i (respectively, column i) have been colored black. Then, we are asked to compute the cardinality of $\Omega - A_1 \cup A_2 \cup A_3 \cup B_1 \cup B_2 \cup B_3$. We observe (explain how) that

$$\begin{aligned} \#\Omega &= 2^9 = 512, \\ \#A_i = \#B_i &= 2^6 = 64 \text{ for all } 1 \le i \le 3, \\ \#(A_i \cap A_j) = \#(B_i \cap B_j) &= 2^3 = 8 \text{ for all pairs } 1 \le i < j \le 3, \\ \#(A_i \cap B_j) &= 2^4 = 16 \text{ for all } 1 \le i, j \le 3, \\ \#(A_i \cap A_j \cap B_k) &= 2^2 = 4 \text{ for all pairs } 1 \le i < j \le 3 \text{ and all } 1 \le k \le 3, \\ \#(A_i \cap B_j \cap B_k) &= 2^2 = 4 \text{ for all } 1 \le i \le 3 \text{ and all pairs } 1 \le j < k \le 3, \\ \#(A_i \cap A_j \cap B_k \cap B_\ell) &= 2 \text{ for all pairs } 1 \le i < j \le 3 \text{ and } 1 \le k < \ell \le 3. \end{aligned}$$

All other intersections of the A_i and the B_j involve all of A_1, A_2, A_3 or all of B_1, B_2, B_3 and thus consist of a single element. Therefore, Theorem 3.1.1 yields that

$$\begin{aligned} \#(\Omega - \cup_{i=1}^3 (A_i \cup B_i)) = 512 - 6 \cdot 64 + 6 \cdot 8 + 9 \cdot 16 - 2 - 18 \cdot 4 + 6 \cdot 1 + \\ 9 \cdot 2 - 6 \cdot 1 + 1 = 265. \end{aligned}$$

Hence, there are 265 colorings with the desired properties.

Let us solve this problem by a slightly easier computation. Let Ω_0 be the set of all colorings of the squares of the 3×3 chessboard with black or white color for which no row has all its squares colored black. For $1 \le i \le 3$, let C_i be the subset of those colorings of Ω_0 for which all squares of column i of the chessboard have been colored black. Then, we need to count the elements of $\Omega_0 - C_1 \cup C_2 \cup C_3$. We have $\#\Omega_0 = 7^3$ since there are $2^3 - 1 = 7$ ways to color the three squares of a row so that at least one square has been colored white. Similarly, we find that $\#C_1 = \#C_2 = \#C_3 = 3^3$, $\#(C_1 \cap C_2) = \#(C_1 \cap C_3) = \#(C_2 \cap C_3) = 1^3$ and $\#(C_1 \cap C_2 \cap C_3) = 0$ and conclude that

$$\#(\Omega_0 - C_1 \cup C_2 \cup C_3) = 7^3 - 3 \cdot 3^3 + 3 \cdot 1^3 - 0 = 265,$$

in agreement with our previous computation.

For the case of $m \times n$ chessboards, see Exercise 10. □

First Proof of Theorem 3.1.1. We set

$$\chi_i(x) = \begin{cases} 1, & \text{if } x \in A_i \\ 0, & \text{otherwise} \end{cases}$$

for $i \in [n]$ and $x \in \Omega$ (the map $\chi_i : \Omega \to \{0, 1\}$ defined in this way is called the *characteristic function* of the subset A_i of Ω). We observe that

$$\prod_{i=1}^{n} (1 - \chi_i(x)) = \begin{cases} 0, & \text{if } x \in \bigcup_{i=1}^{n} A_i \\ 1, & \text{otherwise} \end{cases} \tag{3.7}$$

and conclude that

$$\#\left(\Omega - \bigcup_{i=1}^{n} A_i\right) = \sum_{x \in \Omega} \prod_{i=1}^{n} (1 - \chi_i(x)) = \sum_{x \in \Omega} \sum_{I \subseteq [n]} (-1)^{\#I} \prod_{i \in I} \chi_i(x)$$

$$= \sum_{I \subseteq [n]} (-1)^{\#I} \sum_{x \in \Omega} \prod_{i \in I} \chi_i(x),$$

where in the second equality we have expanded the product $\prod_{i=1}^{n} (1 - \chi_i(x))$ by means of the identity (2.5). We also observe that

$$\prod_{i \in I} \chi_i(x) = \begin{cases} 1, & \text{if } x \in \bigcap_{i \in I} A_i \\ 0, & \text{otherwise,} \end{cases}$$

so that

$$\sum_{x\in\Omega}\prod_{i\in I}\chi_i(x) = \#\bigcap_{i\in I} A_i$$

for $I \subseteq [n]$, and the proof follows. □

Second Proof of Theorem 3.1.1. We set

$$\chi_I(x) = \begin{cases} 1, & \text{if } x \in \bigcap_{i\in I} A_i \\ 0, & \text{otherwise} \end{cases}$$

for $I \subseteq [n]$ and $x \in \Omega$ and observe that

$$\#\bigcap_{i\in I} A_i = \sum_{x\in\Omega}\chi_I(x).$$

Thus, denoting by B the right-hand side of (3.5), we have

$$B = \sum_{I\subseteq[n]}(-1)^{\#I}\sum_{x\in\Omega}\chi_I(x) = \sum_{x\in\Omega} c(x), \tag{3.8}$$

where

$$c(x) = \sum_{I\subseteq[n]}(-1)^{\#I}\chi_I(x)$$

for $x \in \Omega$. Suppose that x belongs to exactly m of the sets $A_1, A_2, \ldots, A_n$, where $0 \le m \le n$. Then, x belongs to exactly $\binom{m}{2}$ of the intersections $A_i \cap A_j$ with $i < j$, to exactly $\binom{m}{3}$ of the intersections $A_i \cap A_j \cap A_r$ with $i < j < r$, and so on. In other words, for $0 \le k \le n$ we have $\chi_I(x) = 1$ for $\binom{m}{k}$ of the k-subsets I of $[n]$ and $\chi_I(x) = 0$ for the rest. As a result,

$$c(x) = \binom{m}{0} - \binom{m}{1} + \binom{m}{2} - \cdots + (-1)^m\binom{m}{m}.$$

This equality and Proposition 2.3.6 (d) yield that $c(x) = 0$ if $m \ge 1$, meaning that $x \in \bigcup_{i=1}^{n} A_i$, and that $c(x) = 1$ if $m = 0$, meaning that $x \in \Omega - \bigcup_{i=1}^{n} A_i$. The proof follows from these facts and (3.8). □

The following section provides applications of the principle of inclusion–exclusion to some interesting enumeration problems. The following examples apply this principle in order to prove two combinatorial identities.

Example 3.1.3. Let's consider the special case of Theorem 3.1.1 in which $\#\Omega = 1$ and $A_i = \Omega$ for every $i \in [n]$. Then,

$$\Omega - \bigcup_{i=1}^{n} A_i = \begin{cases} \Omega, & \text{if } n = 0 \\ \varnothing, & \text{if } n \geq 1 \end{cases}$$

and $\bigcap_{i \in I} A_i = \Omega$ for every $I \subseteq [n]$. Thus, since there are exactly $\binom{n}{k}$ sets $I \subseteq [n]$ such that $\#I = k$, Equation (3.5) specializes to the identity

$$\sum_{k=0}^{n} (-1)^k \binom{n}{k} = \begin{cases} 1, & \text{if } n = 0 \\ 0, & \text{if } n \geq 1 \end{cases}$$

of Proposition 2.3.6 (d). □

Example 3.1.4. Let Ω be the set of all n-subsets of $[2n]$ and for $i \in [n]$, let $A_i \subseteq \Omega$ consist of all $S \in \Omega$ such that $i \in S$. Then, $\{n+1, n+2, \ldots, 2n\}$ is the only element of $\Omega - \bigcup_{i=1}^{n} A_i$. Moreover, for every k-subset I of $[n]$, the intersection $\bigcap_{i \in I} A_i$ consists of all n-subsets of $[2n]$, which contain I and hence,

$$\# \bigcap_{i \in I} A_i = \binom{2n-k}{n-k} = \binom{2n-k}{n}.$$

Hence, Equation (3.5) specializes to the identity

$$\sum_{k=0}^{n} (-1)^k \binom{n}{k} \binom{2n-k}{n} = 1$$

for $n \in \mathbb{N}$. Thus, we have proven the identity of Exercise 57 of Chapter 2 by applying the principle of inclusion–exclusion. □

3.2 Applications

3.2.1 Surjective Maps

We recall that a map $f : X \to Y$ is said to be surjective (or onto) if for every $y \in Y$ there exists $x \in X$ such that $f(x) = y$.

Proposition 3.2.1. *For any positive integers m and n, the number of surjective maps $f : [m] \to [n]$ is equal to*

$$\sum_{k=0}^{n} (-1)^{n-k} \binom{n}{k} k^m.$$

Proof. Let Ω be the set of all maps $f : [m] \to [n]$. For $i \in [n]$, let A_i be the set of maps $f \in \Omega$ for which there is no $x \in [m]$ such that $f(x) = i$. Then, the set of surjective maps $f : [m] \to [n]$ coincides with $\Omega - \cup_{i=1}^{n} A_i$. According to Proposition 2.3.1, we have $\#\Omega = n^m$. More generally, for indices $1 \leq i_1 < \cdots < i_k \leq m$, the same proposition implies that

$$\#\, A_{i_1} \cap \cdots \cap A_{i_k} = \#\, \{f : [m] \to [n] \smallsetminus \{i_1, \ldots, i_k\}\} = (n-k)^m.$$

Theorem 3.1.1 then yields that

$$\#\left(\Omega - \bigcup_{i=1}^{n} A_i\right) = \sum_{k=0}^{n} (-1)^k \binom{n}{k} (n-k)^m,$$

which is equivalent to the proposed formula. □

EXAMPLE. Applying Proposition 3.2.1 for $m = 5$ and $n = 3$, we find that there are $3^5 - 3 \cdot 2^5 + 3 \cdot 1^5 - 0 = 150$ surjective maps $f : [5] \to [3]$.

Remark 3.2.1. The argument given in the proof of Proposition 3.2.1 is valid regardless of whether $m < n$ or $m \geq n$. Moreover, according to Proposition 2.3.2 and Exercises 1 and 3 of Chapter 2, the number of surjective maps $f : [m] \to [n]$ is equal to zero, if $m < n$ and equal to $n!$, if $m = n$. We conclude that

$$\sum_{k=0}^{n} (-1)^{n-k} \binom{n}{k} k^m = \begin{cases} 0, & \text{if } m < n \\ n!, & \text{if } m = n. \end{cases} \tag{3.9}$$

3.2.2 Euler's Function

Two positive integers, say a and b, are said to be *relatively prime* (we also say that a is relatively prime to b and vice versa) if they do not have any common prime divisor. How many of the integers $1, 2, \ldots, m$ are relatively prime to m? Denoting this number by $\varphi(m)$, we get a map $\varphi : \mathbb{Z}_{>0} \to \mathbb{Z}_{>0}$ known as *Euler's totient function*. For the importance of this function in number theory see, for instance, [4, Chapter 3]. For example, $\varphi(9) = 6$ and $\varphi(10) = 4$.

To prove the following proposition, we take for granted the statement from elementary number theory that a given positive integer m is divisible by two distinct prime numbers p and q if and only if m is divisible by their product pq.

Proposition 3.2.2. *We have*

$$\varphi(m) = m \cdot \prod_{i=1}^{n} \left(1 - \frac{1}{p_i}\right), \tag{3.10}$$

for every $m \in \mathbb{Z}_{>0}$, where $p_1, p_2, \ldots, p_n$ are the distinct prime divisors of m. In particular, if m and m' are relatively prime positive integers, then $\varphi(mm') = \varphi(m)\varphi(m')$.

Proof. Since $\varphi(1) = 1$, we may assume that $m, m' \geq 2$. We have defined $\varphi(m)$ to be the number of integers $d \in [m]$, which are not divisible by any of the primes $p_1, p_2, \ldots, p_n$. We set $\Omega = [m]$ and for $i \in [n]$, we denote by A_i the set of elements of Ω, which are divisible by p_i. Then, $\varphi(m) = \#(\Omega - \cup_{i=1}^{n} A_i)$. We now apply the inclusion–exclusion principle. For indices $1 \leq i_1 < i_2 < \cdots < i_k \leq n$, the intersection $A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_k}$ consists of all integers $d \in [m]$, which are divisible by $p_{i_1} p_{i_2} \cdots p_{i_k}$. Hence, $\#(A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_k}) = m/p_{i_1} p_{i_2} \cdots p_{i_k}$ and Theorem 3.1.1 yields that

$$\varphi(m) = \#\left(\Omega - \bigcup_{i=1}^{n} A_i\right) = \sum_{1 \leq i_1 < i_2 < \cdots < i_k \leq n} (-1)^k \frac{m}{p_{i_1} p_{i_2} \cdots p_{i_k}}$$

$$= m \cdot \prod_{i=1}^{n} \left(1 - \frac{1}{p_i}\right).$$

Suppose now that m, m' are relatively prime, and let $p_1, p_2, \ldots, p_n$ be the distinct prime divisors of m and $q_1, q_2, \ldots, q_{n'}$ be those of m'. Then, $p_1, \ldots, p_n, q_1, \ldots, q_{n'}$ are the distinct prime divisors of mm'. Applying (3.10) for m, m' and mm', we get

$$\varphi(mm') = mm' \cdot \prod_{i=1}^{n} \left(1 - \frac{1}{p_i}\right) \prod_{i=1}^{n'} \left(1 - \frac{1}{q_i}\right)$$

$$= m \cdot \prod_{i=1}^{n} \left(1 - \frac{1}{p_i}\right) \cdot m' \cdot \prod_{i=1}^{n'} \left(1 - \frac{1}{q_i}\right) = \varphi(m)\varphi(m')$$

and the proof follows. □

EXAMPLE. The distinct prime divisors of $m = 600$ are 2, 3, and 5, since $600 = 2^3 \cdot 3 \cdot 5^2$. Therefore, $\varphi(600) = 600(1 - 1/2)(1 - 1/3)(1 - 1/5) = 160$.

3.2.3 *Permutations Without Fixed Points*

Let $\sigma = (\sigma_1, \sigma_2, \ldots, \sigma_n)$ be a permutation of the set $[n]$. We say that $i \in [n]$ is a *fixed point* of σ if $\sigma_i = i$. For example, $(4, 2, 1, 3, 5)$ is a permutation of [5] with two fixed points, namely $i = 2$ and $i = 5$, and $(2, 3, 1)$ and $(3, 1, 2)$ are the only permutations of [3] without fixed points.

How many permutations of $[n]$ have a given number k of fixed points? We first consider the important special case $k = 0$ and denote by D_n the number of permutations of $[n]$ without fixed points; such permutations are called *derangements*. Equivalently, in view of the one-to-one correspondence described in Remark 2.3.1, D_n is equal to the number of placements of n nonattacking identical rooks on an $n \times n$ chessboard in which no rook is placed on the main diagonal of the chessboard. We have $D_1 = 0$, $D_2 = 1$, $D_3 = 2$, $D_4 = 9$.

Proposition 3.2.3. *Setting $D_0 := 1$, the following hold for the number D_n of permutations of $[n]$ without fixed points:*

(a) $D_n = n!\left(1 - \frac{1}{1!} + \frac{1}{2!} - \cdots + \frac{(-1)^n}{n!}\right)$ *for every* $n \in \mathbb{N}$.
(b) $D_n = nD_{n-1} + (-1)^n$ *for* $n \geq 1$.
(c) $D_n = (n-1)(D_{n-1} + D_{n-2})$ *for* $n \geq 2$.
(d) *Given* $k \in \{0, 1, \ldots, n\}$, *the number of permutations of* $[n]$, *which have exactly* k *fixed points is equal to* $\binom{n}{k} D_{n-k}$.

Proof. Let Ω be the set of all permutations of $[n]$ and for $i \in [n]$, let A_i be the subset of those permutations $\sigma = (\sigma_1, \sigma_2, \ldots, \sigma_n) \in \Omega$ such that $\sigma_i = i$. Then, D_n is the cardinality of the set $\Omega - \cup_{i=1}^n A_i$, which consists of all derangements of $[n]$.

We observe that for indices $1 \leq i_1 < i_2 < \cdots < i_k \leq n$, the intersection $A_{i_1} \cap \cdots \cap A_{i_k}$ is the set of all permutations of $[n]$ for which $i_1, \ldots, i_k$ are fixed points. The cardinality of this intersection is equal to the number of permutations of the set $[n] \setminus \{i_1, \ldots, i_k\}$ and hence, $\#(A_{i_1} \cap \cdots \cap A_{i_k}) = (n-k)!$. This fact and Theorem 3.1.1 yield that

$$D_n = \#\left(\Omega - \bigcup_{i=1}^n A_i\right) = \sum_{k=0}^n (-1)^k \binom{n}{k} (n-k)! = \sum_{k=0}^n \frac{(-1)^k n!}{k!(n-k)!} (n-k)!$$

$$= n! \sum_{k=0}^n \frac{(-1)^k}{k!} = n!\left(1 - \frac{1}{1!} + \frac{1}{2!} - \cdots + \frac{(-1)^n}{n!}\right).$$

We have proven part (a). Part (b) is a direct consequence of (a), since

$$D_n = n!\left(1 - \frac{1}{1!} + \frac{1}{2!} - \cdots + \frac{(-1)^n}{n!}\right)$$

$$= n!\left(1 - \frac{1}{1!} + \frac{1}{2!} - \cdots + \frac{(-1)^{n-1}}{(n-1)!}\right) + (-1)^n$$

$$= n \cdot (n-1)!\left(1 - \frac{1}{1!} + \frac{1}{2!} - \cdots + \frac{(-1)^{n-1}}{(n-1)!}\right) + (-1)^n$$

$$= nD_{n-1} + (-1)^n$$

for $n \geq 1$. Part (c) is a direct consequence of (b). Indeed, applying the latter twice, we get $(n-1)(D_{n-1}+D_{n-2}) = nD_{n-1} - D_{n-1} + (n-1)D_{n-2} = nD_{n-1} - (-1)^{n-1} = D_n$ for $n \geq 2$. Finally, to choose a permutation $\sigma = (\sigma_1, \sigma_2, \ldots, \sigma_n)$ of $[n]$ with exactly k fixed points, there are $\binom{n}{k}$ ways to choose the fixed points $i_1, i_2, \ldots, i_k$ of σ and for every such choice, there are D_{n-k} ways to permute the remaining $n-k$ elements of $[n]$ so that $\sigma_j \neq j$ for every $j \in [n] \setminus \{i_1, i_2, \ldots, i_k\}$. Thus, part (d) follows from this argument and the multiplication principle. □

EXAMPLE. There are $D_5 = 5!(1 - 1 + 1/2 - 1/6 + 1/24 - 1/120) = 44$ permutations of the set $[5]$ without fixed points. Applying the recursive formula of Proposition 3.2.3 (b), we find that $D_6 = 6D_5 + 1 = 265$, $D_7 = 7D_6 - 1 = 1854$, $D_8 = 8D_7 + 1 = 14833$. □

The next corollary may be restated as follows. Choosing a permutation σ of $[n]$ uniformly at random, the probability that σ has no fixed point converges to e^{-1} when n approaches infinity.

Corollary 3.2.1. *We have*

$$\lim_{n\to\infty} \frac{D_n}{n!} = e^{-1},$$

where e is the base for the natural logarithm.

Proof. Given the formula of Proposition 3.2.3 (a), we compute that

$$\lim_{n\to\infty} \frac{D_n}{n!} = \lim_{n\to\infty} \left(1 - \frac{1}{1!} + \frac{1}{2!} - \cdots + \frac{(-1)^n}{n!}\right) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} = e^{-1}$$

and the proof follows. □

3.2.4 Rook Placements

The question studied in the previous section may be generalized as follows. Let $\mathcal{B}$ be any subset of the set $[n] \times [n]$. How many permutations $\sigma = (\sigma_1, \sigma_2, \ldots, \sigma_n)$ of $[n]$ are such that $(i, \sigma_i) \notin \mathcal{B}$ for every $i \in [n]$? In the special case $\mathcal{B} = \{(1,1), (2,2), \ldots, (n,n)\}$, we are asking to count the permutations of $[n]$ without fixed points.

Let's consider the elements of $[n] \times [n]$ as the squares of an $n \times n$ chessboard numbered in matrix notation, as in Example 2.2.6. Then, $\mathcal{B}$ consists of some of these squares. In view of the one-to-one correspondence between permutations and rook placements described in Remark 2.3.1, the number of permutations of $[n]$ we ask to compute is equal to the number of placements of n nonattacking identical rooks in the squares of the $n \times n$ chessboard, which have the property that no rook has been placed in any of the forbidden squares of $\mathcal{B}$. More generally, for $i \in \{0, 1, \ldots, n\}$, let

us denote by $N_i(\mathcal{B})$ the number of possible placements of n nonattacking identical rooks in the squares of the $n \times n$ chessboard for which exactly i rooks have been placed in squares, which belong to $\mathcal{B}$, and let us consider the polynomial

$$N_{\mathcal{B}}(x) := \sum_{i=0}^{n} N_i(\mathcal{B}) x^i.$$

The constant term $N_{\mathcal{B}}(0) = N_0(\mathcal{B})$ of this polynomial is equal to the number of placements that avoid the squares of $\mathcal{B}$.

Theorem 3.2.1. *We have*

$$N_{\mathcal{B}}(x) = \sum_{k=0}^{n} r_k(\mathcal{B})\,(n-k)!\,(x-1)^k, \tag{3.11}$$

where $r_k(\mathcal{B})$ is defined as the number of possible placements of k nonattacking identical rooks in the squares of $\mathcal{B}$, for $k \in \{0, 1 \ldots, n\}$. In particular,

$$N_{\mathcal{B}}(0) = \sum_{k=0}^{n} (-1)^k r_k(\mathcal{B})\,(n-k)!. \tag{3.12}$$

EXAMPLE. For the subboard $\mathcal{B}$ of the 4×4 chessboard of Figure 3.1, we have $r_0(\mathcal{B}) = 1$, $r_1(\mathcal{B}) = 7$, $r_2(\mathcal{B}) = 15$, $r_3(\mathcal{B}) = 10$, and $r_4(\mathcal{B}) = 1$. Hence,

$$\begin{aligned} N_{\mathcal{B}}(x) &= 24 + 7 \cdot 6(x-1) + 15 \cdot 2(x-1)^2 + 10 \cdot 1(x-1)^3 + (x-1)^4 \\ &= 3 + 8x + 6x^2 + 6x^3 + x^4 \end{aligned}$$

or, equivalently, $N_0(\mathcal{B}) = 3$, $N_1(\mathcal{B}) = 8$, $N_2(\mathcal{B}) = N_3(\mathcal{B}) = 6$, and $N_4(\mathcal{B}) = 1$.

Proof of Theorem 3.2.1. Let Ω be the set of all $n!$ possible placements of n nonattacking identical rooks in the squares of the $n \times n$ chessboard. By definition, we have

$$N_i(\mathcal{B}) = \sum_{X \subseteq \mathcal{B},\, \#X = i} f(X),$$

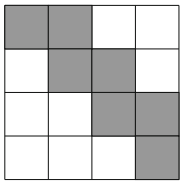


Figure 3.1 A subboard of the 4×4 chessboard

where for $X \subseteq \mathcal{B}$ we have denoted by $f(X)$ the number of placements in Ω for which rooks have been placed in the squares of X, but not in other squares of $\mathcal{B}$. We claim that

$$f(X) = \sum_{X \subseteq Y \subseteq \mathcal{B}} (-1)^{|Y \setminus X|} g(Y)$$

for every $X \subseteq \mathcal{B}$, where $g(Y)$ is the number of placements in Ω for which rooks have been placed in the squares of $Y \subseteq \mathcal{B}$ (and possibly in other squares of $\mathcal{B}$ as well). Indeed, our claim follows if we observe that

$$f(X) = \# \left(\Omega_X \setminus \bigcup_{\tau \in \mathcal{B} \setminus X} A_\tau \right),$$

where Ω_X is the set of placements in Ω for which rooks have been placed in the squares of X (and possibly in other squares of $\mathcal{B}$ as well), and A_τ is the subset of Ω_X consisting of those placements for which a rook has been placed in $\tau \in \mathcal{B} \setminus X$ as well, and we apply Theorem 3.1.1. We also note that

$$\sum_{Y \subseteq \mathcal{B},\, \#Y = k} g(Y) = r_k(\mathcal{B})\, (n-k)!$$

by the multiplication principle, since the sum on the left-hand side is equal to the number of ways to place k nonattacking identical rooks in the squares of $\mathcal{B}$ and extend this placement to a placement of n nonattacking identical rooks in the squares of the $n \times n$ chessboard.

Given these facts, we compute that

$$\begin{aligned} N_i(\mathcal{B}) &= \sum_{X \subseteq \mathcal{B},\, \#X = i} f(X) = \sum_{X \subseteq \mathcal{B},\, \#X = i} \left(\sum_{X \subseteq Y \subseteq \mathcal{B}} (-1)^{|Y \setminus X|} g(Y) \right) \\ &= \sum_{Y \subseteq \mathcal{B}} g(Y) \left(\sum_{X \subseteq Y,\, \#X = i} (-1)^{|Y \setminus X|} \right) \\ &= \sum_{k=0}^{n} \sum_{Y \subseteq \mathcal{B},\, \#Y = k} g(Y) \left(\sum_{X \subseteq Y,\, \#X = i} (-1)^{|Y \setminus X|} \right) \end{aligned}$$

$$= \sum_{k=0}^{n} (-1)^{k-i} \binom{k}{i} \sum_{Y \subseteq \mathcal{B},\, \#Y=k} g(Y) = \sum_{k=0}^{n} (-1)^{k-i} \binom{k}{i} r_k(\mathcal{B})\, (n-k)!$$

$$= [x^i] \sum_{k=0}^{n} r_k(\mathcal{B})\, (n-k)!\, (x-1)^k$$

for every $i \in \{0, 1, \ldots, n\}$. This formula for $N_i(\mathcal{B})$ is equivalent to Equation (3.11). □

Example 3.2.1. Let $\mathcal{B}$ consist of the squares (i, i) for $i \in [n]$ and $(i, i+1)$ for $i \in [n-1]$ (see Figure 3.1 for the case $n = 4$). Suppose that we choose k squares of $\mathcal{B}$ in order to place k identical rooks there (one in every square). For the rooks to be nonattacking, a necessary and sufficient condition is that there are no two successive squares in the linear order

$$(1, 1),\ (1, 2),\ (2, 2),\ (2, 3), \ldots, (n-1, n),\ (n, n)$$

of the squares of $\mathcal{B}$, among those we have chosen. Hence, $r_k(\mathcal{B})$ (as defined in the statement of Theorem 3.2.1) is equal to the number of subsets of $[2n-1]$, which do not contain two consecutive integers. Therefore, according to the result of Exercise 32 of Chapter 2,

$$r_k(\mathcal{B}) = \binom{2n-k}{k}$$

for every $k \in \{0, 1, \ldots, n\}$. From Equations (3.11) and (3.12), we deduce that

$$N_{\mathcal{B}}(x) = \sum_{k=0}^{n} \binom{2n-k}{k} (n-k)!(x-1)^k,$$

$$N_{\mathcal{B}}(0) = \sum_{k=0}^{n} (-1)^k \binom{2n-k}{k} (n-k)!$$

for every $n \in \mathbb{Z}_{>0}$. In particular, the sum on the right-hand side of the last equality expresses the number of permutations $\sigma = (\sigma_1, \sigma_2, \ldots, \sigma_n)$ of $[n]$ such that $\sigma_i \neq i, i+1$ for every $i \in [n]$. □

3.3 A Generalization

This section discusses a more abstract and more general form of the inclusion–exclusion principle. As a mathematical statement, this generalization has no reference to enumeration; it states that a certain system of linear equations, the unknowns

of which correspond to the subsets of a finite set, has a unique explicit solution. This form can be further generalized to the Möbius Inversion Formula [23, Section 3.7], in the setting of partially ordered sets.

We will first state and prove the generalized principle of inclusion–exclusion and then explain why it implies the standard form (Theorem 3.1.1) of the principle. The statement and its proof are valid for maps taking values in any abelian group.

Theorem 3.3.1 (Generalized Principle of Inclusion–Exclusion). *Let I be a finite set. For two maps $f, g : 2^I \to \mathbb{C}$ we have*

$$g(Y) = \sum_{X \subseteq Y} f(X) \tag{3.13}$$

for every $Y \subseteq I$ if and only if

$$f(Y) = \sum_{X \subseteq Y} (-1)^{|Y \setminus X|} g(X) \tag{3.14}$$

for every $Y \subseteq I$.

Proof. Suppose that Equation (3.13) holds for every $Y \subseteq I$. Then, for any given $Y \subseteq I$,

$$\begin{aligned}
\sum_{X \subseteq Y} (-1)^{|Y \setminus X|} g(X) &= \sum_{X \subseteq Y} (-1)^{|Y \setminus X|} \sum_{U \subseteq X} f(U) \\
&= \sum_{U \subseteq Y} f(U) \sum_{U \subseteq X \subseteq Y} (-1)^{|Y \setminus X|} \\
&= \sum_{U \subseteq Y} f(U) \sum_{V \subseteq Y \setminus U} (-1)^{|V|} \\
&= f(Y).
\end{aligned}$$

The last equality holds because, setting $m = |Y \setminus U|$, we have

$$\sum_{V \subseteq Y \setminus U} (-1)^{|V|} = \sum_{i=0}^{m} (-1)^i \binom{m}{i} = \begin{cases} 1, & \text{if } m = 0 \\ 0, & \text{otherwise} \end{cases}$$

$$= \begin{cases} 1, & \text{if } U = Y \\ 0, & \text{otherwise.} \end{cases}$$

Therefore, (3.14) holds for every $Y \subseteq I$. The proof of the converse is similar. □

The following dual form can be deduced from Theorem 3.3.1 or proven in an entirely similar way.

Theorem 3.3.2 (Generalized Principle of Inclusion–Exclusion, Dual Form). *Let I be a finite set. For two maps $f, g : 2^I \to \mathbb{C}$, we have*

$$g(X) = \sum_{X \subseteq Y \subseteq I} f(Y) \tag{3.15}$$

for every $X \subseteq I$ if and only if

$$f(X) = \sum_{X \subseteq Y \subseteq I} (-1)^{|Y \smallsetminus X|} g(Y) \tag{3.16}$$

for every $X \subseteq I$.

Let us now deduce Theorem 3.1.1 from Theorem 3.3.2.

Third Proof of Theorem 3.1.1. We set $I = [n]$ and for $S \subseteq I$, we consider the sets

$$A_S = \bigcap_{i \in S} A_i, \qquad B_S = A_S \smallsetminus \left(\bigcup_{j \in I \smallsetminus S} A_j \right),$$

where $A_\varnothing := \Omega$. Thus, A_S consists of the elements $x \in \Omega$ such that $i \in S \Rightarrow x \in A_i$ and B_S consists of the elements $x \in \Omega$ such that $x \in A_i \Leftrightarrow i \in S$. For example, for $n = 3$ we have $A_{\{1,2\}} = A_1 \cap A_2$ and $B_{\{1,2\}} = (A_1 \cap A_2) \smallsetminus A_3$.

We consider the maps $f, g : 2^I \to \mathbb{C}$ defined by setting $g(S) = \#A_S$ and $f(S) = \#B_S$, for $S \subseteq I$. We observe that every $x \in \Omega$ belongs to exactly one of the sets B_S, namely the one for which $S = \{i \in I : x \in A_i\}$. As a result, Ω is the disjoint union of the sets B_S for $S \subseteq I$. More generally, A_S is the disjoint union of the sets B_T for $T \supseteq S$. Therefore, by the addition principle,

$$g(S) = \sum_{S \subseteq T \subseteq I} f(T)$$

for every $S \subseteq [n]$. Theorem 3.3.2 then implies that

$$f(S) = \sum_{S \subseteq T \subseteq I} (-1)^{|T \smallsetminus S|} g(T)$$

for every $S \subseteq [n]$. Since $f(\varnothing) = \#(\Omega - \cup_{i=1}^n A_i)$, the special case $S = \varnothing$ of this equation yields the desired formula. □

The previous proof suggests that Theorem 3.3.2 can be applied to enumeration problems as follows. Let Ω be a finite set the elements of which may or may not have some properties, indexed by the set I. If $g(S)$ is the number of elements of

Ω, which have all properties corresponding to the elements of $S \subseteq I$ (and possibly more), then $f(S)$ is the number of elements of Ω, which have only the properties corresponding to the elements of S (and no additional property). In particular, $f(\varnothing)$ is the number of elements of Ω, which have none of the properties.

Application: Descent Sets We now provide a typical application of Theorem 3.3.1. Let m, n be positive integers and let $\sigma = (a_1, a_2, \ldots, a_n) \in [m]^n$ be a sequence of length n, with entries in the set $[m]$ (such sequences are also called *words*). An index $i \in [n-1]$ is called a *descent* of σ if $a_i > a_{i+1}$. The *descent set* of σ, denoted by $\mathrm{Des}(\sigma)$, is the set which consists of all descents of σ. For example, if $m = n = 6$ and $\sigma = (4, 2, 2, 6, 3, 1)$, then $\mathrm{Des}(\sigma) = \{1, 4, 5\}$. For a given $S \subseteq [n-1]$, how many words $\sigma \in [m]^n$ with $\mathrm{Des}(\sigma) = S$ are there?

To answer this question, we set $I := [n-1]$ and for $X \subseteq I$, we denote by $f(X)$ and $g(X)$ the number of sequences $\sigma \in [m]^n$ such that $\mathrm{Des}(\sigma) = X$ and $\mathrm{Des}(\sigma) \subseteq X$, respectively. Then, the maps $f, g : 2^I \to \mathbb{C}$ obviously satisfy Equation (3.13) for every $Y \subseteq I$, and hence, they must satisfy Equation (3.14) for every $Y \subseteq I$ as well. The advantage of considering the map g is that it is easier to compute. Indeed, let $X = \{s_1, s_2, \ldots, s_{k-1}\}$, where $1 \leq s_1 < s_2 < \cdots < s_{k-1} < n$, and set $s_0 := 0$ and $s_k := n$. Then, $\sigma = (a_1, a_2, \ldots, a_n) \in [m]^n$ has the property that $\mathrm{Des}(\sigma) \subseteq X$ if and only if

$$a_{s_i+1} \leq a_{s_i+2} \leq \cdots \leq a_{s_{i+1}} \tag{3.17}$$

for all $0 \leq i < k$. For example, if $n = 8$ and $S = \{3, 5\}$, then the conditions are $a_1 \leq a_2 \leq a_3$, $a_4 \leq a_5$, and $a_6 \leq a_7 \leq a_8$. According to Proposition 2.3.10, there are $\binom{m+s_{i+1}-s_i-1}{s_{i+1}-s_i}$ ways to choose $a_{s_i+1}, a_{s_i+2}, \ldots, a_{s_{i+1}} \in [m]$ so that (3.17) holds. The multiplication principle then yields that

$$g(X) = \prod_{i=0}^{k-1} \binom{m + s_{i+1} - s_i - 1}{s_{i+1} - s_i}.$$

Applying Equation (3.14), with S in place of Y, gives the following result.

Proposition 3.3.1. *Let $S = \{s_1, s_2, \ldots, s_{k-1}\}$ be a subset of $[n-1]$, where $1 \leq s_1 < s_2 < \cdots < s_{k-1} < n$, and set $s_0 := 0$ and $s_k := n$. The number of sequences $\sigma \in [m]^n$ with descent set $\mathrm{Des}(\sigma) = S$ is equal to*

$$\sum_{0=i_0<i_1<\cdots<i_{j-1}<i_j=k} (-1)^{k-j} \prod_{\ell=0}^{j-1} \binom{m + s_{i_{\ell+1}} - s_{i_\ell} - 1}{s_{i_{\ell+1}} - s_{i_\ell}},$$

where the sum on the right-hand side ranges over all subsets $\{i_0 < i_1 < \cdots < i_j\}$ of the set $\{0, 1, \ldots, k\}$ with $i_0 = 0$ and $i_j = k$.

EXAMPLE. For $S = \{a, b\} \subseteq [n-1]$ with $a < b$, the formula of Proposition 3.3.1 becomes

$$\begin{aligned}&\binom{m+a-1}{a}\binom{m+b-a-1}{b-a}\binom{m+n-b-1}{n-b}\\ &-\binom{m+a-1}{a}\binom{m+n-a-1}{n-a}\\ &-\binom{m+b-1}{b}\binom{m+n-b-1}{n-b}\\ &+\binom{m+n-1}{n}.\end{aligned}$$

3.4 Exercises

1. We consider subsets A, B, C of the set $\{1, 2, 3, \ldots, 25\}$, each of cardinality 17. Find the minimum and the maximum possible values of $\#(A \cup B \cup C) + \#(A \cap B) + \#(A \cap C) + \#(B \cap C)$.
2. We consider the triples $(a, b, c) \in [5]^3$.
   (a) How many of these are such that $b \neq a + 1$ and $c \neq b + 1$?
   (b) How many are such that $b \neq a + 1$, $c \neq b + 1$ and $a \neq c + 1$?
   (c) How many are such that $b \neq a + 1$, $c \neq b + 1$ and $a \neq c$?
3. (a) How many subsets of $\{1, 2, 3, \ldots, 10\}$ contain neither $\{1, 2, 3\}$ nor $\{3, 4, 5\}$?
   (b) How many of the integers $1, 2, 3, \ldots, 1000$ are divisible by none of 3, 5, 7?
   (c) How many of the integers $1, 2, 3, \ldots, 120$ are not divisible by any square of an integer greater than 1?
   (d) How many 5-subsets of $\{1, 2, 3, \ldots, 12\}$ contain none of $\{1, 2\}$, $\{2, 3, 4\}$, and $\{4, 5\}$?
4. We consider the words with five letters that can be formed from the alphabet {A, B, C, D}, such as DBBAA and ACBDC.
   (a) How many of these words contain at least one A and at least one B?
   (b) How many of these contain at least one A, at least one B and at least one C?
   (c) How many of these do not contain three successive identical letters?
   (d) How many of these do not contain an A, followed by a B?
5. We consider the permutations $(\sigma_1, \sigma_2, \ldots, \sigma_6)$ of the set $[6]$.
   (a) For how many of these $\sigma_1 \notin \{1, 2\}$, $\sigma_2 \notin \{3, 4\}$ and $\sigma_3 \notin \{5, 6\}$?

(b) For how many $\sigma_1 \notin \{1, 2\}$, $\sigma_2 \notin \{2, 3\}$ and $\sigma_3 \notin \{3, 4\}$?
(c) For how many $\{\sigma_1, \sigma_2\} \neq \{1, 2\}$, $\{\sigma_3, \sigma_4\} \neq \{3, 4\}$ and $\{\sigma_5, \sigma_6\} \neq \{5, 6\}$?
(d) For how many $\{\sigma_1, \sigma_2\} \neq \{1, 2\}$, $\{\sigma_2, \sigma_3\} \neq \{2, 3\}$ and $\{\sigma_3, \sigma_4\} \neq \{3, 4\}$?

6. We consider the permutations of the collection $\{1, 1, 1, 1, 2, 2, 3, 3\}$.

   (a) For how many of these at least one entry equal to 1 appears to the left of an entry equal to 2 and at least one entry equal to 1 appears to the left of an entry equal to 3?
   (b) How many do not contain three successive entries equal to 1?
   (c) For how many no entry equal to 2 appears exactly three places to the right of an entry equal to 1?

7. (a) How many compositions of 16 have four parts, all different from 4?
   (b) How many compositions of 16 have four parts, the first three of which are less than 5?
   (c) How many compositions of 20 have four parts, the first three of which are bigger than 2 and less than 7?

8. We consider the paths in $\mathbb{Z}^2$ with initial vertex $(0, 0)$, terminal vertex $(6, 6)$, and steps $(1, 0)$ or $(0, 1)$.

   (a) How many of these paths do not pass through any of the points $(1, 1)$, $(3, 3)$, and $(5, 5)$?
   (b) How many pass through at most one of the points $(3, 2)$ and $(3, 3)$?
   (c) How many pass through at most one of the points $(3, 2)$ and $(3, 3)$ and through at most one of the points $(4, 4)$ and $(4, 5)$?
   (d) How many lie entirely in the half-plane $x \geq y$ and do not pass through any of the points $(2, 2)$ and $(4, 4)$?

9. (a) How many maps $f : [5] \to [6]$ contain 1, 2, and 3 in their image?
   (b) How many injective maps $f : [3] \to [9]$ do not have fixed points, meaning that $f(x) \neq x$ for every $x \in [3]$?

10. Each of the mn squares of an $m \times n$ chessboard is colored either black or white, as in Example 3.1.2. Let $f(m, n)$ be the number of these colorings for which no row or column of the chessboard has all its squares colored black and $g(m, n)$ be the number of these colorings for which no row or column of the chessboard has all its squares colored with the same color.

    (a) Find a formula for $f(m, n)$.
    (b) Compute $g(3, 3)$.
    (c) Find a formula for $g(m, n)$.

11. How many of the integers $1, 2, 3, \ldots, 1000$ are divisible by 4 or 5, but by neither 3 nor 7?
12. Prove the identity

$$\binom{n-r}{m-r} = \sum_{k=0}^{r} (-1)^k \binom{r}{k} \binom{n-k}{m}$$

for integers $0 \le r \le m \le n$ by applying the inclusion–exclusion principle. This result generalizes Example 3.1.4.

13. Given positive integers n, r, and $m \in \mathbb{N}$, we denote by $f(n, r, m)$ the number of ways to distribute mr identical balls in n distinct boxes, so that every box has less than r balls. For example, $f(n, r, 0) = 1$, $f(n, r, m) = 0$ for $m \ge n$, and $f(n, 2, m) = \binom{n}{2m}$.

    (a) Prove that

$$f(n, r, m) = \sum_{k=0}^{n} (-1)^k \binom{n}{k} \binom{(m-k)r + n - 1}{n - 1}.$$

    (b) Prove that $\sum_{m=0}^{n} f(n, r, m) = r^{n-1}$. Which well-known identity comes up in the special case $r = 2$?

14. Prove the identity

$$\binom{m+n}{n} = \sum_{i=0}^{n} (-1)^i \binom{k}{i} \binom{m+n+k-i}{n-i} \tag{3.18}$$

for $m, n, k \in \mathbb{N}$ by applying the inclusion–exclusion principle. Deduce from that the identity of Exercise 51 (a) of Chapter 2.

15. Prove the identity

$$\sum_{k=0}^{n} \binom{m+k}{k} \binom{n}{k} (-2)^k = (-1)^n \sum_{k=0}^{n} \binom{m}{k} \binom{n}{k} 2^k$$

for $m, n \in \mathbb{N}$ (see Exercise 56 of Chapter 2) by applying the inclusion–exclusion principle.

16. Prove the identity

$$\sum_{i=0}^{n} (-1)^i \binom{n}{i} \binom{2n-2i}{k} = \begin{cases} 0, & \text{if } k < n, \\ 2^{2n-k} \binom{n}{k-n}, & \text{otherwise} \end{cases}$$

for $n, k \in \mathbb{N}$ with $k \le 2n$ by applying the inclusion–exclusion principle.

17. Prove the identity

$$\sum_{a_1 + a_2 + \cdots + a_k = n} (-1)^{k-1} \binom{2a_1}{a_1} \binom{2a_2}{a_2} \cdots \binom{2a_k}{a_k} = \frac{2}{n} \binom{2n-2}{n-1}$$

for $n \in \mathbb{Z}_{>0}$, where the sum on the left-hand side runs through all 2^{n-1} compositions $(a_1, a_2, \ldots, a_k)$ of n, by applying the inclusion–exclusion principle.

18. Prove the identity

$$\sum_{k=0}^{n}(-1)^{n-k}\binom{n}{k}k^{n+1} = n(n+1)!/2$$

for $n \in \mathbb{N}$ by counting the surjective maps $f : [n+1] \to [n]$ in two different ways.

19. Given positive integers m and n, we denote by $\gcd(m,n)$ the greatest of the common positive divisors of m and n.

(a) For any positive divisor d of m, prove that there exist exactly $\varphi(m/d)$ numbers $n \in [m]$ such that $\gcd(m,n) = d$.

(b) Deduce that

$$\sum_{d|m}\varphi(d) = m$$

for every $m \in \mathbb{Z}_{>0}$, where the sum runs through all positive divisors of m.

20. Let $m \in \mathbb{Z}_{>0}$. Find the number of pairs $(a,b) \in [m] \times [m]$ for which a, b, and m are relatively prime (meaning, they do not have any common prime divisor). Express your answer as a formula analogous to that of Proposition 3.2.2 for $\varphi(m)$.

21. Prove the identity

$$\sum_{i=0}^{k}(-1)^{k}\binom{k}{i}(n-i)! = \sum_{j=0}^{n-k}\binom{n-k}{j}D_{n-j}$$

for integers $0 \le k \le n$ by counting the elements of a finite set in two different ways. Here D_m stands for the number of permutations of $[m]$ without fixed points.

22. For integers $0 \le k \le n$, we let $f(n,k)$ be the number of permutations of $[n]$, which have exactly k fixed points. Prove that:

(a) $\sum_{k=0}^{n} f(n,k) = n!$ for every $n \ge 1$.

(b) $\sum_{k=0}^{n} kf(n,k) = n!$ for every $n \ge 1$,

(c) $\sum_{k=0}^{n} k(k-1)f(n,k) = n!$ for every $n \ge 2$.

23. For $n \in \mathbb{Z}_{>0}$ and $k \in \{0, 1, \ldots, n\}$ we denote by $g(n, k)$ the number of injective maps $\sigma : [k] \to [n]$, where $g(n, 0) := 1$. Prove that

$$\sum_{k=0}^{n} g(n,k)(x-1)^{n-k} = \sum_{k=0}^{n} f(n,k)x^k,$$

where $f(n, k)$ is the number of permutations of $[n]$, which have exactly k fixed points.

24. Prove the recurrence relation of Proposition 3.2.3 (c) for the number D_n of permutations of $[n]$ without fixed points using only basic principles of enumeration.

25. Let $f(n)$ be the number of permutations $\sigma = (\sigma_1, \sigma_2, \ldots, \sigma_n)$ of $[n]$ such that $\sigma_{i+1} \neq \sigma_i + 1$ for every $i \in [n-1]$. For example, $f(1) = f(2) = 1$, $f(3) = 3$, $f(4) = 11$, and $f(5) = 53$. Prove that:

    (a) $f(n) = \displaystyle\sum_{k=0}^{n-1} (-1)^k \binom{n-1}{k} (n-k)!$ for every $n \in \mathbb{Z}_{>0}$.
    (b) $f(n) = D_n + D_{n-1}$ for $n \geq 1$.
    (c) $f(n) \leq n!/2$ for $n \geq 2$.

26. Let $n \in \mathbb{Z}_{>0}$. We denote by $\mathrm{Fix}(\sigma)$ the set of fixed points of a permutation σ of the set $[n]$. Find a formula for the number of pairs (σ, τ) of permutations of $[n]$ such that $\mathrm{Fix}(\sigma) \cap \mathrm{Fix}(\tau) = \varnothing$.

27. For $n \in \mathbb{N}$, we denote by $E(n)$ the number of permutations $(\sigma_1, \sigma_2, \ldots, \sigma_n)$ of $[n]$ for which the smallest element of the set $\{i \in [n-1] : \sigma_i < \sigma_{i+1}\} \cup \{n\}$ is an even number, where $E(0) := 1$.

    (a) Prove that $E(n) = nE(n-1) + (-1)^n$ for every $n \in \mathbb{Z}_{>0}$.
    (b) Deduce that $E(n)$ is equal to the number of permutations of $[n]$ without fixed points.
    (c) Give a bijective proof of the statement of part (b).

28. For $n \in \mathbb{Z}_{>0}$, we let $\Omega_n = \{1, -1, 2, -2, \ldots, n, -n\}$ and X_n be the set of bijective maps $\sigma : \Omega_n \to \Omega_n$ such that $\sigma(-i) = -\sigma(i)$ for every $i \in [n]$. We denote by $g(n)$ the cardinality of X_n and by $f(n)$ the number of elements $\sigma \in X_n$ such that $\sigma(i) \neq i$ for every $i \in [n]$.

    (a) Prove that $g(n) = 2^n n!$ for $n \geq 1$.
    (b) Prove that

$$f(n) = \sum_{k=0}^{n} (-1)^k \binom{n}{k} 2^{n-k}(n-k)!$$

    for $n \geq 1$.
    (c) Deduce that $\displaystyle\lim_{n\to\infty} \frac{f(n)}{g(n)} = e^{-1/2}$.

29. For those who are familiar with even and odd permutations: If E_n is the number of even permutations of $[n]$ without fixed points, prove that $2E_n = D_n + (-1)^{n-1}(n-1)$ for every $n \in \mathbb{Z}_{>0}$.
30. For a finite set Ω and its subsets $A_1, A_2, \ldots, A_n$, prove the Bonferonni inequalities

$$\sum_{k=p}^{n} (-1)^{k-p} \sum_{1 \le i_1 < \cdots < i_k \le n} \#(A_{i_1} \cap \cdots \cap A_{i_k}) \ge 0 \tag{3.19}$$

for every $p \in \{0, 1, \ldots, n\}$. Equivalently,

$$\# \left(\Omega - \bigcup_{i=1}^{n} A_i \right) \le \sum_{k=0}^{p} (-1)^k \sum_{1 \le i_1 < \cdots < i_k \le n} \#(A_{i_1} \cap \cdots \cap A_{i_k}), \quad p \in 2\mathbb{N} \tag{3.20}$$

$$\ge \sum_{k=0}^{p} (-1)^k \sum_{1 \le i_1 < \cdots < i_k \le n} \#(A_{i_1} \cap \cdots \cap A_{i_k}), \quad p \notin 2\mathbb{N}. \tag{3.21}$$

31. By applying the Bonferonni inequalities of Exercise 30 (or otherwise), prove that $n!/3 + (n-1)!/2 \le f(n) \le n!/2$ for $n \ge 3$, where $f(n)$ is the number of permutations of $[n]$ of Exercise 25.
32. Let $\mathcal{B}$ be the subboard of the $n \times n$ chessboard, which is obtained from that of Example 3.2.1 by adding the square $(n, 1)$.

    (a) Prove that $r_k(\mathcal{B}) = \dfrac{2n}{2n-k} \dbinom{2n-k}{k}$ for every $k \in \{0, 1, \ldots, n\}$.

    (b) Deduce that there are exactly

$$\sum_{k=0}^{n} (-1)^k \frac{2n}{2n-k} \binom{2n-k}{k} (n-k)!$$

    permutations $\sigma = (\sigma_1, \sigma_2, \ldots, \sigma_n)$ of $[n]$ such that $\sigma_i \ne i, i+1 \pmod{n}$ for every $i \in [n]$.

33. Let $a_1, a_2, \ldots, a_n \in \mathbb{R}$ be given. We consider two maps $f, g : 2^{[n]} \to \mathbb{R}$, which satisfy the equivalent conditions of Theorem 3.3.1 for $I = [n]$. If

$$g(S) = \prod_{i \in S} a_i$$

for every $S \subseteq [n]$ (so that $g(\varnothing) = 1$), compute $f(S)$ for $S \subseteq [n]$. Deduce that $f(S) \ge 0$ for every $S \subseteq [n]$, if $a_1, a_2, \ldots, a_n \ge 1$.

34. Let $n \in \mathbb{Z}_{>0}$ and $k \in \{0, 1, \ldots, n\}$.

    (a) Let Ω be a finite set and $A_1, A_2, \ldots, A_n$ be subsets of Ω. Prove that the number of elements of Ω which belong to exactly k of the sets $A_1, A_2, \ldots, A_n$ is equal to

$$\sum_{m=k}^{n} (-1)^{m-k} \binom{m}{k} \sum_{1 \le j_1 < j_2 < \cdots < j_m \le n} \#(A_{j_1} \cap A_{j_2} \cap \cdots \cap A_{j_m}).$$

    (b) Let $p_1, p_2, \ldots, p_n$ be positive divisors of a positive integer m, which are pairwise relatively prime. Deduce that the number of integers among $1, 2, \ldots, m$, which are divisible by exactly k of $p_1, p_2, \ldots, p_n$, is equal to

$$\sum_{r=k}^{n} (-1)^{r-k} \binom{r}{k} \sum_{1 \le j_1 < j_2 < \cdots < j_r \le n} \frac{m}{p_{j_1} p_{j_2} \cdots p_{j_r}}.$$

    (c) How many integers among $1, 2, 3, \ldots, 13860$ are divisible by exactly three of the numbers 4, 5, 7, 9, and 11?

35. Find a simple product formula for the number of sequences $\sigma \in \{1, 2\}^n$ with a given descent set $\{s_1 < s_2 < \cdots < s_{k-1}\} \subseteq [n-1]$. Which identity follows from your answer and the formula of Proposition 3.3.1?

36. Let $f, g : 2^I \to \mathbb{C}$ be as in Theorem 3.3.1. We define $f(Y, U)$ by the formula

$$f(Y, U) = \sum_{U \subseteq X \subseteq Y} (-1)^{|Y \smallsetminus X|} g(X)$$

    for $U \subseteq Y \subseteq I$, so that $f(Y, Y) = g(Y)$ and $f(Y, \varnothing) = f(Y)$ for every $Y \subseteq I$.

    (a) Give a combinatorial interpretation of $f(Y, U)$ in the setting of Proposition 3.3.1, where $I := [n-1]$.

    (b) For given $U \subseteq Y \subseteq I$, prove that

$$g(Y) = \sum_{U \subseteq X \subseteq Y} f(X, U).$$

    (c) Prove that if $f(X)$ is a nonnegative real number for every $X \subseteq I$, then so is $f(Y, U)$ for all $U \subseteq Y \subseteq I$.

Chapter 4
Set Partitions, Equivalence Relations, and Partial Orders



What is the largest number of subsets of an n-element set that can be selected, so that any two of these subsets have at least one element in common? What is the largest number of terms of a sequence of real numbers that does not contain a monotone subsequence with n terms? Questions of this type may be answered by appealing to some simple but extremely useful principles, such as the pigeonhole principle, and by studying fundamental concepts and structures of discrete mathematics, such as set partitions (or equivalence relations) and partial orders. This chapter serves as a brief introduction to these topics.

4.1 The Pigeonhole Principle

We recall that a map $f : X \to Y$ is said to be one-to-one, or injective (respectively, onto, or surjective), if for every $y \in Y$ there exists at most one element (respectively, at least one element) $x \in X$ such that $f(x) = y$. As already noted in Section 3.2.1, the map $f : X \to Y$ is bijective if and only if it is simultaneously injective and surjective. Thus, the following statement is stronger than Proposition 2.2.1.

Proposition 4.1.1. *Let $f : X \to Y$ be a map of finite sets.*

(a) *If f is one-to-one , then $\#X \leq \#Y$.*
(b) *If f is onto , then $\#X \geq \#Y$.*

Proof. Suppose that f is one-to-one. Then, $\#X = \#f(X)$ by Proposition 2.2.1, since the restriction $f_{X,f(X)} : X \to f(X)$ of f is a bijection. We also have $\#f(X) \leq \#Y$, because $f(X)$ is a subset of Y and $\#Y - \#B = \#(Y \setminus B) \geq 0$ for every subset B of Y by Corollary 2.2.2. These facts prove part (a).

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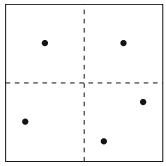


Figure 4.1 Five points in a square

Suppose now that f is surjective. Then, by Exercise 35 (b) of Chapter 2, there exists a one-to-one map $g : Y \to X$. Therefore, $\#Y \leq \#X$ by part (a). This proves part (b). □

Part (a) of Proposition 4.1.1 can be expressed as follows. If $f : X \to Y$ is a map of finite sets and X has more elements than Y, then there exist distinct elements x_1, x_2 of X such that $f(x_1) = f(x_2)$. Here is a way to interpret this result: if $m+1$ or more pigeons (the elements of X) are distributed to m pigeonholes (the elements of Y), then two (or more) pigeons will end up in the same pigeonhole. The usefulness of this result, known as *pigeonhole principle*, or *Dirichlet's drawer principle*, will be evident in the following examples.

Example 4.1.1. Let T be a square in the plane with side lengths equal to 2 (thus, T includes both the interior and the boundary of the square). What is the largest number of points of T that can be chosen, if their pairwise distances apart must all be greater than $\sqrt{2}$?

Clearly, the four vertices of T have this property. To prove that this is the largest possible number of points, we will show that among any five or more points of T, there exist two that are at a distance apart less than or equal to $\sqrt{2}$. Indeed, let us divide T into four squares of side lengths equal to 1 by drawing lines that pass through the center of T and are parallel to its sides (see Figure 4.1). According to the pigeonhole principle, two of the points we have chosen will lie in the same square. Clearly, the distance between these two points cannot exceed $\sqrt{2}$.

For the version of this question in which the pairwise distances apart must be greater than or equal to $\sqrt{2}$, see Exercise 2. □

Example 4.1.2. Let us prove that among any $m + 1$ integers there exist two, the difference of which is divisible by m.

Let X be a set consisting of $m + 1$ integers. For $x \in X$, let $f(x)$ denote the remainder of the Euclidean division of x by m (see Proposition 1.1.4). Then, $f(x) \in \{0, 1, \ldots, m - 1\}$ for every $x \in X$, and hence, we have defined a map $f : X \to \{0, 1, \ldots, m-1\}$. Since $\#X = m+1$, the pigeonhole principle implies that there exist distinct elements x_1, x_2 of X such that $f(x_1) = f(x_2)$. Thus, we have $x_1 = q_1 m + r$ and $x_2 = q_2 m + r$ for some $q_1, q_2 \in \mathbb{Z}$, where $r = f(x_1) = f(x_2)$. It follows that $x_1 - x_2 = m(q_1 - q_2)$ is divisible by m. □

For an unexpected application of the result of Example 4.1.2 to elementary number theory, see Exercise 25.

Example 4.1.3. Let us prove that in a meeting of m random people, there exist two persons who have the same number of acquaintances among the people present at the meeting (we assume that if a is an acquaintance of b, then b is an acquaintance of a and do *not* consider anyone to be an acquaintance of themselves).

Let X be the given set of m people, and let $f(x)$ be the number of acquaintances of $x \in X$ among these people. Clearly, we have $0 \leq f(x) \leq m-1$ for every $x \in X$. We observe that there cannot be two persons $a, b \in X$ such that $f(a) = 0$ and $f(b) = m-1$. Indeed, in that case a would have no acquaintance among the elements of X while b would be acquainted with everyone else. In particular, b would be acquainted with a, and hence, a would also be acquainted with b, contrary to our assumption that $f(a) = 0$. Our observation implies that the map $f : X \to Y$ can be defined for $Y = \{0, 1, \ldots, m-2\}$ or $Y = \{1, 2, \ldots, m-1\}$. In either case, $\#Y = m-1$ and $\#X = m$. Therefore, by the pigeonhole principle, there exist distinct elements $x_1, x_2 \in X$ such that $f(x_1) = f(x_2)$, meaning that x_1 and x_2 have the same number of acquaintances among the people present at the meeting. □

Example 4.1.4. What is the largest number of subsets of $[n]$ that can be selected, so that any two of these subsets have at least one element in common? For example, for $n = 2$, one can choose two subsets with this property, such as $\{1\}$ and $\{1, 2\}$, but no more than two.

We will prove that the largest number in question is equal to 2^{n-1}. Indeed, there are 2^{n-1} subsets of $[n]$ that contain a given element of $[n]$, say $1 \in [n]$, and certainly, any two of these subsets have at least one element in common (since they both contain 1). Thus, it remains to show that among any $2^{n-1}+1$ or more subsets of $[n]$ there are two that have an empty intersection. For this purpose, let us divide the 2^n subsets of $[n]$ into 2^{n-1} (unordered) pairs $\{S, \bar{S}\}$, where $\bar{S} = [n] \smallsetminus S$ is the complement of S in $[n]$. For example, for $n = 2$, we have the pairs $\{\varnothing, \{1, 2\}\}$ and $\{\{1\}, \{2\}\}$. Each of the subsets we have been given belongs to exactly one pair. Since we have been given at least $2^{n-1}+1$ subsets, it follows from the pigeonhole principle that two of these must belong to the same pair. Therefore, among the subsets of $[n]$ we have been given, there are a subset S and its complement $\bar{S}$ and, clearly, $S \cap \bar{S} = \varnothing$. □

Example 4.1.5. Let X be an n-element set and $f : X \to X$ be a bijective map. Following the notation of Exercise 36 of Chapter 1, we denote by $f^i = f \circ f \circ \cdots \circ f : X \to X$ the i-fold composition of f with itself, where $f^0 = \mathrm{id}_X$. We will prove that there exists a positive integer k (necessarily unique) such that $f^k = \mathrm{id}_X$ and the maps $f^0, f^1, \ldots, f^{k-1}$ are pairwise distinct.

Indeed, there are finitely many bijective maps $g : X \to X$. Let q be the number of these maps (specifically, we have $q = n!$ by Proposition 2.3.2 (b)). By the pigeonhole principle, among the $q+1$ maps $f^0, f^1, \ldots, f^q$, there must be two that are equal to each other, namely we have $f^i = f^j$ for some indices $0 \leq i < j \leq q$. Since f^i is injective (explain why), $f^i = f^j$ implies that $f^{j-i} = \mathrm{id}_X$. Thus, we may define k as the minimum positive integer such that $f^k = \mathrm{id}_X$. The argument we have just used and the definition of k show that the maps $f^0, f^1, \ldots, f^{k-1}$ are pairwise distinct.

In the language of algebra, f is a *permutation* of X and k is the *order* of f. For example, the order of $f : \{a, b, c, d\} \to \{a, b, c, d\}$ with $f(a) = b$, $f(b) = c$, $f(c) = d$, and $f(d) = a$ is equal to 4. □

The following statement is a useful generalization of the pigeonhole principle.

Proposition 4.1.2. *Let $f : X \to Y$ be a map of finite sets and $m, n \in \mathbb{Z}_{>0}$. If $\#Y = m$ and $\#X \geq mn + 1$, then there exist $n + 1$ distinct elements $x_1, x_2, \ldots, x_{n+1}$ of X such that $f(x_1) = f(x_2) = \cdots = f(x_{n+1})$.*

Proof. Suppose that there are no such elements of X. Then, $\#f^{-1}(y) \leq n$ for every $y \in Y$. We observe (as in the proof of Proposition 2.2.3) that X is the disjoint union of its subsets $f^{-1}(y)$ for $y \in Y$ and conclude, via the addition principle, that

$$\#X = \#\bigcup_{y \in Y} f^{-1}(y) = \sum_{y \in Y} \#f^{-1}(y) \leq mn.$$

This contradicts our hypothesis that $\#X \geq mn + 1$ and the proof follows. □

EXAMPLE. Applying Proposition 4.1.2 for $m = 3$ and $n = 2$ yields that if $f : X \to \{1, 2, 3\}$ is any map and $\#X = 7$, then there exist distinct elements $x_1, x_2, x_3 \in X$ such that $f(x_1) = f(x_2) = f(x_3)$. □

EXAMPLE. Reasoning as in Example 4.1.2 and applying Proposition 4.1.2 for $n = 2$, we find that among any $2m + 1$ integers one can always choose three, say x_1, x_2, x_3, so that all differences $x_1 - x_2$, $x_1 - x_3$, and $x_2 - x_3$ are divisible by m. □

4.2 Set Partitions

The discussion in Section 4.1 shows that the pigeonhole principle may be restated in the following way: if an $(m + 1)$-element set is partitioned into m or fewer parts, then some part must contain at least two elements. This section studies in detail the concept of a set partition, one that will turn out to be very useful in the sequel, as well as some equivalent ways with which this concept can be described.

Definition 4.2.1. *A partition of a set X is a set $\pi = \{B_1, B_2, \ldots, B_k\}$ consisting of pairwise disjoint nonempty subsets of X, whose union is equal to X. The subsets B_i are called blocks of π.*

EXAMPLE.

(a) Every nonempty set X has the partition $\{X\}$ with only one block and the partition $\{\{x\} : x \in X\}$ into singletons, one for every element of X.
(b) Let $X = 2^{[n]}$ be the powerset of $[n]$. The 2^{n-1} sets of the form $\{S, [n] \setminus S\}$, with $S \subseteq [n]$, considered in Example 4.1.4, are the blocks of a partition π of X. For $n = 3$, we have $\pi = \{\{\varnothing, \{1, 2, 3\}\}, \{\{1\}, \{2, 3\}\}, \{\{2\}, \{1, 3\}\}, \{\{3\}, \{1, 2\}\}\}$.

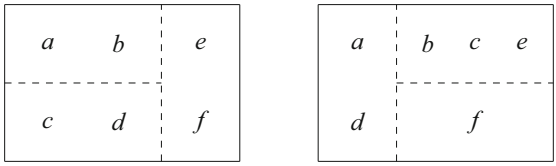


Figure 4.2 Two partitions with three blocks

(c) Let $X = \{a, b, c, d, e, f\}$ be a 6-element set. One partition of X with two blocks is $\{\{a, c, e\}, \{b, d, f\}\}$, while $\{\{a, b\}, \{c, d\}, \{e, f\}\}$ and $\{\{a, d\}, \{b, c, e\}, \{f\}\}$ are partitions of X with three blocks, shown in Figure 4.2. □

A different way to describe the concept of set partition comes from considering that two elements that belong to the same block of the partition are related to each other. The binary relations on a set that arise in this way are called equivalence relations.

Definition 4.2.2. *A binary relation $\sim$ on a set X is called an equivalence relation if the following properties hold for all $a, b, c \in X$:*

(i) $a \sim a$ *(reflexivity),*
(ii) if $a \sim b$*, then* $b \sim a$ *(symmetry),*
(iii) if $a \sim b$ *and* $b \sim c$*, then* $a \sim c$ *(transitivity).*

We say that a is equivalent to b (with respect to the relation $\sim$), if $a \sim b$.

Given a partition π of a set X, we define the binary relation $\sim_\pi$ on X so that for $a, b \in X$ we have $a \sim_\pi b$ if a and b belong to the same block of π. For example, if $X = \{a, b, c, d\}$ and $\pi = \{\{a, c\}, \{b, d\}\}$, then $a \sim_\pi a$, $a \sim_\pi c$, $c \sim_\pi a$, $b \sim_\pi d$, but $a \not\sim_\pi b$, $c \not\sim_\pi d$.

Clearly, $\sim_\pi$ is an equivalence relation on X for every partition π of X. Conversely, as shown by the following proposition, every equivalence relation on X is obtained from a unique partition of X in this way.

Proposition 4.2.1. *Let $\sim$ be an equivalence relation on the set X. Then, there exists a unique partition π of X such that $\sim$ coincides with $\sim_\pi$, meaning for all $a, b \in X$ we have $a \sim b$ if and only if a and b belong to the same block of π.*

The blocks of π are the sets $C_a = \{x \in X : x \sim a\}$ for $a \in X$; they are called equivalence classes *of $\sim$.*

Proof. We claim that the distinct sets among the classes C_a for $a \in X$ are the blocks of a partition π of X. Indeed, since $a \sim a$, we have $a \in C_a$ for every $a \in X$. Therefore, the classes C_a are nonempty, and their union is equal to X. Moreover, two classes C_a and C_b are disjoints, except if $C_a = C_b$. Indeed, let's assume that $C_a \cap C_b \neq \varnothing$. Then, there exists $c \in C_a \cap C_b$, so that $c \sim a$ and $c \sim b$. These relations, in view of the symmetry and transitivity of $\sim$, imply that $a \sim b$. Consider now any element $x \in C_a$. It follows from $x \sim a$ and $a \sim b$ that $x \sim b$, and hence that $x \in C_b$. We have shown that $C_a \subseteq C_b$ and similarly, we find that $C_b \subseteq C_a$. Therefore, $C_a = C_b$. By definition of π, for $a, b \in X$, we have that a and b belong

to the same block of π if and only if $C_a = C_b$, and this happens if and only if $a \sim b$. Clearly, π is the unique partition of X with this property. □

EXAMPLE. Let $X = \{a, b, c, d\}$ and $\sim$ be the binary relation on X having graph

$$\{(a, a), (b, b), (c, c), (d, d), (a, b), (b, a), (c, d), (d, c)\}.$$

Then, $\sim$ is an equivalence relation with two equivalence classes $C_a = C_b = \{a, b\}$ and $C_c = C_d = \{c, d\}$. The corresponding partition of X is $\pi = \{\{a, b\}, \{c, d\}\}$. □

Here is another way in which the concepts of equivalence relation and set partition arise naturally.

Example 4.2.1. Let $f : X \to Y$ be a map, and let's consider the inverse images $f^{-1}(y)$ of f, for $y \in f(X)$. These sets are nonempty subsets X. Moreover, they are mutually disjoint since (by the definition of a map) every $x \in X$ belongs to exactly one of these sets, namely to $f^{-1}(y)$ for $y = f(x)$. We conclude that the sets $f^{-1}(y)$ for $y \in f(X)$ are mutually disjoint nonempty subsets of X and that their union is equal to X. Thus, they are the blocks of a partition π_f of X. For example, if $f : \{a, b, c, d, e\} \to \{1, 2, 3\}$ and $f(a) = f(d) = 1$, $f(b) = 2$, $f(c) = f(e) = 3$, then $\pi_f = \{\{a, d\}, \{b\}, \{c, e\}\}$. The equivalence relation that corresponds to the partition π_f is the binary relation $\sim_f$ on X defined by setting $a \sim_f b$ if $f(a) = f(b)$, for $a, b \in X$.

Conversely, every partition π of X can be obtained from a map $f : X \to Y$ in this way. Indeed, it suffices to set $Y = \pi$ and define as $f(x)$ the unique block of π, which contains x, for $x \in X$. The resulting map $f : X \to \pi$ is surjective, since the blocks of π are nonempty subsets of X; it is referred to as the *canonical map* defined by the partition π. By definition of f, we have $f(a) = f(b)$ if and only if the elements a and b of X belong to the same block of π. Therefore, the partition π_f of X coincides with π.

Example 4.2.2. Let $m \in \mathbb{Z}_{>0}$. For $a, b \in \mathbb{Z}$, we write $a \equiv b \pmod{m}$, and say that *a is equivalent to b* modulo m, if m divides $a - b$, that is, if $a - b = mq$ for some $q \in \mathbb{Z}$. For example, $23 \equiv 2 \pmod{7}$ and $2m + 1 \equiv 1 \pmod{m}$ for every m. The equivalence modulo m is an equivalence relation on $\mathbb{Z}$. Let us verify, for instance, transitivity. Suppose that $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$. Then, there exist $p, q \in \mathbb{Z}$ such that $a - b = mp$ and $b - c = mq$. Adding these equalities we get $a - c = mp + mq = m(p + q)$, where $p + q \in \mathbb{Z}$, and conclude that $a \equiv c \pmod{m}$.

What are the equivalence classes? It follows from Proposition 1.1.4 that for every $n \in \mathbb{Z}$, there exists a unique $r \in \{0, 1, \ldots, m - 1\}$ such that $n \equiv r \pmod{m}$. Hence, n belongs to the equivalence class of unique integer $r \in \{0, 1, \ldots, m - 1\}$. This shows that there are exactly m equivalence classes, namely the classes of $C_0, C_1, \ldots, C_{m-1}$ of $0, 1, \ldots, m-1$, respectively, where $C_r = \{r + mq : q \in \mathbb{Z}\}$ for $r \in \{0, 1, \ldots, m-1\}$. These sets are called *equivalence classes of integers* (mod m) or simply *equivalence classes* (mod m).

4.3 Stirling and Bell Numbers

This section is a digression into the problem of enumeration of set partitions, which is of independent interest. We will denote by $S(n,k)$ the number of partitions of the set $[n]$ (equivalently, of any n-element set) with k blocks. For small values of n, the numbers $S(n,k)$ appear in the infinite matrix

$$(S(n,k)) = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 & \\ 1 & 3 & 1 & 0 & 0 & 0 & 0 & \\ 1 & 7 & 6 & 1 & 0 & 0 & 0 & \cdots \\ 1 & 15 & 25 & 10 & 1 & 0 & 0 & \\ 1 & 31 & 90 & 65 & 15 & 1 & 0 & \\ 1 & 63 & 301 & 350 & 140 & 21 & 1 & \\ & & & \vdots & & & & \end{pmatrix}, \tag{4.1}$$

the nth row of which consists of the numbers $S(n,1), S(n,2), \ldots$ For example, in the third row we have $S(3,1) = 1$, $S(3,2) = 3$, and $S(3,3) = 1$, because there is one partition, namely $\{\{1,2,3\}\}$, of the set $[3]$ with one block, three partitions, namely $\{\{1,2\},\{3\}\}$, $\{\{1,3\},\{2\}\}$, and $\{\{2,3\},\{1\}\}$, with two blocks and one partition, namely $\{\{1\},\{2\},\{3\}\}$, with three blocks. The integers $S(n,k)$ are called *Stirling numbers of the second kind*; they are determined by the following proposition.

Proposition 4.3.1. *Let $a(n,k)$ be the number of surjective maps $f : [n] \to [k]$. Then:*

(a) $S(n,k) = kS(n-1,k) + S(n-1,k-1)$,
(b) $k!S(n,k) = a(n,k)$,
(c) $S(n,k) = \dfrac{1}{k!}\displaystyle\sum_{i=0}^{k}(-1)^{k-i}\binom{k}{i}i^n$

for integers $1 \le k \le n$, where $S(0,0) = 1$ and $S(0,k) = 0$ for $k \ge 1$.

Proof. We observe that out of the $S(n,k)$ partitions of the set $[n]$ with k blocks, $S(n-1,k-1)$ contain $\{n\}$ as a block. The rest are obtained from each of the $S(n-1,k)$ partitions of the set $[n-1]$ with k blocks by including n into one of these blocks in k possible ways. The recurrence of part (a) follows from this argument and the addition and multiplication principles.

We now claim that the surjective maps $f : [n] \to [k]$ are in one-to-one correspondence with the pairs (π,σ), where π is a partition of the set $[n]$ with k blocks and $\sigma = (B_1, B_2, \ldots, B_k)$ is a permutation of the blocks of π (such pairs are called *ordered partitions* of $[n]$). Indeed, every surjective map $f : [n] \to [k]$ gives rise to the pair (π_f, σ_f), where $\pi_f = \{f^{-1}(1), f^{-1}(2), \ldots, f^{-1}(k)\}$ is the partition described in Example 4.2.1 and $\sigma_f = (f^{-1}(1), f^{-1}(2), \ldots, f^{-1}(k))$. Conversely, every such pair (π,σ) is obtained from a unique surjective map $f : [n] \to [k]$,

namely the one with $f^{-1}(i) = B_i$ for $i \in [k]$ (meaning that $f(x) = i \Leftrightarrow x \in B_i$, for $x \in [n]$). For example, if $n = 5$, $k = 3$, $\pi = \{\{1,4\},\{2,5\},\{3\}\}$ and $\sigma = (\{2,5\},\{3\},\{1,4\})$, then $f(2) = f(5) = 1$, $f(3) = 2$, and $f(1) = f(4) = 3$. Since there are $S(n,k)$ partitions of $[n]$ with k blocks and $k!$ ways to permute the blocks of such a partition, it follows from our claim and the multiplication principle that $a(n,k) = k!S(n,k)$. This proves part (b). Part (c) follows from that and Proposition 3.2.1. □

The following proposition provides a useful algebraic interpretation of the Stirling numbers of the second kind. For any positive integer n, the numbers $S(n,k)$ for $k \in [n]$ are the coefficients that arise when the polynomial x^n is expressed as a linear combination of the polynomials $x, x(x-1), x(x-1)(x-2), \ldots, x(x-1)\cdots(x-n+1)$ (the existence and uniqueness of such an expression for every n is guaranteed by basic results of linear algebra).

Proposition 4.3.2. *For every positive integer n,*

$$x^n = \sum_{k=1}^{n} S(n,k)x(x-1)\cdots(x-k+1), \tag{4.2}$$

as polynomials with integer coefficients.

For example, the third and fourth lines of the matrix (4.1) yield the identities

$$x^3 = x + 3x(x-1) + x(x-1)(x-2),$$
$$x^4 = x + 7x(x-1) + 6x(x-1)(x-2) + x(x-1)(x-2)(x-3).$$

Proof of Proposition 4.3.2. The proposed identity expresses an equality of two polynomials in x with real coefficients. To prove such an equality, it suffices to verify it for infinitely many values of the variable x (see Corollary A.2.1 of the appendix). We will verify (4.2) for $x = m$, where m is any positive integer. By Proposition 2.3.1, m^n is equal to the number of maps $f : [n] \to [m]$. Such a map is determined by its image $B = f([n]) \subseteq [m]$ and the surjective map $g : [n] \to B$ with $g(x) = f(x)$ for every $x \in [n]$. There are $\binom{m}{k}$ subsets B of $[m]$ with k elements and for every such subset, according to Proposition 4.3.1 (b), there are $k!S(n,k)$ surjective maps $g : [n] \to B$. Consequently, there are

$$\sum_{k=1}^{m} \binom{m}{k} k!S(n,k) = \sum_{k=1}^{n} \binom{m}{k} k!S(n,k)$$
$$= \sum_{k=1}^{n} S(n,k)\, m(m-1)\cdots(m-k+1)$$

maps $f : [n] \to [m]$ in total. It follows that $m^n = \sum_{k=1}^{n} S(n,k)m(m-1)\cdots(m-k+1)$, which means that (4.2) holds for $x = m$. □

Example 4.3.1. Let's compute the sum $\sum_{i=1}^{m} i^n$ for positive integers m, n, as an application of Proposition 4.3.2. Summing (4.2) for $x \in [m]$ and changing the order of summation in the double sum that comes up, according to Exercise 1 (b) of Chapter 1, we find that

$$\sum_{i=1}^{m} i^n = \sum_{x=1}^{m} x^n = \sum_{x=1}^{m} \sum_{k=1}^{n} k! S(n,k) \binom{x}{k} = \sum_{k=1}^{n} \sum_{x=1}^{m} k! S(n,k) \binom{x}{k}$$

$$= \sum_{k=1}^{n} k! S(n,k) \sum_{x=1}^{m} \binom{x}{k}.$$

From this computation and the identity of Proposition 2.4.2, we get

$$\sum_{i=1}^{m} i^n = \sum_{k=1}^{n} k! S(n,k) \binom{m+1}{k+1}. \tag{4.3}$$

For example, for $n = 3$,

$$\sum_{i=1}^{m} i^3 = 1! S(3,1) \binom{m+1}{2} + 2!\, S(3,2) \binom{m+1}{3} + 3! S(3,3) \binom{m+1}{4}$$

$$= \frac{m(m+1)}{2} + 6 \cdot \frac{m(m+1)(m-1)}{6} + 6 \cdot \frac{m(m+1)(m-1)(m-2)}{24}$$

$$= \frac{m^2(m+1)^2}{4},$$

in agreement with Exercise 2 (e) of Chapter 1. □

We now focus on the number of all partitions of $[n]$. We denote this number by $B(n)$. We have $B(1) = 1$, $B(2) = 2$, $B(3) = 5$, $B(4) = 15$, $B(5) = 52$, $B(6) = 203$, and $B(7) = 877$, where, for example $B(3) = 5$ because there are five partitions of the set $\{1, 2, 3\}$, namely $\{\{1\}, \{2\}, \{3\}\}$, $\{\{1, 2\}, \{3\}\}$, $\{\{1, 3\}, \{2\}\}$, $\{\{2, 3\}, \{1\}\}$, and $\{\{1, 2, 3\}\}$. The numbers $B(n)$ are called *Bell numbers*.

Proposition 4.3.3. *We set $B(0) = 1$. Then:*

(a) $B(n+1) = \sum_{i=0}^{n} \binom{n}{i} B(i)$ *for every* $n \in \mathbb{N}$.

(b) $B(n) = \frac{1}{e} \sum_{k=0}^{\infty} \frac{k^n}{k!}$ *for every* $n \in \mathbb{Z}_{>0}$, *where e is the base for the natural logarithm.*

Proof. Let π be a partition of $[n+1]$, and let $S \subseteq [n+1]$ be the block of π which contains $n+1$. Then, $\#S = i+1$ for some $i \in \{0, 1, \ldots, n\}$. To choose π, there are $\binom{n}{i}$ ways to choose the i elements of S other than $n+1$ and for every such choice, there are $B(n-i)$ ways to partition $[n+1] \setminus S$ and form the other blocks of π. Therefore, there are $\binom{n}{i} B(n-i)$ partitions of $[n+1]$ such that the block that contains $n+1$ has cardinality $i+1$. Summing over all i, we conclude that the number of partitions of $[n+1]$ is equal to $\sum_{i=0}^{n} \binom{n}{i} B(n-i) = \sum_{i=0}^{n} \binom{n}{i} B(i)$. This proves part (a).

To prove the identity of part (b), we observe that $B(n) = \sum_{k \geq 0} S(n,k)$ by the addition principle. This expression and the formula of Proposition 4.3.1 (c), which is valid for every $k \in \mathbb{N}$ (explain why), yield that

$$B(n) = \sum_{k \geq 0} \frac{1}{k!} \sum_{i=0}^{k} (-1)^{k-i} \binom{k}{i} i^n = \sum_{i \geq 0} \frac{i^n}{i!} \sum_{k \geq i} \frac{(-1)^{k-i}}{(k-i)!}$$

$$= \sum_{i \geq 0} \frac{i^n}{i!} \sum_{j \geq 0} \frac{(-1)^j}{j!} = \frac{1}{e} \sum_{i \geq 0} \frac{i^n}{i!}$$

for every positive integer n. □

A different way to determine the sequence $(B(n))_{n \in \mathbb{N}}$ of Bell numbers, by means of its exponential generating function, is described in Section 6.5.

Our terminology for the Stirling numbers of the "second kind" implies that there are Stirling numbers of the "first kind" as well. These numbers $s(n,k)$ are defined via the equality

$$x(x-1)\cdots(x-n+1) = \sum_{k=1}^{n} s(n,k) x^k, \tag{4.4}$$

that is, they are the coefficients that come up when the polynomial $x(x-1)\cdots(x-n+1)$ is expanded as a linear combination of powers of x. For small values of n, the numbers $s(n,1), s(n,2), \ldots$ appear in the nth row of the matrix

$$(s(n,k)) = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 2 & -3 & 1 & 0 & 0 & 0 & 0 \\ -6 & 11 & -6 & 1 & 0 & 0 & 0 \cdots \\ 24 & -50 & 35 & -10 & 1 & 0 & 0 \\ -120 & 274 & -225 & 85 & -15 & 1 & 0 \\ 720 & -1764 & 1624 & -735 & 175 & -21 & 1 \\ & & & \vdots & & & \end{pmatrix}. \tag{4.5}$$

For example, the fourth and fifth rows of this matrix correspond to the identities

$$x(x-1)(x-2)(x-3) = x^4 - 6x^3 + 11x^2 - 6x$$
$$x(x-1)(x-2)(x-3)(x-4) = x^5 - 10x^4 + 35x^3 - 50x^2 + 24x.$$

Readers who are familiar with linear algebra may notice that the matrices (4.1) and (4.5) are inverses of each other. Clearly, the numbers $s(n,k)$ may be both positive and negative and hence do not admit combinatorial interpretations. However, replacing x with $-x$ in (4.4) shows that the numbers $c(n,k) = (-1)^{n-k}s(n,k)$ are determined by the equalities

$$x(x+1)\cdots(x+n-1) = \sum_{k=1}^{n} c(n,k)x^k \tag{4.6}$$

for $n \in \mathbb{Z}_{>0}$ and hence are nonnegative integers. These numbers, called *signless Stirling numbers of the first kind*, are determined by the recurrence of the following proposition.

Proposition 4.3.4. *The following hold for the numbers $c(n,k)$:*

(a) $c(n,k) = c(n-1,k-1) + (n-1)\,c(n-1,k)$ *for* $1 \le k \le n$,

(b) $\sum_{k=1}^{n} c(n,k) = n!$ *for* $n \ge 1$,

where $c(0,0) = 1$ *and* $c(0,k) = 0$ *for* $k \ge 1$.

Proof. Part (b) follows by setting $x = 1$ in (4.6). To prove part (a), we may assume that $n \ge 2$. We observe that

$$\begin{aligned}\sum_{k=1}^{n} c(n,k)x^k &= x(x+1)\cdots(x+n-1) = (x+n-1)\cdot x(x+1)\cdots(x+n-2)\\ &= (x+n-1)\cdot\sum_{k=1}^{n-1} c(n-1,k)x^k\\ &= \sum_{k=1}^{n-1} c(n-1,k)x^{k+1} + (n-1)\sum_{k=1}^{n-1} c(n-1,k)x^k\\ &= \sum_{k=2}^{n} c(n-1,k-1)x^k + (n-1)\sum_{k=1}^{n-1} c(n-1,k)x^k.\end{aligned}$$

We compare the coefficients of x^k in the two extreme sides of this string of equalities and the proof follows. □

For a fundamental combinatorial interpretation of the numbers $c(n,k)$, see [23, Section 1.3]. Based on the material of Chapter 2, we may express $x(x+1)\cdots(x+n-1)$ as a linear combination of the polynomials $x(x-1)\cdots(x-k+1)$ as follows.

Proposition 4.3.5. *For every positive integer n,*

$$x(x+1)\cdots(x+n-1) = \sum_{k=1}^{n} \frac{n!}{k!}\binom{n-1}{k-1} x(x-1)\cdots(x-k+1), \tag{4.7}$$

as polynomials with integer coefficients.

Proof. As argued in the proof of Proposition 4.3.2, it suffices to verify (4.7) for $x = m \in \mathbb{Z}_{>0}$. Then, (4.7) can be written in the equivalent form

$$\binom{m+n-1}{n} = \sum_{k=1}^{n} \binom{n-1}{k-1}\binom{m}{k}.$$

Since $\binom{n-1}{k-1} = \binom{n-1}{n-k}$, this equation follows from the identity of Proposition 2.4.1. □

4.4 Partial Orders

The second important class of binary relations we will study is that of partial orders. These relations provide ways with which one can "compare" the elements of a set to one another.

Definition 4.4.1. *A binary relation, say $\preceq$, on a set P is said to be a partial order if the following properties hold for all $a, b, c \in P$:*

(i) $a \preceq a$ *(reflexivity),*
(ii) *if* $a \preceq b$ *and* $b \preceq a$, *then* $a = b$ *(antisymmetry)*
(iii) *if* $a \preceq b$ *and* $b \preceq c$, *then* $a \preceq c$ *(transitivity).*

Divisibility on a set of positive integers and the inclusion order $\subseteq$ on a collection of subsets of a given set are typical examples of partial orders (see also Examples 4.4.3 and 4.4.4 in the sequel).

Let us introduce some notation and terminology. Let $\preceq$ be a partial order on a set P. The pair $(P, \preceq)$ is called a *partially ordered set* (*poset*, for short) and is simply denoted by P, when it is clear from the context what the partial order $\preceq$ is. The elements a, b of P are called *comparable* if $a \preceq b$ or $b \preceq a$, and *incomparable* otherwise. We write $a \prec b$ (and $b \succ a$) if $a \preceq b$ and $a \neq b$. If $a \prec b$ and there is no $x \in P$ such that $a \prec x \prec b$, then we say that b *covers* a and that the pair (a, b) (and the relation $a \prec b$) is a *cover relation* in P. If P is finite, then $(P, \preceq)$ can be drawn with its *Hasse diagram*, which contains the elements of P as vertices and one edge

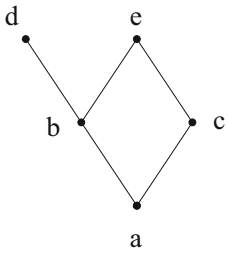


Figure 4.3 A poset with five elements

connecting a and b for every cover relation $a \prec b$ in P, where b appears higher in the diagram then a.

EXAMPLE. Figure 4.3 shows the Hasse diagram of a partial order on the 5-element set $P = \{a, b, c, d, e\}$. The cover relations in P are $a \prec b, a \prec c, b \prec d, b \prec e$, and $c \prec e$. Transitivity implies that $a \prec d$ and $a \prec e$. The sets $\{b, c\}$, $\{c, d\}$, and $\{d, e\}$ are pairs of incomparable elements of P. □

Let $(P, \preceq)$ be a poset, and let $a \in P$. We say that a is a *minimal* element of P if there is no $x \in P$ such that $x \prec a$, and a *maximal* element if there is no $x \in P$ such that $a \prec x$. We also say that a is a *minimum* element of P if $a \preceq x$ for every $x \in P$ and a *maximum* element if $x \preceq a$ for every $x \in P$. We note that P may have at most one minimum and at most one maximum element. Indeed, if $a, b \in P$ are both minimum (or both maximum) elements, then $a \preceq b$ and $b \preceq a$, and hence, $a = b$ by antisymmetry. We also note that a minimum (respectively, maximum) element of P (if such an element exists) is necessarily minimal (respectively, maximal) and that the converse is not true in general. For the example of Figure 4.3, a is the minimum (and hence the unique minimal) element of P and d and e are both maximal elements, whereas P does not have a maximum element.

Let us now discuss other examples of posets.

Example 4.4.1. The natural order $\preceq_{\mathbb{Z}}$ on integers is a partial order on $\mathbb{Z}$, as well as on any subset of $\mathbb{Z}$ (we consider the properties of Definition 4.4.1 as known in this case). With respect to this order, any two elements of $\mathbb{Z}$ are comparable. A partial order having this property is called a *total order*, or *linear order*, or *chain*. For example, there are two chains on the set $\{1, 2\}$, namely $1 \prec 2$ and $2 \prec 1$. More generally (see Exercise 37), every total order on an n-element set P has elements of the form $\sigma_1 \prec \sigma_2 \prec \cdots \prec \sigma_n$ for some permutation $(\sigma_1, \sigma_2, \ldots, \sigma_n)$ of P. As a consequence, there are exactly $n!$ total orders on an n-element set.

Example 4.4.2. Let $\preceq$ be the binary relation on a set P defined by setting $a \preceq b \Leftrightarrow a = b$, for $a, b \in P$. Clearly, this relation is a partial order on P, called an *antichain*. It is the only partial order on P in which there are no two distinct comparable elements. Every element of an antichain is both minimal and maximal.

Example 4.4.3. Consider the binary relation $\mid$ of divisibility on the set $\mathbb{Z}_{>0}$ of positive integers, defined by setting $a \mid b$ if a divides b (see Section 1.1.3).

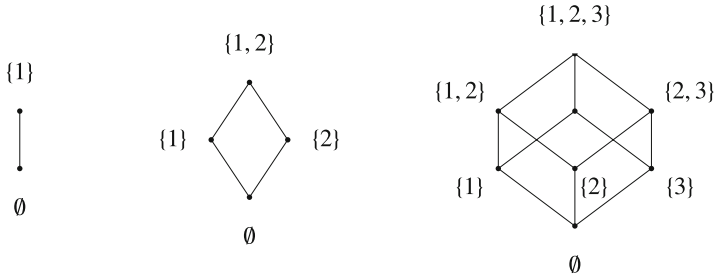


Figure 4.4 The posets B_1, B_2 and B_3

Proposition 1.1.3 implies that this relation is a partial order on $\mathbb{Z}_{>0}$. The number 1 is the minimum element of this order, whereas there are no maximal (or maximum) elements. The elements that cover 1 are those integers greater than 1, which have no divisor other than 1 and themselves; these are exactly the prime numbers. More generally, b covers a in this order if and only if $b = pa$ for some prime number p.

Example 4.4.4. Let B_n be the powerset of $[n]$, endowed with the binary relation $\preceq$ of inclusion, meaning that $S \preceq T \Leftrightarrow S \subseteq T$, for $S, T \subseteq [n]$. The three properties of Definition 4.4.1 are trivially satisfied; for example, antisymmetry is equivalent to the fact that if $S \subseteq T$ and $T \subseteq S$ for some $S, T \subseteq [n]$, then $S = T$. Therefore, $\preceq$ is a partial order. The resulting poset B_n is called the *Boolean algebra* of rank n. The Hasse diagram of B_n is drawn in Figure 4.4 for $n \in \{1, 2, 3\}$. For $S, T \in B_n$, we have that T covers S if and only if $S \subseteq T$ and $\#T = \#S + 1$ (see Exercise 39). The empty set is the minimum element of B_n and $[n]$ is its maximum element.

4.4.1 Chains and Antichains

In this section, P stands for a poset, the partial order of which will be denoted by $\preceq$.

Definition 4.4.2. *A chain in P is defined as any subset of P consisting of pairwise comparable elements. An antichain in P is defined as any subset of P consisting of pairwise incomparable elements.*

We recall (see Exercise 37) that every finite chain in P has the form $\{x_1, x_2, \ldots, x_k\}$, where $x_1 \prec x_2 \prec \cdots \prec x_k$.

Example 4.4.5.

(a) The empty set and all subsets of P of cardinality one are both chains and antichains in P. These are the only subsets of P that have this property.
(b) For the poset of Figure 4.3, $\{a, e\}$, $\{a, b, d\}$, and $\{a, c, e\}$ are chains, whereas $\{b, c\}$, $\{c, d\}$, and $\{d, e\}$ are antichains.

(c) Let's consider the set $\mathbb{Z}_{>0}$ of positive integers, partially ordered by divisibility (see Example 4.4.3). Then, $\{1, 2, 4, \ldots, 2^{n-1}\}$ is an n-element chain and $\{n+1, n+2, \ldots, 2n\}$ is an n-element antichain for every positive integer n. The latter holds because for $a, b \in \mathbb{Z}$ with $n+1 \leq a < b \leq 2n$, we have $1 < b/a < 2n/n = 2$ and hence a does not divide b.

(d) For the poset B_n of Example 4.4.4, a k-element chain is any set of the form $\{S_1, S_2, \ldots, S_k\}$ such that $S_1 \subset S_2 \subset \cdots \subset S_k \subseteq [n]$. An antichain consists of subsets $A_1, A_2, \ldots, A_k$ of $[n]$ none of which is contained in another. Equivalently, $A_i \subseteq A_j$ implies $i = j$. □

The structure of the set of chains in a poset P determines to a considerable extent the structure of P and is connected in nontrivial ways to the structure of the set of antichains in P. We will study two quantities one can define for P, which depend on these sets. We call *height* of P the largest cardinality of a chain in P and *width* of P the largest cardinality of an antichain in P. We will denote by height(P) and width(P) the height and the width of P, respectively. For the poset of Figure 4.3, we have height$(P) = 3$ and width$(P) = 2$.

Lemma 4.4.1. *Let P be a poset.*

(a) *If C is a chain in P and there exists a partition of P into m antichains, then $\#C \leq m$.*

(b) *If A is an antichain in P and there exists a partition of P into m chains, then $\#A \leq m$.*

Proof. Let us prove part (a). Consider a partition of P into m antichains $A_1, A_2, \ldots, A_m$ and a chain C and suppose, by the way of contradiction, that $\#C \geq m+1$. Then (by the pigeonhole principle), at least two elements of C belong to the same antichain, and hence are incomparable, contrary to the fact that C is a chain. This shows that $\#C \leq m$. The proof of part (b) is similar. □

Lemma 4.4.1 yields the following upper bounds for the height and the width of P.

Corollary 4.4.1. *For every finite poset P*

(a) height$(P) \leq a(P)$,

(b) width$(P) \leq c(P)$,

where $a(P)$ and $c(P)$ are the smallest number of antichains and chains, respectively, into which one can partition P.

Proof. This follows from Lemma 4.4.1. □

EXAMPLE. The poset P of Figure 4.3 has a three-element chain, for instance, $\{a, b, d\}$, and a two-element antichain, for instance, $\{b, c\}$. This implies that height$(P) \geq 3$ and width$(P) \geq 2$. Moreover, P has a partition into three antichains, namely $\{a\}$, $\{b, c\}$, and $\{d, e\}$, as well as a partition into two chains, namely $\{a, b, d\}$ and $\{c, e\}$. This implies that $a(P) \leq 3$ and $c(P) \leq 2$ and, in view of Corollary 4.4.1,

that height$(P) \leq a(P) \leq 3$ and width$(P) \leq c(P) \leq 2$. Thus, we conclude that height$(P) = a(P) = 3$ and width$(P) = c(P) = 2$. □

Is height(P) always equal to $a(P)$? Is width(P) always equal to $c(P)$? The following theorem gives an affirmative answer to the first question.

Theorem 4.4.1. *We have* height$(P) = a(P)$ *for every finite poset* P. *Equivalently, the largest number of pairwise comparable elements of* P *is equal to the smallest number of antichains into which one can partition* P.

Proof. Let height$(P) = m$. By Corollary 4.4.1, we have $a(P) \geq m$, so it suffices to show that P can be partitioned into at most m antichains. For any $x \in P$, we denote by $h(x)$ the largest cardinality r of a chain $x_1 \prec x_2 \prec \cdots \prec x_r$ of P having maximum element $x_r = x$. We claim that (i) $1 \leq h(x) \leq m$ for every $x \in P$; and (ii) $x \prec y \Rightarrow h(x) < h(y)$ for all $x, y \in P$. Indeed, (i) is obvious. To verify (ii), we observe that if $x \prec y$ in P and $x_1 \prec x_2 \prec \cdots \prec x_r = x$ is a chain in P of cardinality $r = h(x)$, then the chain $x_1 \prec \cdots \prec x_r \prec x_{r+1} = y$ of P has cardinality $r + 1$ and hence, $h(y) \geq r + 1 > h(x)$.

We now set $B_i = \{x \in P : h(x) = i\}$ for $i \in [m]$. Because of (i), the nonempty sets among the B_i are the blocks of a partition of P. Because of (ii), B_i is an antichain in P for every $i \in [m]$ (in fact, the inequality $a(P) \geq m$ guarantees that all sets B_i are nonempty). Therefore, $\{B_1, B_2, \ldots, B_m\}$ is a partition of P with the required properties. □

Corollary 4.4.2. *Let* P *be a poset of cardinality at least* $mn + 1$. *Then,* P *has a chain of cardinality* $m + 1$ *or an antichain of cardinality* $n + 1$.

Proof. Suppose that there is no chain of cardinality $m + 1$ in P. Then, height$(P) \leq m$ and Theorem 4.4.1 implies that $a(P) \leq m$. Thus, P can be partitioned into at most m antichains A_i. Our hypothesis $\#P \geq mn + 1$ implies that $\#A_i \geq n + 1$ for at least one of these antichains A_i and the proof follows. □

The proof of the following important theorem is given in the solution to Exercise 45.

Theorem 4.4.2 (Dilworth, 1950). *We have* width$(P) = c(P)$ *for every finite poset* P. *Equivalently, the largest number of pairwise incomparable elements of* P *is equal to the smallest number of chains into which one can partition* P.

In the remainder of this section, we give a few typical applications of the previous results.

Example 4.4.6. Let $a_1 < a_2 < \cdots < a_{mn+1}$ be positive integers. Applying Corollary 4.4.2 to the set of these integers, partially ordered by divisibility, we conclude that one may select either $n + 1$ of them, say $a_{i_1} < a_{i_2} < \cdots < a_{i_{n+1}}$, so that every integer divides the following ones in the list, or $m + 1$ of them so that none of these integers is divisible by another.

Example 4.4.7. Let us prove that every sequence $\sigma = (a_1, a_2, \ldots, a_{mn+1})$ of real numbers with $mn + 1$ terms has either an increasing subsequence with $m + 1$ terms,

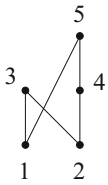


Figure 4.5 A partial order on the set [5]

or a strictly decreasing subsequence with $n+1$ terms; in other words, there exist indices $1 \le i_1 < \cdots < i_{m+1} \le mn+1$ such that $a_{i_1} \le \cdots \le a_{i_{m+1}}$, or indices $1 \le j_1 < \cdots < j_{n+1} \le mn+1$ such that $a_{j_1} > \cdots > a_{j_{n+1}}$. For example, if $m = n = 2$ and $\sigma = (1, 0, 2, 0, 1)$, then there exists an increasing subsequence of σ with three terms, namely $a_2 \le a_4 \le a_5$. This statement is known as the Erdös–Szekeres Theorem.

Let us consider the pair $(P, \preceq)$, where $P = [mn+1]$ and $i \preceq j$ if and only if $i \le j$ and $a_i \le a_j$. We leave it to the reader to verify that $\preceq$ is a partial order on P; its Hasse diagram is shown on Figure 4.5 when $\sigma = (1, 0, 2, 0, 1)$. Corollary 4.4.2 guarantees that P has either a chain $i_1 \prec i_2 \prec \cdots \prec i_{m+1}$, or an antichain $\{j_1, j_2, \ldots, j_{n+1}\}$, where $j_1 < j_2 < \cdots < j_{n+1}$. In the former case, we have $i_1 < i_2 < \cdots < i_{m+1}$ and $a_{i_1} \le a_{i_2} \le \cdots \le a_{i_{m+1}}$ and in the latter, $a_{j_1} > a_{j_2} > \cdots > a_{j_{n+1}}$. For $\sigma = (1, 0, 2, 0, 1)$, the chain $2 \prec 4 \prec 5$ gives rise to the increasing subsequence $a_2 \le a_4 \le a_5$.

Example 4.4.8. Given a positive integer n, what is largest possible number of terms of a sequence of real numbers that has no monotone (meaning, increasing or decreasing) subsequence with more than n terms? Example 4.4.7 shows that such a sequence cannot have more than n^2 terms. An example of a sequence of real numbers with n^2 terms that does not have any monotone subsequences with more than n terms is

$$\sigma_n = (n, n-1, \ldots, 1, 2n, 2n-1, \ldots, n+1, \ldots, n^2, n^2-1, \ldots, n^2-n+1).$$

Indeed, this sequence splits into n segments $\tau_1, \tau_2, \ldots, \tau_n$, each of which is a strictly decreasing subsequence of σ_n with n terms. Moreover, every term of τ_i is smaller than every term of τ_j for $1 \le i < j \le n$. As a result, every increasing subsequence of σ_n contains at most one term from each of $\tau_1, \tau_2, \ldots, \tau_n$ and every decreasing subsequence of σ_n is a subsequence of one of $\tau_1, \tau_2, \ldots, \tau_n$. In particular, all monotone subsequences of σ_n have at most n terms.

In conclusion, the largest number we were asked to compute is equal to n^2.

4.4.2 Sperner's Theorem

We recall that an antichain in the poset B_n is a set consisting of subsets of $[n]$, none of which is contained in another. The width of B_n is the largest number of subsets of $[n]$ with this property.

EXAMPLE. For $n = 3$, there exist two antichains of B_3 with three elements: $\{\{1\}, \{2\}, \{3\}\}$ and $\{\{1, 2\}, \{1, 3\}, \{2, 3\}\}$. Moreover, there exists a partition of B_3 into three chains, for example, $\varnothing \subset \{1\} \subset \{1, 2\} \subset \{1, 2, 3\}$, $\{2\} \subset \{2, 3\}$ and $\{3\} \subset \{1, 3\}$. In view of Corollary 4.4.1, these facts imply that $\operatorname{width}(B_3) = 3$. □

What is the width of B_n for an arbitrary natural number n? The following theorem answers this question.

Theorem 4.4.3 (Sperner, 1928). *We have* $\operatorname{width}(B_n) = \binom{n}{\lfloor n/2 \rfloor}$ *for every* $n \in \mathbb{N}$. *In other words, the largest number of subsets of* $[n]$ *that can be selected, so that none of these subsets is contained in another, is equal to* $\binom{n}{\lfloor n/2 \rfloor}$.

The proof of the theorem is based on the following proposition. The inequality that appears there is known as the LYM inequality (see [2, Section 1.2] for more information); the proof we will provide is due to D. Lubell [15].

Proposition 4.4.1. *For every antichain* $\mathcal{A}$ *in* B_n *we have*

$$\sum_{k=0}^{n} \frac{p_k}{\binom{n}{k}} \leq 1, \tag{4.8}$$

where p_k *is the number of elements* $S \in \mathcal{A}$ *of cardinality* k.

Proof. A subset S of $[n]$ will be called a *prefix* of a permutation $(\sigma_1, \sigma_2, \ldots, \sigma_n)$ of $[n]$ if $S = \varnothing$, or $S = \{\sigma_1, \sigma_2, \ldots, \sigma_k\}$ for some $k \in [n]$. Let m be the number of pairs (S, σ) such that $S \in \mathcal{A}$, σ is a permutation of $[n]$ and S is a prefix of σ. We observe that for a given k-subset $S \subseteq [n]$, there are exactly $k!(n-k)!$ permutations of $[n]$ for which S is a prefix (as many as there are ways one can first list the elements of S in a linear order, followed by those of $[n] \setminus S$ in a linear order, to form a permutation of $[n]$). Hence,

$$m = \sum_{k=0}^{n} p_k \cdot k!(n-k)!.$$

We also observe that any two prefixes of a given permutation of $[n]$ are comparable with respect to the order of B_n (inclusion order). Therefore, every permutation of $[n]$ has at most one prefix that belongs to $\mathcal{A}$ and hence $m \leq n!$. Combining these facts, we get

$$\sum_{k=0}^{n} p_k \cdot k!(n-k)! \leq n!.$$

This inequality is equivalent to (4.8). □

Proof of Theorem 4.4.3. Any two subsets of $[n]$ whose cardinalities are equal are incomparable with respect to the partial order of B_n. As a consequence, the set

$\mathcal{A}_i = \{S \subseteq [n] : \#S = i\}$ of i-subsets of $[n]$ is an antichain in B_n for every $i \in \{0, 1, \ldots, n\}$. Since $\mathcal{A}_i$ has cardinality $\binom{n}{\lfloor n/2 \rfloor}$ for $i = \lfloor n/2 \rfloor$, it follows that $\text{width}(B_n) \geq \binom{n}{\lfloor n/2 \rfloor}$.

To prove the reverse inequality, let us consider an arbitrary antichain $\mathcal{A}$ of B_n and set $p_k = \#\{S \in \mathcal{A} : \#S = k\}$ for $k \in \{0, 1, \ldots, n\}$, as in Proposition 4.4.1. Parts (a) and (e) of Proposition 2.3.6 imply that $\binom{n}{k} \leq \binom{n}{\lfloor n/2 \rfloor}$ for every $k \in \{0, 1, \ldots, n\}$. Thus, by (4.8), we have

$$\sum_{k=0}^{n} \frac{p_k}{\binom{n}{\lfloor n/2 \rfloor}} \leq \sum_{k=0}^{n} \frac{p_k}{\binom{n}{k}} \leq 1$$

and hence,

$$\#\mathcal{A} = \sum_{k=0}^{n} p_k \leq \binom{n}{\lfloor n/2 \rfloor}.$$

Since this holds for every antichain $\mathcal{A}$ in B_n, we have $\text{width}(B_n) \leq \binom{n}{\lfloor n/2 \rfloor}$ and the proof follows. □

A different proof of Theorem 4.4.3, based on Corollary 4.4.1, may be given by constructing a partition of B_n into $\binom{n}{\lfloor n/2 \rfloor}$ chains; see, for instance, [2, Section 3.1]. For other interesting proofs and generalizations of this important result, we refer the reader to the article [22].

4.4.3 The Erdös–Ko–Rado Theorem

We close this section with another important result in combinatorics. We say that a family $\mathcal{A}$ of subsets of a set Ω is *intersecting* if any two elements of $\mathcal{A}$ have a nonempty intersection. We showed in Example 4.1.4 that the largest cardinality of an intersecting family of subsets of $[n]$ is equal to 2^{n-1}. What if we restrict to families of k-subsets of $[n]$ for some fixed $k \in \{0, 1, \ldots, n\}$? In other words, what is the largest number of k-subsets of $[n]$ that can be selected, so that any two of these subsets have a nonempty intersection?

The answer to this question is $\binom{n}{k}$ if $n < 2k$, since then any two k-subsets of $[n]$ have a nonempty intersection. The Erdös–Ko–Rado Theorem asserts that in the nontrivial case that $n \geq 2k$, the answer is $\binom{n-1}{k-1}$, a number achieved by the families consisting of all k-subsets of $[n]$ which contain a fixed element of $[n]$. The proof we will provide, due to G. Katona [12], is similar in spirit to the proof of Sperner's theorem given in Section 4.4.2.

Theorem 4.4.4 (Erdös–Ko–Rado, 1961). *Let n, k be positive integers such that $n \geq 2k$ and $\mathcal{A}$ be an antichain of subsets of $[n]$. If $\mathcal{A}$ is intersecting and $\#A \leq k$ for every $A \in \mathcal{A}$, then*

$$\#\mathcal{A} \leq \binom{n-1}{k-1}.$$

In particular, the largest number of k-subsets of $[n]$ that can be selected, so that any two of these subsets have a nonempty intersection, is equal to $\binom{n-1}{k-1}$.

Proof. Let $\mathcal{A}$ be an intersecting family of subsets of $[n]$, each of cardinality at most k, such that $\mathcal{A}$ is an antichain (no element of $\mathcal{A}$ is contained in another). We say that a cyclic permutation τ of $[n]$ *contains* a subset S of $[n]$ if the elements of S appear successively (in some order) in τ. Let m be the number of pairs (S, τ), where $S \in \mathcal{A}$ and τ is a cyclic permutation of $[n]$, which contains S. For every i-subset S of $[n]$, there are exactly $i!(n-i)!$ cyclic permutations of $[n]$, which contain S. Therefore,

$$m = \sum_{i=0}^{k} p_i \cdot i!(n-i)!,$$

where p_i stands for the number of elements of $\mathcal{A}$ of cardinality i. On the other hand, since $\mathcal{A}$ is intersecting, every cyclic permutation of $[n]$ contains at most k elements of $\mathcal{A}$ (see Exercise 46). This implies that $m \leq (n-1)! \cdot k$ and hence that

$$\sum_{i=0}^{k} p_i \cdot i!(n-i)! \leq k(n-1)!.$$

We rewrite this inequality as

$$\sum_{i=0}^{k} \frac{p_i}{\binom{n}{i}} \leq \frac{k}{n}.$$

As in the proof of Theorem 4.4.3, and given that $k \leq n/2$, we observe that $\binom{n}{i} \leq \binom{n}{k}$ for $0 \leq i \leq k$ by Proposition 2.3.6 (e) and deduce that

$$\#\mathcal{A} = \sum_{i=0}^{k} p_i \leq \frac{k}{n}\binom{n}{k} = \binom{n-1}{k-1}.$$

This completes the proof. □

For other proofs and for various generalizations of the Erdös–Ko–Rado Theorem, see [2, Chapter 5].

4.5 Exercises

1. Let Π be a convex polygon in the plane with n vertices. Prove that the number of points in the interior of Π through which at least two diagonals of Π pass is less than or equal to $\binom{n}{4}$. When does equality hold?
2. Let T be a square in the plane with side lengths equal to 2.
   (a) Find five points in T whose pairwise distances apart are all greater than or equal to $\sqrt{2}$.
   (b) Given any six points of T, prove that there are two among these points that are at a distance smaller than $\sqrt{2}$ apart.
3. Let T be an equilateral triangle in the plane with side lengths equal to 1.
   (a) Find four points in T whose pairwise distances apart are all greater than 1/2.
   (b) Given any five points in T, prove that there are two of these points that are at a distance at most 1/2 apart.
   (c) Find six points in T whose pairwise distances apart are all greater than or equal to 1/2.
   (d) Given any seven points in T, prove that there are two of these points that are at a distance smaller than 1/2 apart.
4. We are given $n^2 + 1$ points in an equilateral triangle with side lengths equal to 1. Prove that there are two of these points that are at a distance at most $1/n$ apart.
5. We are given nineteen points in a square with side lengths equal to 1. Prove that one may choose three of these points, so that the triangle with these points as vertices has area smaller than or equal to 1/18.
6. Find:
   (a) The largest number of integers that can be selected among $1, 2, \ldots, 100$, so that any two of them have at least one common prime divisor.
   (b) The largest number of integers that can be selected among $0, 1, \ldots, 100$, so that no two of them sum to 100.
   (c) The largest number of integers that can be selected among $1, 2, \ldots, 120$, so that any two of them have at least two common prime divisors.
   (d) The largest number of nonempty subsets of $\{1, 2, \ldots, 10\}$ that can be selected, so that any two of them have different minimum elements and different maximum elements.
7. Consider the 64 squares of an 8×8 chessboard.
   (a) Find the largest number of these squares that can be selected, with no two of them having a common vertex.
   (b) Find the largest number of these squares that can be selected, with no three of them having a common vertex.

(c) Find the largest number of these squares that can be selected, with no two of them having a common edge.

8. Two squares of an $n \times n$ chessboard are said to be *adjacent* if they share an edge.
   (a) Twenty five rooks are placed in the squares of a 5×5 chessboard, so that every square contains exactly one rook. Is it possible to move the rooks, each one to a square adjacent to the one originally placed, so that at the end of this process every square of the chessboard contains again exactly one rook?
   (b) What about a 6×6 chessboard?
9. (a) Prove that from n arbitrary integers one can always select some (possibly one or all of them), so that the sum of the selected integers is divisible by n.
   (b) Give an example of a set consisting of $n - 1$ distinct integers that does not have any nonempty subset, the elements of which sum to an integer multiple of n.
10. (a) We are given five points with integer coordinates in the plane. Prove that there are two points among these with the following property: the midpoint of the line segment connecting the two points has also integer coordinates.
    (b) Generalize part (a) for $2^n + 1$ points with integer coordinates in the space $\mathbb{R}^n$.
    (c) Find 2^n points with integer coordinates in $\mathbb{R}^n$ so that no midpoint of a line segment connecting two of these points has integer coordinates (that is, no such midpoint should belong to $\mathbb{Z}^n$).
11. The Greek alphabet has 24 letters. We are given 25 infinite sequences whose terms are letters of this alphabet. Prove that there are two among these sequences, say $(a_1, a_2, \dots)$ and $(b_1, b_2, \dots)$, such that $a_k = b_k$ for infinitely many indices k. Does the same hold for 24 arbitrary infinite sequences?
12. (a) Place 32 rooks in the squares of an 8×8 chessboard (at most one rook in each square), so that the following holds: among any five of the rooks, there are two that lie in the same row or in the same column.
    (b) Prove that if 33 rooks are placed in the squares of an 8×8 chessboard (at most one rook in each square), then one can choose five of them that are pairwise placed in distinct rows and distinct columns.
13. We are given a positive integer n and a set S consisting of $4n$ integers. Prove that there exist four elements a, b, c, d of S such that either all differences $a-b$, $b - c$, and $c - d$ are divisible by $2n - 1$, or the sum $a + b + c + d$ is divisible by $2n - 1$.
14. We are given an $m \times n$ chessboard, where $m, n \geq 2$, with a rook placed in every square. All rooks move, each one to a square that shares exactly one vertex with the square of the original position. What is the smallest possible number of squares that end up empty after such a process?

15. Given integers $a, b \geq 2$, we denote by $X(a, b)$ the set of points (x, y) in the plane with integer coordinates $1 \leq x \leq a$ and $1 \leq y \leq b$. What is the largest possible number of points in $X(a, b)$ that can be selected, so that no three of these points are the vertices of a right-angled triangle?
16. Let A be an $n \times n$ matrix with entries equal to 0 or 1. Suppose that there is no 2×2 submatrix of A (meaning, a matrix formed by two rows and two columns of A) of the form

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Prove that at most

$$n\left(\sqrt{n-\frac{3}{4}}+1/2\right)$$

elements of A are equal to 1 and give an example where equality holds.
17. Given integers $n \geq k \geq 1$, we denote by $\mu_k(n)$ the largest number of subsets of $[n]$ that can be selected, so that the intersection of any two of these sets has at most $k-1$ elements. Prove that

$$\mu_k(n) = \sum_{i=0}^{k} \binom{n}{i}$$

for all n, k.
18. Let $n \in \mathbb{Z}_{>0}$. For $k \in \{0, 1, \ldots, n\}$, we denote by $A(n, k)$ the set of sequences $(\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_n) \in \{-1, 1\}^n$ such that $\varepsilon_1 + \varepsilon_2 + \cdots + \varepsilon_n = -n + 2k$.

    (a) Prove that $A(n, k)$ has exactly $\binom{n}{k}$ elements.
    (b) For $0 \leq k < n/2$, find an injective map $f : A(n, k) \to A(n, k+1)$.
    (c) Deduce that $\binom{n}{k} \leq \binom{n}{k+1}$ for $0 \leq k < n/2$.
    (d) For $0 \leq k < n/2$, find a combinatorial interpretation of $\binom{n}{k+1} - \binom{n}{k}$.

19. Let $n \geq 2$ be an integer. For $k \in \{0, 1, \ldots, n\}$, we denote by $A_n(k)$ the set of sequences $(\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_n) \in \{0, 1\}^n$ such that $\varepsilon_1 + \varepsilon_2 + \cdots + \varepsilon_n = k$.

    (a) Find an injective map $f : A_n(k-1) \times A_n(k+1) \to A_n(k) \times A_n(k)$, for $k \in [n-1]$.
    (b) Deduce that

$$\binom{n}{k}^2 \geq \binom{n}{k-1}\binom{n}{k+1}$$

for every $k \in [n-1]$.

20. Let Y_n be the set of permutations $\sigma = (\sigma_1, \sigma_2, \ldots, \sigma_n)$ of $[n]$ such that $|\sigma_i - \sigma_{i+1}| = 1$ for at least one index $i \in [n-1]$ and let X_n be the set of permutations of $[n]$ that do not belong to Y_n (where $n \geq 2$).

    (a) Find an injective map $f : X_n \to Y_n$.
    (b) Deduce that Y_n has more than $n!/2$ elements.

21. Let $a \geq 2$ and $d \geq 1$ be integers. Prove that d divides at least one of the integers

$$1,\ 1+a,\ 1+a+a^2,\ 1+a+a^2+a^3, \ldots$$

    if and only if d is relatively prime to a (meaning that a and d do not share a common prime divisor). You may use the fact that if $d \mid mn$ and d is relatively prime to m, then $d \mid n$.

22. Given $p \in \mathbb{Z}_{>0}$, we define a sequence $(a(n))_{n\in\mathbb{N}}$ by setting $a(0) = 1, a(1) = p$, and $a(n) = pa(n-1) + a(n-2)$ for $n \geq 2$. For any positive integer m:

    (a) Prove that there exists $r \in \{1, 2, \ldots, m^2\}$ such that $a(n+r) \equiv a(n) \pmod{m}$ for every $n \in \mathbb{N}$.
    (b) Prove that at least one of $a(0), a(1), \ldots, a(m^2-1)$ is divisible by m.

23. We consider a number $\alpha \in \mathbb{R} \setminus \mathbb{Q}$ and set $f(x) = x - \lfloor x \rfloor$ for $x \in \mathbb{R}$.

    (a) Prove that for every $\varepsilon > 0$ there exists an integer $n \neq 0$ such that $f(n\alpha) < \varepsilon$.
    (b) Deduce that for every pair of real numbers $0 \leq c < d \leq 1$ there exists an integer m such that $c < f(m\alpha) < d$.

24. This exercise assumes familiarity with the notion of a field from algebra (see Section A.1). Let F be a finite field with q elements, where q is an odd number.

    (a) Prove that there is no nonzero element $x \in F$ such that $-x = x$.
    (b) If $q \equiv 1 \pmod{4}$, prove that there exists $x \in F$ such that $x^2 = -1$.
    (c) Deduce that for every prime $p \equiv 1 \pmod{4}$, there exists $a \in \mathbb{Z}$ such that $a^2 + 1$ is divisible by p.

25. Apply the result of Example 4.1.2 and that of Exercise 24 (c) to prove that every prime $p \equiv 1 \pmod{4}$ can be expressed as the sum of the squares of two integers.

26. For $S, T \subseteq [n]$ we set $S \equiv T$ if there exists a bijective map $f : S \to T$.

    (a) Prove that $\equiv$ is an equivalence relation on the power set of $[n]$.
    (b) Describe the equivalence classes of $\equiv$. How many are there?

27. Let $X(m, n)$ be the set of all maps $f : [m] \to [n]$. For $f, g \in X(m, n)$, we write $f \sim g$ if there exists $k \in \mathbb{Z}$ such that $g(x) \equiv f(x) + k \pmod{n}$ for every $x \in [m]$.

(a) Prove that $\sim$ is an equivalence relation on $X(m,n)$. How many equivalence classes are there?
(b) We say that two maps $f, g \in X(m,n)$ are compatible if $f(x) = g(x)$ for at least one $x \in [m]$. What is the largest number of pairwise compatible maps that exist in $X(m,n)$?

28. Given an integer $n \geq 2$, prove that the following are equal:

(a) The largest number of pairwise relatively prime integers that can be selected from the set $\{2, 3, \ldots, n\}$.
(b) The smallest number of blocks of a partition π of the set $\{2, 3, \ldots, n\}$, such that no two elements in the same block of π are relatively prime.
(c) The number of primes among $2, 3, \ldots, n$.

29. We say that two permutations $(\sigma_1, \sigma_2, \ldots, \sigma_n)$ and $(\tau_1, \tau_2, \ldots, \tau_n)$ of the set $[n]$ intersect if $\sigma_i = \tau_i$ for at least one index $i \in [n]$. What is the largest number of permutations of $[n]$ one can choose, so that any two of these permutations intersect?
30. Let $a(n,k)$ be the number of surjective maps $f : [n] \to [k]$. Prove that

$$\sum_{k=1}^{n} (-1)^k a(n,k) = (-1)^n$$

for every positive integer n.
31. Prove the following formulas for the Stirling numbers of the second kind

(a) $S(n,2) = 2^{n-1} - 1$,
(b) $S(n,n-1) = \binom{n}{2}$,
(c) $S(n,n-2) = \binom{n}{3} + \frac{1}{2}\binom{n}{2}\binom{n-2}{2}$,

without using Proposition 4.3.1.
32. For the Stirling numbers of the second kind:

(a) Prove that

$$S(n+1,k) = \sum_{m=k-1}^{n} \binom{n}{m} S(m,k-1)$$

for $n \in \mathbb{N}$ and $1 \leq k \leq n+1$.
(b) Deduce that for every $n \geq 2$, there exists $k = k(n) \in [n]$ such that $S(n,1) < S(n,2) < \cdots < S(n,k) > S(n,k+1) > \cdots > S(n,n)$.

33. For the Stirling numbers of the second kind, prove that

$$\binom{n}{k-1} \leq S(n,k) \leq \binom{n-1}{k-1} k^{n-k}$$

for integers $1 \leq k < n$.

34. Find a formula, as simple as possible, for the sum $\sum_{k=1}^{n} k^m \binom{n}{k}$, where $m, n \in \mathbb{Z}_{>0}$.

35. Let $f(n)$ be the number of ways one can choose a partition π of the set $[n]$ and linearly order every block of π. Prove that

$$f(n) = \sum_{k=1}^{n} \frac{n!}{k!} \binom{n-1}{k-1}$$

for every positive integer n.

36. Find all binary relations on a set P that are both equivalence relations and partial orders.

37. (a) Prove that every nonempty finite poset has at least one minimal and at least one maximal element.
    (b) Prove that every nonempty finite poset has a minimal element a and a maximal element b such that $a \preceq b$.
    (c) Prove that for every total order $\preceq$ on an n-element set P, there exists a permutation $(\sigma_1, \sigma_2, \ldots, \sigma_n)$ of P such that $\sigma_1 \prec \sigma_2 \prec \cdots \prec \sigma_n$.

38. At a party, every girl has danced with at least one boy but no boy has danced with all girls. Prove that there are two boys a, a' and two girls b, b' such that the couples ab and $a'b'$ have danced together, but ab' and $a'b$ have not.

39. Consider the Boolean algebra B_n.

    (a) Prove that T covers S in B_n if and only if $S \subseteq T$ and $\#T = \#S + 1$.
    (b) Prove that $\text{height}(B_n) = n + 1$.
    (c) How many $(n+1)$-element chains are there in B_n?

40. Let $\Omega_n = \{1, -1, 2, -2, \ldots, n, -n\}$. We consider the pair $P_n := (\Omega_n, \preceq)$, where $\preceq$ is the binary relation on Ω_n defined by setting $a \preceq b$ if $a = b$ or $|a| < |b|$, for $a, b \in \Omega_n$.

    (a) Prove that $\preceq$ is a partial order on Ω_n.
    (b) Prove that $\text{height}(P_n) = n$. How many n-element chains of P_n are there?
    (c) Find the width of P_n.

41. For elements (a_1, b_1), (a_2, b_2) of $P_n = [n] \times [n]$ we set $(a_1, b_1) \preceq (a_2, b_2)$ if $a_1 \leq a_2$ and $b_1 \leq b_2$ in the usual order of integers.

    (a) Prove that $\preceq$ is a partial order on P_n.
    (b) Draw the Hasse diagram of this order for $n = 4$.

(c) Compute the height of P_n.
(d) Compute the width of P_n.

42. For elements $(a_1, b_1, c_1), (a_2, b_2, c_2)$ of $Q_n = [n] \times [n] \times [n]$ we set $(a_1, b_1, c_1) \preceq (a_2, b_2, c_2)$ if $a_1 \leq a_2$, $b_1 \leq b_2$, and $c_1 \leq c_2$ in the usual order of integers.

(a) Prove that $\preceq$ is a partial order on Q_n.
(b) Compute the height of Q_n.
(c) Prove that $\binom{n+1}{2} \leq \text{width}(Q_n) \leq n^2$ for every positive integer n.

43. Let P_n be the set of all subsets of $\{1, -1, 2, -2, \ldots, n, -n\}$, which contain at most one element of $\{i, -i\}$ for every $i \in [n]$. We consider the inclusion order on P_n, defined by setting $S \preceq T$ if $S \subseteq T$, for $S, T \in P_n$.

(a) What is the cardinality of P_n? For $k \in \{0, 1, \ldots, n\}$, how many elements of P_n have cardinality equal to k?
(b) Draw the Hasse diagram of P_n for $n = 2$.
(c) Compute the height of P_n for every positive integer n.
(d) Prove that $\text{width}(P_n) \geq (3^n - 1)/n$ for every positive integer n. When does equality hold?

44. Let Q_n be the set $[n]$, partially ordered by divisibility.

(a) Draw the Hasse diagram of this poset for $n = 6$ and $n = 9$.
(b) Compute the height of Q_n.
(c) Compute the width of Q_n.

45. Prove Theorem 4.4.2.
46. Prove that, under the assumptions of Theorem 4.4.4, every cyclic permutation of $[n]$ contains at most k elements of $\mathcal{A}$.

Chapter 5
Graphs



The notion of a graph serves as a simple but useful mathematical model with which one can study successfully a wide variety of phenomena in the natural and social sciences. After introducing this concept, the goal of this section is to study various problems in discrete mathematics that can be expressed in the language of graph theory and to acquaint the reader with the methods of this theory. Some of these problems concern special classes of graphs (such as trees, bipartite graphs, and planar graphs), which will be studied in separate sections.

5.1 The Notion of a Graph

We begin with a concrete example.

Example 5.1.1. During a meeting of seven people, some shake hands (we assume that nobody shakes hands with himself or herself). We will prove that somebody has shaken hands an even number of times. □

Our goal is to describe this situation by means of an abstract mathematical structure that takes into account only the information that is essential to solve the problem and ignores all other information. The structure we will use is that of a graph, which considers every person in the meeting as a vertex and a handshake between persons a and b as an edge that connects vertices a and b of the graph. We are asked to prove that there exists a vertex that is an endpoint of an even number of edges. We will return to Example 5.1.1 after explaining the relevant definitions.

Definition 5.1.1. *A graph is defined as a triple $G = (V, E, \varphi)$, where V and E are sets, the elements of which are called vertices and edges, respectively, and*

$$\varphi : E \to \{\{a, b\} : a, b \in V\}$$

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Undergraduate Texts in Mathematics,
https://doi.org/10.1007/978-3-032-14290-0_5

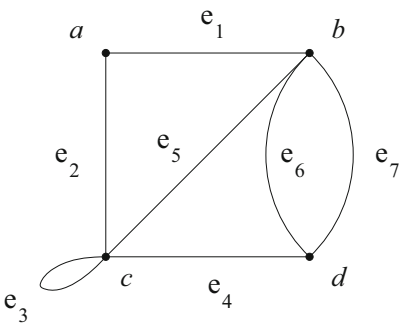


Figure 5.1 A graph with four vertices

is a map from E to the set of subsets of V with one or two elements. If $e \in E$ and $\varphi(e) = \{a, b\}$, then the vertices a, b are called the endpoints of e.

We will study only finite graphs, meaning graphs with finite sets of vertices and edges. Figure 5.1 shows a graph $G = (V, E, \varphi)$ with vertex set $V = \{a, b, c, d\}$ and seven edges such that $\varphi(e_1) = \{a, b\}$, $\varphi(e_2) = \{a, c\}$, $\varphi(e_3) = \{c\}$, and so on. For the graph discussed after Example 5.1.1, V is the set of people present at the meeting, E is the set of handshakes that took place, and $\varphi(e) = \{a, b\}$ if e is a handshake between a and b.

We adopt the following notation and terminology for a graph $G = (V, E, \varphi)$. We say that two vertices $a, b \in V$ are *adjacent*, that they are *neighbors* and that they are *connected by an edge*, in G if there exists $e \in E$ such that $\varphi(e) = \{a, b\}$. We also say that e is *incident* to a and b, and that e is a *loop* if $a = b$. Two edges $e_1, e_2 \in E$ are called *parallel* if $\varphi(e_1) = \varphi(e_2) = \{a, b\}$ for two distinct vertices a, b. The graph G is called *simple* if it has neither loops nor parallel edges. In this case, we may identify any edge $e \in E$ with the set $\{a, b\}$ of its two endpoints a, b and thus consider that $E \subseteq \binom{V}{2}$. The number of edges of G that are not loops and are incident to a given vertex $v \in V$ is called the *degree* of v and is denoted by $\deg_G(v)$. A vertex $v \in V$ is called *isolated* if $\deg_G(v) = 0$, and *even* or *odd* if the degree of v is an even or odd number, respectively. For the graph of Figure 5.1, e_3 is a loop, e_6 and e_7 are parallel edges and vetrices a, b, c, d have degrees 2, 4, 3, 3, respectively. Example 5.1.1 claims that every graph with seven vertices has at least one even vertex.

Proposition 5.1.1. *The sum of the degrees of the vertices of a graph G is equal to $2m$, where m is the number of nonloop edges of G.*

Proof. Let $G = (V, E, \varphi)$, and let F be the set of nonloop edges of G, so that $F \subseteq E$ and $\#F = m$. We consider the set of pairs $(v, e) \in V \times F$, where $v \in V$ is a vertex of G and $e \in F$ is an edge that is incident to v. We observe that for every $e \in F$, there exist exactly two vertices $v \in V$ such that $(v, e) \in S$. This implies that $\#S = 2m$. On the other hand, for every $v \in V$, the number of edges $e \in F$ such that $(v, e) \in S$ is equal to $\deg_G(v)$. This implies that $\#S = \sum_{v \in V} \deg_G(v)$ and the proof follows. □

The following corollary answers the question posed in Example 5.1.1.

Corollary 5.1.1. *The number of odd vertices of an arbitrary graph is an even number. In particular, every graph with an odd number of vertices has an odd number of even vertices.*

Proof. Proposition 5.1.1 implies that the sum of the degrees of the vertices of any graph G is an even number. Equivalently, the number of odd terms in this sum, meaning the number of odd vertices of G, must be even. □

An important notion in graph theory is that of a graph isomorphism. Two graphs are isomorphic if one can be obtained from the other by renaming the vertices and edges. More specifically, we have the following definition.

Definition 5.1.2. *Two graphs $G = (V, E, \varphi)$ and $G' = (V', E', \varphi')$ are called isomorphic if there exist bijections $f : V \to V'$ and $g : E \to E'$ such that for all $a, b \in V$ and $e \in E$,*

$$\varphi(e) = \{a, b\} \Leftrightarrow \varphi'(g(e)) = \{f(a), f(b)\}.$$

In particular, two simple graphs $G = (V, E)$ and $G' = (V', E')$ are isomorphic if there exists a bijection $f : V \to V'$ such that for all $a, b \in V$, we have $\{a, b\} \in E \Leftrightarrow \{f(a), f(b)\} \in E'$. Such a bijection f is called *graph isomorphism*. We leave it to the reader to verify that the relation of isomorphism is an equivalence relation on the set of graphs. The equivalence classes of this relation are called *isomorphism classes* of graphs.

Example 5.1.2. Let V be an n-element set.

(a) The simple graph with vertex set V in which any two vertices are adjacent is called the *complete graph* on (the vertex set) V. Any two complete graphs, each with n vertices, are isomorphic. Their isomorphism class is denoted by K_n.
(b) Let $V = \{v_1, v_2, \ldots, v_n\}$. The simple graph with vertex set V and edges $\{v_i, v_{i+1}\}$ for $i \in [n-1]$ is called a *path of length* $n-1$ (or path with n vertices). The isomorphism class of this graph is denoted by P_n. The graph on the same set of vertices that has one additional edge, namely $\{v_1, v_n\}$, is called an *n-cycle* (or *cycle of length n*, or *cycle with n vertices*). The isomorphism class of this graph is denoted by C_n. Figure 5.2 shows K_n and C_n for $n = 5$.
(c) Let A and B be disjoint sets with cardinalities m and n, respectively. The simple graph with vertex set $A \cup B$ and edges $\{a, b\}$ for $a \in A$ and $b \in B$ is called



Figure 5.2 The graphs K_5 and C_5

Figure 5.3 The graph $K_{3,3}$

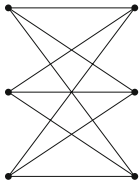


Figure 5.4 The 5×4 grid graph

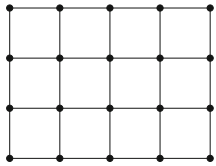
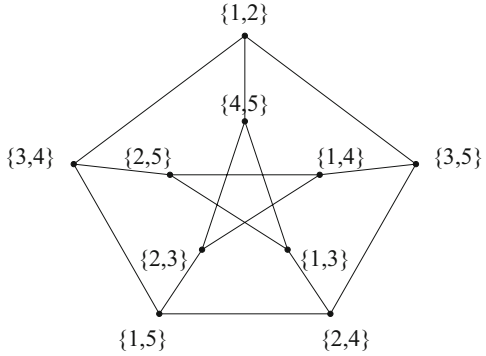


Figure 5.5 Petersen's graph



a *complete bipartite graph* of type (m, n). The isomorphism class of this graph (shown in Figure 5.3 for $m = n = 3$) is denoted by $K_{m,n}$.

(d) The $m \times n$ *grid graph* (see Figure 5.4) is defined as the simple graph with vertex set $[m] \times [n]$ in which two vertices (a_1, a_2) and (b_1, b_2) are adjacent if either $a_1 = b_1$ and $|a_2 - b_2| = 1$, or $a_2 = b_2$ and $|a_1 - b_1| = 1$.

(e) Let Ω be the set of 2-subsets of [5]. The simple graph with vertex set Ω in which two vertices a and b are adjacent if $a \cap b = \varnothing$ is called *Petersen's graph* (see Figure 5.5).

(f) We denote by $\square_n$ the simple graph with vertex set $\{0, 1\}^n$ in which two vertices are adjacent if they differ in exactly one coordinate. For example, $\square_2$ is isomorphic to the 4-cycle and $\square_3$ is shown in Figure 5.6. For reasons that may be apparent to some readers, $\square_n$ is called the *graph of the n-dimensional cube*. □

We close this section with the notion of a subgraph. A graph $G' = (V', E', \varphi')$ is said to be a *subgraph* of $G = (V, E, \varphi)$ (and G is said to be a *supergraph* of G') if $V' \subseteq V$, $E' \subseteq E$, and $\varphi'(e) = \varphi(e)$ for every $e \in E'$. Thus, a subgraph of G is obtained by deleting some of the vertices and edges of G. Every subgraph of G that is isomorphic to a path or a cycle is simply called a *path* or *cycle in G*, respectively.

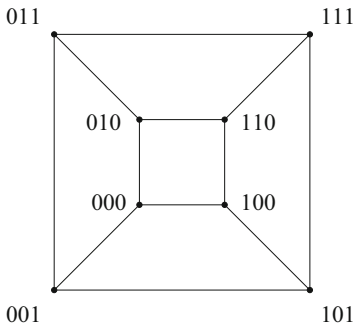


Figure 5.6 The graph $\square_3$

A subgraph $G' = (V', E', \varphi')$ of G is called *induced* on the vertex set V' if G' contains all edges of G whose endpoints belong to V'. Thus, every graph with n vertices has exactly 2^n induced subgraphs (considering the graph with empty vertex set as an induced subgraph).

EXAMPLE. A complete graph G with four vertices has three cycles of length 4 and four cycles of length 3. Which vertices and edges should be deleted from G in order to obtain these subgraphs?

5.2 Walks and Connectivity

The notion of a walk in a graph G aims to describe the ways with which one can reach a given vertex of G from another, by moving (with a finite procedure) along edges of G.

Definition 5.2.1. *Let $G = (V, E, \varphi)$ be a graph, and let $\ell \in \mathbb{N}$. A walk (or tour) of length ℓ in G is defined as any sequence*

$$w = (u_0, e_1, u_1, \ldots, e_\ell, u_\ell)$$

such that $u_i \in V$ for $i \in \{0, 1, \ldots, \ell\}$ and $e_i \in E$ with $\varphi(e_i) = \{u_{i-1}, u_i\}$ for $i \in [\ell]$. Vertices u_0 and u_ℓ are called the endpoints of w; u_0 is called the initial vertex, and u_ℓ is called the terminal vertex.

The walk w is called closed if $u_0 = u_\ell$ and a path if $u_0, u_1, \ldots, u_\ell$ are pairwise distinct.

EXAMPLE. The sequence $(a, e_2, c, e_3, c, e_5, b, e_6, d, e_7, b)$ is a walk of length 5 in the graph of Figure 5.1 with initial vertex a and terminal vertex b. The sequence $(a, e_2, c, e_4, d, e_7, b)$ is a path of length 3. □

Remark 5.2.1.

(a) The sequence $w = (u)$ with only one term $u \in V$ is a closed walk (and a path) of length zero in G.
(b) Suppose that there exists a walk in G with initial vertex u and terminal vertex v. We may choose such a walk of smallest possible length. Since this walk must be a path, we conclude that there exists a path in G with initial vertex u and terminal vertex v. □

Given vertices $u, v \in V$ of a graph $G = (V, E, \varphi)$, we write $u \sim v$ if there exists a walk in G with initial vertex u and terminal vertex v. We observe that for every $u \in V$ there exists a walk of length zero in G with initial and terminal vertex u and that if w is a walk in G with initial vertex u and terminal vertex v, then the sequence obtained from w by reversing the order in which its terms appear is a walk in G with initial vertex v and terminal vertex u. As a result, the binary relation $\sim$ is reflexive and symmetric. This relation is also transitive because if w is a walk in G with initial vertex u and terminal vertex v and w' is a walk in G with initial vertex $u' = v$ and terminal vertex v', then the sequence obtained by listing the terms of w' after those of w and identifying the initial vertex of w' with the terminal vertex of w is a walk in G with initial vertex u and terminal vertex v'. In view of these observations, we conclude that $\sim$ is an equivalence relation on the set V.

The induced subgraphs of G on the equivalence classes of $\sim$ are called *connected components* of G. The graph G is called *connected* if it has a single connected component. We state the following equivalent definition of connectivity.

Definition 5.2.2. *A graph G is called connected if for any two vertices u, v of G there exists a walk in G with initial vertex u and terminal vertex v.*

EXAMPLE.

(a) All graphs of Example 5.1.2 are connected.
(b) Given positive integers m, n with $mn \geq 2$, we let G be the simple graph with vertices the mn squares of an $m \times n$ chessboard in which two such squares are adjacent if they have exactly one common vertex. This graph is *not* connected; in fact, it has exactly two connected components. The vertex sets of the two connected components of G are the squares of the same color in the usual black and white coloring of the chessboard. □

Higher Connectivity Connectivity is a desirable property that a graph might have. Some graphs have stronger properties, which may be described by the concept of higher connectivity.

Definition 5.2.3. *Let G be a graph with vertex set V, and let $k \in \mathbb{N}$. The graph G is called k-connected if it has at least $k + 1$ vertices and for every subset U of V of cardinality less than k, the induced subgraph of G on the vertex set $V \setminus U$ is connected.*

Thus, a graph G is k-connected (we say G is *doubly connected* for $k = 2$) if G is connected, it has at least $k + 1$ vertices and at least k vertices must be deleted from G in order to obtain a disconnected graph.

The smallest natural number k for which a nonempty graph G is k-connected is called the *connectivity degree* of G. This number provides a measure for how well-connected the graph G is.

EXAMPLE.

(a) Every path is connected but not doubly connected. Hence, every path with at least two vertices has connectivity degree 1.
(b) Every cycle C_n is doubly connected but not 3-connected, since either it has only three vertices, or else deleting any two nonadjacent vertices yields a disconnected graph. Hence, C_n has connectivity degree 2. The same holds for the $m \times n$ grid graph, if $m, n \geq 2$ (explain why).
(c) The graph of Figure 5.6 has connectivity degree 3, since it is 3-connected (explain why) and deleting the three vertices that are adjacent to any given vertex yields a graph with an isolated vertex.
(d) The complete graph K_n has connectivity degree $n - 1$. The complete bipartite graph $K_{n,n}$ has connectivity degree n. □

The following theorem characterizes doubly connected (finite) graphs. For the generalization to k-connected graphs for any $k \geq 2$, known as Menger's Theorem, we refer the reader to [8, Section 3.3].

Theorem 5.2.1. *For a graph G with at least three vertices, the following conditions are equivalent:*

(i) *G is doubly connected.*
(ii) *Given any two vertices v, v' of G, there exists a cycle in G that passes through v and v'.*

Proof. Let V be the vertex set of G. First, we assume that (ii) holds, in which case G is certainly connected. We consider an arbitrary vertex $u \in V$ and the induced subgraph, denoted by $G \setminus u$, of G on the vertex set $V \setminus \{u\}$. Given any two vertices v, v' of $G \setminus u$, because of condition (ii), there exist two paths in G with endpoints v and v' that do not have any common vertices other than v and v'. Then, at least one of these paths does not pass through u and hence is a path in $G \setminus u$ with endpoints v and v'. We have shown that any two vertices of $G \setminus u$ are the endpoints of a path in this graph. Therefore, $G \setminus u$ is connected for every $u \in V$ and hence, G is doubly connected. We have proved the implication $(ii) \Rightarrow (i)$.

To prove that $(i) \Rightarrow (ii)$, we assume that G is doubly connected and consider vertices $v, v' \in V$ with $v \neq v'$. We call *distance* between these two vertices the smallest possible length of a path in G with endpoints v and v'. We will prove that (ii) holds by induction on the distance, say k, between v and v' in G.

We first assume that $k = 1$, so that v and v' are the endpoints of an edge, say e, of G. Let u be a vertex of G other than v, v'. Since G is doubly connected, there exists a walk in G with endpoints u and v that does not pass through v' and a walk in G with endpoints u and v' that does not pass through v. It follows that there exists a walk, and hence a path (see Remark 5.2.1), with endpoints v and v' in the graph that

is obtained from G by deleting the edge e. Together with e, this path forms a cycle in G that passes through v and v'.

Finally, we assume that $k \geq 2$ and that (ii) holds for all pairs of vertices of G whose distance is smaller than k. Let $(v_0, e_1, v_1, \ldots, e_k, v_k)$ be a path of length k in G with initial vertex $v_0 = v$ and terminal vertex $v_k = v'$. By the induction hypothesis, there exists a cycle, say C, in G that passes through vertices v and v_{k-1}. We may assume that C does not pass through v' (otherwise, (ii) holds for v and v'). The cycle C consists of two paths, say P and Q, in G, with initial vertex v and terminal vertex v_{k-1}, which do not have common vertices other than v and v_{k-1}. Since G is doubly connected, there exists a path R in G with initial vertex v and terminal vertex v' that does not pass through v_{k-1}. Let u be the last vertex of R that is a vertex of P or Q (possibly, $u = v$). Assuming, without loss of generality, that u is a vertex of P, we find a cycle in G that passes through vertices v and v', namely the one that is formed by the part of P between vertices v and u, the part of R between vertices u and v', the edge e_k that connects vertices v_{k-1} and v' and the path Q. The existence of this cycle completes the proof. □

An important class of 3-connected simple graphs are the graphs of three-dimensional bounded convex polyhedra (convex polytopes) in $\mathbb{R}^3$, such as the tetrahedron, the cube, the triangular prism, and the octahedron (these graphs belong to the class of planar graphs, which we will study in Section 5.6). The vertices of the graph of a three-dimensional convex polytope Π are the vertices of Π and the edges are the pairs of vertices that are connected by an edge of Π. For a generalization to arbitrary dimension, see [31, Section 3.5].

5.3 Trees

What is the smallest number of edges that a connected graph with n vertices can have? The connected graphs on a given vertex set having the smallest possible number of edges are called *trees*. They are especially important in mathematics and theoretical computer science and will be studied in this section.

5.3.1 Definitions and Basic Properties

We begin with the following definition.

Definition 5.3.1. *A tree is a simple graph that is connected and does not contain any cycle. The vertices of degree one of a tree are called leaves.*

Example 5.3.1.

(a) Every path with n vertices is a tree with zero or two leaves, in case $n = 1$ or $n \geq 2$, respectively.

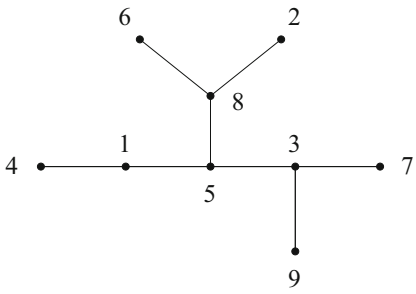


Figure 5.7 A tree on the vertex set [9]

(b) The graph on the vertex set $V = \{v_0, v_1, \ldots, v_n\}$ with edges $\{v_0, v_i\}$ for $i \in [n]$ is a tree that is called a *star* with $n + 1$ vertices. The vertices $v_1, v_2, \ldots, v_n$ are all leaves, while v_0 is a leaf only for $n = 1$.
(c) Figure 5.7 shows a tree on the vertex set [9] with leaves 2, 4, 6, 7, and 9. □

The following statements will help us to prove basic properties of trees.

Lemma 5.3.1. *Every simple graph with at least one edge that does not contain cycles has at least two vertices of degree 1. In particular, every tree with at least two vertices has at least two leaves.*

Proof. Let G be a simple graph with at least one edge that does not contain any cycle. We consider a path $P = (u_0, e_1, u_1, \ldots, u_{\ell-1}, e_\ell, u_\ell)$ in G of maximum possible length ℓ. Since G has at least one edge, $u_0 \neq u_\ell$. We will show that u_0 and u_ℓ are both vertices of degree 1. Suppose, by the way of contradiction, that u_0 is adjacent to a vertex $u \neq u_1$ of G, and let e be the edge connecting u_0 and u. Since G does not contain cycles, u cannot be a vertex of P. As a consequence, $(u, e, u_0, e_1, u_1, \ldots, u_{\ell-1}, e_\ell, u_\ell)$ is a path in G of length larger than ℓ, which is impossible. This contradiction proves that u_0 has degree 1. The same argument applies to u_ℓ. □

REMARK. Lemma 5.3.1 does not hold for graphs with infinitely many vertices. The infinite path with vertex set $\mathbb{Z}$ and edges $\{a, a + 1\}$ for $a \in \mathbb{Z}$ is a counterexample.

Lemma 5.3.2. *Every graph that can be obtained from a connected graph G in one of the following ways:*

(i) *deleting a loop, or an edge of G that is parallel to another edge,*
(ii) *deleting an edge of G that belongs to a cycle,*
(iii) *deleting a vertex v of G of degree 1 and the unique edge of G incident to v,*

is connected as well.

Proof. Let H be a graph that can be obtained from G in such a way. We need to verify that for any two vertices u, v of H, there exists a walk in this graph with endpoints u and v. Since G is connected, we may consider a path P in G with endpoints u and v. In case (iii), P is also a path in H (explain why), while in cases (i) and (ii), one can obtain a walk in H with the same endpoints as P by replacing

(if necessary) the edge e that has been deleted with a parallel edge or with edges that form a cycle together with e. □

A subgraph $G' = (V', E', \varphi')$ of a graph $G = (V, E, \varphi)$ is called *spanning* if $V' = V$. A spanning subgraph of G that is a tree is called a *spanning tree* of G.

EXAMPLE. The graph of Figure 5.1 has 13 spanning trees, with edge sets $\{e_1, e_2, e_4\}$, $\{e_1, e_2, e_6\}$, $\{e_1, e_2, e_7\}$, $\{e_1, e_4, e_5\}$, $\{e_1, e_4, e_6\}$, $\{e_1, e_4, e_7\}$, $\{e_1, e_5, e_6\}$, $\{e_1, e_5, e_7\}$, $\{e_2, e_4, e_5\}$, $\{e_2, e_4, e_6\}$, $\{e_2, e_4, e_7\}$, $\{e_2, e_5, e_6\}$, and $\{e_2, e_5, e_7\}$. □

Proposition 5.3.1. *Every connected graph has a spanning tree.*

Proof. Let G be a connected graph. By the well-ordering principle (Proposition 1.1.1), there exists a connected spanning subgraph T of G with the smallest possible number of edges. This subgraph cannot have loops, parallel edges, or cycles, because in that case, according to Lemma 5.3.2, it would be possible to delete an edge from T (specifically, any loop, or any edge parallel to another, or any edge of a cycle) and obtain a connected spanning subgraph of G with fewer edges than T. We conclude that T is a simple, connected spanning subgraph of G without cycles, meaning a spanning tree of G. □

Proposition 5.3.2. *Let G be a connected graph. Every subgraph of G that is simple and does not contain cycles is a subgraph of a spanning tree of G.*

Proof. Let H be a subgraph of G that is simple and does not contain cycles. By adding isolated vertices to H, if necessary, we may assume that H is spanning. Among all spanning subgraphs of G that contain H as a subgraph and do not have cycles, we may choose one with the largest possible number of edges, say T. This subgraph must be connected, since otherwise it would be possible to add to it any edge of G that connects two vertices from different connected components of T and obtain a spanning subgraph of G with the same properties and one more edge than T. It follows that T is a spanning tree of G. □

Given a graph G with vertex set V and $v \in V$, we will denote by $G \setminus v$ the induced subgraph of G on the vertex set $V \setminus \{v\}$. This is the subgraph of G obtained by deleting v and all edges of G incident to v.

Proposition 5.3.3. *Given a graph G with n vertices, the following conditions are equivalent:*

(i) *The graph G is a tree.*
(ii) *The graph G is connected and has exactly $n - 1$ edges.*
(iii) *The graph G is simple, it does not contain cycles and has exactly $n - 1$ edges.*

In particular, every tree with n vertices has exactly $n - 1$ edges.

Proof. We will prove the implications $(i) \Rightarrow (ii) \Rightarrow (iii) \Rightarrow (i)$. To show that $(i) \Rightarrow (ii)$, we assume that G is a tree and prove by induction on n that it has exactly $n - 1$ edges as follows. Clearly, G has no edges if $n = 1$. Suppose that $n \geq 2$ and

that the statement holds for trees with $n-1$ vertices. According to Lemma 5.3.1, G has at least one leaf, say v. According to Lemma 5.3.2, the graph $G \setminus v$ is also a tree. Since it has $n-1$ vertices, by the induction hypothesis $G \setminus v$ has exactly $n-2$ edges. Therefore, G has exactly $n-1$ edges.

Suppose now that G is connected and has exactly $n-1$ edges. According to Proposition 5.3.1, G has a spanning tree, say T. This tree has n vertices. Therefore, as we have just shown, it has $n-1$ edges, as many as G. It follows that $G = T$. Thus, G is a tree itself and hence is a simple graph without cycles. We have proven that $(ii) \Rightarrow (iii)$.

To prove that $(iii) \Rightarrow (i)$, we assume that G is a simple graph without cycles and that it has exactly $n-1$ edges and follow the proof of $(i) \Rightarrow (ii)$. We apply Lemma 5.3.1 (when $n \geq 2$) to find a vertex of G of degree 1 and show that G is a tree by induction on n. Alternatively, we apply Proposition 5.3.2 to find a tree, say T, which contains G as a spanning subgraph. As in the proof of $(ii) \Rightarrow (iii)$, we observe that T has $n-1$ edges and conclude that $G = T$. The details are left to the reader. □

Corollary 5.3.1.

(a) *Every connected graph with n vertices has at least* $n-1$ *edges.*
(b) *Every simple graph with n vertices that does not contain cycles has at most* $n-1$ *edges.*

Proof. Part (a) follows by combining Proposition 5.3.1 with the fact (see Proposition 5.3.3) that every tree with n vertices has exactly $n-1$ edges. The same argument works for part (b), if we apply Proposition 5.3.2 instead of 5.3.1. □

For a different proof of part (a) of the previous corollary, see Exercise 18.

Example 5.3.2. Given an integer $n \geq 2$, what is the maximum number of edges that a disconnected simple graph with n vertices can have? We will prove that the answer is $(n-1)(n-2)/2$.

Indeed, an example of a disconnected simple graph with n vertices and $(n-1)(n-2)/2$ edges is one obtained from a complete graph K_{n-1} by adding a new isolated vertex. To prove that this number of edges is the maximum possible, we consider any simple graph $G = (V, E)$ with n vertices and more that $(n-1)(n-2)/2$ edges. We will show that G is connected. Let $\bar{G}$ be the simple graph with vertex set V and edges the pairs of vertices that do *not* belong to E. Our assumption for G implies that $\bar{G}$ has fewer than $n-1$ edges. By Corollary 5.3.1, the graph $\bar{G}$ is not connected. Therefore, there exists a partition $V = V_1 \cup V_2$ of V such that no element of V_1 is adjacent to an element of V_2 in $\bar{G}$. Equivalently, every element of V_1 is adjacent to every element of V_2 in G. This implies that any two elements of V can be connected by a path of length at most two in G and hence that G is connected.

5.3.2 The Cayley–Sylvester Theorem

We recall that a simple graph G, such as a tree, with vertex set V may be identified with the pair (V, E), where $E \subseteq \binom{V}{2}$ is the set of edges of G. How many trees with vertex set $[n]$ are there?

Denoting by t_n the number of these trees, we have $t_1 = t_2 = 1$, $t_3 = 3$, $t_4 = 16$, $t_5 = 125$. Figure 5.8 shows the 16 trees with vertex set [4]. The following theorem states that the unexpectedly simple formula $t_n = n^{n-2}$, known as *Cayley's formula*, holds for every positive integer n.

Theorem 5.3.1 (Sylvester, 1857). *The number of trees with vertex set* $[n]$ *is equal to* n^{n-2} *for every* $n \geq 1$.

The proof we will present, given by Prüfer in 1918, is based on a suitable one-to-one correspondence, which we now describe.

For the remainder of this section, we assume that $n \geq 3$. We will denote by $\mathcal{T}_n$ the set of trees with vertex set $[n]$ and by $\mathcal{A}_n$ the set $[n]^{n-2}$ of all sequences $(a_1, a_2, \ldots, a_{n-2})$ with terms in the set $[n]$. We define a map $f : \mathcal{T}_n \to \mathcal{A}_n$ as follows. Let $T \in \mathcal{T}_n$ be a tree. We denote by v_1 the largest leaf of $T_1 = T$ and by a_1 the unique vertex of T_1 that is adjacent to v_1. Deleting the leaf v_1 and the edge $\{v_1, a_1\}$ from T_1 results in a tree T_2 with $n - 1$ vertices. We denote by v_2 the largest leaf of T_2 and by a_2 the unique vertex of T_2 that is adjacent to v_2. We delete the leaf v_2 and the edge $\{v_2, a_2\}$ from T_2 to get a tree T_3 with $n - 2$ vertices and continue

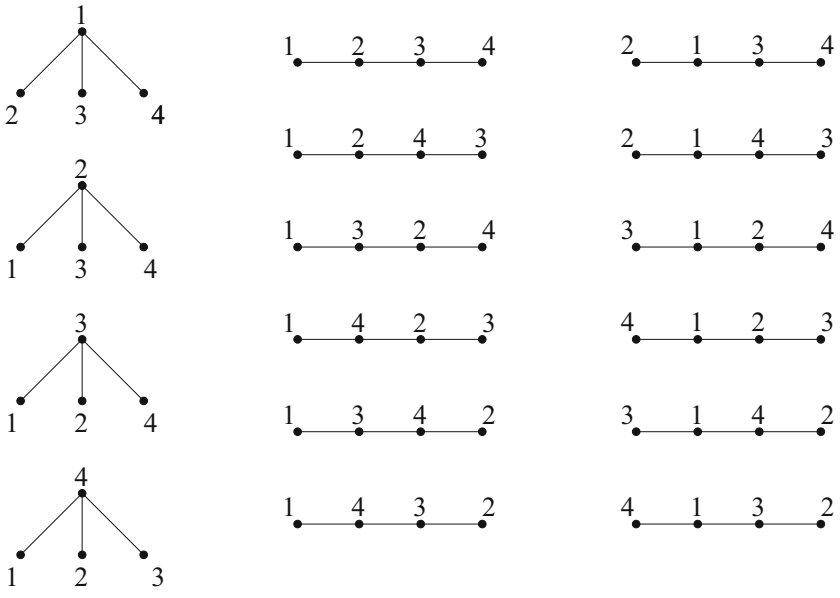


Figure 5.8 The 16 trees with vertices 1, 2, 3, 4

similarly. The process terminates when we reach a tree T_{n-1} with two vertices, the largest of which we denote by v_{n-1}. We set

$$f(T) = (a_1, a_2, \ldots, a_{n-2})$$

and call this sequence the *Prüfer code* of T.

For example, if $n = 9$ and T is the tree of Figure 5.7, then $v_1 = 9$, $v_2 = 7$, $v_3 = 6$, $v_4 = 4$, $v_5 = 3$, $v_6 = 2$, $v_7 = 8$, $v_8 = 5$, and $f(T) = (3, 3, 8, 1, 5, 8, 5)$. The following lemma describes some crucial properties of the map f. These properties will allow us to show that f is bijective.

Lemma 5.3.3. *Setting $v_n = 1$, the following hold:*

(a) $\{v_1, v_2, \ldots, v_n\} = [n]$.
(b) *For every $i \in [n-1]$, there exists a unique $j \in [n]$ with $i < j$ such that $\{v_i, v_j\}$ is an edge of T.*
(c) *For every $j \in [n]$, the following are equivalent:*
   (i) *v_j is a leaf of T.*
   (ii) *$v_j \notin \{a_1, a_2, \ldots, a_{n-2}\}$.*
   (iii) *Either $j \in [n-1]$ and there is no index $i < j$ such that $\{v_i, v_j\}$ is an edge of T, or $j = n$ and there exists a unique index $i < j$ such that $\{v_i, v_j\}$ is an edge of T.*

Proof. To verify part (a) we observe that, by definition, $v_1, v_2, \ldots, v_{n-1}$ are distinct vertices of T. Moreover, since the trees $T_1, T_2, \ldots, T_{n-1}$ have at least two leaves each (see Lemma 5.3.1), their largest leaf is greater than 1. Therefore, $v_1, v_2, \ldots, v_{n-1} \neq 1$ and hence $\{v_1, v_2, \ldots, v_n\} = [n]$.

To prove part (b), we set $a_{n-1} = 1$. Then, $\{v_i, a_i\}$ is an edge of T and $a_i = v_j$ for some $j \in [n]$ by part (a). Since a_i is a vertex of T_{i+1} that has vertices $v_{i+1}, v_{i+2}, \ldots, v_n$, we must have $i < j$. Since (by Proposition 5.3.3) T has exactly $n-1$ edges, there is no other edge $\{v_i, v_k\}$ of T with $i < k$.

To prove part (c), we will show that $(i) \Rightarrow (ii) \Rightarrow (iii) \Rightarrow (i)$. The first implication is valid because each a_i for $i \in [n-2]$ is adjacent to the leaf v_i of T_i and hence cannot be a leaf of T_i itself. Therefore, a_i cannot be a leaf of T either. The second implication is valid because if there exists $i < j$ such that $\{v_i, v_j\}$ is an edge of T and $i \leq n-2$, then we must have $v_j = a_i$ by part (b), whence $v_j \in \{a_1, a_2, \ldots, a_{n-2}\}$. For the third implication, assume that (iii) holds. Then, part (b) implies that v_j has degree 1 in T or, equivalently, that v_j is a leaf of T. □

We leave to the reader to verify the following proposition for $n = 4$ by computing the Prüfer codes of the trees of Figure 5.8. Theorem 5.3.1 follows directly from this statement, since $\#\mathcal{A}_n = n^{n-2}$.

Proposition 5.3.4. *The map $f : \mathcal{T}_n \to \mathcal{A}_n$ is a bijection for every $n \geq 3$.*

Proof. Lemma 5.3.3 implies that the leaf v_1 of the tree T is the largest element of $[n]$ that does not appear in the code $f(T)$. Thus, we may construct an inverse map

$g : \mathcal{A}_n \to \mathcal{T}_n$ as follows. Given $\sigma = (a_1, a_2, \ldots, a_{n-2}) \in \mathcal{A}_n$, we set $a_{n-1} = 1$ and define as $g(\sigma)$ the graph, say T, with vertices $1, 2, \ldots, n$ and edges $\{v_1, a_1\}$, $\{v_2, a_2\}, \ldots, \{v_{n-1}, a_{n-1}\}$, where

$$\begin{aligned} v_1 &:= \max([n] \setminus \{a_1, a_2, \ldots, a_{n-2}\}) \\ v_2 &:= \max([n] \setminus \{v_1, a_2, \ldots, a_{n-2}\}) \\ &\vdots \\ v_{n-1} &:= \max([n] \setminus \{v_1, v_2, \ldots, v_{n-2}\}). \end{aligned}$$

In other words, for $k \in [n-1]$, v_k is the largest element of the set $[n] \setminus \{v_1, \ldots, v_{k-1}\}$ that does not appear in $(a_k, a_{k+1}, \ldots, a_{n-2})$. For example, for $n = 9$ and $\sigma = (3, 3, 8, 1, 5, 8, 5)$ we have $v_1 = 9$, $v_2 = 7$, $v_3 = 6$, $v_4 = 4$, $v_5 = 3$, $v_6 = 2$, $v_7 = 8$, $v_8 = 5$, and T is the tree of Figure 5.7.

We will show that g is well defined, meaning that T is indeed a tree with vertex set $[n]$. Since T has $n - 1$ edges by construction, according to Proposition 5.3.3, it suffices to show that T is connected. We set $v_n = 1$ and claim that (a) $[n] = \{v_1, v_2, \ldots, v_n\}$; and (b) every vertex v_i with $i \in [n-1]$ is adjacent in T with some vertex v_j with $i < j$. Indeed, (a) holds because vertices $v_1, v_2, \ldots, v_{n-1}$ are pairwise distinct and greater than 1 by their definition. Furthermore, for every $i \in [n-1]$, there is the edge $\{v_i, a_i\}$ of T and $a_i \neq v_k$ for $k \leq i$. Therefore, $a_i = v_j$ for some $j > i$. This proves (b). It follows from (a) and (b) that every vertex of T can be connected to $v_n = 1$ by a path in T. This proves that T is connected. We leave it to the reader to verify that g is the inverse map of f. □

Two more short proofs of Theorem 5.3.1 are described in Exercises 26 and 27. As an application of the theorem, we now compute asymptotically the average number of leaves of trees with vertex set $[n]$, when $n \to \infty$.

Proposition 5.3.5. *Let $\varphi(T)$ be the number of leaves of a tree T, and let*

$$a(n) = \frac{1}{n^{n-2}} \sum_{T \in \mathcal{T}_n} \varphi(T)$$

be the average number of leaves of the trees in $\mathcal{T}_n$. Then,

$$\lim_{n \to \infty} \frac{a(n)}{n} = \frac{1}{e}.$$

Equivalently, $a(n) \sim n/e$ when $n \to \infty$.

Proof. We will prove that

$$\sum_{T \in \mathcal{T}_n} \varphi(T) = n(n-1)^{n-2}$$

for every $n \geq 1$. Let $b(n)$ be the sum on the left-hand side of this equality, and let R_n be the set of pairs (x, T), where $T \in \mathcal{T}_n$ and x is a leaf of T. We have $\#R_n = b(n)$, because for every tree $T \in \mathcal{T}_n$, there are exactly $\varphi(T)$ integers $x \in [n]$ such that $(x, T) \in R_n$. Moreover, assuming that $n \geq 2$, for every $x \in [n]$ there are exactly $(n-1)^{n-2}$ trees $T \in \mathcal{T}_n$ such that $(x, T) \in R_n$. This happens because, by Theorem 5.3.1, there are $(n-1)^{n-3}$ trees with vertex set $[n] \setminus \{x\}$ and for every such tree there are $n-1$ ways one can choose an element of $[n] \setminus \{x\}$ to be the unique vertex adjacent to x, so as to obtain a tree with vertex set $[n]$ of which x is a leaf. From these facts we conclude that $b(n) = \#R_n = n(n-1)^{n-2}$ and that $a(n) = b(n)/n^{n-2} = (n-1)^{n-2}/n^{n-3}$. As a result,

$$\lim_{n\to\infty} \frac{a(n)}{n} = \lim_{n\to\infty} \frac{(n-1)^{n-2}}{n^{n-2}} = \lim_{n\to\infty} \frac{1}{\left(1 + \frac{1}{n-1}\right)^{n-2}} = 1/e.$$

□

5.3.3 *Minimum Spanning Trees and Kruskal's Algorithm*

We consider a simple, connected graph $G = (V, E)$ and a function $\mathrm{w} : E \to [0, \infty)$, which assigns to every edge $e \in E$ a nonnegative real number, called the *weight* (or *cost*) of e. We denote by

$$\mathrm{w}(T) = \sum_{e \in F} \mathrm{w}(e) \tag{5.1}$$

the sum of the weights of the edges of a spanning tree $T = (V, F)$ of G and call this sum the *weight* of T. How can one find a spanning tree $T_\circ$ of G with the minimum possible weight, meaning that $\mathrm{w}(T_\circ) \leq \mathrm{w}(T)$ for every spanning tree T of G? We call such a spanning tree *minimum* (with respect to the function w).

Imagine that the vertices of G are cities, that the edges are roads that can be constructed to connect these cities, and that the weight of an edge is the cost of constructing the corresponding road. Then, we are considering the problem to connect the cities by constructing roads with the minimum possible cost.

We will show that one of the most naive strategies that one can think has the desirable outcome: we add the edges of G successively, in the order of increasing weight, ignoring those edges that form cycles with previously added edges, and reach a minimum spanning tree of G. To be more precise, let $(e_1, e_2, \ldots, e_m)$ be any permutation of E such that $\mathrm{w}(e_1) \leq \mathrm{w}(e_2) \leq \cdots \leq \mathrm{w}(e_m)$. We consider the spanning subgraph $T_\circ = (V, F_\circ)$ of G with set of edges defined recursively by setting (a) $e_1 \in F$; and (b) $e_i \in F_\circ$, for $1 < i \leq m$, if the graph $(V, \{e_1, e_2, \ldots, e_{i-1}\} \cap F_\circ)$ has no path with endpoints those of e_i. We then claim that $T_\circ$ is a minimum spanning tree of G. This process, known as *Kruskal's*

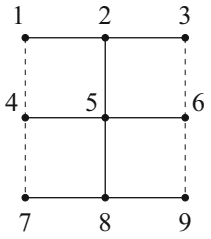


Figure 5.9 A minimum spanning tree

algorithm, is much more effective than computing the weights of all spanning trees of G to find the minimum. As made evident by Theorem 5.3.1, the latter method may be very time-consuming or even practically impossible.

Example 5.3.3. Let $V = [9]$, $m = 12$ and $e_1 = \{1, 2\}$, $e_2 = \{2, 3\}$, $e_3 = \{2, 5\}$, $e_4 = \{4, 5\}$, $e_5 = \{1, 4\}$, $e_6 = \{5, 6\}$, $e_7 = \{3, 6\}$, $e_8 = \{5, 8\}$, $e_9 = \{7, 8\}$, $e_{10} = \{4, 7\}$, $e_{11} = \{8, 9\}$, $e_{12} = \{6, 9\}$ be a permutation of E such that $\mathrm{w}(e_1) \leq \mathrm{w}(e_2) \leq \cdots \leq \mathrm{w}(e_{12})$. Then, the spanning subgraph produced by the algorithm is the spanning tree shown in Figure 5.9 (where dotted lines stand for the edges of G that get ignored by the algorithm and thus do not belong to the tree produced). □

Theorem 5.3.2. *Every spanning subgraph of G obtained by applying Kruskal's algorithm is a minimum spanning tree of G.*

The proof uses the following lemma.

Lemma 5.3.4. *Let $T = (V, F)$ be a spanning tree of a simple graph $G = (V, E)$, and let $e \in E \setminus F$.*

(a) *There exists an edge $f \in F$ such that the graph obtained from T by including e in F and deleting f from F is again a spanning tree of G.*
(b) *Let $T' = (V, F')$ be a spanning tree of G such that $e \in F'$. Then, there exists an edge $f \in F \setminus F'$ such that the graph obtained from T by including e in F and deleting f from F is again a spanning tree of G.*

Proof. Let n be the number of vertices of G. The graph $(V, F \cup \{e\})$ cannot be a tree, by Proposition 5.3.3. Since this graph is connected, it must contain a cycle C. According to Lemma 5.3.2, deleting any edge f of C other than e yields a connected spanning subgraph of G with $n - 1$ edges. By Proposition 5.3.3, this subgraph is a spanning tree of G that has the desired properties. This proves part (a). To prove part (b), we follow the same reasoning and observe that not all edges of C are edges of T'. Therefore, we may choose f so that $f \notin F'$. □

EXAMPLE. Including the edge $\{2, 3\}$ in the tree of Figure 5.7 forms a 4-cycle. Deleting any of the edges $\{2, 8\}$, $\{3, 5\}$, $\{5, 8\}$ of this cycle yields another tree with the same vertices.

Proof of Theorem 5.3.2. By construction, $T_\circ = (V, F_\circ)$ is obtained from the connected graph G by successively deleting edges that belong to cycles. Therefore,

$T_\circ$ is connected by Lemma 5.3.2. Since it does not contain cycles by construction, $T_\circ$ must be a spanning tree of G.

Given any spanning tree $T = (V, F)$ of G, we will prove that $\mathrm{w}(T_\circ) \leq \mathrm{w}(T)$ by induction on the cardinality of $F_\circ \setminus F$. If this cardinality equals zero, then $T_\circ = T$ and the result is trivial. Otherwise, we consider the smallest index $i \in [m]$ such that $e_i \in F_\circ \setminus F$. By Lemma 5.3.4, there exists an edge $e_j \in F \setminus F_\circ$ such that $T' = (V, (F \setminus \{e_j\}) \cup \{e_i\})$ is a spanning tree of G. We claim that $i < j$. Let us assume that, on the contrary, $j < i$. Then, because of our choice of i, we have $\{e_1, e_2, \ldots, e_j\} \cap F_\circ \subseteq F$. From this and the fact that T is a tree, we conclude that no cycle of G has set of edges contained in $\{e_1, e_2, \ldots, e_j\} \cap F_\circ$. By definition of $T_\circ$, it follows that $e_j \in F_\circ$, contrary to our choice of e_j. This contradiction proves our claim that $i < j$. As a result, we have $\mathrm{w}(e_i) \leq \mathrm{w}(e_j)$ and hence $\mathrm{w}(T') \leq \mathrm{w}(T)$. By the induction hypothesis, we also have $\mathrm{w}(T_\circ) \leq \mathrm{w}(T')$. Thus, $\mathrm{w}(T_\circ) \leq \mathrm{w}(T)$ and the proof follows. □

5.3.4 Rooted Trees

This section briefly reviews some of the multitude of interesting variations of the concept of a tree.

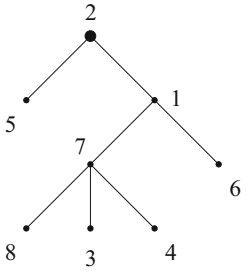
Definition 5.3.2. *A rooted tree is any pair (T, v), where T is a tree and v is a distinguished vertex, called the root.*

Figure 5.10 shows a tree with vertex set [8], rooted at 2. The root of a tree is usually presented with a dot larger that those presenting other vertices.

Let T be a rooted tree with vertex set V and root v. Then, for every $u \in V$, there exists a unique path in T with initial vertex u and terminal vertex v (explain why). Moreover, as explained in the solution of Exercise 18, for every edge $\{a, b\}$ of T, exactly one of the following conditions holds:

- The unique path in T with initial vertex b and terminal vertex v passes through a.
- The unique path in T with initial vertex a and terminal vertex v passes through b.

Figure 5.10 A rooted tree



We say that b is a *successor* (or *child*) of a, and that a is a *predecessor* (or *parent*) of b, in the former case and use the reverse terminology in the latter. A vertex u of T is called a *leaf* if it has no successors; otherwise, u is said to be an *internal vertex*.

REMARK. The notion of a leaf in a rooted tree differs from that of Section 5.3.1 in that the root is considered to be a leaf only if it has degree zero; in other words, only if the rooted tree has a single vertex.

Definition 5.3.3. *A rooted tree is called binary if every internal vertex has exactly two successors.*

EXAMPLE. The rooted tree of Figure 5.10 has five leaves, namely 3, 4, 5, 6, and 8, and three internal vertices, namely 1, 2, and 7. Vertex 7 has successors 3, 4, 8 and the predecessor 1. The tree that can be obtained from that of Figure 5.10 by deleting one of the vertices 3, 4, or 8 and the edge incident to it is binary, with four leaves and three internal vertices.

Proposition 5.3.6. *Every binary rooted tree with n leaves has $n - 1$ internal vertices.*

Proof. We proceed by induction on n. The result is trivial for $n = 1$. Assuming that $n \geq 2$, we consider a binary rooted tree T with n leaves. Let u_1, u_2 be any two leaves of T having a common parent u. Deleting u_1, u_2 and the edges $\{u, u_1\}$ and $\{u, u_2\}$ from T results in a binary rooted tree T' with the same root as T. We observe that u is an internal vertex of T and a leaf of T' and hence that T' has one leaf and one internal vertex less than T. In particular, T' has $n - 1$ leaves. By the induction hypothesis, T' has $n - 2$ internal vertices. Therefore, T has $n - 1$ internal vertices and the proof follows. □

Example 5.3.4. One day Ms Cornelia bought a chicken. The chicken made two eggs and was then cooked for dinner. As is well known, every egg contains either a chicken or a rooster. Every rooster that came out of an egg was immediately cooked for dinner, while every chicken was cooked for dinner after making two eggs. The process continued until only roosters remained and were all cooked for dinner. Given that a total of 2001 roosters were cooked for dinner, how many chickens were cooked?

This process can be described by a binary rooted tree T, the internal vertices of which are the chickens and the leaves of which are the roosters. Since T has 2001 leaves, according to Proposition 5.3.6, it has 2000 internal vertices. Therefore, the number of chickens that were cooked is equal to 2000. □

Another important family of trees is that of plane (or ordered) rooted trees.

Definition 5.3.4. *A plane rooted tree is a rooted tree T, endowed with a permutation of the set of successors of u for every internal vertex u of T.*

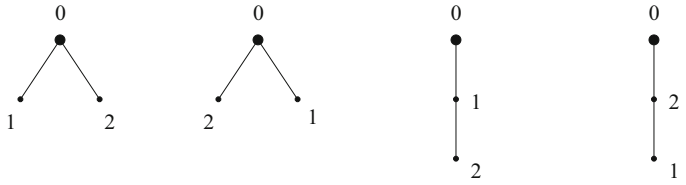


Figure 5.11 Plane trees with vertices 0, 1, 2 and root 0

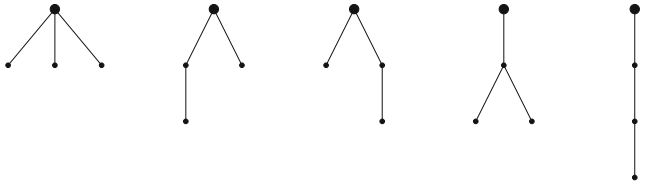


Figure 5.12 Unlabeled plane rooted trees with four vertices

Figure 5.11 shows all four plane rooted trees with vertex set $\{0, 1, 2\}$ and root 0. The successors of an internal vertex u of T appear in the drawing from left to right, according to the permutation specified by T and u.

Two rooted trees $T = (V, E)$ and $T' = (V', E')$ with roots v and v', respectively, are said to be isomorphic (as rooted trees) if there exists a graph isomorphism $f : V \to V'$ such that $f(v) = v'$. For example, the first and third trees from the left shown in Figure 5.11 are isomorphic graphs that are not isomorphic as rooted trees. Similarly, two plane rooted trees $T = (V, E)$ and $T' = (V', E')$ with roots v and v', respectively, are isomorphic (as plane rooted trees) if there exists a graph isomorphism $f : V \to V'$ with $f(v) = v'$ that preserves the order of the successors of all internal vertices of T. For example, the first two (and the last two) trees shown in Figure 5.11 are isomorphic plane rooted trees, where the isomorphism fixes vertex 0 and exchanges vertices 1 and 2.

Clearly, isomorphism is an equivalence relation on the set of rooted (respectively, plane rooted) trees. The equivalence classes are called *unlabeled* rooted (respectively, plane rooted) trees. The equivalence classes that consist of binary trees are the unlabeled binary rooted (respectively, plane rooted) trees. For example, there are two unlabeled plane rooted trees with three vertices (as shown in Figure 5.11), five unlabeled plane rooted trees with four vertices (see Figure 5.12), and five unlabeled binary plane rooted trees with four leaves (see Figure 5.13).

The number of unlabeled plane rooted trees with $n + 1$ vertices, as well as that of unlabeled binary plane rooted trees with $n + 1$ leaves (equivalently, with n internal vertices), are equal to the Catalan number $\frac{1}{n+1}\binom{2n}{n}$, which we studied in Section 2.3.8, for every positive integer n (see Exercise 31).

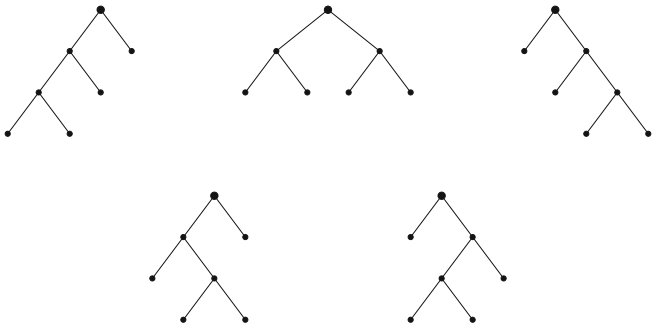


Figure 5.13 Unlabeled binary plane rooted trees with four leaves

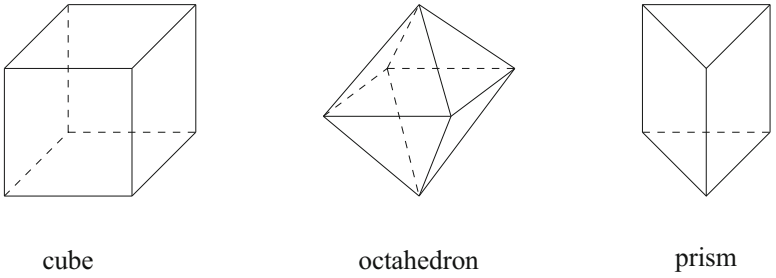


Figure 5.14 Three polyhedra

5.4 Colorings and Matchings

This section introduces some important notions for graphs: those of vertex coloring, chromatic number, and matching. These notions will lead us to the study of bipartite graphs and to the Marriage Theorem, a useful result about the existence of perfect matchings in bipartite graphs.

5.4.1 The Chromatic Number

Suppose that we want to color every one of the six facets of a three-dimensional cube with one of some available colors, so that any two facets that share a common edge have different color. What is the minimum number of colors that are required? What if the cube is replaced by an octahedron, a triangular prism, or another three-dimensional polyhedron (see Figure 5.14)?

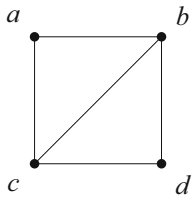


Figure 5.15 A simple graph with four vertices and five edges

These questions can be translated in the language of graph theory by means of the following definition.

Definition 5.4.1. *Let $G = (V, E)$ be a simple graph, and let $q \in \mathbb{Z}_{>0}$.*

(a) A map $\kappa : V \to [q]$ is said to be a coloring of G with colors $1, 2, \ldots, q$. A coloring κ is called proper if $\kappa(u) \neq \kappa(v)$ for every edge $\{u, v\} \in E$.

(b) The smallest positive integer q for which a proper coloring $\kappa : V \to [q]$ of G exists is called chromatic number of G and is denoted by $\chi(G)$.

Thus, a coloring of G is proper if it assigns different colors to the endpoints of every edge. The definition of the chromatic number can be extended to all (not necessarily simple) graphs. Since the definition would not depend on the existence of edges of G that are parallel to another edge, we chose to restrict Definition 5.4.1 to simple graphs.

Example 5.4.1. Let G be the simple graph with set of vertices $V = \{a, b, c, d\}$ and edges $\{a, b\}$, $\{a, c\}$, $\{b, c\}$, $\{b, d\}$, $\{c, d\}$ (see Figure 5.15). A map $\kappa : V \to [q]$ is a proper coloring of G if and only if $\kappa(a) \neq \kappa(b)$, $\kappa(a) \neq \kappa(c)$, $\kappa(b) \neq \kappa(c)$, $\kappa(b) \neq \kappa(d)$, and $\kappa(c) \neq \kappa(d)$.

This graph has chromatic number $\chi(G) = 3$, since on the one hand three colors are required to properly color vertices a, b, c and on the other, there exists a proper coloring κ of G with three colors, for instance, that with $\kappa(a) = \kappa(d) = 1$, $\kappa(b) = 2$, and $\kappa(c) = 3$. □

Example 5.4.2.

(a) Let V be an n-element set. If G is the complete graph with vertex set V, then a coloring $\kappa : V \to [q]$ of G is proper if and only if the map κ is injective. By Proposition 4.1.1 (a), we have $\chi(G) = n$. If G is the graph with vertex set V without edges, than every coloring $\kappa : V \to [q]$ is proper and hence $\chi(G) = 1$.

(b) Let Π be a three-dimensional convex polyhedron. Let $G^\circ(\Pi)$ be the simple graph with vertices the facets of Π and edges the pairs of facets that share a common edge. Then, the minimum number of colors required to color the facets of Π as in the beginning of this section is equal to the chromatic number of $G^\circ(\Pi)$.

(c) Let $H_n = (V, E)$ be the simple graph with vertices the n^2 squares of an $n \times n$ chessboard, in which two such squares are adjacent if they lie in the same row or column of the chessboard. Then, a proper coloring $\kappa : V \to [q]$ of H_n is an

assignment of one of the colors $1, 2, \ldots, q$ to every square of the chessboard, so that squares that lie in the same row or column of the chessboard are assigned different colors. Such assignments with $q = n$ (the smallest possible number of colors) are called *latin squares* of order n. □

The following statements provide bounds for the chromatic number $\chi(G)$ that can be useful for its computation.

Proposition 5.4.1. *If H is a subgraph of a simple graph G, then $\chi(H) \leq \chi(G)$.*

Proof. Let V be the set of vertices of G, and let $\kappa : V \to [q]$ be a proper coloring of G with $q = \chi(G)$ colors. Then, the restriction of κ to the set of vertices of H is a proper coloring of H with q colors and hence $\chi(H) \leq q = \chi(G)$. □

Proposition 5.4.2. *Let G be a simple graph with n vertices.*

(a) *Let $m \in \mathbb{N}$. If there are m pairwise adjacent vertices of G, then $\chi(G) \geq m$.*
(b) *Let $d \in \mathbb{N}$. If $\deg_G(v) \leq d$ for every vertex v of G, then $\chi(G) \leq d + 1$. In particular, $\chi(G) \leq n$.*

Proof.

(a) By assumption, G has a subgraph H that is a complete graph with m vertices. Thus, by Example 5.4.2 (a) and Proposition 5.4.1, $m = \chi(H) \leq \chi(G)$.
(b) We proceed by induction on the number n of vertices of G. The result is trivial for $n = 1$. Assuming that $n \geq 2$, we choose an arbitrary vertex $v \in V$ and consider the graph $G \setminus v$ obtained from G by deleting v and all edges incident to v. Clearly, the degrees of the vertices of $G \setminus v$ do not exceed d. Hence, by the induction hypothesis, there exists a proper coloring κ of $G \setminus v$ with colors $1, 2, \ldots, d + 1$. We observe that there are at most d vertices of G that are adjacent to v. Thus, we may extend κ to a proper coloring of G with colors $1, 2, \ldots, d + 1$ by setting $\kappa(v) = i$ for any $i \in [d + 1]$ that is not the color of any of these vertices. As a result, $\chi(G) \leq d + 1$.

□

EXAMPLE. Proposition 5.4.2 (b) implies that $\chi(G) \leq 3$ for every path or cycle G. As will be noted in the following section, the case of equality holds exactly when G is a cycle of odd length.

Example 5.4.3.

(a) Let $G = G^\circ(\Pi)$ be the graph of Example 5.4.2 (b). If Π is the cube, then G has three pairwise adjacent vertices (three facets of the cube sharing a common vertex) and hence $\chi(G) \geq 3$. Moreover, coloring opposite facets of the cube with the same color gives a proper coloring of G with three colors. This shows that $\chi(G) \leq 3$, whence $\chi(G) = 3$. We leave it to the reader to verify that $\chi(G) = 2$ if Π is the octahedron, and that $\chi(G) = 4$ if Π is the triangular prism.

(b) A similar argument shows that $\chi(H_n) = n$, if H_n is the graph of Example 5.4.2 (c). Indeed, the n squares lying in any row (or column) of the chessboard are pairwise adjacent vertices of H_n. This fact implies that $\chi(H_n) \geq n$. We leave it to the reader to verify that there exist proper colorings (latin squares) of H_n with n colors for every positive integer n (such a coloring appears for $n = 8$ in Figure B.2).

5.4.2 *Bipartite Graphs*

A simple graph $G = (V, E)$ has chromatic number equal to 1 if and only if $E = \varnothing$. This section characterizes simple graphs with chromatic number equal to 2.

Definition 5.4.2. *A graph G is called bipartite if there exists a partition $\{A, B\}$ of the set of vertices of G such that every edge of G connects an element of A to an element of B.*

EXAMPLE.

(a) The complete bipartite graphs described in Example 5.1.2 (c) are bipartite, in the sense of Definition 5.4.2. The same is true for the graph shown in Figure 5.16.
(b) Every grid graph (see Example 5.1.2 (d)) with at least two vertices is bipartite. Indeed, let G be the grid graph with vertex set $V = [m] \times [n]$, where $mn \geq 2$, and let A, B be the subsets of V consisting of those vertices (i, j) for which $i + j$ is even or odd, respectively. Then, $\{A, B\}$ is a partition of V and every edge of G connects an element of A to an element of B.
(c) Similarly, the graph $\square_n$ of the n-dimensional cube (see Example 5.1.2 (f)) is bipartite for every positive integer n. The corresponding partition has blocks the sets of vertices $(\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_n) \in \{0, 1\}^n$ for which the sum $\varepsilon_1 + \varepsilon_2 + \cdots + \varepsilon_n$ is even or odd, respectively. □

The next proposition may be restated as follows: the chromatic number of a simple graph G is equal to 2 if and only if G is bipartite and has at least one edge.

Proposition 5.4.3. *A simple graph G with at least two vertices is bipartite if and only if $\chi(G) \leq 2$.*

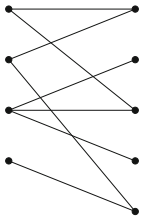


Figure 5.16 A bipartite graph with 9 vertices

Proof. Let G be a bipartite graph, and let $\{A, B\}$ be a partition of its set of vertices V, as in Definition 5.4.2. Then, the map $\kappa : V \to \{1, 2\}$ with

$$\kappa(v) = \begin{cases} 1, & \text{if } v \in A \\ 2, & \text{if } v \in B \end{cases}$$

is a proper coloring of G with two colors. It follows that $\chi(G) \leq 2$. Conversely, let us assume that $\chi(G) \leq 2$. We need to prove that G is bipartite. Indeed, if $\chi(G) = 1$, then G has no edges and the result is trivial. Otherwise, we have $\chi(G) = 2$. We consider a proper coloring $\kappa : V \to \{1, 2\}$ of G and set $A = \kappa^{-1}(1)$ and $B = \kappa^{-1}(2)$. Since $\chi(G) = 2$, the sets A, B are nonempty and form the blocks of a partition of V. Moreover, for every edge $e = \{u, v\}$ of G we have $\kappa(u) \neq \kappa(v)$ and hence one of the vertices u, v belongs to A and the other in B. This shows that G is bipartite. □

Example 5.4.4. Given a cycle G of length n, the following are equivalent: (i) G is a bipartite graph, (ii) $\chi(G) = 2$; and (iii) n is an even number. Indeed, let $V = \{v_1, v_2, \ldots, v_n\}$ be the set of vertices of G and $\{v_{i-1}, v_i\}$ be its edges for $i \in [n]$, where $v_0 = v_n$. A proper coloring of G with two colors must assign one color to all vertices v_i for which i is odd and the other to all vertices v_j for which j is even. Since $v_0 = v_n$, this is possible only when n is even. This shows that $(ii) \Leftrightarrow (iii)$. The equivalence $(i) \Leftrightarrow (ii)$ follows from Proposition 5.4.3. In view of Proposition 5.4.2 (b), we conclude that cycles of odd length have chromatic number equal to 3. □

The following proposition provides another useful necessary and sufficient condition for a graph to be bipartite. A geometric application is given in Exercise 38.

Proposition 5.4.4. *A simple graph G with at least two vertices is bipartite if and only if G does not have cycles of odd length.*

Proof. We assume first that G has a cycle H of odd length. Combining Proposition 5.4.1 with Example 5.4.4 gives that $\chi(G) \geq \chi(H) = 3$. Therefore, by Proposition 5.4.3, G is not bipartite.

Conversely, suppose that G does not have cycles of odd length. We will show that G is bipartite. Since the connected components of G do not have cycles of odd length either and since it suffices to prove that the components with at least two vertices are bipartite, we may assume that G is connected.

Let V be the set of vertices of G and let $v \in V$. Given any vertex $u \in V$, there exist walks in G with initial vertex u and terminal vertex v. We claim that either all such walks have odd length or all have even length. Indeed, suppose on the contrary that there exist walks P and Q in G with initial vertex u and terminal vertex v, one of which has odd length and the other even. Deleting cycles (necessarily of even length) and any successive repeated edges from these walks (if that is necessary), we may assume that P and Q are paths. By induction on the sum of the lengths of P and Q, one shows that the edges of the two paths form paths and cycles in

G the sum of the lengths of which is an odd number (we leave the details of this argument to the reader). This shows that there exists a cycle of odd length in G, contrary to our assumption. This contradiction proves our claim. We now denote by A (respectively, by B) the set of vertices u for which every walk in G with initial vertex u and terminal vertex v has odd (respectively, even) length. Because of our claim, $\{A, B\}$ is a partition of V and clearly, every edge of G connects an element of A with an element of B. □

Corollary 5.4.1. *Every tree with at least two vertices is bipartite.*

Proof. This statement is a direct consequence of Proposition 5.4.4. □

5.4.3 The Marriage Theorem

Let $G = (V, E)$ be a simple graph.

Definition 5.4.3. *A subset F of the set of edges of G is called a matching if every vertex of G is an endpoint of at most one edge that belongs to F. The matching F is called perfect if every vertex of G is an endpoint of exactly one edge that belongs to F.*

EXAMPLE.

(a) Figure 5.17 shows a perfect matching of the 5×4 grid graph of Figure 5.4.
(b) Every path with an even number of vertices has a unique perfect matching. Every cycle of even length has exactly two perfect matchings (the 8-cycle and its two perfect matchings are shown in Figure 5.18). Every cycle of odd length $2n + 1$ has exactly $2n + 1$ matchings with n elements and no perfect matching.

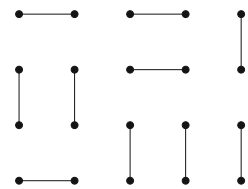


Figure 5.17 A perfect matching of the graph of Figure 5.4

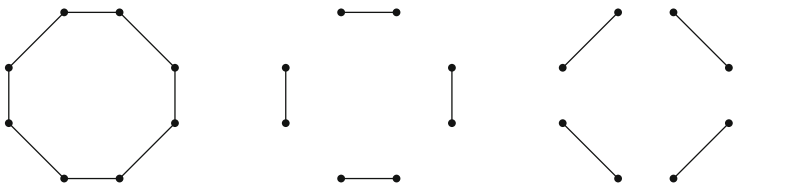


Figure 5.18 The two perfect matchings of the cycle of length 8

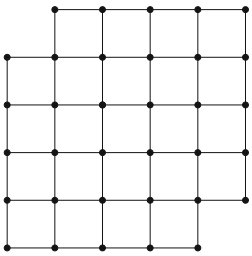


Figure 5.19 A bipartite graph with 34 vertices

(c) The graph G shown in Figure 5.19 has 34 vertices and 56 edges. We claim that this graph has no perfect matching. Indeed, being a subgraph of the 6×6 grid graph, G is bipartite and a partition of the vertex set, as in Definition 5.4.2, has blocks with 16 and 18 elements (explain why). Clearly, a matching of a graph with these properties can have at most 16 elements and hence cannot be perfect. □

Clearly, every graph that has a perfect matching has an even number of vertices. Which graphs have a perfect matching? We will answer this question for bipartite graphs. Let $G = (V, E)$ be a simple bipartite graph, and let $\{A, B\}$ be a partition of V as in Definition 5.4.2. We will first consider the following question: Under what conditions does G have a matching F such that every element of A is an endpoint of an edge which belongs to F? An obvious necessary condition is that $\#A \leq \#B$. More generally, for every subset S of A, say of cardinality r, there must be at least r elements of B each of which is adjacent to at least one element of S in G. Indeed, if such a matching F of G exists, then there are r edges in F that are incident to the elements of S and the endpoints of these edges that belong to B are pairwise distinct and have the required property. For $S \subseteq V$, let us denote by $N_G(S)$ the set of vertices of G that are adjacent to at least one element of S. We have argued that we must have

$$\#N_G(S) \geq \#S$$

for every $S \subseteq A$. This necessary condition is also sufficient, according to the following theorem.

Theorem 5.4.1. *Let G be a simple bipartite graph, and let $\{A, B\}$ be a partition of its vertex set such that every edge of G connects an element of A to an element of B. Then, the following conditions are equivalent:*

(i) *There exists a matching F of G such that every element of A is an endpoint of an edge that belongs to F.*
(ii) *We have*

$$\#N_G(S) \geq \#S$$

for every $S \subseteq A$.

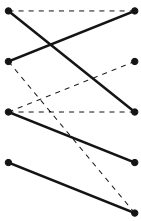


Figure 5.20 A matching of the graph of Figure 5.16

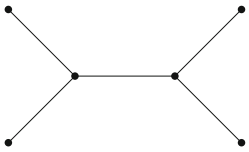


Figure 5.21 A tree without a perfect matching

EXAMPLE.

(a) The bipartite graph shown in Figure 5.16 satisfies condition (ii) of Theorem 5.4.1, where A is the set of four vertices on the left side. Figure 5.20 shows a matching of this graph with four edges, the existence of which is predicted by Theorem 5.4.1.

(b) The tree shown in Figure 5.21 does not satisfy condition (ii) of Theorem 5.4.1 and does not have any perfect matching. Indeed, the set A (as well as the set B) contains two leaves which have a common unique adjacent vertex. This shows that there is a two-element set $S \subseteq A$ such that $\#N_G(S) = 1$.

Proof of Theorem 5.4.1. We have already commented on the validity of the implication $(i) \Rightarrow (ii)$. We need to show that $(ii) \Rightarrow (i)$. We proceed by induction on the cardinality of A. The result is trivial if A is a singleton. We thus assume that $\#A \geq 2$ and distinguish two cases.

(a) We have $\#N_G(S) > \#S$ for every nonempty proper subset S of A. We choose an arbitrary edge $\{a, b\}$ of G, where $a \in A$ and $b \in B$, and denote by H the induced subgraph of G on the vertex set $V \setminus \{a, b\}$. We observe that H is a simple bipartite graph, where the corresponding partition of $V \setminus \{a, b\}$ has blocks $A \setminus \{a\}$ and $B \setminus \{b\}$. By our assumption, $\#N_G(S) \geq r + 1$ for every nonempty r-element subset S of $A \setminus \{a\}$. Since $N_H(S) = N_G(S) \setminus \{b\}$, this inequality shows that $\#N_H(S) \geq r$. Hence, condition (ii) of the theorem holds for H and $A \setminus \{a\}$. Our induction hypothesis implies that there exists a matching F_0 of H such that every element of $A \setminus \{a\}$ is an endpoint of an edge in F_0. We conclude that $F_0 \cup \{\{a, b\}\}$ is a matching of G with the desired property.

(b) There exists a nonempty proper subset S of A such that $\#N_G(S) = \#S$. We set $S' = A \setminus S$, $T = N_G(S)$ and $T' = B \setminus T$ and consider the induced subgraphs H and H' of G on the vertex sets $S \cup T$ and $S' \cup T'$, respectively. Clearly, H and H' are simple bipartite graphs, with corresponding partitions of their vertex sets $\{S, T\}$ and $\{S', T'\}$. We observe that condition (ii) of the theorem holds for H and S, since $N_H(X) = N_G(X)$ and hence

$$\#N_H(X) = \#N_G(X) \geq \#X$$

for every $X \subseteq S$. We claim that it also holds for H' and S'. Let $Y \subseteq S' = A \setminus S$. We know that

$$\#N_G(S \cup Y) \geq \#(S \cup Y) = (\#S) + (\#Y).$$

This inequality and our assumption that $\#T = \#S$ imply that there exist at least $\#Y$ elements of T' each one of which is adjacent in G to at least one element of $S \cup Y$. Since the elements of T' are not adjacent in G to any element of S, it follows that there exist at least $\#Y$ elements of T' each one of which is adjacent in G, hence in H', to at least one element of Y. We have shown that

$$\#N_{H'}(Y) \geq \#Y$$

and conclude that our claim holds for H' and S'. Thus, by the induction hypothesis, there exist matchings F and F' of H and H', respectively, such that every element of S is an endpoint of an edge in F and every element of S' is an endpoint of an edge in F'. Then, $F \cup F'$ is a matching of G and every element of $A = S \cup S'$ is an endpoint of an edge in $F \cup F'$. This completes the induction and the proof follows.

□

Example 5.4.5. Men and women live in a village. We assume that a man a knows a woman b if and only if b knows a (for all a and b) and that for some positive integer k, every man knows at least k women and every woman knows at most k men. We will prove that all men can marry simultaneously, every one with a woman they know.

Indeed, let us consider the bipartite graph with vertices the men and women of the village and edges the pairs of men and women who know each other. According to Theorem 5.4.1, it suffices to prove that if S is any subset of the set of men and T is the set of women who know at least one man in S, then $\#S \leq \#T$. Let m be the number of pairs (a, b) with $a \in S$ and $b \in T$, such that a and b know each other. Since every man in S knows at least k women in T, we have $m \geq (\#S) \cdot k$. We find similarly that $m \leq (\#T) \cdot k$ and conclude that $\#S \leq \#T$. □

The following statement is a direct consequence of Theorem 5.4.1.

Corollary 5.4.2 (Marriage Theorem). *Let G be a simple bipartite graph, and let $\{A, B\}$ be a partition of its vertex set such that every edge of G connects an element of A to an element of B. Then, there exists a perfect matching of G if and only if $\#A = \#B$ and*

$$\#N_G(S) \geq \#S$$

for every $S \subseteq A$. □

Theorem 5.4.1 may be restated as follows. Let $A_1, A_2, \ldots, A_n$ be sets. A sequence $(x_1, x_2, \ldots, x_n)$ of pairwise distinct elements of the union of these sets is called a *system of distinct representatives* for $A_1, A_2, \ldots, A_n$ if $x_i \in A_i$ for every $i \in [n]$. For example, for $n = 5$ the sequence $x_1 = 2$, $x_2 = 4$, $x_3 = 3$, $x_4 = 5$, $x_5 = 1$ is a system of distinct representatives for the sets $A_1 = \{1, 2, 3\}$, $A_2 = \{2, 4, 5\}$, $A_3 = \{3\}$, $A_4 = \{1, 5\}$, and $A_5 = \{1, 4\}$. On the contrary, for $n = 4$, there is no system of distinct representatives for the sets $B_1 = \{1, 2, 3, 4\}$, $B_2 = \{2\}$, $B_3 = \{4\}$, and $B_4 = \{2, 4\}$.

Corollary 5.4.3. *The sets $A_1, A_2, \ldots, A_n$ afford a system of distinct representatives if and only if for every $r \in [n]$, the union of any r of these sets has cardinality at least r; in other words,*

$$\#(A_{i_1} \cup A_{i_2} \cup \cdots \cup A_{i_r}) \geq r \tag{5.2}$$

for every choice of indices $1 \leq i_1 < \cdots < i_r \leq n$.

Proof. We set $A = [n]$ and $B = A_1 \cup A_2 \cup \cdots \cup A_n$. We may assume without loss of generality that A and B are disjoint. We consider the simple bipartite graph with vertex set $A \cup B$ for which $i \in A$ and $b \in B$ are adjacent if and only if $b \in A_i$. We claim that (a) $A_1, A_2, \ldots, A_n$ afford a system of distinct representatives if and only if condition (i) of Theorem 5.4.1 holds for G; and (b) condition (5.2) holds for $A_1, A_2, \ldots, A_n$ if and only if condition (ii) of Theorem 5.4.1 holds for G. Indeed, (a) holds because a sequence $(x_1, x_2, \ldots, x_n)$ is a system of distinct representatives for $A_1, A_2, \ldots, A_n$ if and only if $\{\{i, x_i\} : i \in [n]\}$ is a matching of G. To verify part (b), it suffices to observe that if $S = \{i_1, i_2, \ldots, i_r\} \subseteq A$, then $N_G(S) = A_{i_1} \cup A_{i_2} \cup \cdots \cup A_{i_r}$. The proof follows from claims (a) and (b) and Theorem 5.4.1. □

Application: Magic Squares We close this section with one of many interesting applications of the Marriage Theorem. We call *magic square*, with magic sum k, any square matrix with natural numbers as entries, every row and every column of which has sum equal to k. For example, the matrix

$$A = \begin{pmatrix} 2 & 0 & 2 \\ 1 & 3 & 0 \\ 1 & 1 & 2 \end{pmatrix} \tag{5.3}$$

is a 3×3 magic square with magic sum equal to $k = 4$. The $n \times n$ magic squares with magic sum equal to $k = 1$ are exactly the $n \times n$ permutation matrices, meaning the $n \times n$ matrices every row and every column of which has a single entry equal to 1 and all other entries equal to zero.

Clearly, the sum of two magic squares of the same dimension, with magic sums s and t, is a magic square with magic sum $s + t$. As a consequence, the sum of any number, say k, of $n \times n$ permutation matrices is an $n \times n$ magic square with magic sum equal to k. Is the converse true? In other words, is every magic square equal to a

sum of permutation matrices? For example, the magic square (5.3) can be expressed as the sum of four permutation matrices as follows:

$$A = \begin{pmatrix} 1\,0\,0 \\ 0\,1\,0 \\ 0\,0\,1 \end{pmatrix} + \begin{pmatrix} 1\,0\,0 \\ 0\,1\,0 \\ 0\,0\,1 \end{pmatrix} + \begin{pmatrix} 0\,0\,1 \\ 0\,1\,0 \\ 1\,0\,0 \end{pmatrix} + \begin{pmatrix} 0\,0\,1 \\ 1\,0\,0 \\ 0\,1\,0 \end{pmatrix}.$$

The following proposition gives an affirmative answer to our question.

Proposition 5.4.5. *Every magic square can be expressed as a sum of permutation matrices.*

Proof. Let A be an $n \times n$ magic square with magic sum k. We claim that if $k \geq 1$, then A can be expressed as the sum of an $n \times n$ permutation matrix and an $n \times n$ magic square with magic sum $k-1$. The result follows from the claim by induction on k.

Let us denote by a_{ij} the entry of A which lies in the ith row and the jth column. We set $[n]' = \{1', 2', \ldots, n'\}$ and define a simple bipartite graph G with vertex set $[n] \cup [n]'$ and edges the sets $\{i, j'\}$ for which $a_{ij} \geq 1$. We will show that this graph and the set $[n]$ satisfy condition (ii) of Theorem 5.4.1. Let $S \subseteq [n]$ with $\#S = r$. The sum of the elements of the rows of A indexed by $i \in S$ is equal to rk. Since every column of A sums to k, at least r columns of A have some nonzero element in a row indexed by $i \in S$. This means that $\#N_G(S) \geq r$. Hence, according to Corollary 5.4.2, G has a perfect matching. This means that there exists a permutation $(\sigma_1, \sigma_2, \ldots, \sigma_n)$ of $[n]$ such that $a_{i\,\sigma_i} \geq 1$ for every $i \in [n]$. Therefore, we may write $A = P + B$, where P is the $n \times n$ permutation matrix having nonzero entries in positions (i, σ_i) for $i \in [n]$ and B is an $n \times n$ matrix with natural numbers as entries. Clearly, then, $B = A - P$ is a magic square with magic sum $k-1$. This completes the proofs of the claim and the proposition as well. □

5.5 The Chromatic Polynomial

This section is concerned with the following question: How many proper colorings of a given graph with a given number of q available colors are there? We will prove that, as a function of q, this quantity is a polynomial and we will provide some information about its coefficients. This polynomial has intriguing properties and is called the *chromatic polynomial*.

Definition 5.5.1. *Let G be a simple graph. Given $q \in \mathbb{Z}_{>0}$, we denote by $\chi_G(q)$ the number of proper colorings of G with colors $1, 2, \ldots, q$ and call the resulting function $\chi_G : \mathbb{Z}_{>0} \to \mathbb{N}$ the chromatic function of G.*

Definitions 5.4.1 and 5.5.1 imply that the chromatic number of a simple graph G is equal to the smallest positive integer q such that $\chi_G(q) \neq 0$. The chromatic function χ_G depends only on the isomorphism class of G; in other words, isomorphic graphs have equal chromatic functions (see Exercise 54).

Example 5.5.1. Let G be the simple graph of Example 5.4.1. To choose a proper coloring $\kappa : V \to [q]$ of G there are q choices for the color $\kappa(a)$ and for every such choice, there are $q-1$ choices for $\kappa(b)$ so that $\kappa(b) \neq \kappa(a)$ and then $q-2$ choices for each of the colors $\kappa(c)$ and $\kappa(d)$ so that $\kappa(c) \neq \kappa(a), \kappa(b)$ and $\kappa(d) \neq \kappa(b), \kappa(c)$. This argument and the multiplication principle yield that

$$\chi_G(q) = q(q-1)(q-2)^2 = q^4 - 5q^3 + 8q^2 - 4q$$

for every $q \in \mathbb{Z}_{>0}$.

Example 5.5.2. Let $G = (V, E)$ be a simple graph with n vertices.

(a) If $E = \varnothing$, then every coloring $\kappa : V \to [q]$ of G is proper and hence $\chi_G(q)$ is equal to the number of all maps $\kappa : V \to [q]$. Hence, $\chi_G(q) = q^n$ for every $q \in \mathbb{Z}_{>0}$. If G is a complete graph, then (as discussed in Example 5.4.2) $\chi_G(q)$ is equal to the number of injective maps $\kappa : V \to [q]$. Therefore, according to Proposition 2.3.2 (a), $\chi_G(q) = q(q-1)(q-2)\cdots(q-n+1)$ for every $q \in \mathbb{Z}_{>0}$.

(b) Let $V = \{v_1, v_2, \ldots, v_n\}$, and let G be the path with edges $\{v_{i-1}, v_i\}$ for $2 \leq i \leq n$. To choose a proper coloring $\kappa : V \to [q]$ of G, there are q choices for the color $\kappa(v_1)$ and for every such choice, there are successively $q-1$ choices for each one of the colors $\kappa(v_2), \ldots, \kappa(v_n)$ so that $\kappa(v_i) \neq \kappa(v_{i-1})$ for $2 \leq i \leq n$. By the multiplication principle, $\chi_G(q) = q(q-1)^{n-1}$ for every $q \in \mathbb{Z}_{>0}$.

(c) As will be shown in the sequel, for the cycle C_4 of length 4 we have $\chi_{C_4}(q) = q(q-1)(q^2-3q+3)$ for every $q \in \mathbb{Z}_{>0}$. Thus, $\chi_{C_4}(q)$ cannot be computed by applying the multiplication principle as in the previous examples. □

We will prove (see Corollary 5.5.1) in three different ways that, as already mentioned, the chromatic function $\chi_G(q)$ is a polynomial function of q. In other words, there exists a polynomial in the variable x the value of which at $x = q \in \mathbb{Z}_{>0}$ is equal to the number $\chi_G(q)$ of proper colorings of G with colors $1, 2, \ldots, q$. This polynomial is called the *chromatic polynomial* of G and will also be denoted by $\chi_G(x)$. For example, for the complete graph with n vertices we have $\chi_G(x) = x(x-1)(x-2)\cdots(x-n+1)$.

The first proof is based on the following proposition. A subset W of the vertex set of G is called *independent* (or *stable*) if no two elements of W are adjacent vertices of G.

Proposition 5.5.1. *For every simple graph $G = (V, E)$ with n vertices we have*

$$\chi_G(q) = \sum_{k=1}^{n} c_k(G)\, q(q-1)(q-2)\cdots(q-k+1)$$

for every $q \in \mathbb{Z}_{>0}$, where $c_k(G)$ is the number of partitions of V with k blocks, each of which is independent in G.

Proof. By definition, a proper coloring κ of G with colors $1, 2, \ldots, q$ is a map $\kappa : V \to [q]$ such that the subset $\kappa^{-1}(j)$ of V is independent for every $j \in [q]$. Therefore, the nonempty of these subsets are the blocks of a partition of V, and they are all independent in G. There are $c_k(G)$ ways to choose such a partition with k blocks and $q(q-1)\cdots(q-k+1)$ ways to assign one of the colors $1, 2, \ldots, q$ to each block, so that these colors are pairwise distinct. This proves the proposed formula for $\chi_G(q)$. □

EXAMPLE. For the cycle C_4 of length 4 we have $c_1 = 0$, $c_2 = 1$, $c_3 = 2$, and $c_4 = 1$, where $c_k = c_k(C_4)$. For example, if $\{v_1, v_2\}$, $\{v_2, v_3\}$, $\{v_3, v_4\}$, and $\{v_1, v_4\}$ are the edges of C_4, then the unique partition of the vertex set $\{v_1, v_2, v_3, v_4\}$ into two independent blocks is $\{\{v_1, v_3\}, \{v_2, v_4\}\}$. Hence, Proposition 5.5.1 yields that

$$\begin{aligned}\chi_{C_4}(q) &= q(q-1) + 2q(q-1)(q-2) + q(q-1)(q-2)(q-3)\\ &= q(q-1)(q^2-3q+3) = q^4 - 4q^3 + 6q^2 - 3q\end{aligned}$$

for every $q \in \mathbb{Z}_{>0}$.

Corollary 5.5.1. *Given a simple graph $G = (V, E)$ with n vertices, there exists a unique polynomial $\chi_G(x)$ of degree n with integer coefficients such that $\chi_G(q)$ is equal to the number of proper colorings of G with colors $1, 2, \ldots, q$ for every $q \in \mathbb{Z}_{>0}$. Moreover, if*

$$\chi_G(x) = \sum_{i=0}^{n} a_i x^i, \tag{5.4}$$

then $a_n = 1$, $a_0 = 0$, and $-a_{n-1}$ is equal to the number of edges of G.

Proof. To confirm the first claim, it suffices to define $\chi_G(x)$ as

$$\chi_G(x) = \sum_{k=1}^{n} c_k(G)\, x(x-1)(x-2)\cdots(x-k+1), \tag{5.5}$$

where $c_k(G)$ is as in Proposition 5.5.1. Uniqueness follows from the fact (see Corollary A.2.1 of the appendix) that if two polynomials $p(x)$ and $q(x)$ with real coefficients take equal values for infinitely many values of x, then $p(x) \equiv q(x)$.

Clearly, the right-hand side of (5.5) can be written in the form (5.4) with $a_0 = 0$. Computing the coefficients of x^n and x^{n-1} in the right-hand side of (5.5), we find that $a_n = c_n(G)$ and

$$\begin{aligned} a_{n-1} &= [x^{n-1}] \sum_{k=1}^{n} c_k(G)\, x(x-1)(x-2)\cdots(x-k+1) \\ &= [x^{n-1}]\, c_n(G)\, x(x-1)(x-2)\cdots(x-n+1) + \\ &\quad [x^{n-1}]\, c_{n-1}(G)\, x(x-1)(x-2)\cdots(x-n+2) \\ &= -c_n(G) \cdot \frac{n(n-1)}{2} + c_{n-1}(G), \end{aligned}$$

where we have denoted by $[x^k]\, p(x)$ the coefficient of x^k in a polynomial $p(x)$. Since the definition of $c_k(G)$ implies (explain how) that $c_n(G) = 1$ and $c_{n-1}(G) = \binom{n}{2} - m$, where m is the number of edges of G, we conclude that $a_n = 1$ and $a_{n-1} = -m$. □

Remark 5.5.1. Let V be an n-element set and $G = (V, \varnothing)$ be the graph with vertex set V without edges. Then, every subset of V is independent and hence the number $c_k(G)$ defined in Proposition 5.5.1 is equal to the total number $S(n, k)$ of partitions of an n-element set with k blocks (see Section 4.3). Since $\chi_G(x) = x^n$, formula (5.5) specializes to the identity

$$x^n = \sum_{k=1}^{n} S(n, k)\, x(x-1)(x-2)\cdots(x-k+1) \tag{5.6}$$

of Proposition 4.3.2. □

Our second proof of the polynomiality of the function $\chi_G(q)$ is based on the technique of *deletion–contraction*, an important technique in graph theory. Given a simple graph $G = (V, E)$ and an edge $e \in E$, we say that the graph $(V, E \setminus \{e\})$ is the *deletion* of e from G, and we denote it by $G \setminus e$. We also denote by G/e the graph defined as follows: The set of vertices of G/e is obtained from V by replacing the endpoints of e with a new vertex u. The set of edges of G/e is obtained from E by deleting all edges that have at least one common endpoint with e and including as new edges the sets $\{u, v\}$, where $v \in V$ is not an endpoint of e and is adjacent in G to at least one endpoint of e. We call G/e the *contraction* of e in G. An example of deletion and contraction is shown in Figure 5.22.

Proposition 5.5.2 (Deletion–Contraction Theorem). *For all simple graphs G and edges e of G,*

$$\chi_G(q) = \chi_{G \setminus e}(q) - \chi_{G/e}(q)$$

for every $q \in \mathbb{Z}_{>0}$.

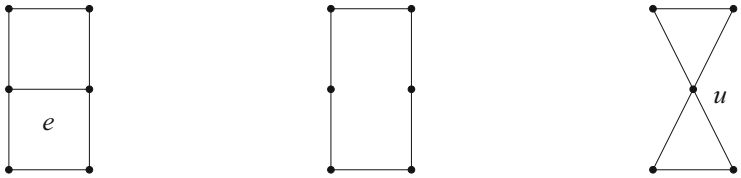


Figure 5.22 Deletion and contraction of e in G

Proof. Let V be the vertex set of G, and let $e = \{u, v\}$. Given a proper coloring $\kappa : V \to [q]$ of $G \smallsetminus e$, we have either $\kappa(u) \neq \kappa(v)$ or $\kappa(u) = \kappa(v)$. The proper colorings for which $\kappa(u) \neq \kappa(v)$ coincide with the proper colorings of G with colors $1, 2, \ldots, q$ and hence their number is equal to $\chi_G(q)$. The proper colorings for which $\kappa(u) = \kappa(v)$ are in one-to-one correspondence with the proper colorings of G/e with colors $1, 2, \ldots, q$ and hence their number is equal to $\chi_{G/e}(q)$. Hence, by the addition principle, $\chi_{G \smallsetminus e}(q) = \chi_G(q) + \chi_{G/e}(q)$ for every $q \in \mathbb{Z}_{>0}$. □

Second Proof of Corollary 5.5.1. We proceed by induction on the number m of edges of G. If $m = 0$, then $\chi_G(q) = q^n$ for every $q \in \mathbb{Z}_{>0}$ (see Example 5.5.2) and the result holds for $\chi_G(x) := x^n$.

Suppose that $m \geq 1$, and let e be any edge of G. The graph $G \smallsetminus e$ has n vertices and $m - 1$ edges, while G/e has $n - 1$ vertices and fewer than m edges. Therefore, by the induction hypothesis, there exist polynomials $\chi_{G \smallsetminus e}(x)$ and $\chi_{G/e}(x)$ having the properties claimed in the corollary. In particular,

$$\chi_{G \smallsetminus e}(x) = \sum_{k=0}^{n} b_k x^k$$

and

$$\chi_{G/e}(x) = \sum_{k=0}^{n-1} c_k x^k,$$

where $b_0 = c_0 = 0$, $b_n = c_{n-1} = 1$, and $b_{n-1} = -m + 1$. These equalities and Proposition 5.5.2 imply that the polynomial

$$\chi_G(x) := \chi_{G \smallsetminus e}(x) - \chi_{G/e}(x) = b_n x^n + \sum_{k=0}^{n-1} (b_k - c_k) x^k \tag{5.7}$$

has the desired properties. □

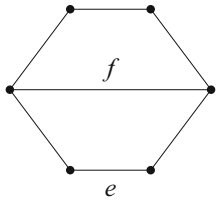


Figure 5.23 A simple graph with six vertices

EXAMPLE. Let G be the simple graph of Example 5.4.1 with set of vertices $V = \{a, b, c, d\}$. We observe that $G = K \smallsetminus e$, where K is the complete graph with set of vertices V and $e = \{a, d\}$, and that K/e is a complete graph with three vertices. Since we have $\chi_K(x) = x(x-1)(x-2)(x-3)$ and $\chi_{K/e}(x) = x(x-1)(x-2)$ by Example 5.5.2, we conclude from Proposition 5.5.2 that

$$\chi_G(x) = \chi_K(x) + \chi_{K/e}(x) = x(x-1)(x-2)^2,$$

in agreement with the computation of Example 5.5.1.

Example 5.5.3. Let's compute the chromatic polynomial of the graph G that is obtained from the cycle C_6 by adding an edge f that connects two opposite vertices (see Figure 5.23).

We will apply Proposition 5.5.2 to the edge e shown in the figure. The enumeration of proper colorings of $G \smallsetminus e$ and G/e according to the multiplication principle shows (explain how) that $\chi_{G \smallsetminus e}(q) = (q-1)^2 \chi_{C_4}(q)$ and $\chi_{G/e}(q) = (q-2)\chi_{C_4}(q)$ for every $q \in \mathbb{Z}_{>0}$, so that $\chi_{G \smallsetminus e}(x) = (x-1)^2 \chi_{C_4}(x)$ and $\chi_{G/e}(x) = (x-2)\chi_{C_4}(x)$. These equations, the deletion–contraction formula of Proposition 5.5.2 and the known formula for the chromatic polynomial of the cycle C_4 yield that

$$\begin{aligned}\chi_G(x) &= \chi_{G \smallsetminus e}(x) - \chi_{G/e}(x) = (x-1)^2 \chi_{C_4}(x) - (x-2)\chi_{C_4}(x) \\ &= (x^2 - 3x + 3)\, \chi_{C_4}(x) = x(x-1)(x^2 - 3x + 3)^2 \\ &= x^6 - 7x^5 + 21x^4 - 33x^3 + 27x^2 - 9x.\end{aligned}$$

The second proof of the polynomiality of the chromatic polynomial, via the deletion–contraction formula of Proposition 5.5.2, yields another important property of $\chi_G(x)$, namely that its coefficients alternate in sign.

Corollary 5.5.2. *Let G be a simple graph with n vertices and chromatic polynomial given by Equation (5.4). Then, $(-1)^{n-k} a_k \geq 0$ for every $k \in \{0, 1, \ldots, n\}$.*

Proof. We proceed by induction on the number m of edges of G, following the second proof of Corollary 5.5.1. The result holds for $m = 0$, since then $\chi_G(x) = x^n$. We assume that $m \geq 1$ and define the graphs $G \smallsetminus e$ and G/e as in that proof. By the induction hypothesis, the coefficients of $\chi_{G \smallsetminus e}(x)$ and $\chi_{G/e}(x)$ satisfy the

inequalities $(-1)^{n-k}b_k \geq 0$ and $(-1)^{n-1-k}c_k \geq 0$ for every k. Since $a_n = 1$ and, as a consequence of (5.7),

$$(-1)^{n-k}a_k = (-1)^{n-k}b_k + (-1)^{n-1-k}c_k$$

for $0 \leq k \leq n-1$, we conclude that $(-1)^{n-k}a_k \geq 0$ for every $k \in \{0, 1, \ldots, n\}$ and the proof follows. □

The nonnegative integers $(-1)^{n-k}a_k$ afford some very interesting combinatorial interpretations for arbitrary graphs; see, for instance, [20, Section 3.8] for more information.

The following proposition provides another nonrecursive formula for the chromatic polynomial. Given a simple graph $G = (V, E)$ and $S \subseteq E$, we denote by G_S the spanning subgraph (V, S) of G with set of edges S and by $c_G(S)$ the number of connected components of G_S. For example, if G is the tree of Figure 5.7 and $S = \{\{1, 5\}, \{2, 8\}, \{3, 7\}, \{3, 9\}, \{6, 8\}\}$, then $c_G(S) = 4$.

Proposition 5.5.3. *Let $G = (V, E)$ be a simple graph. Then*

$$\chi_G(q) = \sum_{S \subseteq E} (-1)^{\#S} q^{c_G(S)}$$

for every $q \in \mathbb{Z}_{>0}$, where $c_G(S)$ is the number of connected components of $G_S = (V, S)$.

Proof. Given a positive integer q, we denote by Ω the set of all colorings $\kappa : V \to [q]$ of G and for $e \in E$ with $e = \{u, v\}$, we denote by A_e the set of those colorings $\kappa \in \Omega$ for which $\kappa(u) = \kappa(v)$. Clearly, the set of proper colorings of G with colors $1, 2, \ldots, q$ is equal to

$$\Omega - \bigcup_{e \in E} A_e.$$

Therefore, by the principle of inclusion–exclusion (Theorem 3.1.1),

$$\chi_G(q) = \sum_{S \subseteq E} (-1)^{\#S} \# \bigcap_{e \in S} A_e.$$

Thus, it suffices to show that the set $\bigcap_{e \in S} A_e$ has cardinality $q^{c_G(S)}$ for every $S \subseteq E$. Indeed, a coloring $\kappa : V \to [q]$ of G belongs to this set if and only if it takes the same value at the endpoints of any edge in S. This happens if and only if κ takes the same value at all vertices of G that belong to the same connected component of G_S. Since there are q choices of color for every connected component of G_S, the number of colorings $\kappa : V \to [q]$ of G with this property is equal to $q^{c_G(S)}$. □

EXAMPLE. Let G be the graph of Example 5.4.1. Then $c_G(\varnothing) = 4$, $c_G(S) = 3$ if S has cardinality 1, $c_G(S) = 2$ if S has cardinality 2, or it consists of the edges of one of the two cycles of G of length 3, and $c_G(S) = 1$ in all other cases. Hence, Proposition 5.5.3 yields that

$$\begin{aligned}\chi_G(q) &= q^4 - 5q^3 + 10q^2 - 2q^2 - 8q + 5q - q \\ &= q^4 - 5q^3 + 8q^2 - 4q \\ &= q(q-1)(q-2)^2\end{aligned}$$

for every $q \in \mathbb{Z}_{>0}$.

Third Proof of Corollary 5.5.1. Setting

$$\chi_G(x) = \sum_{S \subseteq E} (-1)^{\#S} x^{c_G(S)}, \tag{5.8}$$

where $c_G(S)$ is as in Proposition 5.5.3, we get a polynomial that takes the desired values for every $x = q \in \mathbb{Z}_{>0}$. Since $c_G(S) = n$ if and only if $S = \varnothing$, the function $\chi_G(x)$ can be written in the form (5.4) with $a_n = 1$. Moreover, since $c_G(S) \geq 1$ for every $S \subseteq E$, we have $a_0 = 0$. Finally, since $c_G(S) = n - 1$ if and only if $S \subseteq E$ is a singleton, we have $a_{n-1} = -m$, where m is the number of edges of G and the proof follows. □

Acyclic Orientations We close this section with an application of the method of deletion–contraction to acyclic orientations of graphs. Let $G = (V, E, \varphi)$ be a graph and $e \in E$ be an edge with $\varphi(e) = \{a, b\}$. An *orientation* of e is a choice of one of the vertices a, b as the *initial vertex* of e. The edge e is then said to be *oriented* and if a is the initial vertex of e, then b is its *terminal vertex* (so the initial and terminal vertices of e coincide only if e is a loop).

Definition 5.5.2. *An orientation of G is defined as a choice of orientation for every edge of G.*

Let o be an orientation of G. A walk $w = (u_0, e_1, u_1, \ldots, e_\ell, u_\ell)$ in G is called *directed* with respect to o if with the orientation o, the edge e_i has initial vertex u_{i-1} and terminal vertex u_i for $1 \leq i \leq \ell$. A closed walk w in G that is directed with respect to o is called a *cycle* of o. An orientation of G that has no cycles is called *acyclic*. Figure 5.24 shows two orientations of the graph of Figure 5.22, the one on the right being acyclic.



Figure 5.24 Two orientations of a graph

The number of orientations of a graph G is equal to 2^m, where m is the number of edges of G that are not loops. We will denote by $a(G)$ the number of acyclic orientations of G.

Example 5.5.4.

(a) Let G be the complete graph on the n-element vertex set V. We will show that $a(G) = n!$. Given an acyclic orientation o of G, we define the binary relation $\leq_o$ on the set V by setting $a \leq_o b$ if either $a = b$, or a is the initial vertex of the edge $\{a, b\}$ of G with respect to the orientation o. We leave to the reader to verify that $\leq_o$ is a partial order on V (where transitivity holds because o is acyclic). This partial order is in fact a total order (any two elements of V are comparable to each other). Thus, Exercise 37 (c) of Section 1.2 implies that there exists a permutation $f(o) = (v_1, v_2, \ldots, v_n)$ of V such that $v_i <_o v_j \Leftrightarrow i < j$. Clearly, every permutation of V can be expressed as $f(o)$ for a unique acyclic orientation o of G. We conclude that the acyclic orientations of G are in one-to-one correspondence with the permutations of V and hence that $a(G) = n!$.

(b) A cycle G of length n has exactly two orientations (explain which ones) that are not acyclic. Hence, $a(G) = 2^n - 2$.

(c) Let G be a tree with n vertices. Then, every orientation of G is acyclic and hence, by Proposition 5.3.3, $a(G) = 2^{n-1}$. □

The following theorem, which is verified by Example 5.5.4, connects the number $a(G)$ to the chromatic polynomial of G in an unexpected way.

Theorem 5.5.1 (Stanley, 1973). *We have*

$$a(G) = (-1)^n \chi_G(-1) \tag{5.9}$$

for every simple graph G with n vertices, where $\chi_G(-1)$ is the value of the polynomial $\chi_G(x)$ at $x = -1$.

Proof. We will prove (5.9) by induction on the number of edges of G. We set $\psi(G) = (-1)^n \chi_G(-1)$ and denote by $\mathrm{AO}(G)$ the set of acyclic orientations of G. If G has no edges, then $\chi_G(x) = x^n$ and hence $a(G) = \psi(G) = 1$. Otherwise, by Proposition 5.5.2, we have

$$\psi(G) = \psi(G \setminus e) + \psi(G/e)$$

for an arbitrary edge e of G. Since $G \setminus e$ and G/e have fewer edges than G, it suffices to show that $a(G)$ satisfies the same recursion, meaning that

$$a(G) = a(G \setminus e) + a(G/e). \tag{5.10}$$

Given $o \in \mathrm{AO}(G)$, we denote by $f(o)$ the orientation of $G \setminus e$ that is obtained from o by deleting the edge e. Clearly, $f(o) \in \mathrm{AO}(G \setminus e)$ and hence we have defined a map

$$f : \mathrm{AO}(G) \to \mathrm{AO}(G \setminus e).$$

We let $e = \{a, b\}$ and for $o' \in \mathrm{AO}(G \setminus e)$, we denote by o_a (respectively, by o_b) the orientation of G that is obtained from o' by orienting e so that the initial vertex is a (respectively, b). We observe that the inverse image $f^{-1}(o')$ consists of those orientations $o \in \{o_a, o_b\}$ of G which are acyclic. We also observe that o_a (respectively, o_b) is acyclic if and only if there is no directed walk, with respect to o', in $G \setminus e$ with initial vertex b and terminal vertex a (respectively, with initial vertex a and terminal vertex b). Since o' is acyclic, the cardinality of $f^{-1}(o')$ is either one or two, and the latter holds if and only if there is no directed walk, with respect to o', in $G \setminus e$ with initial vertex one of a, b and terminal vertex the other. Since the orientations $o' \in \mathrm{AO}(G \setminus e)$ with this property are in one-to-one correspondence with the acyclic orientations of G/e, it follows that $a(G) - a(G \setminus e) = a(G/e)$. This proves (5.10). □

5.6 Plane Graphs

Loosely speaking, a plane graph is any graph that has points of $\mathbb{R}^2$ as vertices and pairwise noncrossing, non-self-intersecting arcs that connect these points as edges. This section proves a fundamental equation satisfied by the numbers of vertices and edges of such a graph and the number of regions into which $\mathbb{R}^2$ is dissected by these edges. To define all these notions with greater clarity and reach our goal, we will assume that the reader is familiar with the basics of the topology of $\mathbb{R}^2$ (a good intuition will suffice for other readers).

We begin with some terminology. Let $a, b \in \mathbb{R}^2$ be points (not necessarily distinct). *A simple arc* (in $\mathbb{R}^2$) with endpoints a, b is defined as a set of the form $\{\gamma(t) : 0 \le t \le 1\}$, where $\gamma : [0, 1] \to \mathbb{R}^2$ is a continuous map with the following properties: (a) $\{\gamma(0), \gamma(1)\} = \{a, b\}$; and (b) if $0 \le s < t \le 1$ and $\gamma(s) = \gamma(t)$, then $s = 0$ and $t = 1$.

Definition 5.6.1. *A plane graph is any finite graph (V, E, φ) such that:*

(a) $V \subset \mathbb{R}^2$;
(b) every edge $e \in E$ is a simple arc in $\mathbb{R}^2$ with endpoints a and b, where $\{a, b\} = \varphi(e)$, which does not contain elements of V other than a, b; and
(c) if two distinct elements of E have nonempty intersection, then this intersection consists of some of their common endpoints.

Every graph that is isomorphic to a plane graph is called planar.

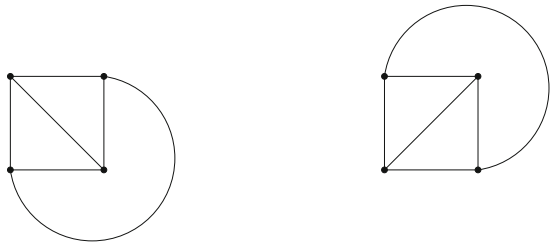


Figure 5.25 Two plane graphs

EXAMPLE.

(a) Plane graphs appear in Figures 5.1, 5.2b, 5.4, 5.7, 5.19 and 5.22.
(b) The complete graph K_4 is planar. Figure 5.25 shows two plane graphs with vertices $(0, 0)$, $(1, 0)$, $(0, 1)$, and $(1, 1)$ that are isomorphic to K_4.
(c) The bipartite graph of Figure 5.16 is planar, although this fact is not obvious from the drawing in that figure. □

Let G be a (finite) plane graph. The set $\mathcal{M}_G$ that is obtained from $\mathbb{R}^2$ by removing the union of the edges of G is a nonempty open subset of $\mathbb{R}^2$. Therefore, $\mathcal{M}_G$ can be written uniquely as a union of pairwise disjoint nonempty connected open subsets of $\mathbb{R}^2$. These subsets are called *regions* of G.

We will denote by $p(G)$, $q(G)$, and $r(G)$ the number of vertices, edges, and regions of G, respectively. We will say that an edge e of G is an edge of a region R if e is contained in the closure of R.

Example 5.6.1.

(a) The plane graph shown in Figure 5.1 has $p(G) = 4$ vertices, $q(G) = 7$ edges and $r(G) = 5$ regions. One of the five regions of G has edges e_1, e_2, e_5 and another has the edges e_6, e_7.
(b) Suppose that G has one vertex and one edge (the endpoints of which coincide). It follows from "Jordan's Curve Theorem" (see, for instance, [18, Chapter 10]), which we assume here as known, that $r(G) = 2$. Moreover, exactly one of the regions of G (the one which is "enclosed" by the unique edge of G) is bounded, and the edge of G is an edge of both regions.
(c) Suppose that the plane graph G is a tree. Applying induction on the number of vertices (explain how), one can prove that $\mathcal{M}_G$ is path connected. As a result, $r(G) = 1$. □

Theorem 5.6.1 (Euler, 1752). *The numbers of vertices, edges, and regions of any connected plane graph G satisfy the equation*

$$p(G) - q(G) + r(G) = 2. \tag{5.11}$$

In particular, any two isomorphic connected plane graphs have the same number of regions.

Proof. We will prove the first assertion by induction on the number $q(G)$ of edges of G. The second follows from the first, since isomorphic graphs have the same number of vertices and the same number of edges.

We observe that if $q(G) = 0$, then $p(G) = r(G) = 1$ and Equation (5.11) holds. Let us assume that $q(G) = m \geq 1$ and that (5.11) holds for every connected plane graph with $m - 1$ edges. We distinguish two cases:

(a) The graph G is a tree. Then, $p(G) = m + 1$ by Proposition 5.3.3 and $r(G) = 1$, as mentioned in Example 5.6.1 (c). Hence, $p(G) - q(G) + r(G) = 2$.
(b) The graph G is not a tree. Then, there exists an edge e of G that is a loop, or parallel to another edge, or is an edge of a cycle of G. We consider the graph $G \smallsetminus e$ obtained from G be deleting e. Clearly, $G \smallsetminus e$ is a connected plane graph with $p(G \smallsetminus e) = p(G)$ vertices and $q(G \smallsetminus e) = q(G) - 1 = m - 1$ edges. Moreover, e is an edge of exactly two regions of G (this claim follows from the Jordan Curve Theorem) and hence $r(G \smallsetminus e) = r(G) - 1$. These facts and the equality $p(G \smallsetminus e) - q(G \smallsetminus e) + r(G \smallsetminus e) = 2$, that is guaranteed by the induction hypothesis, imply Equation (5.11) for G.

□

In the remainder of this section, we apply Theorem 5.6.1 in order to prove that K_5 and $K_{3,3}$ are not planar graphs.

Proposition 5.6.1. *We have $q(G) \leq 3p(G) - 6$ for every simple connected plane graph G with at least three vertices.*

Proof. We let $\mathcal{R}(G)$ denote the set of regions of G. Given $\Omega \in \mathcal{R}(G)$, we denote by $q(\Omega)$ the number of edges of Ω, where we count twice every edge that is not an edge of another region of G. For example, if G is a tree and Ω is the unique region of G, then $q(\Omega) = 2q(G)$. In general, since every edge of G is an edge of one or two regions of G, we have

$$\sum_{\Omega \in \mathcal{R}(G)} q(\Omega) = 2q(G). \tag{5.12}$$

Moreover, since $q(\Omega) \geq 3$ for every $\Omega \in \mathcal{R}(G)$ (the assumption that G has at least three vertices is used at this point), we conclude that

$$2q(G) \geq 3r(G). \tag{5.13}$$

The result follows by replacing $r(G)$ with $q(G) - p(G) + 2$ (by Theorem 5.6.1) in the displayed inequality. □

Corollary 5.6.1. *The complete graph K_5 is not planar.*

Proof. The graph K_5 is simple, connected and has $p = 5$ vertices and $q = 10$ edges. Thus, the result follows from Proposition 5.6.1. □

Proposition 5.6.2. *We have $q(G) \leq 2p(G) - 4$ for every simple connected plane graph G with at least three vertices that does not have cycles of length 3.*

Proof. We follow the proof of Proposition 5.6.1. The assumption that G does not have cycles of length 3 implies that $q(\Omega) \geq 4$ for every $\Omega \in \mathcal{R}(G)$. Therefore, the inequality (5.13) improves to $2q(G) \geq 4r(G)$ and the result follows from Theorem 5.6.1, as in the case of Proposition 5.6.1. □

Corollary 5.6.2. *The complete bipartite graph $K_{3,3}$ is not planar.*

Proof. The graph $K_{3,3}$ is simple and connected, with $p = 6$ vertices and $q = 9$ edges, and does not have cycles of length three. Thus, the result follows from Proposition 5.6.2. □

A theorem of Kuratowski states that a graph is planar if and only if it does not contain any subdivision of K_5 or $K_{3,3}$ as a subgraph. We refer the interested reader to the text [30] for the proof and for the history of another important result in graph theory (the Four-color Theorem), which states that simple plane graphs have chromatic number less than or equal to 4.

5.7 Exercises

1. Given a partially ordered set P, we denote by $\mathrm{inc}(P)$ the simple graph with vertex set P in which two vertices $x, y \in P$ are adjacent if x, y are incomparable elements of P (in other words, we have neither $x \preceq y$ nor $y \preceq x$, where $\preceq$ stands for the partial order of P). For each of the following simple graphs G decide whether or not there exists a partially ordered set P such that $\mathrm{inc}(P) = G$:

   (a) The graph with vertex set $[n]$ and empty set of edges.
   (b) The complete graph K_n.
   (c) The complete bipartite graph $K_{m,n}$.
   (d) The path of length $n - 1$.
   (e) The 6-cycle.

2. Let $f(n)$ be the number of isomorphism classes of simple graphs with n vertices.

   (a) Compute $f(n)$ for $n \leq 4$.
   (b) Prove that $f(n) \leq 2^{\binom{n}{2}}$ for every $n \in \mathbb{Z}_{>0}$.
   (c) Prove that for every $a > 0$, there exists $c > 0$ such that $f(n) \geq c \cdot a^n$ for every $n \in \mathbb{Z}_{>0}$.

3. Among 17 students, some have written letters to some others. Is it possible that every student has written a letter to exactly three other students and has received a letter from exactly three other students, in the following situations:

(a) If a has written a letter to b, then b has written a letter to a?
(b) There is no restriction on who may have written a letter to whom?

4. Is there a simple graph with six vertices of degrees

(a) 5, 4, 2, 2, 1, 0?
(b) 4, 3, 2, 2, 2, 1?
(c) 4, 3, 3, 3, 2, 1?
(d) 4, 4, 4, 2, 1, 1?
(e) 3, 3, 3, 1, 1, 1?

In which cases such a graph exists and is unique up to isomorphism?

5. Given an integer $n \geq 2$, we consider all simple graphs with n vertices, $n-1$ of which have degrees $1, 2, \ldots, n-1$.

(a) Give an example of a graph with these properties.
(b) Prove that any two graphs with these properties are isomorphic.

6. (a) Find a simple connected graph and a simple disconnected graph, each with six vertices of degrees 3, 3, 2, 2, 1, and 1.
(b) Let G be a simple graph with at least two vertices. Prove that there exist two vertices of G with the same degree.
(c) Let G be a simple graph with n vertices such that $\deg_G(u) + \deg_G(v) \geq n-1$ for any two distinct vertices u, v of G. Prove that G is connected.
(d) For every integer $n \geq 2$, prove that there exists a simple disconnected graph with n vertices, each with degree greater than or equal to $\lfloor (n-2)/2 \rfloor$.

7. Let G be a graph without isolated vertices. A walk in G in which every edge of G appears exactly once is called an *Eulerian tour*.

(a) Prove that there exists an Eulerian tour in G if and only if G is connected and has either none or exactly two odd vertices.
(b) Prove that there exists a closed Eulerian tour in G if and only if G is connected and every vertex of G is even.
(c) Prove that there exists a closed walk in G in which every edge of G appears exactly two times if and only if G is connected.

8. Let $G = (V, E, \varphi)$ be a connected graph with $n \geq 2$ vertices.

(a) Prove that the graph $G \setminus v$ is connected for at least two vertices $v \in V$.
(b) For every $n \geq 2$, give an example for which $G \setminus v$ is connected for exactly two vertices $v \in V$.

9. Let $G = (V, E)$ be a connected simple graph with at least three vertices. We say that two distinct vertices of G have distance 2 if they are not adjacent but they have a common neighbor in G.

(a) Suppose that for any two vertices $v, v' \in V$ that have distance 2 in G, there exists a cycle of G that passes through v and v'. Prove that G is doubly connected.

(b) More generally, suppose that G has more than $k \geq 2$ vertices and that for any two vertices $v, v' \in V$ that have distance 2 in G there exist k paths with initial vertex v and terminal vertex v' in G that pairwise have no common vertices other than v and v'. Prove that G is k-connected.

10. Let G be a connected simple graph with n vertices.

(a) Suppose that G is k-connected. Prove that every vertex of G has degree at least k and deduce that G has at least $kn/2$ edges.
(b) Suppose that every vertex of G has degree 2. Prove that G is doubly connected.
(c) Suppose that G is doubly connected and that every vertex has degree 3. Does it follow that G is 3-connected?

11. Consider the graph $\square_n$ of the n-dimensional cube of Example 5.1.2 (f).

(a) How many vertices and how many edges does $\square_n$ have?
(b) Prove that $\square_n$ is n-connected for every positive integer n.

12. A closed walk w in a graph G is called a *Hamiltonian cycle* if every vertex of G appears exactly once in the walk obtained from w by deleting the last vertex and edge. Consider the graph $\square_n$ of Example 5.1.2 (f).

(a) Find a Hamiltonian cycle for $\square_2$ and $\square_3$.
(b) Prove that $\square_n$ has a Hamiltonian cycle for every $n \geq 1$.

13. For a positive integer n, we let G_n be the simple graph with vertices the permutations of $[n]$ in which two permutations are adjacent if one can be obtained from the other by swapping two consecutive terms. For example, $(3, 1, 5, 4, 2)$ and $(3, 5, 1, 4, 2)$ are adjacent in G_5.

(a) How many vertices and how many edges does G_n have?
(b) Draw the graph G_3.
(c) Prove that G_n has a Hamiltonian cycle for every $n \geq 2$.

14. A tree has one vertex of degree 5, two vertices of degree 4, three vertices of degree 3 and four vertices of degree 2, while all other vertices are leaves. How many vertices does this tree have?
15. How many trees with vertex set $[n]$ have:

(a) exactly two leaves?
(b) exactly $n - 2$ leaves?
(c) exactly $n - 1$ leaves?

16. For every integer $n \geq 2$:

(a) Prove that every tree with n vertices has more leaves than vertices of degree greater than 2.
(b) Find the largest possible number of vertices or degree greater than or equal to $n/2$ that a tree with n vertices can have.

17. Let G be a connected simple graph with n vertices and n edges. Prove that G has a unique cycle.
18. Let $G = (V, E, \varphi)$ be a connected graph with n vertices and $v \in V$ be a vertex of G.

    (a) Prove that G has at least $n - 1$ edges by finding an injective map $f : V \setminus \{v\} \to E$.
    (b) Assume that G is a tree. Prove that G has exactly $n - 1$ edges by finding a bijective map $f : V \setminus \{v\} \to E$.

19. Let $T = (V, E)$ be a finite tree. For $J \subseteq V$, let $c_T(J)$ be the number of connected components of the induced subgraph of T on the vertex set J. Prove that the polynomial

$$c_T(x) = \sum_{J \subseteq V} c_T(J) x^{|J|}$$

    depends only on the number of vertices of T.
20. We denote by $\kappa(G)$ the number of spanning trees of a (finite, connected) graph G.

    (a) Prove that $\kappa(G) = 1$ if and only if G is a tree.
    (b) Prove that there is no simple graph G such that $\kappa(G) = 2$.
    (c) Prove that for every integer $n \geq 3$, there exists a simple graph G with n vertices such that $\kappa(G) = 3$.

21. Let $1 \leq m < n$ be integers. We denote by $\mathcal{T}_n$ the set of trees with vertex set $[n]$.

    (a) For how many trees $T \in \mathcal{T}_n$ are all vertices $m + 1, m + 2, \ldots, n$ leaves of T?
    (b) For how many trees $T \in \mathcal{T}_n$ is $\{m + 1, m + 2, \ldots, n\}$ equal to the set of all leaves of T?

22. Let T be a tree with vertex set $[n]$, where $n \geq 3$, set of edges F and Prüfer code $f(T) = (a_1, a_2, \ldots, a_{n-2})$.

    (a) Prove that $\{1, 2\} \in F$ if and only if $a_{n-2} \in \{1, 2\}$.
    (b) How many trees with vertex set $[n]$ contain the edge $\{1, 2\}$?

23. Consider the complete graph K_n on the vertex set $[n]$ and the weight function w for which $\mathrm{w}(\{a, b\}) = \max\{a, b\}$ for every edge $\{a, b\}$ of K_n.

    (a) Prove that $(n^2 + n - 2)/2 \leq \mathrm{w}(T) \leq n^2 - n$ for every spanning tree T of K_n, where $\mathrm{w}(T)$ is the weight of T.
    (b) How many minimum spanning trees are there?

24. A *forest* is a graph every connected component of which is a tree. A *rooted forest* is a forest every connected component of which has a distinguished vertex, called the root.

(a) Prove that there are exactly n^{n-1} rooted trees with vertex set $[n]$.
(b) Prove that there are exactly $(n+1)^{n-1}$ rooted forests with vertex set $[n]$.

25. A forest F (see Exercise 24) with vertex set $[n]$ is called *increasing* if it has the following property: if a, b, c are distinct vertices of the same connected component T of F such that a is the smallest vertex of T and b lies in the unique path in T with endpoints a and c, then $b < c$.

(a) Prove that there are exactly $(n-1)!$ increasing trees with vertex set $[n]$.
(b) Prove that there are exactly $n!$ increasing forests with vertex set $[n]$.

26. Given a rooted forest F (see Exercise 24) with vertex set $[n]$, we denote by $\mathcal{E}(F)$ the set of ordered pairs (a, b), where $\{a, b\}$ is an edge of F and a is the parent of b in the corresponding tree of F. Prove the Cayley–Sylvester formula by counting in two different ways the sequences $(F_0, F_1, \ldots, F_{n-1})$, where the F_i are rooted forests with vertex set $[n]$ and

$$\varnothing = \mathcal{E}(F_0) \subset \mathcal{E}(F_1) \subset \cdots \subset \mathcal{E}(F_{n-1}).$$

27. (a) Describe a one-to-one correspondence from the set of maps $f : [n] \to [n]$ to the set of triples (T, r, s), where T is a tree with vertex set $[n]$ and $r, s \in [n]$.
(b) Deduce the Cayley–Sylvester formula.
28. Let π be a partition of $\{2, 3, \ldots, n\}$ with k blocks.

(a) Prove that there are exactly n^{k-1} trees T on the vertex set $[n]$, rooted at 1, such that elements of $\{2, 3, \ldots, n\}$ that belong to the same block of π have the same parent in T.
(b) Prove that there are exactly $\frac{(n-1)!}{(n-k)!}$ trees T on the vertex set $[n]$, rooted at 1, which have the following property: two elements of $\{2, 3, \ldots, n\}$ have the same parent in T if and only if these elements belong to the same block of π.
(c) Find the number of trees T on the vertex set $[n]$, rooted at 1, which have the following property: the only distinct vertices of T with a common parent are $n-1$ and n.

29. Let $g(n)$ be the number of trees T on the vertex set $[2n]$, rooted at 1, with the following properties:

(i) The even numbers $2, 4, \ldots, 2n$ are all leaves of T.
(ii) There is no $k \in \{2, 3, \ldots, n\}$ such that $2k-2$, $2k-1$ and $2k$ have the same parent in T.

Prove that $g(n) = (n^2-1)^{n-1}$ for every $n \geq 2$.

30. Given integers $n \geq 2$ and $r_0, r_1, \ldots, r_{n-1} \geq 0$, we denote by $\mathcal{T}(r_0, \ldots, r_{n-1})$ the set of rooted trees with vertex set $[n]$ that have exactly r_i vertices with i successors, for $i \in \{0, 1, \ldots, n-1\}$. Provided that $\mathcal{T}(r_0, \ldots, r_{n-1})$ is nonempty, prove that:

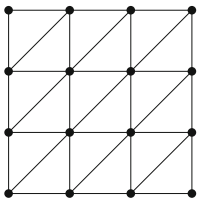


Figure 5.26 A simple graph with 16 vertices

(a) $r_0 + r_1 + \cdots + r_{n-1} = n$,
(b) $r_1 + 2r_2 + \cdots + (n-1)r_{n-1} = n - 1$,
(c)

$$\#\mathcal{T}(r_0, \ldots, r_{n-1}) = \frac{n!(n-1)!}{\prod_{i=0}^{n-1}(i!)^{r_i}\, r_i!}.$$

31. Prove that the Catalan number $\dfrac{1}{n+1}\binom{2n}{n}$ is equal to:
    (a) The number of unlabeled plane rooted trees with $n + 1$ vertices.
    (b) The number of unlabeled binary plane rooted trees with n internal vertices.
32. Let G be a (finite) simple graph. Prove that each vertex of G can be colored either black or white, so that at least half of the vertices that are adjacent to any white (respectively, black) vertex are black (respectively, white).
33. Compute the chromatic number of:
    (a) The graph of Figure 5.26.
    (b) The graph that is obtained from the complete graph K_n by deleting one of its edges.
    (c) The simple graph with set of vertices $[2n]$, where $n \geq 2$, and edges the sets $\{i, j\} \subseteq [n]$ such that $j - i \equiv 1$ or $n \pmod{2n}$.
    (d) The simple graph with set of vertices $[n]$, where $n \geq 3$, and edges the sets $\{i, j\} \subseteq [n]$ such that $j - i \equiv 1$ or $2 \pmod{n}$.
34. For $n \in \mathbb{N}$, compute the chromatic number of the simple graph with vertices the 2^n subsets of $[n]$ and edges the pairs of subsets $\{S, T\}$ such that $S \cap T = \varnothing$.
35. Let $(P, \preceq)$ be a finite partially ordered set.
    (a) Let G_P be the simple graph with set of vertices P and edges the pairs $\{x, y\}$ of elements of P such that $x \prec y$. Prove that $\chi(G_P) = \text{height}(P)$.
    (b) Let G_n be the simple graph with vertices the 2^n subsets of $[n]$ and edges the pairs of subsets $\{S, T\}$ such that $S \subset T$. Deduce that $\chi(G_n) = n + 1$ for every $n \in \mathbb{N}$.
36. Let G be a simple graph with chromatic number $\chi(G) = k$.
    (a) Prove that G has at least $\binom{k}{2}$ edges.
    (b) Give an example of such a graph with exactly $\binom{k}{2}$ edges.

37. Consider a simple bipartite graph with thirteen vertices, one of which has degree 4, another has degree 5, and all other vertices have degree 3.

    (a) Give an example of such a graph.
    (b) Is there an example of such a graph, for which the two vertices of degrees 4 and 5 are adjacent?

38. The plane is dissected into regions by finitely many straight lines. Prove that each of these regions can be colored black or white, so that adjacent regions (meaning, regions that share a common line segment of positive length on their boundary) have different color.

39. Let $G = (V, E)$ be a simple graph with n vertices. We denote by $\alpha(G)$ the largest cardinality of an independent subset of V (meaning, a subset of V that consists of pairwise nonadjacent vertices).

    (a) Prove that $n/\alpha(G) \leq \chi(G) \leq n - \alpha(G) + 1$.
    (b) Given positive integers a, q with $a + q = n + 1$, give an example of a graph G such that $\alpha(G) = a$ and $\chi(G) = q$.
    (c) Given positive integers a, q with $aq = n$, give an example of a graph G such that $\alpha(G) = a$ and $\chi(G) = q$.

40. Let $G = (V, E)$ be a simple graph with n vertices. We denote by $\bar{G} = (V, \bar{E})$ the simple graph on the same vertex set, with the complementary set of edges $\bar{E} = \binom{V}{2} \setminus E$.

    (a) Prove that $\chi(G) \cdot \chi(\bar{G}) \geq n$.
    (b) Prove that $\chi(G) + \chi(\bar{G}) \leq n + 1$.
    (c) What are the possible values of $\chi(G)$, if $\bar{G}$ has two edges?

41. Let n, k be positive integers such that $n \geq 2k$. We denote by $\mathrm{KG}_{n,k}$ the simple graph with vertices the k-subsets of $[n]$ in which two k-subsets S and T are adjacent if $S \cap T = \varnothing$.

    (a) Prove that $\chi(\mathrm{KG}_{n,k}) \leq n - 2k + 2$.
    (b) Prove that $\chi(\mathrm{KG}_{n,k}) = n - 2k + 2$.

42. Compute the number of perfect matchings of:

    (a) The complete graph K_n.
    (b) The complete bipartite graph $K_{m,n}$.
    (c) The graph obtained from $K_{n,n}$ by deleting the edges that belong to a given perfect matching of $K_{n,n}$.
    (d) The simple graph with vertices $0, 1, \ldots, n$ and edges $\{0, i\}$ for $i \in [n]$ and $\{i, i + 1\}$ for $i \in [n - 1]$.

43. How many perfect matchings can:

    (a) a tree have?
    (b) a simple connected graph with as many vertices as edges have?

44. Let n be an even positive integer.
    (a) Give an example of a simple graph with n vertices and $(n-1)(n-2)/2$ edges that does not have a perfect matching.
    (b) Find $(n-1)!$ perfect matchings of the complete graph K_n with the following property: every edge of K_n belongs to exactly $(n-2)!$ of these perfect matchings.
    (c) Prove that every simple graph with n vertices and more than $(n-1)(n-2)/2$ edges has a perfect matching.
45. Let G be a graph with n vertices and let $d \in \mathbb{Z}_{>0}$. If $1 \leq \deg_G(v) \leq d$ for every vertex v of G, prove that there exists a matching of G of cardinality at least $n/(d+1)$.
46. Prove the implication $(ii) \Rightarrow (i)$ of Theorem 5.4.1 as follows. Consider the vertex set $V = A \cup B$ of G partially ordered by declaring that $x \prec y$ if $x \in A$, $y \in B$, and $\{x, y\}$ is an edge of G.
    (a) Assuming condition (ii) of Theorem 5.4.1, prove that $\text{width}(V) = \#B$.
    (b) Apply Dilworth's Theorem (Theorem 4.4.2) to V and show that condition (i) of Theorem 5.4.1 holds as well.
47. In a medieval village, there are n men and m women. A dragon takes one of the women. Let $A_1, A_2, \ldots, A_n$ be the sets of women that the n men know, respectively. Find a necessary and sufficient condition on these sets so that regardless of who the woman taken by the dragon is, the men can marry simultaneously, everyone with one of the remaining women that they know.
48. There are n boys and n girls at a party. We assume that a boy a knows a girl b if and only if b knows a (for all a and b) and that there exists $k \in \{0, 1, \ldots, n\}$ with the following property: every boy knows at least $n-k$ girls and every girl knows at least k boys. Prove that the boys can dance tango simultaneously, every one with a girl they know.
49. Given a positive integer n, find the largest $m \in \mathbb{N}$ with the following property: every graph that can be obtained from the complete bipartite graph $K_{n,n}$ by deleting m or fewer edges has a perfect matching.
50. Let G be a (finite) simple bipartite graph and let $\{A, B\}$ be a partition of the set of vertices, as in Definition 5.4.2. We assume that G has a perfect matching and that every vertex in A has degree at least 2 in G.
    (a) Prove that G has at least two perfect matchings.
    (b) Given an integer $n \geq 2$, give an example of such a graph with $2n$ vertices that has exactly two perfect matchings.
51. Let $m, n \in \mathbb{Z}_{>0}$.
    (a) Prove that there exists a simple bipartite graph with $2n$ vertices and m edges that has exactly one perfect matching if and only if $n \leq m \leq n(n+1)/2$.
    (b) Prove that there exists a simple graph with $2n$ vertices and m edges that has exactly one perfect matching if and only if $n \leq m \leq n^2$.

52. Let $G = (V, E)$ be a simple bipartite graph, and let $\{A, B\}$ be a partition of V, as in Definition 5.4.2. Assume that $\#A = n \leq \#B$.

    (a) If $\deg_G(v) \geq \lfloor (n+1)/2 \rfloor$ for every $v \in V$, prove that G has an n-element matching.
    (b) For $n \in \mathbb{Z}_{>0}$, give an example of such a graph without n-element matchings for which $\deg_G(v) \geq \lfloor (n-1)/2 \rfloor$ for every $v \in V$.

53. Let $G = (V, E)$ be a (finite) simple bipartite graph, and let $\{A, B\}$ be a partition of V, as in Definition 5.4.2. Let $\mathcal{I}(G)$ be the family of subsets $S \subseteq A$ with the following property: S consists of the endpoints of the edges of some matching of G which belong to A. Prove that if $S, T \in \mathcal{I}(G)$ and S has fewer elements than T, then there exists $a \in (T \smallsetminus S)$ such that $S \cup \{a\} \in \mathcal{I}(G)$.
54. Prove the following:

    (a) If two simple graphs are isomorphic, then their chromatic polynomials are equal. Does the converse hold?
    (b) Any two trees with the same number of vertices have equal chromatic polynomials.

55. (a) Give an example of a simple graph G with chromatic polynomial $\chi_G(x) = x(x-1)^3(x-2)^2$.
    (b) Does there exist a simple graph G with chromatic polynomial $\chi_G(x) = x^6 - 10x^5 + 25x^4 - 23x^3 + 11x^2 - 3x$?
    (c) Does there exist a simple graph G with chromatic polynomial $\chi_G(x) = x(x-1)^2(x-2)^2(x-4)$?
    (d) Does there exist a simple graph G with chromatic polynomial $\chi_G(x) = x^5 - 6x^4 + 14x^3 - 15x^2 + 6x$?
56. Which of the following statements are true and which are false?

    (a) There exist two nonisomorphic simple graphs with chromatic polynomial $x(x-1)^2(x-2)$.
    (b) There exist two nonisomorphic simple graphs with chromatic polynomial $x(x-1)^2(x-2)^2$.

57. Compute the chromatic polynomial of the n-cycle for every $n \geq 3$.
58. Let G be a simple graph with n vertices and let $k \in [n]$. If $\chi(G) \geq k$, prove that $\chi_G(x) = x(x-1)\cdots(x-k+1)p_G(x)$ for some polynomial $p_G(x)$ of degree $n-k$ with integer coefficients:

    (a) By applying Proposition 5.5.1.
    (b) By applying standard properties of roots of polynomials.

59. Let $\chi_G(x) = \sum_{i=0}^{n} a_i x^i$ be the chromatic polynomial of a simple graph with n vertices, where $a_0 = 0$ and $a_n = 1$. Prove that G is connected if and only if $a_1 \neq 0$.

60. Which of the following statements are true and which are false?

    (a) If G is a simple graph such that $\chi_G(x) = x(x-1)^{n-1}$ for some $n \in \mathbb{Z}_{>0}$, then G is bipartite.
    (b) If G is a simple graph such that $\chi_G(x) = x(x-1)^{n-1}$ for some $n \in \mathbb{Z}_{>0}$, then G is a tree.

61. A simple graph $G = (V, E)$ is called *chordal* if there is no $U \subseteq V$ such that the induced subgraph of G on the vertex set U is a cycle of length greater than 3. Let n be the number of vertices of G.

    (a) Prove that G is chordal if and only if there exists a permutation $(v_1, v_2, \ldots, v_n)$ of V with the following property: if $1 \leq i < j < k \leq n$ and $\{v_i, v_k\}, \{v_j, v_k\} \in E$, then $\{v_i, v_j\} \in E$.
    (b) Prove that if G is chordal, then there exist nonnegative integers $e_1, e_2, \ldots, e_n$ such that $\chi_G(x) = (x - e_1)(x - e_2) \cdots (x - e_n)$.

62. Let G be a simple graph on the vertex set $[n]$ with m edges. We denote by $a_k(G)$ the number of acyclic orientations of G for which exactly k edges e of G are directed so that the initial vertex of e is smaller than the terminal vertex, in the usual order on the integers. We set

$$a(G, x) = \sum_{k=0}^{m} a_k(G) x^k.$$

    (a) Prove that $a(G, 1)$ is equal to the number of acyclic orientations of G.
    (b) Prove that $a_0(G) = 1$ and that $a_k(G) = a_{m-k}(G)$ for every $k \in \{0, 1, \ldots, m\}$.
    (c) Compute $a(G, x)$, if G is a tree.
    (d) Is it true that $a_0(G) \leq a_1(G) \leq \cdots \leq a_{\lfloor m/2 \rfloor}(G)$ for every graph G?

63. Which of the following statements are true and which are false?

    (a) Every simple planar graph with five vertices and chromatic number 3 contains K_3 as a subgraph.
    (b) Every simple planar graph with six vertices and chromatic number 4 contains K_4 as a subgraph.

64. What is the smallest number of edges that must be deleted from the complete graph K_6 in order to obtain a planar graph?
65. Given a positive integer n and $k \in [n-1]$, we consider the simple graph with set of vertices $[n]$ and edges the pairs $\{a, b\}$ with $a, b \in [n]$ and $a < b \leq a + k$. For which ordered pairs (n, k) is this graph planar?
66. Find all:

    (a) Integers $n \geq 3$ such that the graph obtained from K_n by deleting the edges of an n-cycle is planar.

(b) Positive integers n such the graph obtained from K_{2n} by deleting the edges of a perfect matching is planar.

67. Let $\ell \geq 3$ be an integer.

(a) Let G be a simple connected plane graph with at least $(\ell + 2)/2$ vertices that does not have cycles of length less than ℓ. Prove that

$$q(G) \leq \frac{\ell}{\ell - 2}(p(G) - 2),$$

where $p(G)$ and $q(G)$ are the numbers of vertices and edges of G, respectively.

(b) Is Petersen's graph (see Example 5.1.2 (e)) planar?

68. Prove that every simple planar graph with more than one vertex has at least two vertices of degree less than 6.

Chapter 6
Generating Functions



This chapter briefly presents a powerful tool for solving enumeration problems, the method of generating functions. This is a tool of primary importance in mathematics. We restrict ourselves to applications and examples coming, for the most part, from the enumeration of integer compositions, integer partitions, permutations, words, and set partitions.

6.1 Definitions and Examples

We employ a specific example in order to introduce the method of generating functions.

6.1.1 Fibonacci Numbers

Let us denote by G_n the $n \times 2$ grid graph and by a_n the number of perfect matchings of G_n. For example, $a_1 = 1$, $a_2 = 2$, $a_3 = 3$, $a_4 = 5$, and $a_5 = 8$. Figure 6.1 shows G_n and its five perfect matchings for $n = 4$. We seek to compute a_n, as a function of n.

Let $u_1, \ldots, u_n, v_1, \ldots, v_n$ be the vertices of G_n, numbered so that the pairs $\{u_i, v_i\}$ for $i \in [n]$ and $\{u_i, u_{i+1}\}, \{v_i, v_{i+1}\}$ for $i \in [n-1]$ are the edges of G_n. To compute a_n, we reason as follows. A perfect matching of G_n must contain exactly one of the edges $\{u_n, v_n\}$ and $\{u_{n-1}, u_n\}$. The number of perfect matchings of G_n which contain $\{u_n, v_n\}$ clearly equals a_{n-1}. The number of those perfect matchings that contain $\{u_{n-1}, u_n\}$ is equal to a_{n-2}, since every such matching must contain the

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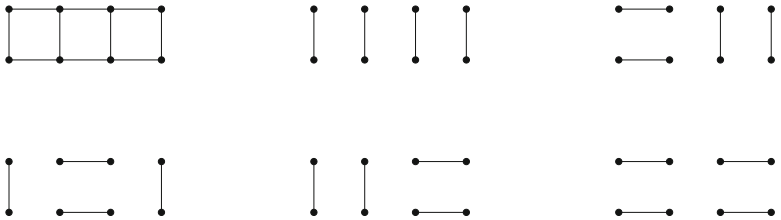


Figure 6.1 Five perfect matchings

edge $\{v_{n-1}, v_n\}$ as well. We conclude that the recurrence

$$a_n = a_{n-1} + a_{n-2} \tag{6.1}$$

holds for $n \geq 2$, where $a_0 = 1$ by convention (so that $(a_n)_{n\in\mathbb{N}}$ is the Fibonacci sequence of Example 1.1.10). The next step constitutes the basic idea of the method of generating functions: we consider the power series

$$F(x) = \sum_{n\geq 0} a_n x^n = a_0 + a_1 x + a_2 x^2 + \cdots \tag{6.2}$$

and attempt to compute it. We observe that

$$\begin{aligned} F(x) &= 1 + x + \sum_{n\geq 2} a_n x^n = 1 + x + \sum_{n\geq 2} (a_{n-1} + a_{n-2}) x^n \\ &= 1 + x + x \sum_{n\geq 2} a_{n-1} x^{n-1} + x^2 \sum_{n\geq 2} a_{n-2} x^{n-2} \\ &= 1 + x + x\,(F(x) - 1) + x^2 F(x) \\ &= 1 + (x + x^2) F(x) \end{aligned}$$

and conclude that

$$F(x) = \frac{1}{1 - x - x^2}. \tag{6.3}$$

We may now expand the right-hand side of Equation (6.3) as a power series in x in various ways and obtain formulas for a_n. For example, expanding the right-hand side as a geometric series and applying the Binomial Theorem, we find that

$$\sum_{n\geq 0} a_n x^n = \frac{1}{1 - (x + x^2)} = \sum_{k\geq 0} (x + x^2)^k = \sum_{k\geq 0} x^k (1 + x)^k = \sum_{k\geq 0} \sum_{i=0}^{k} \binom{k}{i} x^{k+i}.$$

Comparing the coefficients of x^n in the two extreme sides of this string of equalities, we get

$$a_n = \sum_{k+i=n} \binom{k}{i} = \sum_{i=0}^{\lfloor n/2 \rfloor} \binom{n-i}{i}. \tag{6.4}$$

Moreover, writing

$$\frac{1}{1-x-x^2} = \frac{1}{(1-\tau x)(1-\bar{\tau} x)} = \frac{1}{x\sqrt{5}} \left(\frac{1}{1-\tau x} - \frac{1}{1-\bar{\tau} x} \right),$$

where $\tau = (1+\sqrt{5})/2$ and $\bar{\tau} = (1-\sqrt{5})/2$, and expanding $(1-\tau x)^{-1}$ and $(1-\bar{\tau} x)^{-1}$ as geometric series, we find that

$$\sum_{n \geq 0} a_n x^n = \frac{1}{x\sqrt{5}} \sum_{n \geq 0} (\tau^n - \bar{\tau}^n) x^n.$$

As before, this equality yields that

$$a_n = \left(\tau^{n+1} - \bar{\tau}^{n+1}\right) / \sqrt{5} \tag{6.5}$$

for $n \in \mathbb{N}$. For other combinatorial interpretations of a_n, see Exercise 39 of Chapter 2 and Example 6.2.2.

The following section explains the precise context within which the computations made in this example are valid.

6.1.2 *Formal Power Series*

The following definition is implicit in Section 6.1.1.

Definition 6.1.1. *Let $(a_n)_{n \in \mathbb{N}}$ be a sequence of complex numbers. The formal power series*

$$\sum_{n \geq 0} a_n x^n = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots$$

is called the (ordinary) generating function of (a_n).

EXAMPLE. Let a_n be the number of subsets of $[n]$. Then, $a_n = 2^n$ for $n \in \mathbb{N}$ and

$$\sum_{n \geq 0} a_n x^n = \sum_{n \geq 0} 2^n x^n = \frac{1}{1-2x}.$$

□

As the term "formal power series" suggests, we consider power series in an algebraic context and disregard issues of convergence. For example, $\sum_{n\geq 0}(n+1)^n x^n$ is a perfectly defined formal power series, even though it converges only for $x = 0$ as a power series studied in calculus, and thus can be defined only for $x = 0$ as a function. To be more specific, we denote by

$$\mathbb{C}[[x]] = \left\{ \sum_{n\geq 0} a_n x^n : a_n \in \mathbb{C} \right\} \tag{6.6}$$

the set of all power series in x with coefficients from $\mathbb{C}$. For elements $F(x) = \sum_{n\geq 0} a_n x^n$ and $G(x) = \sum_{n\geq 0} b_n x^n$ of $\mathbb{C}[[x]]$, we define that $F(x) = G(x)$ if and only if $a_n = b_n$ for every $n \in \mathbb{N}$. Thus, $F(x)$ is essentially defined as the sequence $(a_n)_{n\in\mathbb{N}}$, which we chose to represent in a different way as an element of $\mathbb{C}[[x]]$. We say that a_n is the *coefficient of* x^n in $F(x)$ and denote it by $[x^n]F(x)$. In particular, $F(0) := a_0$ is the *constant term* of $F(x)$. We define the operations of addition and multiplication in $\mathbb{C}[[x]]$ by setting

$$\left(\sum_{n\geq 0} a_n x^n\right) + \left(\sum_{n\geq 0} b_n x^n\right) = \sum_{n\geq 0}(a_n + b_n)x^n \tag{6.7}$$

and

$$\begin{aligned}\left(\sum_{n\geq 0} a_n x^n\right)\left(\sum_{n\geq 0} b_n x^n\right) &= \left(a_0 + a_1 x + a_2 x^2 + \cdots\right)\left(b_0 + b_1 x + b_2 x^2 + \cdots\right) \\ &= a_0 b_0 + (a_0 b_1 + a_1 b_0)x + (a_0 b_2 + a_1 b_1 + a_2 b_0)x^2 \\ &\quad + (a_0 b_3 + a_1 b_2 + a_2 b_1 + a_3 b_0)x^3 + \cdots \\ &= \sum_{n\geq 0} c_n x^n, \end{aligned} \tag{6.8}$$

where

$$c_n = a_0 b_n + a_1 b_{n-1} + \cdots + a_n b_0 = \sum_{k=0}^{n} a_k b_{n-k}. \tag{6.9}$$

These operations obey the usual laws (associativity, commutativity, distributivity) that are valid for addition and multiplication of polynomials. In the language of algebra (see Appendix A), $\mathbb{C}[[x]]$ is a commutative ring with zero element and unit

$$\begin{aligned} 0 &:= 0 + 0 \cdot x + 0 \cdot x^2 + 0 \cdot x^3 + \cdots \\ 1 &:= 1 + 0 \cdot x + 0 \cdot x^2 + 0 \cdot x^3 + \cdots, \end{aligned}$$

respectively. Moreover, the ring $\mathbb{C}[[x]]$ becomes a $\mathbb{C}$-algebra and $\mathbb{C}$-vector space by setting $\lambda \cdot F(x) = \sum_{n\geq 0}(\lambda a_n)x^n$ for $F(x) = \sum_{n\geq 0} a_n x^n \in \mathbb{C}[[x]]$ and $\lambda \in \mathbb{C}$.

EXAMPLE. If

$$\begin{aligned} F(x) &= 1 + x + x^2 + x^3 + \cdots = \sum_{n\geq 0} x^n, \\ G(x) &= 1 - x + x^2 - x^3 + \cdots = \sum_{n\geq 0} (-1)^n x^n, \\ H(x) &= 1 + x^2 + x^4 + \cdots = \sum_{n\geq 0} x^{2n}, \end{aligned} \tag{6.10}$$

then

$$\begin{aligned} F(x) + G(x) &= 2 + 0 \cdot x + 2 \cdot x^2 + 0 \cdot x^3 + \cdots = 2H(x), \\ F(x)G(x) &= 1 + (-1+1)x + (1-1+1)x^2 + (-1+1-1+1)x^3 + \cdots \\ &= 1 + x^2 + x^4 + \cdots = H(x). \end{aligned}$$

Example 6.1.1. Setting $b_n = 1$ for $n \in \mathbb{N}$ in Equations (6.8) and (6.9), we get

$$\begin{aligned} \sum_{n\geq 0}(a_0 + a_1 + \cdots + a_n)x^n &= \left(\sum_{n\geq 0} x^n\right)\left(\sum_{n\geq 0} a_n x^n\right) \\ &= \frac{1}{1-x} \cdot \left(\sum_{n\geq 0} a_n x^n\right). \end{aligned} \tag{6.11}$$

The calculus of formal power series has many similarities with those of polynomials and of power series, regarded as functions. For example, with mild assumptions, we may divide formal power series and add and multiply the respective quotients as we do with rational functions. We may also differentiate formal power series and apply the usual rules for differentiation. As far as the calculus of formal power series is concerned, in this book we confine ourselves to the following definitions and examples. For an extensive discussion, the reader is referred to [23, Section 1.1].

A formal power series $F(x) \in \mathbb{C}[[x]]$ is called *invertible* if there exists $G(x) \in \mathbb{C}[[x]]$ (called the *inverse* of $F(x)$) such that $F(x)G(x) = 1 := 1 + 0 \cdot x + 0 \cdot x^2 + \cdots$ in $\mathbb{C}[[x]]$. It can be proved (see Exercise 4) that $F(x) \in \mathbb{C}[[x]]$ is invertible if and only if $F(0) \neq 0$. Given $F(x)$, $G(x)$, $H(x) \in \mathbb{C}[[x]]$, we write $F(x)/G(x) = H(x)$ if $F(x) = G(x)H(x)$ in $\mathbb{C}[[x]]$ and $G(x)$ is invertible in $\mathbb{C}[[x]]$. The formal power series of the form $F(x)/G(x)$, when defined, obey the usual rules for addition and multiplication of rational functions of x.

Example 6.1.2. Let $\alpha \in \mathbb{C}$. According to the defining equation (6.8),

$$\begin{aligned}(1-\alpha x)\sum_{n\geq 0}\alpha^n x^n &= (1-\alpha x)\cdot(1+\alpha x+\alpha^2x^2+\cdots)\\ &= 1+(\alpha-\alpha)x+(\alpha^2-\alpha\cdot\alpha)x^2+\cdots = 1\end{aligned}$$

and hence

$$\sum_{n\geq 0}\alpha^n x^n = \frac{1}{1-\alpha x}. \tag{6.12}$$

Similarly, we find that

$$\sum_{n\geq 0}x^{kn} := 1+x^k+x^{2k}+\cdots = \frac{1}{1-x^k} \tag{6.13}$$

for every positive integer k.

EXAMPLE. Let $F(x), G(x), H(x) \in \mathbb{C}[[x]]$ be as in (6.10). Setting $\alpha = 1$ and $\alpha = -1$ in (6.12) we find that $F(x) = 1/(1-x)$ and $G(x) = 1/(1+x)$, respectively. Adding and multiplying these rational function in the usual way, and applying (6.13) for $k = 2$, we verify that

$$\begin{aligned}F(x)+G(x) &= 1/(1-x)+1/(1+x) = 2/(1-x^2)\\ &= 2\,(1+x^2+x^4+\cdots) = 2H(x),\\ F(x)G(x) &= 1/(1-x)(1+x) = 1/(1-x^2) = 1+x^2+x^4+\cdots = H(x).\end{aligned}$$

Example 6.1.3. Let (a_n) be the sequence defined by setting $a_0 = 0$ and $a_{n+1} = 2a_n + (-1)^n$ for $n \geq 0$. For example, $a_n = 0, 1, 1, 3, 5, 11$ for $n = 0, 1, 2, 3, 4, 5$, respectively. In order to compute a_n for any $n \in \mathbb{N}$, we consider the generating function $F(x) = \sum_{n\geq 0} a_n x^n$. Multiplying the equality $a_{n+1} = 2a_n + (-1)^n$ by x^{n+1} and summing for $n \geq 0$, we get

$$\sum_{n\geq 0}a_{n+1}x^{n+1} = 2x\sum_{n\geq 0}a_n x^n + x\sum_{n\geq 0}(-1)^n x^n = 2xF(x)+\frac{x}{1+x}.$$

We observe that $\sum_{n\geq 0} a_{n+1}x^{n+1} = F(x) - a_0 = F(x)$ and conclude that $F(x) = 2xF(x) + x/(1+x)$. Solving this equation for $F(x)$, we find that

$$F(x) = \frac{x}{(1+x)(1-2x)}.$$

Working as in Section 6.1.1, we find that

$$F(x) = \frac{1}{3}\left(\frac{1}{1-2x} - \frac{1}{1+x}\right) = \frac{1}{3}\left(\sum_{n\geq 0} 2^n x^n - \sum_{n\geq 0}(-1)^n x^n\right)$$
$$= \sum_{n\geq 0} \frac{2^n - (-1)^n}{3} x^n$$

and hence that $a_n = (2^n + (-1)^{n-1})/3$ for every $n \in \mathbb{N}$.

Definition 6.1.2. *The derivative of a formal power series* $F(x) = \sum_{n\geq 0} a_n x^n \in \mathbb{C}[[x]]$ *is defined as*

$$F'(x) = \sum_{n\geq 1} n a_n x^{n-1} = a_1 + 2a_2 x + 3a_3 x^2 + \cdots \in \mathbb{C}[[x]]. \tag{6.14}$$

For $F(x), G(x) \in \mathbb{C}[[x]]$, we have the usual rules for differentiation: $(F(x) + G(x))' = F'(x) + G'(x)$, $(F(x)G(x))' = F'(x)G(x) + F(x)G'(x)$, $(1/F(x))' = -F'(x)/(F(x))^2$, if $F(0) \neq 0$ and so on.

EXAMPLE. Let $F(x) = \sum_{n\geq 0} x^n = 1+x+x^2+\cdots$. According to Definition 6.1.2, $F'(x) = \sum_{n\geq 1} n x^{n-1} = 1 + 2x + 3x^2 + \cdots$. One can reach the same result by writing $F(x) = 1/(1-x)$ and using the standard rules for differentiation:

$$F'(x) = 1/(1-x)^2 = (1/(1-x))^2$$
$$= (1 + x + x^2 + \cdots)(1 + x + x^2 + \cdots) = 1 + 2x + 3x^2 + \cdots$$
$$= \sum_{n\geq 0} (n+1)x^n.$$

EXAMPLE. Setting $\alpha = 1$ in Equation (6.12), we get

$$\sum_{n\geq 0} x^n = \frac{1}{1-x}. \tag{6.15}$$

Differentiating with respect to x and multiplying by x, we get

$$\sum_{n\geq 0} n x^n = \frac{x}{(1-x)^2}. \tag{6.16}$$

Repeating this process twice gives that

$$\sum_{n\geq 0} n^2 x^n = \frac{x(1+x)}{(1-x)^3} \tag{6.17}$$

and

$$\sum_{n\geq 0} n^3 x^n = \frac{x(1+4x+x^2)}{(1-x)^4}. \tag{6.18}$$

Example 6.1.4. Differentiating k times Equation (6.15) with respect to x, we get

$$\sum_{n\geq k} n(n-1)\cdots(n-k+1)x^{n-k} = \frac{k!}{(1-x)^{k+1}}.$$

Replacing n by $n+k$ and dividing with $k!$ yields the identity

$$\sum_{n\geq 0} \binom{n+k}{k} x^n = \frac{1}{(1-x)^{k+1}} \tag{6.19}$$

for $k \in \mathbb{N}$. For example, for $k = 1$, we get the identity

$$\sum_{n\geq 0} (n+1)x^n = \frac{1}{(1-x)^2} \tag{6.20}$$

we have also found before.

Example 6.1.5. We are given sequences $(a_n)_{n\in\mathbb{N}}$ and $(b_n)_{n\in\mathbb{N}}$ such that $a_n = (2^n + (-1)^{n-1})/3$ and

$$\sum_{k=0}^{n} a_k b_{n-k} = n2^{n-1} \tag{6.21}$$

for $n \in \mathbb{N}$. In order to compute b_n, we consider the generating functions $F(x) = \sum_{n\geq 0} a_n x^n$ and $G(x) = \sum_{n\geq 0} b_n x^n$ of these sequences. By Equation (6.21) and the definition (6.8) and (6.9) of the product of formal power series, we get

$$F(x)G(x) = \sum_{n\geq 0} n2^{n-1}x^n = \sum_{n\geq 1} n2^{n-1}x^n = x\sum_{n\geq 0}(n+1)2^n x^n = \frac{x}{(1-2x)^2},$$

where the last equality follows from Equation (6.20) by replacing x by $2x$. Moreover,

$$F(x) = \frac{x}{(1+x)(1-2x)}$$

by Example 6.1.3. We conclude that

$$G(x) = \frac{1+x}{1-2x} = (1+x)\sum_{n\geq 0} 2^n x^n = 1 + \sum_{n\geq 1}(2^n + 2^{n-1})x^n$$

and hence that $b_0 = 1$ and $b_n = 3 \cdot 2^{n-1}$ for $n \geq 1$.

Example 6.1.6. Let us compute the sum

$$f(n) = 1^3 + 2^3 + \cdots + n^3 = \sum_{k=0}^{n} k^3$$

for $n \in \mathbb{N}$, using the method of generating functions. Applying Equations (6.11), (6.18), and (6.19), we find that

$$\begin{aligned}\sum_{n\geq 0} f(n)x^n &= \sum_{n\geq 0}\left(\sum_{k=0}^{n} k^3\right)x^n = \frac{1}{1-x}\cdot\sum_{n\geq 0} n^3 x^n \\ &= \frac{x(1+4x+x^2)}{(1-x)^5} = (x+4x^2+x^3)\sum_{n\geq 0}\binom{n+4}{4}x^n \\ &= \sum_{n\geq 0}\binom{n+4}{4}(x^{n+1}+4x^{n+2}+x^{n+3})\end{aligned}$$

and conclude that

$$f(n) = \binom{n+3}{4} + 4\binom{n+2}{4} + \binom{n+1}{4} = n^2(n+1)^2/4$$

for every $n \in \mathbb{N}$, in agreement with Example 4.3.1 of Chapter 4.

Example 6.1.7. Let us compute the sum

$$s_n := \sum_{k=0}^{\lfloor n/2\rfloor}(-1)^k\binom{n-k}{k} = \binom{n}{0} - \binom{n-1}{1} + \binom{n-2}{2} - \cdots$$

for $n \in \mathbb{N}$, by use of the generating function $\sum_{n\geq 0} s_n x^n$. We have

$$\begin{aligned}\sum_{n\geq 0} s_n x^n &= \sum_{n\geq 0}\sum_{k=0}^{\lfloor n/2\rfloor}(-1)^k\binom{n-k}{k}x^n = \sum_{k\geq 0}\sum_{n\geq 2k}(-1)^k\binom{n-k}{k}x^n \\ &= \sum_{k\geq 0}\sum_{m\geq 0}(-1)^k\binom{m+k}{k}x^{m+2k} = \sum_{k\geq 0}(-x^2)^k\sum_{m\geq 0}\binom{m+k}{k}x^m.\end{aligned}$$

Applying Equation (6.19), we find that

$$\begin{aligned}\sum_{n\geq 0} s_n x^n &= \sum_{k\geq 0}(-x^2)^k \frac{1}{(1-x)^{k+1}} = \frac{1}{1-x}\sum_{k\geq 0}\left(\frac{-x^2}{1-x}\right)^k \\ &= \frac{1}{1-x}\cdot\frac{1}{1+x^2/(1-x)} = \frac{1}{1-x+x^2} = \frac{1+x}{1+x^3} \\ &= (1+x)\sum_{m\geq 0}(-1)^m x^{3m} = \sum_{m\geq 0}(-1)^m (x^{3m}+x^{3m+1})\end{aligned}$$

and deduce that $s_{3m} = s_{3m+1} = (-1)^m$ and $s_{3m+2} = 0$ for $m \in \mathbb{N}$.

6.1.3 Linear Recurrence Relations

This section applies the method of generating functions to study sequences defined by recurrences of a special form, thus generalizing the results of Section 6.1.1. We let $\alpha, \beta \in \mathbb{C}$ and consider a sequence $(a_n)_{n\in\mathbb{N}}$ of complex numbers for which

$$a_n = \alpha a_{n-1} + \beta a_{n-2} \tag{6.22}$$

for $n \geq 2$. We say that such a sequence satisfies a *homogeneous linear recurrence relation of second order with constant coefficients*. The equation

$$x^2 - \alpha x - \beta = 0 \tag{6.23}$$

is called the *characteristic equation* of (6.22). The special case $\alpha = \beta = 1$ yields the recurrence (6.1), with characteristic equation $x^2 - x - 1 = 0$.

Theorem 6.1.1. *Let $\alpha, \beta \in \mathbb{C}$ with $\beta \neq 0$, and let $\lambda, \mu \in \mathbb{C}$ be the roots of Equation (6.23). For a sequence $(a_n)_{n\in\mathbb{N}}$ of complex numbers, the following conditions are equivalent:*

(i) *$(a_n)_{n\in\mathbb{N}}$ satisfies the recurrence (6.22).*
(ii) *We have*

$$\sum_{n\geq 0} a_n x^n = \frac{p+qx}{1-\alpha x-\beta x^2} \tag{6.24}$$

for some $p, q \in \mathbb{C}$.

(iii) *There exist* $c, d \in \mathbb{C}$ *such that*

$$a_n = \begin{cases} c\lambda^n + d\mu^n, & \text{if } \lambda \neq \mu \\ (c + dn)\lambda^n, & \text{if } \lambda = \mu \end{cases} \tag{6.25}$$

for every $n \in \mathbb{N}$.

Proof. Suppose first that (i) holds. Then,

$$\begin{aligned}(1 - \alpha x - \beta x^2) \cdot \sum_{n\geq 0} a_n x^n &= (1 - \alpha x - \beta x^2) \cdot (a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots) \\ &= a_0 + (a_1 - \alpha a_0)x + \sum_{n\geq 2} (a_n - \alpha a_{n-1} - \beta a_{n-2}) x^n \\ &= a_0 + (a_1 - \alpha a_0)x.\end{aligned}$$

Hence, (6.24) holds with $p = a_0$ and $q = a_1 - \alpha a_0$. This shows that $(i) \Rightarrow (ii)$. To prove the implication $(ii) \Rightarrow (iii)$, we assume that (6.24) holds and observe that $x^2 - \alpha x - \beta = (x - \lambda)(x - \mu)$, whence $1 - \alpha x - \beta x^2 = (1 - \lambda x)(1 - \mu x)$. In case $\lambda \neq \mu$, we have

$$\sum_{n\geq 0} a_n x^n = \frac{p + qx}{(1 - \lambda x)(1 - \mu x)} = \frac{c}{1 - \lambda x} + \frac{d}{1 - \mu x},$$

where $c = (\lambda p + q)/(\lambda - \mu)$ and $d = (\mu p + q)/(\mu - \lambda)$. By Equation (6.12), we may conclude that

$$\sum_{n\geq 0} a_n x^n = c \cdot \sum_{n\geq 0} \lambda^n x^n + d \cdot \sum_{n\geq 0} \mu^n x^n = \sum_{n\geq 0} (c\lambda^n + d\mu^n) x^n$$

and that $a_n = c\lambda^n + d\mu^n$ for every $n \in \mathbb{N}$. Similarly, if $\lambda = \mu$, then

$$\sum_{n\geq 0} a_n x^n = \frac{p + qx}{(1 - \lambda x)^2} = (p + qx) \cdot \sum_{n\geq 0} (n + 1)\lambda^n x^n.$$

As a result, $a_0 = p$ and $a_n = (n + 1)p\lambda^n + nq\lambda^{n-1}$ for $n \geq 1$. Therefore, a_n can again be expressed by the formula (6.25) for suitable $c, d \in \mathbb{C}$. The implication $(iii) \Rightarrow (i)$ follows by direct verification of the linear recurrence (6.22); the details are left to the reader. □

EXAMPLE. Let $(a_n)_{n\in\mathbb{N}}$ be the sequence for which $a_0 = a_1 = 1$ and $a_n = a_{n-1} - a_{n-2}/4$ for $n \geq 2$, considered in Exercise 17 of Chapter 1. This sequence satisfies a homogeneous linear recurrence relation of second order with constant coefficients. Since its characteristic equation $x^2 - x + 1/4 = 0$ has the double root $\lambda = 1/2$,

according to Theorem 6.1.1 we have $a_n = (c + dn)\,2^{-n}$ for some $c, d \in \mathbb{C}$, for every $n \in \mathbb{N}$. From the initial conditions $a_0 = a_1 = 1$, we find that $c = d = 1$ and conclude that $a_n = (n + 1)/2^n$ for every $n \in \mathbb{N}$. □

Example 6.1.8. We color each one of the integers $1, 2, \ldots, n$ with one of three colors: blue, green, or red. Let us compute the number, say a_n, of those colorings for which no two consecutive integers have been colored blue.

We observe that out of the a_n colorings with the desirable properties, there are exactly $2a_{n-1}$ such that n does not have blue color, since then it must have either green or red color, and exactly $2a_{n-2}$ such that n has blue color, since then $n - 1$ must have either green or red color. It follows that $a_n = 2a_{n-1} + 2a_{n-2}$ for every $n \geq 2$, where $a_0 := 1$. The characteristic equation $x^2 - 2x - 2 = 0$ of this linear recurrence has roots $\lambda = 1 + \sqrt{3}$ and $\mu = 1 - \sqrt{3}$. Given that $a_0 = 1$ and $a_1 = 3$, applying Theorem 6.1.1 and Equation (6.25), we reach the formula (explain how)

$$a_n = \frac{1}{4\sqrt{3}}\left((1 + \sqrt{3})^{n+2} - (1 - \sqrt{3})^{n+2}\right)$$

for $n \in \mathbb{N}$. The same theorem also gives that

$$\sum_{n\geq 0} a_n x^n = \frac{1 + x}{1 - 2x - 2x^2}.$$

Working as in Section 6.1.1, we find that

$$\sum_{n\geq 0} a_n x^n = \frac{1 + x}{1 - 2x - 2x^2} = (1 + x)\sum_{m\geq 0}(2x + 2x^2)^m$$

$$= \sum_{m\geq 0} 2^m x^m (1 + x)^{m+1} = \sum_{m\geq 0}\sum_{k=0}^{m+1} 2^m \binom{m+1}{k} x^{m+k}$$

and conclude that

$$a_n = \sum_{m+k=n} 2^m \binom{m+1}{k} = \sum_{k=0}^{\lfloor (n+1)/2 \rfloor} \binom{n-k+1}{k} 2^{n-k}$$

for every $n \in \mathbb{N}$.

Here is another proof of our second formula for a_n. By the result of Exercise 32 of Chapter 2, there are $\binom{n-k+1}{k}2^{n-k}$ ways to choose k elements of the set $[n]$, no two consecutive, and color them blue and then color each of the remaining $n - k$ elements of $[n]$ green or red. The formula follows by summing over all k. □

More generally, we say that a sequence $(a_n)_{n\in\mathbb{N}}$ of complex numbers satisfies a *linear recurrence relation of order d* if there exist functions $\gamma_1, \gamma_2, \ldots, \gamma_d$ and β of n such that

$$a_n = \gamma_1(n)a_{n-1} + \gamma_2(n)a_{n-2} + \cdots + \gamma_d(n)a_{n-d} + \beta(n) \tag{6.26}$$

for every $n \geq d$. We also say that the recurrence is *homogeneous* if $\beta \equiv 0$ and that it has *constant coefficients* if $\gamma_1, \gamma_2, \ldots, \gamma_d$ and β are constant functions. Theorem 6.1.1 can be generalized to homogeneous linear recurrence relations of order d with constant coefficients as follows. We refer to [23, Section 4.1] for the proof, which is similar to that of Theorem 6.1.1.

Theorem 6.1.2. *Let a positive integer d and $\alpha_1, \alpha_2, \ldots, \alpha_d \in \mathbb{C}$ with $\alpha_d \neq 0$ be given. For a sequence $(a_n)_{n\in\mathbb{N}}$ of complex numbers, the following conditions are equivalent:*

(i) *$(a_n)_{n\in\mathbb{N}}$ satisfies the recurrence*

$$a_n = \alpha_1 a_{n-1} + \alpha_2 a_{n-2} + \cdots + \alpha_d a_{n-d}$$

for every $n \geq d$.

(ii) *We have*

$$\sum_{n\geq 0} a_n x^n = \frac{p_0 + p_1 x + \cdots + p_{d-1}x^{d-1}}{1 - \alpha_1 x - \alpha_2 x^2 - \cdots - \alpha_d x^d}$$

in $\mathbb{C}[[x]]$ for some $p_0, p_1, \ldots, p_{d-1} \in \mathbb{C}$.

(iii) *There exist polynomials $q_1(x), q_2(x), \ldots, q_k(x) \in \mathbb{C}[x]$ of degrees less than $d_1, \ldots, d_k$, respectively, such that*

$$a_n = q_1(n)\xi_1^n + q_2(n)\xi_2^n + \cdots + q_k(n)\xi_k^n$$

for every $n \in \mathbb{N}$, where $d_1, d_2, \ldots, d_k$ are positive integers and $\xi_1, \xi_2, \ldots, \xi_k$ are distinct complex numbers such that $x^d - \alpha_1 x^{d-1} - \alpha_2 x^{d-2} - \cdots - \alpha_d = \prod_{i=1}^{k}(x - \xi_i)^{d_i}$.

Other examples of sequences defined by homogeneous linear recurrence relations with constant coefficients appear in the exercises of this chapter. A sequence that satisfies a nonhomogeneous linear recurrence of the first order has already been studied in Example 6.1.3; another appears in Exercise 24.

6.2 Compositions

This section describes some basic applications of the method of generating functions to the enumeration of compositions of integers.

A *weak composition* (or *weak ordered partition*) of a nonnegative integer n is any sequence $\rho = (r_1, r_2, \ldots, r_k)$ with terms nonnegative integers which sum to n. As is the case with compositions (the terms of which are positive integers), the numbers r_i are called *parts* of ρ.

Proposition 6.2.1. *Let $a_k(n)$ (respectively, $b_k(n)$) be the number of weak compositions (respectively, compositions) of n with k parts. Then,*

$$\sum_{n\geq 0} a_k(n)x^n = \frac{1}{(1-x)^k}, \tag{6.27}$$

$$\sum_{n\geq 0} b_k(n)x^n = \frac{x^k}{(1-x)^k}. \tag{6.28}$$

Moreover, if $b(n)$ is the number of all compositions of n, where $b(0) := 0$, then

$$\sum_{n\geq 0} b(n)x^n = \frac{x}{1-2x}. \tag{6.29}$$

Proof. The numbers $a_k(n)$ can be alternatively defined by the identity

$$\sum_{n\geq 0} a_k(n)x^n = \sum_{r_i\geq 0} x^{r_1+r_2+\cdots+r_k}, \tag{6.30}$$

where the sum on the right-hand side runs through all sequences $(r_1, r_2, \ldots, r_k) \in \mathbb{N}^k$. Indeed, for every $n \in \mathbb{N}$, the coefficient of x^n in this sum is equal to the number of sequences $(r_1, r_2, \ldots, r_k) \in \mathbb{N}^k$ such that $r_1 + r_2 + \cdots + r_k = n$. By definition, this number is equal to $a_k(n)$. Factoring the right-hand side of (6.30), we get

$$\begin{aligned}\sum_{n\geq 0} a_k(n)x^n &= \sum_{r_i\geq 0} x^{r_1}x^{r_2}\cdots x^{r_k}\\ &= \left(\sum_{r_1\geq 0} x^{r_1}\right)\left(\sum_{r_2\geq 0} x^{r_2}\right)\cdots\left(\sum_{r_k\geq 0} x^{r_k}\right)\\ &= (1+x+x^2+\cdots)^k = \frac{1}{(1-x)^k}.\end{aligned}$$

This proves (6.27). A similar computation yields (6.28):

$$\sum_{n\geq 0} b_k(n)x^n = \sum_{r_i\geq 1} x^{r_1+r_2+\cdots+r_k} = \sum_{r_i\geq 1} x^{r_1}x^{r_2}\cdots x^{r_k}$$

$$= \left(\sum_{r_1\geq 1} x^{r_1}\right)\left(\sum_{r_2\geq 1} x^{r_2}\right)\cdots\left(\sum_{r_k\geq 1} x^{r_k}\right)$$

$$= (x + x^2 + x^3 + \cdots)^k = \frac{x^k}{(1-x)^k}.$$

Summing (6.28) over all $k \geq 1$ and observing that $b(n) = \sum_{k\geq 1} b_k(n)$ for $n \geq 1$, we get

$$\sum_{n\geq 0} b(n)x^n = \sum_{k\geq 1} \frac{x^k}{(1-x)^k} = -1 + \sum_{k\geq 0}\left(\frac{x}{1-x}\right)^k$$

$$= -1 + \frac{1}{1 - x/(1-x)} = \frac{x}{1-2x}.$$

This proves (6.29). □

REMARK. Equation (6.29) implies that

$$\sum_{n\geq 0} b(n)x^n = x\sum_{n\geq 0} 2^n x^n = \sum_{n\geq 0} 2^n x^{n+1} = \sum_{n\geq 1} 2^{n-1}x^n$$

and hence that $b(n) = 2^{n-1}$ for every positive integer n. Thus, we have given a new proof of the first claim of Proposition 2.3.9 using the method of generating functions. A similar argument shows, via Equation (6.19), that (6.27) and (6.28) are equivalent to the formulas $a_k(n) = \binom{n+k-1}{k-1}$ and $b_k(n) = \binom{n-1}{k-1}$, respectively (see Propositions 2.3.10 and 2.3.9). □

The method of proof of Proposition 6.2.1 can be applied to more general enumeration problems on compositions, where restrictions on their parts are imposed.

Example 6.2.1. Let a_n be the number of weak compositions (r_1, r_2, r_3, r_4) of n such that r_1 is even, r_2 is divisible by 3, $r_3 \in \{0, 1\}$ and $r_4 \in \{0, 2, 4\}$. For example, $a_4 = 4$ since the corresponding weak compositions of $n = 4$ are $(0, 0, 0, 4)$, $(0, 3, 1, 0)$, $(2, 0, 0, 2)$, and $(4, 0, 0, 0)$.

Let us denote by $\mathcal{R}$ the set of all quadruples $(r_1, r_2, r_3, r_4) \in \mathbb{N}^4$ with the properties just described and by $\mathcal{M}$ the set of quadruples $(m_1, m_2, m_3, m_4) \in \mathbb{N}^4$ such that $m_3 \in \{0, 1\}$ and $m_4 \in \{0, 2, 4\}$. Computing the generating function of the

sequence (a_n) as in the proof of Proposition 6.2.1, we find that

$$\begin{aligned}\sum_{n\geq 0} a_n x^n &= \sum_{(r_1,r_2,r_3,r_4)\in\mathcal{R}} x^{r_1+r_2+r_3+r_4} = \sum_{(m_1,m_2,m_3,m_4)\in\mathcal{M}} x^{2m_1+3m_2+m_3+m_4} \\ &= \left(\sum_{m_1\geq 0} x^{2m_1}\right)\left(\sum_{m_2\geq 0} x^{3m_2}\right)\left(\sum_{m_3\in\{0,1\}} x^{m_3}\right)\left(\sum_{m_4\in\{0,2,4\}} x^{m_4}\right) \\ &= \frac{1}{1-x^2}\cdot\frac{1}{1-x^3}\cdot(1+x)\cdot(1+x^2+x^4) = \frac{1-x+x^2}{(1-x)^2} \\ &= 1+\frac{x}{(1-x)^2} = 1+\sum_{n\geq 1} nx^n\end{aligned}$$

and conclude that $a_0 = 1$ and that $a_n = n$ for every $n \geq 1$. □

Example 6.2.2. Let $c(n)$ be the number of compositions of n with odd parts. We have $c(n) = 0, 1, 1, 2, 3$ for $n = 0, 1, 2, 3, 4$, respectively. For example, the compositions of $n = 4$ with odd parts are $(3, 1)$, $(1, 3)$, and $(1, 1, 1, 1)$. We wish to compute $c(n)$ for any $n \in \mathbb{N}$.

For that reason, we denote by $c_k(n)$ the number of compositions of n with k parts, each of which is odd. Working as in the proof of Proposition 6.2.1, we find that

$$\begin{aligned}\sum_{n\geq 0} c_k(n)x^n &= \sum_{r_i\in\{1,3,5,\dots\}} x^{r_1+r_2+\cdots+r_k} = \sum_{r_i\in\{1,3,5,\dots\}} x^{r_1}x^{r_2}\cdots x^{r_k} \\ &= \left(\sum_{r_1\in\{1,3,5,\dots\}} x^{r_1}\right)\left(\sum_{r_2\in\{1,3,5,\dots\}} x^{r_2}\right)\cdots\left(\sum_{r_k\in\{1,3,5,\dots\}} x^{r_k}\right) \\ &= (x+x^3+x^5+\cdots)^k = x^k(1+x^2+x^4+\cdots)^k \\ &= \frac{x^k}{(1-x^2)^k}.\end{aligned}$$

Summing over all $k \geq 1$ and observing that $c(0) = 0$ and that $c(n) = \sum_{k\geq 1} c_k(n)$ for $n \geq 1$, we get

$$\begin{aligned}\sum_{n\geq 0} c(n)x^n &= \sum_{k\geq 1}\frac{x^k}{(1-x^2)^k} = -1+\sum_{k\geq 0}\left(\frac{x}{1-x^2}\right)^k \\ &= -1+\frac{1}{1-x/(1-x^2)} = \frac{x}{1-x-x^2}.\end{aligned}$$

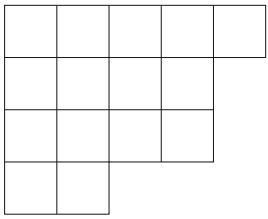


Figure 6.2 The Young diagram of (5, 4, 4, 2)

Comparing this result with Equations (6.2) and (6.3), we deduce that the numbers $c(n)$ are the terms of the Fibonacci sequence. Specifically, we have

$$\sum_{n\geq 0} c(n)x^n = x\sum_{n\geq 0} a_n x^n = \sum_{n\geq 1} a_{n-1}x^n,$$

where a_n is defined as in Section 6.1.1. Hence, $c(n)$ is equal to the number a_{n-1} of perfect matchings of the $(n-1)\times 2$ grid graph, for every $n \geq 1$. □

6.3 Integer Partitions

This section deals with problems on the enumeration of integer partitions. A *partition* of a positive integer n is defined as any sequence $\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_r)$ of positive integers $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_r$ that sum to n. The numbers λ_i are called *parts* of λ. We write $|\lambda| = n$ for their sum.

For example, $(5, 4, 4, 2)$ is a partition of $n = 15$ with four parts. Figure 6.2 shows the *Young diagram* of this partition, one of the most common ways to draw an integer partition. The Young diagram of $\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_r)$ consists of $n = |\lambda|$ unit squares, arranged in r left-justified rows, as shown in the figure. The ith row, counting from top to bottom, consists of λ_i squares.

We denote by $p(n)$ the number of partitions of n. Thus, $p(1) = 1$, $p(2) = 2$, $p(3) = 3$, $p(4) = 5$, $p(5) = 7$, and $p(6) = 11$. For example, the five partitions of $n = 4$ are (4), (3, 1), (2, 2), (2, 1, 1), and (1, 1, 1, 1). While only very complicated formulas for $p(n)$ are known (see, for instance, [3, Chapter 5]), the corresponding generating function can be computed as follows.

Proposition 6.3.1. *Setting $p(0) = 1$, we have*

$$\sum_{n\geq 0} p(n)x^n = \prod_{i\geq 1} \frac{1}{1-x^i} := \frac{1}{(1-x)(1-x^2)(1-x^3)\cdots}. \tag{6.31}$$

REMARK. For Equation (6.31) to make sense as an identity in $\mathbb{C}[[x]]$, we must explain how to define the infinite product that appears there as a formal power series. In other words, we must explain how the coefficient of x^n in this product is defined,

for $n \in \mathbb{N}$. For this purpose, we consider that the terms $(1-x^i)^{-1}$ of this product for $i \geq n+1$ do not contribute to the coefficient of x^n. This coefficient is defined to be the coefficient of x^n in the finite product

$$\frac{1}{(1-x)(1-x^2)\cdots(1-x^n)}.$$

Thus, the coefficients of $1, x, x^2, x^3, x^4$ can be computed as

$$\begin{aligned}\prod_{i\geq 1}\frac{1}{1-x^i} &= \frac{1}{1-x}\cdot\frac{1}{1-x^2}\cdot\frac{1}{1-x^3}\cdot\frac{1}{1-x^4}\cdots \\ &= (1+x+x^2+x^3+x^4+\cdots) \\ &\quad (1+x^2+x^4+x^6+\cdots)(1+x^3+x^6+\cdots)(1+x^4+x^8+\cdots)\cdots \\ &= 1+x+2x^2+3x^3+5x^4+\cdots\end{aligned}$$

Proof of Proposition 6.3.1. Let us denote by Λ_k the set of all partitions $\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_r)$ (including the unique partition without any parts) such that $\lambda_i \leq k$ for every i and by $p_k(n)$ the number of partitions of n that belong to Λ_k. According to our previous remark, it suffices to prove that

$$\sum_{n\geq 0} p_k(n)x^n = \frac{1}{(1-x)(1-x^2)\cdots(1-x^k)} \tag{6.32}$$

for every positive integer k, where $p_k(0) = 1$. Given an integer partition λ, let m_i be the number of parts of λ that are equal to i. For example, for $\lambda = (5,4,4,2)$ we have $m_2 = m_5 = 1$, $m_4 = 2$ and $m_i = 0$ for all other values of i. We observe that that map $\varphi_k : \Lambda_k \to \mathbb{N}^k$ with $\varphi_k(\lambda) = (m_1, m_2, \ldots, m_k)$ for $\lambda \in \Lambda_k$ is bijective and that $|\lambda| = m_1 + 2m_2 + \cdots + km_k$ for every $\lambda \in \Lambda_k$. As a result,

$$\begin{aligned}\sum_{n\geq 0} p_k(n)x^n &= \sum_{\lambda\in\Lambda_k} x^{|\lambda|} = \sum_{m_i\geq 0} x^{m_1+2m_2+\cdots+km_k} = \prod_{i=1}^{k}\left(\sum_{m_i\geq 0} x^{im_i}\right) \\ &= \prod_{i=1}^{k}\frac{1}{1-x^i}\end{aligned}$$

and the proof follows. □

EXAMPLE. According to Equation (6.32), the generating function for the number $p_2(n)$ of partitions of n with parts 1 or 2 can be expressed as

$$\sum_{n\geq 0} p_2(n)x^n = \frac{1}{(1-x)(1-x^2)} = (1+x+x^2+\cdots)\,(1+x^2+x^4+\cdots)$$
$$= 1+x+2x^2+2x^3+3x^4+3x^5+\cdots,$$

that is, $p_2(n) = \lfloor n/2 \rfloor + 1$ for every $n \in \mathbb{N}$. □

The method of proof of Proposition 6.3.1 can be applied to problems on the enumeration of integer partitions λ when restrictions are imposed on the multiplicities m_i with which positive integers appear as parts of λ.

Example 6.3.1. Let $o(n)$ be the number of partitions of n with odd parts and $q(n)$ be the number of partitions of n with distinct parts. Thus, $o(1) = q(1) = 1$, $o(2) = q(2) = 1$, $o(3) = q(3) = 2$, $o(4) = q(4) = 2$, and $o(5) = q(5) = 3$. For example, the three partitions of $n = 5$ with odd parts are (5), (3, 1, 1), and (1, 1, 1, 1, 1) and those with distinct parts are (5), (4, 1), and (3, 2). We will compute the generating functions of $o(n)$ and $q(n)$ and prove that $o(n) = q(n)$ for every $n \in \mathbb{N}$.

Given an integer partition λ, we let m_i be the number of parts of λ equal to i, as in the proof of Proposition 6.3.1, and Γ (respectively, Δ) be the set of integer partitions with odd (respectively, distinct) parts. Then, $\lambda \in \Gamma$ if and only if $m_i = 0$ for every even i, and $\lambda \in \Delta$ if and only if $m_i \in \{0, 1\}$ for every i. We conclude that

$$\sum_{n\geq 0} o(n)x^n = \sum_{\lambda\in\Gamma} x^{|\lambda|} = \sum_{m_i\geq 0} x^{m_1+3m_3+5m_5+\cdots}$$
$$= \left(\sum_{m_1\geq 0} x^{m_1}\right)\left(\sum_{m_3\geq 0} x^{3m_3}\right)\cdots = \frac{1}{1-x}\cdot\frac{1}{1-x^3}\cdots$$
$$= \prod_{j\geq 1}\frac{1}{1-x^{2j-1}}$$

and that

$$\sum_{n\geq 0} q(n)x^n = \sum_{\lambda\in\Delta} x^{|\lambda|} = \sum_{m_i\in\{0,1\}} x^{m_1+2m_2+3m_3+\cdots}$$
$$= \left(\sum_{m_1\in\{0,1\}} x^{m_1}\right)\left(\sum_{m_2\in\{0,1\}} x^{2m_2}\right)\cdots = (1+x)(1+x^2)\cdots$$
$$= \prod_{i\geq 1}(1+x^i),$$

where $o(0) = q(0) = 1$ by convention.

Corollary 6.3.1 (Euler, 1748). *We have $o(n) = q(n)$ for every $n \in \mathbb{N}$.*

Proof. We have

$$\sum_{n\geq 0} q(n)x^n = \prod_{i\geq 1}(1+x^i) = \prod_{i\geq 1}\frac{1-x^{2i}}{1-x^i} = \frac{(1-x^2)(1-x^4)(1-x^6)\cdots}{(1-x)(1-x^2)(1-x^3)\cdots}$$

$$= \frac{1}{(1-x)(1-x^3)(1-x^5)\cdots} = \sum_{n\geq 0} o(n)x^n$$

and hence $q(n) = o(n)$ for every $n \in \mathbb{N}$. □

6.4 Binomial Power Series and Catalan Numbers

The nth Catalan number was defined in Section 2.3.8, for $n \in \mathbb{N}$, as the number C_n of paths in $\mathbb{Z}^2$ with initial vertex $(0, 0)$, terminal vertex (n, n), and steps $(1, 0)$ or $(0, 1)$, which lie entirely in the half-plane $y \leq x$ of $\mathbb{R}^2$. By a simple combinatorial argument, we proved the recurrence (2.12)

$$C_n = \sum_{i=0}^{n-1} C_i C_{n-1-i}$$

for $n \geq 1$. Let us consider the generating function $C(x) = \sum_{n\geq 0} C_n x^n \in \mathbb{C}[[x]]$ of $(C_n)_{n\in\mathbb{N}}$. Multiplying (2.12) with x^n, summing over $n \geq 1$ and taking into account that $C_0 = 1$, we find that

$$C(x) - 1 = \sum_{n\geq 1} C_n x^n = x \sum_{n\geq 1}\left(\sum_{i=0}^{n-1} C_i C_{n-1-i}\right)x^{n-1} = x\sum_{n\geq 0}\left(\sum_{i=0}^{n} C_i C_{n-i}\right)x^n$$

$$= x\left(\sum_{n\geq 0} C_n x^n\right)^2 = x(C(x))^2,$$

that is,

$$x(C(x))^2 - C(x) + 1 = 0. \tag{6.33}$$

Rewriting this equation as

$$(1 - 2xC(x))^2 = 1 - 4x$$

and observing that the constant term of $1 - 2xC(x)$ is equal to 1 we conclude that $C(x) = (1 - \sqrt{1-4x})/2x$, where

$$\sqrt{1-4x} = 1 - 2x - 2x^2 - 4x^3 - 10x^4 - \cdots$$

is the unique element of $\mathbb{C}[[x]]$ with constant term 1, the square of which is equal to $1 - 4x$. Summarizing, we have proved the following statement.

Proposition 6.4.1. *The generating function of the sequence $(C_n)_{n\in\mathbb{N}}$ of Catalan numbers is given by the formula*

$$\sum_{n\geq 0} C_n x^n = \frac{1-\sqrt{1-4x}}{2x}. \tag{6.34}$$

Remark 6.4.1. The practical value of formula (6.34) comes from the fact that the functions $(1+x)^\alpha$, where $\alpha \in \mathbb{R}$, expand as

$$(1+x)^\alpha = 1 + \sum_{n\geq 1} \frac{\alpha(\alpha-1)\cdots(\alpha-n+1)}{n!}\, x^n \tag{6.35}$$

as power series, for $|x| < 1$.

Alternatively, we may consider that Equation (6.35) defines $(1+x)^\alpha$ as an element of $\mathbb{C}[[x]]$. Then, $(1+x)^\alpha$ is said to be a *binomial power series*, and as one can show, it has the basic properties of the corresponding function of calculus. For example, for $\alpha, \beta \in \mathbb{R}$ we have $(1+x)^\alpha(1+x)^\beta = (1+x)^{\alpha+\beta}$ in $\mathbb{C}[[x]]$. For an extensive discussion of this theme, we refer the reader to [24, Chapter 1]. The Binomial Theorem (Theorem 2.3.1) and formula (6.19) are special cases of (6.35).

Setting $\alpha = 1/2$, we find that the unique element of $\mathbb{C}[[x]]$ that has constant term 1 and square equal to $1 + x$ is the formal power series

$$\sqrt{1+x} = 1 + \sum_{n\geq 1} \frac{(1/2)(-1/2)\cdots(-n+3/2)}{n!}\, x^n.$$

Replacing x with $-4x$ in this equality yields that

$$\sqrt{1-4x} = 1 + \sum_{n\geq 1} \frac{(1/2)(-1/2)\cdots(-n+3/2)}{n!}\, (-4)^n x^n.$$

Computing the coefficient of x^n in the right-hand side of (6.34), we find that

$$C_n = -\frac{1}{2} \cdot \frac{(1/2)(-1/2)\cdots(-n+1/2)}{(n+1)!} \cdot (-4)^{n+1}$$

$$= \frac{1 \cdot 1 \cdot 3 \cdots (2n-1)}{2^{n+2} \cdot (n+1)!} \cdot 4^{n+1} = 2^n \cdot \frac{1 \cdot 3 \cdots (2n-1)}{(n+1)!}$$

$$= \frac{2 \cdot 4 \cdots (2n)}{n!} \cdot \frac{1 \cdot 3 \cdots (2n-1)}{(n+1)!} = \frac{(2n)!}{n!(n+1)!}$$

$$= \frac{1}{n+1}\binom{2n}{n},$$

which coincides with formula (2.11). □

Another interesting application of formula (6.35) is given in the following statement.

Proposition 6.4.2. *The identity*

$$\sum_{n\geq 0}\binom{2n}{n}x^n = \frac{1}{\sqrt{1-4x}} \tag{6.36}$$

holds in $\mathbb{C}[[x]]$.

Proof. Applying Equation (6.35) for $\alpha = -1/2$, in which case

$$\frac{\alpha(\alpha-1)\cdots(\alpha-n+1)}{n!} = \frac{1}{n!}\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\cdots\left(-\frac{2n-1}{2}\right)$$

$$= (-1)^n \cdot \frac{1 \cdot 3 \cdots (2n-1)}{2^n n!} = (-1)^n \cdot \frac{(2n)!}{2^{2n} \cdot (n!)^2}$$

$$= (-1/4)^n \binom{2n}{n},$$

gives that $(1+x)^{-1/2} = \sum_{n\geq 0}\binom{2n}{n}(-x/4)^n$. This identity is equivalent to (6.36). □

Example 6.4.1. For $n \in \mathbb{N}$, let us prove the recurrence

$$C_{n+1} = \sum_{k=0}^{\lfloor n/2 \rfloor} 2^{n-2k}\binom{n}{2k} C_k$$

for the Catalan numbers, known as *Touchard's identity*. Let us denote by $a(n)$ the right-hand side of this identity. Then,

$$\sum_{n\geq 0} a(n)x^n = \sum_{n\geq 0}\left(\sum_{k=0}^{\lfloor n/2 \rfloor} 2^{n-2k}\binom{n}{2k} C_k\right)x^n = \sum_{k\geq 0} C_k \sum_{n\geq 2k} 2^{n-2k}\binom{n}{2k} x^n$$

$$= \sum_{k\geq 0} C_k \sum_{m\geq 0} 2^m \binom{m+2k}{2k} x^{m+2k} = \sum_{k\geq 0} C_k x^{2k} \sum_{m\geq 0} \binom{m+2k}{2k} (2x)^m$$

$$= \sum_{k\geq 0} C_k x^{2k} \cdot \frac{1}{(1-2x)^{2k+1}}$$

$$= \frac{1}{1-2x} \cdot C\left(\frac{x^2}{(1-2x)^2}\right),$$

where we have applied identity (6.19) for the fifth equality and $C(x)$ is the generating function for the Catalan numbers, as in the beginning of this section. This result, combined with formula (6.34), yields after routine calculations that

$$\sum_{n\geq 0} a(n)x^n = \frac{1-2x-\sqrt{1-4x}}{2x^2} = \frac{C(x)-1}{x} = \sum_{n\geq 0} C_{n+1}x^n$$

and hence that $a(n) = C_{n+1}$ for every $n \in \mathbb{N}$.

6.5 Exponential Generating Functions

This section introduces exponential generating functions and provides applications to problems on the enumeration of sequences and set partitions. For a thorough treatment of this important topic, we refer the reader to [24, Chapter 5].

Definition 6.5.1. *Let $(a_n)_{n\in\mathbb{N}}$ be a sequence of complex numbers. The formal power series*

$$\sum_{n\geq 0} a_n \frac{x^n}{n!} = a_0 + a_1 x + a_2 \frac{x^2}{2!} + a_3 \frac{x^3}{3!} + \cdots$$

is called the exponential generating function of $(a_n)_{n\in\mathbb{N}}$.

For example, if $a_n = 1$ for every $n \in \mathbb{N}$, then

$$\sum_{n\geq 0} a_n \frac{x^n}{n!} = \sum_{n\geq 0} \frac{x^n}{n!} := e^x.$$

Example 6.5.1. Let a_n be the number of subsets of $[n]$. Then, $a_n = 2^n$ for $n \in \mathbb{N}$ and hence

$$\sum_{n\geq 0} a_n \frac{x^n}{n!} = \sum_{n\geq 0} 2^n \frac{x^n}{n!} = \sum_{n\geq 0} \frac{(2x)^n}{n!} = e^{2x}.$$

Similarly, if b_n is the number of permutations of $[n]$, then $b_n = n!$ for $n \in \mathbb{N}$ and hence

$$\sum_{n\geq 0} b_n \frac{x^n}{n!} = \sum_{n\geq 0} x^n = \frac{1}{1-x}.$$

Finally, let c_n be the number of signed permutations of $[n]$, meaning sequences $\tau = (\tau_1, \tau_2, \ldots, \tau_n)$ such that $\sigma = (|\tau_1|, |\tau_2|, \ldots, |\tau_n|)$ is a permutation of $[n]$. To choose such a signed permutation τ, there are $n!$ ways to choose the permutation σ and two ways to choose the sign of τ_i for every $i \in [n]$. Hence, $c_n = 2^n n!$ by the multiplication principle and

$$\sum_{n\geq 0} c_n \frac{x^n}{n!} = \sum_{n\geq 0} 2^n x^n = \frac{1}{1-2x}.$$

Note that there are no similar closed-form expressions for the *ordinary* generating functions $\sum_{n\geq 0} b_n x^n$ and $\sum_{n\geq 0} c_n x^n$.

Example 6.5.2. Let a_n be the number of ways to first choose a subset S of $[n]$ and then choose an element $a \in S$. We found in Example 2.2.4 that $a_n = n2^{n-1}$ for $n \in \mathbb{N}$ and hence

$$\sum_{n\geq 0} a_n \frac{x^n}{n!} = \sum_{n\geq 1} n2^{n-1} \frac{x^n}{n!} = \sum_{n\geq 1} 2^{n-1} \frac{x^n}{(n-1)!} = \sum_{n\geq 0} 2^n \frac{x^{n+1}}{n!} = xe^{2x}.$$

Example 6.5.3. Let a_n be the number of perfect matchings of the complete graph K_n, say with vertex set $[n]$. There are $n-1$ choices for the unique match of vertex n in a perfect matching of K_n and deleting vertex n and its match from K_n results in a complete graph with $n-2$ vertices. Thus, an easy induction argument (see also Exercise 42 of Chapter 5) shows that

$$a_n = \begin{cases} 1 \cdot 3 \cdot 5 \cdots (2k-1) = \dfrac{(2k)!}{2 \cdot 4 \cdots (2k)} = \dfrac{(2k)!}{2^k k!}, & \text{if } n = 2k \text{ is even} \\ 0, & \text{otherwise.} \end{cases}$$

Therefore,

$$\sum_{n\geq 0} a_n \frac{x^n}{n!} = \sum_{k\geq 0} a_{2k} \frac{x^{2k}}{(2k)!} = \sum_{k\geq 0} \frac{x^{2k}}{2^k k!} = \sum_{k\geq 0} \frac{(x^2/2)^k}{k!} = e^{x^2/2}.$$

□

As suggested by these examples, and as will be demonstrated by the examples in the sequel, certain combinatorial sequences can be better studied in terms of their exponential, rather than ordinary, generating functions (it is mainly through

experience that one learns to decide which kind of generating function to use in each particular case). This study can provide formulas, recurrences, and other useful information about the sequence. The importance of exponential generating functions stems from the fact that algebraic operations on them, such as addition, multiplication, and composition, have interesting combinatorial interpretations. Let's consider addition and multiplication.

By the definition of addition (6.7) and multiplication (6.8)–(6.9) of formal power series, we have

$$\left(\sum_{n\geq 0} a_n \frac{x^n}{n!}\right) + \left(\sum_{n\geq 0} b_n \frac{x^n}{n!}\right) = \sum_{n\geq 0} (a_n + b_n)\frac{x^n}{n!} \tag{6.37}$$

and

$$\left(\sum_{n\geq 0} a_n \frac{x^n}{n!}\right)\left(\sum_{n\geq 0} b_n \frac{x^n}{n!}\right) = \sum_{n\geq 0} c_n \frac{x^n}{n!} \tag{6.38}$$

in $\mathbb{C}[[x]]$, where

$$\frac{c_n}{n!} = \sum_{k=0}^{n} \frac{a_k}{k!} \cdot \frac{b_{n-k}}{(n-k)!},$$

that is,

$$\begin{aligned} c_n &= \sum_{k=0}^{n} \binom{n}{k} a_k b_{n-k} \\ &= a_0 b_n + \binom{n}{1} a_1 b_{n-1} + \binom{n}{2} a_2 b_{n-2} + \cdots + a_n b_0. \end{aligned} \tag{6.39}$$

Thus, it is natural to consider the exponential generating function of a sequence $(c_n)_{n\in\mathbb{N}}$ whenever c_n can be expressed in the form (6.39).

Example 6.5.4. Let

$$c_n = \sum_{k=0}^{n} \frac{n!}{k!} 2^{n-k}$$

for $n \in \mathbb{N}$. We may rewrite this sum in the form

$$c_n = \sum_{k=0}^{n} \binom{n}{k} 2^{n-k}(n-k)!$$

and observe that (6.39) holds with $a_n = 1$ and $b_n = 2^n n!$ for $n \in \mathbb{N}$. Therefore, according to (6.38),

$$\begin{aligned}\sum_{n\geq 0} c_n \frac{x^n}{n!} &= \left(\sum_{n\geq 0} a_n \frac{x^n}{n!}\right)\left(\sum_{n\geq 0} b_n \frac{x^n}{n!}\right) \\ &= \left(\sum_{n\geq 0} \frac{x^n}{n!}\right)\left(\sum_{n\geq 0} 2^n n! \frac{x^n}{n!}\right) \\ &= \frac{e^x}{1-2x}.\end{aligned}$$

Example 6.5.5. Let $a, b \in \mathbb{C}$. Applying (6.39) when $a_n = a^n$ and $b_n = b^n$ for $n \in \mathbb{N}$, we get

$$\begin{aligned}e^{ax} \cdot e^{bx} &= \left(\sum_{n\geq 0} a^n \frac{x^n}{n!}\right)\left(\sum_{n\geq 0} b^n \frac{x^n}{n!}\right) \\ &= \sum_{n\geq 0}\left(\sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k}\right)\frac{x^n}{n!} = \sum_{n\geq 0}(a+b)^n \frac{x^n}{n!} \\ &= e^{(a+b)x},\end{aligned}$$

where the third equality is due to the Binomial Theorem. Thus, the well-known identity $e^{ax} \cdot e^{bx} = e^{(a+b)x}$ from calculus holds as an identity in the ring $\mathbb{C}[[x]]$ of formal power series. In particular, we have $e^x \cdot e^{-x} = 1$ in $\mathbb{C}[[x]]$, so $e^{-x} = \sum_{n\geq 0}(-1)^n x^n/n!$ is the inverse formal power series of $e^x = \sum_{n\geq 0} x^n/n!$, as expected.

Example 6.5.6. Let a_n be the number of bijective maps $\omega : [n] \to [n]$ that are equal to their own inverse; i.e., $\omega = \omega^{-1}$. Such bijective maps are called *involutions*. For instance, $a_1 = 1$, $a_2 = 2$, $a_3 = 4$, and $a_4 = 10$. The number a_n can also be interpreted as:

- the number of partitions of the set $[n]$ with blocks of cardinality one or two,
- the total number of matchings of the complete graph K_n.

Indeed, a map $\omega : [n] \to [n]$ is an involution if and only if $\omega(x) = y \Leftrightarrow \omega(y) = x$ for all $x, y \in [n]$. Hence, to every involution $\omega : [n] \to [n]$, one can associate a partition of $[n]$ with blocks of cardinality at most two, namely the sets $\{x, \omega(x)\}$ where $x \in [n]$, and a matching of the complete graph K_n on the vertex set $[n]$, namely the one with edges $\{x, \omega(x)\}$ where $x \in [n]$ and $\omega(x) \neq x$. For example, if $n = 6$ and $\omega(1) = 4$, $\omega(2) = 2$, $\omega(3) = 6$, $\omega(4) = 1$, $\omega(5) = 5$, and $\omega(6) = 3$,

then the associated partition has blocks $\{1, 4\}$, $\{2\}$, $\{3, 6\}$, and $\{5\}$ and the associated matching has edges $\{1, 4\}$ and $\{3, 6\}$. Clearly, the resulting maps are one-to-one correspondences from the set of involutions of $[n]$ to the set of partitions of $[n]$ with blocks of cardinality at most two and the set of matching of K_n, respectively.

We set $a_0 = 1$ and denote by c_n the number of partitions of $[n]$ with blocks of cardinality two or, equivalently, the number of perfect matchings of K_n. To select a partition π of $[n]$ with k singleton blocks and all other blocks of cardinality two, there are $\binom{n}{k}$ ways to choose the singleton blocks and for every such choice, there are c_{n-k} ways to form the blocks of π of cardinality two. Therefore,

$$a_n = \sum_{k=0}^{n} \binom{n}{k} c_{n-k}$$

for every $n \in \mathbb{N}$, which means that Equation (6.39) holds with $b_n = 1$ for every $n \in \mathbb{N}$ if one interchanges the roles of the sequences $(a_n)_{n\in\mathbb{N}}$ and $(c_n)_{n\in\mathbb{N}}$. In view of the result of Example 6.5.3, we deduce that

$$\sum_{n\geq 0} a_n \frac{x^n}{n!} = \left(\sum_{n\geq 0} \frac{x^n}{n!}\right)\left(\sum_{n\geq 0} c_n \frac{x^n}{n!}\right) = e^x \cdot e^{x^2/2}$$

$$= \exp\left(x + \frac{x^2}{2}\right).$$

Differentiating with respect to x, we find that

$$\sum_{n\geq 0} a_{n+1} \frac{x^n}{n!} = (1 + x) \exp\left(x + \frac{x^2}{2}\right) = (1 + x) \sum_{n\geq 0} a_n \frac{x^n}{n!}.$$

Comparing the coefficients of x^n in the left-hand and right-hand sides yields the recurrence

$$a_{n+1} = a_n + n a_{n-1},$$

valid for $n \geq 1$.

Example 6.5.7. Let D_n be the number of permutations of $[n]$ without fixed points. We claim that

$$\sum_{n\geq 0} D_n \frac{x^n}{n!} = \frac{e^{-x}}{1 - x}, \tag{6.40}$$

where $D_0 := 1$. Indeed, the formula

$$D_n = \sum_{k=0}^{n} \binom{n}{k} (-1)^k (n-k)!$$

of Proposition 3.2.3 shows that Equation (6.39) holds for $c_n = D_n$, $a_n = (-1)^n$, and $b_n = n!$. Hence, we infer from (6.38) that

$$\begin{aligned} \sum_{n\geq 0} D_n \frac{x^n}{n!} &= \left(\sum_{n\geq 0} a_n \frac{x^n}{n!} \right) \left(\sum_{n\geq 0} b_n \frac{x^n}{n!} \right) \\ &= \left(\sum_{n\geq 0} \frac{(-x)^n}{n!} \right) \left(\sum_{n\geq 0} n! \frac{x^n}{n!} \right) \\ &= \frac{e^{-x}}{1-x}. \end{aligned}$$

Here is another way to present this argument. By definition of D_n, we get (see Exercise 22 of Chapter 3) that

$$\sum_{k=0}^{n} \binom{n}{k} D_{n-k} = n!$$

for every $n \in \mathbb{N}$. Equivalently, since (6.38) $\Leftrightarrow$ (6.39),

$$\left(\sum_{n\geq 0} \frac{x^n}{n!} \right) \left(\sum_{n\geq 0} D_n \frac{x^n}{n!} \right) = \sum_{n\geq 0} n! \frac{x^n}{n!}.$$

This equation is equivalent to (6.40).

From Equation (6.40), we may deduce the recurrence $D_n = n D_{n-1} + (-1)^n$, for $n \geq 1$, of Proposition 3.2.3 as follows:

$$\begin{aligned} \sum_{n\geq 0} D_n \frac{x^n}{n!} &= e^{-x} + \frac{x e^{-x}}{1-x} = e^{-x} + x \sum_{n\geq 0} D_n \frac{x^n}{n!} \\ &= \sum_{n\geq 0} \frac{(-x)^n}{n!} + \sum_{n\geq 1} n D_{n-1} \frac{x^n}{n!}. \end{aligned}$$

Comparing the coefficients of $x^n/n!$ in the resulting equation yields the desired formula. Conversely, Equation (6.40) may be derived from the recurrence $D_n = n D_{n-1} + (-1)^n$. Indeed, multiplying the recurrence with $x^n/n!$ and summing over

$n \geq 1$ we get $F(x) - 1 = xF(x) + (e^{-x} - 1)$, where $F(x) = \sum_{n\geq 0} D_n x^n/n!$, from which (6.40) follows. □

The following statement is a special case of the generalized principle of inclusion–exclusion (Theorem 3.3.1). The proof we provide is an application of the multiplication of exponential generating functions.

Proposition 6.5.1. *For two sequences of complex numbers $(a_n)_{n\in\mathbb{N}}$ and $(b_n)_{n\in\mathbb{N}}$, we have*

$$b_n = \sum_{k=0}^{n} \binom{n}{k} a_k \tag{6.41}$$

for every $n \in \mathbb{N}$ if and only if

$$a_n = \sum_{k=0}^{n} (-1)^{n-k} \binom{n}{k} b_k \tag{6.42}$$

for every $n \in \mathbb{N}$.

Proof. Let $F(x) = \sum_{n\geq 0} a_n x^n/n!$ and $G(x) = \sum_{n\geq 0} b_n x^n/n!$ be the exponential generating functions of the two sequences. According to (6.39), Equation (6.41) holds for every $n \in \mathbb{N}$ if and only if

$$\sum_{n\geq 0} b_n \frac{x^n}{n!} = \left(\sum_{n\geq 0} a_n \frac{x^n}{n!}\right)\left(\sum_{n\geq 0} \frac{x^n}{n!}\right).$$

This equation can be rewritten as $G(x) = e^x \cdot F(x)$. Similarly, Equation (6.42) holds for every $n \in \mathbb{N}$ if and only if

$$\sum_{n\geq 0} a_n \frac{x^n}{n!} = \left(\sum_{n\geq 0} b_n \frac{x^n}{n!}\right)\left(\sum_{n\geq 0} (-1)^n \frac{x^n}{n!}\right),$$

which can be rewritten as $F(x) = e^{-x} \cdot G(x)$. Thus, the result follows from the equivalence $G(x) = e^x \cdot F(x) \Leftrightarrow F(x) = e^{-x} \cdot G(x)$. □

Our main goal in this section is to provide a combinatorial interpretation of the product of exponential generating functions. For this reason, we consider a family $\mathcal{A}$ of finite sequences (words), with terms taken from a specific set (alphabet), such that there are finitely many elements of $\mathcal{A}$ of given length n for every $n \in \mathbb{N}$. We define the exponential generating function associated to $\mathcal{A}$ as the formal power series

$$E_{\mathcal{A}}(x) = \sum_{n\geq 0} a_n \frac{x^n}{n!}, \tag{6.43}$$

where a_n is the number of elements of $\mathcal{A}$ of length n. In particular, $a_0 = 1$ if $\mathcal{A}$ contains the unique sequence of length zero (the empty sequence), and $a_0 = 0$ otherwise.

Example 6.5.8. Let us consider the alphabet $\{\alpha, \beta\}$. Let $\mathcal{A}$ (respectively, $\mathcal{B}$) be the family of finite sequences with all terms equal to α (respectively, of nonempty finite sequences with all terms equal to β). Let a_n (respectively, b_n) be the number of elements of $\mathcal{A}$ (respectively, of $\mathcal{B}$) of length n. Then, $a_n = 1$ for every $n \in \mathbb{N}$ (respectively, $b_0 = 0$ and $b_n = 1$ for every $n \geq 1$) and hence

$$E_{\mathcal{A}}(x) = \sum_{n\geq 0} a_n \frac{x^n}{n!} = \sum_{n\geq 0} \frac{x^n}{n!} = e^x$$

and

$$E_{\mathcal{B}}(x) = \sum_{n\geq 0} b_n \frac{x^n}{n!} = \sum_{n\geq 1} \frac{x^n}{n!} = e^x - 1.$$

Let C be the family of all finite sequences with terms equal to α or β, at least one of which is equal to β and c_n be the number of elements of C of length n. Then, $c_n = 2^n - 1$ for every $n \in \mathbb{N}$ and hence

$$E_C(x) = \sum_{n\geq 0} c_n \frac{x^n}{n!} = \sum_{n\geq 0} (2^n - 1)\frac{x^n}{n!} = e^{2x} - e^x.$$

We observe that $E_C(x) = e^x(e^x - 1) = E_{\mathcal{A}}(x)E_{\mathcal{B}}(x)$. □

We now consider two families $\mathcal{A}$ and $\mathcal{B}$ of sequences, as before. We call *shuffle* of $(x_1, x_2, \ldots, x_k) \in \mathcal{A}$ and $(y_1, y_2, \ldots, y_{n-k}) \in \mathcal{B}$ any sequence $(z_1, z_2, \ldots, z_n)$ with the following property: there exists a k-subset $I = \{i_1, i_2, \ldots, i_k\}$ of $[n]$ with $i_1 < i_2 < \cdots < i_k$ such that $z_{i_r} = x_r$ for $1 \leq r \leq k$ and $z_{j_r} = y_r$ for $1 \leq r \leq n-k$, where $\{j_1, j_2, \ldots, j_{n-k}\} = [n] \setminus I$ with $j_1 < j_2 < \cdots < j_{n-k}$. For example, if $n = 5, k = 3$ and $I = \{2, 3, 5\}$, then we have $(z_1, z_2, \ldots, z_n) = (y_1, x_1, x_2, y_2, x_3)$. Less formally, a shuffle of two sequences is obtained by interleaving their terms, while preserving the relative order in which the terms of each sequence appear. For example, the shuffles of (α, β) and (γ, δ) are the sequences $(\alpha, \beta, \gamma, \delta)$, $(\alpha, \gamma, \beta, \delta)$, $(\alpha, \gamma, \delta, \beta)$, $(\gamma, \alpha, \beta, \delta)$, $(\gamma, \alpha, \delta, \beta)$, and $(\gamma, \delta, \alpha, \beta)$.

The family C in Example 6.5.8 consists exactly of all shuffles of the elements of $\mathcal{A}$ with those of $\mathcal{B}$ and $E_C(x) = E_{\mathcal{A}}(x)E_{\mathcal{B}}(x)$. This fact is explained by the following proposition.

Proposition 6.5.2. *Let $\mathcal{A}$ and $\mathcal{B}$ be families of finite sequences, the terms of which belong to two disjoint alphabets. We assume that there are finitely many elements of each family of given length n for every $n \in \mathbb{N}$. If C is the family that consists of all shuffles of the elements of $\mathcal{A}$ with those of $\mathcal{B}$, then $E_C(x) = E_{\mathcal{A}}(x)E_{\mathcal{B}}(x)$.*

Proof. Let a_n (respectively, b_n and c_n) be the number of elements of $\mathcal{A}$ (respectively, $\mathcal{B}$ and C) of length n. Every sequence $(z_1, z_2, \ldots, z_n) \in C$ of length n can be uniquely expressed as the shuffle of two sequences $(x_1, x_2, \ldots, x_k) \in \mathcal{A}$ and $(y_1, y_2, \ldots, y_{n-k}) \in \mathcal{B}$. For a given k, there are $a_k b_{n-k}$ ways to choose these two sequences and $\binom{n}{k}$ ways to choose one of their shuffles (as many as there are k-subsets of $[n]$). We conclude that there is a total of

$$c_n = \sum_{k=0}^{n} \binom{n}{k} a_k b_{n-k}$$

elements of length n of C, meaning that (6.39) holds for every $n \in \mathbb{N}$. This shows that (6.38) holds as well and the proof follows. □

Example 6.5.9. Let us compute the number of sequences of length n with terms in the set $\{\alpha, \beta, \gamma\}$ that have an even number of terms equal to γ.

Let C be the family of all these sequences, and let c_n be the number of elements of C of length n, which we wish to compute. We observe that C consists of exactly the shuffles of the elements of the following two families: the family $\mathcal{A}$ of all finite sequences with terms in the set $\{\alpha, \beta\}$ and the family $\mathcal{B}$ of all sequences of even length with terms equal to γ. We observe that $E_{\mathcal{A}}(x) = e^{2x}$ and that

$$E_{\mathcal{B}}(x) = \sum_{n \geq 0} \frac{x^{2n}}{(2n)!} = \frac{e^x + e^{-x}}{2}.$$

These observations and Proposition 6.5.2 imply that

$$\sum_{n \geq 0} c_n \frac{x^n}{n!} = E_C(x) = E_{\mathcal{A}}(x) E_{\mathcal{B}}(x) = \frac{e^{3x} + e^x}{2} = \sum_{n \geq 0} \frac{(3^n + 1)x^n}{2 \cdot n!}$$

and hence that $c_n = (3^n + 1)/2$ for every $n \in \mathbb{N}$. □

Shuffles can be defined for more than two finite sequences as well. More specifically, for $k \geq 3$, a shuffle of $\sigma_1, \sigma_2, \ldots, \sigma_k$ can be defined recursively as a shuffle of τ and σ_k, where τ is any shuffle of $\sigma_1, \ldots, \sigma_{k-1}$. For example, the sequence $(4, 6, 1, 2, 5, 7, 3)$ is a shuffle of $(1, 2, 3)$, $(4, 5)$, and $(6, 7)$.

Proposition 6.5.3. *Let $\mathcal{A}_1, \mathcal{A}_2, \ldots, \mathcal{A}_k$ be families of finite sequences, the terms of which belong to pairwise disjoint alphabets. We assume that there are finitely many elements of each family of given length n for every $n \in \mathbb{N}$. If $\mathcal{A}$ is the family which consists of all shuffles of sequences $\sigma_1, \sigma_2, \ldots, \sigma_k$ with $\sigma_i \in \mathcal{A}_i$ for $i \in [k]$, then $E_{\mathcal{A}}(x) = E_{\mathcal{A}_1}(x) E_{\mathcal{A}_2}(x) \cdots E_{\mathcal{A}_k}(x)$.*

Proof. The case $k = 1$ is trival and that of $k = 2$ is equivalent to Proposition 6.5.2. The general case follows by induction on k (the details are left to the reader). □

Example 6.5.10. Suppose that $\mathcal{A}_i$ consists of all finite sequences with all terms equal i, for $i \in [k]$ and let $\mathcal{A}$ be as in Proposition 6.5.3. Then, $E_{\mathcal{A}_i}(x) = \sum_{n\geq 0} x^n/n!$ for every $i \in [k]$. Moreover, $\mathcal{A}$ is the family of all finite sequences with terms in the set $[k]$ and hence $E_{\mathcal{A}}(x) = \sum_{n\geq 0} k^n x^n/n!$ (explain why). Thus, Proposition 6.5.3 shows that

$$\left(\sum_{n\geq 0} \frac{x^n}{n!}\right)^k = \sum_{n\geq 0} \frac{k^n x^n}{n!}$$

or, in other words, that the identity $(e^x)^k = e^{kx}$ holds in $\mathbb{C}[x]$. Of course, this identity is a consequence of the result of Example 6.5.5. □

We now present two applications to the Stirling numbers of the second kind and the Bell numbers, studied in Section 4.3.

Proposition 6.5.4. *For $k \in \mathbb{N}$,*

$$\sum_{n\geq 0} S(n,k)\,\frac{x^n}{n!} = \frac{(e^x-1)^k}{k!}. \tag{6.44}$$

Proof. We may assume (explain why) that $k \in \mathbb{Z}_{>0}$. Let us apply Proposition 6.5.4 when $\mathcal{A}_i$ consists of the nonempty finite sequences with all terms equal to i, for $i \in [k]$. Then, $E_{\mathcal{A}_i}(x) = \sum_{n\geq 1} x^n/n! = e^x - 1$ for every $i \in [k]$ and hence $E_{\mathcal{A}}(x) = (e^x-1)^k$. Therefore,

$$\sum_{n\geq 0} a(n,k)\,\frac{x^n}{n!} = (e^x-1)^k,$$

where $a(n,k)$ is the number of elements of $\mathcal{A}$ of length n. By definition, $\mathcal{A}$ consists of all shuffles of elements $\sigma_1, \sigma_2, \ldots, \sigma_k$ of $\mathcal{A}_1, \mathcal{A}_2, \ldots, \mathcal{A}_k$, respectively. These coincide with the sequences $(f(1), f(2), \ldots, f(n))$ with terms in the set $[k]$ such that each $i \in [k]$ appears at least once. Hence, $a(n,k)$ is equal to the number of surjective maps $f : [n] \to [k]$. By Proposition 4.3.1 (b), we have $a(n,k) = k!S(n,k)$ and the proof follows. □

REMARK. Expanding the right-hand side of (6.44) according to the Binomial Theorem as

$$\frac{(e^x-1)^k}{k!} = \frac{1}{k!}\sum_{i=0}^{k}(-1)^{k-i}\binom{k}{i}e^{ix}$$

and computing the coefficient of $x^n/n!$ we reach the formula of Proposition 4.3.1 (c) for $S(n,k)$. Thus, Equation (6.44) is equivalent to this formula.

Corollary 6.5.1.

$$\sum_{n\geq 0} B(n)\,\frac{x^n}{n!} = \exp(e^x - 1), \tag{6.45}$$

where $B(0) = 1$.

Proof. Since $B(n) = \sum_{k\geq 0} S(n,k)$ for $n \in \mathbb{N}$, by (6.44) we have

$$\sum_{n\geq 0} B(n)\,\frac{x^n}{n!} = \sum_{k\geq 0}\sum_{n\geq 0} S(n,k)\,\frac{x^n}{n!} = \sum_{k\geq 0} \frac{(e^x-1)^k}{k!} = e^{e^x-1}$$

and the proof follows. □

6.6 Exercises

1. For $n \in \mathbb{N}$ we let:
   (a) a_n be the number of subsets of $[2n]$ that contain at least one integer greater than n,
   (b) b_n be the number of subsets S of $[2n]$ such that the sum of the elements of S is even.

   Compute the generating functions $\sum_{n\geq 0} a_n x^n$ and $\sum_{n\geq 0} b_n x^n$, where $a_0 = 0$, $b_0 = 1$.
2. Let $f(n)$ be the number of permutations $(\sigma_1, \sigma_2, \ldots, \sigma_n)$ of the set $[n]$ such that $|\sigma_i - i| \leq 1$ for every $i \in [n]$. Compute the generating function $\sum_{n\geq 1} f(n)x^n$.
3. Let $F(x), G(x) \in \mathbb{C}[[x]]$ be formal power series.
   (a) Prove that if $F(x)G(x) = 0$, then $F(x) = 0$ or $G(x) = 0$.
   (b) True or false? If $F(x)$ and $G(x)$ have real coefficients and $(F(x))^2 + (G(x))^2 = 0$, then $F(x) = G(x) = 0$.
4. Prove that a formal power series $F(x) = \sum_{n\geq 0} a_n x^n \in \mathbb{C}[[x]]$ is invertible in $\mathbb{C}[[x]]$ if and only if $a_0 \neq 0$.
5. Let $G(x) \in \mathbb{C}[[x]]$ be a formal power series.
   (a) If $G(0) \neq 0$, prove that the equation $y^2 = G(x)$ has exactly two solutions $y = F(x) \in \mathbb{C}[[x]]$.
   (b) For which $G(x)$ does the equation $y^2 = G(x)$ have a solution $y \in \mathbb{C}[[x]]$?
6. Let $a_{n,k}$ be the number of vectors $(\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_n) \in \{-1, 0, 1\}^n$ that have exactly k coordinates equal to zero.
   (a) Find a simple formula for the generating function $\sum_{k=0}^{n} a_{n,k} x^k$.

(b) How many vectors $(\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_n) \in \{-1, 0, 1\}^n$ have an even number of coordinates equal to zero?

7. For $n \in \mathbb{N}$, let $f(n)$ be the number of ordered pairs (S, T) of subsets of $[n]$, let $g(n)$ be the number of ordered pairs (S, T) of disjoint subsets S, T of $[n]$, and let $h(n)$ be the number of ordered pairs (S, T) of subsets of $[n]$ such that $S \cap T$ has exactly one element. For example, $f(2) = 16$, $g(2) = 9$ and $h(2) = 6$.

   (a) Find simple formulas for $f(n)$, $g(n)$, and $h(n)$.
   (b) Compute the generating functions of $(f(n))_{n\in\mathbb{N}}$, $(g(n))_{n\in\mathbb{N}}$, and $(h(n))_{n\in\mathbb{N}}$.

8. Compute the following generating functions, as rational functions of x:

   (a) $\displaystyle\sum_{n\geq 0} \binom{n}{2} x^n$

   (b) $\displaystyle\sum_{n\geq 0} (n^2 + n + 1)x^n$

   (c) $\displaystyle\sum_{n\geq 0} \left(\binom{2}{2} + \binom{3}{2} + \cdots + \binom{n}{2} \right) x^n$

   (d) $\displaystyle\sum_{n\geq 0} a_n x^n$, where $a_0 = 1$, $a_1 = -2$ and $a_n = 5a_{n-1} - 6a_{n-2}$ for $n \geq 2$.

   In the case of (d), deduce a simple formula for a_n.

9. Let $(a_n)_{n\in\mathbb{N}}$ be the sequence defined by setting $a_0 = 0$, $a_1 = 3$, and $a_n = 5a_{n-1} - 4a_{n-2}$ for $n \geq 2$. Compute the sum $\sum_{k=0}^{n} \binom{n}{k} a_k$ for $n \in \mathbb{N}$.

10. Let $a_n = (2 + \sqrt{3})^n + (2 - \sqrt{3})^n$ for $n \in \mathbb{N}$.

    (a) Prove that a_n is an even positive integer for every $n \in \mathbb{N}$.
    (b) Given $n \in \mathbb{N}$, what is the largest power of 2 that divides a_n?

11. (a) For $n \in \mathbb{N}$, let $f(n)$ be the number of triples $(a, b, c) \in \mathbb{N}^3$ such that $a \leq n$, $b \leq n$ and $a + b + c \leq 3n$. For example, $f(0) = 1$, $f(1) = 12$, and $f(2) = 45$. Find a simple formula for $f(n)$ and prove that

$$\sum_{n\geq 0} f(n)x^n = \frac{1 + 8x + 3x^2}{(1 - x)^4}.$$

    (b) For $n \in \mathbb{N}$, let $g(n)$ be the number of triples $(a, b, c) \in \mathbb{N}^3$ such that $a \leq n$ and $a + b + c \leq 3n$. For example, $g(0) = 1$, $g(1) = 16$, and $g(2) = 64$. Find a formula for $g(n)$ and prove that

$$\sum_{n\geq 0} g(n)x^n = \frac{1 + 12x + 6x^2}{(1 - x)^4}.$$

12. Let a_n be the number of spanning trees of the $n \times 2$ grid graph, where $a_0 = 0$.
    (a) Prove that $a_n = 4a_{n-1} - a_{n-2}$ for $n \geq 2$.
    (b) Deduce that

$$\sum_{n \geq 0} a_n x^n = \frac{x}{1 - 4x + x^2}$$

    and that

$$a_n = \frac{1}{2\sqrt{3}} \left((2 + \sqrt{3})^n - (2 - \sqrt{3})^n \right)$$

    for every $n \in \mathbb{N}$.

13. Prove the identity

$$\sum_{k=0}^{n} k \binom{n}{k} = n 2^{n-1}$$

    of Proposition 2.3.7 using the method of generating functions.

14. Prove the identity

$$\sum_{k=0}^{n} \binom{k}{2} \binom{n-k}{2} = \binom{n+1}{5}$$

    using generating functions.

15. For $m, n, k \in \mathbb{N}$, prove the identities

$$\begin{aligned} \binom{m+n}{n} &= \sum_{i=0}^{n} (-1)^i \binom{k}{i} \binom{m+n+k-i}{n-i} \\ &= \sum_{i=0}^{n} \binom{k+i-1}{i} \binom{m+n-k-i}{n-i} \end{aligned}$$

    of Exercise 45 of Chapter 2 and Exercise 14 of Chapter 3 using generating functions.

16. Consider the natural numbers $f(n, r, m)$ defined in Exercise 13 of Chapter 3.
    (a) Prove that $f(n, r, m)$ is equal to the coefficient of x^{mr} in the polynomial $(1 + x + x^2 + \cdots + x^{r-1})^n$.
    (b) Prove the identity of part (a) of that exercise using generating functions.

17. Let $n \in \mathbb{N}$.

(a) Prove that

$$\sum_{k\geq 1} k\binom{n+k-1}{n}x^{k-1} = \frac{1+nx}{(1-x)^{n+2}}.$$

(b) Deduce from (a) that

$$\sum_{k=1}^{m} k\binom{n+k-1}{n} = \binom{m+n+1}{n+2} + n\binom{m+n}{n+2}$$

for $m \in \mathbb{Z}_{>0}$. Which identities are obtained for $n \in \{0, 1, 2\}$?

18. For $m, n \in \mathbb{N}$ we set

$$f(m,n) = \sum_{k=0}^{m} \binom{m}{k}\binom{n+k}{m}$$

and

$$g(m,n) = \sum_{k=0}^{\max\{m,n\}} \binom{m}{k}\binom{n}{k}2^k$$

and denote by $\tau(m,n)$ the number of solutions $(x_1, x_2, \ldots, x_n) \in \mathbb{Z}^n$ of the inequality $|x_1|+|x_2|+\cdots+|x_n| \leq m$, as in Exercises 53 and 54 of Chapter 2.

(a) Prove that

$$\sum_{m\geq 0} f(m,n)x^m = \sum_{m\geq 0} g(m,n)x^m = \sum_{m\geq 0} \tau(m,n)x^m = \frac{(1+x)^n}{(1-x)^{n+1}}$$

for every $n \in \mathbb{N}$ and that

$$\sum_{m,n\geq 0} f(m,n)x^m y^n = \sum_{m,n\geq 0} g(m,n)x^m y^n = \sum_{m,n\geq 0} \tau(m,n)x^m y^n$$
$$= \frac{1}{1-x-y-xy}.$$

(b) Deduce that $f(m,n) = g(m,n) = \tau(m,n)$ for all $m, n \in \mathbb{N}$.

19. Consider the Fibonacci sequence $(F_n)_{n\in\mathbb{N}}$, with $F_0 = 0$, $F_1 = 1$, and $F_n = F_{n-1} + F_{n-2}$ for $n \geq 2$, and set

$$G_n = \sum_{k=0}^{n} \binom{n}{k} F_k$$

for $n \in \mathbb{N}$. Compute the generating function $\sum_{n\geq 0} G_n x^n$:

(a) By applying formula (6.3) for $\sum_{n\geq 0} F_n x^n = \sum_{n\geq 1} a_{n-1} x^n$.
(b) By applying formula (6.5) for $F_n = a_{n-1}$.

Moreover:

(c) Find a recurrence for the sequence $(G_n)_{n\in\mathbb{N}}$ and a simple formula for G_n.
(d) Deduce that $G_n = F_{2n}$ for every $n \in \mathbb{N}$.

20. We set

$$s_n = \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n-k}{k} 4^{-k}$$

for $n \in \mathbb{N}$.

(a) Compute the generating function $\sum_{n\geq 0} s_n x^n$.
(b) Deduce a simple formula for s_n.

21. Compute the sum

$$t_n = \sum_{k=0}^{n} (-1)^k \binom{2n-k-1}{k} 2^{n-k}$$

for $n \in \mathbb{N}$.

22. Let $F(x) = \sum_{n\geq 0} a_n x^n$ be the formal power series defined by setting $a_0 = a_1 = a_2 = 1$, $a_3 = 0$, and $a_n = a_{n-1} - a_{n-3} + a_{n-4}$ for $n \geq 4$.

(a) Compute $F(x)$ as a rational function of x.
(b) Compute the coefficient of x^9 in $(F(x))^2$.
(c) Compute the coefficients of $(1-x)F(x)$.
(d) Prove that a_n is equal to the number of pairs $(m_1, m_2) \in \mathbb{Z}^2$ such that $m_1 \in \{0, 1, 2\}$, $m_2 \in \{0, 6, 12, \dots\}$ and $m_1 + m_2 = n$. Deduce that $a_n \in \{0, 1\}$ for every $n \in \mathbb{N}$.

23. Let $(a_n)_{n\in\mathbb{N}}$ be the Fibonacci sequence of Section 6.1.1. Prove that

$$a_n^2 = 2a_{n-1}^2 + 2a_{n-2}^2 - a_{n-3}^2$$

for $n \geq 3$ and that

$$\sum_{n\geq 0} a_n^2 x^n = \frac{1-x}{(1+x)(1-3x+x^2)}.$$

24. Let $(a_n)_{n\in\mathbb{N}}$ be a sequence such that $a_0 = 1$ and $a_{n+1} = 2a_n + n$ for $n \geq 0$.

    (a) Prove that

    $$\sum_{n\geq 0} a_n x^n = \frac{1-2x+2x^2}{(1-x)^2(1-2x)}.$$

    (b) Find a simple formula for a_n.

25. Consider the formal power series $F(x) = \sum_{n\geq 0} \left(\frac{x}{1+x^2}\right)^n \in \mathbb{C}[[x]]$.

    (a) Compute $F(x)$ as a rational function of x and prove that its coefficients belong to the set $\{-1, 0, 1\}$.
    (b) Do the same for $F(x)/(1+x)$ and $1/F(x)$.

26. Let $(a_n)_{n\in\mathbb{N}}$ and $(b_n)_{n\in\mathbb{N}}$ be the sequences defined by setting $a_0 = 3$, $b_0 = 5$ and

    $$\begin{aligned} a_n &= 5a_{n-1} - 2b_{n-1} \\ b_n &= a_n + a_{n-1} \end{aligned}$$

    for $n \geq 1$. For example, $a_n = 3, 5, 9, 17$ and $b_n = 5, 8, 14, 26$ for $n = 0, 1, 2, 3$, respectively.

    (a) Compute the generating functions $\sum_{n\geq 0} a_n x^n$ and $\sum_{n\geq 0} b_n x^n$.
    (b) Find simple formulas for a_n and b_n.

27. Let $(a_n)_{n\in\mathbb{N}}$ be the Fibonacci sequence of Section 6.1.1.

    (a) Prove that $5a_n^2 + 4 \cdot (-1)^{n+1} = \left(\tau^{n+1} + \bar{\tau}^{n+1}\right)^2$ for $n \in \mathbb{N}$, where $\tau = (1+\sqrt{5})/2$ and $\bar{\tau} = (1-\sqrt{5})/2$.
    (b) Deduce that $5a_n^2 + 4 \cdot (-1)^{n+1}$ is equal to the square of an integer for every $n \in \mathbb{N}$.

28. Let $n \in \mathbb{N}$.

    (a) Prove that

    $$\sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n}{2k} x^{2k} = \frac{(1+x)^n + (1-x)^n}{2}.$$

(b) Prove that the number

$$\sum_{k=1}^{2m-1} 2^{k-1}\binom{4m-2}{2k}$$

is equal to the square of an integer for every $m \in \mathbb{Z}_{>0}$.

29. We define the integers $a_{m,k}$ by the equality

$$(1-x+x^2)^m = \sum_{k=0}^{2m} a_{m,k}x^k$$

and set $b_n = \sum_{k=0}^{\lfloor 2n/3 \rfloor} a_{n-k,k}$ for $n \in \mathbb{N}$.

(a) Prove that

$$\sum_{n\geq 0} b_n x^n = \frac{1}{1-x+x^2-x^3}.$$

(b) Deduce that $b_n \in \{-1, 0, 1\}$ for every $n \in \mathbb{N}$.

30. Let $a_n(p, q)$ be the number of subsets of $[n]$ that have cardinality p and sum of elements equal to q. Prove that

$$\sum_{p,q\geq 0} a_n(p,q)x^p y^q = \prod_{i=1}^{n}(1+xy^i)$$

for every positive integer n.

31. Prove that

$$\sum_{n\geq k} S(n,k)x^n = \frac{x^k}{(1-x)(1-2x)\cdots(1-kx)}, \tag{6.46}$$

where $S(n, k)$ is the number of partitions of the set $[n]$ with k parts (see Sections 4.3 and 6.5).

32. Ten boxes are numbered with the numbers $1, 2, \ldots, 10$. Let a_n be the number of ways to distribute n identical balls into these boxes, so that every odd-numbered (respectively, even-numbered) box has at most one ball (respectively, at most two balls). Compute the generating function $\sum_{n\geq 0} a_n x^n$ and a_n for $n = 5$.

33. Let $b(n, k)$ be the number of subsets S of $[n]$ of cardinality k such that $|i - j| > 2$ for any two distinct elements i, j of S.

(a) Prove that $b(n, k)$ is equal to the number of compositions $(r_0, r_1, \ldots, r_k)$ of $n + 1$ such that $r_i \geq 3$ for $i \in [k - 1]$.

(b) Compute the generating function $\sum_{n\geq 0} b(n,k)x^n$ and deduce a simple formula for $b(n, k)$.

34. Let a_n be the number of compositions (r_1, r_2, r_3) of n such that r_2 and r_3 are even numbers.

    (a) Prove that

    $$\sum_{n\geq 0} a_n x^n = \frac{1}{(1-x)(1-x^2)^2}.$$

    (b) Deduce a simple formula for a_n.

35. For $n \in \mathbb{Z}_{>0}$, we let

    $$f(n) = \sum_{r_1+r_2+\cdots+r_k=n} (-1)^{n-k}(r_1+1)(r_2+1)\cdots(r_k+1),$$

    where the sum ranges over all compositions $(r_1, r_2, \ldots, r_k)$ of n (with any number of parts). For example, $f(1) = 2$ and $f(2) = 1$.

    (a) Compute the generating function $\sum_{n\geq 1} f(n)x^n$.
    (b) Deduce that $f(n) = 0$ for every $n \geq 3$.

36. Let $q_k(n)$ denote the number of partitions of n with exactly k parts, where $q_0(0) = 1$ by convention. Prove that $q_k(n) = q_{k-1}(n-1) + q_k(n-k)$ for integers $1 \leq k \leq n$.
37. Let $f(n)$ be the number of partitions of n with parts larger than 1. For example $f(6) = 4$, with corresponding partitions (6), (4, 2), (3, 3), and (2, 2, 2).

    (a) Prove that $f(n) = p(n) - p(n-1)$ for every $n \geq 1$, where $p(n)$ is the number of partitions of n.
    (b) Prove that

    $$\sum_{n\geq 0} f(n)x^n = \prod_{i\geq 2} \frac{1}{1-x^i},$$

    where $f(0) := 1$.

38. For $n \in \mathbb{N}$, compute the number of partitions of n, every part of which belongs to the set $\{1, 2, 4\}$.
39. We denote by $r(\lambda)$ the number of parts of an integer partition λ and with $\max(\lambda)$ the largest part of λ. Let Λ_k be the set of integer partitions λ such that $\max(\lambda) \leq k$ and M_k be the set of integer partitions λ such that $r(\lambda) \leq k$. Prove that

    $$\sum_{\lambda\in \mathrm{M}_k} q^{\max(\lambda)} x^{|\lambda|} = \sum_{\lambda\in\Lambda_k} q^{r(\lambda)} x^{|\lambda|}$$
    $$= \frac{1}{(1-qx)(1-qx^2)\cdots(1-qx^k)}.$$

40. Let $\sigma(n)$ be the number of partitions of n having no part divisible by 3, and let $\tau(n)$ be the number of partitions of n no part of which appears more than twice. For example, for $n = 4$ the partitions of the first kind are (4), (2, 2), (2, 1, 1), (1, 1, 1, 1), and those of the second kind are (4), (3, 1), (2, 2), (2, 1, 1). Prove that $\sigma(n) = \tau(n)$ for every positive integer n.
41. Solve Exercise 57 (b) of Chapter 2 with the method of generating functions, applying the identity of Proposition 6.4.2.
42. Let $f(n)$ be the coefficient of x^n in the polynomial $(1 + x + x^2)^n$.

    (a) Prove that $f(n) = \displaystyle\sum_{k=0}^{\lfloor n/2 \rfloor} \binom{2k}{k}\binom{n}{2k}$ for every $n \in \mathbb{N}$.

    (b) Prove that $\sum_{n\geq 0} f(n)x^n = \dfrac{1}{\sqrt{1-2x-3x^2}}$.
43. Two sequences $(a_n)_{n\in\mathbb{N}}$ and $(b_n)_{n\in\mathbb{N}}$ are such that

$$\sum_{k=0}^{n} \binom{n}{k} a_k b_{n-k} = n2^{n-1}$$

    for every $n \in \mathbb{N}$. If

$$\sum_{n\geq 0} b_n \frac{x^n}{n!} = xe^{-x},$$

    compute a_n for $n \in \mathbb{N}$.
44. For $n \in \mathbb{N}$, we let:

    (a) $f(n)$ be the number of ways one can first choose a subset S of $[n]$ and then choose an element of S and an element of $[n]\setminus S$,

    (b) $g(n)$ be the number of surjective maps $\alpha : [n+1] \to [n]$,

    (c) $h(n)$ be the number of perfect matchings of the complete graph K_{2n}.

    Compute the exponential generating functions of $(f(n))_{n\in\mathbb{N}}$, $(g(n))_{n\in\mathbb{N}}$, and $(h(n))_{n\in\mathbb{N}}$, where $f(0) = g(0) = 0$ and $h(0) := 1$.
45. Let $\Omega_n = \{1, -1, 2, -2, \ldots, n, -n\}$, and let b_n be the number of involutions ω on Ω_n (meaning, bijective maps $\omega : \Omega_n \to \Omega_n$ that are equal to their own inverse) such that $\omega(-i) = -\omega(i)$ for every $i \in [n]$.

    (a) Prove that

$$\sum_{n\geq 0} b_n \frac{x^n}{n!} = \exp\left(2x + x^2\right),$$

    where $b_0 := 1$.

    (b) Deduce that $b_{n+1} = 2(b_n + nb_{n-1})$ for $n \geq 1$.

46. For $n \in \mathbb{N}$, let c_n be the number of ways to choose a permutation σ of $[n]$ and color blue some of the fixed points of σ. For example, $c_1 = 2$, $c_2 = 5$, and $c_3 = 16$. We set $c_0 := 1$.

    (a) Prove that

$$c_n = \sum_{k=0}^{n} \binom{n}{k} D_k 2^{n-k}$$

    for every $n \in \mathbb{N}$, where D_n is the number of permutations of $[n]$ without fixed points.

    (b) Prove that

$$\sum_{n \geq 0} c_n \frac{x^n}{n!} = \frac{e^x}{1-x}.$$

    (c) Find a simple recurrence for the sequence $(c_n)_{n \in \mathbb{N}}$ and a formula for c_n, as simple as possible.

47. Consider the sequences $(f(n))_{n \in \mathbb{N}}$ and $(g(n))_{n \in \mathbb{N}}$ defined in Exercise 28 of Chapter 3, where $f(0) = g(0) = 1$.

    (a) Compute their exponential generating functions.
    (b) Find a recurrence for each of these sequences and compute $f(n)$ for $n \leq 5$.

48. In each of the following cases, compute the number of sequences of length n with terms in the set $\{0, 1, 2\}$ that have the given properties:

    (a) The number of terms equal to 0 is even, and the number of terms equal to 1 is odd.
    (b) The numbers of terms equal to 0, to 1, and to 2 are either all even or all odd.

49. Let r be a positive integer. We denote by:

    (a) $f(n)$ the number of ways one can choose a partition of the set $[n]$ and color every element of $[n]$ with one of r given colors,
    (b) $g(n)$ the number of ways one can choose a partition π of the set $[n]$ and color every block of π with one of r given colors.

    Compute the exponential generating functions of $(f(n))_{n \in \mathbb{N}}$ and $(g(n))_{n \in \mathbb{N}}$, if $f(0) = g(0) = 1$.

50. Solve Exercise 30 of Chapter 4 using exponential generating functions.

51. Let $f(n)$ be the number of ordered partitions of the set $[n]$ (see the proof of Proposition 4.3.1), meaning ordered tuples $(B_1, B_2, \ldots, B_k)$ such that $\{B_1, B_2, \ldots, B_k\}$ is a partition of $[n]$, and let $g(n) = \sum_{k=0}^{n} (-1)^{n-k} \binom{n}{k} f(k)$ for $n \in \mathbb{N}$, where $f(0) := 1$.

(a) Prove that

$$\sum_{n\geq 0} f(n)\frac{x^n}{n!} = \frac{1}{2-e^x}$$

and that

$$\sum_{n\geq 0} g(n)\frac{x^n}{n!} = \frac{1}{2e^x - e^{2x}}.$$

(b) By applying Proposition 6.5.1 (or by a different method), find a combinatorial interpretation of $g(n)$ and deduce that $g(n) \geq 0$ for every $n \in \mathbb{N}$.

52. Assuming that $S(n,k) = 0$ unless $0 \leq k \leq n$, prove the identity

$$\binom{a+b}{a} S(n, a+b) = \sum_{k=0}^{n} \binom{n}{k} S(k,a)S(n-k,b) \tag{6.47}$$

for $a, b, n \in \mathbb{N}$:

(a) By using only basic principles of enumeration.
(b) By applying Proposition 6.5.4.

Chapter 7
Discrete Probability



This chapter serves as a brief introduction to the theory of discrete probability, meaning the part of probability theory that can be expressed in the language of discrete mathematics. Our goal is to describe applications of the methods we have developed so far to this theory and, conversely, to point out that probabilistic ideas can be useful to the study of combinatorial problems.

For a thorough treatment of the theory of probability, we refer to other sources in the literature, such as [11].

7.1 Probability Spaces

This section provides basic definitions and examples of (discrete) probability spaces.

7.1.1 Enumeration and Probability

During an experiment, we select arbitrarily an element ω of a finite set Ω of cardinality N. If the selection is made uniformly at random, all elements of Ω are considered "equiprobable" to occur. We then say that every element $\omega \in \Omega$ has probability $1/N$ to be selected. More generally, for every $A \subseteq \Omega$, we call $\#A/N$ the probability of the event that $\omega \in A$ holds and write

$$\mathbf{P}(A) = \frac{\#A}{N}. \tag{7.1}$$

In this way, the problem of enumeration of the elements of A is interpreted as the problem to compute a probability.

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Definition (7.1) is an important special case of the definition of the concept of probability. In the sequel, we provide examples in the framework of this definition, before we present more general definitions in the following sections.

Example 7.1.1. Let n be a given positive integer. We select uniformly at random a subset S of the set $[n]$. What is the probability that the cardinality of S is even?

For this experiment, the broad set Ω from which an element is selected is the powerset of $[n]$, that is, $\Omega = 2^{[n]}$. The event the probability of which we wish to compute corresponds to the set $A \subseteq \Omega$ that consists of all subsets of $[n]$ of even cardinality. From basic results of Chapter 2 (see Example 2.2.2 and Proposition 2.3.6), we have $\#\Omega = 2^n$ and

$$\#A = \binom{n}{0} + \binom{n}{2} + \binom{n}{4} + \cdots = 2^{n-1}.$$

Hence, the probability we were asked to compute is equal to $2^{n-1}/2^n = 1/2$. □

The set Ω, the elements of which are selected, is called the *sample space* of the experiment. The elements of Ω are called *outcomes* , and the subsets of Ω are called *events*.

Example 7.1.2. Let $n \geq 4$ be a given positive integer. We select uniformly at random one of the permutations $\sigma = (\sigma_1, \sigma_2, \ldots, \sigma_n)$ of the set $[n]$. For example, σ could be the permutation obtained when a machine in a casino shuffles a deck of cards $1, 2, \ldots, n$. Which of the following events is the least probable, and which is the most probable to occur?

(a) $\sigma_1 < \sigma_2 < \sigma_3$,
(b) $\sigma_1 < \sigma_2$ and $\sigma_1 < \sigma_3$,
(c) $\sigma_1 < \sigma_2$ and $\sigma_3 < \sigma_4$.

For this experiment, the sample space Ω is the set of $n!$ permutations of $[n]$. The events of interest to us are the subsets, say A, B, and C, which consist of those permutations $\sigma = (\sigma_1, \sigma_2, \ldots, \sigma_n) \in \Omega$ that satisfy the inequalities (a), (b), and (c), respectively. To select $\sigma \in \Omega$ such that $\sigma_1 < \sigma_2 < \sigma_3$, there are $\binom{n}{3}$ ways to choose the subset $\{\sigma_1, \sigma_2, \sigma_3\}$ of $[n]$ and for each of them, there are $(n-3)!$ ways to permute the other elements of $[n]$ so as to form σ. We conclude that

$$\#A = \binom{n}{3}(n-3)! = \frac{n!}{6}.$$

Similarly, we find that

$$\#B = \binom{n}{3} \cdot 2(n-3)! = \frac{n!}{3},$$

$$\#C = \binom{n}{2}\binom{n-2}{2}(n-4)! = \frac{n!}{4}$$

and conclude that $\mathbf{P}(A) = 1/6$, $\mathbf{P}(B) = 1/3$, and $\mathbf{P}(C) = 1/4$. Hence, the least probable event to occur is A, and the most probable is B. □

Example 7.1.3. We are given integers $1 \le k \le n$. We select uniformly at random a k-subset S of the set $[n]$. What is the probability that S contains 1?

For this experiment, the sample space Ω is the set of all k-subsets of $[n]$ and the event A of interest to us is the set of k-subsets S of $[n]$ such that $1 \in S$. We have $\#\Omega = \binom{n}{k}$ and $\#A = \binom{n-1}{k-1}$ and hence

$$\mathbf{P}(A) = \frac{\binom{n-1}{k-1}}{\binom{n}{k}} = \frac{(n-1)!}{(k-1)!\,(n-k)!} \cdot \frac{k!\,(n-k)!}{n!} = \frac{k}{n}.$$

For example, selecting uniformly at random five out of the 52 cards of an ordinary deck of cards, the probability that the ace of spades is among the selected cards is equal to $5/52 \sim 0.096$. □

Example 7.1.4. Let n be a given positive integer. We roll three dice, each with possible outcomes $1, 2, \ldots, n$, and record the results x, y, z. We assume that the dice are fair, so that all possible triples $(x, y, z) \in [n] \times [n] \times [n]$ are equiprobable. What is the probability that $x + y = z$ holds? What is the probability that $x + y \le z$ holds?

For this experiment, the sample space is the set $\Omega = [n]^3$ of all triples (x, y, z) with $x, y, z \in [n]$ and the events, say A and B, of interest to us are the sets of those triples $(x, y, z) \in \Omega$ such that $x + y = z$ and $x + y \le z$, respectively. For each $k \in [n]$, there are exactly $k - 1$ triples $(x, y, z) \in A$ such that $z = k$. Hence,

$$\#A = \sum_{k=1}^{n} (k-1) = n(n-1)/2.$$

Moreover, for each $k \in [n]$, there are exactly $1 + 2 + \cdots + (k-1) = \binom{k}{2}$ triples $(x, y, z) \in B$ such that $z = k$. Hence, according to the identity of Proposition 2.4.2,

$$\#B = \sum_{k=1}^{n} \binom{k}{2} = \binom{n+1}{3}.$$

Since $\#\Omega = n^3$, the probabilities we were asked to compute are $\mathbf{P}(A) = n(n-1)/2n^3 = (n-1)/2n^2$ and

$$\mathbf{P}(B) = \frac{1}{n^3}\binom{n+1}{3} = \frac{n^2-1}{6n^2}.$$

In particular, $\mathbf{P}(B)$ approaches 1/6 when $n \to \infty$. □

Example 7.1.5. We place randomly each of m identical balls into one of n distinct empty boxes $\Delta_1, \Delta_2, \ldots, \Delta_n$, assuming that all possible placements are equiprobable. What is the probability that the last box Δ_n ends up empty? What is the probability that no box ends up being empty?

Let x_i be the number of balls that are placed in Δ_i. Then, $x_i \in \mathbb{N}$ for every $i \in [n]$ and $x_1 + x_2 + \cdots + x_n = m$. Moreover, the set of all possible placements is in one-to-one correspondence with the set, say Ω, of vectors $(x_1, x_2, \ldots, x_n) \in \mathbb{N}^n$ such that $x_1 + x_2 + \cdots + x_n = m$. Thus, we may consider Ω as the sample space for our experiment. The two events, say A and B, that have been described consist of all vectors $(x_1, x_2, \ldots, x_n) \in \Omega$ such that $x_n = 0$ and such that $x_i \geq 1$ for every $i \in [n]$, respectively.

Applying Propositions 2.3.9 and 2.3.10, we find that $\#\Omega = \binom{m+n-1}{n-1}$, $\#A = \binom{m+n-2}{n-2}$ and $\#B = \binom{m-1}{n-1}$ and conclude that

$$\mathbf{P}(A) = \frac{\binom{m+n-2}{n-2}}{\binom{m+n-1}{n-1}} = \frac{(m+n-2)!}{m!\,(n-2)!} \cdot \frac{m!\,(n-1)!}{(m+n-1)!} = \frac{n-1}{m+n-1}$$

and

$$\mathbf{P}(B) = \frac{\binom{m-1}{n-1}}{\binom{m+n-1}{n-1}} = \frac{(m-1)!}{(n-1)!\,(m-n)!} \cdot \frac{m!\,(n-1)!}{(m+n-1)!} = \frac{(m-1)!\,m!}{(m-n)!\,(m+n-1)!},$$

where it is meant that $\mathbf{P}(B) = 0$ for $m < n$. For example, for $m = 8$ and $n = 4$ the probability that all boxes end up being nonempty is equal to $7!8!/4!11! \sim 0.212$. □

Example 7.1.6. Let n be a given positive integer. We select uniformly at random a path p in $\mathbb{Z}^2$ with initial vertex $(0, 0)$, terminal vertex (n, n), and steps $(1, 0)$ or $(0, 1)$, as in Section 2.3.8. What is the probability that p lies in the half-space $x \geq y$ of $\mathbb{R}^2$?

For this experiment, the sample space Ω is the set of all paths in $\mathbb{Z}^2$ with initial vertex $(0, 0)$, terminal vertex (n, n), and steps $(1, 0)$ or $(0, 1)$ and the event A of interest to us is the set consisting of the elements of Ω that lie in the half-space $x \geq y$ of $\mathbb{R}^2$. According to Propositions 2.3.11 and 2.3.12, we have $\#\Omega = \binom{2n}{n}$ and $\#A = \binom{2n}{n}/(n+1)$, respectively. Hence, the probability we wish to compute is given by the unexpectedly simple formula

$$\mathbf{P}(A) = \frac{\#A}{\#\Omega} = \frac{1}{n+1}$$

for every $n \in \mathbb{Z}_{>0}$. □

Example 7.1.7. We are given sets $X = \{x_1, x_2, \ldots, x_m\}$ and $Y = \{y_1, y_2, \ldots, y_n\}$ of cardinalities m and n, respectively. We select uniformly at random a map $f : X \to Y$. For example, the elements of X could be m distinguishable balls, those

of Y could be n distinguishable empty boxes and f could be the map describing a placement of the balls into the boxes, with which ball $x \in X$ is place into box $f(x) \in Y$ for every $x \in X$. What is the probability that:

(a) Balls x_1 and x_2 are placed into the same box?
(b) Box y_n ends up being nonempty?
(c) Box y_i contains exactly m_i balls for every $i \in [n]$?
(d) None of the boxes ends up being empty?

The sample space Ω for this experiment consists of all maps $f : X \to Y$ and the events of interest to us are the subsets of Ω which consist of those maps $f \in \Omega$ such that $f(x_1) = f(x_2)$ in case (a), such that $f(x) = y_n$ for at least one $x \in X$ in case (b), such that $\#f^{-1}(y_i) = m_i$ for every $i \in [n]$ in case (c) and such that $f : X \to Y$ is surjective in case (d). Let us denote by A, B, C, D these four events, respectively. Applying Proposition 2.3.1 and the basic principles of enumeration we find that $\#\Omega = n^m$, $\#A = n^{m-1}$, and $\#B = n^m - (n-1)^m$ and conclude that $\mathbf{P}(A) = 1/n$ and

$$\mathbf{P}(B) = \frac{n^m - (n-1)^m}{n^m} = 1 - \left(1 - \frac{1}{n}\right)^m .$$

Moreover, from the results of Section 2.3.5 and the formula of Proposition 3.2.1 for the number of surjective maps $f : X \to Y$, we deduce that

$$\mathbf{P}(C) = \frac{1}{n^m} \binom{m}{m_1, m_2, \ldots, m_n}$$

$$\mathbf{P}(D) = \frac{1}{n^m} \sum_{k=0}^{n} (-1)^{n-k} \binom{n}{k} k^m .$$

For example, for $m = 6$ and $n = 4$, the probability that boxes y_1, y_2, y_3, and y_4 contain two, one, one, and two balls, respectively, is equal to $(6!/2!1!1!2!)/4^6 \sim 0.043$ and the probability that no box ends up being empty is equal to $1 - 4 \cdot (3/4)^6 + 6 \cdot (2/4)^6 - 4 \cdot (1/4)^6 + 0 \sim 0.38$. □

The problem in the following example, in which the chromatic polynomial of a graph reappears, is known as the *birthday problem*.

Example 7.1.8. We are given a finite set X of cardinality n and a positive integer q. We select uniformly at random a map $f : X \to [q]$. What is the probability that f is one-to-one? For example, X could be a set of n people present in a meeting and $f(x)$ could be the birthdate of $x \in X$, where $q = 365$. Assuming that all birthdates are equiprobable, we are asked to compute the probability that no two persons in the meeting have the same birthdate.

The sample space Ω is the set of all maps $f : X \to [q]$ and the event of interest to us is the subset A of Ω that consists of the injective maps. According to

Propositions 2.3.1 and 2.3.2, we have $\#\Omega = q^n$ and $\#A = q(q-1)(q-2)\cdots(q-n+1)$, whence

$$\mathbf{P}(A) = \begin{cases} 0, & \text{if } n > q \\ \dfrac{q(q-1)(q-2)\cdots(q-n+1)}{q^n}, & \text{if } n \leq q. \end{cases} \tag{7.2}$$

For example, for $n = 25$ and $q = 365$, the probability that no two persons in a meeting of 25 people have the same birthdate is equal to $365 \cdot 364 \cdot 363 \cdots 341/365^{25} \sim 0.431$. Hence, it is more likely that two such persons exist, rather than that they do not.

Let us consider again a set X consisting of n people present in a meeting, and let us now assume that any two of these persons are either acquainted, or not. What is the probability that no two distinct, acquainted with one another, persons in the meeting have the same birthdate? Clearly, the answer depends on the simple graph G with set of vertices X and edges the pairs of acquainted persons in the meeting. As before, the sample space is the set of all maps $f : X \to [q]$, where $q = 365$. The event the probability of which we wish to compute is the set, say A_G, of maps $f \in \Omega$ such that $f(a) \neq f(b)$ for every edge $\{a, b\}$ of G. Since these maps are exactly the proper colorings of G with colors $1, 2, \ldots, q$, we have $\#A_G = \chi_G(q)$ and hence

$$\mathbf{P}(A_G) = \frac{\chi_G(q)}{q^n},$$

where $\chi_G(x)$ is the chromatic polynomial of G studied in Section 5.5.

In the special case in which G is the complete graph, we have $A_G = A$ and $\chi_G(q) = q(q-1)(q-2)\cdots(q-n+1)$, so our answer coincides with that of formula (7.2). □

7.1.2 *Finite Probability Spaces*

The previous section was concerned with experiments with which an element ω of a finite set (sample space) Ω of cardinality N is selected uniformly at random. We defined the probability of an event $A \subseteq \Omega$ by formula (7.1). The resulting function $\mathbf{P} : 2^\Omega \to \mathbb{R}$ has the properties that $\mathbf{P}(A) \geq 0$ for every $A \subseteq \Omega$, $\mathbf{P}(\Omega) = 1$ and

$$\mathbf{P}\left(\bigcup_{i=1}^{n} A_i\right) = \sum_{i=1}^{n} \mathbf{P}(A_i) \tag{7.3}$$

for mutually disjoint subsets $A_1, A_2, \ldots, A_n$ of Ω. The latter property is a direct consequence of the addition principle (Proposition 2.2.2).

However, in some experiments, the possible outcomes $\omega \in \Omega$ are *not* "equiprobable." For example, suppose that we select uniformly at random an element of the collection $\{a, b, b, c, c, c\}$. Then, $\Omega = \{a, b, c\}$ is the sample space of the experiment, but it is natural to assume that a, b, and c have probabilities $1/6$, $2/6 = 1/3$, and $3/6 = 1/2$ to be selected, respectively. The following definition allows the nonuniform selection of the elements of a sample space Ω, in other words, it allows for the probability of selection of $\omega \in \Omega$ to depend on ω.

Definition 7.1.1. *A finite probability space is defined as a pair* $(\Omega, \mathbf{P})$*, where* Ω *is a finite set and* $\mathbf{P} : 2^{\Omega} \to \mathbb{R}$ *is a function such that:*

(i) $\mathbf{P}(A) \geq 0$ *for every* $A \subseteq \Omega$,
(ii) $\mathbf{P}(\Omega) = 1$,
(iii) $\mathbf{P}(A \cup B) = \mathbf{P}(A) + \mathbf{P}(B)$ *for all* $A, B \subseteq \Omega$ *with* $A \cap B = \varnothing$. *(additivity)*

The set Ω *is called sample space, the elements of* Ω *are called outcomes, the subsets of* Ω *are called events , and the function* $\mathbf{P} : 2^{\Omega} \to \mathbb{R}$ *is called probability measure. The number* $\mathbf{P}(A)$ *is the probability of the event* $A \subseteq \Omega$.

EXAMPLE. Let us return to the experiment with which a, b, and c are selected from the set $\Omega = \{a, b, c\}$ with probabilities $1/6$, $1/3$, and $1/2$, respectively. For this experiment, we have the probability space $(\Omega, \mathbf{P})$ with probability measure $\mathbf{P} : 2^{\Omega} \to \mathbb{R}$ such that

$$\begin{array}{ll} \mathbf{P}(\varnothing) = 0 & \mathbf{P}(\{a, b\}) = 1/2 \\ \mathbf{P}(\{a\}) = 1/6 & \mathbf{P}(\{a, c\}) = 2/3 \\ \mathbf{P}(\{b\}) = 1/3 & \mathbf{P}(\{b, c\}) = 5/6 \\ \mathbf{P}(\{c\}) = 1/2 & \mathbf{P}(\{a, b, c\}) = 1. \end{array}$$

For example, the equality $\mathbf{P}(\{a, b\}) = 1/2$ means that the event that either a or b is selected has probability 1/2 and follows from $\mathbf{P}(\{a\}) = 1/6$ and $\mathbf{P}(\{b\}) = 1/3$ since, according to condition (iii) of Definition 7.1.1, $\mathbf{P}(\{a, b\}) = \mathbf{P}(\{a\}) + \mathbf{P}(\{b\}) = 1/6 + 1/3 = 1/2$.

Remark 7.1.1. Given condition (iii), the more general form (7.3) of additivity of the probability measure for mutually disjoint subsets $A_1, A_2, \ldots, A_n$ of Ω follows by induction on n. □

As discussed in the following section, after suitably modified, Definition 7.1.1 can be applied to infinite probability spaces as well. Finite probability spaces or, equivalently, probability measures on finite sets, are characterized by the following statement.

Proposition 7.1.1. *Let Ω be a finite set and $\mathbf{P} : 2^{\Omega} \to \mathbb{R}$ be a function. The pair $(\Omega, \mathbf{P})$ is a (finite) probability space if and only if there exists a function $\mathbf{p} : \Omega \to \mathbb{R}$ with the properties*

(i) $\mathbf{p}(\omega) \geq 0$ *for every* $\omega \in \Omega$,
(ii) $\sum_{\omega \in \Omega} \mathbf{p}(\omega) = 1$,

such that

$$\mathbf{P}(A) = \sum_{x \in A} \mathbf{p}(x) \tag{7.4}$$

for every $A \subseteq \Omega$. The function $\mathbf{p}$ is called the probability distribution of the space $(\Omega, \mathbf{P})$. If $\mathbf{p}$ a constant function, then it is called uniform distribution.

Proof. Let us first assume that (Ω, P) is a finite probability space. We consider the function $\mathbf{p} : \Omega \to \mathbb{R}$ for which $\mathbf{p}(\omega) = \mathbf{P}(\{\omega\})$ for every $\omega \in \Omega$. Then, (7.4) is a direct consequence of (7.3) since

$$\mathbf{P}(A) = \mathbf{P}\left(\bigcup_{x \in A} \{x\}\right) = \sum_{x \in A} \mathbf{P}(\{x\}) = \sum_{x \in A} \mathbf{p}(x)$$

for every $A \subseteq \Omega$. Properties (i) and (ii) of the proposition follow from the corresponding properties of Definition 7.1.1. The verification of the converse is left to the reader. □

Example 7.1.9. Let us cut a deck of cards $1, 2, \ldots, n$, linearly ordered in this way, into two piles. The set of possible outcomes is $\Omega = \{0, 1, \ldots, n\}$, where $k \in \Omega$ is identified with cutting the deck into the first pile $\{1, 2, \ldots, k\}$ and the second pile $\{k+1, k+2, \ldots, n\}$. It seems natural *not* to consider all elements of Ω equiprobable, since in practice one cuts the deck more often toward the middle than toward the extremes. The probability distribution that is usually considered for this experiment is the function $\mathbf{p} : \Omega \to \mathbb{R}$ with

$$\mathbf{p}(k) = \frac{1}{2^n} \binom{n}{k}$$

for $k \in \{0, 1, \ldots, n\}$. By the identity of Proposition 2.3.6 (c), the function $\mathbf{p}$ is indeed a probability distribution on Ω. For example, for $n = 10$, the event that the deck is cut in one of the positions 4, 5, or 6 is equal to

$$\mathbf{P}(\{4, 5, 6\}) = \frac{\mathbf{p}(4) + \mathbf{p}(5) + \mathbf{p}(6)}{2^{10}} = \frac{1}{2^{10}} \left\{ \binom{10}{4} + \binom{10}{5} + \binom{10}{6} \right\} = \frac{672}{1024},$$

that is, ∼0.656. Similarly, the event that the deck is cut in one of the positions 3, 4, 5, 6, or 7 can be computed as approximately equal to 0.978.

Example 7.1.10. Let $n \in \mathbb{N}$. We roll two dice, each with possible outcomes $0, 1, \ldots, n$, and record their sum. The sample space for this experiment is the set $\Omega = \{0, 1, 2, \ldots, 2n\}$. We assume that the dice are fair, meaning that the outcomes $0, 1, \ldots, n$ are equiprobable for each of them. Since for every $k \in \Omega$, the number of pairs $(a, b) \in \{0, 1, \ldots, n\}^2$ such that $a + b = k$ is equal to

$$\min\{k+1, 2n-k+1\} = \begin{cases} k+1, & \text{if } k \leq n \\ 2n-k+1, & \text{if } k \geq n+1, \end{cases}$$

it seems natural to define the probability of the outcome k for this experiment as

$$\mathbf{p}(k) = \begin{cases} \dfrac{k+1}{(n+1)^2}, & \text{if } k \leq n \\ \\ \dfrac{2n-k+1}{(n+1)^2}, & \text{if } k \geq n+1 \end{cases}$$

for $k \in \{0, 1, \ldots, 2n\}$. The function $\mathbf{p} : \Omega \to \mathbb{R}$ is a probability distribution that defines a finite probability space $(\Omega, \mathbf{P})$.

For $n = 5$, the probability that the sum of the dice is an integer multiple of 5 is equal to $\mathbf{P}(\{0, 5, 10\}) = \mathbf{p}(0) + \mathbf{p}(5) + \mathbf{p}(10) = (1 + 6 + 1)/36 = 2/9$.

Example 7.1.11. Given a positive integer n, we select a subset S of the set $[n]$ from the sample space $\Omega = 2^{[n]}$.

In many applications, as was the case in Example 7.1.1, the elements of Ω are *not* considered equiprobable. For example, S can be obtained by coloring each of the numbers $1, 2, \ldots, n$ either black or white and recording those that were colored white, or by tossing n coins, each with possible outcomes heads (H) or tails (T), and recording the set of $i \in [n]$ for which coin i gets H. Then, it is expected that there may be a preference in color, or that the coins used are not fair, and hence, that the probability of the choice of $S \in \Omega$ depends on S.

A widely used theoretical model is the following. Consider real numbers $p, q \geq 0$ such that $p + q = 1$ and set

$$\mathbf{p}(S) = p^{\#S} q^{n-(\#S)} \tag{7.5}$$

for $S \subseteq [n]$. For example, for $n = 3$

$$\begin{aligned} &\mathbf{p}(\varnothing) = q^3 \\ &\mathbf{p}(\{1\}) = \mathbf{p}(\{2\}) = \mathbf{p}(\{3\}) = pq^2 \\ &\mathbf{p}(\{1, 2\}) = \mathbf{p}(\{1, 3\}) = \mathbf{p}(\{2, 3\}) = p^2 q \\ &\mathbf{p}(\{1, 2, 3\}) = p^3. \end{aligned}$$

By the identity (2.5), we have $\sum_{S\subseteq\Omega}\mathbf{p}(S) = (p+q)^n = 1$ and hence, $\mathbf{p} : \Omega \to \mathbb{R}$ is a probability distribution on Ω. For the corresponding probability space $(\Omega, \mathbf{P})$, according to (7.4), the event that the selected set $S \in \Omega$ contains $i \in [n]$ has probability

$$\sum_{S\subseteq[n]\,:\,i\in S} p^{\#S} q^{n-(\#S)} = p \sum_{T\subseteq[n]\setminus\{i\}} p^{\#T} q^{n-1-(\#T)} = p(p+q)^{n-1} = p.$$

In particular, for the coin tossing experiment, the probability that coin i gets H is equal to p for every $i \in [n]$. □

All properties of finite probability spaces in the following proposition are consequences of Definition 7.1.1 and hence are valid for more general probability spaces.

Proposition 7.1.2. *For every finite probability space* $(\Omega, \mathbf{P})$ *we have:*

(a) $\mathbf{P}(\varnothing) = 0$,
(b) $0 \le \mathbf{P}(A) \le 1$ *for every* $A \subseteq \Omega$,
(c) $\mathbf{P}(\Omega\setminus A) = 1 - \mathbf{P}(A)$ *for every* $A \subseteq \Omega$,
(d) $\mathbf{P}(A) \le \mathbf{P}(B)$ *for all* $A, B \subseteq \Omega$ *such that* $A \subseteq B$, *(monotonicity)*
(e) $\mathbf{P}(A \cup B) = \mathbf{P}(A) + \mathbf{P}(B) - \mathbf{P}(A \cap B)$ *for all* $A, B \subseteq \Omega$,
(f) $\mathbf{P}\left(\bigcup_{i=1}^n A_i\right) \le \sum_{i=1}^n \mathbf{P}(A_i)$ *for all* $A_1, A_2, \ldots, A_n \subseteq \Omega$, *(subadditivity)*
(g) *(principle of inclusion–exclusion)*

$$\mathbf{P}\left(\Omega\setminus\bigcup_{i=1}^n A_i\right) = \sum_{k=0}^n (-1)^k \sum_{1\le i_1<\cdots<i_k\le n} \mathbf{P}(A_{i_1}\cap\cdots\cap A_{i_k}) \tag{7.6}$$

$$= \sum_{I\subseteq[n]} (-1)^{\#I}\, \mathbf{P}\left(\bigcap_{i\in I} A_i\right) \tag{7.7}$$

for all $A_1, A_2, \ldots, A_n \subseteq \Omega$, *where* $A_{i_1}\cap\cdots\cap A_{i_k} := \Omega$ *for* $k = 0$.

Proof. By the nonnegativity and the additivity of probability measure, we have $\mathbf{P}(\varnothing) = \mathbf{P}(\varnothing\cup\varnothing) = \mathbf{P}(\varnothing) + \mathbf{P}(\varnothing)$ and $\mathbf{P}(A) \le \mathbf{P}(A) + \mathbf{P}(B\setminus A) = \mathbf{P}(B)$ for $A \subseteq B \subseteq \Omega$. This proves parts (a) and (d). Parts (b) and (c) follow from the special case $B = \Omega$. To prove part (e), we apply the additivity of $\mathbf{P}$ first for A and $B\setminus(A\cap B)$ and then for $A \cap B$ and $B\setminus(A\cap B)$ and find that

$$\mathbf{P}(A\cup B) = \mathbf{P}(A) + \mathbf{P}(B\setminus(A\cap B)) = \mathbf{P}(A) + \mathbf{P}(B) - \mathbf{P}(A\cap B).$$

Part (f) holds as an equality for $n = 1$ and follows from (e) for $n = 2$ and by induction on n for every $n \ge 2$. Part (g) can be proven by induction on n (see Exercise 12) or, for finite probability spaces, as in the second proof of Theorem 3.1.1, setting however

$$\chi_I(x) = \begin{cases} \mathbf{p}(x), & \text{if } x \in \bigcap_{i \in I} A_i \\ 0, & \text{otherwise} \end{cases}$$

for $I \subseteq [n]$ and $x \in \Omega$ (the details are left to the reader). A similar proof is given in Section 7.3.2 as an application of the basic properties of expected values of random variables. □

EXAMPLE. We select uniformly at random a triple $(x, y, z) \in [n]^3$, as in Example 7.1.4. Let A and B be the events that $x + y = z$ and $x + y \leq z$ hold for the triple (x, y, z), respectively. In Example 7.1.4, we computed the probabilities $\mathbf{P}(A)$ and $\mathbf{P}(B)$. From these results, Proposition 7.1.2 (c) and the additivity of probability measure, we conclude that the probability of the events that $x + y > z$, $x + y < z$, and $x + y \geq z$ are equal to $1 - \mathbf{P}(B) = (5n^2 + 1)/6n^2$, to

$$\mathbf{P}(B \setminus A) = \mathbf{P}(B) - \mathbf{P}(A) = \frac{n^2 - 1}{6n^2} - \frac{n - 1}{2n^2} = \frac{(n - 1)(n - 2)}{6n^2}$$

and to $1 - \mathbf{P}(B \setminus A) = (5n^2 + 3n - 2)/6n^2$, respectively.

Example 7.1.12. Let n be a positive integer. We select one of the numbers $1, 2, \ldots, 6n$ with probability proportional to their size. That is, the sample space is the set $\Omega = [6n]$ and the probability distribution $\mathbf{p} : \Omega \to \mathbb{R}$ is given by the formula

$$\mathbf{p}(k) = \frac{k}{1 + 2 + \cdots + 6n} = \frac{k}{3n(6n + 1)}$$

for $k \in [6n]$. What is the probability that the selected number k is divisible by neither 2 nor 3? Let $A = \{k \in \Omega : 2 \mid k\}$ and $B = \{k \in \Omega : 3 \mid k\}$ be the events that the selected number is divisible by 2 or 3, respectively. Then, we are asked to compute the probability of $\Omega \setminus (A \cup B)$. We observe that $A \cap B = \{k \in \Omega : 6 \mid k\}$, and we compute from formula (7.4) and the definition of the function $\mathbf{p}$ that

$$\begin{aligned} \mathbf{P}(A) &= (2 + 4 + \cdots + 6n)/3n(6n + 1) = (3n + 1)/(6n + 1) \\ \mathbf{P}(B) &= (3 + 6 + \cdots + 6n)/3n(6n + 1) = (2n + 1)/(6n + 1) \\ \mathbf{P}(A \cap B) &= (6 + 12 + \cdots + 6n)/3n(6n + 1) = (n + 1)/(6n + 1). \end{aligned}$$

Applying the inclusion–exclusion principle of Proposition 7.1.2 (f) (or combining parts (c) and (e) of that proposition) we conclude that

$$\mathbf{P}(\Omega \setminus (A \cup B)) = 1 - \mathbf{P}(A) - \mathbf{P}(B) + \mathbf{P}(A \cap B) = 2n/(6n + 1).$$

This probability approaches 1/3 as $n \to \infty$. □

Application: Monochromatic Arithmetic Progressions We color each of the numbers $1, 2, \ldots, n$ either black or white. We say that a subset of $[n]$ is *monochromatic* if all its elements have the same color. For given $k \geq 2$, how large should n be so that a monochromatic arithmetic progression with k terms is guaranteed to exist?

Proposition 7.1.3. *Let $2 \leq k \leq n$ be integers. If $n \leq 2^{k/2}$, then it is possible to color each element of the set $[n]$ either black or white, so that there is no monochromatic arithmetic progression with k terms.*

Proof. Let us color each element of $[n]$ either black or white uniformly at random, so that all 2^n possible colorings are equiprobable. Let $\Pi_n(k)$ be the set of all arithmetic progressions with k terms that are contained in $[n]$. For every $S \in \Pi_n(k)$ (more generally, for every k-element subset S of $[n]$), the probability of the event, say A_S, that S is monochromatic is equal to $2 \cdot 2^{-k}$. Applying the subadditivity of probability measure (part (f) of Proposition 7.1.2), we find the upper bound

$$\mathbf{P}\left(\bigcup_{S\in\Pi_n(k)} A_S\right) \leq \sum_{S\in\Pi_n(k)} \mathbf{P}(A_S) = \#\Pi_n(k) \cdot 2^{1-k}$$

for the event $A := \cup_{S\in\Pi_n(k)} A_S$ that at least one monochromatic arithmetic progression with k terms exists.

We now observe that $\#\Pi_n(k) \leq \binom{n}{2} < n^2/2$, since every arithmetic progression with $k \geq 2$ terms is determined by its two smallest terms. Since $n \leq 2^{k/2}$ is assumed, we infer that $\mathbf{P}(A) < n^2 \cdot 2^{-k} \leq 1$. Since $\mathbf{P}(A) < 1$, the complementary event $\Omega \setminus A$ of A in the sample space Ω of 2^n colorings has positive probability. This means that there exists a coloring without monochromatic arithmetic progressions with k terms. □

The previous proof follows the *probabilistic method*, which was pioneered by Paul Erdös. In order to prove the existence of a combinatorial object with desirable properties, according to this method, one constructs a suitable probability space and proves that the set of all objects that satisfy these properties has positive probability and is therefore nonempty. For a thorough treatment of this method, see [1].

7.1.3 Discrete Probability Spaces

Many experiments of mathematical interest have infinite sample spaces. More specifically, in discrete mathematics we are interested in experiments for which the sample space is an infinite countable set.

Definition 7.1.2. *A set X is called countably infinite if there exists a bijection $f : \mathbb{N} \to X$. Then, writing $x_n = f(n)$ for $n \in \mathbb{N}$, we have $X = \{x_0, x_1, x_2, \ldots\}$ where $x_0, x_1, x_2, \ldots$ are pairwise distinct elements of X.*

We recall that $\sum_{k=1}^{\infty} a_k$ denotes the infinite sum of a series $\sum_{k\geq 1} a_k$ with nonnegative real terms. By analogy with finite probability spaces, we have the following definition.

Definition 7.1.3. *Let Ω be a countably infinite set. A probability measure on Ω is a function $\mathbf{P} : 2^{\Omega} \to \mathbb{R}$ with the following properties:*

(i) $\mathbf{P}(A) \geq 0$ *for every* $A \subseteq \Omega$,
(ii) $\mathbf{P}(\Omega) = 1$,
(iii)

$$\mathbf{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbf{P}(A_i) \tag{7.8}$$

for mutually disjoint subsets $A_1, A_2, A_3, \ldots$ of Ω.

The set Ω is called sample space, the elements of Ω are called outcomes, the subsets of Ω are called events , and the function $\mathbf{P} : 2^{\Omega} \to \mathbb{R}$ is called probability measure. The number $\mathbf{P}(A)$ is the probability of the event $A \subseteq \Omega$.

Finite probability spaces and the pairs $(\Omega, \mathbf{P})$, as above, are called discrete probability spaces.

We omit the proof of the following statement, which is similar to that of Proposition 7.1.1

Proposition 7.1.4. *Let Ω be a countably infinite set. A function $\mathbf{P} : 2^{\Omega} \to \mathbb{R}$ is a probability measure on Ω if and only if there exists a function $\mathbf{p} : \Omega \to \mathbb{R}$ with the properties*

(i) $\mathbf{p}(\omega) \geq 0$ *for every* $\omega \in \Omega$,
(ii) $\sum_{\omega\in\Omega} \mathbf{p}(\omega) = 1$,

such that

$$\mathbf{P}(A) = \sum_{x\in A} \mathbf{p}(x) \tag{7.9}$$

for every $A \subseteq \Omega$. The function $\mathbf{p}$ is called probability distribution of the discrete probability space $(\Omega, \mathbf{P})$.

Example 7.1.13. When tossing a fair coin, there are two equiprobable possible outcomes, namely heads (H) and tails (T). We toss successively one such coin until H comes up for the first time and record the stage of the process (first, second, and so on) that this has happened. It seems natural to assume that this happens at stage k with probability $1/2^k$ and that it never happens with probability zero.

This experiment has sample space $\Omega = \mathbb{Z}_{>0} \cup \{\infty\}$ and probability distribution $\mathbf{p} : \Omega \to \mathbb{R}$ given by the formula

$$\mathbf{p}(k) = \begin{cases} 1/2^k, & \text{if } k \in \mathbb{Z}_{>0} \\ 0, & \text{if } k = \infty. \end{cases}$$

Since $\sum_{k=1}^{\infty} 2^{-k} = 1$, $\mathbf{p}$ satisfies the conditions of Proposition 7.1.4. Applying (7.9) we find, for instance, that for given $d \in \mathbb{Z}_{>0}$, the probability that the outcome k of the experiment is an integer multiple of d is equal to

$$\mathbf{P}(d\mathbb{Z}_{>0}) = \sum_{m=1}^{\infty} \mathbf{p}(dm) = \sum_{m=1}^{\infty} 2^{-dm} = \frac{1}{2^d - 1}.$$

Similarly, the probability that k is greater than or equal to d equals

$$\mathbf{P}(\{d, d+1, d+2, \dots\}) = \sum_{m=d}^{\infty} \mathbf{p}(m) = \sum_{m=d}^{\infty} 2^{-m} = \frac{1}{2^{d-1}}.$$

□

Terminology In the sequel, when we refer to probability spaces we will mean discrete probability spaces. Generally speaking, after suitable modification, the basic definitions and propositions that will follow hold for more general infinite probability spaces too.

7.2 Conditional Probability and Independence

This section introduces some important and interrelated concepts, those of conditional probability and dependence or independence of events. As usual, we begin with a concrete example.

We select uniformly at random one of the permutations $(\sigma_1, \sigma_2, \sigma_3, \sigma_4)$ of the set $\{1, 2, 3, 4\}$, represented by the 2×2 matrix

$$\sigma = \begin{pmatrix} \sigma_1 & \sigma_2 \\ \sigma_3 & \sigma_4 \end{pmatrix}.$$

What is the probability that the first row of σ is increasing, that is, $\sigma_1 < \sigma_2$? What is the probability that the second row (respectively, the first column) of σ is increasing? What is the probability that the second row (respectively, the first column) of σ is increasing, if we know that the first row of σ is increasing?

The sample space Ω for this experiment has 24 elements, as many as there are permutations $(\sigma_1, \sigma_2, \sigma_3, \sigma_4)$ of 1, 2, 3, 4. Since $\sigma_1 < \sigma_2$ holds for exactly half

of these, the probability that the first row of $\sigma \in \Omega$ is increasing is equal to 1/2. Similarly, the probability that the second row (respectively, the first column) of σ is increasing is also equal to 1/2. Moreover, out of the 12 matrices $\sigma \in \Omega$ with an increasing first row, the following six

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}, \quad \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}, \quad \begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix}, \quad \begin{pmatrix} 2 & 4 \\ 1 & 3 \end{pmatrix}, \quad \begin{pmatrix} 3 & 4 \\ 1 & 2 \end{pmatrix}$$

have an increasing second row and the following eight

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}, \quad \begin{pmatrix} 1 & 2 \\ 4 & 3 \end{pmatrix}, \quad \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}, \quad \begin{pmatrix} 1 & 3 \\ 4 & 2 \end{pmatrix}, \quad \begin{pmatrix} 1 & 4 \\ 3 & 2 \end{pmatrix},$$

$$\begin{pmatrix} 2 & 3 \\ 4 & 1 \end{pmatrix}, \quad \begin{pmatrix} 2 & 4 \\ 3 & 1 \end{pmatrix}$$

have an increasing first column. Thus, we declare that the probability that the second row (respectively, the first column) of $\sigma \in \Omega$ is increasing, given that its first row is increasing, is equal to 6/12 = 1/2 (respectively, to 8/12 = 2/3).

More generally, we have the following definition.

Definition 7.2.1. *Let $A, B \subseteq \Omega$ be events of a probability space $(\Omega, \mathbf{P})$ such that $\mathbf{P}(A) > 0$. The quotient*

$$\mathbf{P}(B \mid A) := \frac{\mathbf{P}(A \cap B)}{\mathbf{P}(A)}$$

is called conditional probability of B given A.

REMARK. For finite probability spaces $(\Omega, \mathbf{P})$ with a uniform probability distribution, the formula for conditional probability of Definition 7.2.1 can be written equivalently as

$$\mathbf{P}(B \mid A) = \frac{\#(A \cap B)}{\#A}$$

for events $A, B \subseteq \Omega$ with $A \neq \varnothing$.

EXAMPLE. Let us return to the example with the selection of a permutation $(\sigma_1, \sigma_2, \sigma_3, \sigma_4)$ of $\{1, 2, 3, 4\}$. As discussed before, all events

$$\begin{aligned} A_1 &= \{\sigma \in \Omega : \sigma_1 < \sigma_2\}, \\ A_2 &= \{\sigma \in \Omega : \sigma_3 < \sigma_4\}, \\ B_1 &= \{\sigma \in \Omega : \sigma_1 < \sigma_3\}, \\ B_2 &= \{\sigma \in \Omega : \sigma_2 < \sigma_4\} \end{aligned}$$

have probability 1/2. The event $A_1 \cap A_2$ consists of the matrices $\sigma \in \Omega$ with increasing rows and has probability $\mathbf{P}(A_1 \cap A_2) = \#(A_1 \cap A_2)/(\#\Omega) = 6/24 = 1/4$. Similarly, we find that $\mathbf{P}(A_1 \cap B_1) = \#(A_1 \cap B_1)/(\#\Omega) = 8/24 = 1/3$ and verify that

$$\mathbf{P}(A_2 \mid A_1) = \mathbf{P}(A_1 \cap A_2)/\mathbf{P}(A_1) = \#(A_1 \cap A_2)/(\#A_1) = 1/2$$
$$\mathbf{P}(B_1 \mid A_1) = \mathbf{P}(A_1 \cap B_1)/\mathbf{P}(A_1) = \#(A_1 \cap B_1)/(\#A_1) = 2/3.$$

Hence, the probability of A_2 does not change if we assume that A_1 holds, whereas B_1 becomes more probable if we assume that A_1 holds (it is expected, anyway, that the inequality $\sigma_1 < \sigma_3$ becomes more probable under the assumption that $\sigma_1 < \sigma_2$). We will say that the events A_1 and A_2 are independent. The same holds for B_1 and B_2, but not for A_1 and B_1. □

Observing that $\mathbf{P}(B \mid A) = \mathbf{P}(B) \Leftrightarrow \mathbf{P}(A \cap B) = \mathbf{P}(A)\mathbf{P}(B)$ for events $A, B \subseteq \Omega$ with $\mathbf{P}(A) > 0$, we are led to the following definition.

Definition 7.2.2. *Two events $A, B \subseteq \Omega$ of a probability space $(\Omega, \mathbf{P})$ are called independent if*

$$\mathbf{P}(A \cap B) = \mathbf{P}(A)\,\mathbf{P}(B).$$

More generally, the events $A_1, A_2, \ldots, A_n \subseteq \Omega$ are called independent if

$$\mathbf{P}(A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_k}) = \mathbf{P}(A_{i_1})\,\mathbf{P}(A_{i_2}) \cdots \mathbf{P}(A_{i_k}) \tag{7.10}$$

for every $k \in [n]$ and all $1 \le i_1 < i_2 < \cdots < i_k \le n$. Otherwise, $A_1, A_2, \ldots, A_n$ are called dependent.

EXAMPLE. According to Definition 7.2.2, three events $A, B, C \subseteq \Omega$ are independent if and only if

$$\begin{aligned} \mathbf{P}(A \cap B) &= \mathbf{P}(A)\,\mathbf{P}(B), \\ \mathbf{P}(A \cap C) &= \mathbf{P}(A)\,\mathbf{P}(C), \\ \mathbf{P}(B \cap C) &= \mathbf{P}(B)\,\mathbf{P}(C), \\ \mathbf{P}(A \cap B \cap C) &= \mathbf{P}(A)\,\mathbf{P}(B)\,\mathbf{P}(C). \end{aligned} \tag{7.11}$$

Remark 7.2.1. For $n \ge 3$, the fact that the events $A_1, A_2, \ldots, A_n$ are pairwise independent does not imply that $A_1, A_2, \ldots, A_n$ are necessarily independent. For example, let us select uniformly at random one of the four elements of $\Omega = \{a, b, c, d\}$. The events $A = \{a, d\}$, $B = \{b, d\}$, and $C = \{c, d\}$ have all probability 1/2 and $\mathbf{P}(A \cap B) = \mathbf{P}(A \cap C) = \mathbf{P}(B \cap C) = \mathbf{P}(A \cap B \cap C) = \mathbf{P}(\{d\}) = 1/4$. Therefore, A, B, C are not independent but they are pairwise independent.

Let us provide more examples of conditional probability and of dependent or independent events.

Example 7.2.1. We select uniformly at random a cyclic permutation τ of the set $[n]$, where $n \geq 4$. Let A be the event that 1 and 2 appear in adjacent positions in τ and B be the analogous event for 3 and 4. Are A and B independent?

Our sample space consists of all $(n-1)!$ cyclic permutations of $[n]$ (Proposition 2.3.4). Reasoning as in Example 3.1.1, we find that $\#A = \#B = 2(n-2)!$ and $\#(A \cap B) = 2^2(n-3)!$ and conclude that

$$\begin{aligned} \mathbf{P}(A) = \mathbf{P}(B) &= 2(n-2)!/(n-1)! = 2/(n-1), \\ \mathbf{P}(A \cap B) &= 4(n-3)!/(n-1)! = 4/(n-1)(n-2), \\ \mathbf{P}(A \mid B) = \mathbf{P}(B \mid A) &= 2/(n-2). \end{aligned}$$

Hence, according to Definition 7.2.2, the events A and B are dependent for every $n \geq 4$. □

Example 7.2.2. Given real numbers $p, q \geq 0$ such that $p + q = 1$, we select a subset S of $[n]$ according to the probability distribution (7.5) (this distribution is uniform in the special case $p = q = 1/2$). For $i \in [n]$, we let A_i be the event that $i \in S$. For $1 \leq i_1 < i_2 < \cdots < i_k \leq n$, the intersection $A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_k}$ consists of all subsets of $[n]$ that contain $i_1, i_2, \ldots, i_k$. Generalizing the computation of the probability $\mathbf{P}(A_i)$ in Example 7.1.11, we compute that

$$\begin{aligned} \mathbf{P}(A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_k}) &= \sum_{\{i_1, i_2, \ldots, i_k\} \subseteq S \subseteq [n]} p^{\#S} q^{n-(\#S)} = p^k \\ &\quad \times \sum_{T \subseteq [n] \setminus \{i_1, i_2, \ldots, i_k\}} p^{\#T} q^{n-k-(\#T)} \\ &= p^k (p+q)^{n-k} = p^k. \end{aligned}$$

In particular, $\mathbf{P}(A_i) = p$ for every $i \in [n]$ and hence (7.10) holds for every $k \in [n]$ and all $1 \leq i_1 < i_2 < \cdots < i_k \leq n$. As a result, the events $A_1, A_2, \ldots, A_n$ are independent.

In the case that S is obtained from coin tossing, as explained in Example 7.1.11, A_i is the event that coin i gets heads. The tosses in this experiment are often referred to as *n independent Bernoulli trials* with parameter (probability of success at every stage) p. □

Example 7.2.3. Given $k \in [n]$, we select uniformly at random a k-subset S of $[n]$, as in Example 7.1.3. We define the events $A_1, A_2, \ldots, A_n$ as in Example 7.2.2, that is A_i is the set of k-subsets of $[n]$ which contain i. For $i \neq j$ we compute that

$$\mathbf{P}(A_i \mid A_j) = \frac{\#(A_i \cap A_j)}{\#A_j} = \binom{n-2}{k-2} \binom{n-1}{k-1}^{-1} = \frac{k-1}{n-1}.$$

Since $\mathbf{P}(A_i) = k/n$, we conclude that A_i and A_j are independent only in the special case $k = n$.

For example, selecting uniformly at random five cards from an ordinary deck of 52 cards, the conditional probability that the ace of spades is among the selected cards, given that the king of spades has been selected, is equal to $4/51 \sim 0.078$. □

New independent events can be derived from given ones, as the following proposition shows.

Proposition 7.2.1. *Let* $(\Omega, \mathbf{P})$ *be a probability space.*

(a) *If* $A, B_1, \ldots, B_m \subseteq \Omega$ *are independent events, then so are* $\Omega \setminus A, B_1, \ldots, B_m$.
(b) *If* $A, B, C_1, \ldots, C_m \subseteq \Omega$ *are independent events, then so are*

- $A \cap B, C_1, \ldots, C_m$,
- $A \cup B, C_1, \ldots, C_m$,
- $A \setminus B, C_1, \ldots, C_m$.

Proof. See Exercise 19. □

EXAMPLE. We select a subset S of the set $[n]$, as in Example 7.2.2. For $i \in [n]$, we let A_i (respectively, B_i) be the event that $i \in S$ (respectively, $i \notin S$). We observe that $B_i = \Omega \setminus A_i$, whence $\mathbf{P}(B_i) = 1 - \mathbf{P}(A_i) = q$, for every $i \in [n]$. Since $A_1, A_2, \ldots, A_n$ are independent, applying n times part (a) of Proposition 7.2.1, we conclude that $B_1, B_2, \ldots, B_n$ are also independent. More generally, $C_1, C_2, \ldots, C_n$ are independent for every choice $C_i = A_i$ or B_i, for $i \in [n]$. By a suitable application of part (b) we find, for $n \geq 4$, that $A_1 \cup A_2$, that is the event that $1 \in S$ or $2 \in S$, is independent from $B_3 \cap B_4$, which is the event that $3 \in S$ and $4 \in S$, and so on.

Example 7.2.4. Let us generalize the example in the beginning of this section as follows. Given an integer $n \geq 2$, we select uniformly at random one of the permutations $(\sigma_1, \sigma_2, \ldots, \sigma_{2n})$ of the set $[2n]$ and represent it by the $2 \times n$ matrix

$$\sigma = \begin{pmatrix} \sigma_1 & \sigma_2 & \cdots & \sigma_n \\ \sigma_{n+1} & \sigma_{n+2} & \cdots & \sigma_{2n} \end{pmatrix}.$$

We denote by A_i and B_j the events that row i and column j, respectively, of σ is increasing, for $i \in [2]$ and $j \in [n]$. In other words,

$$\begin{aligned} A_1 &= \{\sigma \in \Omega : \sigma_1 < \sigma_2 < \cdots < \sigma_n\}, \\ A_2 &= \{\sigma \in \Omega : \sigma_{n+1} < \sigma_{n+2} < \cdots < \sigma_{2n}\}, \\ B_j &= \{\sigma \in \Omega : \sigma_j < \sigma_{n+j}\} \text{ for } j \in [n], \end{aligned}$$

where Ω is the set of $(2n)!$ matrices that correspond to the permutations of $[2n]$, as described. We claim that events A_1 and A_2 are independent, as are $B_1, B_2, \ldots, B_n$, but that A_1 and B_1 are *not*. Indeed, applying the multiplication principle, or the

results of Section 2.3.5 (explain how), we find that

$$\#A_1 = \#A_2 = \binom{2n}{n} n! = (2n)!/n!,$$

so that $\mathbf{P}(A_1) = \mathbf{P}(A_2) = 1/n!$, and that

$$\#(A_1 \cap A_2) = \binom{2n}{n} = \frac{(2n)!}{(n!)^2},$$

so that $\mathbf{P}(A_1 \cap A_2) = 1/(n!)^2 = \mathbf{P}(A_1)\mathbf{P}(A_2)$. Similarly, we find that

$$\#(B_{j_1} \cap B_{j_2} \cap \cdots \cap B_{j_k}) = \binom{2n}{2}\binom{2n-2}{2} \cdots \binom{2n-2k+2}{2}(2n-2k)! = (2n)!/2^k$$

for $1 \le j_1 < j_2 < \cdots < j_k \le n$, so that $\mathbf{P}(B_j) = 1/2$ for every j and

$$\mathbf{P}(B_{j_1} \cap B_{j_2} \cap \cdots \cap B_{j_k}) = 1/2^k = \mathbf{P}(B_{j_1})\,\mathbf{P}(B_{j_2}) \cdots \mathbf{P}(B_{j_k}).$$

Finally, let us compute the conditional probability $\mathbf{P}(B_1 \mid A_1)$. The event $A_1 \cap B_1$ consists of all matrices $\sigma \in \Omega$ that have an increasing first row and an increasing first column. There are $\binom{2n}{n+1}$ ways to choose the $n+1$ elements in the first row and the first column. For every such choice, the second element σ_{n+1} of the first column can be any of the $n+1$ chosen elements, other than the smallest. Hence, there are n ways to choose σ_{n+1}, a unique way to linearly order the elements chosen for the first row in increasing order and $(n-1)!$ ways to distribute the remaining elements of $[2n]$ in the $n-1$ positions of the second row, other than the first position. These considerations and the multiplication principle imply that

$$\#(A_1 \cap B_1) = \binom{2n}{n+1} \cdot n \cdot (n-1)! = \frac{n\,(2n)!}{(n+1)!},$$

that $\mathbf{P}(A_1 \cap B_1) = \#(A_1 \cap B_1)/(2n)! = n/(n+1)!$ and that

$$\mathbf{P}(B_1 \mid A_1) = \mathbf{P}(A_1 \cap B_1)/\mathbf{P}(A_1) = n/(n+1),$$
$$\mathbf{P}(A_1 \mid B_1) = \mathbf{P}(A_1 \cap B_1)/\mathbf{P}(B_1) = 2n/(n+1)!.$$

Hence, the probability that the first column of σ is increasing gets considerably higher if we assume that the first row of σ is increasing and the probability that the first row of σ is increasing also gets considerably higher if we assume that the first column of σ is increasing. In particular, A_1 and B_1 are dependent events for every $n \ge 2$.

We leave it as an exercise (see Exercise 17) to compute the conditional probabilities $\mathbf{P}(B \mid A)$ and $\mathbf{P}(A \mid B)$, where $A = A_1 \cap A_2$ and $B = B_1 \cap B_2 \cap \cdots \cap B_n$,

that is the conditional probability that $\sigma \in \Omega$ has increasing columns, given that it has increasing rows and vice versa. □

7.3 Random Variables

As happens in other areas of mathematics, the aim of probability theory is not just to study probability spaces but also the functions defined on them. These functions are called *random variables*.

7.3.1 Definitions and Examples

We select uniformly at random a subset S of the set $[2n]$ (see Example 7.1.1). How many even and how many odd numbers does S contain?

Clearly, the answer depends on the selected subset S. Let $\Omega = 2^{[2n]}$ denote the sample space and $X(S)$ and $Y(S)$ denote the number of even and odd elements of $S \in \Omega$, respectively. For example, if $n = 5$ and $S = \{1, 3, 4, 7, 8, 9\}$, then $X(S) = 2$ and $Y(S) = 4$. We have defined two functions $X, Y : \Omega \to \mathbb{R}$, called random variables. These functions have a common image (set of possible values) $T = \{0, 1, \ldots, n\}$. For $k \in T$, there are $\binom{n}{k}2^n$ sets $S \in \Omega$ such that $X(S) = k$. Thus, the probability that $X(S) = k$ is equal to

$$\mathbf{P}(\{S \in \Omega : X(S) = k\}) = \frac{1}{2^{2n}}\binom{n}{k}2^n = \frac{1}{2^n}\binom{n}{k}.$$

The same formula holds for the function Y (for computing the probability that $X(S) = Y(S)$, see Exercise 6). The function $\mathbf{p} : T \to \mathbb{R}$ with $\mathbf{p}(k) = \binom{n}{k}2^{-n}$ for $k \in T$ coincides with that of Example 7.1.9. Setting $\mathbf{p}(t) = 0$ for $t \notin T$, which is natural, we obtain a function defined on $\mathbb{R}$, which is called the probability mass function of the random variables X and Y. These random variables are said to be equidistributed.

In the sequel, we will write $\mathbf{P}(X = t)$ instead of $\mathbf{P}(\{\omega \in \Omega : X(\omega) = t\})$ and similarly, we will adopt the notation $\mathbf{P}(X \geq t)$, $\mathbf{P}(X \leq t)$, $\mathbf{P}(X \in S)$, and so on, for $S \subseteq \mathbb{R}$.

Definition 7.3.1. *Let $(\Omega, \mathbf{P})$ be a (discrete) probability space. A (discrete) random variable on $(\Omega, \mathbf{P})$ is defined as any function $X : \Omega \to \mathbb{R}$. The function $\mathbf{p}_X : \mathbb{R} \to \mathbb{R}$ defined by*

$$\mathbf{p}_X(t) = \mathbf{P}(X = t) \tag{7.12}$$

for $t \in \mathbb{R}$ is called the probability mass function (or just mass function) of X.

Two random variables with equal mass functions are said to be equidistributed.

Remark 7.3.1. Since the events $\{\omega \in \Omega : X(\omega) = t\}$ for $t \in \mathbb{R}$ are pairwise disjoint (and only finitely many, or countably infinitely many of them are nonempty) and their union is equal to Ω, the function $\mathbf{p}_X : \mathbb{R} \to \mathbb{R}$ satisfies conditions (i) and (ii) of Proposition 7.1.1. Consequently, the restriction $\overline{\mathbf{p}}_X : T \to \mathbb{R}$ of the mass function $\mathbf{p}_X$ on the set $T := X(\Omega)$ is a probability distribution of a discrete probability space, namely of $(T, \mathbf{P}_T)$, where the probability measure $\mathbf{P}_T$ is defined as $\mathbf{P}_T(S) = \mathbf{P}(X \in S)$ for $S \subseteq T$. In the sequel, by the term *probability distribution of* X, we will mean the function $\overline{\mathbf{p}}_X$. □

The crucial properties and invariants of a random variable $X : \Omega \to \mathbb{R}$ of interest to us (such as the expected value and the variance of X, to be studied in the following section) depend only on the mass function of X and not necessarily on X itself.

Example 7.3.1. Given $n \in \mathbb{N}$, we set $\Omega := \{0, 1, \ldots, n\} \times \{0, 1, \ldots, n\}$ and select uniformly at random a pair $(a, b) \in \Omega$, for instance, tossing independently two fair dice, each with possible outcomes $0, 1, \ldots, n$.

Let X and Y be the outcomes of the two coin tosses. Then, $X, Y : \Omega \to \mathbb{R}$ are the random variables defined by setting $X(\omega) = a$ and $Y(\omega) = b$ for $\omega = (a, b) \in \Omega$. For every $k \in \{0, 1, \ldots, n\}$, there are exactly $n + 1$ pairs $\omega \in \Omega$ such that $X(\omega) = k$ (respectively, $Y(\omega) = k$). Therefore, X and Y are equidistributed random variables with common probability mass function given by

$$\mathbf{p}_X(k) = \mathbf{p}_Y(k) = \frac{n+1}{\#\Omega} = \frac{1}{n+1}$$

for $k \in \{0, 1, \ldots, n\}$, which is the uniform probability distribution on $\{0, 1, \ldots, n\}$. The sum $Z = X + Y$, meaning the function $Z : \Omega \to \mathbb{R}$ with $Z(\omega) = a + b$ for $\omega = (a, b) \in \Omega$, is also a random variable, the probability distribution of which is the function $\mathbf{p}$ of Example 7.1.10.

Since for every $k \in \{0, 1, \ldots, 2n\}$, there are as many pairs $(a, b) \in \Omega$ with $a + b = k$ as with $a + b = 2n - k$ (explain why), the function $Z' : \Omega \to \mathbb{R}$ with $Z'(\omega) = 2n - a - b$ for $\omega = (a, b) \in \Omega$ is a random variable equidistributed with Z. □

Example 7.3.2. Given $k \in [n]$, we select uniformly at random a k-subset S of the set $[n]$, as in Example 7.1.3. Setting $X(S) = \max(S)$ for every such S, we have defined a random variable $X : \Omega \to \mathbb{R}$ with possible values $k, k + 1, \ldots, n$, where Ω is the sample space of all k-subsets of $[n]$. Since the number of k-subsets of $[n]$ with maximum element t is equal to $\binom{t-1}{k-1}$, the probability mass function of X is given by the formula

$$\mathbf{p}_X(t) = \binom{t-1}{k-1}\binom{n}{k}^{-1} \tag{7.13}$$

for $t \in \{k, k+1, \ldots, n\}$ and $\mathbf{p}_X(t) = 0$ for all other values of $t \in \mathbb{R}$. We note that the equality $\sum_{t=k}^{n} \mathbf{p}_X(t) = 1$ is equivalent to the identity of Proposition 2.4.2 of Chapter 2.

Example 7.3.3. We select a subset S of the set $[n]$ with probability distribution (7.5), as in Example 7.2.2. Let $X(S)$ be the number of elements of S, so that $\#X(S) \in \{0, 1, \ldots, n\}$. Since every k-subset of $[n]$ has probability $p^k q^{n-k}$ to occur, we have defined a random variable $X : \Omega \to \mathbb{R}$ with probability mass function given by the formula

$$\mathbf{p}_X(k) = \binom{n}{k} p^k (1-p)^{n-k} \tag{7.14}$$

for $k \in \{0, 1, \ldots, n\}$ and $\mathbf{p}_X(t) = 0$ for all other values of $t \in \mathbb{R}$, where $\Omega = 2^{[n]}$. The probability distribution defined by this formula is called *binomial distribution* with parameters n and p.

For example, coloring independently either black or white each one of six objects, with probability 2/3 and 1/3, respectively, the probability that exactly four objects will turn out to be black is equal to $\binom{6}{4}(2/3)^4(1/3)^2 = 15 \cdot 2^4/3^6 \sim 0.329$.

Example 7.3.4. Let Ω be the set of vectors $(x_1, x_2, \ldots, x_n) \in \mathbb{N}^n$ such that $x_1 + x_2 + \cdots + x_n = m$. We select uniformly at random one of the elements of Ω, which we may consider as placements of m indistinguishable balls into n distinct empty boxes, as in Example 7.1.5. For $\sigma = (x_1, x_2, \ldots, x_n) \in \mathbb{N}^n$, we denote by $X(\sigma)$ the number of indices $i \in [n]$ such that $x_i = 0$ (this is the number of boxes that remain empty after distributing the balls), so that $\#X(\sigma) \in \{0, 1, \ldots, n\}$. The probability mass function of the random variable $X : \Omega \to \mathbb{R}$ defined this way is given by the formula

$$\mathbf{p}_X(k) = \frac{1}{\#\Omega}\binom{n}{k}\binom{m-1}{n-k-1} = \frac{\binom{n}{k}\binom{m-1}{n-k-1}}{\binom{m+n-1}{n-1}} \tag{7.15}$$

for $k \in \{0, 1, \ldots, n\}$ and $\mathbf{p}_X(t) = 0$ for all other values of $t \in \mathbb{R}$. This is so because to select a placement $\sigma \in \Omega$ such that $X(\sigma) = k$ there are $\binom{n}{k}$ ways to choose the k boxes that remain empty and for every such choice, there are $\binom{m-1}{n-k-1}$ ways to place the m balls into the other $n-k$ boxes so that at least one ball has been placed to each one of those.

The probability distribution defined by (7.15) is called *hypergeometric distribution*.

Example 7.3.5. A deck consists of cards $1, 2, 3, \ldots, 2n$, from which $1, 2, \ldots, n$ are white and the rest are black. We shuffle the deck well, so that a permutation $\sigma = (\sigma_1, \sigma_2, \ldots, \sigma_{2n})$ of the cards arises (all permutations are equiprobable). Let $X(\sigma)$ be the number of white cards among $\sigma_1, \sigma_2, \ldots, \sigma_n$. We have defined a random variable $X : \Omega \to \mathbb{R}$ with possible values $0, 1, \ldots, n$, where Ω is the set of all $(2n)!$ permutations of $1, 2, \ldots, 2n$. What is the probability mass function of X?

For $k \in \{0, 1, \ldots, n\}$, to select a permutation $\sigma \in \Omega$ such that $X(\sigma) = k$, there are $\binom{n}{k}\binom{n}{n-k}$ ways to choose the k white and the $n-k$ black cards that appear in the first n positions of σ and for every such choice, there are $(n!)^2$ ways to permute the chosen cards for the first n positions and separately the rest. We conclude that

$$\mathbf{p}_X(k) = \frac{1}{(2n)!}\binom{n}{k}\binom{n}{n-k}(n!)^2 = \binom{n}{k}^2\binom{2n}{n}^{-1}$$

for $k \in \{0, 1, \ldots, n\}$. For example, for $n = 4$, the probability that there are exactly two white cards in the first four positions in the shuffled deck is equal to $\binom{4}{2}^2\binom{8}{4}^{-1} = 36/70 \sim 0.514$.

The equality $\sum_{t=0}^{n} \mathbf{p}_X(t) = 1$ is equivalent to the identity of Corollary 2.4.1 (a). □

Example 7.3.6. We select uniformly at random a permutation from the set, say Ω, of all $n!$ permutations of $[n]$. We denote by $X(\sigma)$ the number of fixed points of $\sigma \in \Omega$ and by D_n the number of permutations of $[n]$ without fixed points (see Section 3.2.3). Then, $X : \Omega \to \mathbb{R}$ is a random variable with possible values $j \in \{0, 1, \ldots, n\}$ with $j \neq n-1$ (explain why). According to Exercise 22 (a) of Chapter 3, the probability mass function of X is given by the formula

$$\mathbf{p}_X(k) = \frac{1}{n!}\binom{n}{k}D_{n-k}$$

for $k \in \{0, 1, \ldots, n\}$ and $\mathbf{p}_X(t) = 0$ for all other values of $t \in \mathbb{R}$.

For example, for $n = 5$, the probability that a permutation of 1, 2, 3, 4, 5 has exactly one fixed point is equal to $\binom{5}{1}D_4/5! = 9/24 = 0.375$.

7.3.2 Expected Value and Variance

This section studies the expected value and the variance of a random variable X, two quantities that provide useful information for X and depend only on its probability mass function. To compute them in concrete examples, we will apply the methods of computation of combinatorial sums that we have developed in previous chapters.

We recall (and occasionally emphasize) that all probability spaces and random variables we mention are meant to be discrete.

Expected Value The average value of a random variable X, where every value is weighted by its own probability, is called expected value, or expectation, of X. When X takes either finitely many values, or only nonnegative (possibly infinitely many) values, as will be the case in the examples studied in this section, the expected value of X is defined by the formula (7.16) as an element of $\mathbb{R}$ or $[0, \infty]$, respectively.

Definition 7.3.2. *Let* $(\Omega, \mathbf{P})$ *be a (discrete) probability space. The expected value (or expectation) of a random variable* $X : \Omega \to \mathbb{R}$ *with probability mass function* $\mathbf{p}_X : \mathbb{R} \to \mathbb{R}$ *is defined as*

$$E(X) = \sum_{t \in \mathbb{R}} t\, \mathbf{p}_X(t) \tag{7.16}$$

if $X(\omega) \geq 0$ *for every* $\omega \in \Omega$*, or* $X(\omega) \leq 0$ *for every* $\omega \in \Omega$*, and is an element of* $[0, \infty]$ *or* $[-\infty, 0]$*, respectively. In general, we set*

$$E^+(X) = \sum_{t \in \mathbb{R}} t^+ \mathbf{p}_X(t), \tag{7.17}$$

$$E^-(X) = \sum_{t \in \mathbb{R}} t^- \mathbf{p}_X(t), \tag{7.18}$$

where $t^+ = \max\{0, t\}$ *and* $t^- = \min\{0, t\}$ *for* $t \in \mathbb{R}$*, and define that* $E(X) = E^+(X) + E^-(X)$*, except if* $E^+(X) = \infty$ *and* $E^-(X) = -\infty$*. In that case,* $E(X)$ *is not defined.*

REMARK. The sums on the right-hand sides of (7.16), (7.17), and (7.18) have either finitely many, or countably infinitely many nonzero terms. Only those terms with $t > 0$ and $t < 0$ contribute to the sums of (7.17) and (7.18), respectively.

EXAMPLE. Let $X : \Omega \to \mathbb{R}$ be a random variable that takes the values $1, 2, \ldots, n$ for some $n \in \mathbb{Z}_{>0}$.

(a) Suppose that the probability distribution of X is uniform on $[n]$, that is $\mathbf{p}_X(k) = 1/n$ for every $k \in [n]$. Then, $E(X) = \sum_{k=1}^{n} k(1/n) = n(n+1)/2n = (n+1)/2$, where we have applied the identity of Exercise 2 (a) of Chapter 1.

(b) Suppose that the probability distribution of X is such that every $k \in [n]$ has probability proportional to its size, that is,

$$\mathbf{p}_X(k) = \frac{k}{1 + 2 + \cdots + n} = \frac{2k}{n(n+1)}.$$

The expected value of X is then "expected" to range between $(n+1)/2$ and n. Indeed, according to (7.16),

$$E(X) = \sum_{k=1}^{n} k \mathbf{p}_X(k) = \sum_{k=1}^{n} k \cdot \frac{2k}{n(n+1)} = \frac{2}{n(n+1)} \sum_{k=1}^{n} k^2$$

$$= \frac{2}{n(n+1)} \cdot \frac{n(n+1)(2n+1)}{6} = \frac{2n+1}{3},$$

where we have applied the identity of Exercise 2 (c) of Chapter 1 to compute the sum $\sum_{k=1}^{n} k^2$.

EXAMPLE.

(a) Let $X : \Omega \to \mathbb{R}$ be a random variable that takes the values $1, 2, 3, \ldots$ with probability $\mathbf{p}(k) = 1/k(k+1)$ for $k \in \mathbb{Z}_{>0}$. Since

$$\sum_{k=1}^{\infty} \mathbf{p}(k) = \sum_{k=1}^{\infty} \frac{1}{k(k+1)} = \sum_{k=1}^{\infty} \left(\frac{1}{k} - \frac{1}{k+1} \right)$$
$$= \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \cdots = 1,$$

the function $\mathbf{p} : \mathbb{Z}_{>0} \to \mathbb{R}$ is indeed a probability distribution on $\mathbb{Z}_{>0}$ and

$$E(X) = \sum_{k=1}^{\infty} k\, \mathbf{p}(k) = \sum_{k=1}^{\infty} \frac{1}{k+1} = \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots = \infty.$$

(b) Let $Y : \Omega \to \mathbb{R}$ be a random variable that takes the values $-1, 2, -3, 4, \ldots$ with probability $\mathbf{p}((-1)^k k) = 1/k(k+1)$ for $k \in \mathbb{Z}_{>0}$. Then,

$$E^+(Y) = \sum_{k=1}^{\infty} 2k\, \mathbf{p}(2k) = \sum_{k=1}^{\infty} \frac{1}{2k+1} = \frac{1}{3} + \frac{1}{5} + \frac{1}{7} + \cdots = \infty,$$

$$E^-(Y) = -\sum_{k=1}^{\infty} (2k-1)\, \mathbf{p}(2k-1) = -\sum_{k=1}^{\infty} \frac{1}{2k} = -\frac{1}{2} - \frac{1}{4} - \frac{1}{6} - \cdots = -\infty$$

and hence $E(Y)$ is undefined.

Example 7.3.7. Let $(\Omega, \mathbf{P})$ be a probability space. Given an event $A \subseteq \Omega$, the *characteristic function* $\chi_A : \Omega \to \mathbb{R}$ is defined by setting

$$\chi_A(\omega) = \begin{cases} 1, & \text{if } \omega \in A \\ 0, & \text{if } \omega \notin A \end{cases}$$

for $\omega \in \Omega$. The function χ_A is a random variable with probability mass function $\mathbf{p}_A : \mathbb{R} \to \mathbb{R}$ such that

$$\mathbf{p}_A(0) = \mathbf{P}(\chi_A = 0) = \mathbf{P}(\Omega \setminus A) = 1 - \mathbf{P}(A),$$
$$\mathbf{p}_A(1) = \mathbf{P}(\chi_A = 1) = \mathbf{P}(A)$$

and $\mathbf{p}_A(t) = 0$ for $t \in \mathbb{R} \setminus \{0, 1\}$. The expected value of χ_A is computed as $E(\chi_A) = 1 \cdot \mathbf{P}(A) + 0 \cdot (1 - \mathbf{P}(A)) = \mathbf{P}(A)$.

Example 7.3.8. Let us compute the expected value of the random variable X of Example 7.3.2, defined by $X(S) = \max(S)$, where S is selected uniformly at random from the set of all k-subsets of $[n]$.

Applying (7.16) and formula (7.13) for the probability mass function of X, we find that

$$E(X) = \left(\sum_{t=k}^{n} t\binom{t-1}{k-1}\right)\binom{n}{k}^{-1}.$$

We observe that $t\binom{t-1}{k-1} = k\binom{t}{k}$ and apply the identity of Proposition 2.4.2 of Chapter 2 to conclude that

$$E(X) = k\left(\sum_{t=k}^{n}\binom{t}{k}\right)\binom{n}{k}^{-1} = k\binom{n+1}{k+1}\binom{n}{k}^{-1} = \frac{k(n+1)}{k+1}.$$

For $k = 1$, we get the answer $(n+1)/2$, in agreement with a previous example.

Example 7.3.9. Let us generalize the probability distribution of Example 7.1.13. We consider $0 < p \leq 1$ and a random variable X that takes the values $1, 2, 3, \ldots$ with probability mass function

$$\mathbf{p}_X(k) = p(1-p)^{k-1} \tag{7.19}$$

for $k \in \mathbb{Z}_{>0}$ and $\mathbf{p}_X(t) = 0$ for all other values of $t \in \mathbb{R}$.

Since $\sum_{k=1}^{\infty} p(1-p)^{k-1} = 1$, the right-hand side of (7.19) indeed defines a probability distribution on $\mathbb{Z}_{>0}$, known as the *geometric distribution* with parameter p. We compute the expected value of X as

$$E(X) = \sum_{k=1}^{\infty} kp(1-p)^{k-1} = p\sum_{k=0}^{\infty}(k+1)(1-p)^k = \frac{p}{(1-(1-p))^2} = \frac{1}{p},$$

where we have applied (6.19) for $k = 1$ to compute the infinite sum. □

The following proposition describes some useful properties of the expected value. We recall that given functions $X, Y : \Omega \to \mathbb{R}$ and $\lambda, \mu \in \mathbb{R}$, the function $\lambda X + \mu Y : \Omega \to \mathbb{R}$ is defined by setting $(\lambda X + \mu Y)(\omega) = \lambda X(\omega) + \mu Y(\omega)$ for $\omega \in \Omega$.

Proposition 7.3.1. *Suppose that the expected values of $X, Y : \Omega \to \mathbb{R}$ are defined, and let $a, b, \lambda \in \mathbb{R}$.*

(a) *If $a \leq X(\omega) \leq b$ for every $\omega \in \Omega$, then $a \leq E(X) \leq b$.*
(b) *We have*

$$E(X_1 + X_2 + \cdots + X_n) = E(X_1) + E(X_2) + \cdots + E(X_n), \tag{7.20}$$

under the assumption that all random variables $X_1, X_2, \ldots, X_n : \Omega \to \mathbb{R}$ take finitely many values, or that they all take only nonnegative values.

(c) *We have $E(\lambda X) = \lambda E(X)$ for every $\lambda \in \mathbb{R}$, where $0 \cdot (\pm\infty) = 0$ by convention.*

(d) *If $X(\omega) \leq Y(\omega)$ for every $\omega \in \Omega$, then $E(X) \leq E(Y)$.*

(e) *We have $|E(X)| \leq E(|X|)$, where $|X| : \Omega \to \mathbb{R}$ is the random variable defined by $|X|(\omega) = |X(\omega)|$ for $\omega \in \Omega$.*

Proof. Given that the probability mass function $\mathbf{p}_X : \mathbb{R} \to \mathbb{R}$ has properties (i) and (ii) of Proposition 7.1.1 (see Remark 7.3.1), part (a) is a direct consequence of Definition 7.3.2. For example, if $X(\omega) \geq a$ for every $\omega \in \Omega$, then $t \geq a$ for every $t \in \mathbb{R}$ such that $\mathbf{p}_X(t) \neq 0$ and hence

$$E(X) = \sum_{t \in \mathbb{R}} t\, \mathbf{p}_X(t) \geq \sum_{t \in \mathbb{R}} a\, \mathbf{p}_X(t) = a \sum_{t \in \mathbb{R}} \mathbf{p}_X(t) = a.$$

Part (c) is a direct consequence of Definition 7.3.2. Part (d) follows from the equivalent expression

$$E(X) = \sum_{\omega \in \Omega} X(\omega)\mathbf{p}(\omega)$$

for the expected value, where $\mathbf{p} : \Omega \to \mathbb{R}$ is the probability distribution of the discrete probability space $(\Omega, \mathbf{P})$. This expression follows from (7.16) and the fact that $\mathbf{p}_X(t) = \sum_{X(\omega)=t} \mathbf{p}(\omega)$ for $t \in \mathbb{R}$. Alternatively, part (d) follows by combining parts (a) and (b). Part (e) follows by applying part (c) to the inequalities $-|X(\omega)| \leq X(\omega) \leq |X(\omega)|$ for $\omega \in \Omega$.

It remains to prove part (b). For that, it suffices to prove that $E(X + Y) = E(X) + E(Y)$ when $X, Y : \Omega \to \mathbb{R}$ both take finitely many values, or both take only nonnegative values, since (7.20) follows from the special case $n = 2$ by induction on n. We consider the events

$$A_X(u) = \{\omega \in \Omega : X(\omega) = u\}$$
$$A_Y(v) = \{\omega \in \Omega : Y(\omega) = v\}$$

for $u, v \in \mathbb{R}$. Since for $t \in \mathbb{R}$ we have $X(\omega) + Y(\omega) = t$ if and only if $X(\omega) = u$ and $Y(\omega) = v$ for some $u, v \in \mathbb{R}$ with $u + v = t$, the event $\{\omega \in \Omega : (X + Y)(\omega) = t\}$ is equal to the disjoint union of $A_X(u) \cap A_Y(v)$ with $u + v = t$. Therefore, by the additivity of probability measure,

$$\mathbf{p}_{X+Y}(t) = \mathbf{P}(X + Y = t) = \sum_{u+v=t} f(u, v),$$

where the sum ranges over all pairs $(u, v) \in \mathbb{R} \times \mathbb{R}$ such that $u + v = t$, and $f(u, v) = \mathbf{P}(A_X(u) \cap A_Y(v)) = \mathbf{P}(X = u, Y = v)$ is the probability that $X(\omega) = u$ and $Y(\omega) = v$. We observe that $A_X(u)$ is the disjoint union of the events $A_X(u) \cap$

$A_Y(v)$ for $v \in \mathbb{R}$ and hence, by the additivity of probability measure,

$$\mathbf{p}_X(u) = \mathbf{P}(A_X(u)) = \sum_{v\in\mathbb{R}} f(u,v)$$

for every $u \in \mathbb{R}$. Similarly,

$$\mathbf{p}_Y(v) = \mathbf{P}(A_Y(v)) = \sum_{u\in\mathbb{R}} f(u,v)$$

for every $v \in \mathbb{R}$. Applying (7.16) and our previous remarks, we conclude that

$$\begin{aligned} E(X+Y) &= \sum_{t\in\mathbb{R}} t\,\mathbf{p}_{X+Y}(t) = \sum_{t\in\mathbb{R}} t \sum_{u+v=t} f(u,v) \\ &= \sum_{u\in\mathbb{R}}\sum_{v\in\mathbb{R}} (u+v) f(u,v) \\ &= \sum_{u\in\mathbb{R}} u \left(\sum_{v\in\mathbb{R}} f(u,v) \right) + \sum_{v\in\mathbb{R}} v \left(\sum_{u\in\mathbb{R}} f(u,v) \right) \\ &= \left(\sum_{u\in\mathbb{R}} u\,\mathbf{p}_X(u) \right) + \left(\sum_{v\in\mathbb{R}} v\,\mathbf{p}_Y(v) \right) \\ &= E(X) + E(Y) \end{aligned}$$

and the proof follows. □

We now apply Proposition 7.3.1 (b) and other methods, in order to compute the expected value of some of the random variables studied in the previous section.

Example 7.3.10. Suppose that X has the binomial distribution (7.14) with parameters n and p. We will prove that $E(X) = np$. For example, performing n independent tosses of a coin, each with probability 1/3 of getting H, the expected number of tosses with outcome H is equal to $n/3$.
First Proof We apply directly the defining equation (7.16), formula (7.14) for the probability mass function of X, the identity $k\binom{n}{k} = n\binom{n-1}{k-1}$ and the Binomial Theorem and find that

$$\begin{aligned} E(X) &= \sum_{k=0}^{n} k\,\mathbf{p}_X(k) = \sum_{k=0}^{n} k \binom{n}{k} p^k (1-p)^{n-k} = \sum_{k=1}^{n} n \binom{n-1}{k-1} p^k (1-p)^{n-k} \\ &= np \sum_{k=1}^{n} \binom{n-1}{k-1} p^{k-1} (1-p)^{n-k} \\ &= np \cdot (p + (1-p))^{n-1} = np. \end{aligned}$$

Second Proof We may assume that X is the random variable of Example 7.3.3; in other words, $X(S)$ is the cardinality of S when that is selected from the powerset $\Omega = 2^{[n]}$ with probability distribution (7.5). For $i \in [n]$, we define the random variable $X_i : \Omega \to \mathbb{R}$ by setting

$$X_i(S) = \begin{cases} 1, & \text{if } i \in S \\ 0, & \text{if } i \notin S, \end{cases}$$

for $S \subseteq [n]$. Then, X_i is the characteristic function of the event $A_i = \{S \in \Omega : i \in S\}$ and hence $E(X_i) = \mathbf{P}(A_i) = p$ for every $i \in [n]$ (see Examples 7.2.2 and 7.3.7). Moreover, $X = X_1 + X_2 + \cdots + X_n$ and hence $E(X) = E(X_1) + E(X_2) + \cdots + E(X_n) = np$. □

Example 7.3.11. Let us return to Example 7.3.4. Let Ω be the set of placements of m indistinguishable balls into n distinct empty boxes $\Delta_1, \Delta_2, \ldots, \Delta_n$ and let $X(\sigma)$ be the number of boxes that end up being empty with $\sigma \in \Omega$. We will prove that $E(X) = n(n-1)/(m+n-1)$.

First Proof We apply directly the defining equation (7.16), formula (7.15) for the probability mass function of X, the identity $k\binom{n}{k} = n\binom{n-1}{k-1}$ and that of Proposition 2.4.1 and find that

$$\begin{aligned} E(X) &= \sum_{k=0}^{n} k\, \mathbf{p}_X(k) = \sum_{k=0}^{n} k \binom{n}{k} \binom{m-1}{n-k-1} \binom{m+n-1}{n-1}^{-1} \\ &= n \binom{m+n-1}{n-1}^{-1} \sum_{k=1}^{n} \binom{n-1}{k-1} \binom{m-1}{n-k-1} \\ &= n \binom{m+n-1}{n-1}^{-1} \binom{m+n-2}{m} \\ &= \frac{n(n-1)}{m+n-1}. \end{aligned}$$

Second Proof We express X as the sum $X = X_1 + X_2 + \cdots + X_n$, where X_i is the characteristic function of the event, say A_i, that Δ_i ends up being empty. We showed in Example 7.1.5 that $\mathbf{P}(A_i) = (n-1)/(m+n-1)$ for every $i \in [n]$. We conclude that $E(X_i) = (n-1)/(m+n-1)$ for every $i \in [n]$ by Example 7.3.7 and the desired formula for $E(X)$ follows from (7.20). □

Example 7.3.12. Let Ω be the set of the $n!$ permutations of $1, 2, \ldots, n$. We select uniformly at random a permutation $\sigma \in \Omega$ and denote by $X(\sigma)$ the number of fixed points of σ, as in Example 7.3.6. We will show that $E(X) = 1$ for every $n \in \mathbb{Z}_{>0}$, meaning that permutations of $[n]$ have, on the average, one fixed point.

Indeed, for $i \in [n]$, we let $A_i = \{(\sigma_1, \sigma_2, \ldots, \sigma_n) \in \Omega : \sigma_i = i\}$ be the event that i is a fixed point of $\sigma \in \Omega$ and denote by $X_i : \Omega \to \mathbb{R}$ the characteristic function of A_i (see Example 7.3.7). Then, $E(X_i) = \mathbf{P}(A_i) = (n-1)!/n! = 1/n$ for every $i \in [n]$ and hence $E(X) = E(X_1 + X_2 + \cdots + X_n) = (1/n) + (1/n) + \cdots + (1/n) = 1$.

We leave it to the reader to reach the same conclusion by applying Definition 7.3.2 and Exercise 22 (c) of Chapter 3. □

Example 7.3.13. We select uniformly at random a tree from the set, say $\mathcal{T}_n$, of all trees with set of vertices $[n]$. Let $X(\tau)$ be the number of leaves of $\tau \in \mathcal{T}_n$. Then, we showed in Proposition 5.3.5 that $E(X) = (n-1)^{n-2}/n^{n-3}$, and we concluded that $E(X) \sim n/e$ as $n \to \infty$.

A close look in the proof shows that, essentially, we expressed the random variable X as the sum of the characteristic functions of the events that i is a leaf of $\tau \in \mathcal{T}_n$, for $i \in [n]$, and we computed that the probability of each of these events is equal to $(n-1)^{n-2}/n^{n-2}$. □

Application: Principle of Inclusion–Exclusion Let us prove Proposition 7.1.2 (g) by exploiting the properties of the expected value. We consider events $A_1, A_2, \ldots, A_n \subseteq \Omega$, their characteristic functions $\chi_{A_1}, \chi_{A_2}, \ldots, \chi_{A_n} : \Omega \to \mathbb{R}$ (see Example 7.3.7), and the random variable $X : \Omega \to \mathbb{R}$ defined by setting

$$X(\omega) = \prod_{i=1}^{n} (1 - \chi_{A_i}(\omega))$$

for every $\omega \in \Omega$. The definition of χ_{A_i} implies that X is the characteristic function of the event $\Omega \setminus \cup_{i=1}^{n} A_i$, so that $E(X) = \mathbf{P}(\Omega \setminus \cup_{i=1}^{n} A_i)$. Moreover, by the identity (2.5) and the linearity of the expected value, we have

$$E(X) = E\left(\prod_{i=1}^{n} (1 - \chi_{A_i})\right) = E\left(\sum_{k=0}^{n} (-1)^k \sum_{1 \le i_1 < i_2 < \cdots < i_k \le n} \chi_{A_{i_1}} \chi_{A_{i_2}} \cdots \chi_{A_{i_k}}\right)$$

$$= \sum_{k=0}^{n} (-1)^k \sum_{1 \le i_1 < i_2 < \cdots < i_k \le n} E(\chi_{A_{i_1}} \chi_{A_{i_2}} \cdots \chi_{A_{i_k}})$$

$$= \sum_{k=0}^{n} (-1)^k \sum_{1 \le i_1 < i_2 < \cdots < i_k \le n} \mathbf{P}(A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_k}),$$

where for the last equality we have used the fact that $\chi_{A_{i_1}} \chi_{A_{i_2}} \cdots \chi_{A_{i_k}}$ is the characteristic function of $A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_k}$. The result follows by comparing the two expressions we have reached for $E(X)$.

Application: Orientations of Graphs We consider the complete graph K_n with vertex set $[n]$. We recall (see Section 5.5) that an orientation of K_n is a choice of direction $a \to b$ or $b \to a$ for every edge $\{a, b\}$. Given such an orientation, we say

that an ordered triple (a, b, c) of distinct vertices of K_n is *transitive* if the directions $a \to b$, $b \to c$, and $a \to c$ have been chosen. What is the minimum number of transitive triples that an orientation of K_n can have? The following statement gives a partial answer to this question (see also Exercise 32), as an application of the theory of this section.

Proposition 7.3.2. *For every integer $n \geq 3$, there exists an orientation of the complete graph K_n that has at most $n(n-1)(n-2)/8$ transitive triples.*

Proof. Let Ω be the set of all orientations of K_n. We select uniformly at random an element $\omega \in \Omega$ and denote by $X(\omega)$ the number of transitive triples of ω. We will compute the expected value of the random variable $X : \Omega \to \mathbb{R}$.

The probability that a given triple $\sigma = (a, b, c)$ of distinct vertices of K_n is transitive is equal to 1/8. Hence, if X_σ is the characteristic function of this event, then $E(X_\sigma) = 1/8$. Since X is equal to the sum of the X_σ when σ runs through all $n(n-1)(n-2)$ triples of distinct vertices of K_n, we have $E(X) = n(n-1)(n-2)/8$. This fact and Proposition 7.3.1 (a) imply that $X(\omega) \leq n(n-1)(n-2)/8$ for some $\omega \in \Omega$ and the proof follows. □

The reader may naturally wonder whether, in analogy with the rule $E(X+Y) = E(X) + E(Y)$, one also has that $E(XY) = E(X)E(Y)$. This is true under the assumption that X and Y are independent random variables.

Definition 7.3.3. *Let $(\Omega, \mathbf{P})$ be a (discrete) probability space. Two random variables $X, Y : \Omega \to \mathbb{R}$ are called independent if*

$$\mathbf{P}(X = a, Y = b) = \mathbf{P}(X = a)\mathbf{P}(Y = b), \tag{7.21}$$

that is the events $\{\omega \in \Omega : X(\omega) = a\}$ and $\{\omega \in \Omega : Y(\omega) = b\}$ are independent, for all $a \in X(\Omega)$ and $b \in Y(\Omega)$.

More generally, the random variables $X_1, X_2, \ldots, X_n : \Omega \to \mathbb{R}$ are called independent if

$$\mathbf{P}(X_1 = t_1, X_2 = t_2, \ldots, X_n = t_n) = \mathbf{P}(X_1 = t_1)\mathbf{P}(X_2 = t_2) \cdots \mathbf{P}(X_n = t_n) \tag{7.22}$$

for all $t_i \in \mathbb{R}$. Otherwise, $X_1, X_2, \ldots, X_n : \Omega \to \mathbb{R}$ are called dependent.

REMARK. Because of the additivity of $\mathbf{P}$, from Definition 7.3.3 we infer that if $X, Y : \Omega \to \mathbb{R}$ are independent random variables, then

$$\mathbf{P}(X \in S, Y \in T) = \mathbf{P}(X \in S)\mathbf{P}(Y \in T)$$

for all $S, T \subseteq \mathbb{R}$. A similar remark holds for more than two independent random variables (see Exercise 33).

EXAMPLE. Given a finite set $\Omega \subseteq \mathbb{R}^2$, endowed with the uniform probability distribution, we consider the random variables $X, Y : \Omega \to \mathbb{R}$ with $X(\omega) = a$

and $Y(\omega) = b$ for $\omega = (a, b) \in \Omega$. If $\Omega = [m] \times [n]$ for some $m, n \in \mathbb{Z}_{>0}$, then

$$\begin{aligned} \mathbf{P}(X = a, Y = b) &= 1/mn \\ \mathbf{P}(X = a) &= 1/m \\ \mathbf{P}(Y = b) &= 1/n \end{aligned}$$

for all $a \in [m]$ and $b \in [n]$. Therefore, (7.21) holds and X, Y are independent.

If $\Omega = \{(1, 2), (2, 1)\}$, then X and Y are dependent since $\mathbf{P}(X = 1, Y = 2) = 1/2$ but $\mathbf{P}(X = 1)\mathbf{P}(Y = 2) = (1/2)^2 = 1/4$. □

Example 7.3.14. Let $X_1, X_2 \ldots, X_n$ be random variables, each taking only the values 0 and 1 with probability $1 - p$ and p, respectively (such random variables are called *Bernoulli* with parameter p). Suppose that $X_1, X_2 \ldots, X_n$ are independent and let $X = X_1 + X_2 + \cdots + X_n$ be their sum. Clearly, X takes the values $0, 1, \ldots, n$. For $k \in \{0, 1 \ldots, n\}$, let us compute the probability that $X = k$. For $S \subseteq [n]$, let A_S be the event that $X_i = 1$ for every $i \in S$ and $X_j = 0$ for every $j \in [n] \smallsetminus S$. By independence, A_S has probability $p^k(1 - p)^{n-k}$ for every k-element subset S of $[n]$. Since the event that $X = k$ is the disjoint union of all these events A_S, we have $\mathbf{p}_X(k) = \binom{n}{k} p^k(1 - p)^{n-k}$ for every $k \in \{0, 1 \ldots, n\}$. Thus, X has a binomial distribution with parameters n and p. □

We recall that given functions $X_1, X_2, \ldots, X_n : \Omega \to \mathbb{R}$, the product $X_1 X_2 \cdots X_n : \Omega \to \mathbb{R}$ is defined by setting $(X_1 X_2 \cdots X_n)(\omega) = X_1(\omega) X_2(\omega) \cdots X_n(\omega)$ for $\omega \in \Omega$.

Proposition 7.3.3. *Let $X_1, X_2, \ldots, X_n : \Omega \to \mathbb{R}$ be independent random variables that all take only nonnegative values, or all have expected values that are defined as real numbers. Then,*

$$E(X_1 X_2 \cdots X_n) = E(X_1) E(X_2) \cdots E(X_n),$$

where $0 \cdot \infty = 0$ by convention.

Proof. Let $X = X_1 X_2 \cdots X_n$. Applying the additivity of the probability measure $\mathbf{P}$ and the independence of $X_1, X_2, \ldots, X_n$, for $t \in \mathbb{R}$, we have

$$\begin{aligned} \mathbf{p}_X(t) = \mathbf{P}(X_1 X_2 \cdots X_n = t) &= \sum_{t_1 t_2 \cdots t_n = t} \mathbf{P}(X_1 = t_1, X_2 = t_2, \ldots, X_n = t_n) \\ &= \sum_{t_1 t_2 \cdots t_n = t} \mathbf{P}(X_1 = t_1) \mathbf{P}(X_2 = t_2) \cdots \mathbf{P}(X_n = t_n) \\ &= \sum_{t_1 t_2 \cdots t_n = t} \mathbf{p}_{X_1}(t_1) \mathbf{p}_{X_2}(t_2) \cdots \mathbf{p}_{X_n}(t_n), \end{aligned}$$

where the sums run through all n-tuples $(t_1, t_2, \ldots, t_n) \in \mathbb{R}^n$ such that $t_1 t_2 \cdots t_n = t$. We now apply (7.16) and conclude that

$$\begin{aligned} E(X) &= \sum_{t\in\mathbb{R}} t\, \mathbf{p}_X(t) = \sum_{t\in\mathbb{R}} t \sum_{t_1 t_2 \cdots t_n = t} \mathbf{p}_{X_1}(t_1)\mathbf{p}_{X_2}(t_2)\cdots \mathbf{p}_{X_n}(t_n) \\ &= \sum_{t_i\in\mathbb{R}} (t_1 t_2 \cdots t_n)\mathbf{p}_{X_1}(t_1)\mathbf{p}_{X_2}(t_2)\cdots \mathbf{p}_{X_n}(t_n) \\ &= \left(\sum_{t_1\in\mathbb{R}} t_1 \mathbf{p}_{X_1}(t_1)\right)\left(\sum_{t_2\in\mathbb{R}} t_2 \mathbf{p}_{X_2}(t_2)\right)\cdots\left(\sum_{t_n\in\mathbb{R}} t_n \mathbf{p}_{X_n}(t_n)\right) \\ &= E(X_1)E(X_2)\cdots E(X_n). \end{aligned}$$

□

EXAMPLE. Let $\Omega = [m] \times [n]$ for some $m, n \in \mathbb{Z}_{>0}$, endowed with the uniform probability distribution. As already observed, the random variables $X, Y : \Omega \to \mathbb{R}$ defined by $X(\omega) = a$ and $Y(\omega) = b$ for $\omega = (a, b) \in \Omega$ are independent. Since $E(X) = \sum_{k=1}^{m} k(1/m) = (m + 1)/2$ and, similarly, $E(Y) = (n + 1)/2$, we infer from Proposition 7.3.3 that $E(XY) = E(X)E(Y) = (m+1)(n+1)/4$. An analogous result holds more generally for $\Omega = [m_1] \times [m_2] \times \cdots \times [m_k]$.

Variance The variance of a random variable X indicates the extent to which the values of X accumulate around its expected value.

Definition 7.3.4. *Let $(\Omega, \mathbf{P})$ be a (discrete) probability space and $X : \Omega \to \mathbb{R}$ be a random variable, the expected value $E(X) = \mu \in \mathbb{R}$ of which is defined. Then, the variance of X is defined as* $\mathrm{Var}(X) = E((X - \mu)^2)$.

The nonnegative square root of $\mathrm{Var}(X)$ *is called the standard deviation of X.*

REMARK. We have

$$\begin{aligned} \mathrm{Var}(X) &= E((X - \mu)^2) = E(X^2) - 2\mu E(X) + E(\mu^2) = E(X^2) - 2\mu^2 + \mu^2 \\ &= E(X^2) - \mu^2, \end{aligned} \tag{7.23}$$

where $\mu = E(X)$.

EXAMPLE. Suppose that the random variable $X : \Omega \to \mathbb{R}$ takes the values $1, 2, \ldots, n$ for some $n \in \mathbb{Z}_{>0}$, each with probability $\mathbf{p}_X(k) = 1/n$. Then, $E(X) = (n + 1)/2$ and X^2 takes the values $1^2, 2^2, \ldots, n^2$, each with probability $1/n$. As a result,

$$E(X^2) = \sum_{k=1}^{n} k^2(1/n) = n(n + 1)(2n + 1)/6n = (n + 1)(2n + 1)/6$$

and $\mathrm{Var}(X) = E(X^2) - (E(X))^2 = (n^2 - 1)/12$. □

The following statement can be useful for the computation of the expected value of $f(X)$ for any function f.

Proposition 7.3.4. *Let $(\Omega, \mathbf{P})$ be a (discrete) probability space and $X : \Omega \to \mathbb{R}$ be a random variable. Then, for every function $f : \mathbb{R} \to \mathbb{R}$, the random variable $f(X) : \Omega \to \mathbb{R}$ has expected value*

$$E(f(X)) = \sum_{t \in \mathbb{R}} f(t)\, \mathbf{p}_X(t),$$

under the assumption that the right-hand side of this equation is defined. In particular,

$$E(X^2) = \sum_{t \in \mathbb{R}} t^2 \mathbf{p}_X(t).$$

Proof. We first assume that $f(X)$ takes only nonnegative values. By the additivity of the probability measure $\mathbf{P}$,

$$\mathbf{p}_{f(X)}(y) = \mathbf{P}(f(X) = y) = \sum_{t \in \mathbb{R}} \mathbf{P}(f(X) = y, X = t)$$

for every $y \geq 0$. Hence,

$$\begin{aligned} E(f(X)) &= \sum_{y \geq 0} y\, \mathbf{p}_{f(X)}(y) = \sum_{y \geq 0} y \sum_{t \in \mathbb{R}} \mathbf{P}(f(X) = y, X = t) \\ &= \sum_{t \in \mathbb{R}} \sum_{y \geq 0} y\, \mathbf{P}(f(X) = y, X = t) = \sum_{t \in \mathbb{R}} f(t)\, \mathbf{P}(X = t) \\ &= \sum_{t \in \mathbb{R}} f(t)\, \mathbf{p}_X(t). \end{aligned}$$

To treat the general case, we apply the special case to $(f(X))^+$ and $-(f(X))^-$. We find that

$$E^+(f(X)) = \sum_{t \in \mathbb{R}} (f(t))^+ \mathbf{p}_X(t),$$

$$E^-(f(X)) = \sum_{t \in \mathbb{R}} (f(t))^- \mathbf{p}_X(t)$$

and the proof follows. □

Example 7.3.15. Let us compute the variance of the random variable X of Example 7.3.2, defined by $X(S) = \max(S)$, where S is selected uniformly at random from the set of all k-subsets of $[n]$.

We showed in Example 7.3.8 that $E(X) = k(n+1)/(k+1)$. We compute in a similar way that

$$\begin{aligned}\binom{n}{k} E(X^2) &= \binom{n}{k} \sum_{t=k}^{n} t^2 \mathbf{p}_X(t) = \sum_{t=k}^{n} t^2 \binom{t-1}{k-1} = k \sum_{t=k}^{n} t \binom{t}{k} \\ &= k \sum_{t=k}^{n} (t+1) \binom{t}{k} - k \sum_{t=k}^{n} \binom{t}{k} \\ &= k(k+1) \sum_{t=k}^{n} \binom{t+1}{k+1} - k \sum_{t=k}^{n} \binom{t}{k} \\ &= k(k+1) \binom{n+2}{k+2} - k \binom{n+1}{k+1},\end{aligned}$$

where for the second equality we have applied formula (7.13) for the probability mass function of X. After routine computations, we reach the formulas

$$E(X^2) = \frac{k(n+1)(kn+n+k)}{(k+1)(k+2)},$$

$$\mathrm{Var}(X) = E(X^2) - (E(X))^2 = \frac{k(n+1)(n-k)}{(k+1)^2(k+2)}.$$

For $k = 1$, we get the answer $(n^2-1)/12$, in agreement with a previous example. For $k = n$, we get $\mathrm{Var}(X) = 0$ (explain why). □

Example 7.3.16. Let us compute the variance of the geometric distribution of Example 7.3.9, with parameter $0 < p \leq 1$. Let X be a random variable that takes the values $1, 2, 3, \ldots$ and has probability mass function given by formula (7.19). We showed in Example 7.3.9 that $E(X) = 1/p$. By a similar computation we find that

$$\begin{aligned}E(X^2) &= \sum_{k=1}^{\infty} k^2 p(1-p)^{k-1} = p \sum_{k=0}^{\infty} (k+1)^2 (1-p)^k \\ &= p \sum_{k=0}^{\infty} (k+1)(k+2)(1-p)^k - p \sum_{k=0}^{\infty} (k+1)(1-p)^k\end{aligned}$$

$$= \frac{2p}{(1-(1-p))^3} - \frac{p}{(1-(1-p))^2}$$
$$= \frac{2-p}{p^2},$$

where we have applied (6.19) for $k = 1$ and $k = 2$ to compute the infinite sums. It follows that $\mathrm{Var}(X) = E(X^2) - (E(X))^2 = (1-p)/p^2$. □

Proposition 7.3.5. *Let $X_1, X_2, \ldots, X_n : \Omega \to \mathbb{R}$ be pairwise independent random variables, the expected values of which are defined as real numbers. Then,* $\mathrm{Var}(X_1 + X_2 + \cdots + X_n) = \mathrm{Var}(X_1) + \mathrm{Var}(X_2) + \cdots + \mathrm{Var}(X_n)$.

Proof. Let $X = X_1 + X_2 + \cdots + X_n$. According to Proposition 7.3.1 (b), X has expected value $\mu = \mu_1 + \mu_2 + \cdots + \mu_n$, where $\mu_i = E(X_i)$ for $i \in [n]$. Therefore,

$$\mathrm{Var}(X) = E((X-\mu)^2) = E\left(\left(\sum_{i=1}^{n}(X_i - \mu_i)\right)^2\right)$$
$$= \sum_{i=1}^{n} E((X_i - \mu_i)^2) + 2 \sum_{1 \le i < j \le n} E\left((X_i - \mu_i)(X_j - \mu_j)\right)$$
$$= \sum_{i=1}^{n} \mathrm{Var}(X_i) + 2 \sum_{1 \le i < j \le n} E\left((X_i - \mu_i)(X_j - \mu_j)\right)$$
$$= \sum_{i=1}^{n} \mathrm{Var}(X_i).$$

For the last equality, we have used the fact that $X_i - \mu_i$ and $X_j - \mu_j$ are independent random variables for $i < j$, each with expected value zero. According to Proposition 7.3.3, their product $(X_i - \mu_i)(X_j - \mu_j)$ has expected value zero as well. □

Example 7.3.17. Let us compute the variance of the binomial distribution (7.14) with parameters n and p. Let X be a random variable having that distribution. We showed in Example 7.3.10 that $E(X) = np$ by expressing X as a sum of n independent random variables $X_1, X_2, \ldots, X_n$. Each one of them takes the values 1 and 0 with probability p and $1-p$, respectively, and hence

$$E(X_i) = 1 \cdot p + 0 \cdot (1-p) = p$$
$$E(X_i^2) = 1 \cdot p + 0 \cdot (1-p) = p$$
$$\mathrm{Var}(X_i) = E(X_i^2) - (E(X_i))^2 = p(1-p)$$

for every $i \in [n]$. Proposition 7.3.5 then yields that $\mathrm{Var}(X) = \mathrm{Var}(X_1) + \mathrm{Var}(X_2) + \cdots + \mathrm{Var}(X_n) = np(1-p)$.

For given n, as one may expect, the variance is maximized for $p = 1/2$. □

The inequalities that appear in the following proposition are extremely useful in finding bounds for the probabilities of events.

Proposition 7.3.6. *Let $X : \Omega \to \mathbb{R}$ be a random variable the expected value of which is defined.*

(a) (Markov's Inequality) *If $X(\omega) \geq 0$ for every $\omega \in \Omega$, then*

$$\mathbf{P}(X \geq a) \leq \frac{E(X)}{a}$$

for every $a > 0$.

(b) (Chebyshev's Inequality) *If $E(X) = \mu \in \mathbb{R}$, then*

$$\mathbf{P}(|X - \mu| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}$$

for every $a > 0$.

Proof. To prove part (a), we observe that $X(\omega) \geq a\chi_A(\omega)$ for every $\omega \in \Omega$, where $A = \{\omega \in \Omega : X(\omega) \geq a\}$. We apply parts (c) and (d) of Proposition 7.3.1 and conclude that $E(X) \geq E(a\chi_A) = aE(\chi_A) = a\,\mathbf{P}(X \geq a)$.

To prove part (b), we apply part (a) to the random variable $(X-\mu)^2$ and find that

$$\mathbf{P}(|X - \mu| \geq a) = \mathbf{P}(|X - \mu|^2 \geq a^2) \leq \frac{E((X-\mu)^2)}{a^2} = \frac{\mathrm{Var}(X)}{a^2}$$

for every $a > 0$. □

The application of Proposition 7.3.6 provided by the following example is an instance of the weak law of large numbers, one of the important results of probability theory referred to as limit theorems.

Example 7.3.18. Let X be the sum of n independent Bernoulli random variables with parameter $p = 1/2$. By Example 7.3.14, the random variable X has the binomial distribution with parameters n and $p = 1/2$ and hence $E(X) = np = n/2$ and $\mathrm{Var}(X) = np(1-p) = n/4$. By Chebyshev's inequality, for every $\delta > 0$

$$\mathbf{P}\left(\left|X - \frac{n}{2}\right| \geq \delta n\right) \leq \frac{1}{4\delta^2 n}$$

and hence $\lim_{n\to\infty} \mathbf{P}(|X - \frac{n}{2}| \geq \delta n) = 0$. This shows that the values of X accumulate around the expected value when n is sufficiently large.

For small values of n, the bounds are not satisfactory. For example, if $n = 10$, in which case $E(X) = 5$ and $\mathrm{Var}(X) = 5/2$, Chebyshev's inequality yields that

$$\begin{aligned}\mathbf{P}(X \in \{0, 1, 2, 3, 7, 8, 9, 10\}) &= \mathbf{P}(|X - 5| \geq 2) \leq 5/8\\ \mathbf{P}(X \in \{0, 1, 2, 8, 9, 10\}) &= \mathbf{P}(|X - 5| \geq 3) \leq 5/18.\end{aligned}$$

Equivalently, $\mathbf{P}(X \in \{4, 5, 6\}) \geq 3/8$ and $\mathbf{P}(X \in \{3, 4, 5, 6, 7\}) \geq 13/18$. The bound in the second case is closer to the truth (see Example 7.1.9). □

Limit theorems, typically divided into laws of large numbers and central limit theorems, are concerned with the behavior of the sum of large numbers of random variables. Laws of large numbers provide conditions under which the partial sums of a sequence of random variables, appropriately normalized, converge (in some sense) to a constant. Central limit theorems are concerned with the convergence of the distribution of the appropriately centered and normalized partial sums. The reader is referred to other sources for expositions of these important topics in probability theory.

7.4 Exercises

1. Car plates in some European country are decimal numbers with four digits, meaning sequences (a, b, c, d) such that $a, b, c, d \in \{0, 1, \ldots, 9\}$. Assuming that all these sequences are equiprobable, which of the following events is the least probable to occur and which is the most probable?

   (a) The plate number is even.
   (b) Either all digits of the plate number are even, or all are odd.
   (c) The four digits are pairwise distinct.
   (d) The sequence is monotone, meaning $a \leq b \leq c \leq d$ or $a \geq b \geq c \geq d$.

2. A deck consists of cards $1, 2, \ldots, 8$, each having four different suits (so there is a total of 32 cards in the deck). We select uniformly at random a set of five cards from the deck. Compute the probability:

   (a) of a straight (five cards with distinct successive numbers of any suit, but not all of the same suit),
   (b) of a full house (three cards with the same number of any suit and two more cards of the same number, different form the first, of any suit),
   (c) of four of a kind (four cards with the same number).

3. A box contains m indistinguishable black balls and n indistinguishable white balls. We select one of the balls at random and remove it from the box. Then, we select another ball and remove it from the box and continue similarly, until the box gets empty. What is the probability that all white balls have been removed from the box before all black balls have done so?

4. In a country festival, n men meet n women. The first day the men and women are cyclically ordered at random to dance a circle dance, while the second day they are linearly ordered at random to dance a line dance. Which day is the probability that no two people of the same sex are dancing next to each other largest?
5. Given a positive integer n, a nonempty subset S of the set $[n]$ is selected uniformly at random. What is the probability that the largest element of S is an even number? Does the limit of this probability as $n \to \infty$ exist?
6. We select uniformly at random a subset S of the set $[2n]$. Let $p(n)$ be the probability that S contains as many even, as odd numbers.

   (a) Find a simple formula for $p(n)$.
   (b) Prove that $1/(2\sqrt{n}) \leq p(n) < 1/\sqrt{2n}$ for every $n \in \mathbb{Z}_{>0}$.

7. Let m, n be given positive integers. We select subsets $S_1, S_2, \ldots, S_m$ of the set $[n]$ at random, so that all sequences $(S_1, S_2, \ldots, S_m)$ of subsets of $[n]$ are equiprobable. Compute:

   (a) the probability that $S_1, S_2, \ldots, S_m$ are pairwise disjoint,
   (b) the probability that any three of the sets $S_1, S_2, \ldots, S_m$ have empty intersection, if $m \geq 3$.

   Generalize to the intersections of any k of the sets $S_1, S_2, \ldots, S_m$.
8. Let m be a positive integer that has n distinct prime divisors. We select uniformly at random an ordered pair (a, b) of positive divisors of m and denote by $f(m)$ and $g(m)$ the probability of the events that a and b are relatively prime (meaning, they have no common prime divisor) and that a divides b, respectively.

   (a) Prove that $(1/2)^n \leq g(m) \leq (3/4)^n$ for every m.
   (b) Prove that $f(m) \leq g(m)$ for every m.
   (c) Prove that $f(m) = g(m) = (3/4)^n$ if and only if m is square free (meaning, m is not divisible by the square of a prime number).

9. Given $n \in \mathbb{Z}_{>0}$, we select uniformly at random an ordered pair of integers $(x, y) \in [n] \times [n]$. Which one of the events

$$A = \{(x, y) \in [n] \times [n] : x^2 \equiv y^2 \pmod 2\}$$
$$B = \{(x, y) \in [n] \times [n] : x^2 \equiv y^2 \pmod 3\}$$

   is the most probable?
10. We select uniformly at random a permutation $\sigma = (\sigma_1, \sigma_2, \ldots, \sigma_n)$ of the set $[n]$ and denote by $f(\sigma)$ the smallest $i \in [n]$ for which $\sigma_i > \sigma_{i+1}$, where $\sigma_{n+1} := 0$. Let A_n be the event that $f(\sigma)$ is an even number. For example, if $n = 5$ and $\sigma = (2, 4, 5, 1, 3)$, then $f(\sigma) = 3$. Moreover, $A_2 = \{(1, 2)\}$ and $A_3 = \{(1, 3, 2), (2, 3, 1)\}$, so that $\mathbf{P}(A_2) = 1/2$ and $\mathbf{P}(A_3) = 1/3$.

(a) Find the probability $\mathbf{P}(A_n)$ for $n \in \{4, 5\}$.
(b) Prove that $\mathbf{P}(A_n) = \sum_{i=0}^{n} (-1)^i / i!$ and conclude that $\lim_{n\to\infty} \mathbf{P}(A_n) = e^{-1}$.
(c) Deduce that $\#A_n = D_n$ for every n, where D_n is the number of permutations of $[n]$ without fixed points, in agreement with Exercise 27 of Chapter 3.

11. Given $n \in \mathbb{Z}_{>0}$, we select uniformly at random a path p in $\mathbb{Z}^2$ with initial vertex $(0, 0)$, terminal vertex (n, n), and steps $(1, 0)$ or $(0, 1)$, as in Section 2.3.8, which lies in the half-space $x \geq y$ of $\mathbb{R}^2$. Let $f(p)$ be the smallest positive integer k such that (k, k) is a vertex of p. Prove the identity

$$\sum_{k=1}^{n} \frac{\binom{n}{k}\binom{n}{k+1}}{(2k-1)\binom{2n}{2k}} = 2$$

after computing the probability that $f(p) = k$, for $k \in [n]$.
12. Prove part (g) of Proposition 7.1.2 (Principle of inclusion–exclusion) using only the axioms of Definition 7.1.1.
13. We consider the experiment of Example 7.1.7 and denote by $p(m, n)$ the probability that at least one box ends up being empty.

(a) Prove that

$$n\left(1 - \frac{1}{n}\right)^m - \binom{n}{2}\left(1 - \frac{2}{n}\right)^m \leq p(m, n) \leq n\left(1 - \frac{1}{n}\right)^m.$$

(b) Compute the probability $p(m, n)$ up to two decimal places for $m = 50$ and $n = 10$.

14. Let $p(n)$ be the probability that an integer selected uniformly at random out of $1, 2, \ldots, n$ is not divisible by any of 2, 3, and 5. Compute the limit $\lim_{n\to\infty} p(n)$. What is the answer if 2, 3, and 5 are replaced by arbitrary prime numbers $p_1, p_2, \ldots, p_r$?
15. Given $n \in \mathbb{Z}_{>0}$, we select uniformly at random an ordered pair of integers $(a, b) \in [n] \times [n]$ and denote by $q(n)$ the probability that a and b are relatively prime. Compute the limit $\lim_{n\to\infty} q(n)$.
16. We select uniformly at random an ordered triple $(x, y, z) \in [n]^3$, as in Example 7.1.4. Compute the following probabilities:

(a) the probability that $x \leq y \leq z$,
(b) the conditional probability that $x \leq y \leq z$, given that $x \leq z$.

17. Compute the following conditional probabilities for the $2 \times n$ matrix σ of Example 7.2.4:

(a) the probability that it has increasing columns, given that it has increasing rows,
(b) the probability that it has increasing rows, given that it has increasing columns.

18. Let $n \in \mathbb{Z}_{>0}$.

(a) We select uniformly at random an ordered pair (S, T) of subsets of $[n]$. For given $k, \ell \in \{0, 1, \ldots, n\}$, which of the following events are independent one from the other?

- $\#S = k$,
- $\#T = \ell$,
- $S \cap T = \varnothing$.

(b) We select uniformly at random an ordered triple (R, S, T) of subsets of $[n]$. For given $k, \ell \in \{0, 1, \ldots, n\}$, compute the conditional probabilities of the events:

- $\#S = k$,
- $\#S = k$ and $\#T = \ell$,
- $R \cap S = \varnothing$,
- $R \cap S = R \cap T = \varnothing$,

given that $S \cap T = \varnothing$.

19. Prove Proposition 7.2.1.
20. Let n be a positive integer with distinct prime divisors $p_1, p_2, \ldots, p_r$. We select uniformly at random an integer out of $1, 2, \ldots, n$ and for $i \in [r]$, we denote by A_i the event that this integer is divisible by p_i.

(a) Prove that the events $A_1, A_2, \ldots, A_n$ are independent.
(b) Prove Proposition 3.2.2 by applying part (a) and Proposition 7.2.1.

You may consider as known the fact that an integer is divisible by each of the distinct primes $q_1, q_2, \ldots, q_k$ if and only if it is divisible by the product $q_1 q_2 \cdots q_k$.

21. Let $X = \{x_1, x_2, \ldots, x_m\}$ and Y be finite sets of cardinalities m and n, respectively. We select uniformly at random a map $f : X \to Y$, as in Example 7.1.7. For any 2-element subset $e = \{i, j\}$ of $[m]$, we denote by A_e the event that $f(x_i) = f(x_j)$.

(a) Let E denote a set consisting of 2-element subsets of $[m]$. Find a necessary and sufficient condition for the events A_e for $e \in E$ to be independent.
(b) Give an example of n dependent events, any $n-1$ of which are independent.

22. Let Ω be a finite set of cardinality N. For which $n \in \mathbb{Z}_{>0}$ does there exist a probability measure $\mathbf{P} : 2^{\Omega} \to \mathbb{R}$ and independent events $A_1, A_2, \ldots, A_n \subseteq \Omega$ such that $0 < \mathbf{P}(A_i) < 1$ for every $i \in [n]$?

23. We toss a coin independently n times, each with probability p of getting heads (H), where $0 \leq p \leq 1$. Compute the probability that before each appearance of successive results TH (if such exist) there is an appearance of successive results TT. For example, for $n = 3$, the allowable strings of outcomes are TTH, TTT, HTT, HHT, and HHH.
24. We toss a coin independently n times, each with probability p of getting heads (H), where $0 \leq p \leq 1$. Let $a(n)$ be the probability of not getting two successive results H, where $a(0) := 1$.

    (a) Prove that

$$a(n) = \sum_{k=0}^{\lfloor \frac{n+1}{2} \rfloor} \binom{n-k+1}{k} p^k (1-p)^{n-k} \tag{7.24}$$

    for every $n \in \mathbb{N}$.

    (b) Prove that $a(n) = (1-p)a(n-1) + p(1-p)a(n-2)$ for every $n \geq 2$.

    (c) If $p < 1$, find a real number $0 < \lambda \leq 1$ for which the limit $\lim_{n\to\infty} a(n)\lambda^{-n}$ exists and compute this limit.

25. Suppose that the random variable $X : \Omega \to \mathbb{R}$ takes only nonnegative integer values. Prove that

$$E(X) = \sum_{k=1}^{\infty} \mathbf{P}(X \geq k).$$

    Apply this formula to compute the expected value of the geometric distribution of Example 7.3.9 with parameter $0 < p \leq 1$.
26. Let n be a positive integer and $k \in \{0, 1, \ldots, n\}$. We select uniformly at random a k-subset S of the set $[n]$. Compute the expected value of the sum of the elements of S.
27. Let $k, n \in \mathbb{Z}_{>0}$. We select uniformly at random a sequence $(S_1, S_2, \ldots, S_k)$ of subsets of the set $[n]$. Compute the expected value:

    (a) of the cardinality of the union $S_1 \cup S_2 \cup \cdots \cup S_k$,
    (b) of the cardinality of the intersection $S_1 \cap S_2 \cap \cdots \cap S_k$,
    (c) of the sum of the elements of the union $S_1 \cup S_2 \cup \cdots \cup S_k$,
    (d) of the sum of the elements of the intersection $S_1 \cap S_2 \cap \cdots \cap S_k$.

    Verify that the answer in part (a) is equal to that in part (b) multiplied by $2^k - 1$ and similarly for (c) and (d).
28. We toss a fair coin independently n times (so each time, the probability of getting H is 1/2). Let X_n (respectively, Y_n) be the number of times we got H (respectively, T) and let $Z_n = \max\{X_n, Y_n\}$. Find the limit $\lim_{n\to\infty} E(Z_n)/n$.

29. Let $A_1, A_2, \ldots, A_n \subseteq \Omega$ be events. For $k \in [n]$, let B_k be the event that at least k from $A_1, A_2, \ldots, A_n$ hold. In other words,

$$B_k = \bigcup_{1 \le i_1 < i_2 < \cdots < i_k \le n} A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_k}.$$

In particular, B_k is the union (respectively, the intersection) of $A_1, A_2, \ldots, A_n$ for $k = 1$ (respectively, $k = n$).

(a) Prove that $\sum_{i=1}^{n} \mathbf{P}(A_i) = \sum_{k=1}^{n} \mathbf{P}(B_k)$.
(b) Prove that

$$\frac{t-k+1}{n-k+1} \le \mathbf{P}(B_k) \le \frac{t}{k}$$

for every $k \in [n]$, where $t = \sum_{i=1}^{n} \mathbf{P}(A_i)$.

30. Let $A_1, A_2, \ldots, A_n \subseteq \Omega$ be events and let $X_i : \Omega \to \mathbb{R}$ be the characteristic function of A_i, for $i \in [n]$.

(a) If $X = X_1 + X_2 + \cdots + X_n$, prove that

$$E(X(X-1)\cdots(X-k+1)) = k! \sum_{1 \le i_1 < i_2 < \cdots < i_k \le n} \mathbf{P}(A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_k})$$

for every $k \in [n]$. Which well-known identity comes up for $k = 1$?
(b) Compute the expected value $E(X(X-1)\cdots(X-k+1))$ if $X(S) = \#S$ for every $S \subseteq [n]$, where S is selected uniformly at random from the powerset of $[n]$.

31. We color every edge of the complete graph K_n either black or white. Prove that this can be done so that no more than $n(n-1)(n-2)(n-3)/64$ monochromatic cycles (meaning, cycles with edges of the same color) of length 4 are formed.
32. Under the terminology of Proposition 7.3.2, prove that for every $n \in \mathbb{Z}_{>0}$, every orientation of the complete graph K_n:

(a) has at least $(n-1)(n-3)/4$ transitive triples which involve vertex n,
(b) has a total of more than $n^2(n-5)/12$ transitive triples.

33. Let $(\Omega, \mathbf{P})$ be a (discrete) probability space.

(a) Prove that the random variables $X_1, X_2, \ldots, X_n : \Omega \to \mathbb{R}$ are independent if and only if

$$\begin{aligned}&\mathbf{P}(X_1 \in S_1, X_2 \in S_2, \ldots, X_n \in S_n)\\ &\quad = \mathbf{P}(X_1 \in S_1)\mathbf{P}(X_2 \in S_2)\cdots\mathbf{P}(X_n \in S_n)\end{aligned}$$

for all $S_1, S_2, \ldots, S_n \subseteq \mathbb{R}$.

(b) Prove that the events $A_1, A_2, \ldots, A_n \subseteq \Omega$ are independent if and only if the same holds for the characteristic functions $\chi_{A_1}, \chi_{A_2}, \ldots, \chi_{A_n} : \Omega \to \mathbb{R}$.

34. Let $(\Omega, \mathbf{P})$ be a finite probability space and $X, Y : \Omega \to \mathbb{R}$ be independent random variables, each of which takes at least two different values with positive probability. Suppose that $a \leq (X+Y)(\omega) \leq b$ for every $\omega \in \Omega$. Prove that at least one of the probabilities $\mathbf{P}(X+Y=a)$ and $\mathbf{P}(X+Y=b)$ is no larger than 1/4.
35. Let $(\Omega, \mathbf{P})$ be a probability space and $X, Y : \Omega \to \mathbb{R}$ be random variables.

(a) Assuming that $(\Omega, \mathbf{P})$ is finite, prove that $|E(XY)|^2 \leq E(X^2)E(Y^2)$.
(b) Deduce that

$$\left(\sum_{i=1}^{n} a_i b_i\right)^2 \leq \left(\sum_{i=1}^{n} a_i^2\right)\left(\sum_{i=1}^{n} b_i^2\right)$$

for $a_1, a_2, \ldots, a_n, b_1, b_2, \ldots, b_n \in \mathbb{R}$.
(c) Prove that part (a) holds more generally for discrete probability spaces, under the assumption that the expected value of XY is defined.

36. Let $X : \Omega \to \mathbb{R}$ be a random variable for which the expected value $E(X) \in \mathbb{R}$ is defined. Prove that $E((X-t)^2) \geq \mathrm{Var}(X)$ for every $t \in \mathbb{R}$. When does equality hold?
37. We select uniformly at random a nonempty subset S of the set $[n]$ and denote by $X(S)$ the largest element of S. Compute asymptotically the expected value and the variance of X, as $n \to \infty$.
38. Let $X_1, X_2, \ldots, X_n : \Omega \to \mathbb{R}$ be random variables the expected values of which are defined as real numbers.

(a) Prove that

$$\begin{aligned}\mathrm{Var}(X_1 + X_2 + \cdots + X_n) &= \sum_{i=1}^{n} (E(X_i^2) - \mu_i^2) \\ &\quad + 2 \sum_{1 \leq i < j \leq n} (E(X_i X_j) - \mu_i \mu_j) \\ &= \sum_{i=1}^{n} \mathrm{Var}(X_i) + 2 \sum_{1 \leq i < j \leq n} (E(X_i X_j) - \mu_i \mu_j),\end{aligned}$$

where $\mu_i = E(X_i)$ for $i \in [n]$, under the assumption that the right-hand side of this equation is defined (in particular, that the expected values of $X_i X_j$ are defined for $i < j$).

(b) Compute the expected value and the variance of the number of boxes that end up being empty in Example 7.1.7.

39. Compute the variance of the hypergeometric distribution (7.15) of Example 7.3.4.
40. We toss a coin independently n times, each with probability p of getting heads (H). For each possible outcome ω of this experiment, we denote by $X(\omega)$ the number of results HH that appear successively in ω. For example, if $n = 8$ and $\omega = \text{HHTTHHHT}$, then $X(\omega) = 3$.

(a) Compute the expected value and the variance of the random variable X.

(b) Prove that $\lim_{n\to\infty} \mathbf{P}\left(\left|\frac{X}{n-1} - p^2\right| > \varepsilon\right) = 0$ for every $\varepsilon > 0$.

Appendix A
A Glimpse of Algebra

This appendix briefly reviews definitions and background from algebra which will sometimes be helpful to the reader to better understand the text. The emphasis is on notation, terminology, and basic properties of polynomials, since these are often used when proving combinatorial identities, as well as in other circumstances. For a thorough treatment of these topics, we recommend [5].

A.1 Rings and Fields

Let R be a nonempty set, endowed with an operation

$$+ : R \times R \to R$$

called addition (the notion of an operation on a set is discussed in Example 1.2.3).

Definition A.1.1. *The pair $(R, +)$ (or simply, R) is called an abelian group if the following hold for all $a, b, c \in R$:*

(i) $(a + b) + c = a + (b + c)$.
(ii) $a + b = b + a$.
(iii) *There exists an element $0_R \in R$ such that $x + 0_R = x$ for every $x \in R$.*
(iv) *For every $x \in R$, there exists $x' \in R$ such that $x + x' = 0_R$.*

The element 0_R of condition (iii) is unique and called the *zero element* of R. Similarly, the element x' of condition (iv) is unique for every $x \in R$ and called the *negative* of x.

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Suppose now that R is endowed with two operations

$$+ : R \times R \to R$$
$$\times : R \times R \to R$$

called addition and multiplication, respectively. Let us write ab or $a \cdot b$ instead of $a \times b$, for $a, b \in R$.

Definition A.1.2. *The triple $(R, +, \times)$ (or simply, R) is called a ring if $(R, +)$ is an abelian group and the following hold for all $a, b, c \in R$:*

(v) $(ab)c = a(bc)$.
(vi) $a(b + c) = ab + ac$ *and* $(b + c)a = ba + ca$.

The ring R is said to have a unit if:

(vii) *There exists an element $1_R \in R$ such that $x \cdot 1_R = 1_R \cdot x = x$ for every $x \in R$*

and to be commutative if:

(viii) $ab = ba$, *for all $a, b \in R$.*

Let R be a ring with a unit $1_R \in R$. We say that $a \in R$ is *invertible* if there exists $b \in R$ such that $ab = ba = 1_R$. Such an element b, if it exists, is unique, called the *inverse* of a and denoted by a^{-1}.

Definition A.1.3. *A commutative ring F with a unit $1_F \neq 0_F$ is called a field if every nonzero element of F is invertible.*

The smallest field $F = \{0_F, 1_F\}$ has two elements: Addition and multiplication are defined by the rules $x + 0_F = 0_F + x = x$ and $x \cdot 1_F = 1_F \cdot x = x$ for every $x \in F$ and $1_F + 1_F = 0_F$, $0_F \cdot 0_F = 0_F$. Typical examples of fields which appear in this book are the fields $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ of rational, real and complex numbers, respectively. The set $\mathbb{Z}$ of integers, endowed with the usual operations of addition and multiplication, is a commutative ring with a unit which is not a field (1 and -1 are its only invertible elements).

A.2 Polynomials

Throughout this section, F stands for a field or the ring $\mathbb{Z}$ of integers. A *polynomial* (in the variable x) with coefficients from F is an expression of the form

$$f(x) = \sum_{n \geq 0} a_n x^n$$

with $a_n \in F$ for $n \in \mathbb{N}$ for which there exists $N \in \mathbb{N}$ such that $a_n = 0_F$ for every $n > N$. We then write

$$f(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_N x^N \tag{A.1}$$

and denote by $F[x]$ the set of all polynomials in x with coefficients from F. For elements $f(x) = \sum_{n\geq 0} a_n x^n$ and $g(x) = \sum_{n\geq 0} b_n x^n$ of $F[x]$, we define that $f(x) = g(x)$ if and only if $a_n = b_n$ for every $n \in \mathbb{N}$. We say that a_n is the *coefficient of* x^n in $f(x)$ and denote it by $[x^n]f(x)$. In particular, $a_0 = [x^0]f(x)$ is the *constant term* of $f(x)$. The sum and the product of $f(x)$ and $g(x)$ are defined by the rules

$$f(x) + g(x) = \sum_{n\geq 0} (a_n + b_n) x^n$$

and

$$\begin{aligned} f(x)g(x) &= \left(a_0 + a_1 x + a_2 x^2 + \cdots\right)\left(b_0 + b_1 x + b_2 x^2 + \cdots\right) \\ &= a_0 b_0 + (a_0 b_1 + a_1 b_0)x + (a_0 b_2 + a_1 b_1 + a_2 b_0)x^2 \\ &\quad + (a_0 b_3 + a_1 b_2 + a_2 b_1 + a_3 b_0)x^3 + \cdots \\ &= \sum_{n\geq 0} c_n x^n, \end{aligned}$$

where

$$c_n = a_0 b_n + a_1 b_{n-1} + \cdots + a_n b_0 = \sum_{k=0}^{n} a_k b_{n-k}$$

for $n \in \mathbb{N}$. Then, it is routine to verify that $f(x) + g(x)$ and $f(x)g(x)$ are indeed elements of $F[x]$ (that is, the coefficient of x^n in either of them is equal to 0_F for large enough n) and that the following statement holds.

Proposition A.2.1. *The set $F[x]$, endowed with the addition and multiplication of polynomials just defined, is a commutative ring with zero element and unit*

$$\begin{aligned} 0 &:= 0_F + 0_F \cdot x + 0_F \cdot x^2 + 0_F \cdot x^3 + \cdots \\ 1 &:= 1_F + 0_F \cdot x + 0_F \cdot x^2 + 0_F \cdot x^3 + \cdots, \end{aligned}$$

respectively.

The *degree* of a nonzero polynomial $f(x) \in F[x]$ is defined as the largest $n \in \mathbb{N}$ such that $[x^n]f(x) \neq 0_F$ and is denoted by $\deg(f(x))$. Thus, $\deg(f(x)) = N$ if and only if $f(x)$ has the form (A.1) and $a_N \neq 0_F$. We say that $f(x)$ is a *monic*

polynomial, if $a_N = 1_F$. By convention, the degree of the zero polynomial is defined to be $-\infty$. We have

$$\deg(f(x) + g(x)) \leq \max\{\deg(f(x)), \deg(g(x))\}$$
$$\deg(f(x)g(x)) = \deg(f(x)) + \deg(g(x))$$

for all $f(x), g(x) \in F[x]$. Polynomials of degree ≤ 0 are said to be *constant*.

Given $f(x), g(x) \in F[x]$, we say that $f(x)$ *divides* $g(x)$ and write $f(x) \mid g(x)$, if $g(x) = f(x)q(x)$ for some $q(x) \in F[x]$. We omit the proof of the following result, which is similar to that of Proposition 1.1.4.

Proposition A.2.2. *Let $f(x) \in F[x]$ be a nonzero polynomial of degree N and let $g(x) \in F[x]$. Suppose further that $f(x)$ is monic, if $F = \mathbb{Z}$. Then, there exist unique polynomials $q(x), r(x) \in F[x]$ such that $g(x) = f(x)q(x) + r(x)$ and $\deg(r(x)) < N$. Moreover, $f(x)$ divides $g(x)$ if and only if $r(x) = 0$.*

We conclude this appendix with some useful corollaries about roots of polynomials. Let $f(x) \in F[x]$ be as in (A.1). For $\zeta \in F$ we set

$$f(\zeta) = a_0 + a_1\zeta + a_2\zeta^2 + \cdots + a_N\zeta^N \in F$$

and call ζ a *root* of $f(x)$ if $f(\zeta) = 0_F$. For $f(x), g(x) \in F[x]$ we have $p(\zeta) = f(\zeta) + g(\zeta)$ and $q(\zeta) = f(\zeta)g(\zeta)$, if $p(x) = f(x) + g(x)$ and $q(x) = f(x)g(x)$.

Proposition A.2.3. *Let $f(x) \in F[x]$:*

(a) *An element $\zeta \in F$ is a root of $f(x)$ if and only if $f(x) = (x - \zeta)q(x)$ for some $q(x) \in F[x]$.*
(b) *Let $\zeta_1, \zeta_2, \ldots, \zeta_n \in F$ be pairwise distinct. Then, these elements are all roots of $f(x)$ if and only if $f(x) = (x - \zeta_1)(x - \zeta_2)\cdots(x - \zeta_n)q(x)$ for some $q(x) \in F[x]$.*

Proof. To prove part (a), we apply Proposition A.2.2 to $x-\zeta$ and $f(x)$ and conclude that $f(x) = (x-\zeta)q(x)+r(x)$ for some $q(x) \in F[x]$ and some constant polynomial $r(x) = r \in F$. Setting $x = \zeta$, we get $r = 0_F$ and conclude that $f(x) = (x-\zeta)q(x)$. The converse is trivial. Given part (a), part (b) follows by induction on n. □

Corollary A.2.1. *Let $f(x), g(x) \in F[x]$ be polynomials of degree $\leq N$. If $f(\zeta) = g(\zeta)$ for more than N values of $\zeta \in F$, then $f(x) = g(x)$ in $F[x]$.*

Proof. We apply part (b) of Proposition A.2.3 to $f(x) - g(x)$ and the given values of ζ, say $\zeta_1, \zeta_2, \ldots, \zeta_n \in F$, and conclude that

$$f(x) - g(x) = (x - \zeta_1)(x - \zeta_2)\cdots(x - \zeta_n)q(x)$$

for some $n > N$ and some $q(x) \in F[x]$. Comparing the degrees of the two sides of this equation shows that this is possible only if $q(x) = 0$ and the proof follows. □

The previous corollary gives a useful technique to prove that two polynomials $f(x)$ and $g(x)$, say with integer coefficients, are equal: It suffices to show that $f(m) = g(m)$ for infinitely many values of $m \in \mathbb{Z}$, say for every $m \in \mathbb{N}$ (see the proof of Proposition 4.3.2 for such an application).

Appendix B
Hints: Solutions to Selected Exercises

Chapter 1

1. For part (a), apply induction on n. For part (b), let us apply induction on m, in the form of Proposition 1.1.1 (a). We assume that $m \geq 1$ and that the statement holds for $m-1$. Applying definition (1.1) twice and part (a) once, we find that

$$\sum_{i=1}^{m}\left(\sum_{j=1}^{n} a_{ij}\right) = \sum_{i=1}^{m-1}\left(\sum_{j=1}^{n} a_{ij}\right) + \sum_{j=1}^{n} a_{mj} = \sum_{j=1}^{n}\left(\sum_{i=1}^{m-1} a_{ij}\right) + \sum_{j=1}^{n} a_{mj}$$

$$= \sum_{j=1}^{n}\left(\left(\sum_{i=1}^{m-1} a_{ij}\right) + a_{mj}\right) = \sum_{j=1}^{n}\left(\sum_{i=1}^{m} a_{ij}\right).$$

2. Let us prove the identity of part (f). We set

$$S_n = \sum_{k=1}^{n} \frac{(k-1)(k+2)}{k(k+1)} = \frac{0\cdot 3}{1\cdot 2} + \frac{1\cdot 4}{2\cdot 3} + \frac{2\cdot 5}{3\cdot 4} + \cdots + \frac{(n-1)(n+2)}{n(n+1)}$$

and observe that $S_1 = 0$, so the identity is valid for $n = 1$. Assuming that it is valid for $n-1 \in \mathbb{N}$, that is $S_{n-1} = (n-1)(n-2)/n$, we find that

$$S_n = S_{n-1} + \frac{(n-1)(n+2)}{n(n+1)} = \frac{(n-1)(n-2)}{n} + \frac{(n-1)(n+2)}{n(n+1)}$$

$$= \frac{(n-1)(n-2)(n+1) + (n-1)(n+2)}{n(n+1)} = \frac{(n-1)n^2}{n(n+1)} = \frac{n(n-1)}{n+1}.$$

5. We consider the set $S = \{n - m(m-1)/2 : m \in \mathbb{Z}_{>0}\} \cap \mathbb{N}$. We have $S \neq \varnothing$, since $n \in S$, and hence we may consider the minimum element,

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say $a = n - b(b-1)/2$, of S. Clearly, $a \in \mathbb{N}$ and $b \in \mathbb{Z}_{>0}$. Moreover, since $n - b(b+1)/2 < a$, the definition of a implies that $n - b(b+1)/2 \notin S$. As a result, $n - b(b+1)/2 < 0$ and hence

$$a = n - \frac{b(b-1)}{2} < \frac{b(b+1)}{2} - \frac{b(b-1)}{2} = b.$$

This proves part (a). The proof leads to the solution of part (b) as well. Indeed, let (a, b), $(c, d) \in \mathbb{N} \times \mathbb{N}$ be pairs such that $a < b$, $c < d$, and $n = a + b(b-1)/2 = c + d(d-1)/2$. Then,

$$\frac{b(b-1)}{2} \le n = c + \frac{d(d-1)}{2} < d + \frac{d(d-1)}{2} = \frac{d(d+1)}{2},$$

whence it follows that $b < d + 1$, that is, $b \le d$. Similarly, we find that $d \le b$ and conclude that $b = d$. As a result, $a = c$.

6. We first note that (1.11) holds as an equality for $n = 1$. Let us assume that it holds for $n \ge 1$, and let us prove that it holds for $n + 1$ as well, that is,

$$(n+1)^{(n+1)/2} \le (n+1)! \le 2((n+1)/2)^{n+1}.$$

Multiplying (1.11) by $n + 1$ gives that

$$(n+1)n^{n/2} \le (n+1)! \le 2(n+1)(n/2)^n,$$

and hence, it suffices to show that

$$(n+1)^{(n+1)/2} \le (n+1)n^{n/2} \quad \text{and} \quad (n+1)(n/2)^n \le ((n+1)/2)^{n+1}.$$

These inequalities can be written equivalently as $(n+1)^{n-1} \le n^n \le (n+1)^n/2$ and as

$$\left(1 - \frac{1}{n+1}\right)^n \ge \frac{1}{n+1} \quad \text{and} \quad \left(1 + \frac{1}{n}\right)^n \ge 2,$$

respectively. We leave it to the reader to verify that the latter inequalities follow by suitably applying Example 1.1.1 and the remark which follows it.

8. Hint for part (b): $a + 2a^2 + \cdots + na^n = \sum_{k=1}^{n} (a^k + a^{k+1} + \cdots + a^n)$.

10. Let S be the set of positive integers which have this property. We will show that S consists of all positive integers other than 2. From the identity

$$\frac{1}{x} = \frac{1}{x+1} + \frac{1}{x(x+1)},$$

we conclude that if (1.14) holds for positive integers $p_1 \le p_2 \le \cdots \le p_n$, then

$$1 = \frac{1}{p_1} + \cdots + \frac{1}{p_{n-1}} + \frac{1}{q_n} + \frac{1}{q_{n+1}},$$

where $q_n = p_n + 1$ and $q_{n+1} = p_n(p_n + 1)$. Moreover, if $p_n \ge 2$, then $p_n < q_n < q_{n+1}$. For example, this reasoning shows that

$$1 = \frac{1}{2} + \frac{1}{2} = \frac{1}{2} + \frac{1}{3} + \frac{1}{6} = \frac{1}{2} + \frac{1}{3} + \frac{1}{7} + \frac{1}{42} = \cdots$$

We have shown that $n \in S \Rightarrow n + 1 \in S$ for $n \ge 2$ and that $3 \in S$. Since $1 \in S$ and $2 \notin S$ (explain why), we conclude (see Proposition 1.1.2) that $S = \{1, 3, 4, 5, \ldots\}$.

12. We observe that $f(n) - f(n-1) = n!$ and hence that $f(n) - f(n-1) = n \cdot (n-1)! = n(f(n-1) - f(n-2))$. We are thus led to the recurrence $f(n) = (n+1)f(n-1) - nf(n-2)$ for $n \ge 2$, where $f(0) = 0$ and $f(1) = 1$. This exercise appeared as Problem B1 in the William Lowell Putnam Mathematical Competition in 1984.
14. Hint: Apply the principle of mathematical induction, in the form of Proposition 1.1.1 (b), to prove that every $m \in \mathbb{N}$ can be expressed in this way. Since this is trivial for $m \le 1$, assume that $m \ge 2$ and that the statement holds for all natural numbers less than m. Let $p \in \mathbb{N}$ be the largest natural number such that $F_p \le m$ (explain why such p exists). Observe that $F_p \ge 1$, so $m - F_p < m$, apply the induction hypothesis to express $m - F_p$ as a sum of distinct Fibonacci numbers, and conclude that such an expression exists for m as well.
15. Hint for part (c): Show that $F_{2n+1} = 3F_{2n-1} - F_{2n-3}$ for $n \ge 2$.
16. Hint: Express a_n in terms of Fibonacci numbers.
19. Hint for parts (b) and (c): Observe that $a_{k-1}a_{k+2} = a_{k+1}^2 - a_k^2$ for $k \ge 1$.
21. Let us write

$$q_n(x) = (x-1)(x^2-1)\cdots(x^n-1) = \prod_{i=1}^{n} (x^i - 1)$$

for $n \in \mathbb{N}$ and $x \in \mathbb{R}$, where $q_0(x) = 1$. Then, for $1 \le k \le n-1$,

$$\begin{aligned} q_n(x) &= q_{n-1}(x)\,(x^n - 1) = q_{n-1}(x)\,(x^{n-k} - 1 + x^n - x^{n-k}) \\ &= q_{n-1}(x)\,(x^{n-k} - 1) + x^{n-k} q_{n-1}(x)\,(x^k - 1), \end{aligned}$$

and hence,

$$f_{n,k}(x) = \frac{q_n(x)}{q_k(x) \cdot q_{n-k}(x)} = \frac{q_{n-1}(x)\,(x^{n-k} - 1)}{q_k(x) \cdot q_{n-k}(x)} + \frac{x^{n-k} q_{n-1}(x)\,(x^k - 1)}{q_k(x) \cdot q_{n-k}(x)}$$

$$= \frac{q_{n-1}(x)}{q_k(x) \cdot q_{n-k-1}(x)} + x^{n-k} \frac{q_{n-1}(x)}{q_{k-1}(x) \cdot q_{n-k}(x)}$$

$$= f_{n-1,k}(x) + x^{n-k} f_{n-1,k-1}(x).$$

This proves part (a). Since $f_{n,0}(x) = f_{n,n}(x) = 1$, part (b) follows from part (a) by induction on n. In fact a stronger statement follows, namely that the rational function $f_{n,k}(x)$ is a polynomial in x with nonnegative integer coefficients; the degree of this polynomial is equal to $k(n-k)$. For part (c), we take the limit as $x \mapsto 1$ in the recurrence of part (a) and note that

$$\lim_{x \mapsto 1} f_{n,k}(x) = \lim_{x \mapsto 1} \frac{q_n(x)}{q_k(x) \cdot q_{n-k}(x)} = \frac{n!}{k!(n-k)!} = C(n,k).$$

For this reason, the function $f_{n,k}(x)$ is called a *q-binomial* (or *Gaussian*) coefficient.

22. Hint for the first three parts: Factor $a^4 + a^2 + 1$, $a^8 + a^4 + 1$, and $a^5 + a + 1$. We will solve part (d). Let us write $x = a^4 + a^3 + a^2 + a + 1$ and $y = a^{24} + a^{18} + a^{12} + a^6 + 1$. It will suffice to show that $x \mid y - x$ since then, according to parts (a) and (d) of Proposition 1.1.3, we also have $x \mid x + (y - x) = y$. We observe that

$$\begin{aligned} y - x &= (a^{24} - a^4) + (a^{18} - a^3) + (a^{12} - a^2) + (a^6 - a) \\ &= a^4(a^{20} - 1) + a^3(a^{15} - 1) + a^2(a^{10} - 1) + a(a^5 - 1). \end{aligned}$$

Our fundamental identity (1.4) implies that $a^5 - 1$ divides each one of $a^{20} - 1$, $a^{15} - 1$, $a^{10} - 1$, and $a^5 - 1$. Hence, according to Proposition 1.1.3 (d), $a^5 - 1$ also divides $a^4(a^{20} - 1) + a^3(a^{15} - 1) + a^2(a^{10} - 1) + a(a^5 - 1) = y - x$. Another application of (1.4) shows that $a^5 - 1$ is divisible by $a^4 + a^3 + a^2 + a + 1 = x$. Thus, $x \mid a^5 - 1$ and $a^5 - 1 \mid y - x$, and hence, according to Proposition 1.1.3 (c), $x \mid y - x$.

23. Hint: Show that

$$\frac{1 + a^2 + \cdots + a^{2n-2}}{1 + a + \cdots + a^{n-1}} = \frac{a^n + 1}{a + 1} = \frac{a^n - (-1)^n}{a + 1} + \frac{1 + (-1)^n}{a + 1}$$

and conclude that conditions (i) and (ii) are equivalent to each other and to $1 + (-1)^n$ being divisible by $a + 1$.

24. Hint: For part (a), compute $a_{n+1} - 6a_n$. For part (c), guess what the remainder of the division of a_n by 125 is and prove your guess by induction on n.

25. Hint: Prove that the identities $F_{2n-1} = F_n^2 + F_{n-1}^2$ and $F_{2n} = F_n(F_{n+1} + F_{n-1})$ hold for every $n \in \mathbb{Z}_{>0}$ simultaneously, by induction on n. In fact, these identities are special cases of the one of Exercise 26 (a).

26. Part (a) is trivial for $n = 1$. Assuming that it holds for $n \geq 1$ and every $m \in \mathbb{Z}_{>0}$, we find that

$$\begin{aligned} F_{m+(n+1)} &= F_{(m+1)+n} = F_{m+2}F_n + F_{m+1}F_{n-1} \\ &= (F_{m+1} + F_m)F_n + F_{m+1}F_{n-1} = F_{m+1}(F_n + F_{n-1}) + F_m F_n \\ &= F_{m+1}F_{n+1} + F_m F_n \end{aligned}$$

for every $m \in \mathbb{Z}_{>0}$. The result follows by induction, in the form of Proposition 1.1.1 (a). We leave it to the reader to verify (1.16) by induction on n. Substituting $m + n$ for n, we find that

$$\begin{pmatrix} F_{m+n+1} & F_{m+n} \\ F_{m+n} & F_{m+n-1} \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}^{m+n} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}^{m} \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}^{n}$$

$$= \begin{pmatrix} F_{m+1} & F_m \\ F_m & F_{m-1} \end{pmatrix} \begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix}$$

$$= \begin{pmatrix} F_{m+1}F_{n+1} + F_m F_n & F_{m+1}F_n + F_m F_{n-1} \\ F_m F_{n+1} + F_{m-1}F_n & F_m F_n + F_{m-1}F_{n-1} \end{pmatrix}$$

and the identity of part (a) follows. For part (c), you may use part (a) and apply induction on $d \in \mathbb{Z}_{>0}$, where $n = md$.

27. Let us apply the well-ordering principle, as in Example 1.1.6. Let us denote by S the set of integers $n \geq 2$ which *cannot* be expressed as products of prime numbers, and let us assume that $S \neq \varnothing$. According to Proposition 1.1.1 (c), S has a minimum element, say n. Since S clearly does not contain prime numbers, we must have $n = ab$ for some integers $a, b \geq 2$. Then, $a, b < n$ and hence $a, b \notin S$. This means that a and b can be expressed as products of prime numbers and hence so does their product $ab = n$, contrary to our assumption that $n \in S$. This contradiction leads us to the conclusion that $S = \varnothing$, as desired.
28. Hint: To prove the existence of q and r, consider the set $S = \{n - mq : q \in \mathbb{Z}\} \cap \mathbb{N}$ and show that $S \neq \varnothing$. Define r to be the minimum element of S and argue as in the solution of Exercise 5 to show that $r < m$. To prove uniqueness, assume that $n = mq + r$ and $n = mp + s$ are two such expressions for n. Then, observe that $mp \leq n = mq + r < mq + m$ and conclude that $p \leq q$. Argue similarly to prove that $q \leq p$ and conclude that $p = q$ and $r = s$.
29. We proceed by induction on k. The statement is trivial for $k = 0$. Let us assume that $k \geq 1$ and that it holds for $k-1$ (and all integers $m \geq 2$). We will prove that it holds for k as well. Consider an integer $0 \leq n < m^{k+1}$. By Proposition 1.1.4, there exist unique integers q and r such that

$$n = r + mq \tag{B.1}$$

and $0 \leq r < m$. The assumption $n \geq 0$ implies that $q \geq 0$. We also have $qm \leq qm + r = n < m^{k+1}$ and hence $0 \leq q < m^k$. As a result, the induction hypothesis guarantees that there is a unique expression for q of the form

$$q = a_1 + a_2 m + \cdots + a_k m^{k-1} \tag{B.2}$$

with $a_1, \ldots, a_k \in \{0, 1, \ldots, m-1\}$. Setting $r = a_0$, we conclude from (B.1) and (B.2) that n can be expressed in the form (1.10). To prove uniqueness, assume that

$$n = b_0 + b_1 m + b_2 m^2 + \cdots + b_k m^k$$

is another m-ary representation for n. Then, $n = b_0 + tm$ with $b_0 \in \{0, 1, \ldots, m-1\}$ and

$$t = b_1 + b_2 m + \cdots + b_k m^{k-1}. \tag{B.3}$$

Equation (B.1) and the uniqueness part of Proposition 1.1.4 imply that $b_0 = r = a_0$ and $t = q$. The latter equality and the uniqueness of the representation (B.2) imply that $b_i = a_i$ for every $i \in \{1, 2, \ldots, k\}$. This completes the inductive step and the proof of the proposition.

30. Part (a) is left to the reader. By Proposition 1.1.5 applied for $m = 2$ (binary representation), every $n \in \mathbb{N}$ can be uniquely expressed as $n = 2^{a_1} + 2^{a_2} + \cdots + 2^{a_r}$, where $a_1 < a_2 < \cdots < a_r$ are natural numbers. Let $g(n)$ denote the number r of powers of 2 in this representation. Then, $g(0) = 0$, $g(2n) = g(n)$, and $g(2n+1) = g(n)+1$ for $n \in \mathbb{N}$ (explain why). This shows that $g(n) = f(n)$ for every $n \in \mathbb{N}$ and answers part (b). As a consequence, for $1000 \leq n \leq 2000$, we have $f(n) = 2$ if and only if $n = 2^{10} + 2^k$ for some $k \in \{0, 1, \ldots, 9\}$. This answers part (c). For part (d), let us assume that $f(n) = g(n) = n/3$. Then,

$$n \geq 1 + 2 + 2^2 + \cdots + 2^{n/3-1} = 2^{n/3} - 1.$$

Equivalently, $(n+1)^3 \geq 2^n$. As the reader may have verified when solving Exercise 3, the latter inequality holds for $n \leq 10$. We compute $f(n)$ for $n \in \{3, 6, 9\}$ and conclude that $n = 0$ and $n = 6$ are the only values of $n \in \mathbb{N}$ for which $f(n) = n/3$.

32. (a) We proceed by induction on n. The statement is trivial for $n = 1$ and identical to the definition (1.17) for $i = n - 1$ (in particular, it holds for $n = 2$ as well). Assuming that $n \geq 3$ and $i \in [n-2]$, and that the statement holds for $n - 1$, we find that

$$\begin{aligned} a_1 \circ a_2 \circ \cdots \circ a_n &= (a_1 \circ \cdots \circ a_{n-1}) \circ a_n \\ &= ((a_1 \circ \cdots \circ a_i) \circ (a_{i+1} \circ \cdots \circ a_{n-1})) \circ a_n \\ &= (a_1 \circ \cdots \circ a_i) \circ ((a_{i+1} \circ \cdots \circ a_{n-1}) \circ a_n) \\ &= (a_1 \circ \cdots \circ a_i) \circ (a_{i+1} \circ \cdots \circ a_n), \end{aligned}$$

where we have applied successively the definition (1.17), the induction hypothesis, the associative law, and definition (1.17) once again. According to Proposition 1.1.1 (b), the statement holds for every positive integer n.

(b) The first identity is a special case of (1.18). For the second, we observe that if $(a^m)^n = a^{mn}$, then

$$(a^m)^{n+1} = (a^m)^n \circ a^m = a^{mn} \circ a^m = a^{mn+m} = a^{m(n+1)}$$

and proceed by induction on n.

33. Let us verify part (a). We set $C = (\cup_{i=1}^n A_i) \cap B$ and $D = \cup_{i=1}^n (A_i \cap B)$ and assume that $x \in C$, so that $x \in \cup_{i=1}^n A_i$ and $x \in B$. Then, $x \in A_i$ for at least one index $i \in [n]$ and $x \in B$. Therefore, $x \in A_i \cap B$ for at least one $i \in [n]$ and hence $x \in \cup_{i=1}^n (A_i \cap B) = D$. We have shown that $C \subseteq D$. Reversing these steps, we find that $D \subseteq C$ and conclude that $C = D$. We leave it to the reader to apply similar reasoning and verify the other proposed identities.

35. (a) Suppose that f is injective. For $y \in Y$, we define $g(y)$ to be the unique element $x \in X$ for which $f(x) = y$, if $y \in f(X)$, and an arbitrary element of X, otherwise. Then, $g : Y \to X$ is a map with the desired property. Conversely, if such a map g exists, then for $x_1, x_2 \in X$ we have $f(x_1) = f(x_2) \Rightarrow g(f(x_1)) = g(f(x_2)) \Rightarrow x_1 = x_2$ and hence f is injective.

(b) Suppose that f is surjective. For $y \in Y$, we define $g(y)$ to be any element $x \in X$ for which $f(x) = y$ (such an element exists). Then, $g : Y \to X$ is a map with the desired property. The converse is obvious.

Chapter 2

1. We proceed by induction on m. The statement of part (a) holds for $m = 0$, since if $f : \varnothing \to [n]$ is a surjective map, then $n = 0$. Let us assume that $m \geq 1$ and that the statement holds for $m - 1$, and let us consider a bijective map $f : [m] \to [n]$. Then, $f(m) = b$ for some $b \in [n]$, and f restricts to a bijective map $g : [m-1] \to [n] \setminus \{b\}$. Clearly, there exists a bijective map $h : [n] \setminus \{b\} \to [n-1]$, for example, the one with

$$h(x) = \begin{cases} x, & \text{if } x < b \\ x - 1, & \text{otherwise.} \end{cases}$$

Since the composition $h \circ g : [m-1] \to [n-1]$ is also bijective, we conclude from the induction hypothesis that $m - 1 = n - 1$, so, equivalently, $m = n$. This completes the inductive step. We argue similarly for part (b), assuming that f is injective (in which case g and $h \circ g$ have the same property). To prove part (c), we assume that f is surjective, and let $b = f(m)$ and distinguish two cases. If there is no $a \in [m-1]$ such that $f(a) = b$, then we follow the same reasoning as before, since g is defined and is surjective. Otherwise, the restriction $g : [m-1] \to [n]$ of f is surjective, and we conclude by induction that the stronger inequality $m \geq n + 1$ holds.

2. The equivalence (b) $\Leftrightarrow$ (c) is a consequence of Exercise 35 of Chapter 1 (explain why). Thus, it suffices to show that (a) $\Leftrightarrow$ (b). Suppose that S is infinite. Then, we may define an injective map $f : \mathbb{N} \to S$ recursively by choosing $f(0)$ as an arbitrary element of S and $f(n)$ to be any element of S other than $f(0), f(1), \ldots, f(n-1)$, for $n \geq 1$. Conversely, suppose that there exists an injective map $f : \mathbb{N} \to S$. Then, f restricts to an injective map $f|_{[m]} : [m] \to S$ for every $m \in \mathbb{N}$. This fact and Exercise 1 (b) imply that the set S is infinite.
3. It suffices to prove the equivalence (a) $\Leftrightarrow$ (b). Suppose first that f is injective. Then, by Proposition 2.2.1, we have $\#f(X) = \#X$. Since $f(X) \subseteq Y$ and $\#X = \#Y$, it follows from Corollary 2.2.1 that the set $Y \setminus f(X)$ is empty. Therefore, $f(X) = Y$ and f is surjective. Conversely, suppose that f is surjective. Then, the inverse image $f^{-1}(y)$ is a nonempty subset of X for every $y \in Y$. By the addition principle, the sum of the cardinalities of all these subsets is equal to $\#X$. Since there are as many as $\#Y = \#X$ such subsets, we must have $\#f^{-1}(y) = 1$ for every $y \in Y$. Hence, f is bijective.
4. The total number of matches is equal to $n - 1$. Indeed, let X_n be the set of matches played and Y_n be the set of players who have been disqualified during some round of the tournament. Since only one player remains in the end, we have $\#Y_n = n - 1$. Let $\varphi : X_n \to Y_n$ be the map for which $\varphi(x)$ is the loser of the match $x \in X_n$. Since every player $y \in Y_n$ gets disqualified exactly once, the map φ is a bijection. Thus, it follows from Proposition 2.2.1 that $\#X_n = \#Y_n = n - 1$.
5. Hint: For part (a), consider an $(x + y) \times (x + y)$ chessboard which has been divided into unit squares, in the standard way. Count these squares in two different ways. Replace the chessboard with a three-dimensional cube with edge lengths equal to $x + y$ for part (b).
7. Answers: (a) 2^{k-1}; (d) 63.
9. Answers: (a) $(4^n - 1)/3$; (c) $n2^{n-1}$.
14. Hint: For part (a), apply the multiplication principle. Generalize the second method of proof followed in Example 2.3.3 and show that the number of sequences in question is equal to $(k + 1)^n$. For parts (b) and (c), apply the multiplication principle in a similar way.
15. (a) Every such square x is uniquely determined by its side length k and its southwest vertex, which can be any element of the set $[n-k] \times [n-k]$. As a consequence, there are $(n-k)^2$ such squares with side length equal to k and $1^2 + 2^2 + \cdots + (n-1)^2 = n(n-1)(2n-1)/6$ such squares in total (see Exercise 2 of Chapter 1).
    (b) Let X_n be the set of squares in the plane with vertices in the set $[n] \times [n]$, and let Y_n be the set of squares in X_n with edges parallel to one of the lines $x = 0$ or $y = 0$. For $x \in X_n$, let $f(x)$ be the smallest square in Y_n which contains x. An example with $n = 6$ is shown in Figure B.1. As shown in part (a), for every $k \in [n-1]$ there are $(n-k)^2$ squares $y \in Y_n$ with side length k. Moreover, for every $y \in Y_n$ with side length k, there are exactly

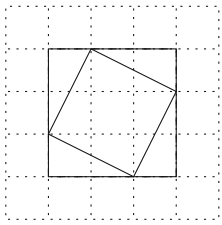


Figure B.1 A square $x \in X_6$ and the square $f(x) \in Y_6$

k squares $x \in X_n$ such that $f(x) = y$. Applying Proposition 2.2.3 and the addition principle (and Exercise 2 of Chapter 1 again), we conclude that

$$\begin{aligned} \#X_n &= \sum_{k=1}^{n-1} (n-k)^2 \cdot k = \sum_{k=1}^{n-1} k^2(n-k) = n \cdot \frac{n(n-1)(2n-1)}{6} \\ &\quad - \frac{n^2(n-1)^2}{4} \\ &= \frac{n^2(n^2-1)}{12}. \end{aligned}$$

A variation of part (b) appeared as Problem 4 in the 1995 Balkan Mathematical Olympiad.

17. Hint for part (c): Choose the elements of the permutation successively in a suitable order, so that you can apply the multiplication principle.
18. Hint: Give a bijective proof that there are exactly 2^{n-1} such permutations.
19. Hint: First choose the index $i \in [n-1]$, or the pair of indices $(i, j) \in [n] \times [n]$.
21. The identity (2.5) is trivial for $n = 1$. Let us assume that it holds for $n - 1 \geq 1$, that is,

$$(x_1 + y_1)(x_2 + y_2) \cdots (x_{n-1} + y_{n-1}) = \sum_{S \subseteq [n-1]} x_S y_{\bar{S}},$$

where $\bar{S} = [n-1] \setminus S$. Multiplying both sides of this equality by $x_n + y_n$, we find that $(x_1 + y_1)(x_2 + y_2) \cdots (x_n + y_n)$ is equal to

$$\begin{aligned} (x_n + y_n) \sum_{S \subseteq [n-1]} x_S y_{\bar{S}} &= \sum_{S \subseteq [n-1]} x_S x_n \, y_{\bar{S}} + \sum_{S \subseteq [n-1]} x_S y_{\bar{S}} y_n \\ &= \sum_{S \subseteq [n-1]} x_{S \cup \{n\}} y_{\bar{S}} + \sum_{S \subseteq [n-1]} x_S y_{[n] \setminus S} \\ &= \sum_{T \subseteq [n]} x_T y_{[n] \setminus T}. \end{aligned}$$

The last equality is due to the fact that every $T \subseteq [n]$ can be written uniquely in the form $T = S \cup \{n\}$ or $T = S$, for some $S \subseteq [n-1]$. It follows by induction that (2.5) holds for every positive integer n.

22. Hint: For part (b), count the pairs (δ, κ), where δ is a diagonal of the convex n-gon Π and κ is one of the two endpoints of δ.
23. Hint: For part (c), first choose the rows and the columns in which the rooks will lie. Then, apply the result of Example 2.2.6 for the $k \times k$ chessboard formed by these rows and columns and the multiplication principle.
27. Hint: To prove the inequality on the left, rewrite the formula (2.3) in the form $\binom{n}{k} = \frac{n}{k} \cdot \frac{n-1}{k-1} \cdots \frac{n-k+1}{1}$ and show that $\frac{n}{k} \leq \frac{n-1}{k-1} \leq \cdots \leq n-k+1$. To prove the one on the right, observe that $\binom{n}{k+1} = \frac{n-k}{k+1}\binom{n}{k} < \frac{n}{k+1}\binom{n}{k}$ and apply induction on k.
28. Hint: Find a simpler formula for p_n first.
29. Hint for part (a): Differentiate the identity $(x+1)^n = \sum_{k=0}^{n} \binom{n}{k} x^k$ to conclude that $nx(x+1)^{n-1} = \sum_{k=0}^{n} k\binom{n}{k} x^k$ and that

$$nx(x+1)^{n-1} - nx(1-x)^{n-1} = 4 \cdot \sum_{k=0}^{\lfloor n/2 \rfloor} k \binom{n}{2k} x^{2k}.$$

31. We claim that the answer is $n!\binom{n-1}{k-1}$. Indeed, such an arrangement can be obtained from a permutation, say $(\sigma_1, \sigma_2, \ldots, \sigma_n)$, of the set $[n]$ by placing barriers in $k-1$ spaces between successive elements σ_i to form k lines. For example, for $n = 10$ and $k = 4$, the arrangement

$$\begin{array}{c} 4\ 9\ 6\ \ 2 \\ 8\ 1\ \ \ 10 \\ \ \ 7\ \ \ \ 3 \\ \ \ 5 \end{array}$$

can be obtained from $(4, 8, 9, 1, 7, 5, 6, 2, 10, 3)$ by placing three barriers as follows:

$$4\ \ 8\ \mid\ 9\ \ \ 1\ \ \ 7\ \ 5\ \mid\ 6\ \mid\ 2\ \ \ 10\ \ \ 3.$$

Since there are $n!$ permutations of $[n]$ and $\binom{n-1}{k-1}$ ways to place the $k-1$ barriers in the $n-1$ spaces between successive elements of σ, the claimed formula follows from the multiplication principle.

32. The answer is $\binom{n-k+1}{k}$, and here is one way to prove it. If $S = \{a_1, a_2, \ldots, a_k\} \subseteq [n]$, say with $a_1 < a_2 < \cdots < a_k$, does not contain two consecutive integers, then

$$1 \leq a_1 < a_2 - 1 < a_3 - 2 < \cdots < a_k - k + 1 \leq n - k + 1$$

and hence $\varphi(S) = \{a_1, a_2 - 1, \ldots, a_k - k + 1\}$ is a k-subset of $[n - k + 1]$. For example, if $n = 20$, $k = 6$, and $S = \{2, 5, 10, 12, 17, 19\}$, then $\varphi(S) = \{2, 4, 8, 9, 13, 14\}$. We leave it to the reader to verify that the resulting map φ is a one-to-one correspondence from the set of k-subsets of $[n]$ which do not contain two consecutive integers to the set of all k-subsets of $[n - k + 1]$. The cardinality of the latter set is equal to $\binom{n-k+1}{k}$.

33. (a) Let us add the lines one by one to the plane, in any order, and consider the number of regions formed at each stage. At the beginning, there is no line and only one region (the plane itself). When adding the ith line, some of the existing regions split into two pieces. Those regions are in one-to-one correspondence with the segments into which the ith line is itself divided by the previous $i - 1$ lines and hence they are as many as i. Therefore, at the last stage, there are $1 + \sum_{i=1}^{n} i = 1 + n + \binom{n}{2}$ regions.

    (b) We argue as in part (a). Adding the ith plane to the previous $i - 1$ planes creates, according to part (a), $1 + (i - 1) + \binom{i-1}{2}$ more three-dimensional regions (as many as the two-dimensional regions into which this plane is dissected by the other $i - 1$ planes). Hence, after adding the last plane, there are

$$1 + \sum_{i=1}^{n} \left\{ 1 + (i - 1) + \binom{i - 1}{2} \right\} = 1 + n + \binom{n}{2} + \binom{n}{3}$$

    three-dimensional regions, where for the summation we applied Proposition 2.4.2 for $k = 2$ and $k = 3$.

    (c) Similar reasoning shows that n straight hyperplanes in general position in Euclidean space $\mathbb{R}^k$ dissect this space into

$$\sum_{i=0}^{k} \binom{n}{i}$$

    open, connected, k-dimensional regions.

36. Answers: (a) $1/n!$; (b) $(n + 2)3^{n-1}/(n - 1)!$, for $n \geq 1$.
38. Answer: $2 \cdot 3^{n-1}$.
39. Hint for part (c): Combine part (b) with the result of Exercise 32.
41. Hint: Deduce the formula of part (a) from Proposition 2.3.10. For part (b), given $x = (x_1, x_2, \ldots, x_n) \in \mathbb{N}^n$ with $x_1 + x_2 + \cdots + x_n \leq m$, let $f(x) = (y_1, y_2, \ldots, y_m) \in \mathbb{N}^m$, where y_k is the number of indices $i \in [n]$ such that $x_1 + x_2 + \cdots + x_i = k - 1$ for $k \in [m]$. Show that $y_1 + y_2 + \cdots + y_m \leq n$ and that the resulting map f has the desired properties.
42. (a) There is a one-to-one correspondence between such polygons Π' and compositions of $n - k$ with k parts: The parts of the composition record the number of vertices of Π between successive vertices of Π', starting with vertex v and continuing in the clockwise direction. Thus, by Proposition 2.3.9, the answer is $\binom{n-k-1}{k-1}$.

(b) Let $p(n,k)$ be the number of polygons Π' we wish to compute. Let us count the pairs (Π', v), where Π' is a convex polygon with properties (a) and (b), stated in the exercise, and v is a vertex of Π'. On the one hand, the number of such pairs is equal to $kp(n,k)$ since every such polygon Π' has exactly k vertices. On the other hand, the number of such pairs is equal to $n\binom{n-k-1}{k-1}$ since, according to part (a), there are exactly $\binom{n-k-1}{k-1}$ polygons Π' with properties (a) and (b) which contain a given vertex v of Π. Hence,

$$p(n,k) = \frac{n}{k}\binom{n-k-1}{k-1} = \frac{n}{n-k}\binom{n-k}{k}.$$

Part (b) of this exercise is known as *Problème des ménages* (see, for instance, [23, Lemma 2.3.4]).

43. The answer is

$$n\binom{m+n-1}{n} = \frac{(m+n-1)!}{(m-1)!(n-1)!}.$$

Indeed, to form a cyclic permutation of the set of boys and girls with the given properties, we fix the cyclic permutation $a_1, a_2, \ldots, a_m$ of the boys and insert the girls in the m spaces between successive boys. This can be done by choosing a cyclic shift $(b_i, b_{i+1}, \ldots, b_n, b_1, \ldots, b_{i-1})$ of the permutation $(b_1, b_2, \ldots, b_n)$ of the girls and by placing them in the m spaces between a_1 and a_2, between a_2 and a_3, and so on, up to that between a_m and a_1, keeping the linear order in which they appear in the cyclic shift. There are n choices for the cyclic shift, and for every such choice, according to Proposition 2.3.10, there are $\left(\!\binom{m}{n}\!\right) = \binom{m+n-1}{n}$ ways to place the girls. This proves the formula claimed above. For example, for $m = n = 4$, if one chooses the cyclic shift (b_3, b_4, b_1, b_2) and places one girl between a_1 and a_2, none between a_2 and a_3, two between a_3 and a_4, and one between a_4 and a_1, then we get the cyclic permutation $(a_1, b_3, a_2, a_3, b_4, b_1, a_4, b_2, a_1)$.

44. Hint: For $0 \le i \le m$ and $0 \le j \le n$, how many paths in $\mathbb{Z}^2$ with initial vertex $(0,0)$, terminal vertex (m,n) and steps $(1,0)$ or $(0,1)$ pass through the point (i,j)?

45. Hint: For $0 \le i \le m-1$ and $0 \le j \le n$, how many paths in $\mathbb{Z}^2$ with initial vertex $(0,0)$, terminal vertex (m,n), and steps $(1,0)$ or $(0,1)$ pass through the points (i,j) and $(i+1,j)$?

46. Hint: Using the recurrence (2.15), show that $f(n,k)$ is equal to the numbers of paths of Proposition 2.3.12 with $n-k$ initial steps $\mathrm{e}_1 = (1,0)$, and that

$$f(n,k) = \frac{n-k+1}{n+1}\binom{n+k}{k}$$

for $0 \le k \le n$. The identity of part (c) is equivalent to the identity of Example 2.4.1. Note: Replacing n by $n-k$, the result of part (a) may be restated

as follows: For $1 \leq k \leq n/2$, the number of paths in $\mathbb{Z}^2$, which have initial vertex $(0, 0)$, terminal vertex $(n-k, k)$, steps $(1, 0)$ or $(0, 1)$ and lie entirely in the half-plane $x \geq y$ of $\mathbb{R}^2$, is equal to

$$\binom{n}{k} - \binom{n}{k-1}.$$

This proves combinatorially the inequalities of part (e) of Proposition 2.3.6.

48. The proposed sum in part (a) is equal to $\binom{n}{k}$ since

$$\begin{aligned}\sum_{i=0}^{k} \frac{n-2i+1}{n-i+1}\binom{n}{i} &= \sum_{i=0}^{k}\left(1 - \frac{i}{n-i+1}\right)\binom{n}{i} \\ &= \sum_{i=0}^{k}\binom{n}{i} - \sum_{i=0}^{k}\frac{i}{n-i+1}\binom{n}{i} \\ &= \sum_{i=0}^{k}\binom{n}{i} - \sum_{i=1}^{k}\binom{n}{i-1} \\ &= \sum_{i=0}^{k}\binom{n}{i} - \sum_{i=0}^{k-1}\binom{n}{i} = \binom{n}{k}.\end{aligned}$$

A proof by induction on k is also possible. Part (b) is left to the reader.

49. (a) We have

$$\begin{aligned}(1+x)^{2n} &= (1+2x+x^2)^n = (1+x(x+2))^n \\ &= \sum_{j=0}^{n}\binom{n}{j}x^j(x+2)^j = \sum_{j=0}^{n}\binom{n}{j}x^j\sum_{i=0}^{j}\binom{j}{i}2^{j-i}x^i \\ &= \sum_{j=0}^{n}\sum_{i=0}^{j}\binom{n}{j}\binom{j}{i}2^{j-i}x^{i+j}.\end{aligned}$$

The proposed identity follows by comparing the coefficients of x^k on the two extreme sides of this string of equalities.

(b) Let $\Omega_n = \{1, -1, 2, -2, \ldots, n, -n\}$. Clearly, the number of k-subsets S of this set is equal to $\binom{2n}{k}$. Let us count these subsets as follows. Let I and J be the sets of numbers $a \in [n]$ such that $\{a, -a\} \subseteq S$ and $\{a, -a\} \cap S \neq \varnothing$, respectively. Let i be the cardinality of I. Then, $J \smallsetminus I$ has $k-2i$ elements, and hence, J has cardinality $k-i$. There are $\binom{n}{k-i}\binom{k-i}{i}$ ways to choose I and J, and for every such choice, there are 2^{k-2i} ways to choose the elements of S whose absolute value belongs to $J \smallsetminus I$. As a

result, the number of k-subsets of Ω_n is also equal to the right-hand side of the proposed identity.

50. Suppose we want to select a committee of n persons out of n men and n women. Let us call a committee *odd* or *even*, if it contains an odd or even number of women, respectively. The number of even committees is equal to

$$\binom{n}{0}\binom{n}{n} + \binom{n}{2}\binom{n}{n-2} + \cdots = \binom{n}{0}^2 + \binom{n}{2}^2 + \cdots,$$

while that of odd committees is equal to

$$\binom{n}{1}\binom{n}{n-1} + \binom{n}{3}\binom{n}{n-3} + \cdots = \binom{n}{1}^2 + \binom{n}{3}^2 + \cdots$$

Hence, we need to show that the number of even committees minus that of odd committees is equal to zero, if n is odd and to $(-1)^{n/2}\binom{n}{n/2}$, if n is even. This will be achieved if we can match the even committees with the odd ones, so that there is no committee without a match, if n is odd, while there are exactly $\binom{n}{n/2}$ committees without a match, if n is even, all of them odd or even according to whether $n/2$ is odd or even, respectively. We may assume that men and women are married in couples, which we enumerate as $1, 2, \ldots, n$. We leave without a match any committee which consists only of couples. There are $\binom{n}{n/2}$ such committees, if n is even, and none, if n is odd. We match any other committee with the one obtained from it as follows: We consider the smallest number i of a couple for which only one of the two spouses belongs to the given committee and replace this person with their spouse. Clearly, even committees are matched in this way with odd committees and vice versa. Moreover, all committees are matched if n is odd, while there are exactly $\binom{n}{n/2}$ committees without a match if n is even, all of them odd, if $n/2$ is odd and all even, if $n/2$ is even.

51. Let us denote by $a(p, n)$ and $b(p, n)$ the left-hand side and right-hand side of the proposed identity of part (a), respectively. For $n = 0$, we have $a(p, 0) = b(p, 0) = 1/p$. For $n \geq 1$, using the basic recurrence of Proposition 2.3.6 (b) for binomial coefficients, we find that

$$\begin{aligned} a(p,n) &= \sum_{i=0}^{n-1} \frac{(-1)^i}{p+i}\binom{n-1}{i} + \sum_{i=1}^{n} \frac{(-1)^i}{p+i}\binom{n-1}{i-1} \\ &= \sum_{i=0}^{n-1} \frac{(-1)^i}{p+i}\binom{n-1}{i} + \sum_{i=0}^{n-1} \frac{(-1)^{i+1}}{p+i+1}\binom{n-1}{i} \\ &= a(p,n-1) - a(p+1,n-1). \end{aligned}$$

We leave it to the reader to verify that $b(p,n) = b(p,n-1) - b(p+1,n-1)$ and to conclude that $a(p,n) = b(p,n)$ for all $p \in \mathbb{Z}_{>0}$ and $n \in \mathbb{N}$, by induction on n. A different solution is provided in Exercise 14 of Chapter 3. For part (b), we denote the given sum by $a(p,q,n)$ and observe that

$$\begin{aligned} a(p,q,n) &= \sum_{i=0}^{n} \frac{(-1)^i}{(p+i)(q-i)} \binom{n}{i} \\ &= \frac{1}{p+q} \sum_{i=0}^{n} (-1)^i \left(\frac{1}{p+i} + \frac{1}{q-i} \right) \binom{n}{i} \\ &= \frac{1}{p+q} \left(\sum_{i=0}^{n} \frac{(-1)^i}{p+i} \binom{n}{i} + \sum_{i=0}^{n} \frac{(-1)^{n-i}}{q-n+i} \binom{n}{n-i} \right) \\ &= \frac{1}{p+q} \left(a(p,n) + (-1)^n a(q-n,n) \right). \end{aligned}$$

Combining this formula with the result of part (a), we get

$$a(p,q,n) = \frac{n!}{p+q} \left(\frac{(p-1)!}{(p+n)!} + (-1)^n \frac{(q-n-1)!}{q!} \right).$$

52. Applying the basic formula (2.3) for binomial coefficients, we find that

$$\frac{\binom{n}{k}}{\binom{2n+1}{k}} = \frac{n!(2n-k+1)!}{(2n+1)!(n-k)!} = \frac{n!(n+1)!}{(2n+1)!} \binom{2n-k+1}{n+1}.$$

Given this expression, we get

$$\begin{aligned} \sum_{k=0}^{n} \frac{\binom{n}{k}}{\binom{2n+1}{k}} &= \frac{n!(n+1)!}{(2n+1)!} \sum_{k=0}^{n} \binom{2n-k+1}{n+1} \\ &= \frac{n!(n+1)!}{(2n+1)!} \sum_{k=0}^{n} \binom{n+k+1}{n+1} = \frac{n!(n+1)!}{(2n+1)!} \binom{2n+2}{n+2} \\ &= \frac{2n+2}{n+2}, \end{aligned}$$

where we have applied the identity of Proposition 2.4.2 to compute the sum in the next to last step.

53. Let X and Y be disjoint sets with cardinalities m and n, respectively. In how many ways can one first choose m elements of $X \cup Y$ and color them white, and then choose some of the remaining elements of X and color them black? This

can be done in $f(m,n)$ ways since, if k is the number of elements of X which will be colored black or white, then there are $\binom{m}{k}$ ways to choose these elements, and for every such choice, there are $\binom{n+k}{m}$ ways to choose the elements of $X \cup Y$, which will be colored white. On the other hand,

$$f(m,n) = \sum_{k=0}^{\min\{m,n\}} \binom{m}{k}\binom{n}{k} 2^k$$

because if $m-k$ is the number of elements of X which will be colored white, then there are $2^k\binom{m}{m-k} = 2^k\binom{m}{k}$ ways to choose the black and white elements of X, and for every such choice, there are $\binom{n}{k}$ ways to choose the white elements of Y. We have proven part (a). Part (b) follows directly from that.

54. Let k be the number of nonzero coordinates of $(x_1, x_2, \ldots, x_n)$. Then, there are $\binom{n}{k}$ ways to choose the positions of these coordinates and 2^k ways to choose their signs. Hence,

$$\tau(m,n) = \sum_{k=0}^{n} 2^k \binom{n}{k} \alpha(m,k),$$

where $\alpha(m,k)$ is the number of positive integer solutions of $y_1 + y_2 + \cdots + y_k \leq m$. As in Exercise 41, we find that $\alpha(m,k) = \binom{m}{k}$ and conclude that

$$\tau(m,n) = \sum_{k=0}^{n} 2^k \binom{m}{k}\binom{n}{k} = f(m,n).$$

Alternatively, let $\beta(m,k) = \binom{m+k}{k}$ be the number of nonnegative integer solutions of the inequality $y_1 + y_2 + \cdots + y_k \leq m$. Choosing first the positions of the coordinates of $(x_1, x_2, \ldots, x_n)$ which are negative, we find that

$$\begin{aligned}\tau(m,n) &= \sum_{k=0}^{n} \binom{n}{k} \beta(m-k,n) = \sum_{k=0}^{n} \binom{n}{k}\binom{m+n-k}{n} \\ &= \sum_{k=0}^{n} \binom{n}{k}\binom{m+k}{n} = f(n,m).\end{aligned}$$

These arguments solve part (a) and give a new solution to Exercise 53 (a). Part (b) follows directly from part (a).

55. Hint for part (b): Find a suitable formula for the sum $\sum_{k=1}^{n}(-1)^{k-1}\binom{n}{k}x^{k-1}$ and integrate with respect to x.

56. It follows from the equality

$$(1+xt)^m(1+yt)^n = \left(\sum_{k=0}^{m}\binom{m}{k}x^{m-k}t^{m-k}\right)\left(\sum_{k=0}^{n}\binom{n}{k}y^k t^k\right)$$

that the left-hand side of the proposed identity is equal to the coefficient of t^m in $(1+xt)^m(1+yt)^n$. We write

$$\begin{aligned}(1+xt)^m(1+yt)^n &= (1+(x-y)t+yt)^m(1+yt)^n\\ &= \left(\sum_{k=0}^{m}\binom{m}{k}(x-y)^{m-k}t^{m-k}(1+yt)^k\right)(1+yt)^n\\ &= \sum_{k=0}^{m}\binom{m}{k}(x-y)^{m-k}t^{m-k}(1+yt)^{n+k}\end{aligned}$$

and conclude that this coefficient is also equal to the right-hand side of the proposed identity. The new formula for $f(m,n)$ follows from that by setting $x=1$ and $y=2$.

57. The identity of part (b) follows from that of part (a) by setting $x=0$. For part (a), we observe that the left-hand side of the proposed identity is equal to the coefficient of y^n in the polynomial

$$(1+y)^n(x+y)^n = \left(\sum_{k=0}^{n}\binom{n}{k}y^k\right)\left(\sum_{k=0}^{n}\binom{n}{k}x^k y^{n-k}\right)$$

and that

$$\begin{aligned}(1+y)^n(x+y)^n &= (1+y)^n\sum_{k=0}^{n}\binom{n}{k}(x-1)^k(1+y)^{n-k}\\ &= \sum_{k=0}^{n}\binom{n}{k}(x-1)^k(1+y)^{2n-k}\\ &= \sum_{k=0}^{n}\binom{n}{k}(x-1)^k\sum_{i=0}^{2n-k}\binom{2n-k}{i}y^i\\ &= \sum_{i=0}^{2n}y^i\sum_{k=0}^{\min(n,2n-i)}\binom{n}{k}\binom{2n-k}{i}(x-1)^k.\end{aligned}$$

58. We observe that $\binom{n}{k}^2$ is the number of pairs (S, T) of k-subsets of the set $[n]$. Let us count these pairs as follows. We set $\#(S \cap T) = j$ and $k - j = i$, so that $i, j \in \mathbb{N}$ with $i + j = k$ and $\#(S \setminus T) = \#(T \setminus S) = i$. Given such i and j, there are $\binom{n}{i}$ ways one can choose $S \setminus T$, and for every such choice, there are $\binom{n-i}{i}$ and $\binom{n-2i}{j}$ ways, respectively, and one can successively choose $T \setminus S$ and $S \cap T$. Therefore, there are

$$\binom{n}{i}\binom{n-i}{i}\binom{n-2i}{j}$$

pairs (S, T) of k-subsets of $[n]$ with $\#(S \cap T) = j$, where $i = k - j$. We sum over all pairs $(i, j) \in \mathbb{N}^2$ with $i + j = k$, and the proof of part (a) follows. For part (b), we deduce that

$$\begin{aligned}
\sum_{k=0}^{n} \binom{n}{k}^2 x^k &= \sum_{k=0}^{n} \sum_{i+j=k} \binom{n}{i}\binom{n-i}{i}\binom{n-2i}{j} x^k \\
&= \sum_{2i+j \le n} \binom{n}{i}\binom{n-i}{i}\binom{n-2i}{j} x^{i+j} \\
&= \sum_{i=0}^{\lfloor n/2 \rfloor} \binom{n}{i}\binom{n-i}{i} x^i \cdot \sum_{j=0}^{n-2i} \binom{n-2i}{j} x^j \\
&= \sum_{i=0}^{\lfloor n/2 \rfloor} \binom{n}{i}\binom{n-i}{i} x^i (1+x)^{n-2i}.
\end{aligned}$$

Setting $x = -1$, we get

$$\sum_{k=0}^{n} (-1)^k \binom{n}{k}^2 = \begin{cases} 0, & \text{if } n \text{ is odd} \\ (-1)^{n/2}\binom{n}{n/2}, & \text{if } n \text{ is even,} \end{cases}$$

meaning the identity (2.13) of Example 2.4.2.

59. Hint: Let

$$a(n,k) = \frac{(-1)^k (n-k)}{(n+2)\binom{n}{k+1}} + \frac{n+1}{n+2}$$

for $k \in \{0, 1, \ldots, n-1\}$. Observe that $a(n, 0) = 1$ and show that

$$a(n,k) - a(n,k-1) = \frac{(-1)^k}{\binom{n}{k}}$$

for $k \in [n-1]$. Conclude that

$$\sum_{k=0}^{n-1} \frac{(-1)^k}{\binom{n}{k}} = a(n, n-1) = \frac{n+1+(-1)^{n-1}}{n+2},$$

which is equivalent to the proposed identity.

60. We present a proof which follows the *involution principle* of Garsia and Milne; see [23, Section 2.6]. Let $A(m,n)$ denote the set of all subsets of $[n]$ with cardinality less than or equal to m. Then, the proposed identity may be rewritten as

$$\sum_{S \in A(m,n)} (-1)^{|S|} = (-1)^m \binom{n-1}{m}. \tag{B.4}$$

To prove this equality, we attempt to match every summand $(-1)^{k-1}$ of the left-hand side with a term equal to $(-1)^k$. For this reason, we consider the map $\tau : A(m,n) \to A(m,n)$ defined by setting

$$\tau(S) = \begin{cases} S \setminus \{n\}, & \text{if } n \in S \\ S \cup \{n\}, & \text{if } n \notin S \text{ and } |S| < m \\ S, & \text{if } n \notin S \text{ and } |S| = m \end{cases}$$

for every $S \in A(m,n)$. We observe that $\tau(\tau(S)) = S$ for every $S \in A(m,n)$, so that τ is bijection whose inverse map is equal to itself (such a map is called an *involution*). We also observe that

$$(-1)^{|S|} + (-1)^{|\tau(S)|} = 0,$$

if $\tau(S) \neq S$. Thus, the summands of the left-hand side of (B.4) with $\tau(S) \neq S$ split into pairs having zero sum each and hence

$$\sum_{S \in A(m,n)} (-1)^{|S|} = \sum_{S \in A(m,n):\, \tau(S) = S} (-1)^{|S|}.$$

Since there are exactly $\binom{n-1}{m}$ sets $S \in A(m,n)$ with $\tau(S) = S$ and each of them has cardinality m, Equation (B.4) follows.

Chapter 3

1. Hint: Use Equation (3.3) to simplify the given quantity.
2. Answers: (a) 88; (b) 74.
3. Answers: (a) 800; (c) 75.
4. Answers: (a) 570; (d) 780.

5. Answers: (b) 216; (c) 592.
6. Answers: (a) 370; (b) 270.
7. Answers: (a) 274; (c) 64.
8. Answers: (b) 724; (d) 84.
9. Answers: (a) 1230; (b) 356.
10. Hint: For part (a), follow either of the two arguments given in Example 3.1.2. For parts (b) and (c), compute the number of colorings for which at least one row or column of the chessboard has all its squares colored black and at least one row or column has all its squares colored white, and apply part (a) and the principle of inclusion–exclusion. Your answer to part (b) should be 102.
11. More generally, consider a finite set Ω and subsets A, B, C, D. How many elements of Ω belong to A or B, but to neither C nor D? We follow the reasoning of the first proof of Theorem 3.1.1. We ask to compute the cardinality of $(A \cup B) \cap (\Omega \smallsetminus C) \cap (\Omega \smallsetminus D)$. This subset of Ω has characteristic function

$$\begin{aligned}\chi_{A\cup B}\,\chi_{\Omega\smallsetminus C}\,\chi_{\Omega\smallsetminus D} &= (\chi_A + \chi_B - \chi_{A\cap B})\,(1-\chi_C)\,(1-\chi_D)\\ &= \chi_A + \chi_B - \chi_{A\cap B} - \chi_A\,\chi_C - \chi_B\,\chi_C + \chi_{A\cap B}\,\chi_C -\\ &\quad \chi_A\,\chi_D - \chi_B\,\chi_D + \chi_{A\cap B}\,\chi_D + \chi_A\,\chi_C\,\chi_D +\\ &\quad \chi_B\,\chi_C\,\chi_D - \chi_{A\cap B}\,\chi_C\,\chi_D\\ &= \chi_A + \chi_B - \chi_{A\cap B} - \chi_{A\cap C} - \chi_{B\cap C} + \chi_{A\cap B\cap C} -\\ &\quad \chi_{A\cap D} - \chi_{B\cap D} + \chi_{A\cap B\cap D} + \chi_{A\cap C\cap D} +\\ &\quad \chi_{B\cap C\cap D} - \chi_{A\cap B\cap C\cap D}.\end{aligned}$$

Summing the values of this function (each equal to 0 or 1) over the elements of Ω, we find that the cardinality of $(A \cup B) \cap (\Omega \smallsetminus C) \cap (\Omega \smallsetminus D)$ is equal to

$$\begin{aligned}&\#A + \#B - \#(A\cap B) - \#(A\cap C) - \#(B\cap C) + \#(A\cap B\cap C) -\\ &\#(A\cap D) - \#(B\cap D) + \#(A\cap B\cap D) + \#(A\cap C\cap D) +\\ &\#(B\cap C\cap D) - \#(A\cap B\cap C\cap D).\end{aligned}$$

The answer to the original question, in which A, B, C, and D consist of the integers among $1, 2, 3, \ldots, 1000$ which are divisible by 4, 5, 3, and 7, respectively, is 229.

13. Hint for part (b): Observe that for every choice of $a_i \in \{0, 1, \ldots, r-1\}$ for $i \in [n-1]$, there exists a unique $a_n \in \{0, 1, \ldots, r-1\}$ such that $a_1+a_2+\cdots+a_n$ is an integer multiple of r.
14. Hint: Apply the inclusion–exclusion principle to the set Ω of n-subsets of $[m+n+k]$ and the sets $A_1, A_2, \ldots, A_k$, where A_i consists of those $S \in \Omega$ that

contain $m+n+i$, for $i \in [k]$. Then, set $m = -n-1$ and $k = p+n$ in (3.18) and deduce that

$$\sum_{i=0}^{n} (-1)^i \binom{p+n}{i} \binom{p+n-i-1}{n-i} = (-1)^n.$$

Finally, verify that this identity is equivalent to that of Exercise 51 (a) of Chapter 2.

15. After changing the index of summation from k to $n-k$, the proposed identity takes the equivalent form

$$\sum_{k=0}^{n} (-1)^k \binom{n}{k} \binom{m+n-k}{n-k} 2^{n-k} = \sum_{k=0}^{n} \binom{m}{k} \binom{n}{k} 2^k. \tag{B.5}$$

We consider disjoint sets X and Y with cardinalities m and n, respectively, and denote by $f(m,n)$ the number of ways to choose an n-subset S of $X \cup Y$ and color each element of S black or white, so that there is no element of $S \cap Y$ with black color. We leave it to the reader to prove, by applying the principle of inclusion–exclusion, that $f(m,n)$ is equal to the left-hand side of (B.5). The fact that $f(m,n)$ is also equal to the right-hand side of (B.5) was shown in the solution of Exercise 53 of Chapter 2.

16. Hint: Count the k-subsets of $[2n]$ which have at least one element in common with each of the sets $\{1,2\}, \{3,4\}, \ldots, \{2n-1, 2n\}$ in two different ways.

17. Hint: Count the paths in $\mathbb{Z}^2$ with initial vertex $(0,0)$, terminal vertex (n,n), and steps $(1,0)$ or $(0,1)$ which do not pass through any of the points (i,i) for $i \in [n-1]$ in two different ways.

19. Hint: For part (a), show that $\gcd(m,n) = d$ if and only if m/d and n/d are relatively prime. Deduce part (b) by applying the addition principle.

20. Answer: $m^2 \prod_{i=1}^{n} \left(1 - \frac{1}{p_i^2}\right)$, where $p_1, p_2, \ldots, p_n$ are the distinct prime divisors of m.

22. Hint: For part (b), count the pairs (σ, i), where σ is a permutation of $[n]$ and i is a fixed point of σ, in two different ways. For part (c), count the triples (σ, i, j), where σ is a permutation of $[n]$ and i, j are distinct fixed points of σ, in two different ways.

24. Let $\mathcal{D}_n$ be the set of permutations of $[n]$ without fixed points, and for $i \in [n-1]$, let $\mathcal{D}_{n,i}$ be the set of those $\sigma = (\sigma_1, \sigma_2, \ldots, \sigma_n) \in \mathcal{D}_n$ such that $\sigma_n = i$. The sets $\mathcal{D}_{n,i}$ for $i \in [n-1]$ are mutually disjoint, and their union equals $\mathcal{D}_n$. Therefore, it suffices to show that $\#\mathcal{D}_{n,i} = D_{n-1} + D_{n-2}$ for every $i \in [n-1]$. Clearly, there are D_{n-2} permutations $\sigma = (\sigma_1, \sigma_2, \ldots, \sigma_n) \in \mathcal{D}_{n,i}$ such that $\sigma_i = n$. Therefore, it suffices to show that there are exactly D_{n-1} permutations $\sigma = (\sigma_1, \sigma_2, \ldots, \sigma_n) \in \mathcal{D}_{n,i}$ such that $\sigma_i \neq n$. Let $C_{n,i}$ be the set of these permutations. Since $\#C_{n,i} = \#C_{n,n-1}$ for every $i \in [n-1]$ (by

exchanging the roles of i and $n-1$), we may assume that $i = n-1$. Given $\sigma = (\sigma_1, \sigma_2, \ldots, \sigma_n) \in C_{n,n-1}$, in which case $\sigma_n = n-1$ and $\sigma_{n-1} \neq n$, we let $\varphi(\sigma)$ be the permutation of $[n-1]$ obtained from $(\sigma_1, \sigma_2, \ldots, \sigma_{n-1})$ by replacing the term equal to n by $n-1$. Then, $\varphi : C_{n,n-1} \to \mathcal{D}_{n-1}$ is a bijection (explain why), and this implies that $\#C_{n,n-1} = D_{n-1}$.

25. Part (a) appears as [6, Theorem 6.5.1]. To prove it, we apply inclusion–exclusion to the set Ω of all permutations of $[n]$ and its subsets $A_1, A_2, \ldots, A_{n-1}$, where A_i is the set of permutations $\sigma = (\sigma_1, \sigma_2, \ldots, \sigma_n) \in \Omega$ such that $\sigma_{i+1} = \sigma_i + 1$, for $i \in [n-1]$. We observe that $\#A_i = (n-1)!$ for every $i \in [n-1]$. Indeed, to choose $\sigma = (\sigma_1, \sigma_2, \ldots, \sigma_n) \in A_i$, there are $n-1$ ways to choose σ_i and $\sigma_{i+1} = \sigma_i + 1$, and for every such choice, there are $(n-2)!$ ways to permute the other elements of $[n]$ to form σ. A similar argument shows that, more generally, $\#(A_{i_1} \cap \cdots \cap A_{i_k}) = (n-k)!$ for all $1 \le i_1 < \cdots < i_k < n$. Given that, we proceed as in the proof of Proposition 3.2.3 to reach the formula of part (a). From that and Proposition 3.2.3 (a), we deduce that

$$\begin{aligned} f(n) &= \sum_{k=0}^{n} (-1)^k \left(\binom{n}{k} - \binom{n-1}{k-1} \right) (n-k)! \\ &= \sum_{k=0}^{n} (-1)^k \binom{n}{k} (n-k)! - \sum_{k=1}^{n} (-1)^k \binom{n-1}{k-1} (n-k)! \\ &= \sum_{k=0}^{n} (-1)^k \binom{n}{k} (n-k)! + \sum_{k=0}^{n-1} (-1)^k \binom{n-1}{k} (n-1-k)! \\ &= D_n + D_{n-1}. \end{aligned}$$

This proves part (b). For part (c), we may assume that $n \ge 5$. Part (b) and Proposition 3.2.3 (c) imply that $f(n) = D_{n+1}/n$. Besides, as a consequence of Proposition 3.2.3 (a), we have $D_{n+1} \le (n+1)!(1/2 - 1/3 + 1/4) = 5(n+1)!/12 \le n(n!/2)$ for $n \ge 5$.

26. Answer: $n! \displaystyle\sum_{k=0}^{n} (-1)^k \frac{(n-k)!}{k!}$.

27. Hint for part (a): Let $\mathcal{E}_n$ be the set of permutations of $[n]$ with the given property, and for $\sigma \in \mathcal{E}_n$, let $f(\sigma)$ be the permutation of $[n-1]$, which is obtained from $\sigma = (\sigma_1, \sigma_2, \ldots, \sigma_n)$ by deleting σ_n and replacing σ_i with $\sigma_i - 1$ for every index i with $\sigma_i > \sigma_n$. Which $\sigma \in \mathcal{E}_n$ satisfy $f(\sigma) \in \mathcal{E}_{n-1}$? For a given $\tau \in \mathcal{E}_{n-1}$, how many $\sigma \in \mathcal{E}_n$ are there such that $f(\sigma) = \tau$? The results of this exercise are due to J. Désarménien [Sém. Lothar. Combin. **8** (1984), 11–16].

28. Hint for parts (b) and (c): Adapt the proofs of Proposition 3.2.3 (a) and Corollary 3.2.3.

29. Hint: Let Ω be the set of even permutations of $[n]$, and for $i \in [n]$, let A_i be the set of permutations $\sigma = (\sigma_1, \sigma_2, \ldots, \sigma_n) \in \Omega$ such that $\sigma_i = i$. Then, $E_n = \#(\Omega - \cup_{i=1}^n A_i)$. Taking into account the fact that the number of even permutations of $[m]$ is equal to $m!/2$, if $m \geq 2$ and to 1 otherwise, follow the proof of Proposition 3.2.3 (a) and conclude that

$$E_n = \sum_{k=0}^{n-2} (-1)^k \binom{n}{k} \frac{(n-k)!}{2} + (-1)^{n-1}(n-1)$$

for $n \geq 2$. Deduce that $2E_n = D_n + (-1)^{n-1}(n-1)$ for every $n \in \mathbb{Z}_{>0}$.

30. Hint: Follow the reasoning in the second proof of Theorem 3.1.1, setting

$$c(x) = \sum_{I \subseteq [n],\, \#I \geq p} (-1)^{\#I} \chi_I(x)$$

for $x \in \Omega$. Use the identity of Proposition 2.4.3 to show that

$$\sum_{k=p}^{m} (-1)^{k-p} \binom{m}{k} = \sum_{k=0}^{p-1} (-1)^{p-1-k} \binom{m}{k} = \binom{m-1}{p-1} \geq 0$$

for all $m \in \mathbb{Z}_{>0}$ and $p \in [m]$ and conclude that $c(x) \geq 0$ for every $x \in \Omega$.

31. Hint: Apply the inequalities (3.20) and (3.21) for $p = 2$ and $p = 3$ to the sets which were defined in the solution of Exercise 25 and show that

$$n! - (n-1)(n-1)! + \tbinom{n-1}{2}(n-2)! - \tbinom{n-1}{3}(n-3)! \leq f(n)$$

$$\leq n! - (n-1)(n-1)! + \tbinom{n-1}{2}(n-2)!$$

for $n \geq 3$.

32. Hint: Follow the reasoning of Example 3.2.1. For part (a), apply the result of Exercise 42 (b) of Chapter 2.

34. Hint for part (a): Follow the reasoning of the third proof of Theorem 3.1.1, given in Section 3.3, and sum the formula for $f(S)$ given there over all k-element subsets of $I = [n]$. Your answer to part (c) should be 359.

36. Hint for part (c): Express $f(Y, U)$ as a nonnegative linear combination of the numbers $f(X)$ for $X \subseteq I$. What does this formula look like in the setting of part (a)?

Chapter 4

1. Hint: Apply Proposition 4.1.1 in a suitable way.
2. We solve part (b). We may assume that the vertices of T are the points $(1, 1)$, $(1, -1)$, $(-1, 1)$, and $(-1, -1)$. We partition T into five blocks as follows. One consists only of the center $(0, 0)$ of T, and the other four blocks are the

intersections of $T \setminus \{(0,0)\}$ with the four regions $\{(x,y) \in \mathbb{R}^2 : x \geq 0, y > 0\}$, $\{(x,y) \in \mathbb{R}^2 : x > 0, y \leq 0\}$, $\{(x,y) \in \mathbb{R}^2 : x \leq 0, y < 0\}$ and $\{(x,y) \in \mathbb{R}^2 : x < 0, y \geq 0\}$ of the plane, respectively. According to the pigeonhole principle, two of the six given points belong to the same block of this partition. We leave it to the reader to verify that these two points are at a distance smaller than $\sqrt{2}$ apart from each other.

3. Hint: Follow the reasoning of Example 4.1.1.
6. Answers: (c) 20; (d) 10.
7. Answers: (a) 16; (c) 32.
8. Hint for part (a): Consider the standard black and white coloring of the chessboard.
9. Hint: Apply the result of Example 4.1.2 in a suitable way.
10. Hint for part (b): First prove that the midpoint of the line segment in $\mathbb{R}^n$ joining two points $(a_1, a_2, \ldots, a_n) \in \mathbb{Z}^n$ and $(b_1, b_2, \ldots, b_n) \in \mathbb{Z}^n$ has integer coordinates if and only if a_i and b_i are both even or both odd, for every $i \in [n]$.
12. Hint for part (b): Color the squares of the chessboard with eight colors, so that no two squares of the same color lie in the same row or same column (such colorings are called 8×8 *latin squares*); see Figure B.2 for an example. Then, apply Proposition 4.1.2 with $m = 8$ and $n = 4$.
13. Hint: Prove that there exist four elements a, b, c, d of S such that either all differences $a - b$, $b - c$, and $c - d$ are divisible by $2n - 1$, or both $a + b$ and $c + d$ are divisible by $2n - 1$.
14. Hint: Color the squares of the first row of the chessboard black, those of the second row white, those of the third row black, and so on, and observe that every rook moves from a black square to a white square and vice versa. The same observation holds if rows are replaced by columns. The answer is zero if both m, n are even, m if m is even and n is odd, n if m is odd and n is even and $\max\{m, n\}$ if both m, n are odd. The special case $m = n$ of this problem was proposed by New Zealand (and was shortlisted) for the 2004 International Mathematical Olympiad.
15. Answer: $a + b - 2$.

Figure B.2 An 8×8 latin square

1	8	7	6	5	4	3	2
2	1	8	7	6	5	4	3
3	2	1	8	7	6	5	4
4	3	2	1	8	7	6	5
5	4	3	2	1	8	7	6
6	5	4	3	2	1	8	7
7	6	5	4	3	2	1	8
8	7	6	5	4	3	2	1

16. Let S_i be the set of $j \in [n]$ such that the (i, j)-element of A is equal to 1, for $i \in [n]$. Let also $\binom{S}{2}$ denote the set of 2-subsets of a set S. According to the assumptions of the problem, the sets $\binom{S_i}{2}$ for $i \in [n]$ are mutually disjoint subsets of $\binom{[n]}{2}$. Setting $a_i = \#S_i$ for $i \in [n]$, we conclude that

$$\sum_{i=1}^{n} \binom{a_i}{2} \leq \binom{n}{2},$$

that is,

$$a_1^2 + a_2^2 + \cdots + a_n^2 \leq a_1 + a_2 + \cdots + a_n + n(n-1).$$

The latter inequality, combined with $(a_1+a_2+\cdots+a_n)^2 \leq n(a_1^2+a_2^2+\cdots+a_n^2)$, implies the proposed upper bound for $a_1 + a_2 + \cdots + a_n$. Equality holds, for instance, for

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}.$$

17. Clearly, the collection of subsets of $[n]$ of cardinality at most k has the required property. This implies that

$$\mu_k(n) \geq \sum_{i=0}^{k} \binom{n}{i}.$$

To prove the reverse inequality, it suffices to show (explain why) that if $\mathcal{A}$ is a collection of subsets of $[n]$, each of cardinality at least k, and if the intersection of any two of these subsets has cardinality at most $k-1$, then $\#\mathcal{A} \leq \binom{n}{k}$. Let us denote by $\binom{S}{k}$ the set of k-subsets of a set S. Our assumption for $\mathcal{A}$ implies that the sets $\binom{S}{k}$, where S runs through all elements of $\mathcal{A}$, are mutually disjoint subsets of $\binom{[n]}{k}$. This fact implies that

$$\sum_{m=k}^{n} p_m \binom{m}{k} \leq \binom{n}{k},$$

where $p_m = \#\{S \in \mathcal{A} : |S| = m\}$. As a consequence,

$$\#\mathcal{A} = \sum_{m=k}^{n} p_m \leq \sum_{m=k}^{n} p_m \binom{m}{k} \leq \binom{n}{k}.$$

18. We answer part (b). Let $\sigma = (\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_n) \in A(n,k)$, where $0 \le k < n/2$. Then, $\varepsilon_1 + \cdots + \varepsilon_n = -n + 2k < 0$, and hence, $\varepsilon_1 + \cdots + \varepsilon_i = -1$ for at least one index $i \in [n]$. We consider the smallest such index and set $f(\sigma) = (-\varepsilon_1, \ldots, -\varepsilon_i, \varepsilon_{i+1}, \ldots, \varepsilon_n)$. For example, if $n = 6$ and $\sigma = (1,-1,-1,-1,1,-1)$, then $i = 3$ and $f(\sigma) = (-1,1,1,-1,1,-1)$. We have $f(\sigma) \in A(n,k+1)$, and thus, we have defined a map $f : A(n,k) \to A(n,k+1)$. This map is injective because if $f(\sigma) = (\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_n) \in A(n,k+1)$, then $\sigma = (-\varepsilon_1, \ldots, -\varepsilon_i, \varepsilon_{i+1}, \ldots, \varepsilon_n)$, where $i \in [n]$ is the smallest index such that $\varepsilon_1 + \cdots + \varepsilon_i = 1$. Part (c) follows from parts (a) and (b) and Proposition 4.1.1 (a). Part (a) and our solution to part (b) yield that $\binom{n}{k+1} - \binom{n}{k}$ is equal to the number of elements of $A(n,k+1)$ which do not belong to the image of f and hence to the number of elements of $A(n,k+1)$ which have nonpositive partial sums (explain why). This answers part (d). This interpretation is equivalent to the one given in the solution of Exercise 46 of Chapter 2.
19. Let $(\sigma,\tau) \in A_n(k-1) \times A_n(k+1)$, where $\sigma = (\delta_1, \delta_2, \ldots, \delta_n)$ and $\tau = (\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_n)$. Clearly, there exists an index $i \in [n-1]$ such that $\varepsilon_1 + \cdots + \varepsilon_i = \delta_1 + \cdots + \delta_i + 1$. We consider the smallest such index and set $f(\sigma,\tau) = (\sigma',\tau')$, where

$$\sigma' = (\varepsilon_1, \ldots, \varepsilon_i, \delta_{i+1}, \ldots, \delta_n),$$
$$\tau' = (\delta_1, \ldots, \delta_i, \varepsilon_{i+1}, \ldots, \varepsilon_n).$$

Then, $\sigma', \tau' \in A_n(k)$, and hence, we have defined a map $f : A_n(k-1) \times A_n(k+1) \to A_n(k) \times A_n(k)$. We leave it to the reader to verify that f is injective. This answers part (a). Part (b) follows from that and Proposition 4.1.1 (a), since $\#A_n(k) = \binom{n}{k}$ for every $k \in \{0,1,\ldots,n\}$.
20. Let $\sigma = (\sigma_1, \sigma_2, \ldots, \sigma_n) \in X_n$ and $i \in [n]$ be the smallest index such that $|\sigma_1 - \sigma_i| = 1$ (since $\sigma \in X_n$, we have $i \ge 3$). We then set

$$f(\sigma) = (\sigma_{i-1}, \ldots, \sigma_2, \sigma_1, \sigma_i, \sigma_{i+1}, \ldots, \sigma_n).$$

For example, if $n = 9$ and $\sigma = (7,1,5,9,3,8,4,6,2) \in X_9$, then $i = 6$ and $f(\sigma) = (3,9,5,1,7,8,4,6,2)$. Clearly, $f(\sigma) \in Y_n$, and i is the unique index $j \ge 2$ such that the $j-1$ and j coordinates of $f(\sigma)$ differ in absolute value by one. The map $f : X_n \to Y_n$ we have defined is injective because if $(\tau_1, \tau_2, \ldots, \tau_n) = f(\sigma)$ for some $\sigma \in X_n$, then there exists a unique index $j \ge 2$ such that $|\tau_{j-1} - \tau_j| = 1$ and $\sigma = (\tau_{j-1}, \ldots, \tau_2, \tau_1, \tau_j, \ldots, \tau_n)$. This answers part (a). In view of Proposition 4.1.1, we conclude that $\#X_n \le \#Y_n$. We also observe that the map $f : X_n \to Y_n$ is not surjective because, for example, the permutations $(\tau_1, \tau_2, \ldots, \tau_n)$ of Y_n such that $|\tau_1 - \tau_2| = 1$ are not in the image of f. Therefore, $\#X_n < \#Y_n$ and, equivalently, $\#Y_n > n!/2$. Part (b) is a variation of Problem 6 of the 1989 International Mathematical Olympiad.

22. Let $f(n) \in \{0, 1, \ldots, m-1\}$ be the remainder of the division of $a(n)$ with m, and let us consider the m^2+1 pairs $(f(k), f(k+1))$ for $0 \le k \le m^2$. Since there are exactly m^2 pairs $(a, b) \in \{0, 1, \ldots, m-1\} \times \{0, 1, \ldots, m-1\}$, according to the pigeonhole principle, two of the pairs we considered must be equal. That is, $(f(i), f(i+1)) = (f(j), f(j+1))$ for some $0 \le i < j \le m^2$ or, equivalently,

$$a(i) \equiv a(j) \pmod{m},$$
$$a(i+1) \equiv a(j+1) \pmod{m}.$$

Moreover, we may assume that $i = 0$ since otherwise, the two congruences above and the definition $a(n-1) = a(n+1) - pa(n)$ of the sequence $a(n)_{n\in\mathbb{N}}$ imply successively (explain how) that

$$a(i-1) \equiv a(j-1) \pmod{m},$$
$$a(i-2) \equiv a(j-2) \pmod{m},$$

and so on. In conclusion, setting $r = j - i$, so that $1 \le r \le m^2$, we have $a(r) \equiv a(0) \pmod{m}$ and $a(r+1) \equiv a(1) \pmod{m}$. It follows by induction on n that $a(r+n) \equiv a(n) \pmod{m}$ for every $n \in \mathbb{N}$, which proves part (a). To deduce part (b), it suffices to observe that $a(r-1) = a(r+1) - pa(r) \equiv a(1) - pa(0) = p - p \cdot 1 = 0 \pmod{m}$. This means that $a(r-1)$ is divisible by m.

23. Hint: Use the pigeonhole principle to show that there exist integers $p \ne q$ such that $|f(p\alpha) - f(q\alpha)| < \varepsilon$.

24. Suppose that there exists a nonzero element $x \in F$ such that $-x = x$. Then, the sets of the form $\{a, a+x\}$, where $a \in F$, are the blocks of a partition of F. This happens because $b = a + x \Rightarrow b + x = a$ in F and shows that the cardinality of F is even, contrary to our hypothesis. This proves part (a). Suppose now that $q \equiv 1 \pmod{4}$. We observe that the sets of the form $\{x, -x, x^{-1}, -x^{-1}\}$, where x runs through the nonzero elements of F, are the blocks of a partition of the set $F \setminus \{0\}$ of nonzero elements of F (explain why). Some of these sets may have fewer than four elements. Since $-x \ne x$ for every $x \in F \setminus \{0\}$, as shown in part (a), this can only happen if $x^{-1} = x$, or $x^{-1} = -x$. In the former case, the set $\{1, -1\}$ is the only possibility, since the equation $x^{-1} = x$ in a field has only the solutions $x = 1$ and $x = -1$. Since the cardinality of $F \setminus \{0\}$ is an integer multiple of 4, the set $\{1, -1\}$ cannot be the only block of the partition with fewer than four elements. As a result, there exists a nonzero element $x \in F$ such that $x^{-1} = -x$, which gives $x^2 = -1$. This proves part (b). Part (c) follows by applying part (b) to the field $\mathbb{Z}_p$ of integers (mod p).

25. Let p be a prime number such that $p \equiv 1 \pmod{4}$. We consider $r \in \mathbb{Z}$ and the ordered pairs $(a, b) \in \mathbb{Z} \times \mathbb{Z}$ such that $0 \le a, b \le \lfloor \sqrt{p} \rfloor$. For every such pair, we consider the integer $a + br$. Since $\lfloor x \rfloor + 1 > x$ for every $x \in \mathbb{R}$, the number of these pairs equals $(\lfloor \sqrt{p} \rfloor + 1)^2 > p$. It follows from the result of

Example 4.1.2 that there are two distinct pairs, say (a, b) and (c, d), for which p divides the difference $(a+br)-(c+dr) = (a-c)+(b-d)r$. Setting $x = a-c$ and $y = b-d$, we have shown that there exist integers x, y, not both equal to zero, having absolute values less than $\sqrt{p}$, such that p divides $x + ry$. Then, p divides $x^2 - r^2y^2$ as well. We now choose r, according to Exercise 24 (c), so that r^2+1 is divisible by p. Then, p divides $x^2 - r^2y^2 + (r^2+1)y^2 = x^2 + y^2$. Since $0 < x^2 + y^2 < 2p$, we must have $x^2 + y^2 = p$.

27. Hint: How many elements do each equivalence class have?
28. Hint: Every integer in $\{2, 3, \ldots, n\}$ is divisible by at least one of the prime numbers in this set.
30. Hint: Combine Proposition 4.3.1 (b) with Proposition 4.3.2.
32. Hint: Adapt the argument in the proof of Proposition 4.3.3 (a) to prove part (a). For part (b), proceed by induction on n to prove that $k(n+1) = k(n)$ or $k(n+1) = k(n) + 1$ works for every $n \geq 2$ as follows. Suppose that the result holds for $n \geq 2$. Apply the recurrence of Proposition 4.3.1 (a) and show first that $S(n+1, k-1) < S(n+1, k)$ for $1 \leq k \leq k(n)$. Then, show that

$$S(n+1, k+1) - S(n+1, k) = \sum_{m=k-1}^{n} \binom{n}{m} (S(m, k) - S(m, k-1)) < 0$$

for $k \geq k(n) + 1$ and conclude that the result holds for $n + 1$ as well.

34. Applying Proposition 4.3.2 with k^m in place of x^n and the identity

$$\begin{aligned} k(k-1)\cdots(k-i+1)\binom{n}{k} &= \frac{k!}{(k-i)!} \cdot \frac{n!}{k!(n-k)!} \\ &= \frac{n!}{(n-i)!} \cdot \frac{(n-i)!}{(k-i)!\,(n-k)!} \\ &= n(n-1)\cdots(n-i+1)\binom{n-i}{k-i}, \end{aligned}$$

we find that

$$\begin{aligned} \sum_{k=1}^{n} k^m \binom{n}{k} &= \sum_{k=1}^{n} \left(\sum_{i=1}^{m} S(m, i)\, k(k-1)\cdots(k-i+1) \right) \binom{n}{k} \\ &= \sum_{i=1}^{m} S(m, i) \sum_{k=i}^{n} k(k-1)\cdots(k-i+1) \binom{n}{k} \\ &= \sum_{i=1}^{m} S(m, i) \sum_{k=i}^{n} n(n-1)\cdots(n-i+1) \binom{n-i}{k-i} \end{aligned}$$

$$= \sum_{i=1}^{m} S(m,i)\, n(n-1)\cdots(n-i+1) \sum_{k=0}^{n-i} \binom{n-i}{k}$$

$$= \sum_{i=1}^{m} S(m,i)\, n(n-1)\cdots(n-i+1)\, 2^{n-i}.$$

35. Hint: Use the result of Exercise 31 of Chapter 2.
37. Suppose there exists a nonempty finite poset $(P, \preceq)$ which does not have a maximal element. Then, for every $x \in P$, there exists $y \in P$ such that $x \prec y$. Thus, starting from an arbitrary element $x_1 \in P$, one can define recursively an infinite sequence $x_1 \prec x_2 \prec x_3 \prec \cdots$ of elements of P. Since P is finite (see Exercise 2 of Chapter 2), the terms of this sequence cannot be pairwise distinct. Hence, there exist indices $i < j$ such that $x_i = x_j$. By transitivity, given that $x_i \prec x_{i+1} \prec \cdots \prec x_j = x_i$, we get $x_i \prec x_{i+1}$ and $x_{i+1} \preceq x_i$. These relations contradict antisymmetry. This contradiction proves that existence of maximal elements. We argue similarly for the existence of minimal elements, and the proof of part (a) follows. To deduce part (b), we consider a nonempty finite poset $(P, \preceq)$ and choose any maximal element $b \in P$. We consider the set $Q = \{x \in P : x \preceq b\}$, partially ordered by the partial order induced from P, meaning that we set $x \preceq y$ in Q if and only if $x \preceq y$ in P, for $x, y \in Q$. The poset Q is nonempty, since $b \in Q$, and certainly finite, since $Q \subseteq P$. By part (a), Q has a minimal element a, and that is also a minimal element of P (explain why). Since $a \in Q$, we have $a \preceq b$ in P by definition of Q. Finally, suppose that $(P, \preceq)$ is totally ordered, of cardinality $n \in \mathbb{Z}_{>0}$. Let us prove part (c) by induction on n. According to part (a), P has at least one minimal element, say σ_1. Any other element $x \in P$ is comparable to σ_1 but does not satisfy $x \prec \sigma_1$. Therefore, we must have $\sigma_1 \prec x$, and hence, σ_1 is the minimum element of P. By induction on n, we may assume that $(\sigma_2, \ldots, \sigma_n)$ is a permutation of the set $P \setminus \{\sigma_1\}$ such that $\sigma_2 \prec \cdots \prec \sigma_n$. Since $\sigma_1 \prec \sigma_2$, by transitivity we conclude that $(\sigma_1, \sigma_2, \ldots, \sigma_n)$ is a permutation of P such that $\sigma_1 \prec \sigma_2 \prec \cdots \prec \sigma_n$.
38. Let S be the set of girls present at the party and let $a_1, a_2, \ldots, a_m$ be the boys. For $i \in [m]$, let A_i be the set of girls who have danced with boy a_i. By the assumptions of the problem, $A_1, A_2, \ldots, A_m$ are proper subsets of S whose union is equal to S. We claim that two of these sets are incomparable with respect to inclusion. This holds because otherwise, according to part (c) of Exercise 37, there would exist a permutation $(\sigma_1, \sigma_2, \ldots, \sigma_m)$ of $[m]$ such that $A_{\sigma_1} \subseteq A_{\sigma_2} \subseteq \cdots \subseteq A_{\sigma_m}$. This would imply that the union of the sets A_i is equal to the proper subset A_{σ_m} of S, contrary to our hypotheses. In conclusion, we have shown that there exist indices $i < j$ such that $A_i \not\subseteq A_j$ and $A_j \not\subseteq A_i$. Choosing elements $b_i \in A_i \setminus A_j$ and $b_j \in A_j \setminus A_i$, we find the couples $a_i b_i$ and $a_j b_j$ having the required properties. This problem appeared as Problem A4 on the 1965 William Lowell Putnam Mathematical Competition.

41. Hint: For part (c), prove that $\mathrm{height}(P_n) = 2n-1$ by finding a $(2n-1)$-element chain in P_n and a partition of P_n into $2n-1$ antichains. For part (d), prove that $\mathrm{width}(P_n) = n$ by finding an n-element antichain in P_n and a partition of P_n into n chains.
42. Hint for part (c): Find an $\binom{n+1}{2}$-element antichain in Q_n and, by applying the pigeonhole principle, prove that every subset of Q_n with more than n^2 elements contains two comparable elements of the form (a, b, c) and (a', b, c).
44. Answers: $\mathrm{height}(Q_n) = 1 + \lfloor \log_2(n) \rfloor$ and $\mathrm{width}(Q_n) = \lfloor (n+1)/2 \rfloor$.
45. For more details on the following proof, see [2, Section 3.2]. Let $m = \mathrm{width}(P)$. By Corollary 4.4.1 we have $m \leq c(P)$. Thus, it suffices to show that $c(P) \leq m$, meaning that P can be partitioned into at most m chains. This is clear if P has only one element. We proceed by induction on the cardinality of P and assume that $\#P \geq 2$ and that the result holds for posets with fewer than $\#P$ elements. We distinguish two cases:

    (a) Every m-element antichain in P contains either all minimal elements of P (and hence no other element) or all maximal elements of P (and hence no other element). By Exercise 37 (b), we may consider a minimal element a and a maximal element b of P such that $a \preceq b$. We also consider the set $Q = P \setminus \{a, b\}$, partially ordered by the order induced from P. We then observe (explain how) that $\mathrm{width}(Q) \leq m-1$ and apply the induction hypothesis on Q to conclude that P can be partitioned into at most m chains, one of which is the chain $\{a, b\}$.

    (b) There exists an m-element antichain $A \subseteq P$ which contains neither all minimal elements, nor all maximal elements of P. We let A^+ (respectively, A^-) be the set which consists of those elements $x \in P$ such that $x \succeq a$ (respectively, $x \preceq a$) for some $a \in A$, endowed with the induced partial order from P. We observe (explain how) A^+ that A^- have fewer elements than P and that $A^+ \cap A^- = A$ and $A^+ \cup A^- = P$. By the induction hypothesis, each of A^+ and A^- can be partitioned into m chains, and this implies that the same holds for P.

46. Let τ be a cyclic permutation of $[n]$, and let $a_1, a_2, \ldots, a_n$ be the elements of $[n]$, listed so that a_i follows a_{i-1} in the clockwise order in τ and $\{a_1, a_2, \ldots, a_r\} \in \mathcal{A}$ for some $r \in \mathbb{N}$, necessarily with $r \leq k$. Since $\mathcal{A}$ is an antichain, for every $j \in [r]$, $\mathcal{A}$ has at most one element of the form $\{a_j, a_{j+1}, \ldots\}$ and at most one element of the form $\{a_j, a_{j-1}, \ldots\}$, where we write $a_i = a_j$ whenever $i, j \in \mathbb{Z}$ and $i \equiv j \pmod{n}$. Since $\mathcal{A}$ is intersecting, all its elements contained in τ are of this form. Finally, since $\mathcal{A}$ is intersecting and $k \leq n/2$, it cannot contain both $\{a_j, a_{j-1}, \ldots\}$ and $\{a_{j+1}, a_{j+2}, \ldots\}$ for some $j \in [r]$. As a result, $\mathcal{A}$ has at most $r \leq k$ elements contained in τ.

Chapter 5

2. Hint for part (c): Observe that every isomorphism class contains at most $n!$ graphs and deduce that $f(n) \geq 2^{\binom{n}{2}}/n!$ for every $n \in \mathbb{Z}_{>0}$.

5. Hint for part (b): Choose two vertices of such a graph G of degrees 1 and $n-1$ and apply induction on the induced subgraph of G on the set of remaining $n-2$ vertices.
7. We leave to the reader to verify that if there exists an Eulerian walk (respectively, closed Eulerian walk) in G, then G is connected and has vertices as in part (a) (respectively, part (b)). Let us prove the converse. Since part (b) follows by applying part (a) to the graph obtained from G by deleting an arbitrary edge, it suffices to prove part (a). For the same reason, we may assume in part (a) that G has exactly two odd vertices, say u and v. We will prove that there exists an Eulerian tour in G with initial vertex u and terminal vertex v. Since G is connected, there exists a walk in G with initial vertex u and terminal vertex v in which every edge of G appears at most once (any path with initial vertex u and terminal vertex v has this property). Among all these walks (which are finitely many), we may choose one, say w, with the largest possible length. We claim that every edge in G appears in w, so that w is an Eulerian tour. Indeed, if not, then because G is connected, there exists an edge e_1 of G which does not appear in w and is incident to a vertex, say v_0, of w. If e_1 is a loop, then we may suitably include the pair (e_1, v_0) in w and get a longer walk in G with initial vertex u and terminal vertex v, contradicting the choice of w. Suppose that e_1 is not a loop, and let v_1 be its endpoint other than v_0. Since u, v have odd degrees and all other vertices of G have even degree, there exists an edge $e_2 \neq e_1$ incident to v_1 which is not a loop and does not appear in w. We call v_2 the endpoint of e_2 other than v_1 and continue similarly until we reach v_0 with a walk which consists of new edges of G. We suitably include this closed walk with initial and terminal vertices v_0 in w and find a longer than w walk in G with initial vertex u and terminal vertex v in which every edge of G appears at most once, in contrast to the choice of w. This contradiction proves our claim. The details of this argument are left to the reader. Parts (a) and (b) are classical theorems of Euler. Part (c) follows by applying part (b) to the graph obtained from G by adding, for every edge e of G, a copy of e with the same endpoints.
8. Hint for part (a): Consider a path P of largest possible length in G and prove that $G \setminus v$ is connected if v is an endpoint of P.
9. (a) Suppose that G is not doubly connected. Then, there exists $u \in V$ such that $G \setminus u$ is disconnected. We choose vertices $v, v' \in V$ which lie in different connected components of $G \setminus u$ and a path $(v_0, e_1, v_1, \ldots, v_{\ell-1}, e_\ell, v_\ell)$ in G with initial vertex v and terminal vertex v'. Finally, we observe that there exists a unique index $1 \leq i < \ell$ such that $v_i = u$ and that v_{i-1} and v_{i+1} have distance 2 in G, but no cycle of G passes through them. This contradicts our hypothesis.
   (b) We follow similar reasoning and replace vertex $u \in V$ with an *inclusion–minimal* set $U \subseteq V$ with fewer than k elements, such that the induced subgraph of G on the vertex set $V \setminus U$ is disconnected. We then choose the path of part (a) so that it passes through a unique vertex which belongs to U; the details are left to the reader. This proposition is known as Liu's

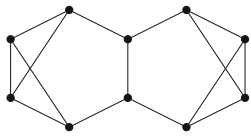


Figure B.3 A two-connected but not three-connected graph

criterion. The proof we described is due to A. Björner and K. Vorwerk [European J. Combin. **31** (2010), 2149–2160].

10. Hint for part (c): Consider the graph shown in Figure B.3.
11. For part (b), we proceed by induction on n. The result is trivial for $n = 1$, so we assume that $n \geq 2$. We consider an arbitrary subset N of the set $\{0, 1\}^n$ of vertices of $\square_n$ of cardinality less than n and denote by $\square_n \setminus N$ the induced subgraph of $\square_n$ on the vertex set $\{0, 1\}^n \setminus N$. We will prove that any two vertices $u = (u_1, \ldots, u_n)$ and $v = (v_1, \ldots, v_n)$ of $\square_n$, which do not belong to N, can be connected by a walk in $\square_n \setminus N$. We distinguish two cases: (i) $u_i \neq v_i$ for $i \in [n]$. Then, one can construct n paths in $\square_n$ with initial vertex u and terminal vertex v which pairwise have no common vertex other than u and v. For example, if $n = 3$ and $u = (0, 0, 0)$, $v = (1, 1, 1)$, such paths are those with successive vertices

$$\begin{array}{l} (0,0,0)\ (1,0,0)\ (1,1,0)\ (1,1,1) \\ (0,0,0)\ (0,1,0)\ (0,1,1)\ (1,1,1) \\ (0,0,0)\ (0,0,1)\ (1,0,1)\ (1,1,1). \end{array}$$

Since N has fewer than n elements, at least one of these paths is a path in $\square_n \setminus N$. (ii) $u_i = v_i$ for some index $i \in [n]$, for instance $u_n = v_n = 0$. We denote by H_n (respectively, $\tilde{H}_n$) the induced subgraph of $\square_n$ on the set of vertices $(\varepsilon_1, \ldots, \varepsilon_n) \in \{0, 1\}^n$ with $\varepsilon_n = 0$ (respectively, $\varepsilon_n = 1$). We observe that H_n (as well as $\tilde{H}_n$) is isomorphic to $\square_{n-1}$. Therefore, the result follows by applying the induction hypothesis to the graph H_n if fewer than $n - 1$ elements of N are vertices of H_n. Otherwise, all elements of N are vertices of H_n. Since both u and v are adjacent in $\square_n$ to a unique vertex of $\tilde{H}_n$, the result in this case follows from the connectivity of $\tilde{H}_n$.

13. Hint for part (c): A Hamiltonian cycle of G_4 is the one with successive vertices

$$\begin{array}{llll} (1,2,3,4) & (1,3,4,2) & (4,3,2,1) & (2,4,3,1) \\ (1,2,4,3) & (1,3,2,4) & (3,4,2,1) & (4,2,3,1) \\ (1,4,2,3) & (3,1,2,4) & (3,2,4,1) & (4,2,1,3) \\ (4,1,2,3) & (3,1,4,2) & (3,2,1,4) & (2,4,1,3) \\ (4,1,3,2) & (3,4,1,2) & (2,3,1,4) & (2,1,4,3) \\ (1,4,3,2) & (4,3,1,2) & (2,3,4,1) & (2,1,3,4) \end{array}$$

and $(1, 2, 3, 4)$, where the columns of this array are read from top to bottom and from left to right. Generalize this construction to arbitrary $n \geq 2$.

14. Answer: 22.
15. Hint: Prove that every tree with exactly two leaves is a path by considering the sum of the degrees of the vertices, or by induction on the number of vertices, or by applying the results of Section 5.3.2.
16. Hint: Combine Propositions 5.1.1 and 5.3.3. For part (a), prove that, in fact, every tree with at least two vertices has at least two more leaves than vertices of degree greater than 2.
18. We set $U = V \setminus \{v\}$. Since G is connected, for every $u \in U$ there exists a walk in G with initial vertex u and terminal vertex v. We consider such a walk $w(u)$ with the smallest possible length, say $\ell(u)$, and denote by $f(u)$ the first edge of $w(u)$. We will prove that the map $f : U \to E$ is one to one. Indeed, suppose that there are distinct elements u_1, u_2 of U such that $f(u_1) = f(u_2) = e$. Then, u_1 and u_2 are the endpoints of e. The walk which is obtained from $w(u_2)$ by deleting the initial vertex u_2 and the first edge e has initial vertex u_1, terminal vertex v, and length $\ell(u_2) - 1$. Our choice of $w(u_1)$ implies that $\ell(u_1) \leq \ell(u_2) - 1$ or, equivalently, that $\ell(u_1) < \ell(u_2)$. The same argument shows that $\ell(u_2) < \ell(u_1)$. This contradiction proves our claim. It follows from Proposition 4.1.1 (a) that $n - 1 = \#U \leq \#E$. This settles part (a). For part (b), we assume that G is a tree and show that f is also onto as follows. We consider any edge e of G with endpoints $u_1, u_2 \in V$ and distinguish two cases. If the unique path, say P, in G with initial vertex u_1 and terminal vertex v passes through vertex u_2, then $u_1 \in U$ and $f(u_1) = e$. Otherwise, the path which is obtained from P adding u_2 as the initial vertex, followed by the edge e, is the unique path in G with initial vertex u_2 and terminal vertex v, and we have $u_2 \in U$ and $f(u_2) = e$. In either case, $e = f(u)$ for some $u \in U$, and hence f is onto. Therefore, $f : U \to E$ is one-to-one and onto and hence $\#E = \#U = n - 1$.
19. Let n be the number of vertices of T, and for $J \subseteq V$, let $T(J)$ be the induced subgraph of T on the vertex set J and $e_T(J)$ be the number of edges of $T(J)$. Since every connected component of $T(J)$ is a tree and hence its number of edges is one less than its number of vertices, we have $c_T(J) = |J| - e_T(J)$. Therefore,

$$c_T(x) = \sum_{J \subseteq V} |J| x^{|J|} - \sum_{J \subseteq V} e_T(J) x^{|J|} = \sum_{k=0}^{n} k \binom{n}{k} x^k - \sum_{J \subseteq V} e_T(J) x^{|J|}$$

$$= n \sum_{k=1}^{n} \binom{n-1}{k-1} x^k - \sum_{J \subseteq V} e_T(J) x^{|J|}$$

$$= nx(x+1)^{n-1} - \sum_{J \subseteq V} e_T(J) x^{|J|}.$$

The last sum is equal to $\sum_{k=0}^{n} e_T(k)x^k$, where

$$e_T(k) := \sum_{J\subseteq V:\,|J|=k} e_T(J)$$

for $0 \le k \le n$. Since T has exactly $n-1$ edges and each one of them is an edge of $T(J)$ for exactly $\binom{n-2}{k-2}$ k-subsets $J \subseteq V$ (explain why), we have

$$e_T(k) = (n-1)\binom{n-2}{k-2}.$$

As a result,

$$\begin{aligned}\sum_{J\subseteq V} e_T(J)x^{|J|} &= \sum_{k=0}^{n} e_T(k)x^k = (n-1)\sum_{k=2}^{n}\binom{n-2}{k-2}x^k \\ &= (n-1)x^2(x+1)^{n-2}\end{aligned}$$

and

$$c_T(x) = nx(x+1)^{n-1} - (n-1)x^2(x+1)^{n-2} = x(x+n)(x+1)^{n-2}$$

for every tree T with n vertices. This result appeared in Section 4.2 of [G. Lehrer, Ann. Inst. Fourier (Grenoble) **58** (2008), 671–688].

21. Hint for part (b): Combine Proposition 5.3.4 with Lemma 5.3.3.
22. We follow the notation introduced in Section 5.3.2. For part (a), we first assume that $\{1,2\} \in F$. Then, 1 and 2 cannot both be leaves of the tree T_i for any $1 \le i \le n-2$. Since each of these trees has at least two leaves, the largest leaf for each of them is greater than 2 and hence $v_1, v_2, \ldots, v_{n-2} \notin \{1,2\}$. As a result, $v_{n-1} = 2$ and $v_n = 1$ and hence $a_{n-2} \in \{v_{n-1}, v_n\} = \{1,2\}$. Conversely, suppose that $\{1,2\} \notin F$. Then, $v_{n-1} \ne 2$, and hence $v_i = 2$ for some $i < n-1$. Since $v_i = 2$ is the largest leaf of the tree T_i, the latter must have exactly two leaves, namely 1 and 2, and hence (by the solution of Exercise 15), it must be a path with successive vertices $2 = v_i, v_{i+1}, \ldots, v_{n-1}, v_n = 1$. Therefore, $a_{n-2} = v_{n-1} \notin \{1,2\}$. Part (a) and Proposition 5.3.4 imply that the trees of part (b) are in one-to-one correspondence with the sequences $(a_1, a_2, \ldots, a_{n-2}) \in [n]^{n-2}$ such that $a_{n-2} \in \{1,2\}$. Hence, there are exactly $2n^{n-3}$ such trees. Here is a proof which does not rely on part (a). Let $t_n = n^{n-2}$ be the total number of trees with vertices $1, 2, \ldots, n$, and for given $a, b \in [n]$ with $a < b$, let $f(n)$ be the number of trees with vertex set $[n]$ for which $\{a,b\}$ is an edge. We observe that $f(n)$ depends only on n, and not on a, b (explain why). Thus, counting the pairs (T, e), where T is a tree with vertex set $[n]$ and e is an edge of T, in two obvious ways, we find that $t_n \cdot (n-1) = \binom{n}{2} f(n)$ and deduce that $f(n) = 2n^{n-3}$.

24. Hint for part (b): Find a bijection from the set of trees with vertex set $\{0, 1, \ldots, n\}$ to the set of rooted forests with vertex set $[n]$.
25. Hint for part (a): Given a map $f : \{2, 3, \ldots, n\} \to [n-1]$ such that $f(x) < x$ for every $x \in \{2, 3, \ldots, n\}$, let T_f be the simple graph with vertex set $[n]$ and edges $\{x, f(x)\}$ for $x \in \{2, 3, \ldots, n\}$. Prove that T_f is an increasing tree on the vertex set $[n]$ and that every increasing tree with vertex set $[n]$ is of the form T_f.
26. Let $r(n)$ be the number of rooted trees on the vertex set $[n]$ and let X_n be the set of sequences $(F_0, F_1, \ldots, F_{n-1})$ described in the exercise. Since a forest with n vertices has at most $n-1$ edges, we have $\#\mathcal{E}(F_i) = i$ for every such sequence and for every $i \in \{0, 1, \ldots, n-1\}$, and hence F_{n-1} is a rooted tree. We observe that $\#X_n = r(n)\,(n-1)!$, because there are $r(n)$ ways one can choose the rooted tree F_{n-1} and $(n-1)!$ ways one can successively delete its $n-1$ edges, so as to obtain $F_{n-2}, \ldots, F_1, F_0$. We also observe that $\mathcal{E}(F_i) = \mathcal{E}(F_{i-1}) \cup \{(a, b)\}$ for every $i \in [n-1]$, where b is necessarily one of the $n-i+1$ roots of F_{i-1} and a is a vertex which does not belong to the same tree as b. Given the forest F_{i-1}, there are n ways one can choose the vertex a and $n-i$ ways one can choose the root b of F_{i-1} (as many as the connected components of F_{i-1} other than the one which contains a) in order to obtain the forest F_i. As a consequence,

$$\#X_n = \prod_{i=1}^{n-1} n(n-i) = n^{n-1}(n-1)!.$$

    We conclude that $r(n) = n^{n-1}$, which is equivalent to Theorem 5.3.1. This proof is due to J. Pitman [J. Combin. Theory Series A **85** (1999), 165–193].
27. A one-to-one correspondence with the properties stated in part (a) has been given by A. Joyal [Adv. Math. **42** (1981), 1–82] (Example 12). This correspondence is also described in the second proof of [24, Proposition 5.3.2].
28. Hint: For part (a), generalize the Prüfer code map to a bijection $\varphi : \mathcal{T}_\pi \to [n]^{k-1}$, where $\mathcal{T}_\pi$ is the set of rooted trees on the vertex set $[n]$ considered in the exercise, as follows. To define $\varphi(T) = (a_1, a_2, \ldots, a_{k-1})$ for $T \in \mathcal{T}_\pi$, among all blocks of π which consist of leaves of T, consider the block B_1 which contains the largest element and denote by a_1 the common parent in T of the elements of B_1. Then, delete the elements of B_1 from T and continue as in the proof of Theorem 5.3.1. Use a similar argument for part (b) and define a one-to-one correspondence from the set of rooted trees considered there to the set of $(k-1)$-partial permutations of $\{2, 3, \ldots, n\}$. Apply part (b) to the partition $\{\{2\}, \ldots, \{n-2\}, \{n-1, n\}\}$ of the set $\{2, 3, \ldots, n\}$ and deduce that the answer to part (c) is $(n-1)!/2$ for $n \geq 3$.
29. This problem is due to J. Propp and the following solution to the author. Let Ω be the set of all trees on the vertex set $[2n]$ which are rooted at 1 and have property (i). We have $\#\Omega = n^{2n-2}$ by Exercise 21. For $i \in \{2, \ldots, n\}$, we

denote by A_i the set of rooted trees $T \in \Omega$ for which $2i-2$, $2i-1$, and $2i$ have a common parent. Clearly,

$$g(n) = \#\left(\Omega - \bigcup_{i=2}^{n} A_i\right).$$

Let $I \subseteq \{2, 3, \ldots, n\}$ have cardinality r and let π_I be the unique partition of the set $\{2, 3, \ldots, 2n\}$ with $2n - 2r - 1$ blocks such that $2i - 2$, $2i - 1$, and $2i$ belong to the same block of π_I for every $i \in I$. A rooted tree $T \in \Omega$ belongs to the intersection $\bigcap_{i \in I} A_i$ if and only if the elements of any block of π_I have a common parent in T. As easy generalization of Exercise 28 (a) (see also Exercise 21) shows that

$$\# \bigcap_{i \in I} A_i = n^{2n-2r-2}.$$

By the inclusion–exclusion principle, we conclude that

$$g(n) = \sum_{r=0}^{n-1} (-1)^r \binom{n-1}{r} n^{2n-2r-2} = \sum_{r=0}^{n-1} \binom{n-1}{r} (-1)^r (n^2)^{n-1-r}$$

$$= (n^2 - 1)^{n-1}.$$

30. Hint for part (b): Double count the pairs (x, y) of vertices of T such that x is the parent of y. Let us outline a proof of part (c). We apply a variant of Proposition 5.3.4, in which the Prüfer code map is replaced by a bijection $f : R_n \to [n]^{n-1}$ from the set R_n of rooted trees with vertex set $[n]$ to the set $[n]^{n-1}$ of all sequences $(a_1, a_2, \ldots, a_{n-1})$ with $a_i \in [n]$ for every i. This differs from the bijection of Theorem 5.3.1 in that we consider rooted trees on the vertex set $[n]$ and their leaves, in the sense of Section 5.3.4, and in that the root appears in the code $f(T) = (a_1, a_2, \ldots, a_{n-1})$ as the coordinate a_{n-1}. We observe that a vertex j has exactly i successors in T if and only if j appears exactly i times as a coordinate of $f(T)$. As a result, $T \in \mathcal{T}(r_0, \ldots, r_{n-1})$ if and only if for $i \in \{0, 1, \ldots, n-1\}$, the number of positive integers which appear exactly i times as coordinates of $f(T)$ is equal to r_i. Therefore, the cardinality of $\mathcal{T}(r_0, \ldots, r_{n-1})$ is equal to the number of sequences in $[n]^{n-1}$ in which exactly r_i positive integers appear each exactly i times, for $i \in \{0, 1, \ldots, n-1\}$. To enumerate these sequences, we choose the r_i elements of $[n]$ which will appear each exactly i times in the sequence in

$$\frac{n!}{r_0! r_1! \cdots r_{n-1}!}$$

ways, and for $i \in \{0, 1, \ldots, n-1\}$, we permute the $r_1+2r_2+\cdots+(n-1)r_{n-1} = n-1$ copies of these integers in

$$\frac{(n-1)!}{(1!)^{r_1}(2!)^{r_2}\cdots((n-1)!)^{r_{n-1}}}$$

ways, so as to form the sequence. It follows that

$$\begin{aligned}\#\mathcal{T}(r_0, \ldots, r_{n-1}) &= \frac{n!}{r_0!r_1!\cdots r_{n-1}!} \cdot \frac{(n-1)!}{(1!)^{r_1}(2!)^{r_2}\cdots((n-1)!)^{r_{n-1}}} \\ &= \frac{n!(n-1)!}{\prod_{i=0}^{n-1}(i!)^{r_i}\, r_i!}.\end{aligned}$$

For more results of this type, we refer the reader to [24, Chapter 5].

31. Let a_n and b_n be the number of trees in parts (a) and (b), respectively, so that $a_0 = b_0 = a_1 = b_1 = 1$. Given an unlabeled plane rooted tree T with $n+1 \geq 3$ vertices, the successors of the root of T come with a specified linear order. Let T_1 be the tree formed by the first successor of the root of T and its own successors and let F be the union (which may be empty) of the trees formed by the other successors of the root of T. Clearly, T_1 is also an unlabeled plane rooted tree, while F is such a tree from which the root has been deleted. Moreover, if T_1 has $k+1$ vertices, then F has $n-k-1$ vertices. There are a_k ways to choose T_1 and a_{n-k-1} ways to choose F, and hence, there are $a_k a_{n-k-1}$ ways to choose the pair (T_1, F). It follows that

$$a_n = \sum_{k=0}^{n-1} a_k a_{n-k-1} \tag{B.6}$$

for $n \geq 2$. A similar reasoning (which considers the two trees defined by the first and second successors of the root of an unlabeled binary plane rooted tree with n internal vertices and their own successors) shows that

$$b_n = \sum_{k=0}^{n-1} b_k b_{n-k-1} \tag{B.7}$$

for $n \geq 2$. It follows from Equations (B.6) and (B.7) and Proposition 2.3.12 that $a_n = b_n = \frac{1}{n+1}\binom{2n}{n}$ for every $n \in \mathbb{N}$.

32. Hint: Given a coloring κ of the vertices of G with two colors, let $e(\kappa)$ denote the number of monochromatic edges of G (meaning edges whose endpoints have the same color). Choose any coloring κ which minimizes $e(\kappa)$ and prove that it has the desired property.

36. Hint for part (a): Consider a proper coloring $\kappa : V \to [k]$ of G and show that for all $1 \leq a < b \leq k$ there exists an edge $\{u, v\}$ of G such that $\kappa(u) = a$ and $\kappa(v) = b$.
38. Hint: Apply Proposition 5.4.4 in a suitable way.
40. Part (a) follows from the result of Exercise 39 (a), since $\chi(\bar{G}) \geq \alpha(G)$ (explain why). For part (b), we proceed by induction on n. We assume that $n \geq 2$ and that the result (which is trivial for $n = 1$) holds for $n - 1$. We choose an arbitrary vertex $v \in V$ and consider the graphs $G \setminus v$ and $\bar{G} \setminus v$. Clearly, $\chi(G) \leq \chi(G \setminus v) + 1$ and $\chi(\bar{G}) \leq \chi(\bar{G} \setminus v) + 1$. If equality holds in both cases, then the proof of Proposition 5.4.2 (b) shows that $\chi(G \setminus v) \leq \deg_G(v)$ and $\chi(\bar{G} \setminus v) \leq \deg_{\bar{G}}(v)$, whence

$$\chi(G) + \chi(\bar{G}) \leq \deg_G(v) + \deg_{\bar{G}}(v) + 2 = n + 1.$$

Otherwise,

$$\chi(G) + \chi(\bar{G}) \leq \chi(G \setminus v) + \chi(\bar{G} \setminus v) + 1,$$

and the result follows from the induction hypothesis. Answer for part (c): The possible values (for $n \geq 4$) are $n - 1$ and $n - 2$.
41. To prove the inequality of part (a), verify that the map $\kappa : \binom{[n]}{k} \to [n - 2k + 2]$ defined by

$$\kappa(S) = \begin{cases} \min(S), & \text{if } \min(S) \leq n - 2k + 1 \\ n - 2k + 2, & \text{if } S \subseteq \{n - 2k + 2, \ldots, n\} \end{cases}$$

for $S \in \binom{[n]}{k}$ is a proper coloring of $\mathrm{KG}_{n,k}$ with $n - 2k + 2$ colors. The equality of part (b) was conjectured by Kneser in 1955 and proven by Lovász [J. Combin. Theory Series A **25** (1978), 319–324] about 23 years later, by use of the Borsuk–Ulam theorem of algebraic topology. Various proofs, each applying some version of the Borsuk–Ulam theorem, as well as generalizations can be found in [J. Matousek, *Using the Borsuk-Ulam Theorem. Lectures on Topological Methods in Combinatorics and Geometry*, Univesitext, Springer, 2003].
44. For part (b), verify that the perfect matchings of the $\frac{1}{2}(n - 1)!$ cycles of K_n of length n have the desired property. For part (c), let $G = (V, E)$ be a simple graph with n vertices and more than $(n - 1)(n - 2)/2$ edges. We observe that the set $\bar{E} := \binom{V}{2} \setminus E$ has at most $n - 2$ elements. Since each of these belongs to exactly $(n - 2)!$ of the perfect matchings of part (b), at most $(n-2)(n-2)!$ of these perfect matchings contain edges in $\bar{E}$. Since the number of these matchings is $(n - 1)!$, at least one of them does not contain edges of $\bar{E}$, and hence it is a perfect matching of G. A different solution can be given by first showing that there exists a partition $\pi = \{F_1, F_2, \ldots, F_{n-1}\}$ of the set of edges of the complete graph with n vertices $v_1, v_2, \ldots, v_n$ into $n - 1$

perfect matchings. For example, one may choose $F_1 = \{\{v_1, v_2\}, \{v_3, v_4\}\}$, $F_2 = \{\{v_1, v_3\}, \{v_2, v_4\}\}$, and $F_3 = \{\{v_1, v_4\}, \{v_2, v_3\}\}$ for $n = 4$, and

$$F_1 = \{\{v_1, v_2\}, \{v_3, v_4\}, \{v_5, v_6\}\}$$
$$F_2 = \{\{v_1, v_6\}, \{v_2, v_3\}, \{v_4, v_5\}\}$$
$$F_3 = \{\{v_1, v_3\}, \{v_2, v_5\}, \{v_4, v_6\}\}$$
$$F_4 = \{\{v_1, v_4\}, \{v_2, v_6\}, \{v_3, v_5\}\}$$
$$F_5 = \{\{v_1, v_5\}, \{v_2, v_4\}, \{v_3, v_6\}\}$$

for $n = 6$. Given π, one can show that every simple graph with vertices $v_1, v_2, \ldots, v_n$ and more than $(n-1)(n-2)/2$ edges has at least one of $F_1, F_2, \ldots, F_{n-1}$ as a perfect matching. For a third solution, consult [14, Theorem 3.4.2].

45. Let $F = \{e_1, e_2, \ldots, e_k\}$ be a matching of G of maximum cardinality k. We will show that $k \geq n/(d+1)$. Let $v_1, v_2, \ldots, v_m$ be the vertices of G which are not endpoints of any edge of F, so that $m = n - 2k$. For $i \in [m]$, let f_i be an edge of G which is not a loop and is incident to v_i. Our definition of k implies (explain why) that f_i connects v_i to one of the $2k$ vertices of F for every $i \in [m]$ and that there are no two edges f_i and f_j which connect v_i and v_j, respectively, with the two endpoints of an edge of F. Since the degrees of the vertices of F are no larger than d, it follows that $n - 2k = m \leq (d-1)k$, that is, $k \geq n/(d+1)$.

49. The largest possible value of m is $n-1$. Indeed, let $\{A, B\}$ be a partition of the vertex set of $K_{n,n}$ as in Definition 5.4.2, where $\#A = \#B = n$. For a fixed $a \in A$, deleting all n edges $\{a, b\}$ of $K_{n,n}$ with $b \in B$ leaves a graph with no perfect matching. Thus, it suffices to prove that every graph G obtained from $K_{n,n}$ by deleting at most $n-1$ edges has a perfect matching. Let us apply Corollary 5.4.2. Given a nonempty k-element subset $S \subseteq A$, we need to show that $\#N_G(S) \geq k$. This is clear for $k = n$, since then $S = A$ and every element of B is adjacent in G to some element of A. Suppose that $k < n$. Since $\{a, b\}$ is not an edge of G for $a \in S$ and $b \in B \setminus N_G(S)$, we must have $\#S \cdot \#(B \setminus N_G(S)) \leq n-1$. Therefore,

$$\#(B \setminus N_G(S)) \leq \frac{n-1}{k} \leq n-k,$$

which implies that $\#N_G(S) \geq k$.

50. Hint for part (a): Let F be a perfect matching of G. Show that G contains a cycle with successive vertices $b_1, a_1, b_2, a_2, \ldots, b_m, a_m$, where $a_i \in A$, $b_i \in B$, and $\{a_i, b_i\} \in F$ for every $i \in [m]$. Replace the edges $\{a_1, b_1\}$, $\{a_2, b_2\}, \ldots, \{a_m, b_m\}$ in F with the remaining m edges of this cycle to find a perfect matching of G other than F.

51. Hint: Let $A = \{u_1, u_2, \ldots, u_n\}$ and $B = \{v_1, v_2, \ldots, v_n\}$ be disjoint n-element sets and $F \subseteq F'$ be edge sets with elements $\{u_i, v_i\}$ for $i \in [n]$ and $\{u_i, v_j\}$ for $i, j \in [n]$ with $i \geq j$, respectively. For part (a), prove first that every simple graph with set of vertices $V = A \cup B$ and set of edges E such that $F \subseteq E \subseteq F'$ has a unique perfect matching (specifically, F). Then, consider a simple bipartite graph $G = (V, E)$, with a partition $\{A, B\}$ of V as in Definition 5.4.2, for which F is the unique perfect matching. By Exercise 50, some element of A has degree one in G. Apply induction on n and show that the number of edges of G does not exceed $n(n+1)/2$. For part (b), include all 2-subsets of B into F' and follow a similar reasoning.
52. Suppose that G does not have any n-element matchings. Theorem 5.4.1 implies that there exists $S \subseteq A$ such that $\#S = k + 1$ and $\#N_G(S) \leq k$ for some $k \in \mathbb{N}$. Then, every element of S (which is nonempty) has degree at most k in G, and every element of $B \setminus N_G(S)$ (which is also nonempty) has degree at most $n - k - 1$ in G (explain why). From these facts, we conclude that $\deg_G(v) \leq (n-1)/2$ for some $v \in V$, contrary to the hypothesis. This proves part (a). Part (b) is left to the reader.
53. The following solution is a good example of an application of linear algebra to combinatorial problems. Let $A = \{u_1, u_2, \ldots, u_n\}$ and $B = \{v_1, v_2, \ldots, v_m\}$. We consider the $m \times n$ matrix $M(G)$, the (j, i)-entry of which is equal to a_{ij} if $\{u_i, v_j\} \in E$ and zero otherwise, where the a_{ij} for $\{u_i, v_j\} \in E$ are algebraically independent variables (so, the rows of $M(G)$ correspond to the elements of B, the columns correspond to the elements of A, and the entries belong to the field of rational functions in the variables a_{ij}). For $S \subseteq A$, we claim that $S \in \mathcal{I}(G)$ if and only if the columns of $M(G)$ which correspond to the elements of S are linearly independent. The result follows from this claim and basic properties of linearly independent subsets of a vector space. To prove the claim, we may assume without loss of generality that $S = A$. Then, it suffices to observe that (a) the columns of $M(G)$ are linearly independent if and only if the matrix formed by some choice of n rows of $M(G)$ has nonzero determinant; (b) such a determinant is nonzero if and only if there is a nonzero term of the form $\pm a_{1\sigma(1)} a_{2\sigma(2)} \cdots a_{n\sigma(n)}$ in its expansion, where $\sigma : [n] \to [m]$ is an injective map; and (c) that the latter holds if and only if there exists a matching of G with edges $\{u_1, v_{\sigma(1)}\}, \{u_2, v_{\sigma(2)}\}, \ldots, \{u_n, v_{\sigma(n)}\}$.
55. Hint for part (d): Observe that $\chi_G(x) = x(x-1)(x-2)(x^2 - 3x + 3)$.
57. Answer: $(x-1)^n + (-1)^n(x-1)$.
59. Hint: Use deletion–contraction to show that $(-1)^{n-1} a_1 \geq 1$ for every connected graph G. To prove the converse, show that $\chi_G(x) = \chi_{G_1}(x)\chi_{G_2}(x)$ if G is the disjoint union of G_1 and G_2.
60. Hint: Apply the implication $(ii) \Rightarrow (i)$ of Proposition 5.3.3 to show that both statements are true.
61. Part (a) is a result of D.R. Fulkerson and O.A. Gross [Pacific J. Math. **15** (1965), 835–855]. Hint for part (b): Apply the multiplication principle to compute $\chi_G(q)$ for $q \in \mathbb{Z}_{>0}$.

62. Answers: (c) $(1+x)^m$; (d) true for graphs with at most four vertices but not for all graphs with five vertices (counterexamples were found by use of computer by I. Eleftheriadis in 2019).
64. Answer: 3.
65. Answer: This graph is planar if and only if $k \leq 3$.
66. Answers: (a) $3 \leq n \leq 6$; (b) $1 \leq n \leq 3$.
67. Hint: To solve part (a), generalize the proofs of Propositions 5.6.1 and 5.6.2. Then, apply this result for $\ell = 5$ to answer part (b).
68. Hint: Combine Propositions 5.1.1 and 5.6.1.

Chapter 6

2. Hint: Find a simple recurrence for $(f(n))_{n\geq 1}$.
4. By Equations (6.8) and (6.9), the formal power series $\sum_{n\geq 0} b_n x^n$ is inverse of $F(x)$ in $\mathbb{C}[[x]]$ if and only if

$$a_0 b_n + a_1 b_{n-1} + \cdots + a_n b_0 = \begin{cases} 1, & \text{if } n = 0 \\ 0, & \text{if } n \geq 1 \end{cases} \tag{B.8}$$

for $n \in \mathbb{N}$. For $n = 0$ we get the equality $a_0 b_0 = 1$, whence $a_0 \neq 0$. Conversely, suppose that $a_0 \neq 0$. Then, (B.8) has a unique solution for $(b_0, b_1, b_2, \dots)$ defined recursively by the equalities $b_0 = a_0^{-1}$, $b_1 = -a_1 b_0/a_0$, $b_2 = -(a_1 b_1 + a_2 b_0)/a_0$, and so on. As a result, $F(x)$ is invertible in $\mathbb{C}[[x]]$, and its inverse is unique.

6. Hint for part (a): First find a simple formula for $a_{n,k}$. Your answer to part (b) should agree with the result of Example 6.5.9.
7. Answer to part (b): $1/(1-4x)$, $1/(1-3x)$, and $x/(1-3x)^2$.
9. Answer: $5^n - 2^n$.
10. Hint: For part (a), apply Theorem 6.1.1 to find a recurrence for $(a_n)_{n\in\mathbb{N}}$. For part (b), use this recurrence to find the remainder of the division of a_n by 8.
11. Hint for part (a): For given $k, \ell \in \{0, 1, \dots, n\}$, find the number of such triples (a, b, c) which satisfy $a = k$ and $b = \ell$. Then, express $f(n)$ as a double sum which ranges over all $k, \ell \in \{0, 1, \dots, n\}$ and compute this sum.
12. Let us adopt the notation of Section 6.1.1 for the $n \times 2$ grid graph G_n and its vertices. Since the spanning trees of G_n do not contain cycles, such a tree does not contain at least one of the edges $\{u_{n-1}, u_n\}$, $\{u_n, v_n\}$, $\{v_{n-1}, v_n\}$, and $\{u_{n-1}, v_{n-1}\}$. The spanning trees of G_n which do not contain the edge $\{u_{n-1}, u_n\}$ necessarily contain $\{u_n, v_n\}$ and $\{v_{n-1}, v_n\}$ and are in one-to-one correspondence (explain why) with the spanning trees of G_{n-1}. The same happens with those which do not contain the edge $\{u_n, v_n\}$, or $\{v_{n-1}, v_n\}$. Therefore, there exist exactly $3a_{n-1}$ spanning trees of G_n which do not contain one of the edges $\{u_{n-1}, u_n\}$, $\{u_n, v_n\}$, and $\{v_{n-1}, v_n\}$. All other spanning trees contain these edges, but not $\{u_{n-1}, v_{n-1}\}$. These trees are in one-to-one correspondence with the spanning trees of G_{n-1} which contain the edge $\{u_{n-1}, v_{n-1}\}$. Reasoning as before, we find that there exist exactly $a_{n-1} - a_{n-2}$

such trees. From these observations we conclude that $a_n = 3a_{n-1} + (a_{n-1} - a_{n-2}) = 4a_{n-1} - a_{n-2}$ for $n \geq 2$. This answers part (a). Part (b) is a direct consequence of part (a) and Theorem 6.1.1.

13. Let a_n denote the left-hand side of the identity. Applying (6.19), we compute that

$$\sum_{n\geq 0} a_n x^n = \sum_{n\geq 0}\sum_{k=0}^{n} k\binom{n}{k}x^n = \sum_{k\geq 0}\sum_{n\geq k} k\binom{n}{k}x^n = \sum_{k\geq 0} k\,\frac{x^k}{(1-x)^{k+1}}$$

$$= \frac{1}{1-x}\cdot\sum_{k\geq 1} k\left(\frac{x}{1-x}\right)^k = \frac{1}{1-x}\cdot\frac{x/(1-x)}{(1-x/(1-x))^2}$$

$$= \frac{x}{(1-2x)^2} = x\sum_{n\geq 0}(n+1)\,2^n x^n = \sum_{n\geq 1} n 2^{n-1}x^n$$

and conclude that $a_n = n\cdot 2^{n-1}$ for every $n \in \mathbb{N}$. We leave it to the reader to give another proof by computing the exponential generating functions of the two sides of the proposed identity.

14. Hint: Square the special case $k = 2$ of Equation (6.19).

15. Hint: For part (a), compare the coefficients of x^n in the two sides of

$$\frac{1}{(1-x)^{m+1}} = (1-x)^k\cdot\frac{1}{(1-x)^{m+k+1}}.$$

Replace k by $-k$ in the first proposed identity to deduce the second.

16. Hint for part (b): Compare the coefficients of x^{mr} in the two sides of

$$(1+x+x^2+\cdots+x^{r-1})^n = (1-x^r)^n\cdot\frac{1}{(1-x)^n}.$$

17. Hint: For part (a), rewrite (6.19) as

$$\sum_{k\geq 1}\binom{n+k-1}{n}x^{k-1} = \frac{1}{(1-x)^{n+1}},$$

multiply by x, and differentiate the resulting equation with respect to x. For part (b), write the identity of part (a) in the form

$$\frac{1}{1-x}\cdot\sum_{k\geq 1} k\binom{n+k-1}{n}x^{k-1} = \frac{1+nx}{(1-x)^{n+3}}$$

and compare the coefficients of x^{m-1} in the two sides of this equation.

18. Since $\tau(m,n)$ is equal to the number of solutions $(r_0, r_1, \dots, r_n) \in \mathbb{N} \times \mathbb{Z}^n$ of the equation $r_0 + |r_1| + \cdots + |r_n| = m$,

$$\begin{aligned}\sum_{m\geq 0} \tau(m,n)x^m &= \sum_{r_0\in\mathbb{N},\, r_i\in\mathbb{Z}} x^{r_0+|r_1|+\cdots+|r_n|} \\ &= \left(\sum_{r_0\in\mathbb{N}} x^{r_0}\right)\left(\sum_{r_1\in\mathbb{Z}} |x^{r_1}|\right)\cdots\left(\sum_{r_n\in\mathbb{Z}} |x^{r_n}|\right) \\ &= (1+x+x^2+\cdots)(1+2x+2x^2+\cdots)^n \\ &= \frac{1}{1-x}\cdot\left(\frac{1+x}{1-x}\right)^n = \frac{(1+x)^n}{(1-x)^{n+1}}.\end{aligned}$$

As a consequence,

$$\begin{aligned}\sum_{m,n\geq 0} \tau(m,n)x^m y^n &= \sum_{n\geq 0}\left(\sum_{m\geq 0}\tau(m,n)x^m\right) y^n = \sum_{n\geq 0}\frac{(1+x)^n}{(1-x)^{n+1}}y^n \\ &= \frac{1}{1-x}\cdot\sum_{n\geq 0}\left(\frac{(1+x)y}{1-x}\right)^n = \frac{1}{1-x}\cdot\left(1-\frac{(1+x)y}{1-x}\right)^{-1} \\ &= \frac{1}{1-x-y-xy}.\end{aligned}$$

The symmetry of this generating function in x and y implies that $\tau(m,n) = \tau(n,m)$ for all $m, n \in \mathbb{N}$. Moreover, applying (6.19), we find that

$$\begin{aligned}\sum_{m,n\geq 0} g(m,n)x^m y^n &= \sum_{m,n\geq 0}\left(\sum_{k\geq 0}\binom{m}{k}\binom{n}{k}2^k\right)x^m y^n \\ &= \sum_{k\geq 0} 2^k\left(\sum_{m,n\geq k}\binom{m}{k}\binom{n}{k}x^m y^n\right) \\ &= \sum_{k\geq 0} 2^k \frac{x^k}{(1-x)^{k+1}}\cdot\frac{y^k}{(1-y)^{k+1}} \\ &= \frac{1}{(1-x)(1-y)}\cdot\sum_{k\geq 0}\frac{2^k x^k y^k}{(1-x)^k(1-y)^k}\end{aligned}$$

$$= \frac{1}{(1-x)(1-y)} \cdot \left(1 - \frac{2xy}{(1-x)(1-y)}\right)^{-1}$$

$$= \frac{1}{1-x-y-xy}$$

and conclude that $g(m,n) = \tau(m,n)$ for all $m,n \in \mathbb{N}$. Similar computations show that $\sum_{n\geq 0} f(m,n)y^n = (1+y)^m/(1-y)^{m+1}$ and that $\sum_{m,n\geq 0} f(m,n)x^m y^n = 1/(1-x-y-xy)$.

19. For part (a), we observe that (6.3) can be written as $\sum_{n\geq 0} F_n x^n = x/(1-x-x^2)$. Applying (6.19), we compute that

$$\sum_{n\geq 0} G_n x^n = \sum_{n\geq 0}\left(\sum_{k=0}^{n}\binom{n}{k}F_k\right)x^n = \sum_{k\geq 0} F_k \sum_{n\geq k}\binom{n}{k}x^n$$

$$= \sum_{k\geq 0} F_k \frac{x^k}{(1-x)^{k+1}} = \frac{1}{1-x}\cdot\sum_{k\geq 0} F_k\left(\frac{x}{1-x}\right)^k$$

$$= \frac{1}{1-x}\cdot\frac{x/(1-x)}{1-x/(1-x)-x^2/(1-x)^2}$$

$$= \frac{x}{1-3x+x^2}.$$

Comparing the coefficients of x^n in the two sides of the resulting equation $(1-3x+x^2)\sum_{n\geq 0} G_n x^n = x$, one gets the recurrence $G_n = 3G_{n-1} - G_{n-2}$ for $n \geq 2$. For part (b), we observe that

$$F_n = (\tau^n - \bar{\tau}^n)/\sqrt{5}$$

for $n \in \mathbb{N}$, where $\tau = (1+\sqrt{5})/2$ and $\bar{\tau} = (1-\sqrt{5})/2$, and compute that

$$G_n = \sum_{k=0}^{n}\binom{n}{k}F_k = \frac{1}{\sqrt{5}}\sum_{k=0}^{n}\binom{n}{k}(\tau^k - \bar{\tau}^k) = \frac{1}{\sqrt{5}}\left((1+\tau)^n - (1+\bar{\tau})^n\right)$$

$$= \frac{1}{\sqrt{5}}\left(\left(\frac{3+\sqrt{5}}{2}\right)^n - \left(\frac{3-\sqrt{5}}{2}\right)^n\right) = \frac{1}{\sqrt{5}}\left(\tau^{2n} - \bar{\tau}^{2n}\right) = F_{2n}.$$

Therefore,

$$\sum_{n\geq 0} G_n x^n = \frac{1}{\sqrt{5}}\left(\sum_{n\geq 0}\tau^{2n}x^n - \sum_{n\geq 0}\bar{\tau}^{2n}x^n\right)$$

$$= \frac{1}{\sqrt{5}} \left(\frac{1}{1-\tau^2 x} - \frac{1}{1-\bar{\tau}^2 x} \right) = \frac{1}{\sqrt{5}} \, \frac{(\tau^2 - \bar{\tau}^2)x}{(1-\tau^2 x)(1-\bar{\tau}^2 x)}$$

$$= \frac{x}{1-3x+x^2}.$$

Parts (c) and (d) have already been answered.

21. We rewrite Equation (6.19) in the form

$$\sum_{k \geq 0} \binom{m+k-1}{k} x^k = \frac{1}{(1-x)^m} \tag{B.9}$$

and compute that

$$\sum_{n \geq 0} t_n x^n = \sum_{n \geq k} \sum_{k \geq 0} (-1)^k \binom{2n-k-1}{k} 2^{n-k} x^n$$

$$= \sum_{m \geq 0} \sum_{k \geq 0} (-1)^k \binom{2m+k-1}{k} 2^m x^{m+k} = \sum_{m \geq 0} (2x)^m \frac{1}{(1+x)^{2m}}$$

$$= \frac{1}{1-2x/(1+x)^2} = \frac{(1+x)^2}{1+x^2} = 1 + 2x \sum_{m \geq 0} (-1)^m x^{2m}$$

$$= 1 + 2\,(x - x^3 + x^5 - \cdots).$$

We conclude that $t_0 = 1$, $t_{2m+1} = 2 \cdot (-1)^m$ for $m \in \mathbb{N}$, and $t_{2m} = 0$ for $m \in \mathbb{Z}_{>0}$.

23. Hint: Square the equalities $a_n = a_{n-1} + a_{n-2}$ and $a_{n-3} = a_{n-1} - a_{n-2}$ and add them up.

28. For part (a), we add up the binomial expansions of $(1+x)^n$ and $(1-x)^n$. For part (b), we apply part (a) twice, for $x = \sqrt{2}$ and for $n \in \{2m-1, 4m-2\}$, to find that

$$\sum_{k=1}^{2m-1} 2^{k-1} \binom{4m-2}{2k} = \frac{(1+\sqrt{2})^{4m-2} + (1-\sqrt{2})^{4m-2} - 2}{4}$$

$$= \left(\frac{(1+\sqrt{2})^{2m-1} + (1-\sqrt{2})^{2m-1}}{2} \right)^2$$

$$= \left(\sum_{k=0}^{m-1} 2^k \binom{2m-1}{2k} \right)^2.$$

29. For part (a) we compute that

$$\begin{aligned}\sum_{n\geq 0} b_n x^n &= \sum_{n\geq 0}\sum_{k=0}^{\lfloor 2n/3\rfloor} a_{n-k,k}x^n = \sum_{m\geq 0}\sum_{k=0}^{2m} a_{m,k}x^{m+k}\\ &= \sum_{m\geq 0} x^m \sum_{k=0}^{2m} a_{m,k}x^k = \sum_{m\geq 0} x^m(1-x+x^2)^m\\ &= \sum_{m\geq 0}(x-x^2+x^3)^m = \frac{1}{1-x+x^2-x^3}.\end{aligned}$$

Given that, we find that

$$\begin{aligned}\sum_{n\geq 0} b_n x^n &= \frac{1+x}{1+x^4} = (1+x)(1-x^4+x^8-x^{12}+\cdots)\\ &= 1+x-x^4-x^5+x^8+x^9-x^{12}-x^{13}+\cdots\end{aligned}$$

and the proof of part (b) follows. Part (b) appeared as Problem B4 in the 1997 William Lowell Putnam Mathematical Competition.

30. Consider the bijection $\varphi_n : 2^{[n]} \to \{0,1\}^n$ defined by setting $\varphi_n(S) = (r_1, r_2, \ldots, r_n)$, where

$$r_i = \begin{cases} 1, & \text{if } i \in S\\ 0, & \text{if } i \notin S,\end{cases}$$

for $S \subseteq [n]$. We observe that $|S| = r_1 + r_2 + \cdots + r_n$ and that the sum, say $\mathrm{sum}(S)$, of the elements of S equals $r_1 + 2r_2 + \cdots + nr_n$, if $S \subseteq [n]$ and $\varphi_n(S) = (r_1, r_2, \ldots, r_n)$. We conclude that

$$\begin{aligned}\sum_{p,q\geq 0} a_n(p,q)x^p y^q &= \sum_{S\subseteq[n]} x^{|S|}y^{\mathrm{sum}(S)} = \sum_{r_i\in\{0,1\}} x^{r_1+r_2+\cdots+r_n}y^{r_1+2r_2+\cdots+nr_n}\\ &= \prod_{i=1}^{n}\left(\sum_{r_i\in\{0,1\}}(xy^i)^{r_i}\right)\\ &= \prod_{i=1}^{n}(1+xy^i).\end{aligned}$$

31. Hint: Let $F_k(x)$ denote the left-hand side of Equation (6.46). Multiply the recurrence $S(n,k) = kS(n-1,k) + S(n-1,k-1)$ of Proposition 4.3.1 by x^n and sum over all $n \geq k$ to express $F_k(x)$ in terms of $F_{k-1}(x)$.

34. Hint for part (b): Observe that $1/(1-x)(1-x^2)^2 = (1+x)/(1-x^2)^3$ and apply Equation (6.19).
38. The number we are asked to compute, say a_n, is equal to the number of triples $(m_1, m_2, m_3) \in \mathbb{N}^3$ such that $m_1 + 2m_2 + 4m_3 = n$. Therefore,

$$\sum_{n\geq 0} a_n x^n = \sum_{m_i \geq 0} x^{m_1+2m_2+4m_3} = \frac{1}{(1-x)(1-x^2)(1-x^4)}.$$

We conclude that

$$\begin{aligned}\sum_{n\geq 0} a_n x^n &= \frac{(1+x+x^2+x^3)(1+x^2)}{(1-x^4)^3} \\ &= (1+x+2x^2+2x^3+x^4+x^5)\sum_{m\geq 0}\binom{m+2}{2}x^{4m}\end{aligned}$$

and hence that $a_{4m} = a_{4m+1} = \binom{m+1}{2} + \binom{m+2}{2}$ and $a_{4m+2} = a_{4m+3} = 2\binom{m+2}{2}$, for $m \in \mathbb{N}$.
39. To prove the first equality, it suffices to find a bijection $\varphi : \Lambda_k \to \mathrm{M}_k$ such that $|\varphi(\lambda)| = |\lambda|$ and $\max(\varphi(\lambda)) = r(\lambda)$ for every $\lambda \in \Lambda_k$ (explain why). For that, one can define as $\varphi(\lambda)$ the partition having Young diagram with column lengths the parts of λ. For example, if $\lambda = (5,4,4,2)$, then $\varphi(\lambda) = (4,4,3,3,1)$. To prove the second equality, we follow the proof of Proposition 6.3.1, noting that $r(\lambda) = m_1 + m_2 + \cdots + m_k$, and find that

$$\begin{aligned}\sum_{\lambda\in\Lambda_k} q^{r(\lambda)}x^{|\lambda|} &= \sum_{m_i\geq 0} q^{m_1+m_2+\cdots+m_k}x^{m_1+2m_2+\cdots+km_k} \\ &= \prod_{i=1}^{k}\left(\sum_{m_i\geq 0} q^{m_i}x^{im_i}\right) = \prod_{i=1}^{k}\frac{1}{1-qx^i}.\end{aligned}$$

40. Hint: Follow the reasoning in Example 6.3.1 and prove that

$$\sum_{n\geq 0}\sigma(n)x^n = \sum_{n\geq 0}\tau(n)x^n = \prod_{i\geq 1}\frac{1-x^{3i}}{1-x^i},$$

where $\sigma(0) = \tau(0) = 1$.
41. We set

$$f(n) = \sum_{k=0}^{n}(-1)^k\binom{n}{k}\binom{2n-k}{n}$$

for $n \in \mathbb{N}$ and compute that

$$\begin{aligned}\sum_{n\geq 0} f(n)x^n &= \sum_{n\geq 0}\sum_{k=0}^{n}(-1)^k\binom{n}{k}\binom{2n-k}{n}x^n \\ &= \sum_{k\geq 0}\sum_{n\geq k}(-1)^k\binom{n}{k}\binom{2n-k}{n}x^n \\ &= \sum_{k\geq 0}\sum_{m\geq 0}(-1)^k\binom{m+k}{k}\binom{2m+k}{m+k}x^{m+k}.\end{aligned}$$

We observe that

$$\binom{m+k}{k}\binom{2m+k}{m+k} = \binom{2m}{m}\binom{2m+k}{k}$$

and apply Equations (6.19) and (6.36) to conclude that

$$\begin{aligned}\sum_{n\geq 0} f(n)x^n &= \sum_{m\geq 0}\binom{2m}{m}x^m\sum_{k\geq 0}\binom{2m+k}{2m}(-x)^k \\ &= \sum_{m\geq 0}\binom{2m}{m}x^m\cdot\frac{1}{(1+x)^{2m+1}} \\ &= \frac{1}{1+x}\cdot\sum_{m\geq 0}\binom{2m}{m}\left(\frac{x}{(1+x)^2}\right)^m \\ &= \frac{1}{1+x}\cdot\left(1-\frac{4x}{(1+x)^2}\right)^{-1/2} = \frac{1}{1-x} = \sum_{n\geq 0}x^n\end{aligned}$$

and hence that $f(n) = 1$ for every $n \in \mathbb{N}$.

42. Hint: For part (a), observe that $(1+x+x^2)^n = \sum_{j=0}^{n}\binom{n}{j}x^{n-j}(1+x^2)^j$. For part (b), work as in the solution of Exercise 41 and apply Equations (6.19) and (6.36).
43. Answer: 3^n.
44. Answer: (a) x^2e^{2x}; (b) $x/(1-x)^3$; (c) $1/\sqrt{1-2x}$.
49. Hint: Observe that $f(n) = r^n B(n)$ and $g(n) = \sum_{k\geq 0} r^k S(n,k)$ for every $n \in \mathbb{N}$ and apply Corollary 6.5.1 and Proposition 6.5.4.
52. Hint: For part (a), observe that the left-hand side of Equation (6.47) is equal to the number of ways to choose a partition of the set $[n]$ with $a+b$ blocks and color a of them white and the others black. Then, count these possibilities in a

different way. For part (b), note that

$$\binom{a+b}{a}\frac{(e^x-1)^{a+b}}{(a+b)!}=\frac{(e^x-1)^a}{a!}\cdot\frac{(e^x-1)^b}{b!}$$

and apply Proposition 6.5.4.

Chapter 7

1. Answer: The least probable event is (b) and the most probable is (c).
3. Answer: $m/(m+n)$.
5. Hint: Let $p(n)$ be the probability in question. Compute $p(2n)$ and $p(2n+1)$ and their limits as $n\to\infty$.
6. Hint: Use the identity of Corollary 2.4.1 (a) to show that $p(n)=\binom{2n}{n}4^{-n}$. Part (b) is a direct consequence of this formula and Exercise 9 of Chapter 1.
7. Answer to part (a): $(m+1)^n/2^{mn}$.
8. Hint: Let us write $m=p_1^{m_1}p_2^{m_2}\cdots p_n^{m_n}$, where $p_1<p_2<\cdots<p_n$ are prime numbers and $m_1,m_2,\ldots,m_n\in\mathbb{Z}_{>0}$. The positive divisors of m are exactly the integers of the form $p_1^{\alpha_1}p_2^{\alpha_2}\cdots p_n^{\alpha_n}$ with $\alpha_i\in\{0,1,\ldots,m_i\}$ for every i. In particular, if $\Delta(m)$ is the set of positive divisors of m, then

$$\begin{aligned}\#\,\Delta(m)&=\textstyle\prod_{i=1}^n(m_i+1)\\ \#\,\Delta(m)\times\Delta(m)&=\textstyle\prod_{i=1}^n(m_i+1)^2.\end{aligned}$$

Moreover, for $a,b\in\Delta(m)$ with $a=p_1^{\alpha_1}p_2^{\alpha_2}\cdots p_n^{\alpha_n}$ and $b=p_1^{\beta_1}p_2^{\beta_2}\cdots p_n^{\beta_n}$, we have that a and b are relatively prime if and only if $\alpha_i\beta_i=0$ for every $i\in[n]$ and $a\mid b$ if and only if $\alpha_i\le\beta_i$ for every $i\in[n]$. Given these remarks, use the basic principles of enumeration to find explicit product formulas for $f(m)$ and $g(m)$ and verify parts (a), (b), and (c).
9. We claim that $P(A)\le P(B)$ and that equality holds only for $n=1$ and $n=3$. Indeed, since $x^2\equiv y^2\pmod 2\Leftrightarrow x\equiv y\pmod 2$, we have

$$\mathbf{P}(A)=\begin{cases}1/2, & \text{if } n \text{ is even},\\ (n^2+1)/2n^2, & \text{if } n \text{ is odd}\end{cases}$$

(explain why). Moreover, $x^2\equiv y^2\pmod 3\Leftrightarrow x\equiv y\pmod 3$, or $x\equiv 1\pmod 3$ and $y\equiv 2\pmod 3$, or $x\equiv 2\pmod 3$ and $y\equiv 1\pmod 3$. Thus, within every set of ordered pairs of the form $\{3k+1,3k+2,3k+3\}\times\{3\ell+1,3\ell+2,3\ell+3\}$, exactly five pairs (x,y) have the property that $x^2\equiv y^2\pmod 3$. If $n\equiv 0\pmod 3$, then the set $[n]\times[n]$ can be partitioned into such sets and hence $\mathbf{P}(B)=5/9$. By a similar reasoning for the cases $n\equiv 1\pmod 3$ and $n\equiv 2\pmod 3$, we find that $\mathbf{P}(B)\ge 5/9$ for every n. Hence, $P(A)<P(B)$ for even numbers n and $P(A)=(1/2)+(1/2n^2)\le 5/9\le P(B)$ for every odd $n\ge 3$, where equality holds only for $n=3$.

10. Hint for part (b): For $1 \leq k \leq n/2$, compute the probability of the event that $\sigma_1 < \sigma_2 < \cdots < \sigma_{2k} > \sigma_{2k+1}$ for a permutation $\sigma = (\sigma_1, \sigma_2, \ldots, \sigma_n)$ of $[n]$.
11. Hint: Consult the proof of the recurrence (2.12) of Proposition 2.3.12 to compute the probability that $f(p) = k$, for $k \in [n]$.
12. Hint: Proceed by induction on n. The case $n = 1$ is trivial, and the case $n = 2$ is equivalent to part (e) of Proposition 7.1.2. Assuming that it holds for $n - 1 \geq 2$, apply first the case $n = 2$ to the events $A = \cup_{i=1}^{n-1} A_i$ and $B = A_n$. Then, apply the induction hypothesis to $A_1, A_2, \ldots, A_{n-1}$ and their intersections with B, taking into account that

$$\left(\bigcup_{i=1}^{n-1} A_i \right) \cap B = \bigcup_{i=1}^{n-1} (A_i \cap B),$$

to complete the proof.
13. Hint for part (a): Observe that $p(m, n) = \mathbf{P}(\cup_{i=1}^{n} A_i)$, where A_i is the event that box Δ_i remains empty. Apply the Bonferonni inequalities of Exercise 30 of Chapter 3 and Proposition 7.1.2 (f) (subadditivity of probability measure) to prove the proposed inequalities. Answer to part (b): $p(50, 10) \sim 0.05$.
14. Hint: Apply the principle of inclusion–exclusion to compute $p(n)$ and the fact that $\lim_{n\to\infty} \frac{1}{n} \lfloor \frac{n}{k} \rfloor = \frac{1}{k}$ for every $k \in \mathbb{Z}_{>0}$ to compute its limit as $n \to \infty$.
15. Let $p_1, p_2, \ldots, p_r$ be the distinct primes which are smaller than or equal to n. We apply the principle of inclusion–exclusion, as in the solution of Exercise 20 of Chapter 3, and find that

$$q(n) = \frac{1}{n^2} \sum_{1 \leq i_1 < i_2 < \cdots < i_k \leq r} (-1)^k \left\lfloor \frac{n^2}{p_{i_1}^2 p_{i_2}^2 \cdots p_{i_k}^2} \right\rfloor.$$

Denoting by $\mathcal{P}$ the set of all prime numbers, we conclude that

$$\lim_{n\to\infty} q(n) = \prod_{p \in \mathcal{P}} \left(1 - \frac{1}{p^2}\right) = \prod_{p \in \mathcal{P}} \left(\frac{p^2}{p^2 - 1}\right)^{-1} = \left(\prod_{p \in \mathcal{P}} \frac{1}{1 - (1/p)^2} \right)^{-1}$$

$$= \left(\sum_{k=1}^{\infty} \frac{1}{k^2} \right)^{-1} = \frac{6}{\pi^2},$$

where we have considered as known the fact that $\sum_{k=1}^{\infty} \frac{1}{k^2} = \pi^2/6$.
17. Hint: Let A (respectively, B) be the set of $2 \times n$ matrices $\sigma \in \Omega$ of Example 7.2.4 which have increasing rows (respectively, columns). To enumerate the elements of $A \cap B$, find a one-to-one correspondence from this set to the set $\mathcal{P}^+(n, n)$ of paths in $\mathbb{Z}^2$ with initial vertex $(0, 0)$, terminal vertex (n, n), and steps $(1, 0)$ or $(0, 1)$ which lie entirely in the half-space $x \geq y$ of $\mathbb{R}^2$ (see Section 2.3.8 and Example 7.1.6).

19. Let us prove this for $m = 1$ (the general case follows by induction on m). For part (a), we assume that $A, B \subseteq \Omega$ are independent. We observe that $(\Omega \setminus A) \cap B = B \setminus (A \cap B)$ and conclude that

$$\begin{aligned}\mathbf{P}((\Omega \setminus A) \cap B) &= \mathbf{P}(B \setminus (A \cap B)) = \mathbf{P}(B) - \mathbf{P}(A \cap B) \\ &= \mathbf{P}(B) - \mathbf{P}(A)\mathbf{P}(B) \\ &= (1 - \mathbf{P}(A))\mathbf{P}(B) = \mathbf{P}(\Omega \setminus A)\mathbf{P}(B).\end{aligned}$$

For part (b), we assume that $A, B, C \subseteq \Omega$ are independent, so (7.11) holds. We observe that $(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$, and, applying Proposition 7.1.2 (e), we conclude that

$$\begin{aligned}\mathbf{P}((A \cup B) \cap C) &= \mathbf{P}((A \cap C) \cup (B \cap C)) \\ &= \mathbf{P}(A \cap C) + \mathbf{P}(B \cap C) - \mathbf{P}(A \cap B \cap C) \\ &= \mathbf{P}(A)\mathbf{P}(C) + \mathbf{P}(B)\mathbf{P}(C) - \mathbf{P}(A \cap B)\mathbf{P}(C) \\ &= (\mathbf{P}(A) + \mathbf{P}(B) - \mathbf{P}(A \cap B))\mathbf{P}(C) \\ &= \mathbf{P}(A \cup B)\mathbf{P}(C).\end{aligned}$$

Hence, $A \cup B$ and C are also independent. Obviously, $A \cap B$ and C have the same property. For $A \setminus B$ and C, we may work as in part (a).

22. Suppose that such a probability measure $\mathbf{P} : 2^{\Omega} \to \mathbb{R}$ and events $A_1, A_2, \ldots, A_n$ exist. We will prove that $2^n \leq N$. We set

$$A^{\varepsilon} = \begin{cases} A, & \text{if } \varepsilon = +, \\ \Omega \setminus A, & \text{if } \varepsilon = -, \end{cases}$$

for $A \subseteq \Omega$ and consider the 2^n events $A_1^{\varepsilon_1} \cap A_2^{\varepsilon_2} \cap \cdots \cap A_n^{\varepsilon_n}$ for $\varepsilon_i \in \{-, +\}$. These events are pairwise disjoint, and their union equals Ω (explain why). The assumption that $A_1, A_2, \ldots, A_n$ are independent, each with positive probability smaller than 1, in view of Proposition 7.2.1 (a), implies that

$$\mathbf{P}\left(\bigcap_{i=1}^{n} A_i^{\varepsilon_i}\right) = \prod_{i=1}^{n} \mathbf{P}(A_i^{\varepsilon_i}) > 0$$

and hence that $A_1^{\varepsilon_1} \cap A_2^{\varepsilon_2} \cap \cdots \cap A_n^{\varepsilon_n} \neq \varnothing$. We have shown that there exists a partition of Ω into 2^n (nonempty) blocks. This implies that $2^n \leq N$. The converse is left to the reader.

23. Hint: Prove that the event, say A, the probability of which we are asked to compute, consists of the outcomes of the form $\mathrm{H} \cdots \mathrm{HTT} \cdots$ (where the number of H in the beginning can range from zero up to $n - 2$ and the results

that succeed the two T are arbitrary), together with the outcomes $\mathrm{HH}\cdots\mathrm{H}$ and $\mathrm{H}\cdots\mathrm{HT}$.

25. Hint: Express $\mathbf{P}(X \geq k)$ as the sum $\sum_{i=k}^{\infty} \mathbf{P}(X = i) = \sum_{i=k}^{\infty} \mathbf{p}_X(i)$ for every $k \in \mathbb{Z}_{>0}$ and change the order of summation to show that $\sum_{k=1}^{\infty} \sum_{i=k}^{\infty} \mathbf{p}_X(i) = E(X)$.
26. Hint: Follow the method of the second proof of Example 7.3.10 or Example 7.3.11.
28. Hint: Compute the probability mass function of Z_n and show that

$$2^n E(Z_n) = \begin{cases} \displaystyle\sum_{k=(n+1)/2}^{n} 2k\binom{n}{k}, & \text{if } n \notin 2\mathbb{N} \\ \\ (n/2)\binom{n}{n/2} + \displaystyle\sum_{k=(n+2)/2}^{n} 2k\binom{n}{k}, & \text{if } n \in 2\mathbb{N}. \end{cases}$$

Compute the sums above to reach the formula

$$\frac{1}{n}E(Z_n) = \begin{cases} \dfrac{1}{2} + \dfrac{1}{2^n}\binom{n-1}{(n-1)/2}, & \text{if } n \notin 2\mathbb{N} \\ \\ \dfrac{1}{2} + \dfrac{1}{2^{n+1}}\binom{n}{n/2}, & \text{if } n \in 2\mathbb{N} \end{cases}$$

and employ Exercise 9 of Chapter 1 to compute $\lim_{n\to\infty} E(Z_n)/n$.

29. We first observe that $B_1 \supseteq B_2 \supseteq \cdots \supseteq B_n$ and set $B_{n+1} := \varnothing$. We consider the characteristic functions χ_{A_i} of the events $A_1, A_2, \ldots, A_n$ and set $X = \chi_{A_1} + \chi_{A_2} + \cdots + \chi_{A_n}$. Then, $E(X) = \mathbf{P}(A_1) + \mathbf{P}(A_2) + \cdots + \mathbf{P}(A_n)$. Moreover, by definition of the B_k, we have $X(\omega) = k \Leftrightarrow \omega \in B_k \smallsetminus B_{k+1}$ for every $\omega \in \Omega$ and every $k \in [n]$. Therefore, $\mathbf{p}_X(k) = \mathbf{P}(B_k \smallsetminus B_{k+1})$ for every $k \in [n]$, and hence

$$E(X) = \sum_{k=1}^{n} k\,\mathbf{p}_X(k) = \sum_{k=1}^{n} k\,\mathbf{P}(B_k \smallsetminus B_{k+1}) = \sum_{k=1}^{n} k(\mathbf{P}(B_k) - \mathbf{P}(B_{k+1}))$$
$$= \sum_{k=1}^{n} \mathbf{P}(B_k).$$

Part (a) follows by comparing the two expressions we have reached for $E(X)$. For part (b), we apply part (a) and the monotonicity of probability measure to find that

$$t = \mathbf{P}(B_1) + \mathbf{P}(B_2) + \cdots + \mathbf{P}(B_n) \geq \mathbf{P}(B_1) + \mathbf{P}(B_2) + \cdots + \mathbf{P}(B_k) \geq k\mathbf{P}(B_k)$$

and that

$$t = \mathbf{P}(B_1) + \mathbf{P}(B_2) + \cdots + \mathbf{P}(B_n) \le (k-1) + (n-k+1)\mathbf{P}(B_k)$$

for every $k \in [n]$.

30. We observe that for $t_1, t_2, \ldots, t_n \in \{0, 1\}$,

$$\binom{t_1 + t_2 + \cdots + t_n}{k} = \sum_{1 \le i_1 < i_2 < \cdots < i_k \le n} t_{i_1} t_{i_2} \cdots t_{i_k}$$

for every $k \in [n]$. This holds because both sides of this equation are equal to $\binom{m}{k}$, where m is the number of $i \in [n]$ such that $t_i = 1$. We set $Y = X(X-1) \cdots (X-k+1)$. Applying Proposition 7.3.4 and the previous remark, we find that

$$\begin{aligned}
\frac{1}{k!} E(Y) &= \frac{1}{k!} \sum_{t=0}^{n} t(t-1)\cdots(t-k+1)\, \mathbf{p}_X(t) \\
&= \sum_{t=0}^{n} \binom{t}{k} \sum_{t_i \in \{0,1\}: \sum t_i = t} \mathbf{P}(X_1 = t_1, X_2 = t_2, \ldots, X_n = t_n) \\
&= \sum_{t_i \in \{0,1\}} \binom{t_1 + t_2 + \cdots + t_n}{k} \mathbf{P}(X_1 = t_1, X_2 = t_2, \ldots, X_n = t_n) \\
&= \sum_{t_i \in \{0,1\}} \left(\sum_{1 \le i_1 < i_2 < \cdots < i_k \le n} t_{i_1} t_{i_2} \cdots t_{i_k} \right) \\
&\quad \times \mathbf{P}(X_1 = t_1, X_2 = t_2, \ldots, X_n = t_n) \\
&= \sum_{1 \le i_1 < i_2 < \cdots < i_k \le n} \mathbf{P}(X_{i_1} = 1, X_{i_2} = 1, \ldots, X_{i_k} = 1) \\
&= \sum_{1 \le i_1 < i_2 < \cdots < i_k \le n} \mathbf{P}(A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_k}),
\end{aligned}$$

where for the second and fifth equalities we have applied the additivity of the measure $\mathbf{P}$. For $k = 1$ we get that $E(X) = \mathbf{P}(A_1) + \mathbf{P}(A_2) + \cdots + \mathbf{P}(A_n)$. Answer to part (b): $n!/2^k(n-k)!$.

31. Hint: Follow the reasoning in the proof of Proposition 7.3.2.

33. Suppose that $X_1, X_2, \ldots, X_n : \Omega \to \mathbb{R}$ are independent, and let us denote by L the left-hand side of the proposed equation. Then, by the additivity of the

probability measure $\mathbf{P}$,

$$L = \sum_{t_i \in S_i} \mathbf{P}(X_1 = t_1, X_2 = t_2, \ldots, X_n = t_n)$$

$$= \sum_{t_i \in S_i} \mathbf{P}(X_1 = t_1)\, \mathbf{P}(X_2 = t_2) \cdots \mathbf{P}(X_n = t_n)$$

$$= \prod_{i=1}^{n} \left(\sum_{t_i \in S_i} \mathbf{P}(X_i = t_i) \right) = \prod_{i=1}^{n} \mathbf{P}(X_i \in S_i).$$

The converse follows by setting $S = \{t_i\}$ for $i \in [n]$. We have proved part (a). For part (b), we assume that $\chi_{A_1}, \chi_{A_2}, \ldots, \chi_{A_n} : \Omega \to \mathbb{R}$ are independent. Then, (7.10) can be verified for $1 \le i_1 < i_2 < \cdots < i_k \le n$ by applying part (a) to $\chi_{A_1}, \chi_{A_2}, \ldots, \chi_{A_n}$ and the sets

$$T_i = \begin{cases} \{1\}, & \text{if } i = i_j \text{ for some } j, \\ \{0, 1\}, & \text{otherwise} \end{cases}$$

for $i \in [n]$. The converse is a consequence of Proposition 7.2.1.

34. Hint: Assume, as one may, that $\mathbf{P}(X + Y = a) > 0$ and $\mathbf{P}(X + Y = b) > 0$. By the assumptions of the problem, we have $s \le X(\omega) \le t$ and $u \le Y(\omega) \le v$ for some $s < t$ and $u < v$ and for every $\omega \in \Omega$, where $p := \mathbf{P}(X = s) > 0$, $q := \mathbf{P}(Y = u) > 0$, $\mathbf{P}(X = t) > 0$, and $\mathbf{P}(Y = v) > 0$.

    Show that $a = s + u$ and $b = t + v$ and that

$$\mathbf{P}(X + Y = a)\, \mathbf{P}(X + Y = b) \le p(1 - p)q(1 - q) \le (1/4)^2$$

and deduce that $\mathbf{P}(X + Y = a) \le 1/4$ or $\mathbf{P}(X + Y = b) \le 1/4$.

35. We apply the linearity of the expected value to observe that $0 \le E((tX+Y)^2) = E(X^2)t^2 + 2E(XY)t + E(Y^2)$ for every $t \in \mathbb{R}$ and the proof of part (a) follows. Part (b) follows by applying part (a) to the random variables $X, Y : [n] \to \mathbb{R}$ which take the values $X(i) = a_i$ and $Y(i) = b_i$ for $i \in [n]$, each with probability $1/n$. The same reasoning applies to part (c), provided one can show that $E(XY)$ is defined as a real number, if $E(X^2), E(Y^2) \in \mathbb{R}$. For that, it suffices to observe that

$$\sum_{t \in \mathbb{R}} |t|\, \mathbf{p}_{XY}(t) = \sum_{t \in \mathbb{R}} |t| \sum_{uv = t} \mathbf{P}(X = u, Y = v) = \sum_{u, v \in \mathbb{R}} |uv|\, \mathbf{P}(X = u, Y = v)$$

$$\le \sum_{u, v \in \mathbb{R}} \frac{u^2 + v^2}{2}\, \mathbf{P}(X = u, Y = v) = \frac{E(X^2) + E(Y^2)}{2} < \infty.$$

36. Hint: Write $X-t=(X-\mu)+(\mu-t)$, where $\mu=E(X)$, square this equation, and apply the linearity of expectation.
37. Answer: $E(X)\sim n-1$ and $\mathrm{Var}(X)\sim 2$, as $n\to\infty$.
39. Hint: We computed in Example 7.3.11 that $E(X)=n(n-1)/(m+n-1)$. According to the formula (7.15),

$$E(X^2)=\sum_{k=0}^{n}k^2\,\mathbf{p}_X(k)=\binom{m+n-1}{n-1}^{-1}\sum_{k=0}^{n}k^2\binom{n}{k}\binom{m-1}{n-k-1}.$$

Work as in Examples 7.3.11 and 7.3.15 and deduce that

$$E(X^2)=\frac{n(n-1)(n^2-2n+m)}{(m+n-1)(m+n-2)},$$

$$\mathrm{Var}(X)=E(X^2)-(E(X))^2=\frac{m(m-1)n(n-1)}{(m+n-1)^2(m+n-2)}.$$

Alternatively, apply Exercise 38 (b) to the random variables $X_1, X_2, \ldots, X_n$ of the second proof of Example 7.3.11.

40. Hint: Observe that $X=X_1+X_2+\cdots+X_{n-1}$, where X_i is the characteristic function of the event that the result of the i and $i+1$ tosses is H. For part (a), apply Proposition 7.3.1 (b) and Exercise 38 (a), with $n-1$ in place of n, to compute $E(X)$ and $\mathrm{Var}(X)$. For part (b), apply Chebyshev's inequality for $a=(n-1)\varepsilon$.

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